

# Answers to Odd-Numbered Exercises

## CHAPTER 1

### Section 1.1

**1.** a) Yes, T b) Yes, F c) Yes, T d) Yes, F e) No f) No **3.** a) Linda is not younger than Sanjay. b) Mei does not make more money than Isabella. c) Moshe is not taller than Monica. d) Abby is not richer than Ricardo. **5.** a) Mei does not have an MP3 player. b) There is pollution in New Jersey. c)  $2 + 1 \neq 3$ . d) The summer in Maine is not hot or it is not sunny. **7.** a) Steve does not have more than 100 GB free disk space on his laptop. b) Zach does not block e-mails from Jennifer, or he does not block texts from Jennifer. c)  $7 \cdot 11 \cdot 13 \neq 999$ . d) Diane did not ride her bike 100 miles on Sunday. **9.** a) F b) T c) T d) T e) T **11.** a) Sharks have not been spotted near the shore. b) Swimming at the New Jersey shore is allowed, and sharks have been spotted near the shore. c) Swimming at the New Jersey shore is not allowed, or sharks have been spotted near the shore. d) If swimming at the New Jersey shore is allowed, then sharks have not been spotted near the shore. e) If sharks have not been spotted near the shore, then swimming at the New Jersey shore is allowed. f) If swimming at the New Jersey shore is not allowed, then sharks have not been spotted near the shore. g) Swimming at the New Jersey shore is allowed if and only if sharks have not been spotted near the shore. h) Swimming at the New Jersey shore is not allowed, and either swimming at the New Jersey shore is allowed or sharks have not been spotted near the shore. (Note that we were able to incorporate the parentheses by using the word “either” in the second half of the sentence.) **13.** a)  $p \wedge q$  b)  $p \wedge \neg q$  c)  $\neg p \wedge \neg q$  d)  $p \vee q$  e)  $p \rightarrow q$  f)  $(p \vee q) \wedge (p \rightarrow \neg q)$  g)  $q \leftrightarrow p$  **15.** a)  $\neg p$  b)  $p \wedge \neg q$  c)  $p \rightarrow q$  d)  $\neg p \rightarrow \neg q$  e)  $p \rightarrow q$  f)  $q \wedge \neg p$  g)  $q \rightarrow p$  **17.** a)  $r \wedge \neg p$  b)  $\neg p \wedge q \wedge r$  c)  $r \rightarrow (q \leftrightarrow \neg p)$  d)  $\neg q \wedge \neg p \wedge r$  e)  $(q \rightarrow (\neg r \wedge \neg p)) \wedge \neg((\neg r \wedge \neg p) \rightarrow q)$  f)  $(p \wedge r) \rightarrow \neg q$  **19.** a) False b) True c) True d) True **21.** a) Exclusive or: You get only one beverage. b) Inclusive or: Long passwords can have any combination of symbols. c) Inclusive or: A student with both courses is even more qualified. d) Either interpretation possible; a traveler might wish to pay with a mixture of the two currencies, or the store may not allow that. **23.** a) Inclusive or: It is allowable to take discrete mathematics if you have had calculus or computer science, or both. Exclusive or: It is allowable to take discrete mathematics if you have had calculus or computer science, but not if you have had both. Most likely the inclusive or is intended. b) Inclusive or: You can take the rebate, or you can get a low-interest loan, or you can get both the rebate and a low-interest loan. Exclusive or: You can take the rebate, or you can get a low-interest loan, but you cannot get both the rebate and a low-interest loan. Most likely the exclusive or is intended. c) Inclusive or: You can order two items from col-

umn A and none from column B, or three items from column B and none from column A, or five items including two from column A and three from column B. Exclusive or: You can order two items from column A or three items from column B, but not both. Almost certainly the exclusive or is intended. d) Inclusive or: More than 2 feet of snow or windchill below  $-100^\circ\text{F}$ , or both, will close school. Exclusive or: More than 2 feet of snow or windchill below  $-100^\circ\text{F}$ , but not both, will close school. Certainly the inclusive or is intended. **25.** a) If the wind blows from the northeast, then it snows. b) If it stays warm for a week, then the apple trees will bloom. c) If the Pistons win the championship, then they beat the Lakers. d) If you get to the top of Long’s Peak, then you must have walked 8 miles. e) If you are world famous, then you will get tenure as a professor. f) If you drive more than 400 miles, then you will need to buy gasoline. g) If your guarantee is good, then you must have bought your CD player less than 90 days ago. h) If the water is not too cold, then Jan will go swimming. i) If people believe in science, then we will have a future. **27.** a) You buy an ice cream cone if and only if it is hot outside. b) You win the contest if and only if you hold the only winning ticket. c) You get promoted if and only if you have connections. d) Your mind will decay if and only if you watch television. e) The train runs late if and only if it is a day I take the train. **29.** a) Converse: “I will ski tomorrow only if it snows today.” Contrapositive: “If I do not ski tomorrow, then it will not have snowed today.” Inverse: “If it does not snow today, then I will not ski tomorrow.” b) Converse: “If I come to class, then there will be a quiz.” Contrapositive: “If I do not come to class, then there will not be a quiz.” Inverse: “If there is not going to be a quiz, then I don’t come to class.” c) Converse: “A positive integer is a prime if it has no divisors other than 1 and itself.” Contrapositive: “If a positive integer has a divisor other than 1 and itself, then it is not prime.” Inverse: “If a positive integer is not prime, then it has a divisor other than 1 and itself.” **31.** a) 2 b) 16 c) 64 d) 16

**33. a)**

| $p$ | $\neg p$ | $p \wedge \neg p$ |
|-----|----------|-------------------|
| T   | F        | F                 |
| F   | T        | F                 |

**b)**

| $p$ | $\neg p$ | $p \vee \neg p$ |
|-----|----------|-----------------|
| T   | F        | T               |
| F   | T        | T               |

**c)**

| $p$ | $q$ | $\neg q$ | $p \vee \neg q$ | $(p \vee \neg q) \rightarrow q$ |
|-----|-----|----------|-----------------|---------------------------------|
| T   | T   | F        | T               | T                               |
| T   | F   | T        | T               | F                               |
| F   | T   | F        | F               | T                               |
| F   | F   | T        | T               | F                               |

d)

| $p$ | $q$ | $p \vee q$ | $p \wedge q$ | $(p \vee q) \rightarrow (p \wedge q)$ |
|-----|-----|------------|--------------|---------------------------------------|
| T   | T   | T          | T            | T                                     |
| T   | F   | T          | F            | F                                     |
| F   | T   | T          | F            | F                                     |
| F   | F   | F          | F            | T                                     |

e)

| $p$ | $q$ | $p \rightarrow q$ | $\neg q$ | $\neg p$ | $\neg q \rightarrow \neg p$ | $(p \rightarrow q) \leftrightarrow (\neg q \rightarrow \neg p)$ |
|-----|-----|-------------------|----------|----------|-----------------------------|-----------------------------------------------------------------|
| T   | T   | T                 | F        | F        | T                           | T                                                               |
| T   | F   | F                 | T        | F        | F                           | T                                                               |
| F   | T   | T                 | F        | T        | T                           | T                                                               |
| F   | F   | T                 | T        | T        | T                           | T                                                               |

f)

| $p$ | $q$ | $p \rightarrow q$ | $q \rightarrow p$ | $(p \rightarrow q) \rightarrow (q \rightarrow p)$ |
|-----|-----|-------------------|-------------------|---------------------------------------------------|
| T   | T   | T                 | T                 | T                                                 |
| T   | F   | F                 | T                 | T                                                 |
| F   | T   | T                 | F                 | F                                                 |
| F   | F   | T                 | T                 | T                                                 |

35. For parts (a), (b), (c), (d), and (f) we have this table.

| $p$ | $q$ | $(p \vee q) \rightarrow (p \oplus q)$ | $(p \oplus q) \rightarrow (p \wedge q)$ | $(p \vee q) \oplus (p \wedge q)$ | $(p \leftrightarrow q) \oplus (\neg p \leftrightarrow q)$ | $(p \oplus q) \rightarrow (p \oplus \neg q)$ |
|-----|-----|---------------------------------------|-----------------------------------------|----------------------------------|-----------------------------------------------------------|----------------------------------------------|
| T   | T   | F                                     | T                                       | F                                | T                                                         | T                                            |
| T   | F   | T                                     | F                                       | T                                | T                                                         | F                                            |
| F   | T   | T                                     | F                                       | T                                | T                                                         | F                                            |
| F   | F   | T                                     | T                                       | F                                | T                                                         | T                                            |

For part (e) we have this table.

| $p$ | $q$ | $r$ | $\neg p$ | $\neg r$ | $p \leftrightarrow q$ | $\neg p \leftrightarrow \neg r$ | $(p \leftrightarrow q) \oplus (\neg p \leftrightarrow \neg r)$ |
|-----|-----|-----|----------|----------|-----------------------|---------------------------------|----------------------------------------------------------------|
| T   | T   | T   | F        | F        | T                     | T                               | F                                                              |
| T   | T   | F   | F        | T        | T                     | F                               | T                                                              |
| T   | F   | T   | F        | F        | F                     | T                               | T                                                              |
| T   | F   | F   | F        | T        | F                     | F                               | F                                                              |
| F   | T   | T   | T        | F        | F                     | F                               | F                                                              |
| F   | T   | F   | T        | T        | F                     | T                               | T                                                              |
| F   | F   | T   | T        | F        | T                     | F                               | T                                                              |
| F   | F   | F   | T        | T        | T                     | T                               | F                                                              |

37.

| $p$ | $q$ | $p \rightarrow \neg q$ | $\neg p \leftrightarrow q$ | $(p \rightarrow q) \vee (\neg p \rightarrow q)$ | $(p \rightarrow q) \wedge (\neg p \rightarrow q)$ | $(p \leftrightarrow q) \vee (\neg p \leftrightarrow q)$ | $(\neg p \leftrightarrow \neg q) \leftrightarrow (p \leftrightarrow q)$ |
|-----|-----|------------------------|----------------------------|-------------------------------------------------|---------------------------------------------------|---------------------------------------------------------|-------------------------------------------------------------------------|
| T   | T   | F                      | F                          | T                                               | T                                                 | T                                                       | T                                                                       |
| T   | F   | T                      | T                          | T                                               | F                                                 | T                                                       | T                                                                       |
| F   | T   | T                      | T                          | T                                               | T                                                 | T                                                       | T                                                                       |
| F   | F   | T                      | F                          | T                                               | F                                                 | T                                                       | T                                                                       |

39.

| $p$ | $q$ | $r$ | $p \rightarrow (\neg q \vee r)$ | $\neg p \rightarrow (q \rightarrow r)$ | $(p \rightarrow q) \vee (\neg p \rightarrow r)$ | $(p \rightarrow q) \wedge (\neg p \rightarrow r)$ | $(p \leftrightarrow q) \vee (\neg q \leftrightarrow r)$ | $(\neg p \leftrightarrow \neg q) \leftrightarrow (q \leftrightarrow r)$ |
|-----|-----|-----|---------------------------------|----------------------------------------|-------------------------------------------------|---------------------------------------------------|---------------------------------------------------------|-------------------------------------------------------------------------|
| T   | T   | T   | T                               | T                                      | T                                               | T                                                 | T                                                       | T                                                                       |
| T   | T   | F   | F                               | T                                      | T                                               | T                                                 | T                                                       | F                                                                       |
| T   | F   | T   | T                               | T                                      | T                                               | F                                                 | T                                                       | T                                                                       |
| T   | F   | F   | T                               | T                                      | T                                               | F                                                 | F                                                       | F                                                                       |
| F   | T   | T   | T                               | T                                      | T                                               | T                                                 | F                                                       | F                                                                       |
| F   | T   | F   | T                               | F                                      | T                                               | F                                                 | T                                                       | T                                                                       |
| F   | F   | T   | T                               | T                                      | T                                               | T                                                 | T                                                       | F                                                                       |
| F   | F   | F   | T                               | T                                      | T                                               | F                                                 | T                                                       | T                                                                       |



41.

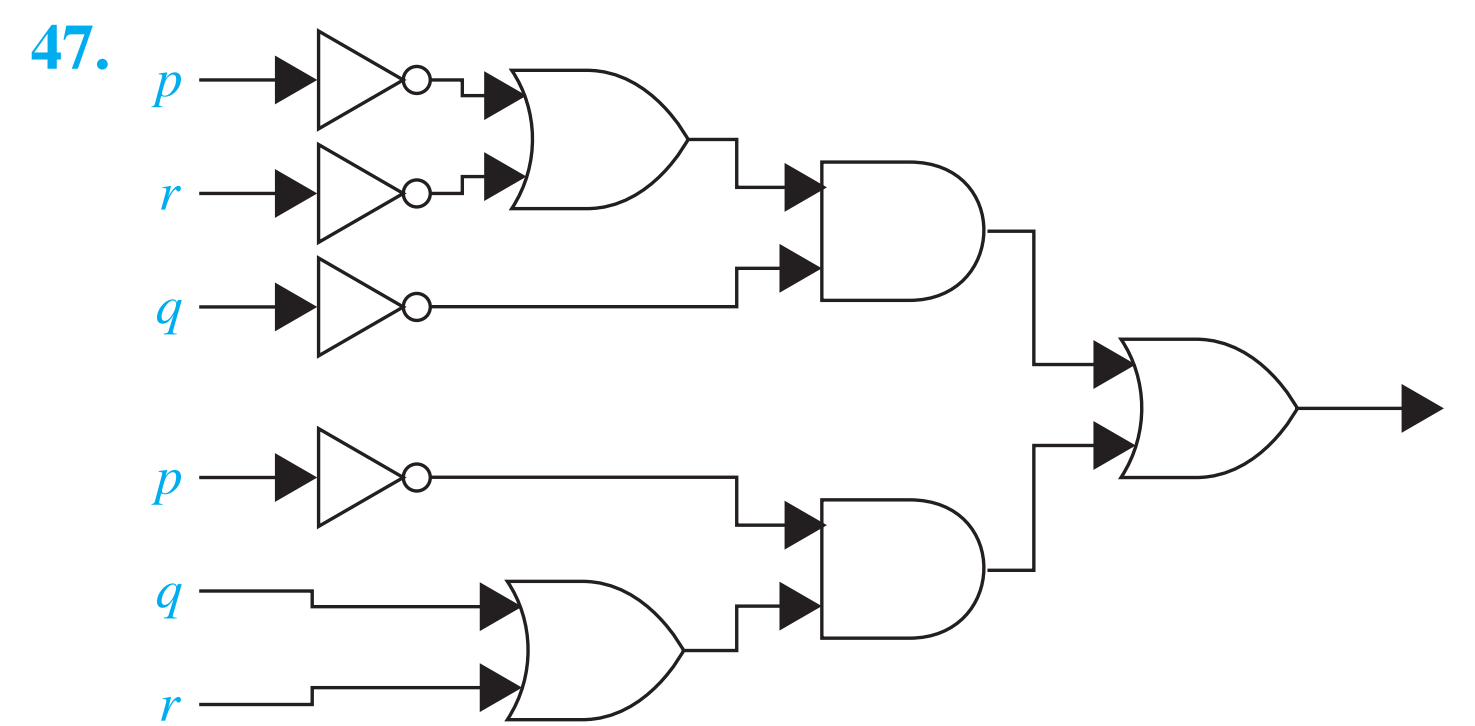
| $p$ | $q$ | $r$ | $s$ | $p \leftrightarrow q$ | $r \leftrightarrow s$ | $(p \leftrightarrow q) \leftrightarrow (r \leftrightarrow s)$ |
|-----|-----|-----|-----|-----------------------|-----------------------|---------------------------------------------------------------|
| T   | T   | T   | T   | T                     | T                     | T                                                             |
| T   | T   | T   | F   | T                     | F                     | F                                                             |
| T   | T   | F   | T   | T                     | F                     | F                                                             |
| T   | T   | F   | F   | T                     | T                     | T                                                             |
| T   | F   | T   | T   | F                     | T                     | F                                                             |
| T   | F   | T   | F   | F                     | F                     | T                                                             |
| T   | F   | F   | T   | F                     | F                     | T                                                             |
| T   | F   | F   | F   | F                     | T                     | F                                                             |
| F   | T   | T   | T   | F                     | T                     | F                                                             |
| F   | T   | T   | F   | F                     | F                     | T                                                             |
| F   | T   | F   | T   | F                     | F                     | T                                                             |
| F   | T   | F   | F   | F                     | T                     | F                                                             |
| F   | F   | T   | T   | T                     | T                     | T                                                             |
| F   | F   | T   | F   | T                     | F                     | F                                                             |
| F   | F   | F   | T   | T                     | F                     | F                                                             |
| F   | F   | F   | F   | T                     | T                     | T                                                             |

43. The first clause is true if and only if at least one of  $p$ ,  $q$ , and  $r$  is true. The second clause is true if and only if at least one of the three variables is false. Therefore, the entire statement is true if and only if there is at least one T and one F among the truth values of the variables, in other words, that they don't all have the same truth value. 45.  $(\bigwedge_{i=1}^{n-1} \bigwedge_{j=i+1}^n (\neg p_i \vee \neg p_j)) \wedge (\bigvee_{i=1}^n p_i)$  47. a) Bitwise OR is 111 1111; bitwise AND is 000 0000; bitwise XOR is 111 1111. b) Bitwise OR is 1111 1010; bitwise AND is 1010 0000; bitwise XOR is 0101 1010. c) Bitwise OR is 10 0111 1001; bitwise AND is 00 0100 0000; bitwise XOR is 10 0011 1001. d) Bitwise OR is 11 1111 1111; bitwise AND is 00 0000 0000; bitwise XOR is 11 1111 1111. 49. 0.2, 0.6 51. 0.8, 0.6 53. a) The 99th statement is true and the rest are false. b) Statements 1 through 50 are all true and statements 51 through 100 are all false. c) This cannot happen; it is a paradox, showing that these cannot be statements.

## Section 1.2

1.  $e \rightarrow a$  3.  $g \rightarrow (r \wedge (\neg m) \wedge (\neg b))$  5.  $e \rightarrow (a \wedge (b \vee p) \wedge r)$  7. a)  $q \rightarrow p$  b)  $q \wedge \neg p$  c)  $q \rightarrow p$  d)  $\neg q \rightarrow \neg p$  9. Not consistent 11. Consistent 13. NEW AND JERSEY AND BEACHES, (JERSEY AND BEACHES) NOT NEW 15. "ETHIOPIAN RESTAURANTS" AND ("NEW YORK" OR "NEW JERSEY") 17. a) Queen cannot say this. b) Queen can say this, but one cannot determine location of treasure. c) Queen can say this; treasure is in Trunk 1. d) Queen cannot say this. 19. "If I were to ask you whether the right branch leads to the ruins, would you answer yes?" 21. If the first professor did not want coffee, then he would know that the answer to the hostess's question was "no." Therefore the hostess and the remaining professors know that the first professor did want coffee. Similarly, the second professor must want coffee. When the third professor said "no," the hostess knows that the third professor does

not want coffee. 23. A is a knight and B is a knave. 25. A is a knight and B is a knight. 27. A is a knave and B is a knight. 29. A is the knight, B is the spy, C is the knave. 31. A is the knight, B is the spy, C is the knave. 33. Any of the three can be the knight, any can be the spy, any can be the knave. 35. No solutions 37. In order of decreasing salary: Fred, Maggie, Janice 39. The detective can determine that the butler and cook are lying but cannot determine whether the gardener is telling the truth or whether the handyman is telling the truth. 41. The Japanese man owns the zebra, and the Norwegian drinks water. 43. One honest, 49 corrupt 45. a)  $\neg(p \wedge (q \vee \neg r))$  b)  $((\neg p) \wedge (\neg q)) \vee (p \wedge r)$



## Section 1.3

1. The equivalences follow by showing that the appropriate pairs of columns of this table agree.

| $p$ | $p \wedge T$ | $p \vee F$ | $p \wedge F$ | $p \vee T$ | $p \vee p$ | $p \wedge p$ |
|-----|--------------|------------|--------------|------------|------------|--------------|
| T   | T            | T          | F            | T          | T          | T            |
| F   | F            | F          | F            | T          | F          | F            |

3. a)

| $p$ | $q$ | $p \vee q$ | $q \vee p$ |
|-----|-----|------------|------------|
| T   | T   | T          | T          |
| T   | F   | T          | T          |
| F   | T   | T          | T          |
| F   | F   | F          | F          |

b)

| $p$ | $q$ | $p \wedge q$ | $q \wedge p$ |
|-----|-----|--------------|--------------|
| T   | T   | T            | T            |
| T   | F   | F            | F            |
| F   | T   | F            | F            |
| F   | F   | F            | F            |

5.

| $p$ | $q$ | $r$ | $q \vee r$ | $p \wedge (q \vee r)$ | $p \wedge q$ | $p \wedge r$ | $(p \wedge q) \vee (p \wedge r)$ |
|-----|-----|-----|------------|-----------------------|--------------|--------------|----------------------------------|
| T   | T   | T   | T          | T                     | T            | T            | T                                |
| T   | T   | F   | T          | T                     | T            | F            | T                                |
| T   | F   | T   | T          | T                     | F            | T            | T                                |
| T   | F   | F   | F          | F                     | F            | F            | F                                |
| F   | T   | T   | T          | F                     | F            | F            | F                                |
| F   | T   | F   | T          | F                     | F            | F            | F                                |
| F   | F   | T   | T          | F                     | F            | F            | F                                |
| F   | F   | F   | F          | F                     | F            | F            | F                                |

7. a) Jan is not rich, or Jan is not happy. b) Carlos will not bicycle tomorrow, and Carlos will not run tomorrow. c) Mei does not walk to class, and Mei does not take the bus to class. d) Ibrahim is not smart, or Ibrahim is not hard working. 9. a)  $\neg p \vee \neg q$  b)  $(p \wedge \neg q) \vee r$  c)  $\neg p \vee \neg q$



11. a)

| $p$ | $q$ | $p \wedge q$ | $(p \wedge q) \rightarrow p$ |
|-----|-----|--------------|------------------------------|
| T   | T   | T            | T                            |
| T   | F   | F            | T                            |
| F   | T   | F            | T                            |
| F   | F   | F            | T                            |

b)

| $p$ | $q$ | $p \vee q$ | $p \rightarrow (p \vee q)$ |
|-----|-----|------------|----------------------------|
| T   | T   | T          | T                          |
| T   | F   | T          | T                          |
| F   | T   | T          | T                          |
| F   | F   | F          | T                          |

c)

| $p$ | $q$ | $\neg p$ | $p \rightarrow q$ | $\neg p \rightarrow (p \rightarrow q)$ |
|-----|-----|----------|-------------------|----------------------------------------|
| T   | T   | F        | T                 | T                                      |
| T   | F   | F        | F                 | T                                      |
| F   | T   | T        | T                 | T                                      |
| F   | F   | T        | T                 | T                                      |

d)

| $p$ | $q$ | $p \wedge q$ | $p \rightarrow q$ | $(p \wedge q) \rightarrow (p \rightarrow q)$ |
|-----|-----|--------------|-------------------|----------------------------------------------|
| T   | T   | T            | T                 | T                                            |
| T   | F   | F            | F                 | T                                            |
| F   | T   | F            | T                 | T                                            |
| F   | F   | F            | T                 | T                                            |

e)

| $p$ | $q$ | $p \rightarrow q$ | $\neg(p \rightarrow q)$ | $\neg(p \rightarrow q) \rightarrow p$ |
|-----|-----|-------------------|-------------------------|---------------------------------------|
| T   | T   | T                 | F                       | T                                     |
| T   | F   | F                 | T                       | T                                     |
| F   | T   | T                 | F                       | T                                     |
| F   | F   | T                 | F                       | T                                     |

f)

| $p$ | $q$ | $p \rightarrow q$ | $\neg(p \rightarrow q)$ | $\neg q$ | $\neg(p \rightarrow q) \rightarrow \neg q$ |
|-----|-----|-------------------|-------------------------|----------|--------------------------------------------|
| T   | T   | T                 | F                       | F        | T                                          |
| T   | F   | F                 | T                       | T        | T                                          |
| F   | T   | T                 | F                       | F        | T                                          |
| F   | F   | T                 | F                       | T        | T                                          |

13. a) If this were not a tautology, then  $p \wedge q$  would be true but  $p$  would be false. This cannot happen, because the truth of  $p \wedge q$  implies the truth of  $p$ . b) If this were not a tautology, then  $p$  would be true but  $p \vee q$  would be false. This cannot happen, because the truth of  $p$  implies the truth of  $p \vee q$ . c) If this were not a tautology, then  $p$  would be false and  $p \rightarrow q$  would be false. This cannot happen, because  $p \rightarrow q$  is true when  $p$  is false. d) If this were not a tautology, then  $p \wedge q$  would be true and  $p \rightarrow q$  would be false. This cannot happen, because  $p \rightarrow q$  is true when both  $p$  and  $q$  are true. e) If this were not a tautology, then  $p \rightarrow q$  would be false and  $p$  would be false. This cannot happen, because  $p \rightarrow q$  is true when  $p$  is false. f) If this were not a tautology, then  $p \rightarrow q$  would be false and  $q$  would be true. This cannot happen, because  $p \rightarrow q$  is true when  $q$  is true. 15. a)  $(p \wedge q) \rightarrow p \equiv \neg(p \wedge q) \vee p \equiv \neg p \vee \neg q \vee p \equiv (p \vee \neg p) \vee \neg q \equiv \mathbf{T} \vee \neg q \equiv \mathbf{T}$  b)  $p \rightarrow (p \vee q) \equiv \neg p \vee (p \vee q) \equiv (\neg p \vee p) \vee q \equiv \mathbf{T} \vee q \equiv \mathbf{T}$  c)  $\neg p \rightarrow (p \rightarrow q) \equiv p \vee (p \rightarrow q) \equiv p \vee (\neg p \vee q) \equiv (p \vee \neg p) \vee q \equiv$

$\mathbf{T} \vee q \equiv \mathbf{T}$  d)  $(p \wedge q) \rightarrow (p \rightarrow q) \equiv \neg(p \wedge q) \vee (\neg p \vee q) \equiv \neg p \vee \neg q \vee \neg p \vee q \equiv (\neg p \vee \neg p) \vee (\neg q \vee q) \equiv \neg p \vee \mathbf{T} \equiv \mathbf{T}$  e)  $\neg(p \rightarrow q) \rightarrow p \equiv (p \rightarrow q) \vee p \equiv \neg p \vee q \vee p \equiv (\neg p \vee p) \vee q \equiv \mathbf{T} \vee q \equiv \mathbf{T}$  f)  $\neg(p \rightarrow q) \rightarrow \neg q \equiv (p \rightarrow q) \vee \neg q \equiv \neg p \vee q \vee \neg q \equiv \neg p \vee \mathbf{T} \equiv \mathbf{T}$  17. That the fourth column of the truth table shown is identical to the first column proves part (a), and that the sixth column is identical to the first column proves part (b).

| $p$ | $q$ | $p \wedge q$ | $p \vee (p \wedge q)$ | $p \vee q$ | $p \wedge (p \vee q)$ |
|-----|-----|--------------|-----------------------|------------|-----------------------|
| T   | T   | T            | T                     | T          | T                     |
| T   | F   | F            | T                     | T          | T                     |
| F   | T   | F            | F                     | T          | F                     |
| F   | F   | F            | F                     | F          | F                     |

19. It is a tautology. 21. Each of these is true precisely when  $p$  and  $q$  have opposite truth values. 23. The proposition  $\neg p \leftrightarrow q$  is true when  $\neg p$  and  $q$  have the same truth values, which means that  $p$  and  $q$  have different truth values. Similarly,  $p \leftrightarrow \neg q$  is true in exactly the same cases. Therefore, these two expressions are logically equivalent. 25. The proposition  $\neg(p \leftrightarrow q)$  is true when  $p \leftrightarrow q$  is false, which means that  $p$  and  $q$  have different truth values. Because this is precisely when  $\neg p \leftrightarrow q$  is true, the two expressions are logically equivalent. 27. For  $(p \rightarrow r) \wedge (q \rightarrow r)$  to be false, one of the two conditional statements must be false, which happens exactly when  $r$  is false and at least one of  $p$  and  $q$  is true. But these are precisely the cases in which  $p \vee q$  is true and  $r$  is false, which is precisely when  $(p \vee q) \rightarrow r$  is false. Because the two propositions are false in exactly the same situations, they are logically equivalent. 29. For  $(p \rightarrow r) \vee (q \rightarrow r)$  to be false, both of the two conditional statements must be false, which happens exactly when  $r$  is false and both  $p$  and  $q$  are true. But this is precisely the case in which  $p \wedge q$  is true and  $r$  is false, which is precisely when  $(p \wedge q) \rightarrow r$  is false. Because the two propositions are false in exactly the same situations, they are logically equivalent. 31. This fact was observed in Section 1 when the biconditional was first defined. Each of these is true precisely when  $p$  and  $q$  have the same truth values. 33. The last column is all Ts.

| $p$ | $q$ | $r$ | $p \rightarrow q$ | $q \rightarrow r$ | $(p \rightarrow q) \wedge (q \rightarrow r)$ | $p \rightarrow r$ | $(p \rightarrow q) \wedge (q \rightarrow r) \rightarrow (p \rightarrow r)$ |
|-----|-----|-----|-------------------|-------------------|----------------------------------------------|-------------------|----------------------------------------------------------------------------|
| T   | T   | T   | T                 | T                 | T                                            | T                 | T                                                                          |
| T   | T   | F   | T                 | F                 | F                                            | F                 | T                                                                          |
| T   | F   | T   | F                 | T                 | F                                            | T                 | T                                                                          |
| T   | F   | F   | F                 | T                 | F                                            | F                 | T                                                                          |
| F   | T   | T   | T                 | T                 | T                                            | T                 | T                                                                          |
| F   | T   | F   | T                 | F                 | F                                            | T                 | T                                                                          |
| F   | F   | T   | T                 | T                 | T                                            | T                 | T                                                                          |
| F   | F   | F   | T                 | T                 | T                                            | T                 | T                                                                          |

35. These are not logically equivalent because when  $p$ ,  $q$ , and  $r$  are all false,  $(p \rightarrow q) \rightarrow r$  is false, but  $p \rightarrow (q \rightarrow r)$  is true. 37. Many answers are possible. If we let  $r$  be true and



$p$ ,  $q$ , and  $s$  be false, then  $(p \rightarrow q) \rightarrow (r \rightarrow s)$  will be false, but  $(p \rightarrow r) \rightarrow (q \rightarrow s)$  will be true. **39. a)**  $p \vee \neg q \vee \neg r$  **b)**  $(p \vee q \vee r) \wedge s$  **c)**  $(p \wedge \mathbf{T}) \vee (q \wedge \mathbf{F})$  **41.** If we take duals twice, every  $\vee$  changes to an  $\wedge$  and then back to an  $\vee$ , every  $\wedge$  changes to an  $\vee$  and then back to an  $\wedge$ , every  $\mathbf{T}$  changes to an  $\mathbf{F}$  and then back to a  $\mathbf{T}$ , every  $\mathbf{F}$  changes to a  $\mathbf{T}$  and then back to an  $\mathbf{F}$ . Hence,  $(s^*)^* = s$ . **43.** Let  $p$  and  $q$  be equivalent compound propositions involving only the operators  $\wedge$ ,  $\vee$ , and  $\neg$ , and  $\mathbf{T}$  and  $\mathbf{F}$ . Note that  $\neg p$  and  $\neg q$  are also equivalent. Use De Morgan's laws as many times as necessary to push negations in as far as possible within these compound propositions, changing  $\vee$ s to  $\wedge$ s, and vice versa, and changing  $\mathbf{T}$ s to  $\mathbf{F}$ s, and vice versa. This shows that  $\neg p$  and  $\neg q$  are the same as  $p^*$  and  $q^*$  except that each atomic proposition  $p_i$  within them is replaced by its negation. From this we can conclude that  $p^*$  and  $q^*$  are equivalent because  $\neg p$  and  $\neg q$  are. **45.**  $(p \wedge q \wedge \neg r) \vee (p \wedge \neg q \wedge r) \vee (\neg p \wedge q \wedge r)$  **47.** Given a compound proposition  $p$ , form its truth table and then write down a proposition  $q$  in disjunctive normal form that is logically equivalent to  $p$ . Because  $q$  involves only  $\neg$ ,  $\wedge$ , and  $\vee$ , this shows that these three operators form a functionally complete set. **49.** By Exercise 47, given a compound proposition  $p$ , we can write down a proposition  $q$  that is logically equivalent to  $p$  and involves only  $\neg$ ,  $\wedge$ , and  $\vee$ . By De Morgan's law we can eliminate all the  $\wedge$ s by replacing each occurrence of  $p_1 \wedge p_2 \wedge \cdots \wedge p_n$  with  $\neg(\neg p_1 \vee \neg p_2 \vee \cdots \vee \neg p_n)$ . **51.**  $\neg(p \wedge q)$  is true when either  $p$  or  $q$ , or both, are false, and is false when both  $p$  and  $q$  are true. Because this was the definition of  $p \mid q$ , the two compound propositions are logically equivalent. **53.**  $\neg(p \vee q)$  is true when both  $p$  and  $q$  are false, and is false otherwise. Because this was the definition of  $p \downarrow q$ , the two are logically equivalent. **55.**  $((p \downarrow p) \downarrow q) \downarrow ((p \downarrow p) \downarrow q)$  **57.** This follows immediately from the truth table or definition of  $p \mid q$ . **59.** 16 **61.** If the database is open, then either the system is in its initial state or the monitor is put in a closed state. **63.** All nine **65. a)** Satisfiable **b)** Not satisfiable **c)** Not satisfiable **67. a)**  $\left(\bigwedge_{i=1}^2 \bigvee_{j=1}^2 p(i, j)\right) \wedge \left(\bigwedge_{i=1}^2 \bigwedge_{j=1}^1 \bigwedge_{k=j+1}^2 (\neg p(i, j) \vee \neg p(i, k))\right) \wedge \left(\bigwedge_{j=1}^2 \bigwedge_{i=1}^1 \bigwedge_{k=i+1}^2 (\neg p(i, j) \vee \neg p(k, j))\right) \wedge \left(\bigwedge_{i=2}^2 \bigwedge_{j=1}^1 \bigwedge_{k=1}^{\min(i-1, 2-j)} (\neg p(i, j) \vee \neg p(i-k, k+j))\right) \wedge \left(\bigwedge_{i=1}^1 \bigwedge_{j=1}^1 \bigwedge_{k=1}^{\min(2-i, 2-j)} (\neg p(i, j) \vee \neg p(i+k, j+k))\right)$ ; No solutions possible. **b)**  $\left(\bigwedge_{i=1}^3 \bigvee_{j=1}^3 p(i, j)\right) \wedge \left(\bigwedge_{i=1}^3 \bigwedge_{j=1}^2 \bigwedge_{k=j+1}^3 (\neg p(i, j) \vee \neg p(i, k))\right) \wedge \left(\bigwedge_{j=1}^3 \bigwedge_{i=1}^2 \bigwedge_{k=i+1}^3 (\neg p(i, j) \vee \neg p(k, j))\right) \wedge \left(\bigwedge_{i=2}^3 \bigwedge_{j=1}^2 \bigwedge_{k=1}^{\min(i-1, 3-j)} (\neg p(i, j) \vee \neg p(i-k, k+j))\right) \wedge \left(\bigwedge_{i=1}^2 \bigwedge_{j=1}^2 \bigwedge_{k=1}^{\min(3-i, 3-j)} (\neg p(i, j) \vee \neg p(i+k, j+k))\right)$ ; No solutions possible. **c)**  $\left(\bigwedge_{i=1}^4 \bigvee_{j=1}^4 p(i, j)\right) \wedge \left(\bigwedge_{i=1}^4 \bigwedge_{j=1}^3 \bigwedge_{k=j+1}^4 (\neg p(i, j) \vee \neg p(i, k))\right) \wedge \left(\bigwedge_{j=1}^4 \bigwedge_{i=1}^3 \bigwedge_{k=i+1}^4 (\neg p(i, j) \vee \neg p(k, j))\right) \wedge \left(\bigwedge_{i=2}^4 \bigwedge_{j=1}^3 \bigwedge_{k=1}^{\min(i-1, 4-j)} (\neg p(i, j) \vee \neg p(i-k, k+j))\right) \wedge \left(\bigwedge_{i=1}^3 \bigwedge_{j=1}^3 \bigwedge_{k=1}^{\min(4-i, 4-j)} (\neg p(i, j) \vee \neg p(i+k, k+j))\right)$

$\neg p(i+k, j+k))$ ; (1, 2), (2, 4), (3, 1), (4, 3) or (1, 3), (2, 1), (3, 4), (4, 2) **69.** Use the same propositions as were given in the text for a  $9 \times 9$  Sudoku puzzle, with the variables indexed from 1 to 4, instead of from 1 to 9, and with a similar change for the propositions for the  $2 \times 2$  blocks:  $\bigwedge_{r=0}^1 \bigwedge_{s=0}^1 \bigwedge_{n=1}^4 \bigvee_{i=1}^2 \bigvee_{j=1}^2 p(2r+i, 2s+j, n)$  **71.**  $\bigvee_{i=1}^9 p(i, j, n)$  asserts that column  $j$  contains the number  $n$ , so  $\bigwedge_{n=1}^9 \bigvee_{i=1}^9 p(i, j, n)$  asserts that column  $j$  contains all 9 numbers; therefore  $\bigwedge_{j=1}^9 \bigwedge_{n=1}^9 \bigvee_{i=1}^9 p(i, j, n)$  asserts that every column contains every number.

## Section 1.4

**1. a)** T **b)** T **c)** F **3. a)** T **b)** F **c)** F **d)** F **5. a)** There is a student who spends more than 5 hours every weekday in class. **b)** Every student spends more than 5 hours every weekday in class. **c)** There is a student who does not spend more than 5 hours every weekday in class. **d)** No student spends more than 5 hours every weekday in class. **7. a)** Every comedian is funny. **b)** Every person is a funny comedian. **c)** There exists a person such that if she or he is a comedian, then she or he is funny. **d)** Some comedians are funny. **9. a)**  $\exists x(P(x) \wedge Q(x))$  **b)**  $\exists x(P(x) \wedge \neg Q(x))$  **c)**  $\forall x(P(x) \vee Q(x))$  **d)**  $\forall x\neg(P(x) \vee Q(x))$  **11. a)** T **b)** T **c)** F **d)** F **e)** T **f)** F **13. a)** T **b)** T **c)** T **d)** T **15. a)** T **b)** F **c)** T **d)** F **17. a)**  $P(0) \vee P(1) \vee P(2) \vee P(3) \vee P(4)$  **b)**  $P(0) \wedge P(1) \wedge P(2) \wedge P(3) \wedge P(4)$  **c)**  $\neg P(0) \vee \neg P(1) \vee \neg P(2) \vee \neg P(3) \vee \neg P(4)$  **d)**  $\neg P(0) \wedge \neg P(1) \wedge \neg P(2) \wedge \neg P(3) \wedge \neg P(4)$  **e)**  $\neg(P(0) \vee P(1) \vee P(2) \vee P(3) \vee P(4))$  **f)**  $\neg(P(0) \wedge P(1) \wedge P(2) \wedge P(3) \wedge P(4))$  **19. a)**  $P(1) \vee P(2) \vee P(3) \vee P(4) \vee P(5)$  **b)**  $P(1) \wedge P(2) \wedge P(3) \wedge P(4) \wedge P(5)$  **c)**  $\neg(P(1) \vee P(2) \vee P(3) \vee P(4) \vee P(5))$  **d)**  $\neg(P(1) \wedge P(2) \wedge P(3) \wedge P(4) \wedge P(5))$  **e)**  $(P(1) \wedge P(2) \wedge P(4) \wedge P(5)) \vee (\neg P(1) \vee \neg P(2) \vee \neg P(3) \vee \neg P(4) \vee \neg P(5))$  **21.** Many answers are possible. **a)** All students in your discrete mathematics class; all students in the world **b)** All United States senators; all college football players **c)** George W. Bush and Jeb Bush; all politicians in the United States **d)** Bill Clinton and George W. Bush; all politicians in the United States **23.** Let  $C(x)$  be the propositional function "x is in your class." **a)**  $\exists xH(x)$  and  $\exists x(C(x) \wedge H(x))$ , where  $H(x)$  is "x can speak Hindi" **b)**  $\forall xF(x)$  and  $\forall x(C(x) \rightarrow F(x))$ , where  $F(x)$  is "x is friendly" **c)**  $\exists x\neg B(x)$  and  $\exists x(C(x) \wedge \neg B(x))$ , where  $B(x)$  is "x was born in California" **d)**  $\exists xM(x)$  and  $\exists x(C(x) \wedge M(x))$ , where  $M(x)$  is "x has been in a movie" **e)**  $\forall x\neg L(x)$  and  $\forall x(C(x) \rightarrow \neg L(x))$ , where  $L(x)$  is "x has taken a course in logic programming" **25.** Let  $P(x)$  be "x is perfect"; let  $F(x)$  be "x is your friend"; and let the domain be all people. **a)**  $\forall x \neg P(x)$  **b)**  $\neg \forall x P(x)$  **c)**  $\forall x(F(x) \rightarrow P(x))$  **d)**  $\exists x(F(x) \wedge P(x))$  **e)**  $\forall x(F(x) \wedge P(x))$  or  $(\forall x F(x)) \wedge (\forall x P(x))$  **f)**  $(\neg \forall x F(x)) \vee (\exists x \neg P(x))$  **27.** Let  $Y(x)$  be the propositional function that  $x$  is in your school or class, as appropriate. **a)** If we let  $V(x)$  be "x has lived in Vietnam," then we have  $\exists xV(x)$  if the domain is just your schoolmates, or  $\exists x(Y(x) \wedge V(x))$  if the domain is all people. If we let  $D(x, y)$  mean that person  $x$  has lived in country  $y$ , then we can rewrite this last one



as  $\exists x(Y(x) \wedge D(x, \text{Vietnam}))$ . **b)** If we let  $H(x)$  be “ $x$  can speak Hindi,” then we have  $\exists x \neg H(x)$  if the domain is just your schoolmates, or  $\exists x(Y(x) \wedge \neg H(x))$  if the domain is all people. If we let  $S(x, y)$  mean that person  $x$  can speak language  $y$ , then we can rewrite this last one as  $\exists x(Y(x) \wedge \neg S(x, \text{Hindi}))$ . **c)** If we let  $J(x)$ ,  $P(x)$ , and  $C(x)$  be the propositional functions asserting  $x$ ’s knowledge of Java, Prolog, and C++, respectively, then we have  $\exists x(J(x) \wedge P(x) \wedge C(x))$  if the domain is just your schoolmates, or  $\exists x(Y(x) \wedge J(x) \wedge P(x) \wedge C(x))$  if the domain is all people. If we let  $K(x, y)$  mean that person  $x$  knows programming language  $y$ , then we can rewrite this last one as  $\exists x(Y(x) \wedge K(x, \text{Java}) \wedge K(x, \text{Prolog}) \wedge K(x, \text{C++}))$ . **d)** If we let  $T(x)$  be “ $x$  enjoys Thai food,” then we have  $\forall x T(x)$  if the domain is just your classmates, or  $\forall x(Y(x) \rightarrow T(x))$  if the domain is all people. If we let  $E(x, y)$  mean that person  $x$  enjoys food of type  $y$ , then we can rewrite this last one as  $\forall x(Y(x) \rightarrow E(x, \text{Thai}))$ . **e)** If we let  $H(x)$  be “ $x$  plays hockey,” then we have  $\exists x \neg H(x)$  if the domain is just your classmates, or  $\exists x(Y(x) \wedge \neg H(x))$  if the domain is all people. If we let  $P(x, y)$  mean that person  $x$  plays game  $y$ , then we can rewrite this last one as  $\exists x(Y(x) \wedge \neg P(x, \text{hockey}))$ . **29.** Let  $T(x)$  mean that  $x$  is a tautology and  $C(x)$  mean that  $x$  is a contradiction. **a)**  $\exists x T(x)$  **b)**  $\forall x(C(x) \rightarrow T(\neg x))$  **c)**  $\exists x \exists y(\neg T(x) \wedge \neg C(x) \wedge \neg T(y) \wedge \neg C(y) \wedge T(x \vee y))$  **d)**  $\forall x \forall y((T(x) \wedge T(y)) \rightarrow T(x \wedge y))$  **31. a)**  $Q(0, 0, 0) \wedge Q(0, 1, 0)$  **b)**  $Q(0, 1, 1) \vee Q(1, 1, 1) \vee Q(2, 1, 1)$  **c)**  $\neg Q(0, 0, 0) \vee \neg Q(0, 0, 1)$  **d)**  $\neg Q(0, 0, 1) \vee \neg Q(1, 0, 1) \vee \neg Q(2, 0, 1)$  **33. a)** Let  $T(x)$  be the predicate that  $x$  can learn new tricks, and let the domain be old dogs. Original is  $\exists x T(x)$ . Negation is  $\forall x \neg T(x)$ : “No old dogs can learn new tricks.” **b)** Let  $C(x)$  be the predicate that  $x$  knows calculus, and let the domain be rabbits. Original is  $\neg \exists x C(x)$ . Negation is  $\exists x C(x)$ : “There is a rabbit that knows calculus.” **c)** Let  $F(x)$  be the predicate that  $x$  can fly, and let the domain be birds. Original is  $\forall x F(x)$ . Negation is  $\exists x \neg F(x)$ : “There is a bird who cannot fly.” **d)** Let  $T(x)$  be the predicate that  $x$  can talk, and let the domain be dogs. Original is  $\neg \exists x T(x)$ . Negation is  $\exists x T(x)$ : “There is a dog that talks.” **e)** Let  $F(x)$  and  $R(x)$  be the predicates that  $x$  knows French and knows Russian, respectively, and let the domain be people in this class. Original is  $\neg \exists x(F(x) \wedge R(x))$ . Negation is  $\exists x(F(x) \wedge R(x))$ : “There is someone in this class who knows French and Russian.” **35. a)**  $\exists x(x \leq 1)$  **b)**  $\exists x(x > 2)$  **c)**  $\forall x(x < 4)$  **d)**  $\forall x(x \geq 0)$  **e)**  $\exists x((x \geq -1) \wedge (x \leq 2))$  **f)**  $\forall x((x \geq 4) \wedge (x \leq 7))$  **37. a)** There is no counterexample. **b)**  $x = 0$  **c)**  $x = 2$  **39. a)**  $\forall x((F(x, 25,000) \vee S(x, 25)) \rightarrow E(x))$ , where  $E(x)$  is “Person  $x$  qualifies as an elite flyer in a given year,”  $F(x, y)$  is “Person  $x$  flies more than  $y$  miles in a given year,” and  $S(x, y)$  is “Person  $x$  takes more than  $y$  flights in a given year” **b)**  $\forall x(((M(x) \wedge T(x, 3)) \vee (\neg M(x) \wedge T(x, 3.5))) \rightarrow Q(x))$ , where  $Q(x)$  is “Person  $x$  qualifies for the marathon,”  $M(x)$  is “Person  $x$  is a man,” and  $T(x, y)$  is “Person  $x$  has run the marathon in less than  $y$  hours” **c)**  $M \rightarrow ((H(60) \vee (H(45) \wedge T)) \wedge \forall y G(B, y))$ , where  $M$  is the proposition “The student received a masters degree,”  $H(x)$  is “The student took at least  $x$  course hours,”  $T$  is the proposition “The student wrote a thesis,” and  $G(x, y)$  is “The person got grade  $x$  or higher in

course  $y$ ” **d)**  $\exists x((T(x, 21) \wedge G(x, 4.0))$ , where  $T(x, y)$  is “Person  $x$  took more than  $y$  credit hours” and  $G(x, p)$  is “Person  $x$  earned grade point average  $p$ ” (we assume that we are talking about one given semester) **41. a)** If there is a printer that is both out of service and busy, then some job has been lost. **b)** If every printer is busy, then there is a job in the queue. **c)** If there is a job that is both queued and lost, then some printer is out of service. **d)** If every printer is busy and every job is queued, then some job is lost. **43. a)**  $(\exists x F(x, 10)) \rightarrow \exists x S(x)$ , where  $F(x, y)$  is “Disk  $x$  has more than  $y$  kilobytes of free space,” and  $S(x)$  is “Mail message  $x$  can be saved” **b)**  $(\exists x A(x)) \rightarrow \forall x(Q(x) \rightarrow T(x))$ , where  $A(x)$  is “Alert  $x$  is active,”  $Q(x)$  is “Message  $x$  is queued,” and  $T(x)$  is “Message  $x$  is transmitted” **c)**  $\forall x((x \neq \text{main console}) \rightarrow T(x))$ , where  $T(x)$  is “The diagnostic monitor tracks the status of system  $x$ ” **d)**  $\forall x(\neg L(x) \rightarrow B(x))$ , where  $L(x)$  is “The host of the conference call put participant  $x$  on a special list” and  $B(x)$  is “Participant  $x$  was billed” **45.** They are not equivalent. Let  $P(x)$  be any propositional function that is sometimes true and sometimes false, and let  $Q(x)$  be any propositional function that is always false. Then  $\forall x(P(x) \rightarrow Q(x))$  is false but  $\forall x P(x) \rightarrow \forall x Q(x)$  is true. **47.** Both statements are true precisely when at least one of  $P(x)$  and  $Q(x)$  is true for at least one value of  $x$  in the domain. **49. a)** If  $A$  is true, then both sides are logically equivalent to  $\forall x P(x)$ . If  $A$  is false, the left-hand side is clearly false. Furthermore, for every  $x$ ,  $P(x) \wedge A$  is false, so the right-hand side is false. Hence, the two sides are logically equivalent. **b)** If  $A$  is true, then both sides are logically equivalent to  $\exists x P(x)$ . If  $A$  is false, the left-hand side is clearly false. Furthermore, for every  $x$ ,  $P(x) \wedge A$  is false, so  $\exists x(P(x) \wedge A)$  is false. Hence, the two sides are logically equivalent. **51.** We can establish these equivalences by arguing that one side is true if and only if the other side is true. **a)** Suppose that  $A$  is true. Then for each  $x$ ,  $P(x) \rightarrow A$  is true; therefore, the left-hand side is always true in this case. By similar reasoning the right-hand side is always true in this case. Therefore, the two propositions are logically equivalent when  $A$  is true. On the other hand, suppose that  $A$  is false. There are two subcases. If  $P(x)$  is false for every  $x$ , then  $P(x) \rightarrow A$  is vacuously true, so the left-hand side is vacuously true. The same reasoning shows that the right-hand side is also true, because in this subcase  $\exists x P(x)$  is false. For the second subcase, suppose that  $P(x)$  is true for some  $x$ . Then for that  $x$ ,  $P(x) \rightarrow A$  is false, so the left-hand side is false. The right-hand side is also false, because in this subcase  $\exists x P(x)$  is true but  $A$  is false. Thus, in all cases, the two propositions have the same truth value. **b)** If  $A$  is true, then both sides are trivially true, because the conditional statements have true conclusions. If  $A$  is false, then there are two subcases. If  $P(x)$  is false for some  $x$ , then  $P(x) \rightarrow A$  is vacuously true for that  $x$ , so the left-hand side is true. The same reasoning shows that the right-hand side is true, because in this subcase  $\forall x P(x)$  is false. For the second subcase, suppose that  $P(x)$  is true for every  $x$ . Then for every  $x$ ,  $P(x) \rightarrow A$  is false, so the left-hand side is false (there is no  $x$  making the conditional statement true). The right-hand side is also false, because it is a conditional statement with a true hypothesis and a false



conclusion. Thus, in all cases, the two propositions have the same truth value. **53.** To show these are not logically equivalent, let  $P(x)$  be the statement “ $x$  is positive,” and let  $Q(x)$  be the statement “ $x$  is negative” with domain the set of integers. Then  $\exists x P(x) \wedge \exists x Q(x)$  is true, but  $\exists x(P(x) \wedge Q(x))$  is false. **55. a)** True **b)** False, unless the domain consists of just one element or the hypothesis is false. **c)** True **57. a)** Yes **b)** No **c)** juana, kiko **d)** math273, cs301 **e)** juana, kiko **59. sibling( $X, Y$ ): - mother( $M, X$ ), mother( $M, Y$ ), father( $F, X$ ), father( $F, Y$ )** **61. a)**  $\forall x(P(x) \rightarrow \neg Q(x))$  **b)**  $\forall x(Q(x) \rightarrow R(x))$  **c)**  $\forall x(P(x) \rightarrow \neg R(x))$  **d)** The conclusion does not follow. There may be vain professors, because the premises do not rule out the possibility that there are other vain people besides ignorant ones. **63. a)**  $\forall x(P(x) \rightarrow \neg Q(x))$  **b)**  $\forall x(R(x) \rightarrow \neg S(x))$  **c)**  $\forall x(\neg Q(x) \rightarrow S(x))$  **d)**  $\forall x(P(x) \rightarrow \neg R(x))$  **e)** The conclusion follows. Suppose  $x$  is a baby. Then, by the first premise,  $x$  is illogical, so by the third premise,  $x$  is despised. The second premise says that if  $x$  could manage a crocodile, then  $x$  would not be despised. Therefore,  $x$  cannot manage a crocodile.

## Section 1.5

**1. a)** For every real number  $x$  there exists a real number  $y$  such that  $x$  is less than  $y$ . **b)** For every real number  $x$  and real number  $y$ , if  $x$  and  $y$  are both nonnegative, then their product is nonnegative. **c)** For every real number  $x$  and real number  $y$ , there exists a real number  $z$  such that  $xy = z$ . **3. a)** There is some student in your class who has sent a message to some student in your class. **b)** There is some student in your class who has sent a message to every student in your class. **c)** Every student in your class has sent a message to at least one student in your class. **d)** There is a student in your class who has been sent a message by every student in your class. **e)** Every student in your class has been sent a message from at least one student in your class. **f)** Every student in the class has sent a message to every student in the class. **5. a)** Sarah Smith has visited www.att.com. **b)** At least one person has visited www.imdb.org. **c)** Jose Orez has visited at least one website. **d)** There is a website that both Ashok Puri and Cindy Yoon have visited. **e)** There is a person besides David Belcher who has visited all the websites that David Belcher has visited. **f)** There are two different people who have visited exactly the same websites. **7. a)** Abdallah Hussein does not like Japanese cuisine. **b)** Some student at your school likes Korean cuisine, and everyone at your school likes Mexican cuisine. **c)** There is some cuisine that either Monique Arsenault or Jay Johnson likes. **d)** For every pair of distinct students at your school, there is some cuisine that at least one of them does not like. **e)** There are two students at your school who like exactly the same set of cuisines. **f)** For every pair of students at your school, there is some cuisine about which they have the same opinion (either they both like it or they both do not like it). **9. a)**  $\forall x L(x, \text{Jerry})$  **b)**  $\forall x \exists y L(x, y)$  **c)**  $\exists y \forall x L(x, y)$  **d)**  $\forall x \exists y \neg L(x, y)$  **e)**  $\exists x \neg L(\text{Lydia}, x)$  **f)**  $\exists x \forall y \neg L(y, x)$  **g)**  $\exists x (\forall y L(y, x) \wedge \forall z ((\forall w L(w, z)) \rightarrow z = x))$  **h)**  $\exists x \exists y (x \neq$

$y \wedge L(\text{Lynn}, x) \wedge L(\text{Lynn}, y) \wedge \forall z (L(\text{Lynn}, z) \rightarrow (z = x \vee z = y)))$  **i)**  $\forall x L(x, x)$  **j)**  $\exists x \forall y (L(x, y) \leftrightarrow x = y)$  **11. a)**  $A(\text{Lois}, \text{Professor Michaels})$  **b)**  $\forall x (S(x) \rightarrow A(x, \text{Professor Gross}))$  **c)**  $\forall x (F(x) \rightarrow (A(x, \text{Professor Miller}) \vee A(\text{Professor Miller}, x)))$  **d)**  $\exists x (S(x) \wedge \forall y (F(y) \rightarrow \neg A(x, y)))$  **e)**  $\exists x (F(x) \wedge \forall y (S(y) \rightarrow \neg A(y, x)))$  **f)**  $\forall y (F(y) \rightarrow \exists x (S(x) \vee A(x, y)))$  **g)**  $\exists x (F(x) \wedge \forall y ((F(y) \wedge (y \neq x)) \rightarrow A(x, y)))$  **h)**  $\exists x (S(x) \wedge \forall y (F(y) \rightarrow \neg A(y, x)))$  **13. a)**  $\neg M(\text{Chou}, \text{Koko})$  **b)**  $\neg M(\text{Arlene}, \text{Sarah}) \wedge \neg T(\text{Arlene}, \text{Sarah})$  **c)**  $\neg M(\text{Deborah}, \text{Jose})$  **d)**  $\forall x M(x, \text{Ken})$  **e)**  $\forall x \neg T(x, \text{Nina})$  **f)**  $\forall x (T(x, \text{Avi}) \vee M(x, \text{Avi}))$  **g)**  $\exists x \forall y (y \neq x \rightarrow M(x, y))$  **h)**  $\exists x \forall y (y \neq x \rightarrow (M(x, y) \vee T(x, y)))$  **i)**  $\exists x \exists y (x \neq y \wedge M(x, y) \wedge M(y, x))$  **j)**  $\exists x M(x, x)$  **k)**  $\exists x \forall y (x \neq y \rightarrow (\neg M(x, y) \wedge \neg T(y, x)))$  **l)**  $\forall x (\exists y (x \neq y \wedge (M(y, x) \vee T(y, x))))$  **m)**  $\exists x \exists y (x \neq y \wedge M(x, y) \wedge T(y, x))$  **n)**  $\exists x \exists y (x \neq y \wedge \forall z ((z \neq x \wedge z \neq y) \rightarrow (M(x, z) \vee M(y, z) \vee T(x, z) \vee T(y, z))))$  **15. a)**  $\forall x P(x)$ , where  $P(x)$  is “ $x$  needs a course in discrete mathematics” and the domain consists of all computer science students **b)**  $\exists x P(x)$ , where  $P(x)$  is “ $x$  owns a personal computer” and the domain consists of all students in this class **c)**  $\forall x \exists y P(x, y)$ , where  $P(x, y)$  is “ $x$  has taken  $y$ ,” the domain for  $x$  consists of all students in this class, and the domain for  $y$  consists of all computer science classes **d)**  $\exists x \exists y P(x, y)$ , where  $P(x, y)$  and domains are the same as in part (c) **e)**  $\forall x \forall y P(x, y)$ , where  $P(x, y)$  is “ $x$  has been in  $y$ ,” the domain for  $x$  consists of all students in this class, and the domain for  $y$  consists of all buildings on campus **f)**  $\exists x \exists y \forall z (P(z, y) \rightarrow Q(x, z))$ , where  $P(z, y)$  is “ $z$  is in  $y$ ” and  $Q(x, z)$  is “ $x$  has been in  $z$ ”; the domain for  $x$  consists of all students in the class, the domain for  $y$  consists of all buildings on campus, and the domain of  $z$  consists of all rooms. **g)**  $\forall x \forall y \exists z (P(z, y) \wedge Q(x, z))$ , with same environment as in part (f) **17. a)**  $\forall u \exists m (A(u, m) \wedge \forall n (n \neq m \rightarrow \neg A(u, n)))$ , where  $A(u, m)$  means that user  $u$  has access to mailbox  $m$  **b)**  $\exists p \forall e (H(e) \wedge S(p, \text{running})) \rightarrow S(\text{kernel}, \text{working correctly})$ , where  $H(e)$  means that error condition  $e$  is in effect and  $S(x, y)$  means that the status of  $x$  is  $y$  **c)**  $\forall u \forall s (E(s, \text{.edu}) \rightarrow A(u, s))$ , where  $E(s, x)$  means that website  $s$  has extension  $x$ , and  $A(u, s)$  means that user  $u$  can access website  $s$  **d)**  $\exists x \exists y (x \neq y \wedge \forall z ((\forall s M(z, s)) \leftrightarrow (z = x \vee z = y)))$ , where  $M(a, b)$  means that system  $a$  monitors remote server  $b$  **19. a)**  $\forall x \forall y ((x < 0) \wedge (y < 0) \rightarrow (x + y < 0))$  **b)**  $\neg \forall x \forall y ((x > 0) \wedge (y > 0) \rightarrow (x - y > 0))$  **c)**  $\forall x \forall y (x^2 + y^2 \geq (x + y)^2)$  **d)**  $\forall x \forall y (|xy| = |x||y|)$  **21. a)**  $\forall x \exists a \exists b \exists c \exists d ((x > 0) \rightarrow x = a^2 + b^2 + c^2 + d^2)$ , where the domain consists of all integers **23. a)**  $\forall x \forall y ((x < 0) \wedge (y < 0) \rightarrow (xy > 0))$  **b)**  $\forall x (x - x = 0)$  **c)**  $\forall x \exists a \exists b (a \neq b \wedge \forall c (c^2 = x \leftrightarrow (c = a \vee c = b)))$  **d)**  $\forall x ((x < 0) \rightarrow \neg \exists y (x = y^2))$  **25. a)** There is a multiplicative identity for the real numbers. **b)** The product of two negative real numbers is always a positive real number. **c)** There exist real numbers  $x$  and  $y$  such that  $x^2$  exceeds  $y$  but  $x$  is less than  $y$ . **d)** The real numbers are closed under the operation of addition. **27. a)** True **b)** True **c)** True **d)** True **e)** True **f)** False **g)** False **h)** True **i)** False **29. a)**  $P(1, 1) \wedge P(1, 2) \wedge P(1, 3) \wedge P(2, 1) \wedge P(2, 2) \wedge P(2, 3) \wedge P(3, 1) \wedge P(3, 2) \wedge P(3, 3)$  **b)**  $P(1, 1) \vee P(1, 2) \vee P(1, 3) \vee P(2, 1) \vee P(2, 2) \vee P(2, 3) \vee P(3, 1) \vee P(3, 2) \vee P(3, 3)$  **c)**  $(P(1, 1) \wedge P(1, 2) \wedge P(1, 3)) \vee$



$(P(2, 1) \wedge P(2, 2) \wedge P(2, 3)) \vee (P(3, 1) \wedge P(3, 2) \wedge P(3, 3))$   
**d)**  $(P(1, 1) \vee P(2, 1) \vee P(3, 1)) \wedge (P(1, 2) \vee P(2, 2) \vee P(3, 2)) \wedge (P(1, 3) \vee P(2, 3) \vee P(3, 3))$     **31. a)**  $\exists x \forall y \exists z \neg T(x, y, z)$     **b)**  $\exists x \forall y \neg P(x, y) \wedge \exists x \forall y \neg Q(x, y)$     **c)**  $\exists x \forall y (\neg P(x, y) \vee \forall z \neg R(x, y, z))$     **d)**  $\exists x \forall y (P(x, y) \wedge \neg Q(x, y))$   
**33. a)**  $\exists x \exists y \neg P(x, y)$     **b)**  $\exists y \forall x \neg P(x, y)$     **c)**  $\exists y \exists x (\neg P(x, y) \wedge \neg Q(x, y))$     **d)**  $(\forall x \forall y P(x, y)) \vee (\exists x \exists y \neg Q(x, y))$   
**e)**  $\exists x (\forall y \exists z \neg P(x, y, z) \vee \forall z \exists y \neg P(x, y, z))$     **35.** Any domain with four or more members makes the statement true; any domain with three or fewer members makes the statement false.    **37. a)** There is someone in this class such that for every two different math courses, these are not the two and only two math courses this person has taken. **b)** Every person has either visited Libya or has not visited a country other than Libya. **c)** Someone has climbed every mountain in the Himalayas. **d)** There is someone who has neither been in a movie with Kevin Bacon nor has been in a movie with someone who has been in a movie with Kevin Bacon.    **39. a)**  $x = 2, y = -2$     **b)**  $x = -4$     **c)**  $x = 17, y = -1$     **41.**  $\forall x \forall y \forall z ((x \cdot y) \cdot z = x \cdot (y \cdot z))$   
**43.**  $\forall m \forall b (m \neq 0 \rightarrow \exists x (mx + b = 0 \wedge \forall w (mw + b = 0 \rightarrow w = x)))$     **45. a)** True    **b)** False    **c)** True  
**47.**  $\neg(\exists x \forall y P(x, y)) \leftrightarrow \forall x (\neg \forall y P(x, y)) \leftrightarrow \forall x \exists y \neg P(x, y)$   
**49. a)** Suppose that  $\forall x P(x) \wedge \exists x Q(x)$  is true. Then  $P(x)$  is true for all  $x$  and there is an element  $y$  for which  $Q(y)$  is true. Because  $P(x) \wedge Q(y)$  is true for all  $x$  and there is a  $y$  for which  $Q(y)$  is true,  $\forall x \exists y (P(x) \wedge Q(y))$  is true. Conversely, suppose that the second proposition is true. Let  $x$  be an element in the domain. There is a  $y$  such that  $Q(y)$  is true, so  $\exists x Q(x)$  is true. Because  $\forall x P(x)$  is also true, it follows that the first proposition is true. **b)** Suppose that  $\forall x P(x) \vee \exists x Q(x)$  is true. Then either  $P(x)$  is true for all  $x$ , or there exists a  $y$  for which  $Q(y)$  is true. In the former case,  $P(x) \vee Q(y)$  is true for all  $x$ , so  $\forall x \exists y (P(x) \vee Q(y))$  is true. In the latter case,  $Q(y)$  is true for a particular  $y$ , so  $P(x) \vee Q(y)$  is true for all  $x$  and consequently  $\forall x \exists y (P(x) \vee Q(y))$  is true. Conversely, suppose that the second proposition is true. If  $P(x)$  is true for all  $x$ , then the first proposition is true. If not,  $P(x)$  is false for some  $x$ , and for this  $x$  there must be a  $y$  such that  $P(x) \vee Q(y)$  is true. Hence,  $Q(y)$  must be true, so  $\exists y Q(y)$  is true. It follows that the first proposition must hold.    **51.** We will show how an expression can be put into prenex normal form (PNF) if subexpressions in it can be put into PNF. Then, working from the inside out, any expression can be put in PNF. (To formalize the argument, it is necessary to use the method of structural induction that will be discussed in Section 5.3.) By Exercise 49 of Section 1.3, we can assume that the proposition uses only  $\vee$  and  $\neg$  as logical connectives. Now note that any proposition with no quantifiers is already in PNF. (This is the basis case of the argument.) Now suppose that the proposition is of the form  $QxP(x)$ , where  $Q$  is a quantifier. Because  $P(x)$  is a shorter expression than the original proposition, we can put it into PNF. Then  $Qx$  followed by this PNF is again in PNF and is equivalent to the original proposition. Next, suppose that the proposition is of the form  $\neg P$ . If  $P$  is already in PNF, we slide the negation sign past all the quantifiers using the equivalences in Table 2 in Section 1.4. Finally, assume that

proposition is of the form  $P \vee Q$ , where each of  $P$  and  $Q$  is in PNF. If only one of  $P$  and  $Q$  has quantifiers, then we can use Exercise 48 in Section 1.4 to bring the quantifier in front of both. If both  $P$  and  $Q$  have quantifiers, we can use Exercise 47 in Section 1.4, Exercise 48, or part (b) of Exercise 49 to rewrite  $P \vee Q$  with two quantifiers preceding the disjunction of a proposition of the form  $R \vee S$ , and then put  $R \vee S$  into PNF.

## Section 1.6

**1.** Modus ponens; valid; the conclusion is true, because the hypotheses are true.    **3. a)** Addition    **b)** Simplification  
**c)** Modus ponens    **d)** Modus tollens    **e)** Hypothetical syllogism    **5.** Let  $w$  be “Randy works hard,” let  $d$  be “Randy is a dull boy,” and let  $j$  be “Randy will get the job.” The hypotheses are  $w, w \rightarrow d$ , and  $d \rightarrow \neg j$ . Using modus ponens and the first two hypotheses,  $d$  follows. Using modus ponens and the last hypothesis,  $\neg j$ , which is the desired conclusion, “Randy will not get the job,” follows.    **7.** Universal instantiation is used to conclude that “If Socrates is a man, then Socrates is mortal.” Modus ponens is then used to conclude that Socrates is mortal.    **9. a)** Valid conclusions are “I did not take Tuesday off,” “I took Thursday off,” and “It rained on Thursday.”    **b)** “I did not eat spicy foods and it did not thunder” is a valid conclusion.    **c)** “I am clever” is a valid conclusion.    **d)** “Ralph is not a CS major” is a valid conclusion.    **e)** “That you buy lots of stuff is good for the U.S. and is good for you” is a valid conclusion.    **f)** “Mice gnaw their food” and “Rabbits are not rodents” are valid conclusions.  
**11.** Suppose that  $p_1, p_2, \dots, p_n$  are true. We want to establish that  $q \rightarrow r$  is true. If  $q$  is false, then we are done, vacuously. Otherwise,  $q$  is true, so by the validity of the given argument form (that whenever  $p_1, p_2, \dots, p_n, q$  are true, then  $r$  must be true), we know that  $r$  is true.    **13. a)** Let  $c(x)$  be “ $x$  is in this class,”  $j(x)$  be “ $x$  knows how to write programs in JAVA,” and  $h(x)$  be “ $x$  can get a high-paying job.” The premises are  $c(\text{Doug}), j(\text{Doug}), \forall x (j(x) \rightarrow h(x))$ . Using universal instantiation and the last premise,  $j(\text{Doug}) \rightarrow h(\text{Doug})$  follows. Applying modus ponens to this conclusion and the second premise,  $h(\text{Doug})$  follows. Using conjunction and the first premise,  $c(\text{Doug}) \wedge h(\text{Doug})$  follows. Finally, using existential generalization, the desired conclusion,  $\exists x (c(x) \wedge h(x))$  follows.    **b)** Let  $c(x)$  be “ $x$  is in this class,”  $w(x)$  be “ $x$  enjoys whale watching,” and  $p(x)$  be “ $x$  cares about ocean pollution.” The premises are  $\exists x (c(x) \wedge w(x))$  and  $\forall x (w(x) \rightarrow p(x))$ . From the first premise,  $c(y) \wedge w(y)$  for a particular person  $y$ . Using simplification,  $w(y)$  follows. Using the second premise and universal instantiation,  $w(y) \rightarrow p(y)$  follows. Using modus ponens,  $p(y)$  follows, and by conjunction,  $c(y) \wedge p(y)$  follows. Finally, by existential generalization, the desired conclusion,  $\exists x (c(x) \wedge p(x))$ , follows.    **c)** Let  $c(x)$  be “ $x$  is in this class,”  $p(x)$  be “ $x$  owns a PC,” and  $w(x)$  be “ $x$  can use a word-processing program.” The premises are  $c(\text{Zeke}), \forall x (c(x) \rightarrow p(x))$ , and



$\forall x(p(x) \rightarrow w(x))$ . Using the second premise and universal instantiation,  $c(\text{Zeke}) \rightarrow p(\text{Zeke})$  follows. Using the first premise and modus ponens,  $p(\text{Zeke})$  follows. Using the third premise and universal instantiation,  $p(\text{Zeke}) \rightarrow w(\text{Zeke})$  follows. Finally, using modus ponens,  $w(\text{Zeke})$ , the desired conclusion, follows. **d)** Let  $j(x)$  be “ $x$  is in New Jersey,”  $f(x)$  be “ $x$  lives within 50 miles of the ocean,” and  $s(x)$  be “ $x$  has seen the ocean.” The premises are  $\forall x(j(x) \rightarrow f(x))$  and  $\exists x(j(x) \wedge \neg s(x))$ . The second hypothesis and existential instantiation imply that  $j(y) \wedge \neg s(y)$  for a particular person  $y$ . By simplification,  $j(y)$  for this person  $y$ . Using universal instantiation and the first premise,  $j(y) \rightarrow f(y)$ , and by modus ponens,  $f(y)$  follows. By simplification,  $\neg s(y)$  follows from  $j(y) \wedge \neg s(y)$ . So  $f(y) \wedge \neg s(y)$  follows by conjunction. Finally, the desired conclusion,  $\exists x(f(x) \wedge \neg s(x))$ , follows by existential generalization.

**15. a)** Correct, using universal instantiation and modus ponens **b)** Invalid; fallacy of affirming the conclusion **c)** Invalid; fallacy of denying the hypothesis **d)** Correct, using universal instantiation and modus tollens **17.** We know that *some*  $x$  exists that makes  $H(x)$  true, but we cannot conclude that Lola is one such  $x$ . **19. a)** Fallacy of affirming the conclusion **b)** Valid argument using modus tollens **c)** Fallacy of denying the hypothesis **21.** By the second premise, there is some lion that does not drink coffee. Let Leo be such a creature. By simplification we know that Leo is a lion. By modus ponens we know from the first premise that Leo is fierce. Hence, Leo is fierce and does not drink coffee. By the definition of the existential quantifier, there exist fierce creatures that do not drink coffee, that is, some fierce creatures do not drink coffee. **23.** The error occurs in step (5), because we cannot assume, as is being done here, that the  $c$  that makes  $P$  true is the same as the  $c$  that makes  $Q$  true. **25.** We are given the premises  $\forall x(P(x) \rightarrow Q(x))$  and  $\neg Q(a)$ . We want to show  $\neg P(a)$ . Suppose, to the contrary, that  $\neg P(a)$  is not true. Then  $P(a)$  is true. Therefore, by universal modus ponens, we have  $Q(a)$ . But this contradicts the given premise  $\neg Q(a)$ . Therefore, our supposition must have been wrong, and so  $\neg P(a)$  is true, as desired.

| 27. Step                                            | Reason                                  |
|-----------------------------------------------------|-----------------------------------------|
| 1. $\forall x(P(x) \wedge R(x))$                    | Premise                                 |
| 2. $P(a) \wedge R(a)$                               | Universal instantiation from (1)        |
| 3. $P(a)$                                           | Simplification from (2)                 |
| 4. $\forall x(P(x) \rightarrow (Q(x) \wedge S(x)))$ | Premise                                 |
| 5. $Q(a) \wedge S(a)$                               | Universal modus ponens from (3) and (4) |
| 6. $S(a)$                                           | Simplification from (5)                 |
| 7. $R(a)$                                           | Simplification from (2)                 |
| 8. $R(a) \wedge S(a)$                               | Conjunction from (7) and (6)            |
| 9. $\forall x(R(x) \wedge S(x))$                    | Universal generalization from (5)       |

| 29. Step                                   | Reason                                 |
|--------------------------------------------|----------------------------------------|
| 1. $\exists x \neg P(x)$                   | Premise                                |
| 2. $\neg P(c)$                             | Existential instantiation from (1)     |
| 3. $\forall x(P(x) \vee Q(x))$             | Premise                                |
| 4. $P(c) \vee Q(c)$                        | Universal instantiation from (3)       |
| 5. $Q(c)$                                  | Disjunctive syllogism from (4) and (2) |
| 6. $\forall x(\neg Q(x) \vee S(x))$        | Premise                                |
| 7. $\neg Q(c) \vee S(c)$                   | Universal instantiation from (6)       |
| 8. $S(c)$                                  | Disjunctive syllogism from (5) and (7) |
| 9. $\forall x(R(x) \rightarrow \neg S(x))$ | Premise                                |
| 10. $R(c) \rightarrow \neg S(c)$           | Universal instantiation from (9)       |
| 11. $\neg R(c)$                            | Modus tollens from (8) and (10)        |
| 12. $\exists x \neg R(x)$                  | Existential generalization from (11)   |

**31.** Let  $p$  be “It is raining”; let  $q$  be “Yvette has her umbrella”; let  $r$  be “Yvette gets wet.” Assumptions are  $\neg p \vee q$ ,  $\neg q \vee \neg r$ , and  $p \vee \neg r$ . Resolution on the first two gives  $\neg p \vee \neg r$ . Resolution on this and the third assumption gives  $\neg r$ , as desired.

**33.** Assume that this proposition is satisfiable. Using resolution on the first two clauses enables us to conclude  $q \vee q$ ; in other words, we know that  $q$  has to be true. Using resolution on the last two clauses enables us to conclude  $\neg q \vee \neg q$ ; in other words, we know that  $\neg q$  has to be true. This is a contradiction. So this proposition is not satisfiable. **35.** Valid

## Section 1.7

**1.** Let  $n = 2k + 1$  and  $m = 2l + 1$  be odd integers. Then  $n + m = 2(k + l + 1)$  is even. **3.** Suppose that  $n$  is even. Then  $n = 2k$  for some integer  $k$ . Therefore,  $n^2 = (2k)^2 = 4k^2 = 2(2k^2)$ . Because we have written  $n^2$  as 2 times an integer, we conclude that  $n^2$  is even. **5.** Direct proof: Suppose that  $m + n$  and  $n + p$  are even. Then  $m + n = 2s$  for some integer  $s$  and  $n + p = 2t$  for some integer  $t$ . If we add these, we get  $m + p + 2n = 2s + 2t$ . Subtracting  $2n$  from both sides and factoring, we have  $m + p = 2s + 2t - 2n = 2(s + t - n)$ . Because we have written  $m + p$  as 2 times an integer, we conclude that  $m + p$  is even. **7.** Because  $n$  is odd, we can write  $n = 2k + 1$  for some integer  $k$ . Then  $(k + 1)^2 - k^2 = k^2 + 2k + 1 - k^2 = 2k + 1 = n$ . **9.** Suppose that  $r$  is rational and  $i$  is irrational and  $s = r + i$  is rational. Then by Example 8,  $s + (-r) = i$  is rational, which is a contradiction. **11.** Because  $\sqrt{2} \cdot \sqrt{2} = 2$  is rational and  $\sqrt{2}$  is irrational, the product of two irrational numbers is not necessarily irrational. **13.** Proof by contraposition: If  $1/x$  were rational, then by definition  $1/x = p/q$  for some integers  $p$  and  $q$  with  $q \neq 0$ . Because  $1/x$  cannot be 0 (if it were, then we’d have the contradiction  $1 = x \cdot 0$  by multiplying both sides by  $x$ ), we know that  $p \neq 0$ . Now  $x = 1/(1/x) = 1/(p/q) = q/p$  by the usual rules of algebra and arithmetic. Hence,  $x$  can be written as the quotient of two integers with the denominator nonzero. Thus, by definition,  $x$  is rational. **15.** Assume that  $\sqrt{x}$  were rational. Then, because the product of two rational numbers



is rational,  $(\sqrt{x})^2 = x$  is also rational. This contradicts the hypothesis that  $x$  is irrational. **17.** Assume that it is not true that  $x \geq 1$  or  $y \geq 1$ . Then  $x < 1$  and  $y < 1$ . Adding these two inequalities, we obtain  $x + y < 2$ , which is the negation of  $x + y \geq 2$ . **19. a)** Assume that  $n$  is odd, so  $n = 2k + 1$  for some integer  $k$ . Then  $n^3 + 5 = 2(4k^3 + 6k^2 + 3k + 3)$ . Because  $n^3 + 5$  is two times some integer, it is even. **b)** Suppose that  $n^3 + 5$  is odd and  $n$  is odd. Because  $n$  is odd and the product of two odd numbers is odd, it follows that  $n^2$  is odd and then that  $n^3$  is odd. But then  $5 = (n^3 + 5) - n^3$  would have to be even because it is the difference of two odd numbers. Therefore, the supposition that  $n^3 + 5$  and  $n$  were both odd is wrong. **21.** The proposition is vacuously true because 0 is not a positive integer. Vacuous proof. **23.**  $P(1)$  is true because  $(a+b)^1 = a+b \geq a^1 + b^1 = a+b$ . Direct proof. **25.** If we chose 9 or fewer days on each day of the week, this would account for at most  $9 \cdot 7 = 63$  days. But we chose 64 days. This contradiction shows that at least 10 of the days we chose must be on the same day of the week. **27.** Suppose by way of contradiction that  $a/b$  is a rational root, where  $a$  and  $b$  are integers and this fraction is in lowest terms (that is,  $a$  and  $b$  have no common divisor greater than 1). Plug this proposed root into the equation to obtain  $a^3/b^3 + a/b + 1 = 0$ . Multiply through by  $b^3$  to obtain  $a^3 + ab^2 + b^3 = 0$ . If  $a$  and  $b$  are both odd, then the left-hand side is the sum of three odd numbers and therefore must be odd. If  $a$  is odd and  $b$  is even, then the left-hand side is odd + even + even, which is again odd. Similarly, if  $a$  is even and  $b$  is odd, then the left-hand side is even + even + odd, which is again odd. Because the fraction  $a/b$  is in simplest terms, it cannot happen that both  $a$  and  $b$  are even. Thus, in all cases, the left-hand side is odd, and therefore cannot equal 0. This contradiction shows that no such root exists. **29.** First, assume that  $n$  is odd, so that  $n = 2k + 1$  for some integer  $k$ . Then  $5n + 6 = 5(2k + 1) + 6 = 10k + 11 = 2(5k + 5) + 1$ . Hence,  $5n + 6$  is odd. To prove the converse, suppose that  $n$  is even, so that  $n = 2k$  for some integer  $k$ . Then  $5n + 6 = 10k + 6 = 2(5k + 3)$ , so  $5n + 6$  is even. Hence,  $n$  is odd if and only if  $5n + 6$  is odd. **31.** This proposition is true. Suppose that  $m$  is neither 1 nor  $-1$ . Then  $mn$  has a factor  $m$  larger than 1. On the other hand,  $mn = 1$ , and 1 has no such factor. Hence,  $m = 1$  or  $m = -1$ . In the first case  $n = 1$ , and in the second case  $n = -1$ , because  $n = 1/m$ . **33.** We prove that all these are equivalent to  $x$  being even. If  $x$  is even, then  $x = 2k$  for some integer  $k$ . Therefore,  $3x + 2 = 3 \cdot 2k + 2 = 6k + 2 = 2(3k + 1)$ , which is even, because it has been written in the form  $2t$ , where  $t = 3k + 1$ . Similarly,  $x + 5 = 2k + 5 = 2k + 4 + 1 = 2(k + 2) + 1$ , so  $x + 5$  is odd; and  $x^2 = (2k)^2 = 2(2k^2)$ , so  $x^2$  is even. For the converses, we will use a proof by contraposition. So assume that  $x$  is not even; thus,  $x$  is odd and we can write  $x = 2k + 1$  for some integer  $k$ . Then  $3x + 2 = 3(2k + 1) + 2 = 6k + 5 = 2(3k + 2) + 1$ , which is odd (i.e., not even), because it has been written in the form  $2t + 1$ , where  $t = 3k + 2$ . Similarly,  $x + 5 = 2k + 1 + 5 = 2(k + 3)$ , so  $x + 5$  is even (i.e., not odd). That  $x^2$  is odd was already proved in Example 1. **35.** We give proofs by contraposition of  $(i) \rightarrow (ii)$ ,  $(ii) \rightarrow (i)$ ,  $(i) \rightarrow (iii)$ , and  $(iii) \rightarrow (i)$ . For the first of these, suppose that  $3x + 2$  is rational, namely, equal to

$p/q$  for some integers  $p$  and  $q$  with  $q \neq 0$ . Then we can write  $x = ((p/q) - 2)/3 = (p - 2q)/(3q)$ , where  $3q \neq 0$ . This shows that  $x$  is rational. For the second conditional statement, suppose that  $x$  is rational, namely, equal to  $p/q$  for some integers  $p$  and  $q$  with  $q \neq 0$ . Then we can write  $3x + 2 = (3p + 2q)/q$ , where  $q \neq 0$ . This shows that  $3x + 2$  is rational. For the third conditional statement, suppose that  $x/2$  is rational, namely, equal to  $p/q$  for some integers  $p$  and  $q$  with  $q \neq 0$ . Then we can write  $x = 2p/q$ , where  $q \neq 0$ . This shows that  $x$  is rational. And for the fourth conditional statement, suppose that  $x$  is rational, namely, equal to  $p/q$  for some integers  $p$  and  $q$  with  $q \neq 0$ . Then we can write  $x/2 = p/(2q)$ , where  $2q \neq 0$ . This shows that  $x/2$  is rational. **37.** No **39.** Suppose that  $p_1 \rightarrow p_4 \rightarrow p_2 \rightarrow p_5 \rightarrow p_3 \rightarrow p_1$ . To prove that one of these propositions implies any of the others, just use hypothetical syllogism repeatedly. **41.** We will give a proof by contradiction. Suppose that  $a_1, a_2, \dots, a_n$  are all less than  $A$ , where  $A$  is the average of these numbers. Then  $a_1 + a_2 + \dots + a_n < nA$ . Dividing both sides by  $n$  shows that  $A = (a_1 + a_2 + \dots + a_n)/n < A$ , which is a contradiction. **43.** We will show that the four statements are equivalent by showing that  $(i)$  implies  $(ii)$ ,  $(ii)$  implies  $(iii)$ ,  $(iii)$  implies  $(iv)$ , and  $(iv)$  implies  $(i)$ . First, assume that  $n$  is even. Then  $n = 2k$  for some integer  $k$ . Then  $n + 1 = 2k + 1$ , so  $n + 1$  is odd. This shows that  $(i)$  implies  $(ii)$ . Next, suppose that  $n + 1$  is odd, so  $n + 1 = 2k + 1$  for some integer  $k$ . Then  $3n + 1 = 2n + (n + 1) = 2(n + k) + 1$ , which shows that  $3n + 1$  is odd, showing that  $(ii)$  implies  $(iii)$ . Next, suppose that  $3n + 1$  is odd, so  $3n + 1 = 2k + 1$  for some integer  $k$ . Then  $3n = (2k + 1) - 1 = 2k$ , so  $3n$  is even. This shows that  $(iii)$  implies  $(iv)$ . Finally, suppose that  $n$  is not even. Then  $n$  is odd, so  $n = 2k + 1$  for some integer  $k$ . Then  $3n = 3(2k + 1) = 6k + 3 = 2(3k + 1) + 1$ , so  $3n$  is odd. This completes a proof by contraposition that  $(iv)$  implies  $(i)$ .

## Section 1.8

**1.**  $1^2 + 1 = 2 \geq 2 = 2^1$ ;  $2^2 + 1 = 5 \geq 4 = 2^2$ ;  $3^2 + 1 = 10 \geq 8 = 2^3$ ;  $4^2 + 1 = 17 \geq 16 = 2^4$ . **3.** We must show that for all positive integers  $x$  it is not true that  $x^3 = 100$ . *Case (i):* If  $x \leq 4$ , then  $x^3 \leq 64$ , so  $x^3 \neq 100$ . *Case (ii):* If  $x \geq 5$ , then  $x^3 \geq 125$ , so  $x^3 \neq 100$ . **5.** If  $x \leq y$ , then  $\max(x, y) + \min(x, y) = y + x = x + y$ . If  $x \geq y$ , then  $\max(x, y) + \min(x, y) = x + y$ . Because these are the only two cases, the equality always holds. **7.** Because  $|x - y| = |y - x|$ , the values of  $x$  and  $y$  are interchangeable. Therefore, without loss of generality, we can assume that  $x \geq y$ . Then  $(x + y - (x - y))/2 = (x + y - x + y)/2 = 2y/2 = y = \min(x, y)$ . Similarly,  $(x + y + (x - y))/2 = (x + y + x - y)/2 = 2x/2 = x = \max(x, y)$ . **9.** There are four cases. *Case 1:*  $x \geq 0$  and  $y \geq 0$ . Then  $|x| + |y| = x + y = |x + y|$ . *Case 2:*  $x < 0$  and  $y < 0$ . Then  $|x| + |y| = -x + (-y) = -(x + y) = |x + y|$  because  $x + y < 0$ . *Case 3:*  $x \geq 0$  and  $y < 0$ . Then  $|x| + |y| = x + (-y)$ . If  $x \geq -y$ , then  $|x + y| = x + y$ . But because  $y < 0$ ,  $-y > y$ , so  $|x| + |y| = x + (-y) > x + y = |x + y|$ . If  $x < -y$ , then  $|x + y| = -(x + y) = -x + (-y)$ . But because  $x \geq 0$ ,  $x \geq -x$ ,



so  $|x| + |y| = x + (-y) \geq -x + (-y) = |x + y|$ . *Case 4:*  $x < 0$  and  $y \geq 0$ . Identical to Case 3 with the roles of  $x$  and  $y$  reversed. **11.** 10,001, 10,002, ..., 10,100 are all nonsquares, because  $100^2 = 10,000$  and  $101^2 = 10,201$ ; constructive. **13.**  $8 = 2^3$  and  $9 = 3^2$  **15.** Let  $x = 2$  and  $y = \sqrt{2}$ . If  $x^y = 2^{\sqrt{2}}$  is irrational, we are done. If not, then let  $x = 2^{\sqrt{2}}$  and  $y = \sqrt{2}/4$ . Then  $x^y = (2^{\sqrt{2}})^{\sqrt{2}/4} = 2^{\sqrt{2} \cdot (\sqrt{2}/4)} = 2^{1/2} = \sqrt{2}$ . **17. a)** This statement asserts the existence of  $x$  with a certain property. If we let  $y = x$ , then we see that  $P(x)$  is true. If  $y$  is anything other than  $x$ , then  $P(x)$  is not true. Thus,  $x$  is the unique element that makes  $P$  true. **b)** The first clause here says that there is an element that makes  $P$  true. The second clause says that whenever two elements both make  $P$  true, they are in fact the same element. Together these say that  $P$  is satisfied by exactly one element. **c)** This statement asserts the existence of an  $x$  that makes  $P$  true and has the further property that whenever we find an element that makes  $P$  true, that element is  $x$ . In other words,  $x$  is the unique element that makes  $P$  true. **19.** The equation  $|a - c| = |b - c|$  is equivalent to the disjunction of two equations:  $a - c = b - c$  or  $a - c = -b + c$ . The first of these is equivalent to  $a = b$ , which contradicts the assumptions made in this problem, so the original equation is equivalent to  $a - c = -b + c$ . By adding  $b + c$  to both sides and dividing by 2, we see that this equation is equivalent to  $c = (a + b)/2$ . Thus, there is a unique solution. Furthermore, this  $c$  is an integer, because the sum of the odd integers  $a$  and  $b$  is even. **21.** We are being asked to solve  $n = (k - 2) + (k + 3)$  for  $k$ . Using the usual, reversible, rules of algebra, we see that this equation is equivalent to  $k = (n - 1)/2$ . In other words, this is the one and only value of  $k$  that makes our equation true. Because  $n$  is odd,  $n - 1$  is even, so  $k$  is an integer. **23.** If  $x$  is itself an integer, then we can take  $n = x$  and  $\epsilon = 0$ . No other solution is possible in this case, because if the integer  $n$  is greater than  $x$ , then  $n$  is at least  $x + 1$ , which would make  $\epsilon \geq 1$ . If  $x$  is not an integer, then round it up to the next integer, and call that integer  $n$ . Let  $\epsilon = n - x$ . Clearly  $0 \leq \epsilon < 1$ ; this is the only  $\epsilon$  that will work with this  $n$ , and  $n$  cannot be any larger, because  $\epsilon$  is constrained to be less than 1. **25.** The harmonic mean of distinct positive real numbers  $x$  and  $y$  is always less than their geometric mean. To prove  $2xy/(x + y) < \sqrt{xy}$ , multiply both sides by  $(x + y)/(2\sqrt{xy})$  to obtain the equivalent inequality  $\sqrt{xy} < (x + y)/2$ , which is proved in Example 14. **27.** The parity (oddness or evenness) of the sum of the numbers written on the board never changes, because  $j + k$  and  $|j - k|$  have the same parity (and at each step we reduce the sum by  $j + k$  but increase it by  $|j - k|$ ). Therefore the integer at the end of the process must have the same parity as  $1 + 2 + \dots + (2n) = n(2n + 1)$ , which is odd because  $n$  is odd. **29.** Without loss of generality we can assume that  $n$  is nonnegative, because the fourth power of an integer and the fourth power of its negative are the same. We divide an arbitrary positive integer  $n$  by 10, obtaining a quotient  $k$  and remainder  $l$ , whence  $n = 10k + l$ , and  $l$  is an integer between 0 and 9, inclusive. Then we compute  $n^4$  in each of these 10 cases. We get the following values, where  $X$  is some integer that is a multiple of 10, whose exact value we

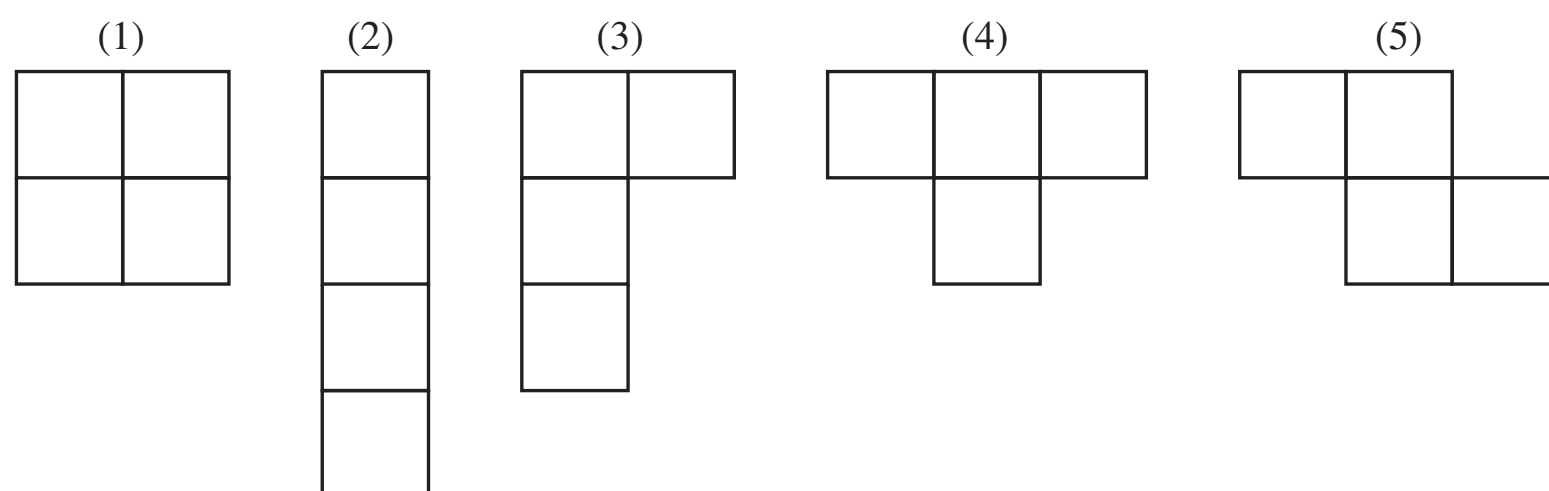
do not care about.  $(10k + 0)^4 = 10,000k^4 = 10,000k^4 + 0$ ,  $(10k + 1)^4 = 10,000k^4 + X \cdot k^3 + X \cdot k^2 + X \cdot k + 1$ ,  $(10k + 2)^4 = 10,000k^4 + X \cdot k^3 + X \cdot k^2 + X \cdot k + 16$ ,  $(10k + 3)^4 = 10,000k^4 + X \cdot k^3 + X \cdot k^2 + X \cdot k + 81$ ,  $(10k + 4)^4 = 10,000k^4 + X \cdot k^3 + X \cdot k^2 + X \cdot k + 256$ ,  $(10k + 5)^4 = 10,000k^4 + X \cdot k^3 + X \cdot k^2 + X \cdot k + 625$ ,  $(10k + 6)^4 = 10,000k^4 + X \cdot k^3 + X \cdot k^2 + X \cdot k + 1296$ ,  $(10k + 7)^4 = 10,000k^4 + X \cdot k^3 + X \cdot k^2 + X \cdot k + 2401$ ,  $(10k + 8)^4 = 10,000k^4 + X \cdot k^3 + X \cdot k^2 + X \cdot k + 4096$ ,  $(10k + 9)^4 = 10,000k^4 + X \cdot k^3 + X \cdot k^2 + X \cdot k + 6561$ . Because each coefficient indicated by  $X$  is a multiple of 10, the corresponding term has no effect on the ones digit of the answer. Therefore, the ones digits are 0, 1, 6, 1, 6, 5, 6, 1, 6, 1, respectively, so it is always a 0, 1, 5, or 6. **31.** Because  $n^3 > 100$  for all  $n > 4$ , we need only note that  $n = 1$ ,  $n = 2$ ,  $n = 3$ , and  $n = 4$  do not satisfy  $n^2 + n^3 = 100$ . **33.** Because  $5^4 = 625$ , both  $x$  and  $y$  must be less than 5. Then  $x^4 + y^4 \leq 4^4 + 4^4 = 512 < 625$ . **35.** If it is not true that  $a \leq \sqrt[3]{n}$ ,  $b \leq \sqrt[3]{n}$ , or  $c \leq \sqrt[3]{n}$ , then  $a > \sqrt[3]{n}$ ,  $b > \sqrt[3]{n}$ , and  $c > \sqrt[3]{n}$ . Multiplying these inequalities of positive numbers together we obtain  $abc < (\sqrt[3]{n})^3 = n$ , which implies the negation of our hypothesis that  $n = abc$ . **37.** By finding a common denominator, we can assume that the given rational numbers are  $a/b$  and  $c/b$ , where  $b$  is a positive integer and  $a$  and  $c$  are integers with  $a < c$ . In particular,  $(a + 1)/b \leq c/b$ . Thus,  $x = (a + \frac{1}{2}\sqrt{2})/b$  is between the two given rational numbers, because  $0 < \sqrt{2} < 2$ . Furthermore,  $x$  is irrational, because if  $x$  were rational, then  $2(bx - a) = \sqrt{2}$  would be as well, in violation of Example 11 in Section 1.7. **39. a)** Without loss of generality, we can assume that the  $x$  sequence is already sorted into nondecreasing order, because we can relabel the indices. There are only a finite number of possible orderings for the  $y$  sequence, so if we can show that we can increase the sum (or at least keep it the same) whenever we find  $y_i$  and  $y_j$  that are out of order (i.e.,  $i < j$  but  $y_i > y_j$ ) by switching them, then we will have shown that the sum is largest when the  $y$  sequence is in nondecreasing order. Indeed, if we perform the swap, then we have added  $x_i y_j + x_j y_i$  to the sum and subtracted  $x_i y_i + x_j y_j$ . The net effect is to have added  $x_i y_j + x_j y_i - x_i y_i - x_j y_j = (x_j - x_i)(y_i - y_j)$ , which is nonnegative by our ordering assumptions. **b)** Similar to part (a) **41. a)**  $6 \rightarrow 3 \rightarrow 10 \rightarrow 5 \rightarrow 16 \rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 1$  **b)**  $7 \rightarrow 22 \rightarrow 11 \rightarrow 34 \rightarrow 17 \rightarrow 52 \rightarrow 26 \rightarrow 13 \rightarrow 40 \rightarrow 20 \rightarrow 10 \rightarrow 5 \rightarrow 16 \rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 1$  **c)**  $17 \rightarrow 52 \rightarrow 26 \rightarrow 13 \rightarrow 40 \rightarrow 20 \rightarrow 10 \rightarrow 5 \rightarrow 16 \rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 1$  **d)**  $21 \rightarrow 64 \rightarrow 32 \rightarrow 16 \rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 1$  **43.** Without loss of generality, assume that the upper left and upper right corners of the board are removed. Place three dominoes horizontally to fill the remaining portion of the first row, and fill each of the other seven rows with four horizontal dominoes. **45.** Because there is an even number of squares in all, either there is an even number of squares in each row or there is an even number of squares in each column. In the former case, tile the board in the obvious way by placing the dominoes horizontally, and in the latter case, tile the board in the obvious way by placing the dominoes vertically. **47.** We can rotate



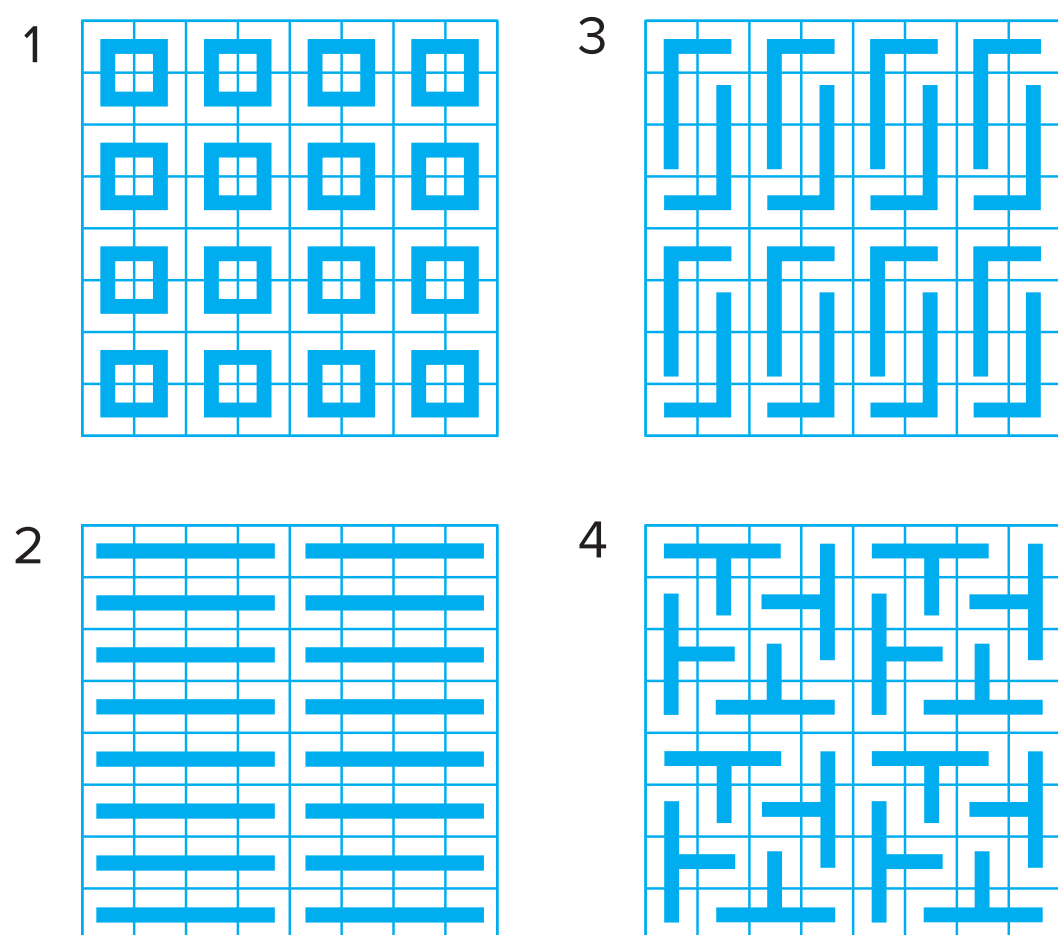
the board if necessary to make the removed squares be 1 and 16. Square 2 must be covered by a domino. If that domino is placed to cover squares 2 and 6, then the following domino placements are forced in succession: 5-9, 13-14, and 10-11, at which point there is no way to cover square 15. Otherwise, square 2 must be covered by a domino placed at 2-3. Then the following domino placements are forced: 4-8, 11-12, 6-7, 5-9, and 10-14, and again there is no way to cover square 15.

**49.** Remove the two black squares adjacent to a white corner, and remove two white squares other than that corner. Then no domino can cover that white corner.

**51. a)**



**b)** The picture shows tilings for the first four patterns.



To show that pattern 5 cannot tile the checkerboard, label the squares from 1 to 64, one row at a time from the top, from left to right in each row. Thus, square 1 is the upper left corner, and square 64 is the lower right. Suppose we did have a tiling. By symmetry and without loss of generality, we may suppose that the tile is positioned in the upper left corner, covering squares 1, 2, 10, and 11. This forces a tile to be adjacent to it on the right, covering squares 3, 4, 12, and 13. Continue in this manner and we are forced to have a tile covering squares 6, 7, 15, and 16. This makes it impossible to cover square 8. Thus, no tiling is possible.

## Supplementary Exercises

**1. a)**  $q \rightarrow p$  **b)**  $q \wedge p$  **c)**  $\neg q \vee \neg p$  **d)**  $q \leftrightarrow p$  **3. a)** The proposition cannot be false unless  $\neg p$  is false, so  $p$  is true. If  $p$  is true and  $q$  is true, then  $\neg q \wedge (p \rightarrow q)$  is false, so the conditional statement is true. If  $p$  is true and  $q$  is false, then  $p \rightarrow q$  is false, so  $\neg q \wedge (p \rightarrow q)$  is false and the conditional statement is true. **b)** The proposition cannot be false unless  $q$  is false. If  $q$  is false and  $p$  is true, then  $(p \vee q) \wedge \neg p$  is false, and the conditional statement is true. If  $q$  is false and  $p$  is

false, then  $(p \vee q) \wedge \neg p$  is false, and the conditional statement is true. **5.**  $\neg q \rightarrow \neg p$ ;  $p \rightarrow q$ ;  $\neg p \rightarrow \neg q$  **7.**  $(p \wedge q \wedge r \wedge \neg s) \vee (p \wedge q \wedge \neg r \wedge s) \vee (p \wedge \neg q \wedge r \wedge s) \vee (\neg p \wedge q \wedge r \wedge s)$  **9.** Translating these statements into symbols, using the obvious letters, we have  $\neg t \rightarrow \neg g$ ,  $\neg g \rightarrow \neg q$ ,  $r \rightarrow q$ , and  $\neg t \wedge r$ . Assume the statements are consistent. The fourth statement tells us that  $\neg t$  must be true. Therefore, by modus ponens with the first statement, we know that  $\neg g$  is true, hence (from the second statement), that  $\neg q$  is true. Also, the fourth statement tells us that  $r$  must be true, and so again modus ponens (third statement) makes  $q$  true. This is a contradiction:  $q \wedge \neg q$ . Thus, the statements are inconsistent. **11.** Reject-accept-reject-accept, accept-accept-accept-accept, accept-accept-reject-accept, reject-reject-reject-reject, reject-reject-accept-reject, and reject-accept-accept-accept **13.** Aaron is a knave and Crystal is a knight; it cannot be determined what Bohan is. **15.** Brenda **17.** The premises cannot both be true, because they are contradictory. Therefore, it is (vacuously) true that whenever all the premises are true, the conclusion is also true, which by definition makes this a valid argument. Because the premises are not both true, we cannot conclude that the conclusion is true. **19.** Use the same propositions as were given in Section 1.3 for a  $9 \times 9$  Sudoku puzzle, with the variables indexed from 1 to 16, instead of from 1 to 9, and with a similar change for the propositions for the  $4 \times 4$  blocks:  $\bigwedge_{r=0}^3 \bigwedge_{s=0}^3 \bigwedge_{n=1}^{16} \bigvee_{i=1}^4 \bigvee_{j=1}^4 p(4r+i, 4s+j, n)$ . **21. a)** F **b)** T **c)** F **d)** T **e)** F **f)** T **23.** Many answers are possible. One example is United States senators. **25.**  $\forall x \exists y \exists z (y \neq z \wedge \forall w (P(w, x) \leftrightarrow (w = y \vee w = z)))$  **27. a)**  $\neg \exists x P(x)$  **b)**  $\exists x (P(x) \wedge \forall y (P(y) \rightarrow y = x))$  **c)**  $\exists x_1 \exists x_2 (P(x_1) \wedge P(x_2) \wedge x_1 \neq x_2 \wedge \forall y (P(y) \rightarrow (y = x_1 \vee y = x_2)))$  **d)**  $\exists x_1 \exists x_2 \exists x_3 (P(x_1) \wedge P(x_2) \wedge P(x_3) \wedge x_1 \neq x_2 \wedge x_1 \neq x_3 \wedge x_2 \neq x_3 \wedge \forall y (P(y) \rightarrow (y = x_1 \vee y = x_2 \vee y = x_3)))$  **29.** Suppose that  $\exists x (P(x) \rightarrow Q(x))$  is true. Then either  $Q(x_0)$  is true for some  $x_0$ , in which case  $\forall x P(x) \rightarrow \exists x Q(x)$  is true; or  $P(x_0)$  is false for some  $x_0$ , in which case  $\forall x P(x) \rightarrow \exists x Q(x)$  is true. Conversely, suppose that  $\exists x (P(x) \rightarrow Q(x))$  is false. That means that  $\forall x (P(x) \wedge \neg Q(x))$  is true, which implies  $\forall x P(x)$  and  $\forall x (\neg Q(x))$ . This latter proposition is equivalent to  $\neg \exists x Q(x)$ . Thus,  $\forall x P(x) \rightarrow \exists x Q(x)$  is false. **31.** No **33.**  $\forall x \forall z \exists y T(x, y, z)$ , where  $T(x, y, z)$  is the statement that student  $x$  has taken class  $y$  in department  $z$ , where the domains are the set of students in the class, the set of courses at this university, and the set of departments in the school of mathematical sciences **35.**  $\exists! x \exists! y T(x, y)$  and  $\exists x \forall z ((\exists y \forall w (T(z, w) \leftrightarrow w = y)) \leftrightarrow z = x)$ , where  $T(x, y)$  means that student  $x$  has taken class  $y$  and the domain is all students in this class **37.**  $P(a) \rightarrow Q(a)$  and  $Q(a) \rightarrow R(a)$  by universal instantiation; then  $\neg Q(a)$  by modus tollens and  $\neg P(a)$  by modus tollens **39.** This is not true. Let  $x = 2^{1/3}$ . Then  $x^2 = 4^{1/3}$  is irrational (the proof of this is very similar to the proof in Example 11 in Section 1.7), but  $x^3 = 2$  is rational. **41.** We can give a constructive proof by letting  $m = 10^{500} + 1$ . Then  $m^2 = (10^{500} + 1)^2 > (10^{500})^2 = 10^{1000}$ . **43.** 23 cannot be written as the sum of eight cubes. **45.** 223 cannot be written as the sum of 36 fifth powers.

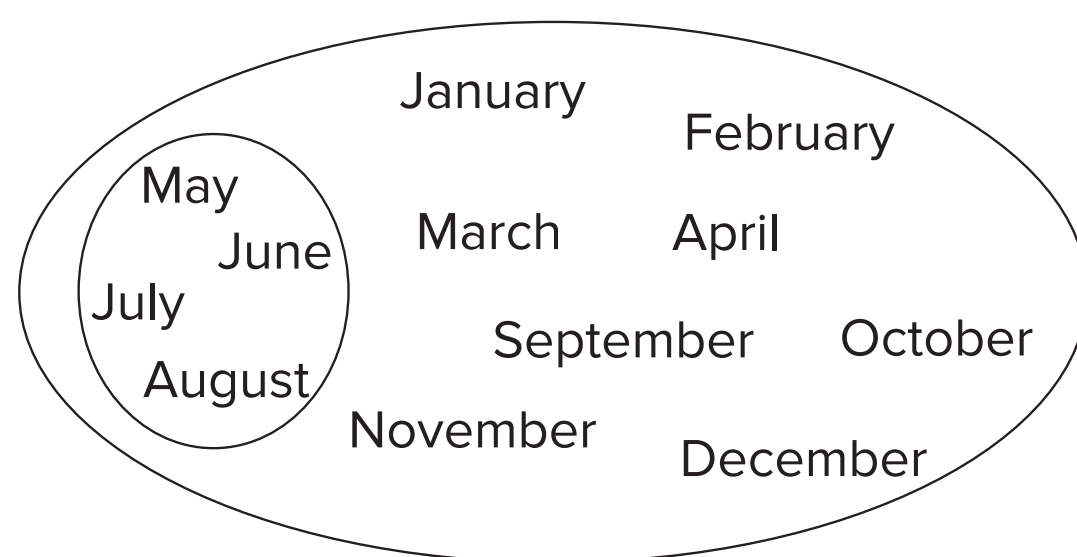


## CHAPTER 2

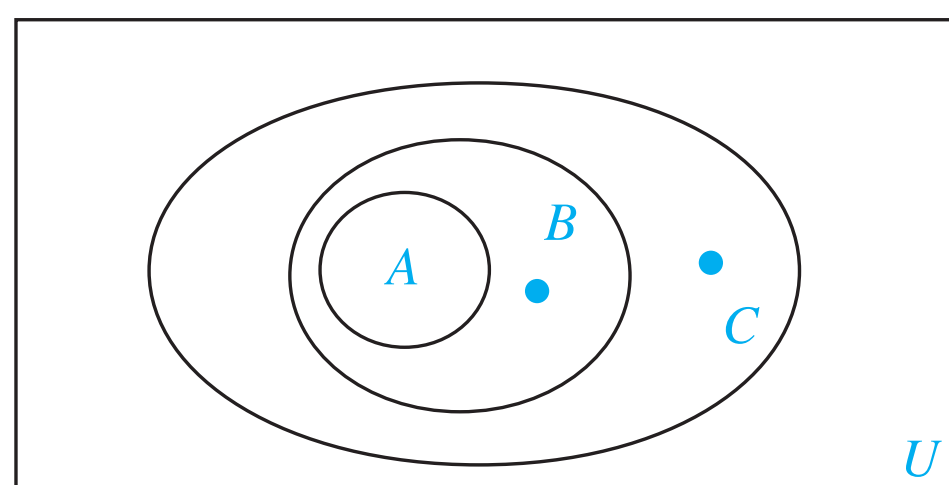
### Section 2.1

1. a)  $\{-1, 1\}$  b)  $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}$  c)  $\{0, 1, 4, 9, 16, 25, 36, 49, 64, 81\}$  d)  $\emptyset$  3. a)  $[0, 5)$ ,  $[0, 5]$  b)  $(0, 5)$ ,  $(0, 5]$ ,  $[0, 5)$ ,  $[0, 5]$  c)  $(0, 5)$ ,  $(0, 5]$ ,  $[0, 5)$ ,  $[0, 5]$ ,  $(1, 4]$ ,  $[2, 3]$  d)  $(0, 5)$ ,  $(0, 5]$ ,  $[0, 5)$ ,  $[0, 5]$ ,  $(1, 4]$ ,  $[2, 3]$  e)  $(0, 5)$ ,  $(0, 5]$ ,  $[0, 5)$ ,  $[0, 5]$ ,  $(1, 4]$  f)  $(0, 5]$ ,  $[0, 5]$  5. a) The second is a subset of the first, but the first is not a subset of the second. b) Neither is a subset of the other. c) The first is a subset of the second, but the second is not a subset of the first. 7. a) Yes b) No c) No 9. a) Yes b) No c) Yes d) No e) No f) No 11. a) False b) False c) False d) True e) False f) False g) True 13. a) True b) True c) False d) True e) True f) False

15.



17. The dots in certain regions indicate that those regions are not empty.



19. Suppose that  $x \in A$ . Because  $A \subseteq B$ , this implies that  $x \in B$ . Because  $B \subseteq C$ , we see that  $x \in C$ . Because  $x \in A$  implies that  $x \in C$ , it follows that  $A \subseteq C$ . 21. a) 1 b) 1 c) 2 d) 3 23. a)  $\{\emptyset, \{a\}\}$  b)  $\{\emptyset, \{a\}, \{b\}, \{a, b\}\}$  c)  $\{\emptyset, \{\emptyset\}, \{\{\emptyset\}\}, \{\emptyset, \{\emptyset\}\}\}$  25. a) 8 b) 16 c) 2 27. For the “if” part, given  $A \subseteq B$ , we want to show that  $\mathcal{P}(A) \subseteq \mathcal{P}(B)$ , i.e., if  $C \subseteq A$  then  $C \subseteq B$ . But this follows directly from Exercise 19. For the “only if” part, given that  $\mathcal{P}(A) \subseteq \mathcal{P}(B)$ , we want to show that  $A \subseteq B$ . Suppose  $a \in A$ . Then  $\{a\} \subseteq A$ , so  $\{a\} \in \mathcal{P}(A)$ . Since  $\mathcal{P}(A) \subseteq \mathcal{P}(B)$ , it follows that  $\{a\} \in \mathcal{P}(B)$ , which means that  $\{a\} \subseteq B$ . But this implies  $a \in B$ , as desired. 29. a)  $\{(a, y), (b, y), (c, y), (d, y), (a, z), (b, z), (c, z), (d, z)\}$  b)  $\{(y, a), (y, b), (y, c), (y, d), (z, a), (z, b), (z, c), (z, d)\}$  31. The set of triples  $(a, b, c)$ , where  $a$  is an airline and  $b$  and  $c$  are cities. A useful subset of this set is the set of triples  $(a, b, c)$  for which  $a$  flies between  $b$  and  $c$ . 33.  $\emptyset \times A = \{(x, y) \mid x \in \emptyset \text{ and } y \in A\} = \emptyset = \{(x, y) \mid x \in A \text{ and } y \in \emptyset\} = A \times \emptyset$  35. a)  $\{(0, 0), (0, 1), (0, 3), (1, 0), (1, 1), (1, 3), (3, 0), (3, 1), (3, 3)\}$  b)  $\{(1, 1), (1, 2), (1, a), (1, b), (2, 1), (2, 2), (2, a), (2, b), (a, 1), (a, 2), (a, a), (a, b), (b, 1), (b, 2), (b, a), (b, b)\}$  37.  $mn$  39.  $m^n$  41. The elements of  $A \times B \times C$  consist of 3-tuples  $(a, b, c)$ , where  $a \in A$ ,  $b \in B$ , and  $c \in C$ , whereas the elements of  $(A \times B) \times C$  look

like  $((a, b), c)$ —ordered pairs, the first coordinate of which is again an ordered pair. 43. This is not true. The simplest counterexample is to let  $A = B = \emptyset$ . Then  $A \times B = \emptyset$  and  $\mathcal{P}(A \times B) = \{\emptyset\}$ , whereas  $\mathcal{P}(A) = \mathcal{P}(B) = \{\emptyset\}$  and  $\mathcal{P}(A) \times \mathcal{P}(B) = \{(\emptyset, \emptyset)\}$ . Thus,  $\mathcal{P}(A \times B) \neq \mathcal{P}(A) \times \mathcal{P}(B)$ . 45. a) The square of a real number is never  $-1$ . True b) There exists an integer whose square is 2. False c) The square of every integer is positive. False d) There is a real number equal to its own square. True 47. a)  $\{-1, 0, 1\}$  b)  $\mathbb{Z} - \{0, 1\}$  c)  $\emptyset$  49. We must show that  $\{\{a\}, \{a, b\}\} = \{\{c\}, \{c, d\}\}$  if and only if  $a = c$  and  $b = d$ . The “if” part is immediate. So assume these two sets are equal. First, consider the case when  $a \neq b$ . Then  $\{\{a\}, \{a, b\}\}$  contains exactly two elements, one of which contains one element. Thus,  $\{\{c\}, \{c, d\}\}$  must have the same property, so  $c \neq d$  and  $\{c\}$  is the element containing exactly one element. Hence,  $\{a\} = \{c\}$ , which implies that  $a = c$ . Also, the two-element sets  $\{a, b\}$  and  $\{c, d\}$  must be equal. Because  $a = c$  and  $a \neq b$ , it follows that  $b = d$ . Second, suppose that  $a = b$ . Then  $\{\{a\}, \{a, b\}\} = \{\{a\}\}$ , a set with one element. Hence,  $\{\{c\}, \{c, d\}\}$  has only one element, which can happen only when  $c = d$ , and the set is  $\{\{c\}\}$ . It then follows that  $a = c$  and  $b = d$ . 51. Let  $S = \{a_1, a_2, \dots, a_n\}$ . Represent each subset of  $S$  with a bit string of length  $n$ , where the  $i$ th bit is 1 if and only if  $a_i \in S$ . To generate all subsets of  $S$ , list all  $2^n$  bit strings of length  $n$  (for instance, in increasing order), and write down the corresponding subsets.

### Section 2.2

1. a) The set of students who live within one mile of school and walk to classes b) The set of students who live within one mile of school or walk to classes (or do both) c) The set of students who live within one mile of school but do not walk to classes d) The set of students who walk to classes but live more than one mile away from school 3. a)  $\{0, 1, 2, 3, 4, 5, 6\}$  b)  $\{3\}$  c)  $\{1, 2, 4, 5\}$  d)  $\{0, 6\}$  5.  $\overline{\overline{A}} = \{x \mid \neg(x \in \overline{A})\} = \{x \mid \neg(\neg(x \in A))\} = \{x \mid x \in A\} = A$  7. a)  $A \cup U = \{x \mid x \in A \vee x \in U\} = \{x \mid x \in A \vee \mathbf{T}\} = \{x \mid \mathbf{T}\} = U$  b)  $A \cap \emptyset = \{x \mid x \in A \wedge x \in \emptyset\} = \{x \mid x \in A \wedge \mathbf{F}\} = \{x \mid \mathbf{F}\} = \emptyset$  9. a)  $A \cup \overline{A} = \{x \mid x \in A \vee x \notin A\} = U$  b)  $A \cap \overline{A} = \{x \mid x \in A \wedge x \notin A\} = \emptyset$  11. a)  $A \cup B = \{x \mid x \in A \vee x \in B\} = \{x \mid x \in B \vee x \in A\} = B \cup A$  b)  $A \cap B = \{x \mid x \in A \wedge x \in B\} = \{x \mid x \in B \wedge x \in A\} = B \cap A$  13. Suppose  $x \in A \cap (A \cup B)$ . Then  $x \in A$  and  $x \in A \cup B$  by the definition of intersection. Because  $x \in A$ , we have proved that the left-hand side is a subset of the right-hand side. Conversely, let  $x \in A$ . Then by the definition of union,  $x \in A \cup B$  as well. Therefore,  $x \in A \cap (A \cup B)$  by the definition of intersection, so the right-hand side is a subset of the left-hand side. 15. a)  $x \in \overline{A \cup B} \equiv x \notin A \cup B \equiv \neg(x \in A \vee x \in B) \equiv \neg(x \in A) \wedge \neg(x \in B) \equiv x \notin A \wedge x \notin B \equiv x \in \overline{A} \wedge x \in \overline{B} \equiv x \in \overline{A} \cap \overline{B}$



b)

| A | B | $A \cup B$ | $\overline{A \cup B}$ | $\bar{A}$ | $\bar{B}$ | $\overline{A \cap B}$ |
|---|---|------------|-----------------------|-----------|-----------|-----------------------|
| 1 | 1 | 1          | 0                     | 0         | 0         | 0                     |
| 1 | 0 | 1          | 0                     | 0         | 1         | 0                     |
| 0 | 1 | 1          | 0                     | 1         | 0         | 0                     |
| 0 | 0 | 0          | 1                     | 1         | 1         | 1                     |

**17.** Suppose  $A \subseteq B$ . We must show that every element  $x$  of  $U$  is an element of  $\overline{A \cup B}$ . Either  $x \in \bar{A}$  or  $x \in A$ , and if  $x \in A$  then  $x \in B$ . Thus,  $x \in \overline{A \cup B}$  in all cases. Conversely, suppose that  $\overline{A \cup B} = U$ , and let  $x \in A$ . Then  $x \notin \bar{A}$ , so it must be that  $x \in B$ . This shows that  $A \subseteq B$ , and the proof is complete.

**19. a)**  $x \in \overline{A \cap B \cap C} \equiv x \notin A \cap B \cap C \equiv x \notin A \vee x \notin B \vee x \notin C \equiv x \in \bar{A} \vee x \in \bar{B} \vee x \in \bar{C} \equiv x \in \overline{A \cup B \cup C}$

b)

| A | B | C | $A \cap B \cap C$ | $\overline{A \cap B \cap C}$ | $\bar{A}$ | $\bar{B}$ | $\bar{C}$ | $\overline{A \cup B \cup C}$ |
|---|---|---|-------------------|------------------------------|-----------|-----------|-----------|------------------------------|
| 1 | 1 | 1 | 1                 | 0                            | 0         | 0         | 0         | 0                            |
| 1 | 1 | 0 | 0                 | 1                            | 0         | 0         | 1         | 1                            |
| 1 | 0 | 1 | 0                 | 1                            | 0         | 1         | 0         | 1                            |
| 1 | 0 | 0 | 0                 | 1                            | 0         | 1         | 1         | 1                            |
| 0 | 1 | 1 | 0                 | 1                            | 1         | 0         | 0         | 1                            |
| 0 | 1 | 0 | 0                 | 1                            | 1         | 0         | 1         | 1                            |
| 0 | 0 | 1 | 0                 | 1                            | 1         | 1         | 0         | 1                            |
| 0 | 0 | 0 | 0                 | 1                            | 1         | 1         | 1         | 1                            |

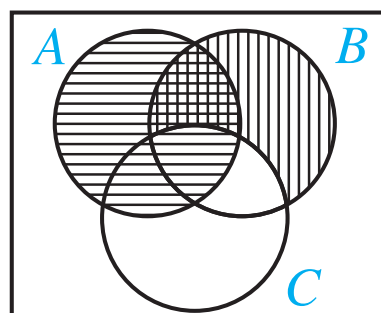
**21. a)** Both sides equal  $\{x \mid x \in A \wedge x \notin B\}$ . **b)**  $A = A \cap U = A \cap (B \cup \bar{B}) = (A \cap B) \cup (A \cap \bar{B})$

**23.**  $x \in A \cup (B \cup C) \equiv (x \in A) \vee (x \in (B \cup C)) \equiv (x \in A) \vee (x \in B \vee x \in C) \equiv (x \in A \vee x \in B) \vee (x \in C) \equiv x \in (A \cup B) \cup C$

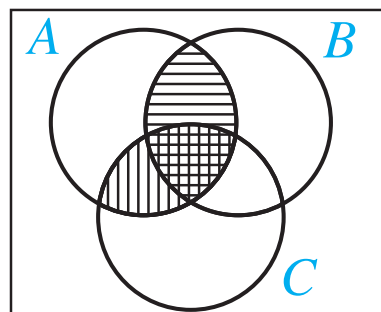
**25.**  $x \in A \cup (B \cap C) \equiv (x \in A) \vee (x \in (B \cap C)) \equiv (x \in A) \vee (x \in B \wedge x \in C) \equiv (x \in A \vee x \in B) \wedge (x \in A \vee x \in C) \equiv x \in (A \cup B) \cap (A \cup C)$

**27. a)**  $\{4, 6\}$  **b)**  $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$  **c)**  $\{4, 5, 6, 8, 10\}$

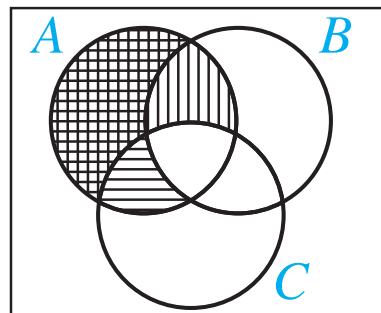
**d)**  $\{0, 2, 4, 5, 6, 7, 8, 9, 10\}$  **29. a)** The double-shaded portion is the desired set.



**b)** The desired set is the entire shaded portion.



**c)** The desired set is the entire shaded portion.



**31. a)**  $B \subseteq A$  **b)**  $A \subseteq B$  **c)**  $A \cap B = \emptyset$  **d)** Nothing, because this is always true **e)**  $A = B$  **33.**  $A \subseteq B \equiv \forall x(x \in A \rightarrow x \in B) \equiv \forall x(x \notin B \rightarrow x \notin A) \equiv \forall x(x \in \bar{B} \rightarrow x \in \bar{A}) \equiv \bar{B} \subseteq \bar{A}$  **35.** By De Morgan's law, the left-hand side equals  $(\bar{A} \cap \bar{B}) \cap (\bar{B} \cap \bar{C}) \cap (\bar{A} \cap \bar{C})$ . By the commutative, associative, and idempotent laws, this simplifies to the

right-hand side. **37. a)** Let  $(x, y) \in A \times (B - C)$ , which means that  $x \in A$  and  $y$  is an element of  $B$  but not  $C$ . Thus,  $(x, y) \in A \times B$  and  $(x, y) \notin A \times C$ , so by the definition of set difference,  $(x, y) \in (A \times B) - (A \times C)$ . Conversely, let  $(x, y) \in (A \times B) - (A \times C)$ . Then  $(x, y) \in A \times B$  and  $(x, y) \notin A \times C$ . Thus,  $x \in A$  and  $y \in B$ , and because  $x \in A$ , it must be that  $y \notin C$ . This implies that  $y \in B - C$ , so indeed  $(x, y) \in A \times (B - C)$ . **b)** Note that the complement in the right-hand side must mean with respect to  $U \times U$ . This is not true. For example, let  $U = \{a, b\}$ ,  $A = \{a\}$ ,  $B = \{b\}$ , and  $C = \emptyset$ . Then the left-hand side is  $\{(b, a)\}$ , whereas the right-hand side is  $\{(a, a), (b, a), (b, b)\}$ .

**39.** The set of students who are computer science majors but not mathematics majors or who are mathematics majors but not computer science majors **41.** An element is in  $(A \cup B) - (A \cap B)$  if it is in the union of  $A$  and  $B$  but not in the intersection of  $A$  and  $B$ , which means that it is in either  $A$  or  $B$  but not in both  $A$  and  $B$ . This is exactly what it means for an element to belong to  $A \oplus B$ . **43. a)**  $A \oplus A = (A - A) \cup (A - A) = \emptyset \cup \emptyset = \emptyset$  **b)**  $A \oplus \emptyset = (A - \emptyset) \cup (\emptyset - A) = A \cup \emptyset = A$  **c)**  $A \oplus U = (A - U) \cup (U - A) = \emptyset \cup \bar{A} = \bar{A}$  **d)**  $A \oplus \bar{A} = (A - \bar{A}) \cup (\bar{A} - A) = A \cup \bar{A} = U$  **45.**  $B = \emptyset$  **47.** Yes **49.** Yes

**51.** If  $A \cup B$  were finite, then it would have  $n$  elements for some natural number  $n$ . But  $A$  already has more than  $n$  elements, because it is infinite, and  $A \cup B$  has all the elements that  $A$  has, so  $A \cup B$  has more than  $n$  elements. This contradiction shows that  $A \cup B$  must be infinite. **53. a)**  $\{1, 2, 3, \dots, n\}$  **b)**  $\{1\}$

**55. a)**  $A_n$  **b)**  $\{0, 1\}$  **57. a)**  $\mathbb{Z}$ ,  $\{-1, 0, 1\}$  **b)**  $\mathbb{Z} - \{0\}$ ,  $\emptyset$  **c)**  $\mathbb{R}$ ,  $[-1, 1]$  **d)**  $[1, \infty)$ ,  $\emptyset$  **59. a)**  $\{1, 2, 3, 4, 7, 8, 9, 10\}$  **b)**  $\{2, 4, 5, 6, 7\}$  **c)**  $\{1, 10\}$

**61.** The bit in the  $i$ th position of the bit string of the difference of two sets is 1 if the  $i$ th bit of the first string is 1 and the  $i$ th bit of the second string is 0, and is 0 otherwise. **63. a)** 11 1110 0000 0000 0000 0000 0000 0100 0101 0000 = 11 1110 1000 0000 0100 0101 0000, representing  $\{a, b, c, d, e, g, p, t, v\}$

**b)** 11 1110 0000 0000 0000 0000 0000 0100 0101 0000 = 01 1100 1000 0000 0100 0101 0000 = 01 1100 0000 0000 0000 0000 0000, representing  $\{b, c, d\}$

**c)**  $(11 1110 0000 0000 0000 0000 0000 0000 \vee 00 0110 0110 0001 1000 0110 0110) \wedge (01 1100 1000 0000 0100 0101 0000 \vee 00 1010 0010 0000 1000 0010 0111) =$

$11 1110 0110 0001 1000 0110 0110 \wedge 01 1110 1010 0000 1100 0111 0111 = 01 1110 0010 0000 1000 0110 0110$ , representing  $\{b, c, d, e, i, o, t, u, x, y\}$

**d)**  $11 1110 0000 0000 0000 0000 0000 0000 \vee 01 1100 1000 0000 0100 0101 0000 \vee 00 1010 0010 0000 1000 0010 0111 \vee 00 0110 0110 0001 1000 0110 0110 =$

$11 1110 1110 0001 1100 0111 0111$ , representing  $\{a, b, c, d, e, g, h, i, n, o, p, t, u, v, x, y, z\}$

**65. a)**  $\{1, 2, 3, \{1, 2, 3\}\}$  **b)**  $\{\emptyset\}$  **c)**  $\{\emptyset, \{\emptyset\}\}$  **d)**  $\{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}$  **67. a)**  $\{3 \cdot a, 3 \cdot b, 1 \cdot c, 4 \cdot d\}$  **b)**  $\{2 \cdot a, 2 \cdot b\}$  **c)**  $\{1 \cdot a, 1 \cdot c\}$  **d)**  $\{1 \cdot b, 4 \cdot d\}$

**e)**  $\{5 \cdot a, 5 \cdot b, 1 \cdot c, 4 \cdot d\}$  **69.** These all follow from the definitions of the multiset operations. **a)** False, because the union is actually  $\{a, a, a\}$  **b)** False, because the union is actually  $\{a, a, a\}$  **c)** True **d)** False, because the correct intersection is as stated in part (c) **e)** True

**71. a)** 0, 1 **b)**  $1/3, 2/3$  **c)** 1, 0 **d)**  $1/6, 5/6$  **73.**  $\bar{F} = \{0.4 \text{ Alice}, 0.1 \text{ Brian}, 0.6 \text{ Fred}, 0.9 \text{ Oscar}, 0.5 \text{ Rita}\}$ ,  $\bar{R} = \{0.6 \text{ Alice}, 0.2 \text{ Brian}, 0.8 \text{ Fred}, 0.1 \text{ Oscar},$

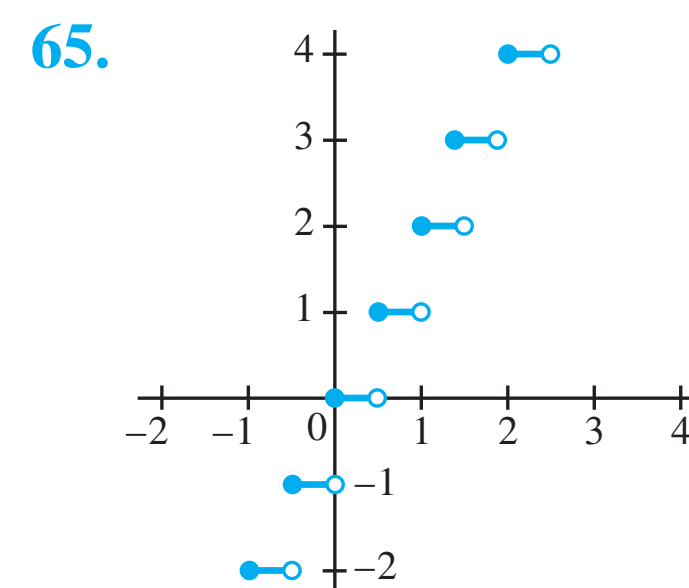


0.3 Rita} **75.** {0.4 Alice, 0.8 Brian, 0.2 Fred, 0.1 Oscar, 0.5 Rita}

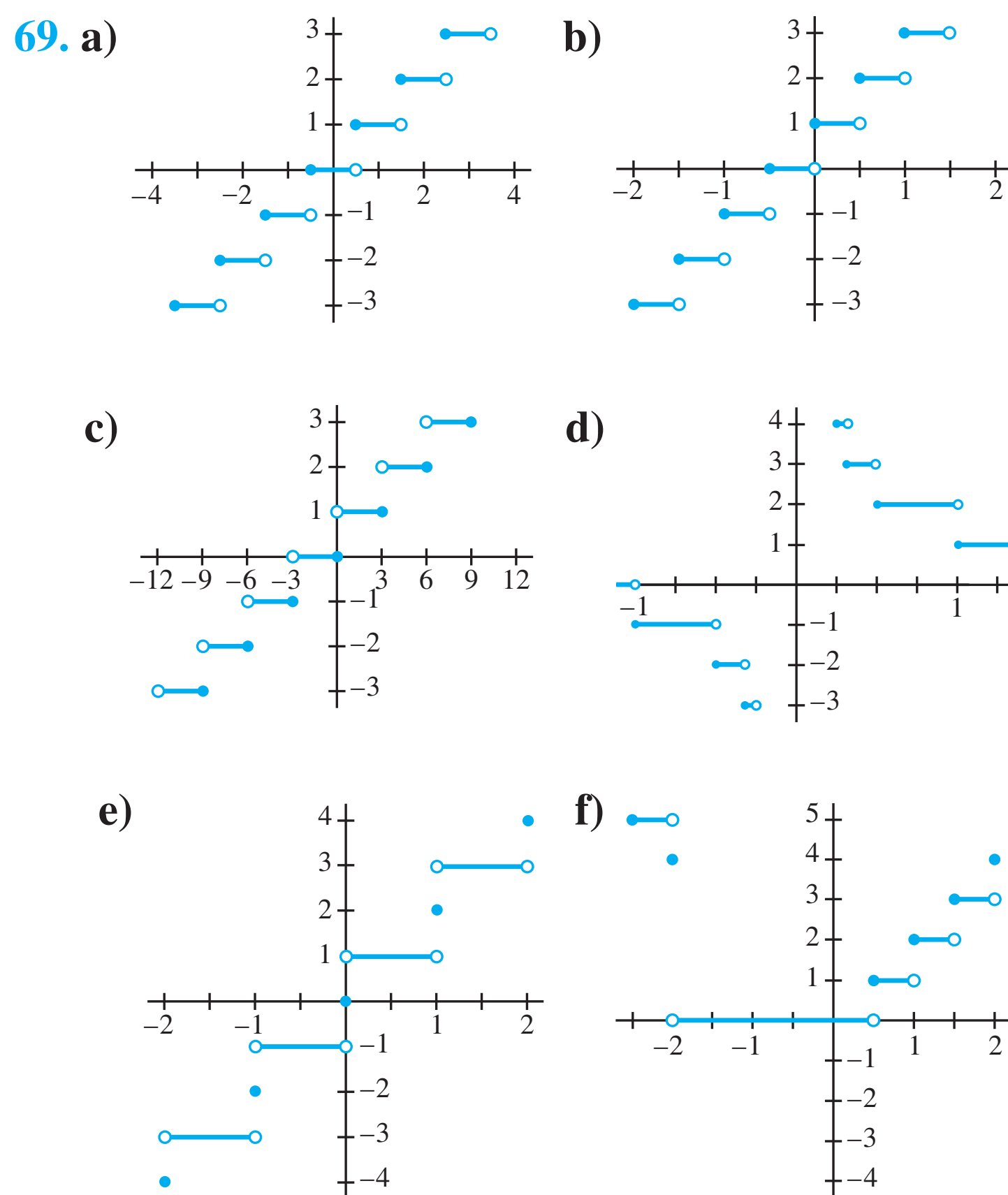
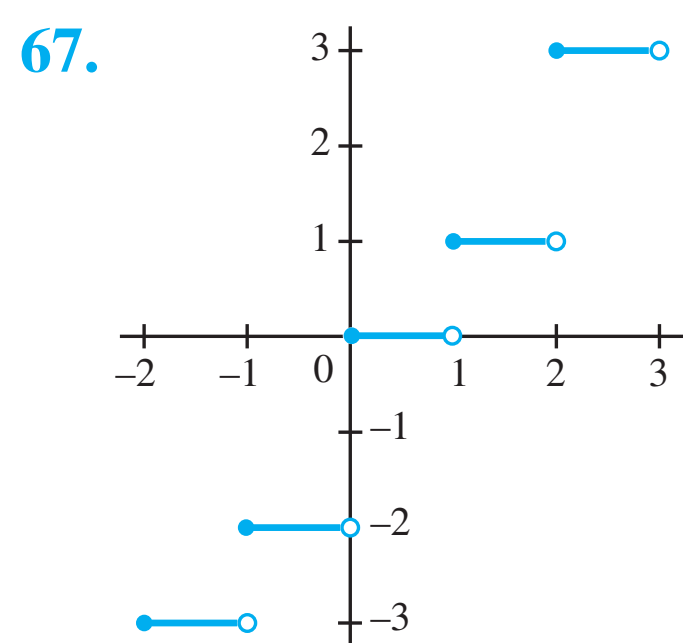
## Section 2.3

**1. a)**  $f(0)$  is not defined. **b)**  $f(x)$  is not defined for  $x < 0$ . **c)**  $f(x)$  is not well defined because there are two distinct values assigned to each  $x$ . **3. a)** Not a function **b)** A function **c)** Not a function **5. a)** Domain the set of bit strings; range the set of integers **b)** Domain the set of bit strings; range the set of even nonnegative integers **c)** Domain the set of bit strings; range the set of nonnegative integers not exceeding 7 **d)** Domain the set of positive integers; range the set of squares of positive integers =  $\{1, 4, 9, 16, \dots\}$  **7. a)** Domain  $\mathbf{Z}^+ \times \mathbf{Z}^+$ ; range  $\mathbf{Z}^+$  **b)** Domain  $\mathbf{Z}^+$ ; range  $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$  **c)** Domain the set of bit strings; range  $\mathbf{N}$  **d)** Domain the set of bit strings; range  $\mathbf{N}$  **9. a)** 1 **b)** 0 **c)** 0 **d)** -1 **e)** 3 **f)** -1 **g)** 2 **h)** 1 **11.** Only the function in part (a) **13.** Only the functions in parts (a) and (d) **15. a)** Onto **b)** Not onto **c)** Onto **d)** Not onto **e)** Onto **17. a)** Depends on whether teachers share offices **b)** One-to-one assuming only one teacher per bus **c)** Most likely not one-to-one, especially if salary is set by a collective bargaining agreement **d)** One-to-one **19.** Answers will vary. **a)** Set of offices at the school; probably not onto **b)** Set of buses going on the trip; onto, assuming every bus gets a teacher chaperone **c)** Set of real numbers; not onto **d)** Set of strings of nine digits with hyphens after third and fifth digits; not onto **21. a)** The function  $f(x)$  with  $f(x) = 3x + 1$  when  $x \geq 0$  and  $f(x) = -3x + 2$  when  $x < 0$  **b)**  $f(x) = |x| + 1$  **c)** The function  $f(x)$  with  $f(x) = 2x + 1$  when  $x \geq 0$  and  $f(x) = -2x$  when  $x < 0$  **d)**  $f(x) = x^2 + 1$  **23. a)** Yes **b)** No **c)** Yes **d)** No **25.** Suppose that  $f$  is strictly decreasing. This means that  $f(x) > f(y)$  whenever  $x < y$ . To show that  $g$  is strictly increasing, suppose that  $x < y$ . Then  $g(x) = 1/f(x) < 1/f(y) = g(y)$ . Conversely, suppose that  $g$  is strictly increasing. This means that  $g(x) < g(y)$  whenever  $x < y$ . To show that  $f$  is strictly decreasing, suppose that  $x < y$ . Then  $f(x) = 1/g(x) > 1/g(y) = f(y)$ . **27. a)** Let  $f$  be a given strictly decreasing function from  $\mathbf{R}$  to itself. If  $a < b$ , then  $f(a) > f(b)$ ; if  $a > b$ , then  $f(a) < f(b)$ . Thus, if  $a \neq b$ , then  $f(a) \neq f(b)$ . **b)** Answers will vary; for example,  $f(x) = 0$  for  $x < 0$  and  $f(x) = -x$  for  $x \geq 0$ . **29.** The function is not one-to-one, so it is not invertible. On the restricted domain, the function is the identity function on the nonnegative real numbers,  $f(x) = x$ , so it is its own inverse. **31. a)**  $f(S) = \{0, 1, 3\}$  **b)**  $f(S) = \{0, 1, 3, 5, 8\}$  **c)**  $f(S) = \{0, 8, 16, 40\}$  **d)**  $f(S) = \{1, 12, 33, 65\}$  **33. a)** Let  $x$  and  $y$  be distinct elements of  $A$ . Because  $g$  is one-to-one,  $g(x)$  and  $g(y)$  are distinct elements of  $B$ . Because  $f$  is one-to-one,  $f(g(x)) = (f \circ g)(x)$  and  $f(g(y)) = (f \circ g)(y)$  are distinct elements of  $C$ . Hence,  $f \circ g$  is one-to-one. **b)** Let  $y \in C$ . Because  $f$  is onto,  $y = f(b)$  for some  $b \in B$ . Now because  $g$  is onto,  $b = g(x)$  for some  $x \in A$ . Hence,  $y = f(b) = f(g(x)) = (f \circ g)(x)$ . It follows that  $f \circ g$  is onto. **35.** Let  $A = \{a\}$ ,  $B = \{b_1, b_2\}$ ,  $C = \{c\}$ ,  $g(a) = b_1$ , and  $f(b_1) = f(b_2) = c$ . **37.** No. For example, suppose that  $A = \{a\}$ ,  $B = \{b, c\}$ , and  $C = \{d\}$ .

Let  $g(a) = b$ ,  $f(b) = d$ , and  $f(c) = d$ . Then  $f$  and  $f \circ g$  are onto, but  $g$  is not. **39.**  $(f + g)(x) = x^2 + x + 3$ ,  $(fg)(x) = x^3 + 2x^2 + x + 2$  **41.**  $f$  is one-to-one because  $f(x_1) = f(x_2) \rightarrow ax_1 + b = ax_2 + b \rightarrow ax_1 = ax_2 \rightarrow x_1 = x_2$ .  $f$  is onto because  $f((y - b)/a) = y$ .  $f^{-1}(y) = (y - b)/a$ . **43. a)**  $A = B = \mathbf{R}$ ,  $S = \{x \mid x > 0\}$ ,  $T = \{x \mid x < 0\}$ ,  $f(x) = x^2$  **b)** It suffices to show that  $f(S) \cap f(T) \subseteq f(S \cap T)$ . Let  $y \in B$  be an element of  $f(S) \cap f(T)$ . Then  $y \in f(S)$ , so  $y = f(x_1)$  for some  $x_1 \in S$ . Similarly,  $y = f(x_2)$  for some  $x_2 \in T$ . Because  $f$  is one-to-one, it follows that  $x_1 = x_2$ . Therefore,  $x_1 \in S \cap T$ , so  $y \in f(S \cap T)$ . **45. a)**  $\{x \mid 0 \leq x < 1\}$  **b)**  $\{x \mid -1 \leq x < 2\}$  **c)**  $\emptyset$  **47.**  $f^{-1}(\bar{S}) = \{x \in A \mid f(x) \notin S\} = \overline{\{x \in A \mid f(x) \in S\}} = \overline{f^{-1}(S)}$  **49.** Let  $x = [x] + \epsilon$ , where  $\epsilon$  is a real number with  $0 \leq \epsilon < 1$ . If  $\epsilon < \frac{1}{2}$ , then  $[x] - 1 < x - \frac{1}{2} < [x]$ , so  $[x - \frac{1}{2}] = [x]$  and this is the integer closest to  $x$ . If  $\epsilon > \frac{1}{2}$ , then  $[x] < x - \frac{1}{2} < [x] + 1$ , so  $[x - \frac{1}{2}] = [x] + 1$  and this is the integer closest to  $x$ . If  $\epsilon = \frac{1}{2}$ , then  $[x - \frac{1}{2}] = [x]$ , which is the smaller of the two integers that surround  $x$  and are the same distance from  $x$ . **51.** Write the real number  $x$  as  $[x] + \epsilon$ , where  $\epsilon$  is a real number with  $0 \leq \epsilon < 1$ . Because  $\epsilon = x - [x]$ , it follows that  $0 \leq -[x] < 1$ . The first two inequalities,  $x - 1 < [x]$  and  $[x] \leq x$ , follow directly. For the other two inequalities, write  $x = [x] - \epsilon'$ , where  $0 \leq \epsilon' < 1$ . Then  $0 \leq [x] - x < 1$ , and the desired inequality follows. **53. a)** If  $x < n$ , because  $[x] \leq x$ , it follows that  $[x] < n$ . Suppose that  $x \geq n$ . By the definition of the floor function, it follows that  $[x] \geq n$ . This means that if  $[x] < n$ , then  $x < n$ . **b)** If  $n < x$ , then because  $x \leq [x]$ , it follows that  $n \leq [x]$ . Suppose that  $n \geq x$ . By the definition of the ceiling function, it follows that  $[x] \leq n$ . This means that if  $n < [x]$ , then  $n < x$ . **55.** If  $n$  is even, then  $n = 2k$  for some integer  $k$ . Thus,  $[n/2] = [k] = k = n/2$ . If  $n$  is odd, then  $n = 2k + 1$  for some integer  $k$ . Thus,  $[n/2] = [k + \frac{1}{2}] = k = (n - 1)/2$ . **57.** Assume that  $x \geq 0$ . The left-hand side is  $\lceil -x \rceil$  and the right-hand side is  $-[x]$ . If  $x$  is an integer, then both sides equal  $-x$ . Otherwise, let  $x = n + \epsilon$ , where  $n$  is a natural number and  $\epsilon$  is a real number with  $0 \leq \epsilon < 1$ . Then  $\lceil -x \rceil = \lceil -n - \epsilon \rceil = -n$  and  $-[x] = -[n + \epsilon] = -n$  also. When  $x < 0$ , the equation also holds because it can be obtained by substituting  $-x$  for  $x$ . **59.**  $[b] - [a] - 1$  **61. a)** 1 **b)** 3 **c)** 126 **d)** 3600 **63. a)** 100 **b)** 256 **c)** 1030 **d)** 30,200







g) See part (a). **71.**  $f^{-1}(y) = (y - 1)^{1/3}$  **73. a)**  $f_{A \cap B}(x) = 1 \leftrightarrow x \in A \cap B \leftrightarrow x \in A \text{ and } x \in B \leftrightarrow f_A(x) = 1 \text{ and } f_B(x) = 1 \leftrightarrow f_A(x)f_B(x) = 1$  **b)**  $f_{A \cup B}(x) = 1 \leftrightarrow x \in A \cup B \leftrightarrow x \in A \text{ or } x \in B \leftrightarrow f_A(x) = 1 \text{ or } f_B(x) = 1 \leftrightarrow f_A(x) + f_B(x) - f_A(x)f_B(x) = 1$  **c)**  $f_{\bar{A}}(x) = 1 \leftrightarrow x \in \bar{A} \leftrightarrow x \notin A \leftrightarrow f_A(x) = 0 \leftrightarrow 1 - f_A(x) = 1$  **d)**  $f_{A \oplus B}(x) = 1 \leftrightarrow x \in A \oplus B \leftrightarrow (x \in A \text{ and } x \notin B) \text{ or } (x \notin A \text{ and } x \in B) \leftrightarrow f_A(x) + f_B(x) - 2f_A(x)f_B(x) = 1$  **75. a)** True; because  $\lfloor x \rfloor$  is already an integer,  $\lceil \lfloor x \rfloor \rceil = \lfloor x \rfloor$ . **b)** False;  $x = \frac{1}{2}$  is a counterexample. **c)** True; if  $x$  or  $y$  is an integer, then by property 4b in Table 1, the difference is 0. If neither  $x$  nor  $y$  is an integer, then  $x = n + \epsilon$  and  $y = m + \delta$ , where  $n$  and  $m$  are integers and  $\epsilon$  and  $\delta$  are positive real numbers less than 1. Then  $m + n < x + y < m + n + 2$ , so  $\lceil x + y \rceil$  is either  $m + n + 1$  or  $m + n + 2$ . Therefore, the given expression is either  $(n + 1) + (m + 1) - (m + n + 1) = 1$  or  $(n + 1) + (m + 1) - (m + n + 2) = 0$ , as desired. **d)** False;  $x = \frac{1}{4}$  and  $y = 3$  is a counterexample. **e)** False;  $x = \frac{1}{2}$  is a counterexample. **77. a)** If  $x$  is a positive integer, then the two sides are equal. So suppose that  $x = n^2 + m + \epsilon$ , where  $n^2$  is the largest perfect square less than  $x$ ,  $m$  is a nonnegative integer, and  $0 < \epsilon \leq 1$ . Then both  $\sqrt{x}$  and  $\sqrt{\lfloor x \rfloor} = \sqrt{n^2 + m}$  are between  $n$  and  $n + 1$ , so both sides equal  $n$ . **b)** If  $x$  is a positive integer, then the two sides are equal. So suppose that  $x = n^2 - m - \epsilon$ , where  $n^2$  is the smallest perfect square

greater than  $x$ ,  $m$  is a nonnegative integer, and  $\epsilon$  is a real number with  $0 < \epsilon \leq 1$ . Then both  $\sqrt{x}$  and  $\sqrt{\lceil x \rceil} = \sqrt{n^2 - m}$  are between  $n - 1$  and  $n$ . Therefore, both sides of the equation equal  $n$ . **79. a)** Domain is  $\mathbf{Z}$ ; codomain is  $\mathbf{R}$ ; domain of definition is the set of nonzero integers; the set of values for which  $f$  is undefined is  $\{0\}$ ; not a total function. **b)** Domain is  $\mathbf{Z}$ ; codomain is  $\mathbf{Z}$ ; domain of definition is  $\mathbf{Z}$ ; set of values for which  $f$  is undefined is  $\emptyset$ ; total function. **c)** Domain is  $\mathbf{Z} \times \mathbf{Z}$ ; codomain is  $\mathbf{Q}$ ; domain of definition is  $\mathbf{Z} \times (\mathbf{Z} - \{0\})$ ; set of values for which  $f$  is undefined is  $\mathbf{Z} \times \{0\}$ ; not a total function. **d)** Domain is  $\mathbf{Z} \times \mathbf{Z}$ ; codomain is  $\mathbf{Z}$ ; domain of definition is  $\mathbf{Z} \times \mathbf{Z}$ ; set of values for which  $f$  is undefined is  $\emptyset$ ; total function. **e)** Domain is  $\mathbf{Z} \times \mathbf{Z}$ ; codomain is  $\mathbf{Z}$ ; domain of definitions is  $\{(m, n) \mid m > n\}$ ; set of values for which  $f$  is undefined is  $\{(m, n) \mid m \leq n\}$ ; not a total function. **81. a)** By definition, to say that  $S$  has cardinality  $m$  is to say that  $S$  has exactly  $m$  distinct elements. Therefore we can assign the first object to 1, the second to 2, and so on. This provides the one-to-one correspondence. **b)** By part (a), there is a bijection  $f$  from  $S$  to  $\{1, 2, \dots, m\}$  and a bijection  $g$  from  $T$  to  $\{1, 2, \dots, m\}$ . Then the composition  $g^{-1} \circ f$  is the desired bijection from  $S$  to  $T$ .

## Section 2.4

**1. a)** 3 **b)** -1 **c)** 787 **d)** 2639 **3. a)**  $a_0 = 2, a_1 = 3, a_2 = 5, a_3 = 9$  **b)**  $a_0 = 1, a_1 = 4, a_2 = 27, a_3 = 256$  **c)**  $a_0 = 0, a_1 = 0, a_2 = 1, a_3 = 1$  **d)**  $a_0 = 0, a_1 = 1, a_2 = 2, a_3 = 3$  **5. a)** 2, 5, 8, 11, 14, 17, 20, 23, 26, 29 **b)** 1, 1, 1, 2, 2, 2, 3, 3, 3, 4 **c)** 1, 1, 3, 3, 5, 5, 7, 7, 9, 9 **d)** -1, -2, -2, 8, 88, 656, 4912, 40064, 362368, 3627776 **e)** 3, 6, 12, 24, 48, 96, 192, 384, 768, 1536 **f)** 2, 4, 6, 10, 16, 26, 42, 68, 110, 178 **g)** 1, 2, 2, 3, 3, 3, 3, 4, 4, 4 **h)** 3, 3, 5, 4, 4, 3, 5, 5, 4, 3 **7.** Each term could be twice the previous term; the  $n$ th term could be obtained from the previous term by adding  $n - 1$ ; the terms could be the positive integers that are not multiples of 3; there are infinitely many other possibilities. **9. a)** 2, 12, 72, 432, 2592 **b)** 2, 4, 16, 256, 65, 536 **c)** 1, 2, 5, 11, 26 **d)** 1, 1, 6, 27, 204 **e)** 1, 2, 0, 1, 3 **11. a)** 6, 17, 49, 143, 421 **b)**  $49 = 5 \cdot 17 - 6 \cdot 6, 143 = 5 \cdot 49 - 6 \cdot 17, 421 = 5 \cdot 143 - 6 \cdot 49$  **c)**  $5a_{n-1} - 6a_{n-2} = 5(2^{n-1} + 5 \cdot 3^{n-1}) - 6(2^{n-2} + 5 \cdot 3^{n-2}) = 2^{n-2}(10 - 6) + 3^{n-2}(75 - 30) = 2^{n-2} \cdot 4 + 3^{n-2} \cdot 9 \cdot 5 = 2^n + 3^n \cdot 5 = a_n$  **13. a)** Yes **b)** No **c)** No **d)** Yes **e)** Yes **f)** Yes **g)** No **h)** No **15. a)**  $a_{n-1} + 2a_{n-2} + 2n - 9 = -(n - 1) + 2 + 2[-(n - 2) + 2] + 2n - 9 = -n + 2 = a_n$  **b)**  $a_{n-1} + 2a_{n-2} + 2n - 9 = 5(-1)^{n-1} - (n - 1) + 2 + 2[5(-1)^{n-2} - (n - 2) + 2] + 2n - 9 = 5(-1)^{n-2}(-1 + 2) - n + 2 = a_n$  **c)**  $a_{n-1} + 2a_{n-2} + 2n - 9 = 3(-1)^{n-1} + 2^{n-1} - (n - 1) + 2 + 2[3(-1)^{n-2} + 2^{n-2} - (n - 2) + 2] + 2n - 9 = 3(-1)^{n-2}(-1 + 2) + 2^{n-2}(2 + 2) - n + 2 = a_n$  **d)**  $a_{n-1} + 2a_{n-2} + 2n - 9 = 7 \cdot 2^{n-1} - (n - 1) + 2 + 2[7 \cdot 2^{n-2} - (n - 2) + 2] + 2n - 9 = 2^{n-2}(7 \cdot 2 + 2 \cdot 7) - n + 2 = a_n$  **17. a)**  $a_n = 2 \cdot 3^n$  **b)**  $a_n = 2n + 3$  **c)**  $a_n = 1 + n(n + 1)/2$  **d)**  $a_n = n^2 + 4n + 4$  **e)**  $a_n = 1$  **f)**  $a_n = (3^{n+1} - 1)/2$  **g)**  $a_n = 5n!$  **h)**  $a_n = 2^n n!$  **19. a)**  $a_n = 3a_{n-1}$  **b)** 5,904,900 **21. a)**  $a_n = n + a_{n-1}, a_0 = 0$  **b)**  $a_{12} = 78$  **c)**  $a_n = n(n + 1)/2$



**23.**  $B(k) = [1 + (0.07/12)]B(k-1) - 100$ , with  $B(0) = 5000$   
**25. a)** One 1 and one 0, followed by two 1s and two 0s, followed by three 1s and three 0s, and so on; 1, 1, 1 **b)** The positive integers are listed in increasing order with each even positive integer listed twice; 9, 10, 10. **c)** The terms in odd-numbered locations are the successive powers of 2; the terms in even-numbered locations are all 0; 32, 0, 64.  
**d)**  $a_n = 3 \cdot 2^{n-1}$ ; 384, 768, 1536 **e)**  $a_n = 15 - 7(n-1) = 22 - 7n$ ; -34, -41, -48 **f)**  $a_n = (n^2 + n + 4)/2$ ; 57, 68, 80 **g)**  $a_n = 2n^3$ ; 1024, 1458, 2000 **h)**  $a_n = n! + 1$ ; 362881, 3628801, 39916801 **27.** Among the integers 1, 2, ...,  $a_n$ , where  $a_n$  is the  $n$ th positive integer not a perfect square, the nonsquares are  $a_1, a_2, \dots, a_n$  and the squares are  $1^2, 2^2, \dots, k^2$ , where  $k$  is the integer with  $k^2 < n+k < (k+1)^2$ . Consequently,  $a_n = n+k$ , where  $k^2 < a_n < (k+1)^2$ . To find  $k$ , first note that  $k^2 < n+k < (k+1)^2$ , so  $k^2 + 1 \leq n+k \leq (k+1)^2 - 1$ . Hence,  $(k - \frac{1}{2})^2 + \frac{3}{4} = k^2 - k + 1 \leq n \leq k^2 + k = (k + \frac{1}{2})^2 - \frac{1}{4}$ . It follows that  $k - \frac{1}{2} < \sqrt{n} < k + \frac{1}{2}$ , so  $k = \{\sqrt{n}\}$  and  $a_n = n+k = n + \{\sqrt{n}\}$ . **29. a)** 20 **b)** 11 **c)** 30 **d)** 511  
**31. a)** 1533 **b)** 510 **c)** 4923 **d)** 9842 **33. a)** 21 **b)** 78  
**c)** 18 **d)** 18 **35.**  $\sum_{j=1}^n (a_j - a_{j-1}) = a_n - a_0$  **37. a)**  $n^2$   
**b)**  $n(n+1)/2$  **39.** 15150 **41.** 34320 **43.**  $\frac{n(n+1)(2n+1)}{3} + \frac{n(n+1)}{2} + (n+1)(m - (n+1)^2 + 1)$ , where  $n = \lfloor \sqrt{m} \rfloor - 1$   
**45. a)** 0 **b)** 1680 **c)** 1 **d)** 1024 **47.** 34

## Section 2.5

**1. a)** Countably infinite, -1, -2, -3, -4, ... **b)** Countably infinite, 0, 2, -2, 4, -4, ... **c)** Countably infinite, 99, 98, 97, ... **d)** Uncountable **e)** Finite **f)** Countably infinite, 0, 7, -7, 14, -14, ... **3. a)** Countable: match  $n$  with the string of  $n$  1s. **b)** Countable. To find a correspondence, follow the path in Example 4, but omit fractions in the top three rows (as well as continuing to omit fractions not in lowest terms). **c)** Uncountable **d)** Uncountable **5.** Suppose  $m$  new guests arrive at the fully occupied hotel. Move the guest in Room  $n$  to Room  $m+n$  for  $n = 1, 2, 3, \dots$ ; then the new guests can occupy rooms 1 to  $m$ . **7.** For  $n = 1, 2, 3, \dots$ , put the guest currently in Room  $2n$  into Room  $n$ , and the guest currently in Room  $2n-1$  into Room  $n$  of the new building. **9.** Move the guest currently in Room  $i$  to Room  $2i+1$  for  $i = 1, 2, 3, \dots$ . Put the  $j$ th guest from the  $k$ th bus into Room  $2^k(2j+1)$ . **11. a)**  $A = [1, 2]$  (closed interval of real numbers from 1 to 2),  $B = [3, 4]$  **b)**  $A = [1, 2] \cup \mathbf{Z}^+$ ,  $B = [3, 4] \cup \mathbf{Z}^+$  **c)**  $A = [1, 3]$ ,  $B = [2, 4]$  **13.** Suppose that  $A$  is countable. Then either  $A$  has cardinality  $n$  for some nonnegative integer  $n$ , in which case there is a one-to-one function from  $A$  to a subset of  $\mathbf{Z}^+$  (the range is the first  $n$  positive integers), or there exists a one-to-one correspondence  $f$  from  $A$  to  $\mathbf{Z}^+$ ; in either case we have satisfied Definition 2. Conversely, suppose that  $|A| \leq |\mathbf{Z}^+|$ . By definition, this means that there is a one-to-one function from  $A$  to  $\mathbf{Z}^+$ , so  $A$  has the same cardinality as a subset of  $\mathbf{Z}^+$  (namely the range of that function). By Exercise 16 we conclude that  $A$  is countable. **15.** Assume that  $B$  is countable. Then the elements of

$B$  can be listed as  $b_1, b_2, b_3, \dots$ . Because  $A$  is a subset of  $B$ , taking the subsequence of  $\{b_n\}$  that contains the terms that are in  $A$  gives a listing of the elements of  $A$ . Because  $A$  is uncountable, this is impossible. **17.** Assume that  $A - B$  is countable. Then, because  $A = (A - B) \cup (A \cap B)$ , the elements of  $A$  can be listed in a sequence by alternating elements of  $A - B$  and elements of  $A \cap B$ . This contradicts the uncountability of  $A$ . **19.** We are given bijections  $f$  from  $A$  to  $B$  and  $g$  from  $C$  to  $D$ . Then the function from  $A \times C$  to  $B \times D$  that sends  $(a, c)$  to  $(f(a), g(c))$  is a bijection. **21.** By the definition of  $|A| \leq |B|$ , there is a one-to-one function  $f: A \rightarrow B$ . Similarly, there is a one-to-one function  $g: B \rightarrow C$ . By Exercise 33 in Section 2.3, the composition  $g \circ f: A \rightarrow C$  is one-to-one. Therefore, by definition  $|A| \leq |C|$ . **23.** Using the Axiom of Choice from set theory, choose distinct elements  $a_1, a_2, a_3, \dots$  of  $A$  one at a time (this is possible because  $A$  is infinite). The resulting set  $\{a_1, a_2, a_3, \dots\}$  is the desired infinite subset of  $A$ . **25.** The set of finite strings of characters over a finite alphabet is countably infinite, because we can list these strings in alphabetical order by length. Therefore, the infinite set  $S$  can be identified with an infinite subset of this countable set, which by Exercise 16 is also countably infinite. **27.** Suppose that  $A_1, A_2, A_3, \dots$  are countable sets. Because  $A_i$  is countable, we can list its elements in a sequence as  $a_{i1}, a_{i2}, a_{i3}, \dots$ . The elements of the set  $\bigcup_{i=1}^n A_i$  can be listed by listing all terms  $a_{ij}$  with  $i+j=2$ , then all terms  $a_{ij}$  with  $i+j=3$ , then all terms  $a_{ij}$  with  $i+j=4$ , and so on. **29.** There are a finite number of bit strings of length  $m$ , namely,  $2^m$ . The set of all bit strings is the union of the sets of bit strings of length  $m$  for  $m = 0, 1, 2, \dots$ . Because the union of a countable number of countable sets is countable (see Exercise 27), there are a countable number of bit strings. **31.** It is clear from the formula that the range of values the function takes on for a fixed value of  $m+n$ , say  $m+n=x$ , is  $(x-2)(x-1)/2 + 1$  through  $(x-2)(x-1)/2 + (x-1)$ , because  $m$  can assume the values 1, 2, 3, ...,  $(x-1)$  under these conditions, and the first term in the formula is a fixed positive integer when  $m+n$  is fixed. To show that this function is one-to-one and onto, we merely need to show that the range of values for  $x+1$  picks up precisely where the range of values for  $x$  left off, i.e., that  $f(x-1, 1) + 1 = f(1, x)$ . We have  $f(x-1, 1) + 1 = \frac{(x-2)(x-1)}{2} + (x-1) + 1 = \frac{x^2 - x + 2}{2} = \frac{(x-1)x}{2} + 1 = f(1, x)$ . **33.** By the Schröder-Bernstein theorem, it suffices to find one-to-one functions  $f: (0, 1) \rightarrow [0, 1]$  and  $g: [0, 1] \rightarrow (0, 1)$ . Let  $f(x) = x$  and  $g(x) = (x+1)/3$ . **35.** Each element  $A$  of the power set of the set of positive integers (i.e.,  $A \subseteq \mathbf{Z}^+$ ) can be represented uniquely by the bit string  $a_1 a_2 a_3 \dots$ , where  $a_i = 1$  if  $i \in A$  and  $a_i = 0$  if  $i \notin A$ . Assume there were a one-to-one correspondence  $f: \mathbf{Z}^+ \rightarrow \mathcal{P}(\mathbf{Z}^+)$ . Form a new bit string  $s = s_1 s_2 s_3 \dots$  by setting  $s_i$  to be 1 minus the  $i$ th bit of  $f(i)$ . Then because  $s$  differs in the  $i$  bit from  $f(i)$ ,  $s$  is not in the range of  $f$ , a contradiction. **37.** For any finite alphabet there are a finite number of strings of length  $n$ , whenever  $n$  is a positive integer. It follows by the result of Exercise 27 that there are only a countable number of strings from any given finite alphabet. Because the set of



all computer programs in a particular language is a subset of the set of all strings of a finite alphabet, which is a countable set by the result from Exercise 16, it is itself a countable set. **39.** Exercise 37 shows that there are only a countable number of computer programs. Consequently, there are only a countable number of computable functions. Because, as Exercise 38 shows, there are an uncountable number of functions, not all functions are computable. **41. a)** Note that if  $x$  is in the chain generated by  $y$ , then by the way the chains are generated,  $y$  is in the chain generated by  $x$ , so these two chains are the same. Thus, if  $x$  is in both the chain generated by  $y_1$  and the chain generated by  $y_2$ , then the chain generated by  $y_1$  and the chain generated by  $y_2$  are both the same as the chain generated by  $x$  and are therefore the same chain. **b)** A moment's reflection will show that by the way the chains are constructed, this is true. **c)** Again, this is clear from the construction. **d)** Because the chains are disjoint and every element of  $A$  appears in exactly one chain and every element of  $B$  appears in exactly one chain, the function  $h$  cannot map two different elements of  $A$  to the same element of  $B$ . **e)** If an element  $b$  of  $B$  appears in a chain of types 1, 2, or 3, then it is preceded by an element of  $A$ , which maps to it under  $h$ . If  $b$  appears in a chain of type 4, then it is followed by an element of  $A$ , which maps to it under  $h$ .

## Section 2.6

**1. a)**  $3 \times 4$       **b)**  $\begin{bmatrix} 1 \\ 4 \\ 3 \end{bmatrix}$       **c)**  $\begin{bmatrix} 2 & 0 & 4 & 6 \end{bmatrix}$       **d)** 1

**e)**  $\begin{bmatrix} 1 & 2 & 1 \\ 1 & 0 & 1 \\ 1 & 4 & 3 \\ 3 & 6 & 7 \end{bmatrix}$       **3. a)**  $\begin{bmatrix} 1 & 11 \\ 2 & 18 \end{bmatrix}$       **b)**  $\begin{bmatrix} 2 & -2 & -3 \\ 1 & 0 & 2 \\ 9 & -4 & 4 \end{bmatrix}$

**c)**  $\begin{bmatrix} -4 & 15 & -4 & 1 \\ -3 & 10 & 2 & -3 \\ 0 & 2 & -8 & 6 \\ 1 & -8 & 18 & -13 \end{bmatrix}$       **5.**  $\begin{bmatrix} 9/5 & -6/5 \\ -1/5 & 4/5 \end{bmatrix}$

**7.**  $\mathbf{0} + \mathbf{A} = [0 + a_{ij}] = [a_{ij} + 0] = \mathbf{0} + \mathbf{A}$       **9.**  $\mathbf{A} + (\mathbf{B} + \mathbf{C}) = [a_{ij} + (b_{ij} + c_{ij})] = [(a_{ij} + b_{ij}) + c_{ij}] = (\mathbf{A} + \mathbf{B}) + \mathbf{C}$       **11.** The number of rows of  $\mathbf{A}$  equals the number of columns of  $\mathbf{B}$ , and the number of columns of  $\mathbf{A}$  equals the number of rows of  $\mathbf{B}$ .

**13.**  $\mathbf{A}(\mathbf{BC}) = [\sum_q a_{iq} (\sum_r b_{qr} c_{rl})] = [\sum_q \sum_r a_{iq} b_{qr} c_{rl}] = [\sum_r \sum_q a_{iq} b_{qr} c_{rl}] = [\sum_r (\sum_q a_{iq} b_{qr}) c_{rl}] = (\mathbf{AB})\mathbf{C}$

**15.**  $\mathbf{A}^n = \begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix}$       **17. a)** Let  $\mathbf{A} = [a_{ij}]$  and  $\mathbf{B} = [b_{ij}]$ . Then  $\mathbf{A} + \mathbf{B} = [a_{ij} + b_{ij}]$ . We have  $(\mathbf{A} + \mathbf{B})^t = [a_{ji} + b_{ji}] = [a_{ji}] + [b_{ji}] = \mathbf{A}^t + \mathbf{B}^t$ . **b)** Using the same notation as in part (a), we have  $\mathbf{B}^t \mathbf{A}^t = [\sum_q b_{qi} a_{jq}] = [\sum_q a_{jq} b_{qi}] = (\mathbf{AB})^t$ , because the  $(i, j)$ th entry is the  $(j, i)$ th entry of  $\mathbf{AB}$ . **19.** The result

follows because  $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \begin{bmatrix} ad - bc & 0 \\ 0 & ad - bc \end{bmatrix} = (ad - bc)\mathbf{I}_2 = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ . **21.**  $\mathbf{A}^n (\mathbf{A}^{-1})^n = \mathbf{A}(\mathbf{A} \dots (\mathbf{A}(\mathbf{A}\mathbf{A}^{-1})\mathbf{A}^{-1}) \dots \mathbf{A}^{-1})\mathbf{A}^{-1}$  by the associative law. Because  $\mathbf{AA}^{-1} = \mathbf{I}$ , working from the inside shows that  $\mathbf{A}^n (\mathbf{A}^{-1})^n = \mathbf{I}$ . Similarly  $(\mathbf{A}^{-1})^n \mathbf{A}^n = \mathbf{I}$ . Therefore,  $(\mathbf{A}^n)^{-1} = (\mathbf{A}^{-1})^n$ . **23.** The  $(i, j)$ th entry of  $\mathbf{A} + \mathbf{A}^t$  is  $a_{ij} + a_{ji}$ , which equals  $a_{ji} + a_{ij}$ , the  $(j, i)$ th entry of  $\mathbf{A} + \mathbf{A}^t$ , so by definition  $\mathbf{A} + \mathbf{A}^t$  is symmetric. **25.**  $x_1 = 1, x_2 = -1, x_3 = -2$

**27. a)**  $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$       **b)**  $\begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$       **c)**  $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$

**29. a)**  $\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$       **b)**  $\begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$       **c)**  $\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

**31. a)**  $\mathbf{A} \vee \mathbf{B} = [a_{ij} \vee b_{ij}] = [b_{ij} \vee a_{ij}] = \mathbf{B} \vee \mathbf{A}$       **b)**  $\mathbf{A} \wedge \mathbf{B} = [a_{ij} \wedge b_{ij}] = [b_{ij} \wedge a_{ij}] = \mathbf{B} \wedge \mathbf{A}$       **33. a)**  $\mathbf{A} \vee (\mathbf{B} \wedge \mathbf{C}) = [a_{ij} \vee (b_{ij} \wedge c_{ij})] = [(a_{ij} \vee b_{ij}) \wedge (a_{ij} \vee c_{ij})] = [a_{ij} \vee b_{ij}] \wedge [a_{ij} \vee c_{ij}] = (\mathbf{A} \vee \mathbf{B}) \wedge (\mathbf{A} \vee \mathbf{C})$       **b)**  $\mathbf{A} \wedge (\mathbf{B} \vee \mathbf{C}) = [a_{ij} \wedge (b_{ij} \vee c_{ij})] = [(a_{ij} \wedge b_{ij}) \vee (a_{ij} \wedge c_{ij})] = [a_{ij} \wedge b_{ij}] \vee [a_{ij} \wedge c_{ij}] = (\mathbf{A} \wedge \mathbf{B}) \vee (\mathbf{A} \wedge \mathbf{C})$       **35.**  $\mathbf{A} \odot (\mathbf{B} \odot \mathbf{C}) = [\bigvee_q a_{iq} \wedge (\bigvee_r (b_{qr} \wedge c_{rl}))] = [\bigvee_q \bigvee_r (a_{iq} \wedge b_{qr} \wedge c_{rl})] = [\bigvee_r \bigvee_q (a_{iq} \wedge b_{qr} \wedge c_{rl})] = [\bigvee_r (\bigvee_q (a_{iq} \wedge b_{qr})) \wedge c_{rl}] = (\mathbf{A} \odot \mathbf{B}) \odot \mathbf{C}$

## Supplementary Exercises

**1. a)**  $\bar{A}$       **b)**  $A \cap B$       **c)**  $A - B$       **d)**  $\bar{A} \cap \bar{B}$       **e)**  $A \oplus B$

**3.** Yes      **5.**  $A - (A - B) = A - (A \cap \bar{B}) = A \cap (A \cap \bar{B}) = A \cap (\bar{A} \cup B) = (A \cap \bar{A}) \cup (A \cap B) = \emptyset \cup (A \cap B) = A \cap B$       **7.** Let  $A = \{1\}, B = \emptyset, C = \{1\}$ . Then  $(A - B) - C = \emptyset$ , but  $A - (B - C) = \{1\}$ . **9.** No. For example, let  $A = B = \{a, b\}, C = \emptyset$ , and  $D = \{a\}$ . Then  $(A - B) - (C - D) = \emptyset - \emptyset = \emptyset$ , but  $(A - C) - (B - D) = \{a, b\} - \{b\} = \{a\}$ . **11. a)**  $|\emptyset| \leq |A \cap B| \leq |A| \leq |A \cup B| \leq |U|$       **b)**  $|\emptyset| \leq |A - B| \leq |A \oplus B| \leq |A \cup B| \leq |A| + |B|$       **13. a)** Yes, no      **b)** Yes, no      **c)**  $f$  has inverse with  $f^{-1}(a) = 3, f^{-1}(b) = 4, f^{-1}(c) = 2, f^{-1}(d) = 1$ ;  $g$  has no inverse. **15.** If  $f$  is one-to-one, then  $f$  provides a bijection between  $S$  and  $f(S)$ , so they have the same cardinality. If  $f$  is not one-to-one, then there exist elements  $x$  and  $y$  in  $S$  such that  $f(x) = f(y)$ . Let  $S = \{x, y\}$ . Then  $|S| = 2$  but  $|f(S)| = 1$ . **17.** Let  $x \in A$ . Then  $S_f(\{x\}) = \{f(y) \mid y \in \{x\}\} = \{f(x)\}$ . By the same reasoning,  $S_g(\{x\}) = \{g(x)\}$ . Because  $S_f = S_g$ , we can conclude that  $\{f(x)\} = \{g(x)\}$ , and so necessarily  $f(x) = g(x)$ . **19.** The equation is true if and only if the sum of the fractional parts of  $x$  and  $y$  is less than 1. **21.** The equation is true if and only if either both  $x$  and  $y$  are integers, or  $x$  is not an integer but the sum of the fractional parts of  $x$  and  $y$  is less than or equal to 1. **23.** If  $x$  is an integer, then  $\lfloor x \rfloor + \lfloor m - x \rfloor = x + m - x = m$ . Otherwise, write  $x$  in terms of its integer and fractional parts:  $x = n + \epsilon$ , where  $n = \lfloor x \rfloor$  and  $0 < \epsilon < 1$ . In this case  $\lfloor x \rfloor +$



$$\lfloor m - x \rfloor = \lfloor n + \epsilon \rfloor + \lfloor m - n - \epsilon \rfloor = n + m - n - 1 = m - 1.$$

**25.** Write  $n = 2k + 1$  for some integer  $k$ . Then  $n^2 = 4k^2 + 4k + 1$ , so  $n^2/4 = k^2 + k + \frac{1}{4}$ . Therefore,  $\lceil n^2/4 \rceil = k^2 + k + 1$ . But  $(n^2 + 3)/4 = (4k^2 + 4k + 1 + 3)/4 = k^2 + k + 1$ . **27.** Let  $x = n + (r/m) + \epsilon$ , where  $n$  is an integer,  $r$  is a nonnegative integer less than  $m$ , and  $\epsilon$  is a real number with  $0 \leq \epsilon < 1/m$ . The left-hand side is  $\lfloor nm + r + m\epsilon \rfloor = nm + r$ . On the right-hand side, the terms  $\lfloor x \rfloor$  through  $\lfloor x + (m + r - 1)/m \rfloor$  are all just  $n$  and the terms from  $\lfloor x + (m - r)/m \rfloor$  on are all  $n + 1$ . Therefore, the right-hand side is  $(m - r)n + r(n + 1) = nm + r$ , as well. **29.** 101 **31.**  $a_1 = 1$ ;  $a_{2n+1} = n \cdot a_{2n}$  for all  $n > 0$ ; and  $a_{2n} = n + a_{2n-1}$  for all  $n > 0$ . The next four terms are 5346, 5353, 37, 471, and 37, 479. **33.** If each  $f^{-1}(j)$  is countable, then  $S = f^{-1}(1) \cup f^{-1}(2) \cup \dots$  is the countable union of countable sets and is therefore countable by Exercise 27 in Section 2.5. **35.** Because there is a one-to-one correspondence between  $\mathbf{R}$  and the open interval  $(0, 1)$  (given by  $f(x) = 2 \arctan(x)/\pi$ ), it suffices to show that  $|(0, 1) \times (0, 1)| = |(0, 1)|$ . By the Schröder-Bernstein theorem it suffices to find injective functions  $f : (0, 1) \rightarrow (0, 1) \times (0, 1)$  and  $g : (0, 1) \times (0, 1) \rightarrow (0, 1)$ . Let  $f(x) = (x, \frac{1}{2})$ . For  $g$  we follow the hint. Suppose  $(x, y) \in (0, 1) \times (0, 1)$ , and represent  $x$  and  $y$  with their decimal expansions  $x = 0.x_1x_2x_3\dots$  and  $y = 0.y_1y_2y_3\dots$ , never choosing the expansion of any number that ends in an infinite string of 9s. Let  $g(x, y)$  be the decimal expansion obtained by interweaving these two strings, namely

$$0.x_1y_1x_2y_2x_3y_3\dots \quad \mathbf{37.} \mathbf{A}^{4n} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \mathbf{A}^{4n+1} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix},$$

$$\mathbf{A}^{4n+2} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}, \mathbf{A}^{4n+3} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \text{ for } n \geq 0$$

**39.** Suppose that  $\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ . Let  $\mathbf{B} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ . Because  $\mathbf{AB} = \mathbf{BA}$ , it follows that  $c = 0$  and  $a = d$ . Let  $\mathbf{B} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$ . Because  $\mathbf{AB} = \mathbf{BA}$ , it follows that  $b = 0$ .

Hence,  $\mathbf{A} = \begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix} = a\mathbf{I}$ . **41. a)** Let  $\mathbf{A} \odot \mathbf{0} = [b_{ij}]$ . Then  $b_{ij} = (a_{i1} \wedge 0) \vee \dots \vee (a_{ip} \wedge 0) = 0$ . Hence,  $\mathbf{A} \odot \mathbf{0} = \mathbf{0}$ . Similarly,  $\mathbf{0} \odot \mathbf{A} = \mathbf{0}$ . **b)**  $\mathbf{A} \vee \mathbf{0} = [a_{ij} \vee 0] = [a_{ij}] = \mathbf{A}$ . Hence,  $\mathbf{A} \vee \mathbf{0} = \mathbf{A}$ . Similarly,  $\mathbf{0} \vee \mathbf{A} = \mathbf{A}$ . **c)**  $\mathbf{A} \wedge \mathbf{0} = [a_{ij} \wedge 0] = [0] = \mathbf{0}$ . Hence,  $\mathbf{A} \wedge \mathbf{0} = \mathbf{0}$ . Similarly,  $\mathbf{0} \wedge \mathbf{A} = \mathbf{0}$ .

## CHAPTER 3

### Section 3.1

**1.**  $max := 1, i := 2, max := 8, i := 3, max := 12, i := 4, i := 5, i := 6, i := 7, max := 14, i := 8, i := 9, i := 10, i := 11$

**3. procedure** *AddUp*( $a_1, \dots, a_n$ : integers)

$sum := a_1$

**for**  $i := 2$  **to**  $n$

$sum := sum + a_i$

**return**  $sum$

**5. procedure** *duplicates*( $a_1, a_2, \dots, a_n$ : integers in nondecreasing order)

$k := 0$  {this counts the duplicates}

$j := 2$

**while**  $j \leq n$

**if**  $a_j = a_{j-1}$  **then**

$k := k + 1$

$c_k := a_j$

**while**  $j \leq n$  and  $a_j = c_k$

$j := j + 1$

$j := j + 1$

{ $c_1, c_2, \dots, c_k$  is the desired list}

**7. procedure** *last even location*( $a_1, a_2, \dots, a_n$ : integers)

$k := 0$

**for**  $i := 1$  **to**  $n$

**if**  $a_i$  is even **then**  $k := i$

**return**  $k$  { $k = 0$  if there are no evens}

**9. procedure** *palindrome check*( $a_1 a_2 \dots a_n$ : string)

$answer := \text{true}$

**for**  $i := 1$  **to**  $\lfloor n/2 \rfloor$

**if**  $a_i \neq a_{n+1-i}$  **then**  $answer := \text{false}$

**return**  $answer$

**11. procedure** *interchange*( $x, y$ : real numbers)

$z := x$

$x := y$

$y := z$

The minimum number of assignments needed is three.

**13.** Linear search:  $i := 1, i := 2, i := 3, i := 4, i := 5, i := 6, i := 7, location := 7$ ; binary search:  $i := 1, j := 8, m := 4, i := 5, m := 6, i := 7, m := 7, j := 7, location := 7$

**15. procedure** *insert*( $x, a_1, a_2, \dots, a_n$ : integers)

{the list is in order:  $a_1 \leq a_2 \leq \dots \leq a_n$ }

$a_{n+1} := x + 1$

$i := 1$

**while**  $x > a_i$

$i := i + 1$

**for**  $j := 0$  **to**  $n - i$

$a_{n-j+1} := a_{n-j}$

$a_i := x$

{ $x$  has been inserted into correct position}

**17. procedure** *first largest*( $a_1, \dots, a_n$ : integers)

$max := a_1$

$location := 1$

**for**  $i := 2$  **to**  $n$

**if**  $max < a_i$  **then**

$max := a_i$

$location := i$

**return**  $location$

**19. procedure** *mean-median-max-min*( $a, b, c$ : integers)

$mean := (a + b + c)/3$

{the six different orderings of  $a, b, c$  with respect to  $\geq$  will be handled separately}



**if**  $a \geq b$  **then**  
     **if**  $b \geq c$  **then**  $median := b; max := a; min := c$   
     :  
 (The rest of the algorithm is similar.)

**21. procedure** *first-three*( $a_1, a_2, \dots, a_n$ : integers)  
     **if**  $a_1 > a_2$  **then** interchange  $a_1$  and  $a_2$   
     **if**  $a_2 > a_3$  **then** interchange  $a_2$  and  $a_3$   
     **if**  $a_1 > a_2$  **then** interchange  $a_1$  and  $a_2$

**23. procedure** *onto*( $f$ : function from  $A$  to  $B$ , where  
      $A = \{a_1, \dots, a_n\}$ ,  $B = \{b_1, \dots, b_m\}$ ,  $a_1, \dots, a_n$ ,  
      $b_1, \dots, b_m$  are integers)  
     **for**  $i := 1$  **to**  $m$   
          $hit(b_i) := 0$   
      $count := 0$   
     **for**  $j := 1$  **to**  $n$   
         **if**  $hit(f(a_j)) = 0$  **then**  
              $hit(f(a_j)) := 1$   
              $count := count + 1$   
     **if**  $count = m$  **then return true** **else return false**

**25. procedure** *ones*( $a$ : bit string,  $a = a_1 a_2 \dots a_n$ )  
      $count := 0$   
     **for**  $i := 1$  **to**  $n$   
         **if**  $a_i := 1$  **then**  
              $count := count + 1$   
     **return count**

**27. procedure** *ternary search*( $s$ : integer,  $a_1, a_2, \dots, a_n$ :  
     increasing integers)  
      $i := 1$   
      $j := n$   
     **while**  $i < j - 1$   
          $l := \lfloor (i + j)/3 \rfloor$   
          $u := \lfloor 2(i + j)/3 \rfloor$   
         **if**  $x > a_u$  **then**  $i := u + 1$   
         **else if**  $x > a_l$  **then**  
              $i := l + 1$   
              $j := u$   
         **else**  $j := l$   
     **if**  $x = a_i$  **then**  $location := i$   
     **else if**  $x = a_j$  **then**  $location := j$   
     **else**  $location := 0$   
     **return location** {0 if not found}

**29. procedure** *find a mode*( $a_1, a_2, \dots, a_n$ : nondecreasing  
     integers)  
      $modecount := 0$   
      $i := 1$   
     **while**  $i \leq n$   
          $value := a_i$   
          $count := 1$   
         **while**  $i \leq n$  and  $a_i = value$   
              $count := count + 1$   
              $i := i + 1$   
         **if**  $count > modecount$  **then**  
              $modecount := count$   
              $mode := value$   
     **return mode**

**31.** Assume the input is strings  $a_1 a_2 \dots a_n$  and  $b_1 b_2 \dots b_n$ , where each character is a letter, A through Z. Assume also that a function *index* is available, such that  $index(x)$  is the position of the letter  $x$  in the alphabet ( $index('A') = 1, \dots, index('Z') = 26$ ). **a)** Initialize *a-count* and *b-count* to be lists of length 26 with all values equal to 0. For  $i$  from 1 to  $n$  increment  $a-count(index(a_i))$  and  $b-count(index(b_i))$ . If  $a-count(i) = b-count(i)$  for all  $i$  from 1 to 26, then return “true”; otherwise return “false.” **b)** Sort both strings into alphabetical order. Then the two original strings were anagrams if and only if the sorted strings are identical.

**33. procedure** *find duplicate*( $a_1, a_2, \dots, a_n$ : integers)  
      $location := 0$   
      $i := 2$   
     **while**  $i \leq n$  and  $location = 0$   
          $j := 1$   
         **while**  $j < i$  and  $location = 0$   
             **if**  $a_i = a_j$  **then**  $location := i$   
             **else**  $j := j + 1$   
          $i := i + 1$   
     **return location**  
     {*location* is the subscript of the first value that repeats a previous value in the sequence}

**35. procedure** *find decrease*( $a_1, a_2, \dots, a_n$ : positive  
     integers)  
      $location := 0$   
      $i := 2$   
     **while**  $i \leq n$  and  $location = 0$   
         **if**  $a_i < a_{i-1}$  **then**  $location := i$   
         **else**  $i := i + 1$   
     **return location**  
     {*location* is the subscript of the first value less than the immediately preceding one}

**37.** At the end of the first pass: 1, 3, 5, 4, 7; at the end of the second pass: 1, 3, 4, 5, 7; at the end of the third pass: 1, 3, 4, 5, 7; at the end of the fourth pass: 1, 3, 4, 5, 7

**39. procedure** *better bubblesort*( $a_1, \dots, a_n$ : integers)  
      $i := 1$ ;  $done := \text{false}$   
     **while**  $i < n$  and  $done = \text{false}$   
          $done := \text{true}$   
         **for**  $j := 1$  **to**  $n - i$   
             **if**  $a_j > a_{j+1}$  **then**  
                 interchange  $a_j$  and  $a_{j+1}$   
                  $done := \text{false}$   
          $i := i + 1$   
     { $a_1, \dots, a_n$  is in increasing order}

**41.** At the end of the first, second, and third passes: 1, 3, 5, 7, 4; at the end of the fourth pass: 1, 3, 4, 5, 7   **43. a)** 1, 5, 4, 3, 2; 1, 2, 4, 3, 5; 1, 2, 3, 4, 5; 1, 2, 3, 4, 5   **b)** 1, 4, 3, 2, 5; 1, 2, 3, 4, 5; 1, 2, 3, 4, 5; 1, 2, 3, 4, 5   **c)** 1, 2, 3, 4, 5; 1, 2, 3, 4, 5; 1, 2, 3, 4, 5; 1, 2, 3, 4, 5   **45.** We carry out the linear search algorithm given as Algorithm 2 in this section, except that we replace  $x \neq a_i$  by  $x < a_i$ , and we replace the **else** clause with **else**  $location := n + 1$ .   **47.**  $2 + 3 + 4 + \dots + n = (n^2 + n - 2)/2$    **49.** Find the location for the 2 in the list 3 (one



comparison), and insert it in front of the 3, so the list now reads 2, 3, 4, 5, 1, 6. Find the location for the 4 (compare it to the 2 and then the 3), and insert it, leaving 2, 3, 4, 5, 1, 6. Find the location for the 5 (compare it to the 3 and then the 4), and insert it, leaving 2, 3, 4, 5, 1, 6. Find the location for the 1 (compare it to the 3 and then the 2 and then the 2 again), and insert it, leaving 1, 2, 3, 4, 5, 6. Find the location for the 6 (compare it to the 3 and then the 4 and then the 5), and insert it, giving the final answer 1, 2, 3, 4, 5, 6.

**51. procedure** *binary insertion sort*( $a_1, a_2, \dots, a_n$ :  
real numbers with  $n \geq 2$ )  
**for**  $j := 2$  **to**  $n$   
  {binary search for insertion location  $i$ }  
   $left := 1$   
   $right := j - 1$   
  **while**  $left < right$   
     $middle := \lfloor (left + right)/2 \rfloor$   
    **if**  $a_j > a_{middle}$  **then**  $left := middle + 1$   
    **else**  $right := middle$   
  **if**  $a_j < a_{left}$  **then**  $i := left$  **else**  $i := left + 1$   
  {insert  $a_j$  in location  $i$  by moving  $a_i$  through  $a_{j-1}$   
  toward back of list}  
   $m := a_j$   
  **for**  $k := 0$  **to**  $j - i - 1$   
     $a_{j-k} := a_{j-k-1}$   
   $a_i := m$   
  { $a_1, a_2, \dots, a_n$  are sorted}

**53.** The variation from Exercise 52    **55.**  $m = 3, n = 8; s = 0, j = 1$ , no match;  $s = 1, j = 1, j = 2, j = 3$ , no match;  $s = 2, j = 1$ , no match;  $s = 3, j = 1, j = 2$ , no match;  $s = 4, j = 1$ , no match;  $s = 5, j = 1, j = 2, j = 3, j = 4$ , “5 is a valid shift”    **57. a)** Two quarters, one penny    **b)** Two quarters, one dime, one nickel, four pennies    **c)** Three quarters, one penny    **d)** Two quarters, one dime    **59.** Cashier’s algorithm uses fewest coins in parts (a), (c), and (d).    **a)** Two quarters, one penny    **b)** Two quarters, one dime, nine pennies    **c)** Three quarters, one penny    **d)** Two quarters, one dime  
**61.** The 9:00–9:45 talk, the 9:50–10:15 talk, the 10:15–10:45 talk, the 11:00–11:15 talk    **63. a)** Order the talks by starting time. Number the lecture halls 1, 2, 3, and so on. For each talk, assign it to lowest numbered lecture hall that is currently available.    **b)** If this algorithm uses  $n$  lecture halls, then at the point the  $n$ th hall was first assigned, it *had* to be used (otherwise a lower-numbered hall would have been assigned), which means that  $n$  talks were going on simultaneously (this talk just assigned and the  $n - 1$  talks currently in halls 1 through  $n - 1$ ).    **65.** Here we assume that the men are the suitors and the women the suitees.

**procedure** *stable*( $M_1, M_2, \dots, M_s, W_1, W_2, \dots, W_s$ :  
preference lists)  
**for**  $i := 1$  **to**  $s$   
  mark man  $i$  as rejected  
**for**  $i := 1$  **to**  $s$   
  set man  $i$ ’s rejection list to be empty  
**for**  $j := 1$  **to**  $s$   
  set woman  $j$ ’s proposal list to be empty

**while** rejected men remain

**for**  $i := 1$  **to**  $s$

**if** man  $i$  is marked rejected **then** add  $i$  to the  
    proposal list for the woman  $j$  who ranks highest  
    on his preference list but does not appear on his  
    rejection list, and mark  $i$  as not rejected

**for**  $j := 1$  **to**  $s$

**if** woman  $j$ ’s proposal list is nonempty **then**  
    remove from  $j$ ’s proposal list all men  $i$   
    except the man  $i_0$  who ranks highest on her  
    preference list, and for each such man  $i$  mark  
    him as rejected and add  $j$  to his rejection list

**for**  $j := 1$  **to**  $s$

  match  $j$  with the one man on  $j$ ’s proposal list

{This matching is stable.}

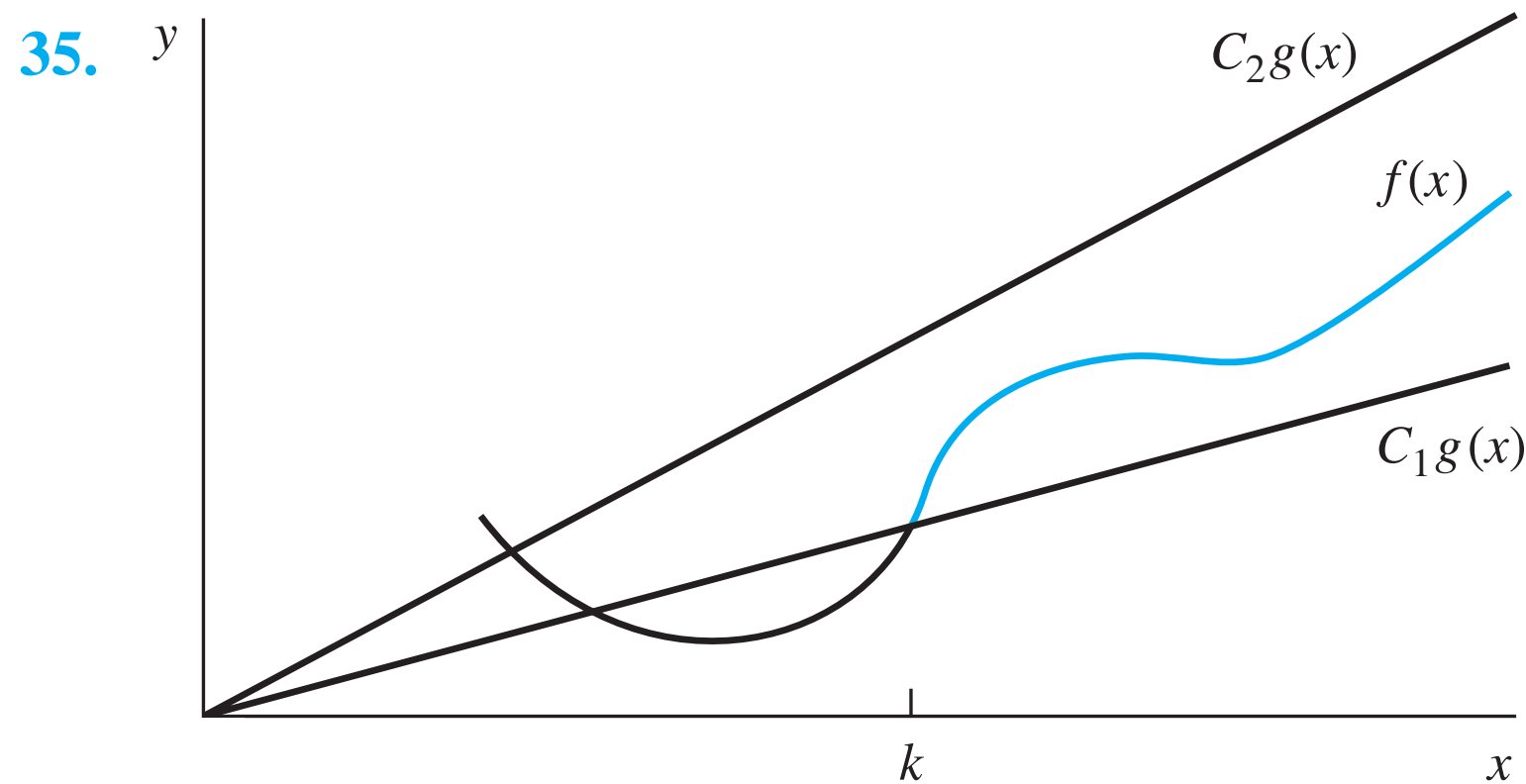
**67.** If the assignment is not stable, then there is a man  $m$  and a woman  $w$  such that  $m$  prefers  $w$  to the woman  $w'$  with whom he is matched, and  $w$  prefers  $m$  to the man with whom she is matched. But  $m$  must have proposed to  $w$  before he proposed to  $w'$ , because he prefers the former. Because  $m$  did not end up matched with  $w$ , she must have rejected him. Women reject a suitor only when they get a better proposal, and they eventually get matched with a pending suitor, so the woman with whom  $w$  is matched must be better in her eyes than  $m$ , contradicting our original assumption. Therefore, the marriage is stable.    **69. a)** (adapted from Wikipedia) Let  $m$  be the majority element. Let  $n$  be a number defined at any step of the algorithm to be either the counter, if the majority candidate is  $m$ , or the negative of the counter otherwise. Then at each step in which the algorithm encounters a value equal to  $m$ , the value of  $n$  will increase by 1, and at each step at which it encounters a different value, the value of  $n$  may either increase or decrease by one. If  $m$  truly is the majority element, there will be more increases than decreases, and  $n$  will be positive at the end of the algorithm. But this can be true only when the final stored element is  $m$ , the majority element.    **b)** A counterexample is ABABC.    **71.** Run the two programs on their inputs concurrently and report which one halts.

## Section 3.2

**1.** The choices of  $C$  and  $k$  are not unique. **a)**  $C = 1, k = 10$     **b)**  $C = 4, k = 7$     **c)** No    **d)**  $C = 5, k = 1$     **e)**  $C = 1, k = 0$     **f)**  $C = 1, k = 2$     **3.**  $x^4 + 9x^3 + 4x + 7 \leq 4x^4$  for all  $x > 9$ ; witnesses  $C = 4, k = 9$     **5.**  $(x^2+1)/(x+1) = x-1+2/(x+1) < x$  for all  $x > 1$ ; witnesses  $C = 1, k = 1$     **7.** The choices of  $C$  and  $k$  are not unique.    **a)**  $n = 3, C = 3, k = 1$     **b)**  $n = 3, C = 4, k = 1$     **c)**  $n = 1, C = 2, k = 1$     **d)**  $n = 0, C = 2, k = 1$     **9.**  $x^2 + 4x + 17 \leq 3x^3$  for all  $x > 17$ , so  $x^2 + 4x + 17$  is  $O(x^3)$ , with witnesses  $C = 3, k = 17$ . However, if  $x^3$  were  $O(x^2 + 4x + 17)$ , then  $x^3 \leq C(x^2 + 4x + 17) \leq 3Cx^2$  for some  $C$ , for all sufficiently large  $x$ , which implies that  $x \leq 3C$  for all sufficiently large  $x$ , which is impossible. Hence,  $x^3$  is not  $O(x^2 + 4x + 17)$ .    **11.**  $3x^4 + 1 \leq 4x^4 = 8(x^4/2)$  for all  $x > 1$ , so  $3x^4 + 1$  is  $O(x^4/2)$ , with witnesses  $C = 8, k = 1$ . Also  $x^4/2 \leq 3x^4 + 1$  for all  $x > 0$ , so  $x^4/2$  is



$O(3x^4 + 1)$ , with witnesses  $C = 1, k = 0$ . **13.** Because  $2^n \leq 3^n$  for all  $n > 0$ , it follows that  $2^n$  is  $O(3^n)$ , with witnesses  $C = 1, k = 0$ . However, if  $3^n$  were  $O(2^n)$ , then for some  $C$ ,  $3^n \leq C \cdot 2^n$  for all sufficiently large  $n$ . This says that  $C \geq (3/2)^n$  for all sufficiently large  $n$ , which is impossible. Hence,  $3^n$  is not  $O(2^n)$ . **15.** All functions for which there exist real numbers  $k$  and  $C$  with  $|f(x)| \leq C$  for  $x > k$ . These are the functions  $f(x)$  that are bounded for all sufficiently large  $x$ . **17.** There are constants  $C_1, C_2, k_1$ , and  $k_2$  such that  $|f(x)| \leq C_1|g(x)|$  for all  $x > k_1$  and  $|g(x)| \leq C_2|h(x)|$  for all  $x > k_2$ . Hence, for  $x > \max(k_1, k_2)$  it follows that  $|f(x)| \leq C_1|g(x)| \leq C_1C_2|h(x)|$ . This shows that  $f(x)$  is  $O(h(x))$ . **19.**  $2^{n+1}$  is  $O(2^n)$ ;  $2^{2n}$  is not. **21.**  $1000 \log n$ ;  $\sqrt{n}$ ;  $n \log n$ ;  $n^2/1,000,000$ ;  $2^n$ ;  $3^n$ ;  $2n!$  **23.** The algorithm that uses  $n \log n$  operations **25. a)**  $O(n^3)$  **b)**  $O(n^5)$  **c)**  $O(n^3 \cdot n!)$  **27. a)**  $O(n^2 \log n)$  **b)**  $O(n^2(\log n)^2)$  **c)**  $O(n^{2^n})$  **29. a)** Neither  $\Theta(x^2)$  nor  $\Omega(x^2)$  **b)**  $\Theta(x^2)$  and  $\Omega(x^2)$  **c)** Neither  $\Theta(x^2)$  nor  $\Omega(x^2)$  **d)**  $\Omega(x^2)$ , but not  $\Theta(x^2)$  **e)**  $\Omega(x^2)$ , but not  $\Theta(x^2)$  **f)**  $\Omega(x^2)$  and  $\Theta(x^2)$  **31.** If  $f(x)$  is  $\Theta(g(x))$ , then there exist constants  $C_1$  and  $C_2$  with  $C_1|g(x)| \leq |f(x)| \leq C_2|g(x)|$ . It follows that  $|f(x)| \leq C_2|g(x)|$  and  $|g(x)| \leq (1/C_1)|f(x)|$  for  $x > k$ . Thus,  $f(x)$  is  $O(g(x))$  and  $g(x)$  is  $O(f(x))$ . Conversely, suppose that  $f(x)$  is  $O(g(x))$  and  $g(x)$  is  $O(f(x))$ . Then there are constants  $C_1, C_2, k_1$ , and  $k_2$  such that  $|f(x)| \leq C_1|g(x)|$  for  $x > k_1$  and  $|g(x)| \leq C_2|f(x)|$  for  $x > k_2$ . We can assume that  $C_2 > 0$  (we can always make  $C_2$  larger). Then we have  $(1/C_2)|g(x)| \leq |f(x)| \leq C_1|g(x)|$  for  $x > \max(k_1, k_2)$ . Hence,  $f(x)$  is  $\Theta(g(x))$ . **33.** If  $f(x)$  is  $\Theta(g(x))$ , then  $f(x)$  is both  $O(g(x))$  and  $\Omega(g(x))$ . Hence, there are positive constants  $C_1, k_1, C_2$ , and  $k_2$  such that  $|f(x)| \leq C_2|g(x)|$  for all  $x > k_2$  and  $|f(x)| \geq C_1|g(x)|$  for all  $x > k_1$ . It follows that  $C_1|g(x)| \leq |f(x)| \leq C_2|g(x)|$  whenever  $x > k$ , where  $k = \max(k_1, k_2)$ . Conversely, if there are positive constants  $C_1, C_2$ , and  $k$  such that  $C_1|g(x)| \leq |f(x)| \leq C_2|g(x)|$  for  $x > k$ , then taking  $k_1 = k_2 = k$  shows that  $f(x)$  is both  $O(g(x))$  and  $\Theta(g(x))$ .



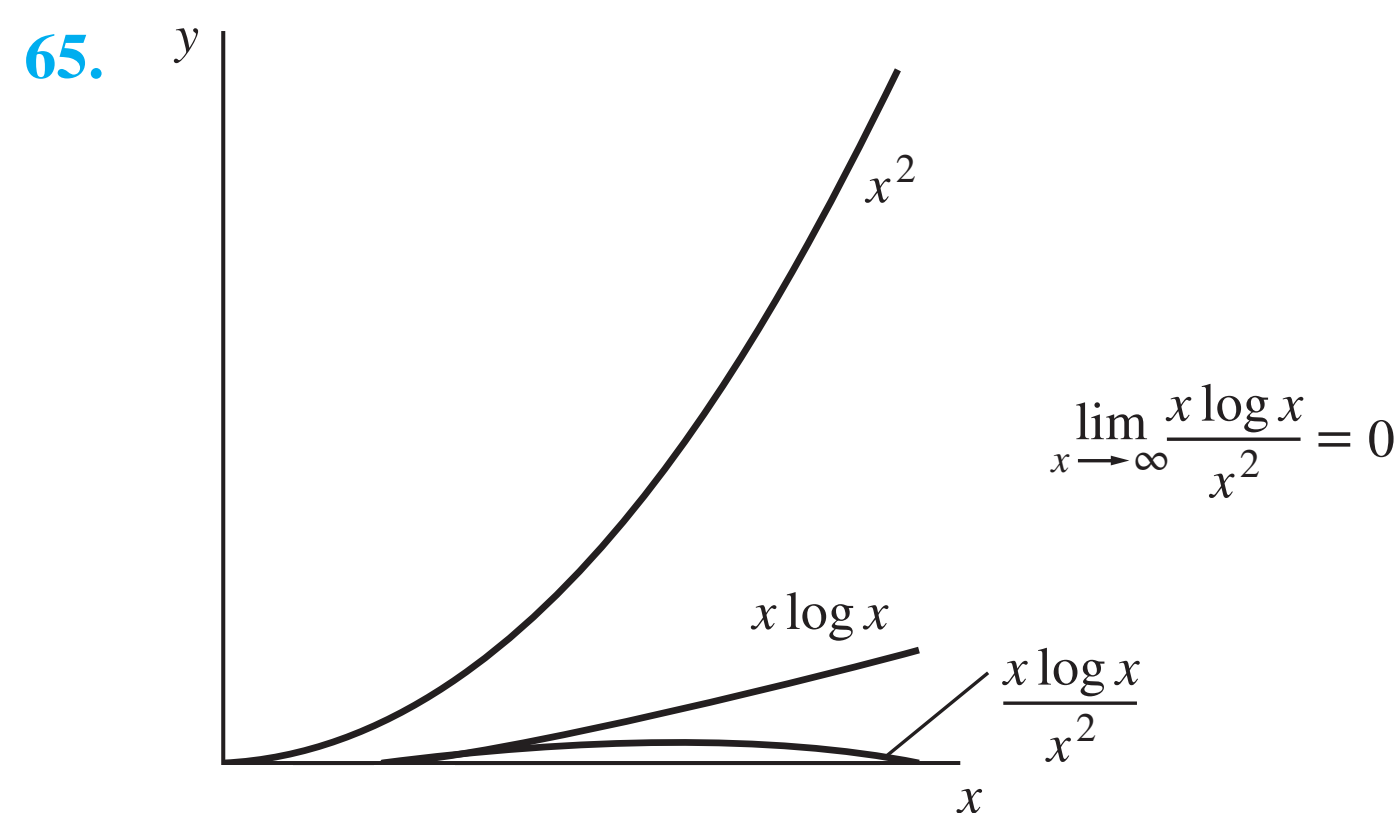
**37.** If  $f(x)$  is  $\Theta(1)$ , then  $|f(x)|$  is bounded between positive constants  $C_1$  and  $C_2$ . In other words,  $f(x)$  cannot grow larger than a fixed bound or smaller than the negative of this bound and must not get closer to 0 than some fixed bound. **39.** Because  $f(x)$  is  $O(g(x))$ , there are constants  $C$  and  $k$  such that  $|f(x)| \leq C|g(x)|$  for  $x > k$ . Hence,  $|f^n(x)| \leq C^n|g^n(x)|$  for  $x > k$ , so  $f^n(x)$  is  $O(g^n(x))$  by taking the constant to be  $C^n$ . **41.** Because  $f(x)$  and  $g(x)$  are increasing and unbounded, we can assume  $f(x) \geq 1$  and  $g(x) \geq 1$  for sufficiently large  $x$ .

There are constants  $C$  and  $k$  with  $f(x) \leq Cg(x)$  for  $x > k$ . This implies that  $\log f(x) \leq \log C + \log g(x) < 2 \log g(x)$  for sufficiently large  $x$ . Hence,  $\log f(x)$  is  $O(\log g(x))$ . **43.** By definition there are positive constraints  $C_1, C'_1, C_2, C'_2, k_1, k'_1, k_2$ , and  $k'_2$  such that  $f_1(x) \geq C_1|g(x)|$  for all  $x > k_1$ ,  $f_1(x) \leq C'_1|g(x)|$  for all  $x > k'_1$ ,  $f_2(x) \geq C_2|g(x)|$  for all  $x > k_2$ , and  $f_2(x) \leq C'_2|g(x)|$  for all  $x > k'_2$ . Adding the first and third inequalities shows that  $f_1(x) + f_2(x) \geq (C_1 + C_2)|g(x)|$  for all  $x > k$  where  $k = \max(k_1, k_2)$ . Adding the second and fourth inequalities shows that  $f_1(x) + f_2(x) \leq (C'_1 + C'_2)|g(x)|$  for all  $x > k'$  where  $k' = \max(k'_1, k'_2)$ . Hence,  $f_1(x) + f_2(x)$  is  $\Theta(g(x))$ . This is no longer true if  $f_1$  and  $f_2$  can assume negative values. **45.** This is false. Let  $f_1 = x^2 + 2x, f_2(x) = x^2 + x$ , and  $g(x) = x^2$ . Then  $f_1(x)$  and  $f_2(x)$  are both  $O(g(x))$ , but  $(f_1 - f_2)(x)$  is not. **47.** Take  $f(n)$  to be the function with  $f(n) = n$  if  $n$  is an odd positive integer and  $f(n) = 1$  if  $n$  is an even positive integer and  $g(n)$  to be the function with  $g(n) = 1$  if  $n$  is an odd positive integer and  $g(n) = n$  if  $n$  is an even positive integer. **49.** There are positive constants  $C_1, C_2, C'_1, C'_2, k_1, k'_1, k_2$ , and  $k'_2$  such that  $|f_1(x)| \geq C_1|g_1(x)|$  for all  $x > k_1$ ,  $|f_1(x)| \leq C'_1|g_1(x)|$  for all  $x \geq k'_1$ ,  $|f_2(x)| > C_2|g_2(x)|$  for all  $x > k_2$ , and  $|f_2(x)| \leq C'_2|g_2(x)|$  for all  $x > k'_2$ . Because  $f_2$  and  $g_2$  are never zero, the last two inequalities can be rewritten as  $|1/f_2(x)| \leq (1/C_2)|1/g_2(x)|$  for all  $x > k_2$  and  $|1/f_2(x)| \geq (1/C'_2)|1/g_2(x)|$  for all  $x > k'_2$ . Multiplying the first and rewritten fourth inequalities shows that  $|f_1(x)/f_2(x)| \geq (C_1/C'_2)|g_1(x)/g_2(x)|$  for all  $x > \max(k_1, k'_2)$ , and multiplying the second and rewritten third inequalities gives  $|f_1(x)/f_2(x)| \leq (C'_1/C_2)|g_1(x)/g_2(x)|$  for all  $x > \max(k'_1, k_2)$ . It follows that  $f_1/f_2$  is big-Theta of  $g_1/g_2$ . **51.** There exist positive constants  $C_1, C_2, k_1, k_2, k'_1, k'_2$  such that  $|f(x, y)| \leq C_1|g(x, y)|$  for all  $x > k_1$  and  $y > k_2$  and  $|f(x, y)| \geq C_2|g(x, y)|$  for all  $x > k'_1$  and  $y > k'_2$ . **53.**  $(x^2 + xy + x \log y)^3 < (3x^2y^3) = 27x^6y^3$  for  $x > 1$  and  $y > 1$ , because  $x^2 < x^2y, xy < x^2y$ , and  $x \log y < x^2y$ . Hence,  $(x^2 + xy + x \log y)^3$  is  $O(x^6y^3)$ . **55.** For all positive real numbers  $x$  and  $y$ ,  $\lfloor xy \rfloor \leq xy$ . Hence,  $\lfloor xy \rfloor$  is  $O(xy)$  from the definition, taking  $C = 1$  and  $k_1 = k_2 = 0$ . **57.** Clearly  $n^d < n^c$  for all  $n \geq 2$ ; therefore,  $n^d$  is  $O(n^c)$ . The ratio  $n^d/n^c = n^{d-c}$  is unbounded so there is no constant  $C$  such that  $n^d \leq Cn^c$  for large  $n$ . **59.** If  $f$  and  $g$  are positive-valued functions such that  $\lim_{n \rightarrow \infty} f(x)/g(x) = C < \infty$ , then  $f(x) < (C + 1)g(x)$  for large enough  $x$ , so  $f(n)$  is  $O(g(n))$ . If that limit is  $\infty$ , then  $f(n)$  is not  $O(g(n))$ . Here repeated applications of L'Hôpital's rule shows that  $\lim_{x \rightarrow \infty} x^d/b^x = 0$  and  $\lim_{x \rightarrow \infty} b^x/x^d = \infty$ . **61.** To show that  $c^n$  is  $O(n!)$ , assume WLOG that  $c$  is an integer greater than 1. Claim that if  $n \geq c^2 + c$ , then  $n! \geq c^n$ . Both  $n!$  and  $c^n$  have  $n$  factors. Replacing each of the factors from  $c^2 + 1$  to  $c^2 + c$  in  $n!$  by  $c^2$  only makes the product smaller. But then each of these factors  $c^2$  can be factored as  $c \cdot c$  and one of those factors of  $c$  moved to pair with the factors 1 through  $c$  in  $n!$ . At this point every factor of  $n!$  is greater than or equal to  $c$ , so the product is greater than or equal to  $c^n$ . To show that  $n!$  is not  $O(c^n)$ , look at the ratio  $n!/c^n$  and write this as  $(c!/c^c) \cdot ((c+1)/c) \cdot ((c+2)/c) \cdot ((c+3)/c) \cdots ((2c)/c) \cdot ((2c+1)/c) \cdots (n/c)$ . Treat the product of the



first  $c + 1$  factors as a constant and note that every other factor is greater than 2. Thus, the product is greater than an arbitrarily large power of 2 as  $n \rightarrow \infty$ , which therefore approaches  $\infty$ . Therefore,  $n!$  cannot be bounded by a constant times  $c^n$ .

**63. a)**  $\lim_{x \rightarrow \infty} x^2/x^3 = \lim_{x \rightarrow \infty} 1/x = 0$  **b)**  $\lim_{x \rightarrow \infty} \frac{x \log x}{x^2} = \lim_{x \rightarrow \infty} \frac{\log x}{x} = \lim_{x \rightarrow \infty} \frac{1}{x \ln 2} = 0$  (using L'Hôpital's rule) **c)**  $\lim_{x \rightarrow \infty} \frac{x^2}{2^x} = \lim_{x \rightarrow \infty} \frac{2x}{2^x \cdot \ln 2} = \lim_{x \rightarrow \infty} \frac{2}{2^x \cdot (\ln 2)^2} = 0$  (using L'Hôpital's rule) **d)**  $\lim_{x \rightarrow \infty} \frac{x^2+x+1}{x^2} = \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x} + \frac{1}{x^2}\right) = 1 \neq 0$



**67.** No. Take  $f(x) = 1/x^2$  and  $g(x) = 1/x$ . **69. a)** Because  $\lim_{x \rightarrow \infty} f(x)/g(x) = 0$ ,  $|f(x)|/|g(x)| < 1$  for sufficiently large  $x$ . Hence,  $|f(x)| < |g(x)|$  for  $x > k$  for some constant  $k$ . Therefore,  $f(x)$  is  $O(g(x))$ . **b)** Let  $f(x) = g(x) = x$ . Then  $f(x)$  is  $O(g(x))$ , but  $f(x)$  is not  $o(g(x))$  because  $f(x)/g(x) = 1$ . **71.** Because  $f_2(x)$  is  $o(g(x))$ , from Exercise 69(a) it follows that  $f_2(x)$  is  $O(g(x))$ . By Corollary 1, we have  $f_1(x) + f_2(x)$  is  $O(g(x))$ . **73.** We can show that  $(n - i)(i + 1) \geq n$  for  $i = 0, 1, \dots, n - 1$ . Hence,  $(n!)^2 = (n \cdot 1)((n - 1) \cdot 2) \cdot ((n - 2) \cdot 3) \cdots (2 \cdot (n - 1)) \cdot (1 \cdot n) \geq n^n$ . Therefore,  $2 \log n! \geq n \log n$ . **75.** Compute that  $\log 5! \approx 6.9$  and  $(5 \log 5)/4 \approx 2.9$ , so the inequality holds for  $n = 5$ . Assume  $n \geq 6$ . Because  $n!$  is the product of all the integers from  $n$  down to 1, we have  $n! > n(n - 1)(n - 2) \cdots \lceil n/2 \rceil$  (because at least the term 2 is missing). Note that there are more than  $n/2$  terms in this product, and each term is at least as big as  $n/2$ . Therefore, the product is greater than  $(n/2)^{(n/2)}$ . Taking the log of both sides of the inequality, we have  $\log n! > \log \left(\frac{n}{2}\right)^{n/2} = \frac{n}{2} \log \frac{n}{2} = \frac{n}{2}(\log n - 1) > (n \log n)/4$ , because  $n > 4$  implies  $\log n - 1 > (\log n)/2$ . **77.** All are not asymptotic.

### Section 3.3

**1.**  $O(1)$  **3.**  $O(n^2)$  **5.**  $2n - 1$  **7.** Linear **9.**  $O(n)$   
**11. a) procedure disjointpair( $S_1, S_2, \dots, S_n$  :**  
     subsets of  $\{1, 2, \dots, n\}$ )  
     **answer := false**  
     **for**  $i := 1$  **to**  $n$   
         **for**  $j := i + 1$  **to**  $n$   
             **disjoint := true**

**for**  $k := 1$  **to**  $n$   
     **if**  $k \in S_i$  and  $k \in S_j$  **then** **disjoint := false**  
     **if** **disjoint** **then** **answer := true**  
**return answer**

**b)**  $O(n^3)$  **13. a)**  $power := 1, y := 1; i := 1, power := 2, y := 3; i := 2, power := 4, y := 15$  **b)**  $2n$  multiplications and  $n$  additions **15. a)**  $2^{10^9} \approx 10^{3 \times 10^8}$  **b)**  $10^9$  **c)**  $3.96 \times 10^7$  **d)**  $3.16 \times 10^4$  **e)** 29 **f)** 12 **17. a)**  $2^{2^{60 \cdot 10^{12}}}$  **b)**  $2^{60 \cdot 10^{12}}$  **c)**  $\lfloor 2^{\sqrt{60 \cdot 10^6}} \rfloor \approx 2 \times 10^{2331768}$  **d)** 60,000,000 **e)** 7,745,966 **f)** 45 **g)** 6 **19. a)** 36 years **b)** 13 days **c)** 19 minutes **21. a)** Less than 1 millisecond more **b)** 100 milliseconds more **c)**  $2n + 1$  milliseconds more **d)**  $3n^2 + 3n + 1$  milliseconds more **e)** Twice as much time **f)**  $2^{2n+1}$  times as many milliseconds **g)**  $n + 1$  times as many milliseconds **23.** The average number of comparisons is  $(3n + 4)/2$ . **25.**  $O(\log n)$  **27.**  $O(n)$  **29.**  $O(n^2)$  **31.**  $O(n)$  **33.**  $O(n)$  **35.**  $O(\log n)$  comparisons;  $O(n^2)$  swaps **37.**  $O(m(n - m))$  **39. a)**  $O(n^2)$  **b)**  $O(n \log n)$  **41.**  $O(n^2 2^n)$  **43. a)** doubles **b)** increases by 1 **45.** Use Algorithm 1, where **A** and **B** are now  $n \times n$  upper triangular matrices, by replacing  $m$  by  $n$  in line 1, and having  $q$  iterate only from  $i$  to  $j$ , rather than from 1 to  $k$ . **47.**  $n(n + 1)(n + 2)/6$  **49.** **A((BC)D)**

### Supplementary Exercises

**1. a) procedure last max( $a_1, \dots, a_n$ : integers)**

**max :=  $a_1$**   
**last := 1**  
 **$i := 2$**   
**while**  $i \leq n$   
     **if**  $a_i \geq \text{max}$  **then**  
         **max :=  $a_i$**   
         **last :=  $i$**   
      **$i := i + 1$**   
**return last**

**b)**  $2n - 1 = O(n)$  comparisons

**3. a) procedure pair zeros( $b_1 b_2 \dots b_n$ : bit string,  $n \geq 2$ )**

**$x := b_1$**   
 **$y := b_2$**   
 **$k := 2$**   
**while**  $k < n$  and  $(x \neq 0$  or  $y \neq 0)$   
      **$k := k + 1$**   
      **$x := y$**   
      **$y := b_k$**   
**if**  $x = 0$  and  $y = 0$  **then** print "YES"  
**else** print "NO"

**b)**  $O(n)$

**5. a) and b)**

**procedure smallest and largest( $a_1, a_2, \dots, a_n$ : integers)**

**min :=  $a_1$**   
**max :=  $a_1$**



**for**  $i := 2$  **to**  $n$   
     **if**  $a_i < \min$  **then**  $\min := a_i$   
     **if**  $a_i > \max$  **then**  $\max := a_i$   
 { $\min$  is the smallest integer among the input, and  $\max$  is the largest}

c)  $2n - 2$

7. Before any comparisons are done, there is a possibility that each element could be the maximum and a possibility that it could be the minimum. This means that there are  $2n$  different possibilities, and  $2n - 2$  of them have to be eliminated through comparisons of elements, because we need to find the unique maximum and the unique minimum. We classify comparisons of two elements as “virgin” or “nonvirgin,” depending on whether or not both elements being compared have been in any previous comparison. A virgin comparison eliminates the possibility that the larger one is the minimum and that the smaller one is the maximum; thus, each virgin comparison eliminates two possibilities, but it clearly cannot do more. A nonvirgin comparison must be between two elements that are still in the running to be the maximum or two elements that are still in the running to be the minimum, and at least one of these elements must *not* be in the running for the other category. For example, we might be comparing  $x$  and  $y$ , where all we know is that  $x$  has been eliminated as the minimum. If we find that  $x > y$  in this case, then only one possibility has been ruled out—we now know that  $y$  is not the maximum. Thus, in the worst case, a nonvirgin comparison eliminates only one possibility. (The cases of other nonvirgin comparisons are similar.) Now there are at most  $\lfloor n/2 \rfloor$  comparisons of elements that have not been compared before, each removing two possibilities; they remove  $2\lfloor n/2 \rfloor$  possibilities altogether. Therefore, we need  $2n - 2 - 2\lfloor n/2 \rfloor$  more comparisons that, as we have argued, can remove only one possibility each, in order to find the answers in the worst case, because  $2n - 2$  possibilities have to be eliminated. This gives us a total of  $2n - 2 - 2\lfloor n/2 \rfloor + \lfloor n/2 \rfloor$  comparisons in all. But  $2n - 2 - 2\lfloor n/2 \rfloor + \lfloor n/2 \rfloor = 2n - 2 - \lfloor n/2 \rfloor = 2n - 2 + \lceil -n/2 \rceil = \lceil 2n - n/2 \rceil - 2 = \lceil 3n/2 \rceil - 2$ , as desired. 9. The following algorithm has worst-case complexity  $O(n^4)$ .

**procedure** *equal sums*( $a_1, a_2, \dots, a_n$ )  
**for**  $i := 1$  **to**  $n$   
     **for**  $j := i + 1$  **to**  $n$  {since we want  $i < j$ }  
         **for**  $k := 1$  **to**  $n$   
             **for**  $l := k + 1$  **to**  $n$  {since we want  $k < l$ }  
                 **if**  $a_i + a_j = a_k + a_l$  and  $(i, j) \neq (k, l)$   
                     **then** output these pairs

11. At end of first pass: 3, 1, 4, 5, 2, 6; at end of second pass: 1, 3, 2, 4, 5, 6; at end of third pass: 1, 2, 3, 4, 5, 6; fourth pass finds nothing to exchange and algorithm terminates 13. There are possibly as many as  $n$  passes through the list, and each pass uses  $O(n)$  comparisons. Thus, there are  $O(n^2)$  comparisons in all. 15. Because  $\log n < n$ , we have  $(n \log n + n^2)^3 \leq (n^2 + n^2)^3 \leq (2n^2)^3 = 8n^6$  for all  $n > 0$ . This proves that  $(n \log n + n^2)^3$  is  $O(n^6)$ , with witnesses  $C = 8$  and  $k = 0$ . 17.  $O(x^2 2^x)$  19. Note that  $\frac{n!}{2^n} = \frac{n}{2} \cdot \frac{n-1}{2} \cdots \frac{3}{2} \cdot \frac{2}{2} \cdot \frac{1}{2} >$

$\frac{n}{2} \cdot 1 \cdot 1 \cdots 1 \cdot \frac{1}{2} = \frac{n}{4}$ . 21. All of these functions are of the same order. 23.  $2^{107}$  25.  $(\log n)^2$ ,  $2^{\sqrt{\log_2 n}}$ ,  $n(\log n)^{1001}$ ,  $n^{1.0001}$ ,  $1.0001^n$ ,  $n^n$  27. For example,  $f(n) = n^{2\lfloor n/2 \rfloor + 1}$  and  $g(n) = n^{2\lceil n/2 \rceil}$

29. a) **procedure** *brute*( $a_1, a_2, \dots, a_n$  : integers)  
     **for**  $i := 1$  **to**  $n - 1$   
         **for**  $j := i + 1$  **to**  $n$   
             **for**  $k := 1$  **to**  $n$   
                 **if**  $a_i + a_j = a_k$  **then return true**  
             **else return false**

b)  $O(n^3)$

31. For  $m_1$ :  $w_1$  and  $w_2$ ; for  $m_2$ :  $w_1$  and  $w_3$ ; for  $m_3$ :  $w_2$  and  $w_3$ ; for  $w_1$ :  $m_1$  and  $m_2$ ; for  $w_2$ :  $m_1$  and  $m_3$ ; for  $w_3$ :  $m_2$  and  $m_3$  33. A matching in which each woman is assigned her valid partner ranking highest on her preference list is female optimal; a matching in which each man is assigned his valid partner ranking lowest on his preference list is male pessimal. 35. a) Modify the preamble to Exercise 64 in Section 3.1 so that there are  $s$  men  $m_1, m_2, \dots, m_s$  and  $t$  women  $w_1, w_2, \dots, w_t$ . A matching will contain  $\min(s, t)$  marriages. The definition of “stable marriage” is the same, with the understanding that each person prefers any mate to being unmatched. b) Create  $|s - t|$  fictitious people (men or women, whichever is in shorter supply) so that the number of men and the number of women become the same, and put these fictitious people at the bottom of everyone’s preference lists. c) This follows immediately from Exercise 67 in Section 3.1. 37. 5; 15 39. The first situation in Exercise 37 41. a) For each subset  $S$  of  $\{1, 2, \dots, n\}$ , compute  $\sum_{j \in S} w_j$ . Keep track of the subset giving the largest such sum that is less than or equal to  $W$ , and return that subset as the output of the algorithm. b) The food pack and the portable stove 43. a) The makespan is always at least as large as the load on the processor assigned to do the lengthiest job, which must be at least  $\max_{j=1,2,\dots,n} t_j$ . Therefore, the minimum makespan satisfies this inequality. b) The total amount of time the processors need to spend working on the jobs (the total load) is  $\sum_{j=1}^n t_j$ . Therefore, the average load per processor is  $\frac{1}{p} \sum_{j=1}^n t_j$ . The maximum load cannot be any smaller than the average, so the minimum makespan is always at least this large. 45. Processor 1: jobs 1, 4; processor 2: job 2; processor 3: jobs 3, 5

## CHAPTER 4

### Section 4.1

1. a) Yes b) No c) Yes d) No 3. Suppose that  $a \mid b$ . Then there exists an integer  $k$  such that  $ka = b$ . Because  $a(ck) = bc$  it follows that  $a \mid bc$ . 5. If  $a \mid b$  and  $b \mid a$ , there are integers  $c$  and  $d$  such that  $b = ac$  and  $a = bd$ . Hence,  $a = acd$ . Because  $a \neq 0$  it follows that  $cd = 1$ . Thus, either  $c = d = 1$  or  $c = d = -1$ . Hence, either  $a = b$  or  $a = -b$ . 7. Because  $ac \mid bc$  there is an integer  $k$  such



that  $ack = bc$ . Hence,  $ak = b$ , so  $a \mid b$ . **9.** It is given that  $b = m \cdot a$  for some integer  $m$ . If  $a$  is not odd, then  $a = 2k$  for some integer  $k$ , whence  $b = 2km$ . This means by definition that  $b$  is even. **11.** If  $a$  is an integer that is not divisible by 3, then it must leave a remainder of either 1 or 2 when divided by 3. In the first case  $a = 3k + 1$  for some integer  $k$ , so  $(a+1)(a+2) = (3k+2)(3k+3) = 3(3k+2)(k+1)$  and so is divisible by 3. In the second case  $a = 3k + 2$  for some integer  $k$ , so  $(a+1)(a+2) = (3k+3)(3k+4) = 3(k+1)(3k+4)$  and so is divisible by 3. **13. a)** 2, 5 **b)** -11, 10 **c)** 34, 7 **d)** 77, 0 **e)** 0, 0 **f)** 0, 3 **g)** -1, 2 **h)** 4, 0 **15. a)** 7:00 **b)** 8:00 **c)** 10:00 **17. a)** 10 **b)** 8 **c)** 0 **d)** 9 **e)** 6 **f)** 11 **19.** If  $d \mid a$ , then  $a = md$  for some integer  $m$ , and it follows that  $-a = (-m)d$  as well. Therefore,  $-(a \text{ div } d) = -m$  and  $(-a) \text{ div } d = -m$ , as desired. Conversely, if  $d$  does not divide  $a$ , then  $a = qd + r$  for some  $r$  with  $0 < r < d$ . Here  $q = a \text{ div } d$ . It follows that  $-a = (-q)d - r = (-q-1)d + (d-r)$ . Note that  $0 < d-r < d$ , so by definition  $(-a) \text{ div } d = -q-1$ . But  $-(a \text{ div } d) = -q$ , so  $(-a) \text{ div } d \neq -(a \text{ div } d)$ . **21.** If  $a \bmod m = b \bmod m$ , then  $a$  and  $b$  have the same remainder when divided by  $m$ . Hence,  $a = q_1m + r$  and  $b = q_2m + r$ , where  $0 \leq r < m$ . It follows that  $a - b = (q_1 - q_2)m$ , so  $m \mid (a - b)$ . It follows that  $a \equiv b \pmod{m}$ . **23.** There is some  $b$  with  $(b-1)k < n \leq bk$ . Hence,  $(b-1)k \leq n-1 < bk$ . Divide by  $k$  to obtain  $b-1 < n/k \leq b$  and  $b-1 \leq (n-1)/k < b$ . Hence,  $\lceil n/k \rceil = b$  and  $\lfloor (n-1)/k \rfloor = b-1$ . **25.**  $x \bmod m$  if  $x \bmod m \leq \lceil m/2 \rceil$  and  $(x \bmod m) - m$  if  $x \bmod m > \lceil m/2 \rceil$ . **27. a)** 1 **b)** 2 **c)** 3 **d)** 9 **29. a)** 1, 109 **b)** 40, 89 **c)** -31, 222 **d)** -21, 38259 **31. a)** -15 **b)** -7 **c)** 140 **33.** -1, -26, -51, -76, 24, 49, 74, 99 **35. a)** No **b)** No **c)** Yes **d)** No **37. a)** 13 **a)** 6 **39. a)** 9 **b)** 4 **c)** 25 **d)** 0 **41.** Let  $m = tn$ . Because  $a \equiv b \pmod{m}$  there exists an integer  $s$  such that  $a = b + sm$ . Hence,  $a = b + (st)n$ , so  $a \equiv b \pmod{n}$ . **43. a)** Let  $m = c = 2$ ,  $a = 0$ , and  $b = 1$ . Then  $0 = ac \equiv bc = 2 \pmod{2}$ , but  $0 = a \not\equiv b = 1 \pmod{2}$ . **b)** Let  $m = 5$ ,  $a = b = 3$ ,  $c = 1$ , and  $d = 6$ . Then  $3 \equiv 3 \pmod{5}$  and  $1 \equiv 6 \pmod{5}$ , but  $3^1 = 3 \not\equiv 4 \equiv 729 = 3^6 \pmod{5}$ . **45.** By Exercise 44 the sum of two squares must be either  $0 + 0 = 0$ ,  $0 + 1 = 1$ , or  $1 + 1 = 2$ , modulo 4, never 3, and therefore not of the form  $4k + 3$ . **47.** Because  $a \equiv b \pmod{m}$ , there exists an integer  $s$  such that  $a = b + sm$ , so  $a - b = sm$ . Then  $a^k - b^k = (a - b)(a^{k-1} + a^{k-2}b + \cdots + ab^{k-2} + b^{k-1})$ ,  $k \geq 2$ , is also a multiple of  $m$ . It follows that  $a^k \equiv b^k \pmod{m}$ . **49.** To prove closure, note that  $a \cdot_m b = (a \cdot b) \bmod m$ , which by definition is an element of  $\mathbf{Z}_m$ . Multiplication is associative because  $(a \cdot_m b) \cdot_m c$  and  $a \cdot_m (b \cdot_m c)$  both equal  $(a \cdot b \cdot c) \bmod m$  and multiplication of integers is associative. Similarly, multiplication in  $\mathbf{Z}_m$  is commutative because multiplication in  $\mathbf{Z}$  is commutative, and 1 is the multiplicative identity for  $\mathbf{Z}_m$  because 1 is the multiplicative identity for  $\mathbf{Z}$ . **51.**  $0+_50 = 0$ ,  $0+_51 = 1$ ,  $0+_52 = 2$ ,  $0+_53 = 3$ ,  $0+_54 = 4$ ;  $1+_51 = 2$ ,  $1+_52 = 3$ ,  $1+_53 = 4$ ,  $1+_54 = 0$ ;  $2+_52 = 4$ ,  $2+_53 = 0$ ,  $2+_54 = 1$ ;  $3+_53 = 1$ ,  $3+_54 = 2$ ;  $4+_54 = 3$  and  $0 \cdot_50 = 0$ ,  $0 \cdot_51 = 0$ ,  $0 \cdot_52 = 0$ ,  $0 \cdot_53 = 0$ ,  $0 \cdot_54 = 0$ ;  $1 \cdot_51 = 1$ ,  $1 \cdot_52 = 2$ ,  $1 \cdot_53 = 3$ ,  $1 \cdot_54 = 4$ ;  $2 \cdot_52 = 4$ ,  $2 \cdot_53 = 1$ ,  $2 \cdot_54 =$

$3$ ;  $3 \cdot_53 = 4$ ,  $3 \cdot_54 = 2$ ;  $4 \cdot_54 = 1$  **53.**  $f$  is onto but not one-to-one (unless  $d = 1$ );  $g$  is neither.

## Section 4.2

**1. a)** 1110 0111 **b)** 1 0001 1011 0100 **c)** 1 0111 11010110 1100 **3. a)** 31 **b)** 513 **c)** 341 **d)** 26,896 **5. a)** 1 0111 1010 **b)** 11 1000 0100 **c)** 1 0001 0011 **d)** 101 0000 1111 **7. a)** 1000 0000 1110 **b)** 1 0011 0101 1010 1011 **c)** 10101011 1011 1010 **d)** 1101 1110 1111 1010 11001110 1101 **9.** 1010 1011 1100 1101 1110 1111 **11.**  $(B7B)_{16}$  **13.** Adding up to three leading 0s if necessary, write the binary expansion as  $(\dots b_{23}b_{22}b_{21}b_{20}b_{13}b_{12}b_{11}b_{10}b_{03}b_{02}b_{01}b_{00})_2$ . The value of this numeral is  $b_{00} + 2b_{01} + 4b_{02} + 8b_{03} + 2^4b_{10} + 2^5b_{11} + 2^6b_{12} + 2^7b_{13} + 2^8b_{20} + 2^9b_{21} + 2^{10}b_{22} + 2^{11}b_{23} + \cdots$ , which we can rewrite as  $b_{00} + 2b_{01} + 4b_{02} + 8b_{03} + (b_{10} + 2b_{11} + 4b_{12} + 8b_{13}) \cdot 2^4 + (b_{20} + 2b_{21} + 4b_{22} + 8b_{23}) \cdot 2^8 + \cdots$ . Now  $(b_{13}b_{12}b_{11}b_{10})_2$  translates into the hexadecimal digit  $h_i$ . So our number is  $h_0 + h_1 \cdot 2^4 + h_2 \cdot 2^8 + \cdots = h_0 + h_1 \cdot 16 + h_2 \cdot 16^2 + \cdots$ , which is the hexadecimal expansion  $(\dots h_1h_1h_0)_{16}$ . **15.** Adding up to two leading 0s if necessary, write the binary expansion as  $(\dots b_{22}b_{21}b_{20}b_{12}b_{11}b_{10}b_{02}b_{01}b_{00})_2$ . The value of this numeral is  $b_{00} + 2b_{01} + 4b_{02} + 2^3b_{10} + 2^4b_{11} + 2^5b_{12} + 2^6b_{20} + 2^7b_{21} + 2^8b_{22} + \cdots$ , which we can rewrite as  $b_{00} + 2b_{01} + 4b_{02} + (b_{10} + 2b_{11} + 4b_{12}) \cdot 2^3 + (b_{20} + 2b_{21} + 4b_{22}) \cdot 2^6 + \cdots$ . Now  $(b_{12}b_{11}b_{10})_2$  translates into the octal digit  $h_i$ . So our number is  $h_0 + h_1 \cdot 2^3 + h_2 \cdot 2^6 + \cdots = h_0 + h_1 \cdot 8 + h_2 \cdot 8^2 + \cdots$ , which is the octal expansion  $(\dots h_1h_1h_0)_8$ . **17.** 1 1101 1100 1010 1101 0001, 1273)<sub>8</sub> **19.** Convert the given octal numeral to binary, then convert from binary to hexadecimal using Example 7. **21. a)** 1011 1110, 10 0001 0000 0001 **b)** 1 1010 1100, 1011 0000 0111 0011 **c)** 100 1001 1010, 101 0010 1001 0110 0000 **d)** 110 0000 0000, 1000 0000 0001 1111 1111 **23. a)** 1132, 144,305 **b)** 6273, 2,134,272 **c)** 2110, 1,107,667 **d)** 57,777, 237,326,216 **25.** 436 **27.** 27 **29.** The binary expansion of the integer is the unique such sum. **31.** Let  $a = (a_{n-1}a_{n-2} \cdots a_1a_0)_{10}$ . Then  $a = 10^{n-1}a_{n-1} + 10^{n-2}a_{n-2} + \cdots + 10a_1 + a_0 \equiv a_{n-1} + a_{n-2} + \cdots + a_1 + a_0 \pmod{3}$ , because  $10^j \equiv 1 \pmod{3}$  for all nonnegative integers  $j$ . It follows that  $3 \mid a$  if and only if 3 divides the sum of the decimal digits of  $a$ . **33.** Let  $a = (a_{n-1}a_{n-2} \cdots a_1a_0)_2$ . Then  $a = a_0 + 2a_1 + 2^2a_2 + \cdots + 2^{n-1}a_{n-1} \equiv a_0 - a_1 + a_2 - a_3 + \cdots \pm a_{n-1} \pmod{3}$ . It follows that  $a$  is divisible by 3 if and only if the sum of the binary digits in the even-numbered positions minus the sum of the binary digits in the odd-numbered positions is divisible by 3. **35. a)**  $n$  is divisible by 4 if and only if the two rightmost digits of the decimal expansion, viewed as a two-digit integer (or the single digit if it's a one-digit integer), is divisible by 4. **b)**  $n$  is divisible by 25 if and only if the two rightmost digits are 00 (or 0 if it's a one-digit integer), 25, 50, or 75. **c)**  $n$  is divisible by 20 if and only if the rightmost digit of the decimal expansion is 0 and the digit in the tens place (if it's not a one-digit integer)



is 0, 2, 4, 6, or 8. **37.** The base  $b$  representation of  $n$  has  $k$  digits iff  $b^{k-1} \leq n < b^k$  iff  $k-1 \leq \log_b n < k$  iff  $\lfloor \log_b n \rfloor + 1 = k$ . **39.**  $5(8^n - 1)/7$  **41. a)** -6 **b)** 13 **c)** -14 **d)** 0 **43.** The one's complement of the sum is found by adding the one's complements of the two integers except that a carry in the leading bit is used as a carry to the last bit of the sum. **45.** If  $m \geq 0$ , then the leading bit  $a_{n-1}$  of the one's complement expansion of  $m$  is 0 and the formula reads  $m = \sum_{i=0}^{n-2} a_i 2^i$ . This is correct because the right-hand side is the binary expansion of  $m$ . When  $m$  is negative, the leading bit  $a_{n-1}$  of the one's complement expansion of  $m$  is 1. The remaining  $n-1$  bits can be obtained by subtracting  $-m$  from  $111 \dots 1$  (where there are  $n-1$  1s), because subtracting a bit from 1 is the same as complementing it. Hence, the bit string  $a_{n-2} \dots a_0$  is the binary expansion of  $(2^{n-1} - 1) - (-m)$ . Solving the equation  $(2^{n-1} - 1) - (-m) = \sum_{i=0}^{n-2} a_i 2^i$  for  $m$  gives the desired equation because  $a_{n-1} = 1$ . **47. a)** -7 **b)** 13 **c)** -15 **d)** -1 **49.** To obtain the two's complement representation of the sum of two integers, add their two's complement representations (as binary integers are added) and ignore any carry out of the leftmost column. However, the answer is invalid if an overflow has occurred. This happens when the leftmost digits in the two's complement representation of the two terms agree and the leftmost digit of the answer differs. **51.** If  $m \geq 0$ , then the leading bit  $a_{n-1}$  is 0 and the formula reads  $m = \sum_{i=0}^{n-2} a_i 2^i$ . This is correct because the right-hand side is the binary expansion of  $m$ . If  $m < 0$ , its two's complement expansion has 1 as its leading bit and the remaining  $n-1$  bits are the binary expansion of  $2^{n-1} - (-m)$ . This means that  $(2^{n-1}) - (-m) = \sum_{i=0}^{n-2} a_i 2^i$ . Solving for  $m$  gives the desired equation because  $a_{n-1} = 1$ . **53.**  $4n$

**55. procedure** *Cantor*( $x$ : positive integer)

```

 $n := 1; f := 1$ 
while  $(n+1) \cdot f \leq x$ 
     $n := n+1$ 
     $f := f \cdot n$ 
 $y := x$ 
while  $n > 0$ 
     $a_n := \lfloor y/f \rfloor$ 
     $y := y - a_n \cdot f$ 
     $f := f/n$ 
     $n := n-1$ 
 $\{x = a_n n! + a_{n-1}(n-1)! + \dots + a_1 1!\}$ 

```

**57.** First step:  $c = 0, d = 0, s_0 = 1$ ; second step:  $c = 0, d = 1, s_1 = 0$ ; third step:  $c = 1, d = 1, s_2 = 0$ ; fourth step:  $c = 1, d = 1, s_3 = 0$ ; fifth step:  $c = 1, d = 1, s_4 = 1$ ; sixth step:  $c = 1, s_5 = 1$

**59. procedure** *subtract*( $a, b$ : positive integers,  $a > b$ ,

```

 $a = (a_{n-1} a_{n-2} \dots a_1 a_0)_2,$ 
 $b = (b_{n-1} b_{n-2} \dots b_1 b_0)_2$ 
 $B := 0$  { $B$  is the borrow}
for  $j := 0$  to  $n-1$ 
    if  $a_j \geq b_j + B$  then
         $s_j := a_j - b_j - B$ 
         $B := 0$ 

```

**else**

```

 $s_j := a_j + 2 - b_j - B$ 
 $B := 1$ 

```

{ $(s_{n-1} s_{n-2} \dots s_1 s_0)_2$  is the difference}

**61. procedure** *compare*( $a, b$ : positive integers,

```

 $a = (a_n a_{n-1} \dots a_1 a_0)_2, b = (b_n b_{n-1} \dots b_1 b_0)_2$ 
 $k := n$ 
while  $a_k = b_k$  and  $k > 0$ 
     $k := k-1$ 
if  $a_k = b_k$  then print " $a$  equals  $b$ "
if  $a_k > b_k$  then print " $a$  is greater than  $b$ "
if  $a_k < b_k$  then print " $a$  is less than  $b$ "

```

**63.**  $O(\log n)$  **65.** The only time-consuming part of the algorithm is the **while** loop, which is iterated  $q$  times. The work done inside is a subtraction of integers no bigger than  $a$ , which has  $\log a$  bits. The result now follows from Example 9.

### Section 4.3

**1.** 29, 71, 97 prime; 21, 111, 143 not prime **3. a)**  $2^3 \cdot 11$  **b)**  $2 \cdot 3^2 \cdot 7$  **c)**  $3^6$  **d)**  $7 \cdot 11 \cdot 13$  **e)**  $11 \cdot 101$  **f)**  $2 \cdot 3^3 \cdot 5 \cdot 7 \cdot 13 \cdot 37$  **5.**  $2^8 \cdot 3^4 \cdot 5^2 \cdot 7$

**7. procedure** *primetester*( $n$ : integer greater than 1)

```

 $isprime := \text{true}$ 
 $d := 2$ 
while  $isprime$  and  $d \leq \sqrt{n}$ 
    if  $n \bmod d = 0$  then  $isprime := \text{false}$ 
    else  $d := d+1$ 
return  $isprime$ 

```

**9.** Write  $n = rs$ , where  $r > 1$  and  $s > 1$ . Then  $2^n - 1 = 2^{rs} - 1 = (2^r)^s - 1 = (2^r - 1)((2^r)^{s-1} + (2^r)^{s-2} + \dots + 1)$ . The first factor is at least  $2^2 - 1 = 3$  and the second factor is at least  $2^2 + 1 = 5$ . This provides a factoring of  $2^n - 1$  into two factors greater than 1, so  $2^n - 1$  is composite.

**11.** Suppose that  $\log_2 3 = a/b$  where  $a, b \in \mathbb{Z}^+$  and  $b \neq 0$ . Then  $2^{a/b} = 3$ , so  $2^a = 3^b$ . This violates the fundamental theorem of arithmetic. Hence,  $\log_2 3$  is irrational. **13.** 3, 5, and 7 are primes of the desired form. **15.** 1, 7, 11, 13, 17, 19, 23, 29

**17. a)** Yes **b)** No **c)** Yes **d)** Yes **19.** Suppose that  $n$  is not prime, so that  $n = ab$ , where  $a$  and  $b$  are integers greater than 1. Because  $a > 1$ , by the identity in the hint,  $2^a - 1$  is a factor of  $2^n - 1$  that is greater than 1, and the second factor in this identity is also greater than 1. Hence,  $2^n - 1$  is not prime.

**21. a)** 2 **b)** 4 **c)** 12 **23.**  $\phi(p^k) = p^k - p^{k-1}$  **25. a)**  $3^5 \cdot 5^3$  **b)** 1 **c)**  $23^{17}$  **d)**  $41 \cdot 43 \cdot 53$  **e)** 1 **f)** 1111 **27. a)**  $2^{11} \cdot 3^7 \cdot 5^9 \cdot 7^3$  **b)**  $2^9 \cdot 3^7 \cdot 5^5 \cdot 7^3 \cdot 11 \cdot 13 \cdot 17$  **c)**  $23^{31}$  **d)**  $41 \cdot 43 \cdot 53$  **e)**  $2^{12} 3^{13} 5^{17} 7^{21}$  **f)** Undefined **29.**  $\gcd(92928, 123552) = 1056$ ;  $\text{lcm}(92928, 123552) = 10,872,576$ ; both products are 11,481,440,256.

**31.** Because  $\min(x, y) + \max(x, y) = x + y$ , the exponent of  $p_i$  in the prime factorization of  $\gcd(a, b) \cdot \text{lcm}(a, b)$  is the sum of the exponents of  $p_i$  in the prime factorizations of  $a$  and  $b$ . **33. a)** 6 **b)** 3 **c)** 11 **d)** 3 **e)** 40 **f)** 12 **35.** 9 **37.** By Exercise 36 it follows that  $\gcd(2^b - 1, (2^a - 1) \bmod (2^b - 1)) = \gcd(2^b - 1, 2^{a \bmod b} - 1)$ .



Because the exponents involved in the calculation are  $b$  and  $a \bmod b$ , the same as the quantities involved in computing  $\gcd(a, b)$ , the steps used by the Euclidean algorithm to compute  $\gcd(2^a - 1, 2^b - 1)$  run in parallel to those used to compute  $\gcd(a, b)$  and show that  $\gcd(2^a - 1, 2^b - 1) = 2^{\gcd(a, b)} - 1$ .

- 39. a)**  $1 = (-1) \cdot 10 + 1 \cdot 11$  **b)**  $1 = 21 \cdot 21 + (-10) \cdot 44$   
**c)**  $12 = (-1) \cdot 36 + 48$  **d)**  $1 = 13 \cdot 55 + (-21) \cdot 34$   
**e)**  $3 = 11 \cdot 213 + (-20) \cdot 117$  **f)**  $223 = 1 \cdot 0 + 1 \cdot 223$  **g)**  $1 = 37 \cdot 2347 + (-706) \cdot 123$  **h)**  $2 = 1128 \cdot 3454 + (-835) \cdot 4666$  **i)**  $1 = 2468 \cdot 9999 + (-2221) \cdot 11111$  **41.**  $(-3) \cdot 26 + 1 \cdot 91 = 13$   
**43.**  $34 \cdot 144 + (-55) \cdot 89 = 1$

**45. procedure** *extended Euclidean*( $a, b$ : positive integers)

```

 $x := a$ 
 $y := b$ 
 $oldolds := 1$ 
 $olds := 0$ 
 $oldoldt := 0$ 
 $oldt := 1$ 
while  $y \neq 0$ 
   $q := x \text{ div } y$ 
   $r := x \bmod y$ 
   $x := y$ 
   $y := r$ 
   $s := oldolds - q \cdot olds$ 
   $t := oldoldt - q \cdot oldt$ 
   $oldolds := olds$ 
   $oldoldt := oldt$ 
   $olds := s$ 
   $oldt := t$ 
 $\{\gcd(a, b) \text{ is } x, \text{ and } (oldolds)a + (oldoldt)b = x\}$ 

```

- 47. a)**  $a_n = 1$  if  $n$  is prime and  $a_n = 0$  otherwise. **b)**  $a_n$  is the smallest prime factor of  $n$  with  $a_1 = 1$ . **c)**  $a_n$  is the number of positive divisors of  $n$ . **d)**  $a_n = 1$  if  $n$  has no divisors that are perfect squares greater than 1 and  $a_n = 0$  otherwise. **e)**  $a_n$  is the largest prime less than or equal to  $n$ . **f)**  $a_n$  is the product of the first  $n - 1$  primes. **49.** Because every second integer is divisible by 2, the product is divisible by 2. Because every third integer is divisible by 3, the product is divisible by 3. Therefore, the product has both 2 and 3 in its prime factorization and is therefore divisible by  $3 \cdot 2 = 6$ . **51.**  $n = 1601$  is a counterexample. **53.** Setting  $k = a + b + 1$  will produce the composite number  $a(a + b + 1) + b = a^2 + ab + a + b = (a + 1)(a + b)$ . **55.** Suppose that there are only finitely many primes of the form  $4k + 3$ , namely,  $q_1, q_2, \dots, q_n$ , where  $q_1 = 3, q_2 = 7$ , and so on. Let  $Q = 4q_1q_2 \cdots q_n - 1$ . Note that  $Q$  is of the form  $4k + 3$  (where  $k = q_1q_2 \cdots q_n - 1$ ). If  $Q$  is prime, then we have found a prime of the desired form different from all those listed. If  $Q$  is not prime, then  $Q$  has at least one prime factor not in the list  $q_1, q_2, \dots, q_n$ , because the remainder when  $Q$  is divided by  $q_j$  is  $q_j - 1$ , and  $q_j - 1 \neq 0$ . Because all odd primes are either of the form  $4k + 1$  or of the form  $4k + 3$ , and the product of primes of the form  $4k + 1$  is also of this form (because  $(4k + 1)(4m + 1) = 4(4km + k + m) + 1$ ), there must be a factor of  $Q$  of the form  $4k + 3$  different from the primes we listed. **57.** Given a positive integer  $x$ , we show that there

is exactly one positive rational number  $m/n$  (in lowest terms) such that  $K(m/n) = x$ . From the prime factorization of  $x$ , read off the  $m$  and  $n$  such that  $K(m/n) = x$ . The primes that occur to even powers are the primes that occur in the prime factorization of  $m$ , with the exponents being half the corresponding exponents in  $x$ ; and the primes that occur to odd powers are the primes that occur in the prime factorization of  $n$ , with the exponents being half of one more than the exponents in  $x$ .

## Section 4.4

- 1.**  $15 \cdot 7 = 105 \equiv 1 \pmod{26}$  **3.** 7 **5. a)** 7 **b)** 52 **c)** 34 **d)** 73 **7.** Suppose that  $b$  and  $c$  are both inverses of  $a$  modulo  $m$ . Then  $ba \equiv 1 \pmod{m}$  and  $ca \equiv 1 \pmod{m}$ . Hence,  $ba \equiv ca \pmod{m}$ . Because  $\gcd(a, m) = 1$  it follows by Theorem 7 in Section 4.3 that  $b \equiv c \pmod{m}$ . **9.** 8 **11. a)** 67 **b)** 88 **c)** 146 **13.** 3 and 6 **15.** Let  $m' = m / \gcd(c, m)$ . Because all the common factors of  $m$  and  $c$  are divided out of  $m$  to obtain  $m'$ , it follows that  $m'$  and  $c$  are relatively prime. Because  $m$  divides  $ac - bc = (a - b)c$ , it follows that  $m'$  divides  $(a - b)c$ . By Lemma 2 in Section 4.3, we see that  $m'$  divides  $a - b$ , so  $a \equiv b \pmod{m'}$ . **17.** Suppose that  $x^2 \equiv 1 \pmod{p}$ . Then  $p$  divides  $x^2 - 1 = (x + 1)(x - 1)$ . By Lemma 3 in Section 4.3 it follows that  $p \mid x + 1$  or  $p \mid x - 1$ , so  $x \equiv -1 \pmod{p}$  or  $x \equiv 1 \pmod{p}$ . **19. a)** Suppose that  $ia \equiv ja \pmod{p}$ , where  $1 \leq i < j < p$ . Then  $p$  divides  $ja - ia = a(j - i)$ . By Lemma 3 in Section 4.3, because  $a$  is not divisible by  $p$ ,  $p$  divides  $j - i$ , which is impossible because  $j - i$  is a positive integer less than  $p$ . **b)** By part (a), because no two of  $a, 2a, \dots, (p - 1)a$  are congruent modulo  $p$ , each must be congruent to a different number from 1 to  $p - 1$ . It follows that  $a \cdot 2a \cdot 3a \cdots (p - 1)a \equiv 1 \cdot 2 \cdot 3 \cdots (p - 1) \pmod{p}$ . It follows that  $(p - 1)! \cdot a^{p-1} \equiv p - 1 \pmod{p}$ . **c)** By Wilson's theorem and part (b), if  $p$  does not divide  $a$ , it follows that  $(-1) \cdot a^{p-1} \equiv -1 \pmod{p}$ . Hence,  $a^{p-1} \equiv 1 \pmod{p}$ . **d)** If  $p \mid a$ , then  $p \mid a^p$ . Hence,  $a^p \equiv a \equiv 0 \pmod{p}$ . If  $p$  does not divide  $a$ , then  $a^{p-1} \equiv a \pmod{p}$ , by part (c). Multiplying both sides of this congruence by  $a$  gives  $a^p \equiv a \pmod{p}$ . **21.** All integers of the form  $323 + 330k$ , where  $k$  is an integer **23.** All integers of the form  $53 + 60k$ , where  $k$  is an integer **25. procedure** *chinese*( $m_1, m_2, \dots, m_n$ : relatively prime positive integers;  $a_1, a_2, \dots, a_n$ : integers)  
 $m := 1$   
**for**  $k := 1$  **to**  $n$   
 $m := m \cdot m_k$   
**for**  $k := 1$  **to**  $n$   
 $M_k := m / m_k$   
 $y_k := M_k^{-1} \bmod m_k$   
 $x := 0$   
**for**  $k := 1$  **to**  $n$   
 $x := x + a_k M_k y_k$   
**while**  $x \geq m$   
 $x := x - m$   
**return**  $x$  {the smallest solution to the system  
 $\{x \equiv a_k \pmod{m_k}, k = 1, 2, \dots, n\}$ }



**27.** All integers of the form  $16 + 252k$ , where  $k$  is an integer

**29.** Suppose that  $p$  is a prime appearing in the prime factorization of  $m_1 m_2 \cdots m_n$ . Because the  $m_i$ s are relatively prime,  $p$  is a factor of exactly one of the  $m_i$ s, say  $m_j$ . Because  $m_j$  divides  $a - b$ , it follows that  $a - b$  has the factor  $p$  in its prime factorization to a power at least as large as the power to which it appears in the prime factorization of  $m_j$ . It follows that  $m_1 m_2 \cdots m_n$  divides  $a - b$ , so  $a \equiv b \pmod{m_1 m_2 \cdots m_n}$ .

**31.**  $x \equiv 1 \pmod{6}$

**33.** 7    **35.**  $a^{p-2} \cdot a = a \cdot a^{p-2} = a^{p-1} \equiv 1 \pmod{p}$     **37. a)** By Fermat's little theorem, we have  $2^{10} \equiv 1 \pmod{11}$ . Hence,  $2^{340} = (2^{10})^{34} \equiv 1^{34} = 1 \pmod{11}$ .    **b)** Because  $32 \equiv 1 \pmod{31}$ , it follows that  $2^{340} = (2^5)^{68} = 32^{68} \equiv 1^{68} = 1 \pmod{31}$ .    **c)** Because 11 and 31 are relatively prime, and  $11 \cdot 31 = 341$ , it follows by parts (a) and (b) and Exercise 29 that  $2^{340} \equiv 1 \pmod{341}$ .

**39. a)** 3, 4, 8    **b)** 983

**41.** Suppose that  $q$  is an odd prime with  $q \mid 2^p - 1$ . By Fermat's little theorem,  $q \mid 2^{q-1} - 1$ . From Exercise 37 in Section 4.3,  $\gcd(2^p - 1, 2^{q-1} - 1) = 2^{\gcd(p, q-1)} - 1$ . Because  $q$  is a common divisor of  $2^p - 1$  and  $2^{q-1} - 1$ ,  $\gcd(2^p - 1, 2^{q-1} - 1) > 1$ . Hence,  $\gcd(p, q - 1) = p$ , because the only other possibility, namely,  $\gcd(p, q - 1) = 1$ , gives us  $\gcd(2^p - 1, 2^{q-1} - 1) = 1$ . Hence,  $p \mid q - 1$ , and therefore there is a positive integer  $m$  such that  $q - 1 = mp$ . Because  $q$  is odd,  $m$  must be even, say,  $m = 2k$ , and so every prime divisor of  $2^p - 1$  is of the form  $2kp + 1$ . Furthermore, the product of numbers of this form is also of this form. Therefore, all divisors of  $2^p - 1$  are of this form.

**43.**  $M_{11}$  is not prime;  $M_{17}$  is prime.

**45.** First,  $2047 = 23 \cdot 89$  is composite. Write  $2047 - 1 = 2046 = 2 \cdot 1023$ , so  $s = 1$  and  $t = 1023$  in the definition. Then  $2^{1023} = (2^{11})^{93} = 2048^{93} \equiv 1^{93} = 1 \pmod{2047}$ , as desired.

**47.** We must show that  $b^{2820} \equiv 1 \pmod{2821}$  for all  $b$  relatively prime to 2821. Note that  $2821 = 7 \cdot 13 \cdot 31$ , and if  $\gcd(b, 2821) = 1$ , then  $\gcd(b, 7) = \gcd(b, 13) = \gcd(b, 31) = 1$ . Using Fermat's little theorem we find that  $b^6 \equiv 1 \pmod{7}$ ,  $b^{12} \equiv 1 \pmod{13}$ , and  $b^{30} \equiv 1 \pmod{31}$ . It follows that  $b^{2820} \equiv (b^6)^{470} \equiv 1 \pmod{7}$ ,  $b^{2820} \equiv (b^{12})^{235} \equiv 1 \pmod{13}$ , and  $b^{2820} \equiv (b^{30})^{94} \equiv 1 \pmod{31}$ . By Exercise 29 (or the Chinese remainder theorem) it follows that  $b^{2820} \equiv 1 \pmod{2821}$ , as desired.

**49. a)** If we multiply out this expression, we get  $n = 1296m^3 + 396m^2 + 36m + 1$ . Clearly  $6m \mid n - 1$ ,  $12m \mid n - 1$ , and  $18m \mid n - 1$ . Therefore, the conditions of Exercise 48 are met, and we conclude that  $n$  is a Carmichael number.    **b)** Letting  $m = 51$  gives  $n = 172,947,529$ .

**51.**  $0 = (0, 0)$ ,  $1 = (1, 1)$ ,  $2 = (2, 2)$ ,  $3 = (0, 3)$ ,  $4 = (1, 4)$ ,  $5 = (2, 0)$ ,  $6 = (0, 1)$ ,  $7 = (1, 2)$ ,  $8 = (2, 3)$ ,  $9 = (0, 4)$ ,  $10 = (1, 0)$ ,  $11 = (2, 1)$ ,  $12 = (0, 2)$ ,  $13 = (1, 3)$ ,  $14 = (2, 4)$

**53.** We have  $m_1 = 99$ ,  $m_2 = 98$ ,  $m_3 = 97$ , and  $m_4 = 95$ , so  $m = 99 \cdot 98 \cdot 97 \cdot 95 = 89,403,930$ . We find that  $M_1 = m/m_1 = 903,070$ ,  $M_2 = m/m_2 = 912,285$ ,  $M_3 = m/m_3 = 921,690$ , and  $M_4 = m/m_4 = 941,094$ . Using the Euclidean algorithm, we compute that  $y_1 = 37$ ,  $y_2 = 33$ ,  $y_3 = 24$ , and  $y_4 = 4$  are inverses of  $M_k$  modulo  $m_k$  for  $k = 1, 2, 3, 4$ , respectively. It follows that the solution is  $65 \cdot 903,070 \cdot 37 + 2 \cdot 912,285 \cdot 33 + 51 \cdot 921,690 \cdot 24 + 10 \cdot 941,094 \cdot 4 = 3,397,886,480 \equiv 537,140 \pmod{89,403,930}$ .

**55.**  $\log_2 5 = 16$ ,  $\log_2 6 = 14$     **57.**  $\log_3 1 = 0$ ,  $\log_3 2 = 14$ ,  $\log_3 3 = 1$ ,  $\log_3 4 = 12$ ,  $\log_3 5 = 5$ ,  $\log_3 6 = 15$ ,

$\log_3 7 = 11$ ,  $\log_3 8 = 10$ ,  $\log_3 9 = 2$ ,  $\log_3 10 = 3$ ,  $\log_3 11 = 7$ ,  $\log_3 12 = 13$ ,  $\log_3 13 = 4$ ,  $\log_3 14 = 9$ ,  $\log_3 15 = 6$ ,  $\log_3 16 = 8$

**59.** Assume that  $s$  is a solution of  $x^2 \equiv a \pmod{p}$ . Then because  $(-s)^2 = s^2$ ,  $-s$  is also a solution. Furthermore,  $s \not\equiv -s \pmod{p}$ . Otherwise,  $p \mid 2s$ , which implies that  $p \mid s$ , and this implies, using the original assumption, that  $p \mid a$ , which is a contradiction. Furthermore, if  $s$  and  $t$  are incongruent solutions modulo  $p$ , then because  $s^2 \equiv t^2 \pmod{p}$ ,  $p \mid s^2 - t^2$ . This implies that  $p \mid (s + t)(s - t)$ , and by Lemma 3 in Section 4.3,  $p \mid s - t$  or  $p \mid s + t$ , so  $s \equiv t \pmod{p}$  or  $s \equiv -t \pmod{p}$ . Hence, there are at most two solutions.

**61.** The value of  $\left(\frac{a}{p}\right)$  depends only on whether  $a$  is a quadratic residue modulo  $p$ , that is, whether  $x^2 \equiv a \pmod{p}$  has a solution. Because this depends only on the equivalence class of  $a$  modulo  $p$ , it follows that  $\left(\frac{a}{p}\right) = \left(\frac{b}{p}\right)$  if  $a \equiv b \pmod{p}$ .

**63.** By Exercise 62,  $\left(\frac{a}{p}\right)\left(\frac{b}{p}\right) = a^{(p-1)/2} b^{(p-1)/2} = (ab)^{(p-1)/2} \equiv \left(\frac{ab}{p}\right) \pmod{p}$ .

**65.**  $x \equiv 8, 13, 22$ , or  $27 \pmod{35}$     **67.** Compute  $r^e \pmod{p}$  for  $e = 0, 1, 2, \dots, p - 2$  until we get the answer  $a$ . Worst case and average case time complexity are  $O(p \log p)$ .

## Section 4.5

**1.** 91, 57, 21, 5    **3. a)** 7, 19, 7, 7, 18, 0    **b)** Take the next available space **mod** 31.    **5.** 1, 5, 4, 1, 5, 4, 1, 5, 4, ...    **7.** 2, 6, 7, 10, 8, 2, 6, 7, 10, 8, ...    **9.** 2357, 5554, 8469, 7239, 4031, 2489, 1951, 8064    **11.** 2, 1, 1, 1, ...    **13.** Only string (d)

**15.** 4    **17.** Correctly, of course    **19. a)** Not valid    **b)** Valid    **c)** Valid    **d)** Not valid    **21. a)** No    **b)** 5    **c)** 7    **d)** 8

**23.** Transposition errors involving the last digit    **25. a)** Yes    **b)** No    **c)** Yes    **d)** No    **27.** Transposition errors will be detected if and only if the transposed digits are an odd number of positions apart and do not differ by 5.    **29. a)** Valid    **b)** Not valid    **c)** Valid    **d)** Valid

**31.** Yes, as long as the two digits do not differ by 7    **33. a)** Not valid    **b)** Valid    **c)** Valid    **d)** Not valid

**35.** The given congruence is equivalent to  $3d_1 + 4d_2 + 5d_3 + 6d_4 + 7d_5 + 8d_6 + 9d_7 + 10d_8 \equiv 0 \pmod{11}$ . Transposing adjacent digits  $x$  and  $y$  (with  $x$  on the left) causes the left-hand side to increase by  $x - y$ . Because  $x \not\equiv y \pmod{11}$ , the congruence will no longer hold. Therefore, errors of this type are always detected.

## Section 4.6

**1. a)** GR QRW SDVV JR    **b)** QB ABG CNFF TB    **c)** QX UXM AHJJ ZX    **3. a)** KOHQV MCIF GHSD    **b)** RVBXP TJPZ NBZX    **c)** DBYNE PHRM FYZA

**5. a)** SURRENDER NOW    **b)** BE MY FRIEND    **c)** TIME FOR FUN    **7.** TO SLEEP PERCHANCE TO DREAM    **9.** ANY SUFFICIENTLY ADVANCED TECHNOLOGY IS INDISTINGUISHABLE FROM MAGIC    **11.**  $p = 7c + 13 \pmod{26}$     **13.**  $a = 18$ ,  $b = 5$     **15.** BEWARE OF MARTIANS    **17.** Presumably something like an affine cipher    **19.** HURRICANE    **21.** The length of the key may well



be the greatest common divisor of the distances between the starts of the repeated string (or a factor of the gcd). **23.** Suppose we know both  $n = pq$  and  $(p - 1)(q - 1)$ . To find  $p$  and  $q$ , first note that  $(p - 1)(q - 1) = pq - p - q + 1 = n - (p + q) + 1$ . From this we can find  $s = p + q$ . Because  $q = s - p$ , we have  $n = p(s - p)$ . Hence,  $p^2 - ps + n = 0$ . We now can use the quadratic formula to find  $p$ . Once we have found  $p$ , we can find  $q$  because  $q = n/p$ . **25.** 2545 2757 1211 **27.** SILVER **29.** Alice sends  $5^8 \bmod 23 = 16$  to Bob. Bob sends  $5^5 \bmod 23 = 20$  to Alice. Alice computes  $20^8 \bmod 23 = 6$  and Bob computes  $16^5 \bmod 23 = 6$ . The shared key is 6. **31.** 2186 2087 1279 1251 0326 0816 1948 **33.** Alice can decrypt the first part of Cathy's message to learn the key, and Bob can decrypt the second part of Cathy's message, which Alice forwarded to him, to learn the key. No one else besides Cathy can learn the key, because all of these communications use secure private keys. **35.** Working modulo  $n^2$ ,  $E(m_1 + m_2) = g^{m_1+m_2}(r_1 r_2)^n = (g^{m_1} r_1^n)(g^{m_2} r_2^n) = E(m_1) \cdot E(m_2)$ .

## Supplementary Exercises

**1.** The actual number of miles driven is  $46,518 + 100,000k$  for some natural number  $k$ . **3.** 5, 22, -12, -29 **5.** Because  $ac \equiv bc \pmod{m}$ , there is an integer  $k$  such that  $ac = bc + km$ . Hence,  $a - b = km/c$ . Because  $a - b$  is an integer,  $c \mid km$ . Letting  $d = \gcd(m, c)$ , write  $c = de$ . Because no factor of  $e$  divides  $m/d$ , it follows that  $d \mid m$  and  $e \mid k$ . Thus,  $a - b = (k/e)(m/d)$ , where  $k/e \in \mathbb{Z}$  and  $m/d \in \mathbb{Z}$ . Therefore,  $a \equiv b \pmod{m/d}$ . **7.** Proof of the contrapositive: If  $n$  is odd, then  $n = 2k + 1$  for some integer  $k$ . Therefore,  $n^2 + 1 = (2k + 1)^2 + 1 = 4k^2 + 4k + 2 \equiv 2 \pmod{4}$ . But perfect squares of even numbers are congruent to 0 modulo 4 (because  $(2m)^2 = 4m^2$ ), and perfect squares of odd numbers are congruent to 1 or 3 modulo 4, so  $n^2 + 1$  is not a perfect square. **9.**  $n$  is divisible by 8 if and only if the binary expansion of  $n$  ends with 000. **11.** We assume that someone has chosen a positive integer less than  $2^n$ , which we are to guess. We ask the person to write the number in binary, using leading 0s if necessary to make it  $n$  bits long. We then ask, "Is the first bit a 1?", "Is the second bit a 1?", "Is the third bit a 1?", and so on. After we know the answers to these  $n$  questions, we will know the number, because we will know its binary expansion. **13.**  $(a_n a_{n-1} \dots a_1 a_0)_{10} = \sum_{k=0}^n 10^k a_k \equiv \sum_{k=0}^n a_k \pmod{9}$  because  $10^k \equiv 1 \pmod{9}$  for every nonnegative integer  $k$ . **15.** Because for all  $k \leq n$ , when  $Q_n$  is divided by  $k$  the remainder will be 1, it follows that no prime number less than or equal to  $n$  is a factor of  $Q_n$ . Thus, by the fundamental theorem of arithmetic,  $Q_n$  must have a prime factor greater than  $n$ . **17.** Take  $a = 10$  and  $b = 1$  in Dirichlet's theorem. **19.** Every number greater than 11 can be written as either  $8 + 2n$  or  $9 + 2n$  for some  $n \geq 2$ . **21.** Assume that every even integer greater than 2 is the sum of two primes, and let  $n$  be an integer greater than 5. If  $n$  is odd, write  $n = 3 + (n - 3)$  and decompose  $n - 3 = p + q$  into the sum of two primes; if  $n$  is even, then write  $n = 2 + (n - 2)$  and decompose  $n - 2 = p + q$  into

the sum of two primes. For the converse, assume that every integer greater than 5 is the sum of three primes, and let  $n$  be an even integer greater than 2. Write  $n + 2$  as the sum of three primes, one of which is necessarily 2, so  $n + 2 = 2 + p + q$ , whence  $n = p + q$ . **23.** Recall that a nonconstant polynomial can take on the same value only a finite number of times. Thus,  $f$  can take on the values 0 and  $\pm 1$  only finitely many times, so if there is not some  $y$  such that  $f(y)$  is composite, then there must be some  $x_0$  such that  $\pm f(x_0)$  is prime, say  $p$ . Look at  $f(x_0 + kp)$ . When we plug  $x_0 + kp$  in for  $x$  in the polynomial and multiply it out, every term will contain a factor of  $p$  except for the terms that form  $f(x_0)$ . Therefore,  $f(x_0 + kp) = f(x_0) + mp = (m \pm 1)p$  for some integer  $m$ . As  $k$  varies, this value can be 0,  $p$ , or  $-p$  only finitely many times; therefore, it must be a composite number for some values of  $k$ . **25.** 1 **27.** 1 **29.** If not, then suppose that  $q_1, q_2, \dots, q_n$  are all the primes of the form  $6k + 5$ . Let  $Q = 6q_1 q_2 \dots q_n - 1$ . Note that  $Q$  is of the form  $6k + 5$ , where  $k = q_1 q_2 \dots q_n - 1$ . Let  $Q = p_1 p_2 \dots p_t$  be the prime factorization of  $Q$ . No  $p_i$  is 2, 3, or any  $q_j$ , because the remainder when  $Q$  is divided by 2 is 1, by 3 is 2, and by  $q_j$  is  $q_j - 1$ . All odd primes other than 3 are of the form  $6k + 1$  or  $6k + 5$ , and the product of primes of the form  $6k + 1$  is also of this form. Therefore, at least one of the  $p_i$ s must be of the form  $6k + 5$ , a contradiction. **31.** The product of numbers of the form  $4k + 1$  is of the form  $4k + 1$ , but numbers of this form might have numbers not of this form as their only prime factors. For example,  $49 = 4 \cdot 12 + 1$ , but the prime factorization of 49 is  $7 \cdot 7 = (4 \cdot 1 + 3)(4 \cdot 1 + 3)$ . **33. a)** Not mutually relatively prime **b)** Mutually relatively prime **c)** Mutually relatively prime **d)** Mutually relatively prime **35.** 1 **37.**  $x \equiv 28 \pmod{30}$  **39.** By the Chinese remainder theorem, it suffices to show that  $n^9 - n \equiv 0 \pmod{2}$ ,  $n^9 - n \equiv 0 \pmod{3}$ , and  $n^9 - n \equiv 0 \pmod{5}$ . Each in turn follows from applying Fermat's little theorem. **41.** By Fermat's little theorem,  $p^{q-1} \equiv 1 \pmod{q}$  and clearly  $q^{p-1} \equiv 0 \pmod{q}$ . Therefore,  $p^{q-1} + q^{p-1} \equiv 1 + 0 = 1 \pmod{q}$ . Similarly,  $p^{q-1} + q^{p-1} \equiv 1 \pmod{p}$ . It follows from the Chinese remainder theorem that  $p^{q-1} + q^{p-1} \equiv 1 \pmod{pq}$ . **43.** If  $a_i$  is changed from  $x$  to  $y$ , then the change in the left-hand side of the congruence is either  $y - x$  or  $3(y - x)$ , modulo 10, neither of which can be 0 because 1 and 3 are relatively prime to 10. Therefore, the sum can no longer be 0 modulo 10. **45.** Working modulo 10, solve for  $d_9$ . The check digit for 11100002 is 5. **47.** PLEASE SEND MONEY **49. a)** QAL HUVEM AT WVESGB **b)** QXB EVZZL ZEZZRFS

## CHAPTER 5

### Section 5.1

**1.** Let  $P(n)$  be the statement that the train stops at station  $n$ . *Basis step:* We are told that  $P(1)$  is true. *Inductive step:* We are told that  $P(n)$  implies  $P(n + 1)$  for each  $n \geq 1$ . Therefore,



by the principle of mathematical induction,  $P(n)$  is true for all positive integers  $n$ . **3. a)**  $1^2 = 1 \cdot 2 \cdot 3 / 6$  **b)** Both sides of  $P(1)$  shown in part (a) equal 1. **c)**  $1^2 + 2^2 + \dots + k^2 = k(k+1)(2k+1)/6$  **d)** For each  $k \geq 1$  that  $P(k)$  implies  $P(k+1)$ ; in other words, that assuming the inductive hypothesis [see part (c)] we can show  $1^2 + 2^2 + \dots + k^2 + (k+1)^2 = (k+1)(k+2)(2k+3)/6$  **e)**  $(1^2 + 2^2 + \dots + k^2) + (k+1)^2 = [k(k+1)(2k+1)/6] + (k+1)^2 = [(k+1)/6][k(2k+1) + 6(k+1)] = [(k+1)/6](2k^2 + 7k + 6) = [(k+1)/6](k+2)(2k+3) = (k+1)(k+2)(2k+3)/6$  **f)** We have completed both the basis step and the inductive step, so by the principle of mathematical induction, the statement is true for every positive integer  $n$ . **5.** Let  $P(n)$  be " $1^2 + 3^2 + \dots + (2n+1)^2 = (n+1)(2n+1)(2n+3)/3$ ." *Basis step:*  $P(0)$  is true because  $1^2 = 1 = (0+1)(2 \cdot 0+1)(2 \cdot 0+3)/3$ . *Inductive step:* Assume that  $P(k)$  is true. Then  $1^2 + 3^2 + \dots + (2k+1)^2 + [2(k+1)+1]^2 = (k+1)(2k+1)(2k+3)/3 + (2k+3)^2 = (2k+3)[(k+1)(2k+1)/3 + (2k+3)] = (2k+3)(2k^2 + 9k + 10)/3 = (2k+3)(2k+5)(k+2)/3 = [(k+1)+1][2(k+1)+1][2(k+1)+3]/3$ . **7.** Let  $P(n)$  be " $\sum_{j=0}^n 3 \cdot 5^j = 3(5^{n+1} - 1)/4$ ." *Basis step:*  $P(0)$  is true because  $\sum_{j=0}^0 3 \cdot 5^j = 3 = 3(5^1 - 1)/4$ . *Inductive step:* Assume that  $\sum_{j=0}^k 3 \cdot 5^j = 3(5^{k+1} - 1)/4$ . Then  $\sum_{j=0}^{k+1} 3 \cdot 5^j = (\sum_{j=0}^k 3 \cdot 5^j) + 3 \cdot 5^{k+1} = 3(5^{k+1} - 1)/4 + 3 \cdot 5^{k+1} = 3(5^{k+1} + 4 \cdot 5^{k+1} - 1)/4 = 3(5^{k+2} - 1)/4$ . **9. a)**  $2 + 4 + 6 + \dots + 2n = n(n+1)$  **b)** *Basis step:*  $2 = 1 \cdot (1+1)$  is true. *Inductive step:* Assume that  $2 + 4 + 6 + \dots + 2k = k(k+1)$ . Then  $(2 + 4 + 6 + \dots + 2k) + 2(k+1) = k(k+1) + 2(k+1) = (k+1)(k+2)$ . **11. a)**  $\sum_{j=1}^n 1/2^j = (2^n - 1)/2^n$  **b)** *Basis step:*  $P(1)$  is true because  $\frac{1}{2} = (2^1 - 1)/2^1$ . *Inductive step:* Assume that  $\sum_{j=1}^k 1/2^j = (2^k - 1)/2^k$ . Then  $\sum_{j=1}^{k+1} \frac{1}{2^j} = (\sum_{j=1}^k \frac{1}{2^j}) + \frac{1}{2^{k+1}} = \frac{2^k - 1}{2^k} + \frac{1}{2^{k+1}} = \frac{2^{k+1} - 2 + 1}{2^{k+1}} = \frac{2^{k+1} - 1}{2^{k+1}}$ . **13.** Let  $P(n)$  be " $1^2 - 2^2 + 3^2 - \dots + (-1)^{n-1} n^2 = (-1)^{n-1} n(n+1)/2$ ." *Basis step:*  $P(1)$  is true because  $1^2 = 1 = (-1)^0 1^2$ . *Inductive step:* Assume that  $P(k)$  is true. Then  $1^2 - 2^2 + 3^2 - \dots + (-1)^{k-1} k^2 + (-1)^k (k+1)^2 = (-1)^{k-1} k(k+1)/2 + (-1)^k (k+1)^2 = (-1)^k (k+1)[-k/2 + (k+1)] = (-1)^k (k+1)[(k/2) + 1] = (-1)^k (k+1)(k+2)/2$ . **15.** Let  $P(n)$  be " $1 \cdot 2 + 2 \cdot 3 + \dots + n(n+1) = n(n+1)(n+2)/3$ ." *Basis step:*  $P(1)$  is true because  $1 \cdot 2 = 2 = 1(1+1)(1+2)/3$ . *Inductive step:* Assume that  $P(k)$  is true. Then  $1 \cdot 2 + 2 \cdot 3 + \dots + k(k+1) + (k+1)(k+2) = [k(k+1)(k+2)/3] + (k+1)(k+2) = (k+1)(k+2)[(k/3) + 1] = (k+1)(k+2)(k+3)/3$ . **17.** Let  $P(n)$  be the statement that  $1^4 + 2^4 + 3^4 + \dots + n^4 = n(n+1)(2n+1)(3n^2 + 3n - 1)/30$ .  $P(1)$  is true because  $1 \cdot 2 \cdot 3 \cdot 5/30 = 1$ . Assume that  $P(k)$  is true. Then  $(1^4 + 2^4 + 3^4 + \dots + k^4) + (k+1)^4 = k(k+1)(2k+1)(3k^2 + 3k - 1)/30 + (k+1)^4 = [(k+1)/30][k(2k+1)(3k^2 + 3k - 1) + 30(k+1)^3] = [(k+1)/30](6k^4 + 39k^3 + 91k^2 + 89k + 30) = [(k+1)/30](k+2)(2k+3)[3(k+1)^2 + 3(k+1) - 1]$ . This demonstrates that  $P(k+1)$  is true. **19. a)**  $1 + \frac{1}{4} < 2 - \frac{1}{2}$  **b)** This is true because  $5/4$  is less than  $6/4$ . **c)**  $1 + \frac{1}{4} + \dots + \frac{1}{k^2} < 2 - \frac{1}{k}$  **d)** For each  $k \geq 2$  that  $P(k)$  implies  $P(k+1)$ ; in other words, we want to

show that assuming the inductive hypothesis [see part (c)] we can show  $1 + \frac{1}{4} + \dots + \frac{1}{k^2} + \frac{1}{(k+1)^2} < 2 - \frac{1}{k+1}$  **e)**  $1 + \frac{1}{4} + \dots + \frac{1}{k^2} + \frac{1}{(k+1)^2} < 2 - \frac{1}{k} + \frac{1}{(k+1)^2} = 2 - [\frac{1}{k} - \frac{1}{(k+1)^2}] = 2 - [\frac{k^2 + 2k + 1 - k}{k(k+1)^2}] = 2 - \frac{k^2 + k}{k(k+1)^2} - \frac{1}{k(k+1)^2} = 2 - \frac{1}{k+1} - \frac{1}{k(k+1)^2} < 2 - \frac{1}{k+1}$  **f)** We have completed both the basis step and the inductive step, so by the principle of mathematical induction, the statement is true for every integer  $n$  greater than 1. **21.** Let  $P(n)$  be " $2^n > n^2$ ." *Basis step:*  $P(5)$  is true because  $2^5 = 32 > 25 = 5^2$ . *Inductive step:* Assume that  $P(k)$  is true, that is,  $2^k > k^2$ . Then  $2^{k+1} = 2 \cdot 2^k > k^2 + k^2 > k^2 + 4k \geq k^2 + 2k + 1 = (k+1)^2$  because  $k > 4$ . **23.** By inspection we find that the inequality  $2n + 3 \leq 2^n$  does not hold for  $n = 0, 1, 2, 3$ . Let  $P(n)$  be the proposition that this inequality holds for the positive integer  $n$ .  $P(4)$ , the basis case, is true because  $2 \cdot 4 + 3 = 11 \leq 16 = 2^4$ . For the inductive step assume that  $P(k)$  is true. Then, by the inductive hypothesis,  $2(k+1) + 3 = (2k+3) + 2 < 2^k + 2$ . But because  $k \geq 1$ ,  $2^k + 2 \leq 2^k + 2^k = 2^{k+1}$ . This shows that  $P(k+1)$  is true. **25.** Let  $P(n)$  be " $1 + nh \leq (1+h)^n$ ,  $h > -1$ ." *Basis step:*  $P(0)$  is true because  $1 + 0 \cdot h = 1 \leq 1 = (1+h)^0$ . *Inductive step:* Assume  $1 + kh \leq (1+h)^k$ . Then because  $(1+h) > 0$ ,  $(1+h)^{k+1} = (1+h)(1+h)^k \geq (1+h)(1+kh) = 1 + (k+1)h + kh^2 \geq 1 + (k+1)h$ . **27.** Let  $P(n)$  be " $1/\sqrt{1} + 1/\sqrt{2} + 1/\sqrt{3} + \dots + 1/\sqrt{n} > 2(\sqrt{n+1} - 1)$ ." *Basis step:*  $P(1)$  is true because  $1 > 2(\sqrt{2} - 1)$ . *Inductive step:* Assume that  $P(k)$  is true. Then  $1 + 1/\sqrt{2} + \dots + 1/\sqrt{k} + 1/\sqrt{k+1} > 2(\sqrt{k+1} - 1) + 1/\sqrt{k+1}$ . If we show that  $2(\sqrt{k+1} - 1) + 1/\sqrt{k+1} > 2(\sqrt{k+2} - 1)$ , it follows that  $P(k+1)$  is true. This inequality is equivalent to  $2(\sqrt{k+2} - \sqrt{k+1}) < 1/\sqrt{k+1}$ , which is equivalent to  $2(\sqrt{k+2} - \sqrt{k+1})(\sqrt{k+2} + \sqrt{k+1}) < \sqrt{k+1}/\sqrt{k+1} + \sqrt{k+2}/\sqrt{k+1}$ . This is equivalent to  $2 < 1 + \sqrt{k+2}/\sqrt{k+1}$ , which is clearly true. **29.** Let  $P(n)$  be " $H_{2^n} \leq 1 + n$ ." *Basis step:*  $P(0)$  is true because  $H_{2^0} = H_1 = 1 \leq 1 + 0$ . *Inductive step:* Assume that  $H_{2^k} \leq 1 + k$ . Then  $H_{2^{k+1}} = H_{2^k} + \sum_{j=2^k+1}^{2^{k+1}} \frac{1}{j} \leq 1 + k + 2^k \left(\frac{1}{2^{k+1}}\right) < 1 + k + 1 = 1 + (k+1)$ . **31.** *Basis step:*  $1^2 + 1 = 2$  is divisible by 2. *Inductive step:* Assume the inductive hypothesis, that  $k^2 + k$  is divisible by 2. Then  $(k+1)^2 + (k+1) = k^2 + 2k + 1 + k + 1 = (k^2 + k) + 2(k+1)$ , the sum of a multiple of 2 (by the inductive hypothesis) and a multiple of 2 (by definition), hence, divisible by 2. **33.** Let  $P(n)$  be " $n^5 - n$  is divisible by 5." *Basis step:*  $P(0)$  is true because  $0^5 - 0 = 0$  is divisible by 5. *Inductive step:* Assume that  $P(k)$  is true, that is,  $k^5 - 5$  is divisible by 5. Then  $(k+1)^5 - (k+1) = (k^5 + 5k^4 + 10k^3 + 10k^2 + 5k + 1) - (k+1) = (k^5 - k) + 5(k^4 + 2k^3 + 2k^2 + k)$  is also divisible by 5, because both terms in this sum are divisible by 5. **35.** Let  $P(n)$  be the proposition that  $(2n-1)^2 - 1$  is divisible by 8. The basis case  $P(1)$  is true because  $8 \mid 0$ . Now assume that  $P(k)$  is true. Because  $[(2(k+1)-1)^2 - 1] = [(2k-1)^2 - 1] + 8k$ ,  $P(k+1)$  is true because both terms on the right-hand side are divisible



by 8. This shows that  $P(n)$  is true for all positive integers  $n$ , so  $m^2 - 1$  is divisible by 8 whenever  $m$  is an odd positive integer.

**37. Basis step:**  $11^{1+1} + 12^{2 \cdot 1 - 1} = 121 + 12 = 133$  *Inductive step:* Assume the inductive hypothesis, that  $11^{n+1} + 12^{2n-1}$  is divisible by 133. Then  $11^{(n+1)+1} + 12^{2(n+1)-1} = 11 \cdot 11^{n+1} + 144 \cdot 12^{2n-1} = 11 \cdot 11^{n+1} + (11 + 133) \cdot 12^{2n-1} = 11(11^{n+1} + 12^{2n-1}) + 133 \cdot 12^{2n-1}$ . The expression in parentheses is divisible by 133 by the inductive hypothesis, and obviously the second term is divisible by 133, so the entire quantity is divisible by 133, as desired.

**39. Basis step:**  $A_1 \subseteq B_1$  tautologically implies that  $\bigcap_{j=1}^1 A_j \subseteq \bigcap_{j=1}^1 B_j$ .

*Inductive step:* Assume the inductive hypothesis that if  $A_j \subseteq B_j$  for  $j = 1, 2, \dots, k$ , then  $\bigcap_{j=1}^k A_j \subseteq \bigcap_{j=1}^k B_j$ . We want to show that if  $A_j \subseteq B_j$  for  $j = 1, 2, \dots, k+1$ , then  $\bigcap_{j=1}^{k+1} A_j \subseteq \bigcap_{j=1}^{k+1} B_j$ .

Let  $x$  be an arbitrary element of  $\bigcap_{j=1}^{k+1} A_j = (\bigcap_{j=1}^k A_j) \cap A_{k+1}$ . Because  $x \in \bigcap_{j=1}^k A_j$ , we know by the inductive hypothesis that  $x \in \bigcap_{j=1}^k B_j$ ; because  $x \in A_{k+1}$ , we know from the given fact that  $A_{k+1} \subseteq B_{k+1}$  that  $x \in B_{k+1}$ . Therefore,  $x \in (\bigcap_{j=1}^k B_j) \cap B_{k+1} = \bigcap_{j=1}^{k+1} B_j$ .

**41.** Let  $P(n)$  be “ $(A_1 \cup A_2 \cup \dots \cup A_n) \cap B = (A_1 \cap B) \cup (A_2 \cap B) \cup \dots \cup (A_n \cap B)$ .”

*Basis step:*  $P(1)$  is trivially true. *Inductive step:* Assume that  $P(k)$  is true. Then  $(A_1 \cup A_2 \cup \dots \cup A_k \cup A_{k+1}) \cap B = [(A_1 \cup A_2 \cup \dots \cup A_k) \cup A_{k+1}] \cap B = [(A_1 \cup A_2 \cup \dots \cup A_k) \cap B] \cup (A_{k+1} \cap B) = [(A_1 \cap B) \cup (A_2 \cap B) \cup \dots \cup (A_k \cap B)] \cup (A_{k+1} \cap B) = (A_1 \cap B) \cup (A_2 \cap B) \cup \dots \cup (A_k \cap B) \cup (A_{k+1} \cap B)$ .

**43.** Let  $P(n)$  be “ $\bigcup_{k=1}^n A_k = \bigcap_{k=1}^n \overline{A_k}$ .” *Basis step:*  $P(1)$  is trivially true. *Inductive step:* Assume that  $P(k)$  is true. Then

$$\bigcup_{j=1}^{k+1} A_j = \left( \bigcup_{j=1}^k A_j \right) \cup A_{k+1} = \left( \bigcup_{j=1}^k A_j \right) \cap \overline{A_{k+1}} = \left( \bigcap_{j=1}^k \overline{A_j} \right) \cap \overline{A_{k+1}} = \bigcap_{j=1}^{k+1} \overline{A_j}.$$

**45.** Let  $P(n)$  be the statement that a set with  $n$  elements has  $n(n-1)/2$  two-element subsets.  $P(2)$ , the basis case, is true, because a set with two elements has one subset with two elements—namely, itself—and  $2(2-1)/2 = 1$ . Now assume that  $P(k)$  is true. Let  $S$  be a set with  $k+1$  elements. Choose an element  $a$  in  $S$  and let  $T = S - \{a\}$ . A two-element subset of  $S$  either contains  $a$  or does not. Those subsets not containing  $a$  are the subsets of  $T$  with two elements; by the inductive hypothesis there are  $k(k-1)/2$  of these. There are  $k$  subsets of  $S$  with two elements that contain  $a$ , because such a subset contains  $a$  and one of the  $k$  elements in  $T$ . Hence, there are  $k(k-1)/2 + k = (k+1)k/2$  two-element subsets of  $S$ . This completes the inductive proof.

**47.** Reorder the locations if necessary so that  $x_1 \leq x_2 \leq x_3 \leq \dots \leq x_d$ . Place the first tower at position  $t_1 = x_1 + 1$ . Assume tower  $k$  has been placed at position  $t_k$ . Then place tower  $k+1$  at position  $t_{k+1} = x + 1$ , where  $x$  is the smallest  $x_i$  greater than  $t_k + 1$ .

**49.** The two sets do not overlap if  $n+1 = 2$ . In fact, the conditional statement  $P(1) \rightarrow P(2)$  is false.

**51.** The mistake is in applying the inductive hypothesis to look at  $\max(x-1, y-1)$ , because even though  $x$  and  $y$  are positive integers,  $x-1$  and  $y-1$  need not be (one or both could be 0).

**53.** For the basis step ( $n = 2$ ) the first person cuts the cake into two portions that she thinks are each  $1/2$  of the cake, and the second person chooses the portion he thinks is at least  $1/2$  of the cake (at

least one of the pieces must satisfy that condition). For the inductive step, suppose there are  $k+1$  people. By the inductive hypothesis, we can suppose that the first  $k$  people have divided the cake among themselves so that each person is satisfied that he got at least a fraction  $1/k$  of the cake. Each of them now cuts his or her piece into  $k+1$  pieces of equal size. The last person gets to choose one piece from each of the first  $k$  people's portions. After this is done, each of the first  $k$  people is satisfied that she still has  $(1/k)(k/(k+1)) = 1/(k+1)$  of the cake. To see that the last person is satisfied, suppose that he thought that the  $i$ th person ( $1 \leq i \leq k$ ) had a portion  $p_i$  of the cake, where  $\sum_{i=1}^k p_i = 1$ . By choosing what he thinks is the largest piece from each person, he is satisfied that he has at least  $\sum_{i=1}^k p_i / (k+1) = (1/(k+1)) \sum_{i=1}^k p_i = 1/(k+1)$  of the cake.

**55.** We use the notation  $(i, j)$  to mean the square in row  $i$  and column  $j$  and use induction on  $i+j$  to show that every square can be reached by the knight. *Basis step:* There are six base cases, for the cases when  $i+j \leq 2$ . The knight is already at  $(0, 0)$  to start, so the empty sequence of moves reaches that square. To reach  $(1, 0)$ , the knight moves  $(0, 0) \rightarrow (2, 1) \rightarrow (0, 2) \rightarrow (1, 0)$ . Similarly, to reach  $(0, 1)$ , the knight moves  $(0, 0) \rightarrow (1, 2) \rightarrow (2, 0) \rightarrow (0, 1)$ . Note that the knight has reached  $(2, 0)$  and  $(0, 2)$  in the process. For the last basis step there is  $(0, 0) \rightarrow (1, 2) \rightarrow (2, 0) \rightarrow (0, 1) \rightarrow (2, 2) \rightarrow (0, 3) \rightarrow (1, 1)$ . *Inductive step:* Assume the inductive hypothesis, that the knight can reach any square  $(i, j)$  for which  $i+j = k$ , where  $k$  is an integer greater than 1. We must show how the knight can reach each square  $(i, j)$  when  $i+j = k+1$ . Because  $k+1 \geq 3$ , at least one of  $i$  and  $j$  is at least 2. If  $i \geq 2$ , then by the inductive hypothesis, there is a sequence of moves ending at  $(i-2, j+1)$ , because  $i-2+j+1 = i+j-1 = k$ ; from there it is just one step to  $(i, j)$ ; similarly, if  $j \geq 2$ .

**57. Basis step:** The base cases  $n = 0$  and  $n = 1$  are true because the derivative of  $x^0$  is 0 and the derivative of  $x^1 = x$  is 1. *Inductive step:* Using the product rule, the inductive hypothesis, and the basis step shows that  $\frac{d}{dx} x^{k+1} = \frac{d}{dx} (x \cdot x^k) = x \cdot \frac{d}{dx} x^k + x^k \frac{d}{dx} x = x \cdot kx^{k-1} + x^k \cdot 1 = kx^k + x^k = (k+1)x^k$ .

**59. Basis step:** For  $k = 0, 1 \equiv 1 \pmod{m}$ . *Inductive step:* Suppose that  $a \equiv b \pmod{m}$  and  $a^k \equiv b^k \pmod{m}$ ; we must show that  $a^{k+1} \equiv b^{k+1} \pmod{m}$ . By Theorem 5 from Section 4.1,  $a \cdot a^k \equiv b \cdot b^k \pmod{m}$ , which by definition says that  $a^{k+1} \equiv b^{k+1} \pmod{m}$ .

**61.** Let  $P(n)$  be “ $[(p_1 \rightarrow p_2) \wedge (p_2 \rightarrow p_3) \wedge \dots \wedge (p_{n-1} \rightarrow p_n)] \rightarrow [(p_1 \wedge \dots \wedge p_{n-1}) \rightarrow p_n]$ .”

*Basis step:*  $P(2)$  is true because  $(p_1 \rightarrow p_2) \rightarrow (p_1 \rightarrow p_2)$  is a tautology. *Inductive step:* Assume  $P(k)$  is true. To show

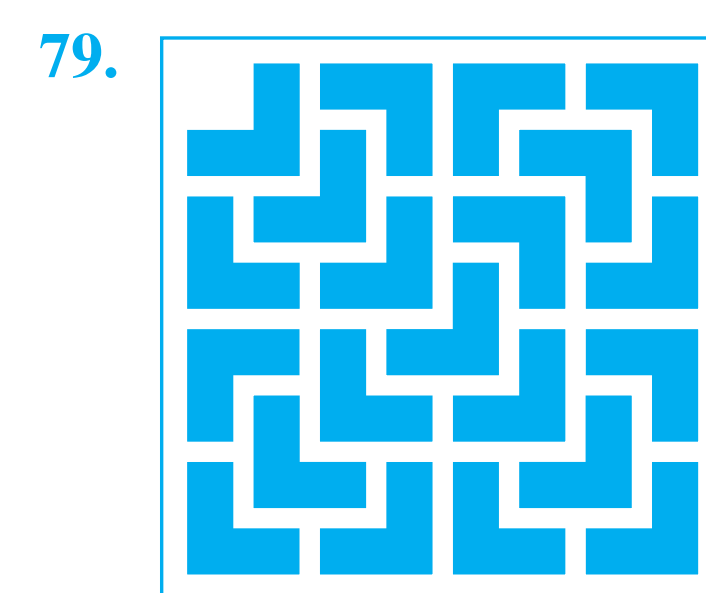
$[(p_1 \rightarrow p_2) \wedge \dots \wedge (p_{k-1} \rightarrow p_k) \wedge (p_k \rightarrow p_{k+1})] \rightarrow [(p_1 \wedge \dots \wedge p_{k-1} \wedge p_k) \rightarrow p_{k+1}]$  is a tautology, assume that the hypothesis of this conditional statement is true. Because both the hypothesis and  $P(k)$  are true, it follows that  $(p_1 \wedge \dots \wedge p_{k-1}) \rightarrow p_k$  is true. Because this is true, and because  $p_k \rightarrow p_{k+1}$  is true (it is part of the assumption) it follows by hypothetical syllogism that  $(p_1 \wedge \dots \wedge p_{k-1}) \rightarrow p_{k+1}$  is true. The weaker statement  $(p_1 \wedge \dots \wedge p_{k-1} \wedge p_k) \rightarrow p_{k+1}$  follows from this.

**63.** We will first prove the result when  $n$  is a power of 2, that is, if  $n = 2^k$ ,  $k = 1, 2, \dots$ . Let  $P(k)$  be the statement  $A \geq G$ , where  $A$  and  $G$  are the arithmetic and geometric means, respectively, of a set



of  $n = 2^k$  positive real numbers. *Basis step:*  $k = 1$  and  $n = 2^1 = 2$ . Note that  $(\sqrt{a_1} - \sqrt{a_2})^2 \geq 0$ . Expanding this shows that  $a_1 - 2\sqrt{a_1 a_2} + a_2 \geq 0$ , that is,  $(a_1 + a_2)/2 \geq (a_1 a_2)^{1/2}$ . *Inductive step:* Assume that  $P(k)$  is true, with  $n = 2^k$ . We will show that  $P(k + 1)$  is true. We have  $2^{k+1} = 2n$ . Now  $(a_1 + a_2 + \dots + a_{2n})/(2n) = [(a_1 + a_2 + \dots + a_n)/n + (a_{n+1} + a_{n+2} + \dots + a_{2n})/n]/2$  and similarly  $(a_1 a_2 \dots a_{2n})^{1/(2n)} = [(a_1 \dots a_n)^{1/n} (a_{n+1} \dots a_{2n})^{1/n}]^{1/2}$ . To simplify the notation, let  $A(x, y, \dots)$  and  $G(x, y, \dots)$  denote the arithmetic mean and geometric mean of  $x, y, \dots$ , respectively. Also, if  $x \leq x'$ ,  $y \leq y'$ , and so on, then  $A(x, y, \dots) \leq A(x', y', \dots)$  and  $G(x, y, \dots) \leq G(x', y', \dots)$ . Hence,  $A(a_1, \dots, a_{2n}) = A(A(a_1, \dots, a_n), A(a_{n+1}, \dots, a_{2n})) \geq A(G(a_1, \dots, a_n), G(a_{n+1}, \dots, a_{2n})) \geq G(G(a_1, \dots, a_n), G(a_{n+1}, \dots, a_{2n})) = G(a_1, \dots, a_{2n})$ . This finishes the proof for powers of 2. Now if  $n$  is not a power of 2, let  $m$  be the next higher power of 2, and let  $a_{n+1}, \dots, a_m$  all equal  $A(a_1, \dots, a_n) = \bar{a}$ . Then we have  $[(a_1 a_2 \dots a_n) \bar{a}^{m-n}]^{1/m} \leq A(a_1, \dots, a_m)$ , because  $m$  is a power of 2. Because  $A(a_1, \dots, a_m) = \bar{a}$ , it follows that  $(a_1 \dots a_n)^{1/m} \bar{a}^{1-n/m} \leq \bar{a}^{n/m}$ . Raising both sides to the  $(m/n)$ th power gives  $G(a_1, \dots, a_n) \leq A(a_1, \dots, a_n)$ . **65. Basis step:** For  $n = 1$ , the left-hand side is just  $\frac{1}{1}$ , which is 1. For  $n = 2$ , there are three nonempty subsets  $\{1\}$ ,  $\{2\}$ , and  $\{1, 2\}$ , so the left-hand side is  $\frac{1}{1} + \frac{1}{2} + \frac{1}{1 \cdot 2} = 2$ . *Inductive step:* Assume that the statement is true for  $k$ . The set of the first  $k + 1$  positive integers has many nonempty subsets, but they fall into three categories: a nonempty subset of the first  $k$  positive integers together with  $k + 1$ , a nonempty subset of the first  $k$  positive integers, or just  $\{k + 1\}$ . By the inductive hypothesis, the sum of the first category is  $k$ . For the second category, we can factor out  $1/(k + 1)$  from each term of the sum and what remains is just  $k$  by the inductive hypothesis, so this part of the sum is  $k/(k + 1)$ . Finally, the third category simply yields  $1/(k + 1)$ . Hence, the entire summation is  $k + k/(k + 1) + 1/(k + 1) = k + 1$ . **67. Basis step:** If  $A_1 \subseteq A_2$ , then  $A_1$  satisfies the condition of being a subset of each set in the collection; otherwise  $A_2 \subseteq A_1$ , so  $A_2$  satisfies the condition. *Inductive step:* Assume the inductive hypothesis, that the conditional statement is true for  $k$  sets, and suppose we are given  $k + 1$  sets that satisfy the given conditions. By the inductive hypothesis, there must be a set  $A_i$  for some  $i \leq k$  such that  $A_i \subseteq A_j$  for  $1 \leq j \leq k$ . If  $A_i \subseteq A_{k+1}$ , then we are done. Otherwise, we know that  $A_{k+1} \subseteq A_i$ , and this tells us that  $A_{k+1}$  satisfies the condition of being a subset of  $A_j$  for  $1 \leq j \leq k + 1$ . **69.**  $G(1) = 0$ ,  $G(2) = 1$ ,  $G(3) = 3$ ,  $G(4) = 4$  **71.** To show that  $2n - 4$  calls are sufficient to exchange all the gossip, select persons 1, 2, 3, and 4 to be the central committee. Every person outside the central committee calls one person on the central committee. At this point the central committee members *as a group* know all the scandals. They then exchange information among themselves by making the calls 1-2, 3-4, 1-3, and 2-4 in that order. At this point, *every* central committee member knows all the scandals. Finally, again every person outside the central committee calls one person on the central committee, at which point everyone knows all the scandals. [The total num-

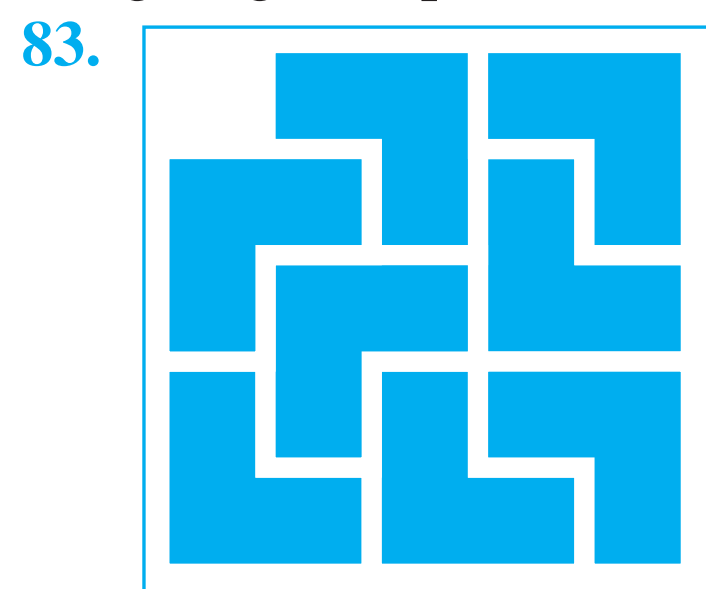
ber of calls is  $(n - 4) + 4 + (n - 4) = 2n - 4$ .] That this cannot be done with fewer than  $2n - 4$  calls is much harder to prove; see Sandra M. Hedetniemi, Stephen T. Hedetniemi, and Arthur L. Liestman, "A survey of gossiping and broadcasting in communication networks," *Networks* **18** (1988), no. 4, 319–349, for details. **73.** We prove this by mathematical induction. The basis step ( $n = 2$ ) is true tautologically. For  $n = 3$ , suppose that the intervals are  $(a, b)$ ,  $(c, d)$ , and  $(e, f)$ , where without loss of generality we can assume that  $a \leq c \leq e$ . Because  $(a, b) \cap (e, f) \neq \emptyset$ , we must have  $e < b$ ; for a similar reason,  $e < d$ . It follows that the number halfway between  $e$  and the smaller of  $b$  and  $d$  is common to all three intervals. Now for the inductive step, assume that whenever we have  $k$  intervals that have pairwise nonempty intersections then there is a point common to all the intervals, and suppose that we are given intervals  $I_1, I_2, \dots, I_{k+1}$  that have pairwise nonempty intersections. For each  $i$  from 1 to  $k$ , let  $J_i = I_i \cap I_{k+1}$ . We claim that the collection  $J_1, J_2, \dots, J_k$  satisfies the inductive hypothesis, that is, that  $J_{i_1} \cap J_{i_2} \neq \emptyset$  for each choice of subscripts  $i_1$  and  $i_2$ . This follows from the  $n = 3$  case proved above, using the sets  $I_{i_1}, I_{i_2}$ , and  $I_{k+1}$ . We can now invoke the inductive hypothesis to conclude that there is a number common to all of the sets  $J_i$  for  $i = 1, 2, \dots, k$ , which perforce is in the intersection of all the sets  $I_i$  for  $i = 1, 2, \dots, k + 1$ . **75. a)** The basis step is  $\sum_{j=1}^1 j/(j + 1)! < 1$ , which is true because the left-hand side equals  $1/2$ . It does not follow from  $\sum_{j=1}^k j/(j + 1)! < 1$  that  $\sum_{j=1}^{k+1} j/(j + 1)! < 1$  because the amount by which  $\sum_{j=1}^k j/(j + 1)!$  is less than 1 could be less than  $(k + 1)/(k + 2)!$ , the new term being added in. **b)** Basis step is the true statement that  $1/2! \leq 1 - 1/2!$ . Assuming inductive hypothesis  $P(k)$  that  $\sum_{j=1}^k j/(j + 1)! \leq 1 - 1/(k + 1)!$ , it follows that  $\sum_{j=1}^{k+1} j/(j + 1)! = \sum_{j=1}^k j/(j + 1)! + (k + 1)/(k + 2)! \leq 1 - 1/(k + 1)! + (k + 1)/(k + 2)! = 1 - (k + 2)/(k + 2)! + (k + 1)/(k + 2)! = 1 - 1/(k + 2)!$ , which is  $P(k + 1)$ . **77.** Pair up the people. Have the people stand at mutually distinct small distances from their partners but far away from everyone else. Then each person throws a pie at his or her partner, so everyone gets hit.



**81.** Let  $P(n)$  be the statement that every  $2^n \times 2^n \times 2^n$  checkerboard with a  $1 \times 1 \times 1$  cube removed can be covered by tiles that are  $2 \times 2 \times 2$  cubes each with a  $1 \times 1 \times 1$  cube removed. The basis step,  $P(1)$ , holds because one tile coincides with the solid to be tiled. Now assume that  $P(k)$  holds. Now consider a  $2^{k+1} \times 2^{k+1} \times 2^{k+1}$  cube with a  $1 \times 1 \times 1$  cube removed. Split this object into eight pieces using planes parallel to its faces and running through its center. The missing  $1 \times 1 \times 1$  piece occurs in one of these eight pieces. Now position one tile with



its center at the center of the large object so that the missing  $1 \times 1 \times 1$  cube lies in the octant in which the large object is missing a  $1 \times 1 \times 1$  cube. This creates eight  $2^k \times 2^k \times 2^k$  cubes, each missing a  $1 \times 1 \times 1$  cube. By the inductive hypothesis we can fill each of these eight objects with tiles. Putting these tilings together produces the desired tiling.



85. Let  $Q(n)$  be  $P(n + b - 1)$ . The statement that  $P(n)$  is true for  $n = b, b + 1, b + 2, \dots$  is the same as the statement that  $Q(m)$  is true for all positive integers  $m$ . We are given that  $P(b)$  is true [i.e., that  $Q(1)$  is true], and that  $P(k) \rightarrow P(k + 1)$  for all  $k \geq b$  [i.e., that  $Q(m) \rightarrow Q(m + 1)$  for all positive integers  $m$ ]. Therefore, by the principle of mathematical induction,  $Q(m)$  is true for all positive integers  $m$ .

## Section 5.2

1. *Basis step:* We are told we can run one mile, so  $P(1)$  is true. *Inductive step:* Assume the inductive hypothesis, that we can run any number of miles from 1 to  $k$ . We must show that we can run  $k + 1$  miles. If  $k = 1$ , then we are already told that we can run two miles. If  $k > 1$ , then the inductive hypothesis tells us that we can run  $k - 1$  miles, so we can run  $(k - 1) + 2 = k + 1$  miles. 3. **a)**  $P(8)$  is true, because we can form 8 cents of postage with one 3-cent stamp and one 5-cent stamp.  $P(9)$  is true, because we can form 9 cents of postage with three 3-cent stamps.  $P(10)$  is true, because we can form 10 cents of postage with two 5-cent stamps. **b)** The statement that using just 3-cent and 5-cent stamps we can form  $j$  cents postage for all  $j$  with  $8 \leq j \leq k$ , where we assume that  $k \geq 10$ . **c)** Assuming the inductive hypothesis, we can form  $k + 1$  cents postage using just 3-cent and 5-cent stamps. **d)** Because  $k \geq 10$ , we know that  $P(k - 2)$  is true, that is, that we can form  $k - 2$  cents of postage. Put one more 3-cent stamp on the envelope, and we have formed  $k + 1$  cents of postage. **e)** We have completed both the basis step and the inductive step, so by the principle of strong induction, the statement is true for every integer  $n$  greater than or equal to 8. 5. **a)** 4, 8, 11, 12, 15, 16, 19, 20, 22, 23, 24, 26, 27, 28, and all values greater than or equal to 30. **b)** Let  $P(n)$  be the statement that we can form  $n$  cents of postage using just 4-cent and 11-cent stamps. We want to prove that  $P(n)$  is true for all  $n \geq 30$ . For the basis step,  $30 = 11 + 11 + 4 + 4$ . Assume that we can form  $k$  cents of postage (the inductive hypothesis); we will show how to form  $k + 1$  cents of postage. If the  $k$  cents included an 11-cent stamp, then replace it by three 4-cent stamps. Otherwise,  $k$  cents was formed from just 4-cent stamps. Because  $k \geq 30$ , there must be at least eight 4-cent stamps involved.

Replace eight 4-cent stamps by three 11-cent stamps, and we have formed  $k + 1$  cents in postage. **c)**  $P(n)$  is the same as in part (b). To prove that  $P(n)$  is true for all  $n \geq 30$ , we check for the basis step that  $30 = 11 + 11 + 4 + 4$ ,  $31 = 11 + 4 + 4 + 4 + 4 + 4$ ,  $32 = 4 + 4 + 4 + 4 + 4 + 4 + 4 + 4$ , and  $33 = 11 + 11 + 11$ . For the inductive step, assume the inductive hypothesis, that  $P(j)$  is true for all  $j$  with  $30 \leq j \leq k$ , where  $k$  is an arbitrary integer greater than or equal to 33. We want to show that  $P(k + 1)$  is true. Because  $k - 3 \geq 30$ , we know that  $P(k - 3)$  is true, that is, that we can form  $k - 3$  cents of postage. Put one more 4-cent stamp on the envelope, and we have formed  $k + 1$  cents of postage. In this proof, our inductive hypothesis was that  $P(j)$  was true for all values of  $j$  between 30 and  $k$  inclusive, rather than just that  $P(30)$  was true. 7. We can form all amounts except \$1 and \$3. Let  $P(n)$  be the statement that we can form  $n$  dollars using just 2-dollar and 5-dollar bills. We want to prove that  $P(n)$  is true for all  $n \geq 5$ . (It is clear that \$1 and \$3 cannot be formed and that \$2 and \$4 can be formed.) For the basis step, note that  $5 = 5$  and  $6 = 2 + 2 + 2$ . Assume the inductive hypothesis, that  $P(j)$  is true for all  $j$  with  $5 \leq j \leq k$ , where  $k$  is an arbitrary integer greater than or equal to 6. We want to show that  $P(k + 1)$  is true. Because  $k - 1 \geq 5$ , we know that  $P(k - 1)$  is true, that is, that we can form  $k - 1$  dollars. Add another 2-dollar bill, and we have formed  $k + 1$  dollars. 9. Let  $P(n)$  be the statement that there is no positive integer  $b$  such that  $\sqrt{2} = n/b$ . *Basis step:*  $P(1)$  is true because  $\sqrt{2} > 1 \geq 1/b$  for all positive integers  $b$ . *Inductive step:* Assume that  $P(j)$  is true for all  $j \leq k$ , where  $k$  is an arbitrary positive integer; we prove that  $P(k + 1)$  is true by contradiction. Assume that  $\sqrt{2} = (k + 1)/b$  for some positive integer  $b$ . Then  $2b^2 = (k + 1)^2$ , so  $(k + 1)^2$  is even, and hence,  $k + 1$  is even. So write  $k + 1 = 2t$  for some positive integer  $t$ , whence  $2b^2 = 4t^2$  and  $b^2 = 2t^2$ . By the same reasoning as before,  $b$  is even, so  $b = 2s$  for some positive integer  $s$ . Then  $\sqrt{2} = (k + 1)/b = (2t)/(2s) = t/s$ . But  $t \leq k$ , so this contradicts the inductive hypothesis, and our proof of the inductive step is complete. 11. *Basis step:* There are four base cases. If  $n = 1 = 4 \cdot 0 + 1$ , then clearly the second player wins. If there are two, three, or four matches ( $n = 4 \cdot 0 + 2$ ,  $n = 4 \cdot 0 + 3$ , or  $n = 4 \cdot 1$ ), then the first player can win by removing all but one match. *Inductive step:* Assume the strong inductive hypothesis, that in games with  $k$  or fewer matches, the first player can win if  $k \equiv 0, 2$ , or  $3 \pmod{4}$  and the second player can win if  $k \equiv 1 \pmod{4}$ . Suppose we have a game with  $k + 1$  matches, with  $k \geq 4$ . If  $k + 1 \equiv 0 \pmod{4}$ , then the first player can remove three matches, leaving  $k - 2$  matches for the other player. Because  $k - 2 \equiv 1 \pmod{4}$ , by the inductive hypothesis, this is a game that the second player at that point (who is the first player in our game) can win. Similarly, if  $k + 1 \equiv 2 \pmod{4}$ , then the first player can remove one match; and if  $k + 1 \equiv 3 \pmod{4}$ , then the first player can remove two matches. Finally, if  $k + 1 \equiv 1 \pmod{4}$ , then the first player must leave  $k$ ,  $k - 1$ , or  $k - 2$  matches for the other player. Because  $k \equiv 0 \pmod{4}$ ,  $k - 1 \equiv 3 \pmod{4}$ , and  $k - 2 \equiv 2 \pmod{4}$ , by the inductive hypothesis, this is a game that the first player at that point (who is the second player in our game) can win. 13. Let  $P(n)$  be the statement that



exactly  $n - 1$  moves are required to assemble a puzzle with  $n$  pieces. Now  $P(1)$  is trivially true. Assume that  $P(j)$  is true for all  $j \leq k$ , and consider a puzzle with  $k + 1$  pieces. The final move must be the joining of two blocks, of size  $j$  and  $k + 1 - j$  for some integer  $j$  with  $1 \leq j \leq k$ . By the inductive hypothesis, it required  $j - 1$  moves to construct the one block, and  $k + 1 - j - 1 = k - j$  moves to construct the other. Therefore,  $1 + (j - 1) + (k - j) = k$  moves are required in all, so  $P(k + 1)$  is true. **15.** Let the Chomp board have  $n$  rows and  $n$  columns. We claim that the first player can win the game by making the first move to leave just the top row and leftmost column. Let  $P(n)$  be the statement that if a player has presented his opponent with a Chomp configuration consisting of just  $n$  cookies in the top row and  $n$  cookies in the leftmost column, then he can win the game. We will prove  $\forall n P(n)$  by strong induction. We know that  $P(1)$  is true, because the opponent is forced to take the poisoned cookie at his first turn. Fix  $k \geq 1$  and assume that  $P(j)$  is true for all  $j \leq k$ . We claim that  $P(k + 1)$  is true. It is the opponent's turn to move. If she picks the poisoned cookie, then the game is over and she loses. Otherwise, assume she picks the cookie in the top row in column  $j$ , or the cookie in the left column in row  $j$ , for some  $j$  with  $2 \leq j \leq k + 1$ . The first player now picks the cookie in the left column in row  $j$ , or the cookie in the top row in column  $j$ , respectively. This leaves the position covered by  $P(j - 1)$  for his opponent, so by the inductive hypothesis, he can win. **17.** Let  $P(n)$  be the statement that if a simple polygon with  $n$  sides is triangulated, then at least two of the triangles in the triangulation have two sides that border the exterior of the polygon. We will prove  $\forall n \geq 4 P(n)$ . The statement is true for  $n = 4$ , because there is only one diagonal, leaving two triangles with the desired property. Fix  $k \geq 4$  and assume that  $P(j)$  is true for all  $j$  with  $4 \leq j \leq k$ . Consider a polygon with  $k + 1$  sides, and some triangulation of it. Pick one of the diagonals in this triangulation. First suppose that this diagonal divides the polygon into one triangle and one polygon with  $k$  sides. Then the triangle has two sides that border the exterior. Furthermore, the  $k$ -gon has, by the inductive hypothesis, two triangles that have two sides that border the exterior of that  $k$ -gon, and only one of these triangles can fail to be a triangle that has two sides that border the exterior of the original polygon. The only other case is that this diagonal divides the polygon into two polygons with  $j$  sides and  $k + 3 - j$  sides for some  $j$  with  $4 \leq j \leq k - 1$ . By the inductive hypothesis, each of these two polygons has two triangles that have two sides that border their exterior, and in each case only one of these triangles can fail to be a triangle that has two sides that border the exterior of the original polygon. **19.** Let  $P(n)$  be the statement that the area of a simple polygon with  $n$  sides and vertices all at lattice points is given by  $I(P) + B(P)/2 - 1$ . We will prove  $P(n)$  for all  $n \geq 3$ . We begin with an additivity lemma: If  $P$  is a simple polygon with all vertices at lattice points, divided into polygons  $P_1$  and  $P_2$  by a diagonal, then  $I(P) + B(P)/2 - 1 = [I(P_1) + B(P_1)/2 - 1] + [I(P_2) + B(P_2)/2 - 1]$ . To prove this, suppose there are  $k$  lattice points on the diagonal, not counting its endpoints. Then  $I(P) = I(P_1) + I(P_2) + k$  and

$B(P) = B(P_1) + B(P_2) - 2k - 2$ ; and the result follows by simple algebra. What this says in particular is that if Pick's formula gives the correct area for  $P_1$  and  $P_2$ , then it must give the correct formula for  $P$ , whose area is the sum of the areas for  $P_1$  and  $P_2$ ; and similarly if Pick's formula gives the correct area for  $P$  and one of the  $P_i$ 's, then it must give the correct formula for the other  $P_i$ . Next we prove the theorem for rectangles whose sides are parallel to the coordinate axes. Such a rectangle necessarily has vertices at  $(a, b)$ ,  $(a, c)$ ,  $(d, b)$ , and  $(d, c)$ , where  $a, b, c$ , and  $d$  are integers with  $b < c$  and  $a < d$ . Its area is  $(c - b)(d - a)$ . Also,  $B = 2(c - b + d - a)$  and  $I = (c - b - 1)(d - a - 1) = (c - b)(d - a) - (c - b) - (d - a) + 1$ . Therefore,  $I + B/2 - 1 = (c - b)(d - a) - (c - b) - (d - a) + 1 + (c - b + d - a) - 1 = (c - b)(d - a)$ , which is the desired area. Next consider a right triangle whose legs are parallel to the coordinate axes. This triangle is half a rectangle of the type just considered, for which Pick's formula holds, so by the additivity lemma, it holds for the triangle as well. (The values of  $B$  and  $I$  are the same for each of the two triangles, so if Pick's formula gave an answer that was either too small or too large, then it would give a correspondingly wrong answer for the rectangle.) For the next step, consider an arbitrary triangle with vertices at lattice points that is not of the type already considered. Embed it in as small a rectangle as possible. There are several possible ways this can happen, but in any case (and adding one more edge in one case), the rectangle will have been partitioned into the given triangle and two or three right triangles with sides parallel to the coordinate axes. Again by the additivity lemma, we are guaranteed that Pick's formula gives the correct area for the given triangle. This completes the proof of  $P(3)$ , the basis step in our strong induction proof. For the inductive step, given an arbitrary polygon, use Lemma 1 in the text to split it into two polygons. Then by the additivity lemma above and the inductive hypothesis, we know that Pick's formula gives the correct area for this polygon. **21. a)** In the left figure  $\angle abp$  is smallest, but  $bp$  is not an interior diagonal. **b)** In the right figure  $\overline{bd}$  is not an interior diagonal. **c)** In the right figure  $\overline{bd}$  is not an interior diagonal. **23. a)** When we try to prove the inductive step and find a triangle in each subpolygon with at least two sides bordering the exterior, it may happen in each case that the triangle we are guaranteed in fact borders the diagonal (which is part of the boundary of that polygon). This leaves us with no triangles guaranteed to touch the boundary of the *original* polygon. **b)** We proved the stronger statement  $\forall n \geq 4 T(n)$  in Exercise 17. **25. a)** The inductive step here allows us to conclude that  $P(3), P(5), \dots$  are all true, but we can conclude nothing about  $P(2), P(4), \dots$ . **b)**  $P(n)$  is true for all positive integers  $n$ , using strong induction. **c)** The inductive step here enables us to conclude that  $P(2), P(4), P(8), P(16), \dots$  are all true, but we can conclude nothing about  $P(n)$  when  $n$  is not a power of 2. **d)** This is mathematical induction; we can conclude that  $P(n)$  is true for all positive integers  $n$ . **27.** Suppose, for a proof by contradiction, that there is some positive integer  $n$  such that  $P(n)$  is not true. Let  $m$  be the smallest positive integer greater than  $n$  for which  $P(m)$  is true; we know that such an  $m$  exists



because  $P(m)$  is true for infinitely many values of  $m$ . But we know that  $P(m) \rightarrow P(m-1)$ , so  $P(m-1)$  is also true. Thus,  $m-1$  cannot be greater than  $n$ , so  $m-1 = n$  and  $P(n)$  is in fact true. This contradiction shows that  $P(n)$  is true for all  $n$ .

**29.** The error is in going from the base case  $n = 0$  to the next case,  $n = 1$ ; we cannot write 1 as the sum of two smaller natural numbers.

**31.** Assume that the well-ordering property holds. Suppose that  $P(1)$  is true and that the conditional statement  $[P(1) \wedge P(2) \wedge \cdots \wedge P(n)] \rightarrow P(n+1)$  is true for every positive integer  $n$ . Let  $S$  be the set of positive integers  $n$  for which  $P(n)$  is false. We will show  $S = \emptyset$ . Assume that  $S \neq \emptyset$ . Then by the well-ordering property there is a least integer  $m$  in  $S$ . We know that  $m$  cannot be 1 because  $P(1)$  is true. Because  $n = m$  is the least integer such that  $P(n)$  is false,  $P(1), P(2), \dots, P(m-1)$  are true, and  $m-1 \geq 1$ . Because  $[P(1) \wedge P(2) \wedge \cdots \wedge P(m-1)] \rightarrow P(m)$  is true, it follows that  $P(m)$  must also be true, which is a contradiction. Hence,  $S = \emptyset$ .

**33.** In each case, give a proof by contradiction based on a “smallest counterexample,” that is, values of  $n$  and  $k$  such that  $P(n, k)$  is not true and  $n$  and  $k$  are smallest in some sense.

**a)** Choose a counterexample with  $n + k$  as small as possible. We cannot have  $n = 1$  and  $k = 1$ , because we are given that  $P(1, 1)$  is true. Therefore, either  $n > 1$  or  $k > 1$ . In the former case, by our choice of counterexample, we know that  $P(n-1, k)$  is true. But the inductive step then forces  $P(n, k)$  to be true, a contradiction. The latter case is similar. So our supposition that there is a counterexample must be wrong, and  $P(n, k)$  is true in all cases.

**b)** Choose a counterexample with  $n$  as small as possible. We cannot have  $n = 1$ , because we are given that  $P(1, k)$  is true for all  $k$ . Therefore,  $n > 1$ . By our choice of counterexample, we know that  $P(n-1, k)$  is true. But the inductive step then forces  $P(n, k)$  to be true, a contradiction.

**c)** Choose a counterexample with  $k$  as small as possible. We cannot have  $k = 1$ , because we are given that  $P(n, 1)$  is true for all  $n$ . Therefore,  $k > 1$ . By our choice of counterexample, we know that  $P(n, k-1)$  is true. But the inductive step then forces  $P(n, k)$  to be true, a contradiction.

**35.** Let  $P(n)$  be the statement that if  $x_1, x_2, \dots, x_n$  are  $n$  distinct real numbers, then  $n-1$  multiplications are used to find the product of these numbers no matter how parentheses are inserted in the product. We will prove that  $P(n)$  is true using strong induction. The basis case  $P(1)$  is true because  $1-1=0$  multiplications are required to find the product of  $x_1$ , a product with only one factor. Suppose that  $P(k)$  is true for  $1 \leq k \leq n$ . The last multiplication used to find the product of the  $n+1$  distinct real numbers  $x_1, x_2, \dots, x_n, x_{n+1}$  is a multiplication of the product of the first  $k$  of these numbers for some  $k$  and the product of the last  $n+1-k$  of them. By the inductive hypothesis,  $k-1$  multiplications are used to find the product of  $k$  of the numbers, no matter how parentheses were inserted in the product of these numbers, and  $n-k$  multiplications are used to find the product of the other  $n+1-k$  of them, no matter how parentheses were inserted in the product of these numbers. Because one more multiplication is required to find the product of all  $n+1$  numbers, the total number of multiplications used equals  $(k-1) + (n-k) + 1 = n$ . Hence,

$P(n+1)$  is true.

**37.** Assume that  $a = dq + r = dq' + r'$  with  $0 \leq r < d$  and  $0 \leq r' < d$ . Then  $d(q - q') = r' - r$ . It follows that  $d$  divides  $r' - r$ . Because  $-d < r' - r < d$ , we have  $r' - r = 0$ . Hence,  $r' = r$ . It follows that  $q = q'$ .

**39.** This is a paradox caused by self-reference. The answer is “no.” There are a finite number of English words, so only a finite number of strings of 15 words or fewer; therefore, only a finite number of positive integers can be so described, not all of them.

**41.** Suppose that the well-ordering property were false. Let  $S$  be a nonempty set of nonnegative integers that has no least element. Let  $P(n)$  be the statement “ $i \notin S$  for  $i = 0, 1, \dots, n$ .”  $P(0)$  is true because if  $0 \in S$  then  $S$  has a least element, namely, 0. Now suppose that  $P(n)$  is true. Thus,  $0 \notin S, 1 \notin S, \dots, n \notin S$ . Now,  $n+1$  cannot be in  $S$ , for if it were, it would be its least element. Thus,  $P(n+1)$  is true. So by the principle of mathematical induction,  $n \notin S$  for all nonnegative integers  $n$ . Thus,  $S = \emptyset$ , a contradiction.

**43.** Strong induction implies the principle of mathematical induction, for if one has shown that  $P(k) \rightarrow P(k+1)$  is true, then one has also shown that  $[P(1) \wedge \cdots \wedge P(k)] \rightarrow P(k+1)$  is true. By Exercise 41, the principle of mathematical induction implies the well-ordering property. Therefore, by assuming strong induction as an axiom, we can prove the well-ordering property.

## Section 5.3

**1. a)**  $f(1) = 3, f(2) = 5, f(3) = 7, f(4) = 9$  **b)**  $f(1) = 3, f(2) = 9, f(3) = 27, f(4) = 81$  **c)**  $f(1) = 2, f(2) = 4, f(3) = 16, f(4) = 65, 536$  **d)**  $f(1) = 3, f(2) = 13, f(3) = 183, f(4) = 33, 673$

**3. a)**  $f(2) = -1, f(3) = 5, f(4) = 2, f(5) = 17$  **b)**  $f(2) = -4, f(3) = 32, f(4) = -4096, f(5) = 536, 870, 912$  **c)**  $f(2) = 8, f(3) = 176, f(4) = 92, 672, f(5) = 25, 764, 174, 848$  **d)**  $f(2) = -\frac{1}{2}, f(3) = -4, f(4) = \frac{1}{8}, f(5) = -32$

**5. a)** Not valid **b)**  $f(n) = 1 - n$ . *Basis step:*  $f(0) = 1 = 1 - 0$ . *Inductive step:* if  $f(k) = 1 - k$ , then  $f(k+1) = f(k) - 1 = 1 - k - 1 = 1 - (k+1)$ . **c)**  $f(n) = 4 - n$  if  $n > 0$ , and  $f(0) = 2$ . *Basis step:*  $f(0) = 2$  and  $f(1) = 3 = 4 - 1$ . *Inductive step* (with  $k \geq 1$ ):  $f(k+1) = f(k) - 1 = (4 - k) - 1 = 4 - (k+1)$ . **d)**  $f(n) = 2^{\lfloor (n+1)/2 \rfloor}$ . *Basis step:*  $f(0) = 1 = 2^{\lfloor (0+1)/2 \rfloor}$  and  $f(1) = 2 = 2^{\lfloor (1+1)/2 \rfloor}$ . *Inductive step* (with  $k \geq 1$ ):  $f(k+1) = 2f(k-1) = 2 \cdot 2^{\lfloor (k-1)/2 \rfloor} = 2^{\lfloor (k-1)/2 \rfloor + 1} = 2^{\lfloor ((k+1)+1)/2 \rfloor}$ . **e)**  $f(n) = 3^n$ . *Basis step:* Trivial. *Inductive step:* For odd  $n$ ,  $f(n) = 3f(n-1) = 3 \cdot 3^{n-1} = 3^n$ ; and for even  $n > 1$ ,  $f(n) = 9f(n-2) = 9 \cdot 3^{n-2} = 3^n$ .

**7.** There are many possible correct answers. We will supply relatively simple ones.

**a)**  $a_{n+1} = a_n + 6$  for  $n \geq 1$  and  $a_1 = 6$  **b)**  $a_{n+1} = a_n + 2$  for  $n \geq 1$  and  $a_1 = 3$  **c)**  $a_{n+1} = 10a_n$  for  $n \geq 1$  and  $a_1 = 10$  **d)**  $a_{n+1} = a_n$  for  $n \geq 1$  and  $a_1 = 5$

**9.**  $F(0) = 0, F(n) = F(n-1) + n$  for  $n \geq 1$

**11.**  $P_m(0) = 0, P_m(n+1) = P_m(n) + m$

**13.** Let  $P(n)$  be “ $f_1 + f_3 + \cdots + f_{2n-1} = f_{2n}$ .” *Basis step:*  $P(1)$  is true because  $f_1 = 1 = f_2$ . *Inductive step:* Assume that  $P(k)$  is true. Then  $f_1 + f_3 + \cdots + f_{2k-1} + f_{2k+1} = f_{2k} + f_{2k+1} = f_{2k+2} + f_{2(k+1)}$ .

**15.** *Basis step:*  $f_0 f_1 + f_1 f_2 = 0 \cdot 1 + 1 \cdot 1 = 1^2 = f_2^2$ . *Inductive step:* Assume that  $f_0 f_1 + f_1 f_2 + \cdots + f_{2k-1} f_{2k} = f_{2k}^2$ . Then  $f_0 f_1 + f_1 f_2 + \cdots + f_{2k-1} f_{2k} + f_{2k} f_{2k+1} + f_{2k+1} f_{2k+2} =$



$f_{2k}^2 + f_{2k}f_{2k+1} + f_{2k+1}f_{2k+2} = f_{2k}(f_{2k} + f_{2k+1}) + f_{2k+1}f_{2k+2} = f_{2k}f_{2k+2} + f_{2k+1}f_{2k+2} = (f_{2k} + f_{2k+1})f_{2k+2} = f_{2k+2}^2$ . **17.** The number of divisions used by the Euclidean algorithm to find  $\gcd(f_{n+1}, f_n)$  is 0 for  $n = 0$ , 1 for  $n = 1$ , and  $n - 1$  for  $n \geq 2$ . To prove this result for  $n \geq 2$  we use mathematical induction. For  $n = 2$ , one division shows that  $\gcd(f_3, f_2) = \gcd(2, 1) = \gcd(1, 0) = 1$ . Now assume that  $k - 1$  divisions are used to find  $\gcd(f_{k+1}, f_k)$ . To find  $\gcd(f_{k+2}, f_{k+1})$ , first divide  $f_{k+2}$  by  $f_{k+1}$  to obtain  $f_{k+2} = 1 \cdot f_{k+1} + f_k$ . After one division we have  $\gcd(f_{k+2}, f_{k+1}) = \gcd(f_{k+1}, f_k)$ . By the inductive hypothesis it follows that exactly  $k - 1$  more divisions are required. This shows that  $k$  divisions are required to find  $\gcd(f_{k+2}, f_{k+1})$ , finishing the inductive proof. **19.**  $|A| = -1$ . Hence,  $|A^n| = (-1)^n$ . It follows that  $f_{n+1}f_{n-1} - f_n^2 = (-1)^n$ . **21. a)** Proof by induction. *Basis step:* For  $n = 1$ ,  $\max(-a_1) = -a_1 = -\min(a_1)$ . For  $n = 2$ , there are two cases. If  $a_2 \geq a_1$ , then  $-a_1 \geq -a_2$ , so  $\max(-a_1, -a_2) = -a_1 = -\min(a_1, a_2)$ . If  $a_2 < a_1$ , then  $-a_1 < -a_2$ , so  $\max(-a_1, -a_2) = -a_2 = -\min(a_1, a_2)$ . *Inductive step:* Assume true for  $k$  with  $k \geq 2$ . Then  $\max(-a_1, -a_2, \dots, -a_k, -a_{k+1}) = \max(\max(-a_1, \dots, -a_k), -a_{k+1}) = \max(-\min(a_1, \dots, a_k), -a_{k+1}) = -\min(\min(a_1, \dots, a_k), a_{k+1}) = -\min(a_1, \dots, a_{k+1})$ . **b)** Proof by mathematical induction. *Basis step:* For  $n = 1$ , the result is the identity  $a_1 + b_1 = a_1 + b_1$ . For  $n = 2$ , first consider the case in which  $a_1 + b_1 \geq a_2 + b_2$ . Then  $\max(a_1 + b_1, a_2 + b_2) = a_1 + b_1$ . Also note that  $a_1 \leq \max(a_1, a_2)$  and  $b_1 \leq \max(b_1, b_2)$ , so  $a_1 + b_1 \leq \max(a_1, a_2) + \max(b_1, b_2)$ . Therefore,  $\max(a_1 + b_1, a_2 + b_2) = a_1 + b_1 \leq \max(a_1, a_2) + \max(b_1, b_2)$ . The case with  $a_1 + b_1 < a_2 + b_2$  is similar. *Inductive step:* Assume that the result is true for  $k$ . Then  $\max(a_1 + b_1, a_2 + b_2, \dots, a_k + b_k, a_{k+1} + b_{k+1}) = \max(\max(a_1 + b_1, a_2 + b_2, \dots, a_k + b_k), a_{k+1} + b_{k+1}) \leq \max(\max(a_1, a_2, \dots, a_k) + \max(b_1, b_2, \dots, b_k), a_{k+1} + b_{k+1}) \leq \max(\max(a_1, a_2, \dots, a_k), a_{k+1}) + \max(\max(b_1, b_2, \dots, b_k), b_{k+1}) = \max(a_1, a_2, \dots, a_k, a_{k+1}) + \max(b_1, b_2, \dots, b_k, b_{k+1})$ . **c)** Same as part (b), but replace every occurrence of “max” by “min” and invert each inequality. **23.**  $5 \in S$ , and  $x + y \in S$  if  $x, y \in S$ . **25. a)**  $0 \in S$ , and if  $x \in S$ , then  $x + 2 \in S$  and  $x - 2 \in S$ . **b)**  $2 \in S$ , and if  $x \in S$ , then  $x + 3 \in S$ . **c)**  $1 \in S, 2 \in S, 3 \in S, 4 \in S$ , and if  $x \in S$ , then  $x + 5 \in S$ . **27. a)** *Basis step:*  $5 \equiv 5 \pmod{10}$ . *Inductive step:* If  $n \equiv 5 \pmod{10}$ , then  $3n \equiv 3 \cdot 5 = 15 \equiv 5 \pmod{10}$  and  $n^2 \equiv 5^2 = 25 \equiv 5 \pmod{10}$ . **b)**  $35 \notin S$  because 35 is not a multiple of 3 nor a perfect square. **29. a)**  $(0, 1), (1, 1), (2, 1); (0, 2), (1, 2), (2, 2), (3, 2), (4, 2); (0, 3), (1, 3), (2, 3), (3, 3), (4, 3), (5, 3), (6, 3); (0, 4), (1, 4), (2, 4), (3, 4), (4, 4), (5, 4), (6, 4), (7, 4), (8, 4)$  **b)** Let  $P(n)$  be the statement that  $a \leq 2b$  whenever  $(a, b) \in S$  is obtained by  $n$  applications of the recursive step. *Basis step:*  $P(0)$  is true, because the only element of  $S$  obtained with no applications of the recursive step is  $(0, 0)$ , and indeed  $0 \leq 2 \cdot 0$ . *Inductive step:* Assume that  $a \leq 2b$  whenever  $(a, b) \in S$  is obtained by  $k$  or fewer applications of the recursive step, and consider an element obtained with  $k + 1$  applications of the recursive step. Because the final application of the recursive step to an element  $(a, b)$  must be applied to an element obtained with fewer applica-

tions of the recursive step, we know that  $a \leq 2b$ . Add  $0 \leq 2, 1 \leq 2$ , and  $2 \leq 2$ , respectively, to obtain  $a \leq 2(b + 1), a + 1 \leq 2(b + 1)$ , and  $a + 2 \leq 2(b + 1)$ , as desired. **c)** This holds for the basis step, because  $0 \leq 0$ . If this holds for  $(a, b)$ , then it also holds for the elements obtained from  $(a, b)$  in the recursive step, because adding  $0 \leq 2, 1 \leq 2$ , and  $2 \leq 2$ , respectively, to  $a \leq 2b$  yields  $a \leq 2(b + 1), a + 1 \leq 2(b + 1)$ , and  $a + 2 \leq 2(b + 1)$ . **31. a)** Define  $S$  by  $(1, 1) \in S$ , and if  $(a, b) \in S$ , then  $(a + 2, b) \in S, (a, b + 2) \in S$ , and  $(a + 1, b + 1) \in S$ . All elements put in  $S$  satisfy the condition, because  $(1, 1)$  has an even sum of coordinates, and if  $(a, b)$  has an even sum of coordinates, then so do  $(a + 2, b), (a, b + 2)$ , and  $(a + 1, b + 1)$ . Conversely, we show by induction on the sum of the coordinates that if  $a + b$  is even, then  $(a, b) \in S$ . If the sum is 2, then  $(a, b) = (1, 1)$ , and the basis step put  $(a, b)$  into  $S$ . Otherwise the sum is at least 4, and at least one of  $(a - 2, b), (a, b - 2)$ , and  $(a - 1, b - 1)$  must have positive integer coordinates whose sum is an even number smaller than  $a + b$ , and therefore must be in  $S$ . Then one application of the recursive step shows that  $(a, b) \in S$ . **b)** Define  $S$  by  $(1, 1), (1, 2)$ , and  $(2, 1)$  are in  $S$ , and if  $(a, b) \in S$ , then  $(a + 2, b)$  and  $(a, b + 2)$  are in  $S$ . To prove that our definition works, we note first that  $(1, 1), (1, 2)$ , and  $(2, 1)$  all have an odd coordinate, and if  $(a, b)$  has an odd coordinate, then so do  $(a + 2, b)$  and  $(a, b + 2)$ . Conversely, we show by induction on the sum of the coordinates that if  $(a, b)$  has at least one odd coordinate, then  $(a, b) \in S$ . If  $(a, b) = (1, 1)$  or  $(a, b) = (1, 2)$  or  $(a, b) = (2, 1)$ , then the basis step put  $(a, b)$  into  $S$ . Otherwise either  $a$  or  $b$  is at least 3, so at least one of  $(a - 2, b)$  and  $(a, b - 2)$  must have positive integer coordinates whose sum is smaller than  $a + b$ , and therefore must be in  $S$ . Then one application of the recursive step shows that  $(a, b) \in S$ . **c)**  $(1, 6) \in S$  and  $(2, 3) \in S$ , and if  $(a, b) \in S$ , then  $(a + 2, b) \in S$  and  $(a, b + 6) \in S$ . To prove that our definition works, we note first that  $(1, 6)$  and  $(2, 3)$  satisfy the condition, and if  $(a, b)$  satisfies the condition, then so do  $(a + 2, b)$  and  $(a, b + 6)$ . Conversely we show by induction on the sum of the coordinates that if  $(a, b)$  satisfies the condition, then  $(a, b) \in S$ . For sums 5 and 7, the only points are  $(1, 6)$ , which the basis step put into  $S$ ,  $(2, 3)$ , which the basis step put into  $S$ , and  $(4, 3) = (2 + 2, 3)$ , which is in  $S$  by one application of the recursive definition. For a sum greater than 7, either  $a \geq 3$ , or  $a \leq 2$  and  $b \geq 9$ , in which case either  $(a - 2, b)$  or  $(a, b - 6)$  must have positive integer coordinates whose sum is smaller than  $a + b$  and satisfy the condition for being in  $S$ . Then one application of the recursive step shows that  $(a, b) \in S$ . **33.** If  $x$  is a set or a variable representing a set, then  $x$  is a well-formed formula. If  $x$  and  $y$  are well-formed formulae, then so are  $\bar{x}, (x \cup y), (x \cap y)$ , and  $(x - y)$ . **35. a)** If  $x \in D = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$ , then  $m(x) = x$ ; if  $s = tx$ , where  $t \in D^*$  and  $x \in D$ , then  $m(s) = \min(m(s), x)$ . **b)** Let  $t = wx$ , where  $w \in D^*$  and  $x \in D$ . If  $w = \lambda$ , then  $m(st) = m(sx) = \min(m(s), x) = \min(m(s), m(x))$  by the recursive step and the basis step of the definition of  $m$ . Otherwise,  $m(st) = m((sw)x) = \min(m(sw), x)$  by the definition of  $m$ . Now  $m(sw) = \min(m(s), m(w))$  by the inductive hypothesis of the structural induction, so  $m(st) =$



$\min(\min(m(s), m(w)), x) = \min(m(s), \min(m(w), x))$  by the meaning of  $\min$ . But  $\min(m(w), x) = m(wx) = m(t)$  by the recursive step of the definition of  $m$ . Thus,  $m(st) = \min(m(s), m(t))$ . **37.**  $\lambda^R = \lambda$  and  $(ux)^R = xu^R$  for  $x \in \Sigma$ ,  $u \in \Sigma^*$ . **39.**  $w^0 = \lambda$  and  $w^{n+1} = ww^n$ . **41.** When the string consists of  $n$  0s followed by  $n$  1s for some nonnegative integer  $n$ . **43.** Let  $P(i)$  be " $l(w^i) = i \cdot l(w)$ ."  $P(0)$  is true because  $l(w^0) = 0 = 0 \cdot l(w)$ . Assume  $P(i)$  is true. Then  $l(w^{i+1}) = l(ww^i) = l(w) + l(w^i) = l(w) + i \cdot l(w) = (i+1) \cdot l(w)$ . **45. Basis step:** For the full binary tree consisting of just a root the result is true because  $n(T) = 1$  and  $h(T) = 0$ , and  $1 \geq 2 \cdot 0 + 1$ . **Inductive step:** Assume that  $n(T_1) \geq 2h(T_1) + 1$  and  $n(T_2) \geq 2h(T_2) + 1$ . By the recursive definitions of  $n(T)$  and  $h(T)$ , we have  $n(T) = 1 + n(T_1) + n(T_2)$  and  $h(T) = 1 + \max(h(T_1), h(T_2))$ . Therefore,  $n(T) = 1 + n(T_1) + n(T_2) \geq 1 + 2h(T_1) + 1 + 2h(T_2) + 1 \geq 1 + 2 \cdot \max(h(T_1), h(T_2)) + 2 = 1 + 2(\max(h(T_1), h(T_2)) + 1) = 1 + 2h(T)$ . **47. Basis step:**  $a_{0,0} = 0 = 0 + 0$ . **Inductive step:** Assume that  $a_{m',n'} = m' + n'$  whenever  $(m', n')$  is less than  $(m, n)$  in the lexicographic ordering of  $\mathbf{N} \times \mathbf{N}$ . If  $n = 0$  then  $a_{m,n} = a_{m-1,n} + 1 = m - 1 + n + 1 = m + n$ . If  $n > 0$ , then  $a_{m,n} = a_{m,n-1} + 1 = m + n - 1 + 1 = m + n$ . **49. a)**  $P_{m,m} = P_m$  because a number exceeding  $m$  cannot be used in a partition of  $m$ . **b)** Because there is only one way to partition 1, namely,  $1 = 1$ , it follows that  $P_{1,n} = 1$ . Because there is only one way to partition  $m$  into 1s,  $P_{m,1} = 1$ . When  $n > m$  it follows that  $P_{m,n} = P_{m,m}$  because a number exceeding  $m$  cannot be used.  $P_{m,m} = 1 + P_{m,m-1}$  because one extra partition, namely,  $m = m$ , arises when  $m$  is allowed in the partition.  $P_{m,n} = P_{m,n-1} + P_{m-n,n}$  if  $m > n$  because a partition of  $m$  into integers not exceeding  $n$  either does not use any  $n$ s and hence, is counted in  $P_{m,n-1}$  or else uses an  $n$  and a partition of  $m - n$ , and hence, is counted in  $P_{m-n,n}$ . **c)**  $P_5 = 7$ ,  $P_6 = 11$ . **51.** Let  $P(n)$  be " $A(n, 2) = 4$ ." **Basis step:**  $P(1)$  is true because  $A(1, 2) = A(0, A(1, 1)) = A(0, 2) = 2 \cdot 2 = 4$ . **Inductive step:** Assume that  $P(n)$  is true, that is,  $A(n, 2) = 4$ . Then  $A(n+1, 2) = A(n, A(n+1, 1)) = A(n, 2) = 4$ . **53. a)** 16 **b)** 65,536 **55.** Use a double induction argument to prove the stronger statement:  $A(m, k) > A(m, l)$  when  $k > l$ . **Basis step:** When  $m = 0$  the statement is true because  $k > l$  implies that  $A(0, k) = 2k > 2l = A(0, l)$ . **Inductive step:** Assume that  $A(m, x) > A(m, y)$  for all nonnegative integers  $x$  and  $y$  with  $x > y$ . We will show that this implies that  $A(m+1, k) > A(m+1, l)$  if  $k > l$ . **Basis steps:** When  $l = 0$  and  $k > 0$ ,  $A(m+1, l) = 0$  and either  $A(m+1, k) = 2$  or  $A(m+1, k) = A(m, A(m+1, k-1))$ . If  $m = 0$ , this is  $2A(1, k-1) = 2^k$ . If  $m > 0$ , this is greater than 0 by the inductive hypothesis. In all cases,  $A(m+1, k) > 0$ , and in fact,  $A(m+1, k) \geq 2$ . If  $l = 1$  and  $k > 1$ , then  $A(m+1, l) = 2$  and  $A(m+1, k) = A(m, A(m+1, k-1))$ , with  $A(m+1, k-1) \geq 2$ . Hence, by the inductive hypothesis,  $A(m, A(m+1, k-1)) \geq A(m, 2) > A(m, 1) = 2$ . **Inductive step:** Assume that  $A(m+1, r) > A(m+1, s)$  for all  $r > s$ ,  $s = 0, 1, \dots, l$ . Then if  $k+1 > l+1$  it follows that  $A(m+1, k+1) = A(m, A(m+1, k)) > A(m, A(m+1, l)) = A(m+1, l+1)$ . **57.** From Exercise 56 it follows that  $A(i, j) \geq A(i-1, j) \geq \dots \geq A(0, j) = 2j \geq j$ . **59.** Let  $P(n)$  be " $F(n)$  is

well defined." Then  $P(0)$  is true because  $F(0)$  is specified. Assume that  $P(k)$  is true for all  $k < n$ . Then  $F(n)$  is well defined at  $n$  because  $F(n)$  is given in terms of  $F(0), F(1), \dots, F(n-1)$ . So  $P(n)$  is true for all integers  $n$ . **61. a)** The value of  $F(1)$  is ambiguous. **b)**  $F(2)$  is not defined because  $F(0)$  is not defined. **c)**  $F(3)$  is ambiguous and  $F(4)$  is not defined because  $F(\frac{4}{3})$  makes no sense. **d)** The definition of  $F(1)$  is ambiguous because both the second and third clause seem to apply. **e)**  $F(2)$  cannot be computed because trying to compute  $F(2)$  gives  $F(2) = 1 + F(F(1)) = 1 + F(2)$ . **63. a)** 1 **b)** 2 **c)** 3 **d)** 3 **e)** 4 **f)** 4 **g)** 5 **65.**  $f_0^*(n) = \lceil n/a \rceil$  **67.**  $f_2^*(n) = \lceil \log \log n \rceil$  for  $n \geq 2$ ,  $f_2^*(1) = 0$

## Section 5.4

**1.** First, we use the recursive step to write  $5! = 5 \cdot 4!$ . We then use the recursive step repeatedly to write  $4! = 4 \cdot 3!$ ,  $3! = 3 \cdot 2!$ ,  $2! = 2 \cdot 1!$ , and  $1! = 1 \cdot 0!$ . Inserting the value of  $0! = 1$ , and working back through the steps, we see that  $1! = 1 \cdot 1 = 1$ ,  $2! = 2 \cdot 1! = 2 \cdot 1 = 2$ ,  $3! = 3 \cdot 2! = 3 \cdot 2 = 6$ ,  $4! = 4 \cdot 3! = 4 \cdot 6 = 24$ , and  $5! = 5 \cdot 4! = 5 \cdot 24 = 120$ . **3.** With this input, the algorithm uses the **else** clause to find that  $\gcd(8, 13) = \gcd(13 \bmod 8, 8) = \gcd(5, 8)$ . It uses this clause again to find that  $\gcd(5, 8) = \gcd(8 \bmod 5, 5) = \gcd(3, 5)$ , then to get  $\gcd(3, 5) = \gcd(5 \bmod 3, 3) = \gcd(2, 3)$ , then  $\gcd(2, 3) = \gcd(3 \bmod 2, 2) = \gcd(1, 2)$ , and once more to get  $\gcd(1, 2) = \gcd(2 \bmod 1, 1) = \gcd(0, 1)$ . Finally, to find  $\gcd(0, 1)$  it uses the first step with  $a = 0$  to find that  $\gcd(0, 1) = 1$ . Consequently, the algorithm finds that  $\gcd(8, 13) = 1$ . **5.** First, because  $n = 11$  is odd, we use the **else** clause to see that  $\text{mpower}(3, 11, 5) = (\text{mpower}(3, 5, 5))^2 \bmod 5 \cdot 3 \bmod 5 \bmod 5$ . We next use the **else** clause again to see that  $\text{mpower}(3, 5, 5) = (\text{mpower}(3, 2, 5))^2 \bmod 5 \cdot 3 \bmod 5 \bmod 5$ . Then we use the **else if** clause to see that  $\text{mpower}(3, 2, 5) = \text{mpower}(3, 1, 5)^2 \bmod 5$ . Using the **else** clause again, we have  $\text{mpower}(3, 1, 5) = (\text{mpower}(3, 0, 5))^2 \bmod 5 \cdot 3 \bmod 5 \bmod 5$ . Finally, using the **if** clause, we see that  $\text{mpower}(3, 0, 5) = 1$ . Working backward it follows that  $\text{mpower}(3, 1, 5) = (1^2 \bmod 5 \cdot 3 \bmod 5) \bmod 5 = 3$ ,  $\text{mpower}(3, 2, 5) = 3^2 \bmod 5 = 4$ ,  $\text{mpower}(3, 5, 5) = (4^2 \bmod 5 \cdot 3 \bmod 5) \bmod 5 = 3$ , and finally  $\text{mpower}(3, 11, 5) = (3^2 \bmod 5 \cdot 3 \bmod 5) \bmod 5 = 2$ . We conclude that  $3^{11} \bmod 5 = 2$ .

**7. procedure** *mult*( $n$ : positive integer,  $x$ : integer)

**if**  $n = 1$  **then return**  $x$

**else return**  $x + \text{mult}(n-1, x)$

**9. procedure** *sum of odds*( $n$ : positive integer)

**if**  $n = 1$  **then return** 1

**else return** *sum of odds*( $n-1$ ) +  $2n-1$

**11. procedure** *smallest*( $a_1, \dots, a_n$ : integers)

**if**  $n = 1$  **then return**  $a_1$

**else return**

$\min(\text{smallest}(a_1, \dots, a_{n-1}), a_n)$

**13. procedure** *modfactorial*( $n, m$ : positive integers)

**if**  $n = 1$  **then return** 1



**else return**

$(n \cdot \text{modfactorial}(n-1, m)) \bmod m$

**15. procedure** *gcd*( $a, b$ : nonnegative integers)

$\{a < b$  assumed to hold $\}$

**if**  $a = 0$  **then return**  $b$

**else if**  $a = b - a$  **then return**  $a$

**else if**  $a < b - a$  **then return** *gcd*( $a, b - a$ )

**else return** *gcd*( $b - a, a$ )

**17. procedure** *multiply*( $x, y$ : nonnegative integers)

**if**  $y = 0$  **then return** 0

**else if**  $y$  is even **then**

**return**  $2 \cdot \text{multiply}(x, y/2)$

**else return**  $2 \cdot \text{multiply}(x, (y-1)/2) + x$

**19.** We use strong induction on  $a$ . *Basis step:* If  $a = 0$ , we know that  $\text{gcd}(0, b) = b$  for all  $b > 0$ , and that is precisely what the **if** clause does. *Inductive step:* Fix  $k > 0$ , assume the inductive hypothesis—that the algorithm works correctly for all values of its first argument less than  $k$ —and consider what happens with input  $(k, b)$ , where  $k < b$ . Because  $k > 0$ , the **else** clause is executed, and the answer is whatever the algorithm gives as output for inputs  $(b \bmod k, k)$ . Because  $b \bmod k < k$ , the input pair is valid. By our inductive hypothesis, this output is in fact  $\text{gcd}(b \bmod k, k)$ , which equals  $\text{gcd}(k, b)$  by Lemma 1 in Section 4.3. **21.** If  $n = 1$ , then  $nx = x$ , and the algorithm correctly returns  $x$ . Assume that the algorithm correctly computes  $kx$ . To compute  $(k+1)x$  it recursively computes the product of  $k+1-1 = k$  and  $x$ , and then adds  $x$ . By the inductive hypothesis, it computes that product correctly, so the answer returned is  $kx + x = (k+1)x$ , which is correct.

**23. procedure** *square*( $n$ : nonnegative integer)

**if**  $n = 0$  **then return** 0

**else return**  $\text{square}(n-1) + 2(n-1) + 1$

Let  $P(n)$  be the statement that this algorithm correctly computes  $n^2$ . Because  $0^2 = 0$ , the algorithm works correctly (using the **if** clause) if the input is 0. Assume that the algorithm works correctly for input  $k$ . Then for input  $k+1$ , it gives as output (because of the **else** clause) its output when the input is  $k$ , plus  $2(k+1-1) + 1$ . By the inductive hypothesis, its output at  $k$  is  $k^2$ , so its output at  $k+1$  is  $k^2 + 2(k+1-1) + 1 = k^2 + 2k + 1 = (k+1)^2$ , as desired.

**25.**  $n$  multiplications versus  $2^n$     **27.**  $O(\log n)$  versus  $n$

**29. procedure** *a*( $n$ : nonnegative integer)

**if**  $n = 0$  **then return** 1

**else if**  $n = 1$  **then return** 2

**else return**  $a(n-1) \cdot a(n-2)$

**31.** Iterative

**33. procedure** *iterative*( $n$ : nonnegative integer)

**if**  $n = 0$  **then**  $z := 1$

**else if**  $n = 1$  **then**  $z := 2$

**else**

$x := 1$

$y := 2$

$z := 3$

**for**  $i := 1$  **to**  $n-2$

$w := x + y + z$

$x := y$

$y := z$

$z := w$

**return**  $z$   $\{z$  is the  $n$ th term of the sequence $\}$

**35.** We first give a recursive procedure and then an iterative procedure.

**procedure** *r*( $n$ : nonnegative integer)

**if**  $n < 3$  **then return**  $2n + 1$

**else return**  $r(n-1) \cdot (r(n-2))^2 \cdot (r(n-3))^3$

**procedure** *i*( $n$ : nonnegative integer)

**if**  $n = 0$  **then**  $z := 1$

**else if**  $n = 1$  **then**  $z := 3$

**else**

$x := 1$

$y := 3$

$z := 5$

**for**  $i := 1$  **to**  $n-2$

$w := z \cdot y^2 \cdot x^3$

$x := y$

$y := z$

$z := w$

**return**  $z$   $\{z$  is the  $n$ th term of the sequence $\}$

The iterative version is more efficient.

**37. procedure** *reverse*( $w$ : bit string)

$n := \text{length}(w)$

**if**  $n \leq 1$  **then return**  $w$

**else return**

$\text{substr}(w, n, n) \text{reverse}(\text{substr}(w, 1, n-1))$

$\{\text{substr}(w, a, b)$  is the substring of  $w$  consisting of the symbols in the  $a$ th through  $b$ th positions $\}$

**39.** The procedure correctly gives the reversal of  $\lambda$  as  $\lambda$  (basis step), and because the reversal of a string consists of its last character followed by the reversal of its first  $n-1$  characters (see Exercise 37 in Section 5.3), the algorithm behaves correctly when  $n > 0$  by the inductive hypothesis. **41.** The algorithm implements the idea of Example 14 in Section 5.1. If  $n = 1$  (basis step), place the one right triomino so that its armpit corresponds to the hole in the  $2 \times 2$  board. If  $n > 1$ , then divide the board into four boards, each of size  $2^{n-1} \times 2^{n-1}$ , notice which quarter the hole occurs in, position one right triomino at the center of the board with its armpit in the quarter where the missing square is (see Figure 7 in Section 5.1), and invoke the algorithm recursively four times—once on each of the  $2^{n-1} \times 2^{n-1}$  boards, each of which has one square missing (either because it was missing to begin with, or because it is covered by the central triomino).

**43. procedure** *A*( $m, n$ : nonnegative integers)

**if**  $m = 0$  **then return**  $2n$

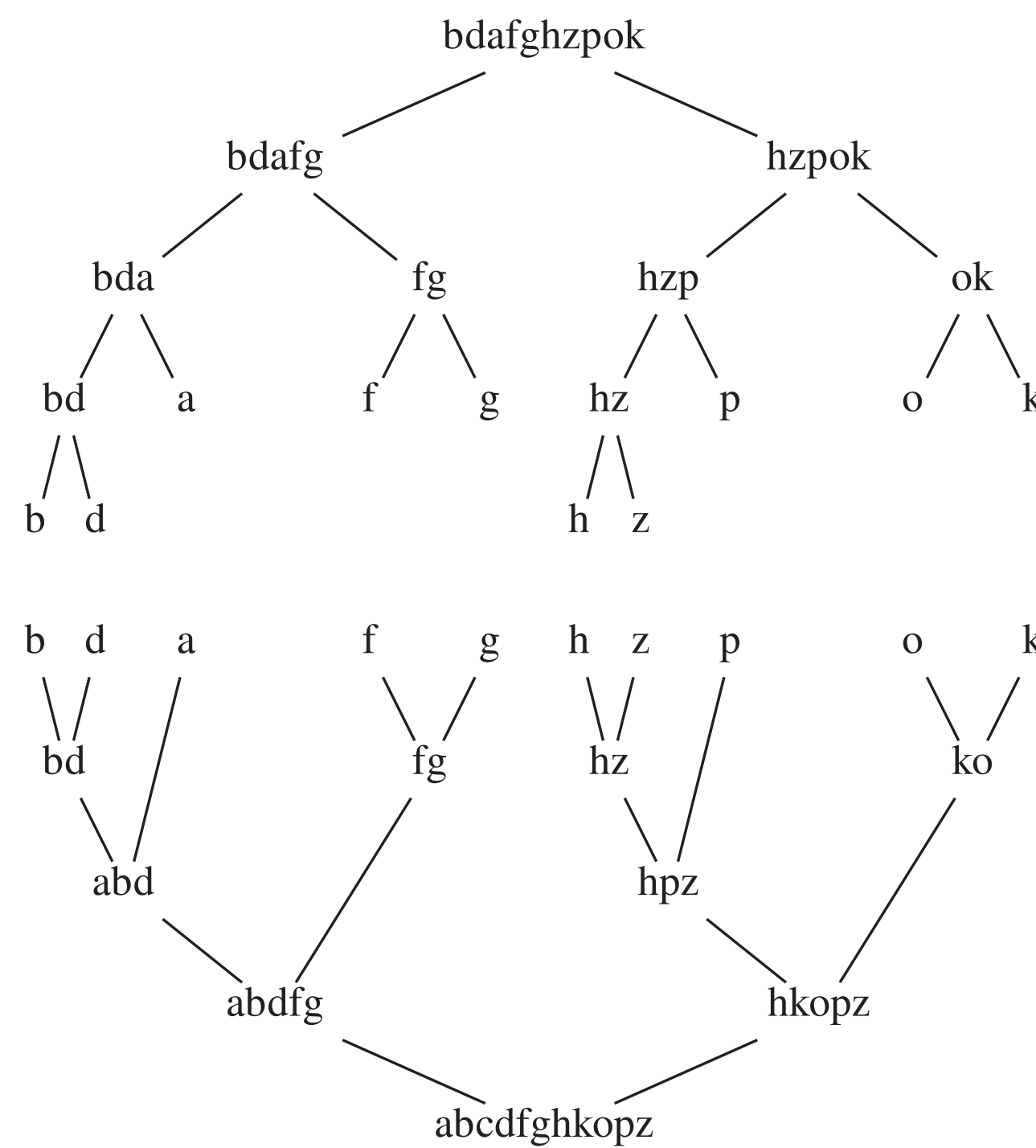
**else if**  $n = 0$  **then return** 0

**else if**  $n = 1$  **then return** 2

**else return**  $A(m-1, A(m, n-1))$



45.



47. Let the two lists be  $1, 2, \dots, m-1, m+n-1$  and  $m, m+1, \dots, m+n-2, m+n$ , respectively. 49. If  $n = 1$ , then the algorithm does nothing, which is correct because a list with one element is already sorted. Assume that the algorithm works correctly for  $n = 1$  through  $n = k$ . If  $n = k+1$ , then the list is split into two lists,  $L_1$  and  $L_2$ . By the inductive hypothesis, *mergesort* correctly sorts each of these sublists; furthermore, *merge* correctly merges two sorted lists into one because with each comparison the smallest element in  $L_1 \cup L_2$  not yet put into  $L$  is put there. 51.  $O(n)$  53. 6 55.  $O(n^2)$

## Section 5.5

1. Suppose that  $x = 0$ . The program segment first assigns the value 1 to  $y$  and then assigns the value  $x + y = 0 + 1 = 1$  to  $z$ . 3. Suppose that  $y = 3$ . The program segment assigns the value 2 to  $x$  and then assigns the value  $x + y = 2 + 3 = 5$  to  $z$ . Because  $y = 3 > 0$  it then assigns the value  $z + 1 = 5 + 1 = 6$  to  $z$ .

5. 
$$\frac{(p \wedge \text{condition1})\{S_1\}q \quad (p \wedge \neg \text{condition1} \wedge \text{condition2})\{S_2\}q \quad \vdots \quad (p \wedge \neg \text{condition1} \wedge \neg \text{condition2} \quad \cdots \wedge \neg \text{condition}(n-1))\{S_n\}q}{\therefore p\{\text{if condition1 then } S_1; \quad \text{else if condition2 then } S_2; \dots; \text{else } S_n\}q}$$

7. We will show that  $p = \text{"power} = x^{i-1} \text{ and } i \leq n+1\text{"}$  is a loop invariant. Note that  $p$  is true initially, because before the loop starts,  $i = 1$  and  $\text{power} = 1 = x^0 = x^{1-1}$ . Next, we must show that if  $p$  is true and  $i \leq n$  after an execution of the loop, then  $p$  remains true after one more execution. The loop increments  $i$  by 1. Hence, because  $i \leq n$  before this pass,  $i \leq n+1$  after this pass. Also the loop assigns  $\text{power} \cdot x$  to  $\text{power}$ . By the

inductive hypothesis we see that  $\text{power}$  is assigned the value  $x^{i-1} \cdot x = x^i$ . Hence,  $p$  remains true. Furthermore, the loop terminates after  $n$  traversals of the loop with  $i = n+1$  because  $i$  is assigned the value 1 prior to entering the loop, is incremented by 1 on each pass, and the loop terminates when  $i > n$ . Consequently, at termination  $\text{power} = x^n$ , as desired.

9. Suppose that  $p$  is " $m$  and  $n$  are integers." Then if the condition  $n < 0$  is true,  $a = -n = |n|$  after  $S_1$  is executed. If the condition  $n < 0$  is false, then  $a = n = |n|$  after  $S_1$  is executed. Hence,  $p\{S_1\}q$  is true where  $q$  is  $p \wedge (a = |n|)$ . Because  $S_2$  assigns the value 0 to both  $k$  and  $x$ , it is clear that  $q\{S_2\}r$  is true where  $r$  is  $q \wedge (k = 0) \wedge (x = 0)$ . Suppose that  $r$  is true. Let  $P(k)$  be " $x = mk$  and  $k \leq a$ ." We can show that  $P(k)$  is a loop invariant for the loop in  $S_3$ .  $P(0)$  is true because before the loop is entered  $x = 0 = m \cdot 0$  and  $0 \leq a$ . Now assume  $P(k)$  is true and  $k < a$ . Then  $P(k+1)$  is true because  $x$  is assigned the value  $x + m = mk + m = m(k+1)$ . The loop terminates when  $k = a$ , and at that point  $x = ma$ . Hence,  $r\{S_3\}s$  is true where  $s$  is " $a = |n|$  and  $x = ma$ ." Now assume that  $s$  is true. Then if  $n < 0$  it follows that  $a = -n$ , so  $x = -mn$ . In this case  $S_4$  assigns  $-x = mn$  to *product*. If  $n > 0$  then  $x = ma = mn$ , so  $S_4$  assigns  $mn$  to *product*. Hence,  $s\{S_4\}t$  is true.

11. Suppose that the initial assertion  $p$  is true. Then because  $p\{S\}q_0$  is true,  $q_0$  is true after the segment  $S$  is executed. Because  $q_0 \rightarrow q_1$  is true, it also follows that  $q_1$  is true after  $S$  is executed. Hence,  $p\{S\}q_1$  is true. 13. We will use the proposition  $p$ , " $\text{gcd}(a, b) = \text{gcd}(x, y)$  and  $y \geq 0$ ," as the loop invariant. Note that  $p$  is true before the loop is entered, because at that point  $x = a$ ,  $y = b$ , and  $y$  is a positive integer, using the initial assertion. Now assume that  $p$  is true and  $y > 0$ ; then the loop will be executed again. Inside the loop,  $x$  and  $y$  are replaced by  $y$  and  $x \bmod y$ , respectively. By Lemma 1 of Section 4.3,  $\text{gcd}(x, y) = \text{gcd}(y, x \bmod y)$ . Therefore, after execution of the loop, the value of  $\text{gcd}(x, y)$  is the same as it was before. Moreover, because  $y$  is the remainder, it is at least 0. Hence,  $p$  remains true, so it is a loop invariant. Furthermore, if the loop terminates, then  $y = 0$ . In this case, we have  $\text{gcd}(x, y) = x$ , the final assertion. Therefore, the program, which gives  $x$  as its output, has correctly computed  $\text{gcd}(a, b)$ . Finally, we can prove the loop must terminate, because each iteration causes the value of  $y$  to decrease by at least 1. Therefore, the loop can be iterated at most  $b$  times.

## Supplementary Exercises

1. Let  $P(n)$  be the statement that this equation holds. *Basis step*:  $P(1)$  says  $2/3 = 1 - (1/3^1)$ , which is true. *Inductive step*: Assume that  $P(k)$  is true. Then  $2/3 + 2/9 + 2/27 + \dots + 2/3^n + 2/3^{n+1} = 1 - 1/3^n + 2/3^{n+1}$  (by the inductive hypothesis), and this equals  $1 - 1/3^{n+1}$ , as desired. 3. Let  $P(n)$  be " $1 \cdot 1 + 2 \cdot 2 + \dots + n \cdot 2^{n-1} = (n-1)2^n + 1$ ." *Basis step*:  $P(1)$  is true because  $1 \cdot 1 = 1 = (1-1)2^1 + 1$ . *Inductive step*: Assume that  $P(k)$  is true. Then  $1 \cdot 1 + 2 \cdot 2 + \dots + k \cdot 2^{k-1} + (k+1) \cdot 2^k = (k-1)2^k + 1 + (k+1)2^k = 2k \cdot 2^k + 1 = [(k+1)-1]2^{k+1} + 1$ . 5. Let  $P(n)$  be " $1/(1 \cdot 4) + \dots + 1/[(3n-2)(3n+1)] = n/(3n+1)$ ." *Basis step*:  $P(1)$  is true because  $1/(1 \cdot 4) = 1/4$ . *Inductive*



*step:* Assume  $P(k)$  is true. Then  $1/(1 \cdot 4) + \cdots + 1/[(3k - 2)(3k + 1)] + 1/[(3k + 1)(3k + 4)] = k/(3k + 1) + 1/[(3k + 1)(3k + 4)] = [k(3k + 4) + 1]/[(3k + 1)(3k + 4)] = [(3k + 1)(k + 1)]/[(3k + 1)(3k + 4)] = (k + 1)/(3k + 4)$ .

**7.** Let  $P(n)$  be “ $2^n > n^3$ .” *Basis step:*  $P(10)$  is true because  $1024 > 1000$ . *Inductive step:* Assume  $P(k)$  is true. Then  $(k + 1)^3 = k^3 + 3k^2 + 3k + 1 \leq k^3 + 9k^2 \leq k^3 + k^3 = 2k^3 < 2 \cdot 2^k = 2^{k+1}$ .

**9.** Let  $P(n)$  be “ $a - b$  is a factor of  $a^n - b^n$ .” *Basis step:*  $P(1)$  is trivially true. Assume  $P(k)$  is true. Then  $a^{k+1} - b^{k+1} = a^{k+1} - ab^k + ab^k - b^{k+1} = a(a^k - b^k) + b^k(a - b)$ . Then because  $a - b$  is a factor of  $a^k - b^k$  and  $a - b$  is a factor of  $a - b$ , it follows that  $a - b$  is a factor of  $a^{k+1} - b^{k+1}$ .

**11.** *Basis step:* When  $n = 1$ ,  $6^{n+1} + 7^{2n-1} = 36 + 7 = 43$ . *Inductive step:* Assume the inductive hypothesis, that 43 divides  $6^{n+1} + 7^{2n-1}$ ; we must show that 43 divides  $6^{n+2} + 7^{2n+1}$ . We have  $6^{n+2} + 7^{2n+1} = 6 \cdot 6^{n+1} + 49 \cdot 7^{2n-1} = 6 \cdot 6^{n+1} + 6 \cdot 7^{2n-1} + 43 \cdot 7^{2n-1} = 6(6^{n+1} + 7^{2n-1}) + 43 \cdot 7^{2n-1}$ . By the inductive hypothesis the first term is divisible by 43, and the second term is divisible by 43; therefore, the sum is divisible by 43.

**13.** Let  $P(n)$  be “ $a + (a + d) + \cdots + (a + nd) = (n + 1)(2a + nd)/2$ .” *Basis step:*  $P(1)$  is true because  $a + (a + d) = 2a + d = 2(2a + d)/2$ . *Inductive step:* Assume that  $P(k)$  is true. Then  $a + (a + d) + \cdots + (a + kd) + [a + (k + 1)d] = (k + 1)(2a + kd)/2 + a + (k + 1)d = \frac{1}{2}(2ak + 2a + k^2d + kd + 2a + 2kd + 2d) = \frac{1}{2}(2ak + 4a + k^2d + 3kd + 2d) = \frac{1}{2}(k + 2)[2a + (k + 1)d]$ .

**15.** *Basis step:* This is true for  $n = 1$  because  $5/6 = 10/12$ . *Inductive step:* Assume that the equation holds for  $n = k$ , and consider  $n = k + 1$ . Then  $\sum_{i=1}^{k+1} \frac{i+4}{i(i+1)(i+2)} = \sum_{i=1}^k \frac{i+4}{i(i+1)(i+2)} + \frac{k+5}{(k+1)(k+2)(k+3)} = \frac{k(3k+7)}{2(k+1)(k+2)} + \frac{(k+1)(k+2)(k+3)}{(k+1)(k+2)(k+3)} = \frac{1}{(k+1)(k+2)} \cdot \left( \frac{k(3k+7)}{2} + \frac{k+5}{k+3} \right) = \frac{1}{2(k+1)(k+2)(k+3)} \cdot [k(3k+7)(k+3) + 2(k+5)] = \frac{1}{2(k+1)(k+2)(k+3)} \cdot (3k^3 + 16k^2 + 23k + 10) = \frac{1}{2(k+1)(k+2)(k+3)} \cdot (3k+10)(k+1)^2 = \frac{1}{2(k+2)(k+3)} \cdot (3k+10)(k+1) = \frac{(k+1)(3(k+1)+7)}{2((k+1)+1)((k+1)+2)}$ , as desired.

**17.** *Basis step:* The statement is true for  $n = 1$  because the derivative of  $g(x) = xe^x$  is  $x \cdot e^x + e^x = (x + 1)e^x$  by the product rule. *Inductive step:* Assume that the statement is true for  $n = k$ , i.e., the  $k$ th derivative is given by  $g^{(k)} = (x + k)e^x$ . Differentiating by the product rule gives the  $(k + 1)$ st derivative:  $g^{(k+1)} = (x + k)e^x + e^x = [x + (k + 1)]e^x$ , as desired.

**19.** We will use strong induction to show that  $f_n$  is even if  $n \equiv 0 \pmod{3}$  and is odd otherwise. *Basis step:* This follows because  $f_0 = 0$  is even and  $f_1 = 1$  is odd. *Inductive step:* Assume that if  $j \leq k$ , then  $f_j$  is even if  $j \equiv 0 \pmod{3}$  and is odd otherwise. Now suppose  $k + 1 \equiv 0 \pmod{3}$ . Then  $f_{k+1} = f_k + f_{k-1}$  is even because  $f_k$  and  $f_{k-1}$  are both odd. If  $k + 1 \equiv 1 \pmod{3}$ , then  $f_{k+1} = f_k + f_{k-1}$  is odd because  $f_k$  is even and  $f_{k-1}$  is odd. Finally, if  $k + 1 \equiv 2 \pmod{3}$ , then  $f_{k+1} = f_k + f_{k-1}$  is odd because  $f_k$  is odd and  $f_{k-1}$  is even.

**21.** Let  $P(n)$  be the statement that  $f_k f_n + f_{k+1} f_{n+1} = f_{n+k+1}$  for every nonnegative integer  $k$ . *Basis step:* This consists of showing that  $P(0)$  and  $P(1)$  both hold.  $P(0)$  is true because  $f_k f_0 + f_{k+1} f_1 = f_{k+1} \cdot 0 + f_{k+1} \cdot 1 = f_1$ . Because  $f_k f_1 + f_{k+1} f_2 = f_k + f_{k+1} = f_{k+2}$ , it follows that  $P(1)$  is true. *Inductive step:* Now assume that  $P(j)$  holds. Then, by the inductive hypoth-

esis and the recursive definition of the Fibonacci numbers, it follows that  $f_{k+1} f_{j+1} + f_{k+2} f_{j+2} = f_k (f_{j-1} + f_j) + f_{k+1} (f_j + f_{j+1}) = (f_k f_{j-1} + f_{k+1} f_j) + (f_k f_j + f_{k+1} f_{j+1}) = f_{j-1+k+1} + f_{j+k+1} = f_{j+k+2}$ . This shows that  $P(j + 1)$  is true.

**23.** Let  $P(n)$  be the statement  $l_0^2 + l_1^2 + \cdots + l_n^2 = l_n l_{n+1} + 2$ . *Basis step:*  $P(0)$  and  $P(1)$  both hold because  $l_0^2 = 2^2 = 2 \cdot 1 + 2 = l_0 l_1 + 2$  and  $l_0^2 + l_1^2 = 2^2 + 1^2 = 1 \cdot 3 + 2 = l_1 l_3 + 2$ . *Inductive step:* Assume that  $P(k)$  holds. Then by the inductive hypothesis  $l_0^2 + l_1^2 + \cdots + l_k^2 + l_{k+1}^2 = l_k l_{k+1} + 2 + l_{k+1}^2 = l_{k+1} (l_k + l_{k+1}) + 2 = l_{k+1} l_{k+2} + 2$ . This shows that  $P(k + 1)$  holds.

**25.** Let  $P(n)$  be the statement that the identity holds for the integer  $n$ . *Basis step:*  $P(1)$  is obviously true. *Inductive step:* Assume that  $P(k)$  is true. Then  $\cos((k + 1)x) + i \sin((k + 1)x) = \cos(kx + x) + i \sin(kx + x) = \cos kx \cos x - \sin kx \sin x + i(\sin kx \cos x + \cos kx \sin x) = \cos x(\cos kx + i \sin kx)(\cos x + i \sin x) = (\cos x + i \sin x)^k (\cos x + i \sin x) = (\cos x + i \sin x)^{k+1}$ . It follows that  $P(k + 1)$  is true.

**27.** Rewrite the right-hand side as  $2^{n+1}(n^2 - 2n + 3) - 6$ . For  $n = 1$  we have  $2 = 4 \cdot 2 - 6$ . Assume that the equation holds for  $n = k$ , and consider  $n = k + 1$ . Then  $\sum_{j=1}^{k+1} j^2 2^j = \sum_{j=1}^k j^2 2^j + (k + 1)^2 2^{k+1} = 2^{k+1}(k^2 - 2k + 3) - 6 + (k^2 + 2k + 1)2^{k+1}$  (by the inductive hypothesis)  $= 2^{k+1}(2k^2 + 4) - 6 = 2^{k+2}(k^2 + 2) - 6 = 2^{k+2}[(k + 1)^2 - 2(k + 1) + 3] - 6$ .

**29.** Let  $P(n)$  be the statement that this equation holds. *Basis step:* In  $P(2)$  both sides reduce to  $1/3$ . *Inductive step:* Assume that  $P(k)$  is true. Then  $\sum_{j=1}^{k+1} 1/(j^2 - 1) = \left( \sum_{j=1}^k 1/(j^2 - 1) \right) + 1/[(k + 1)^2 - 1] = (k - 1)(3k + 2)/[4k(k + 1)] + 1/[(k + 1)^2 - 1]$  by the inductive hypothesis. This simplifies to  $(k - 1)(3k + 2)/[4k(k + 1)] + 1/(k^2 + 2k) = (3k^3 + 5k^2)/[4k(k + 1)(k + 2)] = \{[(k + 1) - 1][3(k + 1) + 2]\}/[4(k + 1)(k + 2)]$ , which is exactly what  $P(k + 1)$  asserts.

**31.** Let  $P(n)$  be the assertion that at least  $n + 1$  lines are needed to cover the lattice points in the given triangular region. *Basis step:*  $P(0)$  is true, because we need at least one line to cover the one point at  $(0, 0)$ . *Inductive step:* Assume the inductive hypothesis, that at least  $k + 1$  lines are needed to cover the lattice points with  $x \geq 0$ ,  $y \geq 0$ , and  $x + y \leq k$ . Consider the triangle of lattice points defined by  $x \geq 0$ ,  $y \geq 0$ , and  $x + y \leq k + 1$ . By way of contradiction, assume that  $k + 1$  lines could cover this set. Then these lines must cover the  $k + 2$  points on the line  $x + y = k + 1$ . But only the line  $x + y = k + 1$  itself can cover more than one of these points, because two distinct lines intersect in at most one point. Therefore, none of the  $k + 1$  lines that are needed (by the inductive hypothesis) to cover the set of lattice points within the triangle but not on this line can cover more than one of the points on this line, and this leaves at least one point uncovered. Therefore, our assumption that  $k + 1$  lines could cover the larger set is wrong, and our proof is complete.

**33.** Let  $P(n)$  be  $\mathbf{B}^k = \mathbf{M}\mathbf{A}^k\mathbf{M}^{-1}$ . *Basis step:* Part of the given conditions. *Inductive step:* Assume the inductive hypothesis. Then  $\mathbf{B}^{k+1} = \mathbf{B}\mathbf{B}^k = \mathbf{M}\mathbf{A}\mathbf{M}^{-1}\mathbf{B}^k = \mathbf{M}\mathbf{A}\mathbf{M}^{-1}\mathbf{M}\mathbf{A}^k\mathbf{M}^{-1}$  (by the inductive hypothesis)  $= \mathbf{M}\mathbf{A}\mathbf{I}\mathbf{A}^k\mathbf{M}^{-1} = \mathbf{M}\mathbf{A}\mathbf{A}^k\mathbf{M}^{-1} = \mathbf{M}\mathbf{A}^{k+1}\mathbf{M}^{-1}$ .

**35.** We prove by mathematical induction the following stronger statement: For every  $n \geq 3$ , we can write  $n!$  as the sum of  $n$  of its distinct positive divisors, one of which



is 1. That is, we can write  $n! = a_1 + a_2 + \cdots + a_n$ , where each  $a_i$  is a divisor of  $n!$ , the divisors are listed in strictly decreasing order, and  $a_n = 1$ . *Basis step:*  $3! = 3 + 2 + 1$ . *Inductive step:* Assume that we can write  $k!$  as a sum of the desired form, say  $k! = a_1 + a_2 + \cdots + a_k$ , where each  $a_i$  is a divisor of  $n!$ , the divisors are listed in strictly decreasing order, and  $a_n = 1$ . Consider  $(k+1)!$ . Then we have  $(k+1)! = (k+1)k! = (k+1)(a_1 + a_2 + \cdots + a_k) = (k+1)a_1 + (k+1)a_2 + \cdots + (k+1)a_k = (k+1)a_1 + (k+1)a_2 + \cdots + k \cdot a_k + a_k$ . Because each  $a_i$  was a divisor of  $k!$ , each  $(k+1)a_i$  is a divisor of  $(k+1)!$ . Furthermore,  $k \cdot a_k = k$ , which is a divisor of  $(k+1)!$ , and  $a_k = 1$ , so the new last summand is again 1. (Notice also that our list of summands is still in strictly decreasing order.) Thus, we have written  $(k+1)!$  in the desired form. **37.** When  $n = 1$  the statement is vacuously true. Assume that the statement is true for  $n = k$ , and consider  $k+1$  people standing in a line, with a woman first and a man last. If the  $k$ th person is a woman, then we have that woman standing in front of the man at the end. If the  $k$ th person is a man, then the first  $k$  people in line satisfy the conditions of the inductive hypothesis for the first  $k$  people in line, so again we can conclude that there is a woman directly in front of a man somewhere in the line. **39.** *Basis step:* When  $n = 1$  there is one circle, and we can color the inside blue and the outside red to satisfy the conditions. *Inductive step:* Assume the inductive hypothesis that if there are  $k$  circles, then the regions can be 2-colored such that no regions with a common boundary have the same color, and consider a situation with  $k+1$  circles. Remove one of the circles, producing a picture with  $k$  circles, and invoke the inductive hypothesis to color it in the prescribed manner. Then replace the removed circle and change the color of every region inside this circle. The resulting figure satisfies the condition, because if two regions have a common boundary, then either that boundary involved the new circle, in which case the regions on either side used to be the same region and now the inside portion is different from the outside, or else the boundary did not involve the new circle, in which case the regions are colored differently because they were colored differently before the new circle was restored.

**41.** If  $n = 1$  then the equation reads  $1 \cdot 1 = 1 \cdot 2/2$ , which is true. Assume that the equation is true for  $n$  and consider it for  $n+1$ . Then  $\sum_{j=1}^{n+1} (2j-1) \left( \sum_{k=j}^{n+1} \frac{1}{k} \right) = \sum_{j=1}^n (2j-1) \left( \sum_{k=j}^{n+1} \frac{1}{k} \right) + [2(n+1)-1] \cdot \frac{1}{n+1} = \sum_{j=1}^n (2j-1) \left( \frac{1}{n+1} + \sum_{k=j}^n \frac{1}{k} \right) + \frac{2n+1}{n+1} = \left( \frac{1}{n+1} \sum_{j=1}^n (2j-1) \right) + \left( \sum_{j=1}^n (2j-1) \sum_{k=j}^n \frac{1}{k} \right) + \frac{2n+1}{n+1} = \left( \frac{1}{n+1} \cdot n^2 \right) + \frac{n(n+1)}{2} + \frac{2n+1}{n+1}$  (by the inductive hypothesis)  $= \frac{2n^2 + n(n+1)^2 + (4n+2)}{2(n+1)} = \frac{2(n+1)^2 + n(n+1)^2}{2(n+1)} = \frac{(n+1)(n+2)}{2}$ .

**43.** Let  $T(n)$  be the statement that the sequence of towers of 2 is eventually constant modulo  $n$ . We use strong induction to prove that  $T(n)$  is true for all positive integers  $n$ . *Basis step:* When  $n = 1$  (and  $n = 2$ ), the sequence of towers of 2 modulo  $n$  is the sequence of all 0s. *Inductive step:* Suppose that  $k$  is an integer with  $k \geq 2$ . Suppose that  $T(j)$  is true for  $1 \leq j \leq k-1$ . In the proof of the inductive step we denote the  $r$ th term of the

sequence modulo  $n$  by  $a_r$ . First suppose  $k$  is even. Let  $k = 2^s q$  where  $s \geq 1$  and  $q < k$  is odd. When  $j$  is large enough,  $a_{j-2} \geq s$ , and for such  $j$ ,  $a_j = 2^{2^{j-2}}$  is a multiple of  $2^s$ . It follows that for sufficiently large  $j$ ,  $a_j \equiv 0 \pmod{2^s}$ . Hence, for large enough  $i$ ,  $2^s$  divides  $a_{i+1} - a_i$ . By the inductive hypothesis  $T(q)$  is true, so the sequence  $a_1, a_2, a_3, \dots$  is eventually constant modulo  $q$ . This implies that for large enough  $i$ ,  $q$  divides  $a_{i+1} - a_i$ . Because  $\gcd(q, 2^s) = 1$  and for sufficiently large  $i$  both  $q$  and  $2^s$  divide  $a_{i+1} - a_i$ ,  $k = 2^s q$  divides  $a_{i+1} - a_i$  for sufficiently large  $i$ . Hence, for sufficiently large  $i$ ,  $a_{i+1} - a_i \equiv 0 \pmod{k}$ . This means that the sequence is eventually constant modulo  $k$ . Finally, suppose  $k$  is odd. Then  $\gcd(2, k) = 1$ , so by Euler's theorem (found in elementary number theory books, such as [Ro10]), we know that  $2^{\phi(k)} \equiv 1 \pmod{k}$ . Let  $r = \phi(k)$ . Because  $r < k$ , by the inductive hypothesis  $T(r)$ , the sequence  $a_1, a_2, a_3, \dots$  is eventually constant modulo  $r$ , say equal to  $c$ . Hence, for large enough  $i$ , for some integer  $t_i$ ,  $a_i = t_i r + c$ . Hence,  $a_{i+1} = 2^{a_i} = 2^{t_i r + c} = (2^r)^{t_i} 2^c \equiv 2^c \pmod{k}$ . This shows that  $a_1, a_2, \dots$  is eventually constant modulo  $k$ . **45. a)** 92 **b)** 91 **c)** 91 **d)** 91 **e)** 91 **f)** 91

**47.** The basis step is incorrect because  $n \neq 1$  for the sum shown. **49.** Let  $P(n)$  be “the plane is divided into  $n^2 - n + 2$  regions by  $n$  circles if every two of these circles have two common points but no three have a common point.” *Basis step:*  $P(1)$  is true because a circle divides the plane into  $2 = 1^2 - 1 + 2$  regions. *Inductive step:* Assume that  $P(k)$  is true, that is,  $k$  circles with the specified properties divide the plane into  $k^2 - k + 2$  regions. Suppose that a  $(k+1)$ st circle is added. This circle intersects each of the other  $k$  circles in two points, so these points of intersection form  $2k$  new arcs, each of which splits an old region. Hence, there are  $2k$  regions split, which shows that there are  $2k$  more regions than there were previously. Hence,  $k+1$  circles satisfying the specified properties divide the plane into  $k^2 - k + 2 + 2k = (k^2 + 2k + 1) - (k+1) + 2 = (k+1)^2 - (k+1) + 2$  regions. **51.** Suppose  $\sqrt{2}$  were rational. Then  $\sqrt{2} = a/b$ , where  $a$  and  $b$  are positive integers. It follows that the set  $S = \{n\sqrt{2} \mid n \in \mathbf{N}\} \cap \mathbf{N}$  is a nonempty set of positive integers, because  $b\sqrt{2} = a$  belongs to  $S$ . Let  $t$  be the least element of  $S$ , which exists by the well-ordering property. Then  $t = s\sqrt{2}$  for some integer  $s$ . We have  $t - s = s\sqrt{2} - s = s(\sqrt{2} - 1)$ , so  $t - s$  is a positive integer because  $\sqrt{2} > 1$ . Hence,  $t - s$  belongs to  $S$ . This is a contradiction because  $t - s = s\sqrt{2} - s < s$ . Hence,  $\sqrt{2}$  is irrational.

**53. a)** Let  $d = \gcd(a_1, a_2, \dots, a_n)$ . Then  $d$  is a divisor of each  $a_i$  and so must be a divisor of  $\gcd(a_{n-1}, a_n)$ . Hence,  $d$  is a common divisor of  $a_1, a_2, \dots, a_{n-2}$ , and  $\gcd(a_{n-1}, a_n)$ . To show that it is the greatest common divisor of these numbers, suppose that  $c$  is a common divisor of them. Then  $c$  is a divisor of  $a_i$  for  $i = 1, 2, \dots, n-2$  and a divisor of  $\gcd(a_{n-1}, a_n)$ , so it is a divisor of  $a_{n-1}$  and  $a_n$ . Hence,  $c$  is a common divisor of  $a_1, a_2, \dots, a_{n-1}$ , and  $a_n$ . Hence, it is a divisor of  $d$ , the greatest common divisor of  $a_1, a_2, \dots, a_n$ . It follows that  $d$  is the greatest common divisor, as claimed. **b)** If  $n = 2$ , apply the Euclidean algorithm. Otherwise, apply the Euclidean algorithm to  $a_{n-1}$  and  $a_n$ , obtaining  $d = \gcd(a_{n-1}, a_n)$ , and then apply the algorithm recursively to  $a_1, a_2, \dots, a_{n-2}, d$ . **55.**  $f(n) = n^2$ . Let  $P(n)$  be “ $f(n) = n^2$ .” *Basis step:*  $P(1)$



is true because  $f(1) = 1 = 1^2$ , which follows from the definition of  $f$ . *Inductive step:* Assume  $f(n) = n^2$ . Then  $f(n+1) = f((n+1) - 1) + 2(n+1) - 1 = f(n) + 2n + 1 = n^2 + 2n + 1 = (n+1)^2$ . **57. a)**  $\lambda, 0, 1, 00, 01, 11, 000, 001, 011, 111, 0000, 0001, 0011, 0111, 1111, 00000, 00001, 00011, 00111, 01111, 11111$  **b)**  $S = \{\alpha\beta \mid \alpha \text{ is a string of } m \text{ 0s and } \beta \text{ is a string of } n \text{ 1s, } m \geq 0, n \geq 0\}$  **59.** Apply the first recursive step to  $\lambda$  to get  $() \in B$ . Apply the second recursive step to this string to get  $()() \in B$ . Apply the first recursive step to this string to get  $((())) \in B$ . By Exercise 62,  $((()))$  is not in  $B$  because the number of left parentheses does not equal the number of right parentheses. **61.**  $\lambda, (), (()), ()()$  **63. a)** 0 **b)** -2 **c)** 2 **d)** 0

**65.**

**procedure** *generate*( $n$ : nonnegative integer)

**if**  $n$  is odd **then**

$S := S(n-1)$  {the  $S$  constructed by *generate*( $n-1$ )}

$T := T(n-1)$  {the  $T$  constructed by *generate*( $n-1$ )}

**else if**  $n = 0$  **then**

$S := \emptyset$

$T := \{\lambda\}$

**else**

$S' := S(n-2)$  {the  $S$  constructed by *generate*( $n-2$ )}

$T' := T(n-2)$  {the  $T$  constructed by *generate*( $n-2$ )}

$T := T' \cup \{(x) \mid x \in T' \cup S' \wedge \text{length}(x) = n-2\}$

$S := S' \cup \{xy \mid x \in T' \wedge y \in T' \cup S' \wedge \text{length}(xy) = n\}$

{ $T \cup S$  is the set of balanced strings of length at most  $n$ }

**67.** If  $x \leq y$  initially, then  $x := y$  is not executed, so  $x \leq y$  is a true final assertion. If  $x > y$  initially, then  $x := y$  is executed, so  $x \leq y$  is again a true final assertion.

**69. procedure** *zerocount*( $a_1, a_2, \dots, a_n$ : list of integers)

**if**  $n = 1$  **then**

**if**  $a_1 = 0$  **then return** 1

**else return** 0

**else**

**if**  $a_n = 0$  **then return** *zerocount*( $a_1, a_2, \dots, a_{n-1}$ ) + 1

**else return** *zerocount*( $a_1, a_2, \dots, a_{n-1}$ )

**71.** We will prove that  $a(n)$  is a natural number and  $a(n) \leq n$ . This is true for the base case  $n = 0$  because  $a(0) = 0$ . Now assume that  $a(n-1)$  is a natural number and  $a(n-1) \leq n-1$ . Then  $a(a(n-1))$  is  $a$  applied to a natural number less than or equal to  $n-1$ . Hence,  $a(a(n-1))$  is also a natural number less than or equal to  $n-1$ . Therefore,  $n - a(a(n-1))$  is  $n$  minus some natural number less than or equal to  $n-1$ , which is a natural number less than or equal to  $n$ . **73.** From Exercise 72,  $a(n) = \lfloor (n+1)\mu \rfloor$  and  $a(n-1) = \lfloor n\mu \rfloor$ . Because  $\mu < 1$ , these two values are equal or they differ by 1. First suppose that  $\mu n - \lfloor \mu n \rfloor < 1 - \mu$ . This is equivalent to  $\mu(n+1) < 1 + \lfloor \mu n \rfloor$ . If this is true, then  $\lfloor \mu(n+1) \rfloor = \lfloor \mu n \rfloor$ . On the other hand, if  $\mu n - \lfloor \mu n \rfloor \geq 1 - \mu$ , then  $\mu(n+1) \geq 1 + \lfloor \mu n \rfloor$ , so  $\lfloor \mu(n+1) \rfloor = \lfloor \mu n \rfloor + 1$ , as desired. **75.**  $f(0) = 1, m(0) = 0$ ;  $f(1) = 1, m(1) = 0$ ;  $f(2) = 2, m(2) = 1$ ;  $f(3) = 2, m(3) = 2$ ;  $f(4) = 3, m(4) = 2$ ;  $f(5) = 3, m(5) = 3$ ;  $f(6) = 4, m(6) = 4$ ;  $f(7) = 5, m(7) = 4$ ;  $f(8) = 5, m(8) = 5$ ;  $f(9) = 6, m(9) = 6$  **77.** The last occurrence of  $n$  is in the position for which the total number of 1s, 2s,  $\dots$ ,  $n$ s all together is that

position number. But because  $a_k$  is the number of occurrences of  $k$ , this is just  $\sum_{k=1}^n a_k$ , as desired. Because  $f(n)$  is the sum of the first  $n$  terms of the sequence,  $f(f(n))$  is the sum of the first  $f(n)$  terms of the sequence. But because  $f(n)$  is the last term whose value is  $n$ , this means that the sum is the sum of all terms of the sequence whose value is at most  $n$ . Because there are  $a_k$  terms of the sequence whose value is  $k$ , this sum is  $\sum_{k=1}^n k \cdot a_k$ , as desired.

## CHAPTER 6

### Section 6.1

**1. a)** 5850 **b)** 343 **3. a)**  $4^{10}$  **b)**  $5^{10}$  **5.** 42 **7.**  $26^3$   
**9.** 676 **11.**  $2^8$  **13.**  $n + 1$  (counting the empty string)  
**15.** 475,255 (counting the empty string) **17.** 1,321,368,961  
**19. a)** 729 **b)** 256 **c)** 1024 **d)** 64 **21. a)** Seven: 56, 63, 70, 77, 84, 91, 98 **b)** Five: 55, 66, 77, 88, 99 **c)** One: 77 **23. a)** 128 **b)** 450 **c)** 9 **d)** 675 **e)** 450 **f)** 450  
**g)** 225 **h)** 75 **25. a)** 990 **b)** 500 **c)** 27 **27.**  $3^{50}$   
**29.** 52,457,600 **31.** 20,077,200 **33. a)** 37,822,859,361  
**b)** 8,204,716,800 **c)** 40,159,050,880 **d)** 12,113,640,000  
**e)** 171,004,205,215 **f)** 72,043,541,640 **g)** 6,230,721,635  
**h)** 223,149,655 **35. a)** 0 **b)** 120 **c)** 720 **d)** 2520 **37. a)** 2  
if  $n = 1$ , 2 if  $n = 2$ , 0 if  $n \geq 3$  **b)**  $2^{n-2}$  for  $n > 1$ ;  
1 if  $n = 1$  **c)**  $2(n-1)$  **39.**  $(n+1)^m$  **41.** If  $n$  is even,  
 $2^{n/2}$ ; if  $n$  is odd,  $2^{(n+1)/2}$  **43. a)** 175 **b)** 248 **c)** 232 **d)** 84  
**45.** 40 **47.** 60 **49. a)** 240 **b)** 480 **c)** 360 **51.** 352  
**53.** 147 **55.** 33 **57. a)** 9,920,671,339,261,325,541,376  $\approx$   
 $9.9 \times 10^{21}$  **b)** 6,641,514,961,387,068,437,760  $\approx$   $6.6 \times$   
 $10^{21}$  **c)** About 314,000 years **59.**  $54(64^{65536} - 1)/63$   
**61.** 7,104,000,000,000 **63.**  $16^{10} + 16^{26} + 16^{58}$  **65.** 666,667  
**67.** 18 **69.** 17 **71.** 22 **73.**  $2n-2, 2n-1$  **75.** Let  $P(m)$   
be the sum rule for  $m$  tasks. For the basis case take  $m = 2$ .  
This is just the sum rule for two tasks. Now assume that  $P(m)$   
is true. Consider  $m+1$  tasks,  $T_1, T_2, \dots, T_m, T_{m+1}$ , which can  
be done in  $n_1, n_2, \dots, n_m, n_{m+1}$  ways, respectively, such that  
no two of these tasks can be done at the same time. To do one  
of these tasks, we can either do one of the first  $m$  of these or  
do task  $T_{m+1}$ . By the sum rule for two tasks, the number of  
ways to do this is the sum of the number of ways to do one  
of the first  $m$  tasks, plus  $n_{m+1}$ . By the inductive hypothesis,  
this is  $n_1 + n_2 + \dots + n_m + n_{m+1}$ , as desired. **77.**  $n(n-3)/2$

### Section 6.2

**1.** Because there are six classes, but only five weekdays, the pigeonhole principle shows that at least two classes must be held on the same day. **3. a)** 3 **b)** 14 **5.** 85 **7.** Because there are four possible remainders when an integer is divided by 4, the pigeonhole principle implies that given five integers, at least two have the same remainder. **9.** Let  $a, a+1, \dots$ ,



$a + n - 1$  be the integers in the sequence. The integers  $(a + i) \bmod n, i = 0, 1, 2, \dots, n - 1$ , are distinct, because  $0 < (a + j) - (a + k) < n$  whenever  $0 \leq k < j \leq n - 1$ . Because there are  $n$  possible values for  $(a + i) \bmod n$  and there are  $n$  different integers in the set, each of these values is taken on exactly once. It follows that there is exactly one integer in the sequence that is divisible by  $n$ . **11.** 4951

**13.** The midpoint of the segment joining the points  $(a, b, c)$  and  $(d, e, f)$  is  $((a + d)/2, (b + e)/2, (c + f)/2)$ . It has integer coefficients if and only if  $a$  and  $d$  have the same parity,  $b$  and  $e$  have the same parity, and  $c$  and  $f$  have the same parity. Because there are eight possible triples of parity [such as  $(\text{even}, \text{odd}, \text{even})$ ], by the pigeonhole principle at least two of the nine points have the same triple of parities. The midpoint of the segment joining two such points has integer coefficients. **15. a)** Group the first eight positive integers into four subsets of two integers each so that the integers of each subset add up to 9:  $\{1, 8\}, \{2, 7\}, \{3, 6\}$ , and  $\{4, 5\}$ . If five integers are selected from the first eight positive integers, by the pigeonhole principle at least two of them come from the same subset. Two such integers have a sum of 9, as desired. **b)** No. Take  $\{1, 2, 3, 4\}$ , for example.

**17.** 4 **19.** 21,251 **21. a)** If there were fewer than 9 freshmen, fewer than 9 sophomores, and fewer than 9 juniors in the class, there would be no more than 8 with each of these three class standings, for a total of at most 24 students, contradicting the fact that there are 25 students in the class. **b)** If there were fewer than 3 freshmen, fewer than 19 sophomores, and fewer than 5 juniors, then there would be at most 2 freshmen, at most 18 sophomores, and at most 4 juniors, for a total of at most 24 students. This contradicts the fact that there are 25 students in the class. **23.** 4, 3, 2, 1, 8, 7, 6, 5, 12, 11, 10, 9, 16, 15, 14, 13 **25.** Number the seats around the table from 1 to 50, and think of seat 50 as being adjacent to seat 1. There are 25 seats with odd numbers and 25 seats with even numbers.

If no more than 12 boys occupied the odd-numbered seats, then at least 13 boys would occupy the even-numbered seats, and vice versa. Without loss of generality, assume that at least 13 boys occupy the 25 odd-numbered seats. Then at least two of those boys must be in consecutive odd-numbered seats, and the person sitting between them will have boys as both of his or her neighbors.

**27. procedure** *long*( $a_1, \dots, a_n$ : positive integers)

{first find longest increasing subsequence}

$max := 0$ ;  $set := 00 \dots 00$  { $n$  bits}

**for**  $i := 1$  **to**  $2^n$

$last := 0$ ;  $count := 0$ ,  $OK := \text{true}$

**for**  $j := 1$  **to**  $n$

**if**  $set(j) = 1$  **then**

**if**  $a_j > last$  **then**  $last := a_j$

$count := count + 1$

**else**  $OK := \text{false}$

**if**  $count > max$  **then**

$max := count$

$best := set$

$set := set + 1$  (binary addition)

{ $max$  is length and  $best$  indicates the sequence}

{repeat for decreasing subsequence with only

changes being  $a_j < last$  instead of  $a_j > last$

and  $last := \infty$  instead of  $last := 0$ }

**29.** By symmetry we need prove only the first statement. Let  $A$  be one of the people. Either  $A$  has at least four friends, or  $A$  has at least six enemies among the other nine people (because  $3 + 5 < 9$ ). Suppose, in the first case, that  $B, C, D$ , and  $E$  are all  $A$ 's friends. If any two of these are friends with each other, then we have found three mutual friends. Otherwise  $\{B, C, D, E\}$  is a set of four mutual enemies. In the second case, let  $\{B, C, D, E, F, G\}$  be a set of enemies of  $A$ . By Example 13, among  $B, C, D, E, F$ , and  $G$  there are either three mutual friends or three mutual enemies, who form, with  $A$ , a set of four mutual enemies. **31.** We need to show two things: that if we have a group of  $n$  people, then among them we must find either a pair of friends or a subset of  $n$  of them all of whom are mutual enemies; and that there exists a group of  $n - 1$  people for which this is not possible. For the first statement, if there is any pair of friends, then the condition is satisfied, and if not, then every pair of people are enemies, so the second condition is satisfied. For the second statement, if we have a group of  $n - 1$  people all of whom are enemies of each other, then there is neither a pair of friends nor a subset of  $n$  of them all of whom are mutual enemies. **33.** There are 6,432,816 possibilities for the three initials and a birthday. So, by the generalized pigeonhole principle, there are at least  $\lceil 39,000,000/6,432,816 \rceil = 7$  people who share the same initials and birthday. **35.** Because  $800,001 > 200,000$ , the pigeonhole principle guarantees that there are at least two Parisians with the same number of hairs on their heads. The generalized pigeonhole principle guarantees that there are at least  $\lceil 800,001/200,000 \rceil = 5$  Parisians with the same number of hairs on their heads. **37.** 18 **39.** Because there are six computers, the number of other computers a computer is connected to is an integer between 0 and 5, inclusive. However, 0 and 5 cannot both occur. To see this, note that if some computer is connected to no others, then no computer is connected to all five others, and if some computer is connected to all five others, then no computer is connected to no others. Hence, by the pigeonhole principle, because there are at most five possibilities for the number of computers a computer is connected to, there are at least two computers in the set of six connected to the same number of others. **41.** Label the computers  $C_1$  through  $C_{100}$ , and label the printers  $P_1$  through  $P_{20}$ . If we connect  $C_k$  to  $P_k$  for  $k = 1, 2, \dots, 20$  and connect each of the computers  $C_{21}$  through  $C_{100}$  to *all* the printers, then we have used a total of  $20 + 80 \cdot 20 = 1620$  cables. This is sufficient, because if computers  $C_1$  through  $C_{20}$  need printers, then they can use the printers with the same subscripts, and if any computers with higher subscripts need a printer instead of one or more of these, then they can use the printers that are not being used, because they are connected to all the printers. Now we must show that 1619 cables is not enough. Because there are 1619 cables and 20 printers, the average number of computers per printer is  $1619/20$ , which is less than 81. Therefore,



some printer must be connected to fewer than 81 computers. That means it is connected to 80 or fewer computers, so there are 20 computers that are not connected to it. If those 20 computers all needed a printer simultaneously, then they would be out of luck, because they are connected to at most the 19 other printers.

**43.** Let  $a_i$  be the number of matches completed by hour  $i$ . Then  $1 \leq a_1 < a_2 < \dots < a_{75} \leq 125$ . Also  $25 \leq a_1 + 24 < a_2 + 24 < \dots < a_{75} + 24 \leq 149$ . There are 150 numbers  $a_1, \dots, a_{75}, a_1 + 24, \dots, a_{75} + 24$ . By the pigeonhole principle, at least two are equal. Because all the  $a_i$ s are distinct and all the  $(a_i + 24)$ s are distinct, it follows that  $a_i = a_j + 24$  for some  $i > j$ . Thus, in the period from the  $(j + 1)$ st to the  $i$ th hour, there are exactly 24 matches.

**45.** Use the generalized pigeonhole principle, placing the  $|S|$  objects  $f(s)$  for  $s \in S$  in  $|T|$  boxes, one for each element of  $T$ .

**47.** Let  $d_j$  be  $jx - N(jx)$ , where  $N(jx)$  is the integer closest to  $jx$  for  $1 \leq j \leq n$ . Each  $d_j$  is an irrational number between  $-1/2$  and  $1/2$ . We will assume that  $n$  is even; the case where  $n$  is odd is messier. Consider the  $n$  intervals  $\{x \mid j/n < x < (j+1)/n\}$ ,  $\{x \mid -(j+1)/n < x < -j/n\}$  for  $j = 0, 1, \dots, (n/2) - 1$ . If  $d_j$  belongs to the interval  $\{x \mid 0 < x < 1/n\}$  or to the interval  $\{x \mid -1/n < x < 0\}$  for some  $j$ , we are done. If not, because there are  $n-2$  intervals and  $n$  numbers  $d_j$ , the pigeonhole principle tells us that there is an interval  $\{x \mid (k-1)/n < x < k/n\}$  containing  $d_r$  and  $d_s$  with  $r < s$ . The proof can be finished by showing that  $(s-r)x$  is within  $1/n$  of its nearest integer.

**49. a)** Assume that  $i_k \leq n$  for all  $k$ . Then by the generalized pigeonhole principle, at least  $\lceil (n^2 + 1)/n \rceil = n + 1$  of the numbers  $i_1, i_2, \dots, i_{n^2+1}$  are equal. **b)** If  $a_{k_j} < a_{k_{j+1}}$ , then the subsequence consisting of  $a_{k_j}$  followed by the increasing subsequence of length  $i_{k_{j+1}}$  starting at  $a_{k_{j+1}}$  contradicts the fact that  $i_{k_j} = i_{k_{j+1}}$ . Hence,  $a_{k_j} > a_{k_{j+1}}$ . **c)** If there is no increasing subsequence of length greater than  $n$ , then parts (a) and (b) apply. Therefore, we have  $a_{k_{n+1}} > a_{k_n} > \dots > a_{k_2} > a_{k_1}$ , a decreasing sequence of length  $n + 1$ .

## Section 6.3

1.  $abc, acb, bac, bca, cab, cba$  3. 720 5. a) 120 b) 720  
c) 8 d) 6720 e) 40,320 f) 3,628,800 7. 15,120 9. 1320  
11. a) 210 b) 386 c) 848 d) 252 13.  $2(n!)^2$  15. 65,780  
17.  $2^{100} - 5051$  19. a) 1024 b) 45 c) 176 d) 252 21. a) 120  
b) 24 c) 120 d) 24 e) 6 f) 0 23. 609,638,400 25. a) 17,280  
b) 14,400 27. a) 94,109,400 b) 941,094 c) 3,764,376  
d) 90,345,024 e) 114,072 f) 2328 g) 24 h) 79,727,040  
i) 3,764,376 j) 109,440 29. a) 12,650 b) 303,600  
31. a) 37,927 b) 18,915 33. a) 122,523,030 b) 72,930,375  
c) 223,149,655 d) 100,626,625 35. 54,600 37. 45  
39. 912 41. 11,232,000 43.  $\frac{n!}{r(n-r)!}$  45. 13 47. 873

## Section 6.4

1.  $x^4 + 4x^3y + 6x^2y^2 + 4xy^3 + y^4$  3.  $x^6 + 6x^5y + 15x^4y^2 + 20x^3y^3 + 15x^2y^4 + 6xy^5 + y^6$  5. 101 7.  $-2^{10} \binom{19}{9} = -94,595,072$   
9.  $-2^{101} 3^{99} \binom{200}{99}$  11.  $\sum_{j=0}^5 \binom{5}{j} (3x^4)^{5-j} (-2y^3)^j = 243x^{20} -$

- $810x^{16}y^3 + 1080x^{12}y^6 - 720x^8y^9 + 240x^4y^{12} - 32y^{15}$   
13. a) 71,680 b) 0 c)  $-16,384$  d)  $-35,840$  e)  $-1,792$   
15.  $(-1)^{(200-k)/3} \binom{100}{(200-k)/3}$  if  $k \equiv 2 \pmod{3}$  and  $-100 \leq k \leq 200$ ; 0 otherwise 17. 1 9 36 84 126 126 84 36  
9 1 19. The sum of all the positive numbers  $\binom{n}{k}$ , as  $k$  runs from 0 to  $n$ , is  $2^n$ , so each one of them is no bigger than this sum. 21.  $\binom{n}{k} = \frac{n(n-1)(n-2)\dots(n-k+1)}{k(k-1)(k-2)\dots 2} \leq \frac{n \cdot n \cdot \dots \cdot n}{2 \cdot 2 \cdot \dots \cdot 2} = n^k / 2^{k-1}$   
23.  $\binom{n}{k-1} + \binom{n}{k} = \frac{n!}{(k-1)!(n-k+1)!} + \frac{n!}{k!(n-k)!} = \frac{n!}{k!(n-k+1)!} \cdot [k + (n-k+1)] = \frac{(n+1)!}{k!(n+1-k)!} = \binom{n+1}{k}$  25. a) We show that each side counts the number of ways to choose from a set with  $n$  elements a subset with  $k$  elements and a distinguished element of that set. For the left-hand side, first choose the  $k$ -set (this can be done in  $\binom{n}{k}$  ways) and then choose one of the  $k$  elements in this subset to be the distinguished element (this can be done in  $k$  ways). For the right-hand side, first choose the distinguished element out of the entire  $n$ -set (this can be done in  $n$  ways), and then choose the remaining  $k-1$  elements of the subset from the remaining  $n-1$  elements of the set (this can be done in  $\binom{n-1}{k-1}$  ways). b)  $k \binom{n}{k} = k \cdot \frac{n!}{k!(n-k)!} = \frac{n \cdot (n-1)!}{(k-1)!(n-k)!} = n \binom{n-1}{k-1}$   
27.  $\binom{n+1}{k} = \frac{(n+1)!}{k!(n+1-k)!} = \frac{(n+1)}{k} \frac{n!}{(k-1)!(n-k)!} = (n+1) \binom{n}{k-1} / k$ . This identity together with  $\binom{n}{0} = 1$  gives a recursive definition. 29.  $\binom{2n}{n+1} + \binom{2n}{n} = \binom{2n+1}{n+1} = \frac{1}{2} \left[ \binom{2n+1}{n+1} + \binom{2n+1}{n+1} \right] = \frac{1}{2} \left[ \binom{2n+1}{n+1} + \binom{2n+1}{n} \right] = \frac{1}{2} \binom{2n+2}{n+1}$  31. a)  $\binom{n+r+1}{r}$  counts the number of ways to choose a sequence of  $r$  0s and  $n+1$  1s by choosing the positions of the 0s. Alternately, suppose that the  $(j+1)$ st term is the last term equal to 1, so that  $n \leq j \leq n+r$ . Once we have determined where the last 1 is, we decide where the 0s are to be placed in the  $j$  spaces before the last 1. There are  $n$  1s and  $j-n$  0s in this range. By the sum rule it follows that there are  $\sum_{j=n}^{n+r} \binom{j}{j-n} = \sum_{k=0}^r \binom{n+k}{k}$  ways to do this. b) Let  $P(r)$  be the statement to be proved. The basis step is the equation  $\binom{n}{0} = \binom{n+1}{0}$ , which is just  $1 = 1$ . Assume that  $P(r)$  is true. Then  $\sum_{k=0}^{r+1} \binom{n+k}{k} = \sum_{k=0}^r \binom{n+k}{k} + \binom{n+r+1}{r+1} = \binom{n+r+1}{r} + \binom{n+r+1}{r+1} = \binom{n+r+2}{r+1}$ , using the inductive hypothesis and Pascal's identity. 33. We can choose the leader first in  $n$  different ways. We can then choose the rest of the committee in  $2^{n-1}$  ways. Hence, there are  $n2^{n-1}$  ways to choose the committee and its leader. Meanwhile, the number of ways to select a committee with  $k$  people is  $\binom{n}{k}$ . Once we have chosen a committee with  $k$  people, there are  $k$  ways to choose its leader. Hence, there are  $\sum_{k=1}^n k \binom{n}{k}$  ways to choose the committee and its leader. Hence,  $\sum_{k=1}^n k \binom{n}{k} = n2^{n-1}$ . 35. Let the set have  $n$  elements. From Corollary 2 we have  $\binom{n}{0} - \binom{n}{1} + \binom{n}{2} - \dots + (-1)^n \binom{n}{n} = 0$ . It follows that  $\binom{n}{0} + \binom{n}{2} + \binom{n}{4} + \dots = \binom{n}{1} + \binom{n}{3} + \binom{n}{5} + \dots$ . The left-hand side gives the number of subsets with an even number of elements, and the right-hand side gives the number of subsets with an odd number of elements. 37. a) A path of the desired type consists of  $m$  moves to the right and  $n$  moves up. Each such path can be represented by a bit string of length  $m+n$  with  $m$  0s and  $n$  1s, where a 0 represents a move to the right and a 1 a move up. b) The number of bit strings of length  $m+n$



containing exactly  $n$  1s equals  $\binom{m+n}{n} = \binom{m+n}{m}$  because such a string is determined by specifying the positions of the  $n$  1s or by specifying the positions of the  $m$  0s. **39.** By Exercise 37 the number of paths of length  $n$  of the type described in that exercise equals  $2^n$ , the number of bit strings of length  $n$ . On the other hand, a path of length  $n$  of the type described in Exercise 37 must end at a point that has  $n$  as the sum of its coordinates, say  $(n-k, k)$  for some  $k$  between 0 and  $n$ , inclusive. By Exercise 37, the number of such paths ending at  $(n-k, k)$  equals  $\binom{n-k+k}{k} = \binom{n}{k}$ . Hence,  $\sum_{k=0}^n \binom{n}{k} = 2^n$ . **41.** By Exercise 37 the number of paths from  $(0, 0)$  to  $(n+1, r)$  of the type described in that exercise equals  $\binom{n+r+1}{r}$ . But such a path starts by going  $j$  steps vertically for some  $j$  with  $0 \leq j \leq r$ . The number of these paths beginning with  $j$  vertical steps equals the number of paths of the type described in Exercise 37 that go from  $(1, j)$  to  $(n+1, r)$ . This is the same as the number of such paths that go from  $(0, 0)$  to  $(n, r-j)$ , which by Exercise 37 equals  $\binom{n+r-j}{r-j}$ . Because  $\sum_{j=0}^r \binom{n+r-j}{r-j} = \sum_{k=0}^r \binom{n+k}{k}$ , it follows that  $\sum_{k=1}^r \binom{n+k}{k} = \binom{n+r-1}{r}$ . **43. a)**  $\binom{n+1}{2}$  **b)**  $\binom{n+2}{3}$  **c)**  $\binom{2n-2}{n-1}$  **d)**  $\binom{n-1}{\lfloor (n-1)/2 \rfloor}$  **e)** Largest odd entry in  $n$ th row of Pascal's triangle **f)**  $\binom{3n-3}{n-1}$

## Section 6.5

**1.** 243 **3.**  $26^6$  **5.** 125 **7.** 35 **9. a)** 1716 **b)** 50,388  
**c)** 2,629,575 **d)** 330 **11.** 9 **13.** 4,504,501 **15. a)** 10,626  
**b)** 1,365 **c)** 11,649 **d)** 106 **17.** 2,520 **19.** 302,702,400  
**21. a)** 169 **b)** 156 **c)** 78 **d)** 91 **23.** 3003 **25.** 7,484,400  
**27.** 30,492 **29.**  $C(59, 50)$  **31.** 35 **33.** 83,160 **35.** 63  
**37.** 19,635 **39.** 210 **41.** 27,720 **43.**  $52!/(7!^5 17!)$   
**45.** Approximately  $6.5 \times 10^{32}$  **47. a)**  $C(k+n-1, n)$   
**b)**  $(k+n-1)!/(k-1)!$  **49.** There are  $C(n, n_1)$  ways to choose  $n_1$  objects for the first box. Once these objects are chosen, there are  $C(n-n_1, n_2)$  ways to choose objects for the second box. Similarly, there are  $C(n-n_1-n_2, n_3)$  ways to choose objects for the third box. Continue in this way until there is  $C(n-n_1-n_2-\dots-n_{k-1}, n_k) = C(n_k, n_k) = 1$  way to choose the objects for the last box (because  $n_1+n_2+\dots+n_k=n$ ). By the product rule, the number of ways to make the entire assignment is  $C(n, n_1)C(n-n_1, n_2)C(n-n_1-n_2, n_3)\dots C(n-n_1-n_2-\dots-n_{k-1}, n_k)$ , which equals  $n!/(n_1!n_2!\dots n_k!)$ , as straightforward simplification shows. **51. a)** Because  $x_1 \leq x_2 \leq \dots \leq x_r$ , it follows that  $x_1+0 < x_2+1 < \dots < x_r+r-1$ . The inequalities are strict because  $x_j+j-1 < x_{j+1}+j$  as long as  $x_j \leq x_{j+1}$ . Because  $1 \leq x_j \leq n+r-1$ , this sequence is made up of  $r$  distinct elements from  $T$ . **b)** Suppose that  $1 \leq x_1 < x_2 < \dots < x_r \leq n+r-1$ . Let  $y_k = x_k - (k-1)$ . Then it is not hard to see that  $y_k \leq y_{k+1}$  for  $k=1, 2, \dots, r-1$  and that  $1 \leq y_k \leq n$  for  $k=1, 2, \dots, r$ . It follows that  $\{y_1, y_2, \dots, y_r\}$  is an  $r$ -combination with repetitions allowed of  $S$ . **c)** From parts (a) and (b) it follows that there is a one-to-one correspondence of  $r$ -combinations with repetitions allowed of  $S$  and  $r$ -combinations of  $T$ , a

set with  $n+r-1$  elements. We conclude that there are  $C(n+r-1, r)$   $r$ -combinations with repetitions allowed of  $S$ . **53.** 65 **55.** 65 **57.** 2 **59.** 3 **61. a)** 150 **b)** 25 **c)** 6 **d)** 2  
**63.** 90,720 **65.** The terms in the expansion are of the form  $x_1^{n_1} x_2^{n_2} \dots x_m^{n_m}$ , where  $n_1+n_2+\dots+n_m=n$ . Such a term arises from choosing the  $x_1$  in  $n_1$  factors, the  $x_2$  in  $n_2$  factors,  $\dots$ , and the  $x_m$  in  $n_m$  factors. This can be done in  $C(n; n_1, n_2, \dots, n_m)$  ways, because a choice is a permutation of  $n_1$  labels "1,"  $n_2$  labels "2,"  $\dots$ , and  $n_m$  labels " $m$ ." **67.** 2520

## Section 6.6

**1.** 14532, 15432, 21345, 23451, 23514, 31452, 31542, 43521, 45213, 45321 **3.** AAA1, AAA2, AAB1, AAB2, AAC1, AAC2, ABA1, ABA2, ABB1, ABB2, ABC1, ABC2, ACA1, ACA2, ACB1, ACB2, ACC1, ACC2, BAA1, BAA2, BAB1, BAB2, BAC1, BAC2, BBA1, BBA2, BBB1, BBB2, BBC1, BBC2, BCA1, BCA2, BCB1, BCB2, BCC1, BCC2, CAA1, CAA2, CAB1, CAB2, CAC1, CAC2, CBA1, CBA2, CBB1, CBB2, CBC1, CBC2, CCA1, CCA2, CCB1, CCB2, CCC1, CCC2 **5. a)** 2134 **b)** 54132 **c)** 12534 **d)** 45312 **7.** 1234, 1243, 1324, 1342, 1423, 1432, 2134, 2143, 2314, 2341, 2413, 2431, 3124, 3142, 3214, 3241, 3412, 3421, 4123, 4132, 4213, 4231, 4312, 4321 **9.**  $\{1, 2, 3\}$ ,  $\{1, 2, 4\}$ ,  $\{1, 2, 5\}$ ,  $\{1, 3, 4\}$ ,  $\{1, 3, 5\}$ ,  $\{1, 4, 5\}$ ,  $\{2, 3, 4\}$ ,  $\{2, 3, 5\}$ ,  $\{2, 4, 5\}$ ,  $\{3, 4, 5\}$  **11.** The bit string representing the next larger  $r$ -combination must differ from the bit string representing the original one in position  $i$  because positions  $i+1, \dots, r$  are occupied by the largest possible numbers. Also  $a_i+1$  is the smallest possible number we can put in position  $i$  if we want a combination greater than the original one. Then  $a_i+2, \dots, a_i+r-i+1$  are the smallest allowable numbers for positions  $i+1$  to  $r$ . Thus, we have produced the next  $r$ -combination. **13.** 123, 132, 213, 231, 312, 321, 124, 142, 214, 241, 412, 421, 125, 152, 215, 251, 512, 521, 134, 143, 314, 341, 413, 431, 135, 153, 315, 351, 513, 531, 145, 154, 415, 451, 514, 541, 234, 243, 324, 342, 423, 432, 235, 253, 325, 352, 523, 532, 245, 254, 425, 452, 524, 542, 345, 354, 435, 453, 534, 543 **15.** We will show that it is a bijection by showing that it has an inverse. Given a positive integer less than  $n!$ , let  $a_1, a_2, \dots, a_{n-1}$  be its Cantor digits. Put  $n$  in position  $n-a_{n-1}$ ; then  $a_{n-1}$  is the number of integers less than  $n$  that follow  $n$  in the permutation. Then put  $n-1$  in free position  $(n-1)-a_{n-2}$ , where we have numbered the free positions  $1, 2, \dots, n-1$  (excluding the position that  $n$  is already in). Continue until 1 is placed in the only free position left. Because we have constructed an inverse, the correspondence is a bijection.

**17. procedure** *Cantor permutation*( $n, i$ : integers with

$n \geq 1$  and  $0 \leq i < n!$ )

$x := n$

**for**  $j := 1$  **to**  $n$

$p_j := 0$

**for**  $k := 1$  **to**  $n-1$

$c := \lfloor x/(n-k)! \rfloor$ ;  $x := x - c(n-k)!;$   $h := n$

**while**  $p_h \neq 0$



```

    h := h - 1
  for j := 1 to c
    h := h - 1
    while ph ≠ 0
      h := h - 1
    ph := n - k + 1
    h := 1
    while ph ≠ 0
      h := h + 1
    ph := 1
  {p1p2 ... pn is the permutation corresponding to i}

```

## Supplementary Exercises

1. a) 151,200 b) 1,000,000 c) 210 d) 5005 3.  $3^{100}$   
 5. 24,600 7. a) 4060 b) 2688 c) 25,009,600 9. a) 192  
 b) 301 c) 300 d) 300 11. 639 13. The maximum possible sum is 240, and the minimum possible sum is 15. So the number of possible sums is 226. Because there are 252 subsets with five elements of a set with 10 elements, by the pigeonhole principle it follows that at least two have the same sum.  
 15. a) 50 b) 50 c) 14 d) 17 17. Let  $a_1, a_2, \dots, a_m$  be the integers, and let  $d_i = \sum_{j=1}^i a_j$ . If  $d_i \equiv 0 \pmod{m}$  for some  $i$ , we are done. Otherwise  $d_1 \bmod m, d_2 \bmod m, \dots, d_m \bmod m$  are  $m$  integers with values in  $\{1, 2, \dots, m-1\}$ . By the pigeonhole principle  $d_k = d_l$  for some  $1 \leq k < l \leq m$ . Then  $\sum_{j=k+1}^l a_j = d_l - d_k \equiv 0 \pmod{m}$ .  
 19. The decimal expansion of the rational number  $a/b$  can be obtained by division of  $b$  into  $a$ , where  $a$  is written with a decimal point and an arbitrarily long string of 0s following it. The basic step is finding the next digit of the quotient, namely,  $\lfloor r/b \rfloor$ , where  $r$  is the remainder with the next digit of the dividend brought down. The current remainder is obtained from the previous remainder by subtracting  $b$  times the previous digit of the quotient. Eventually the dividend has nothing but 0s to bring down. Furthermore, there are only  $b$  possible remainders. Thus, at some point, by the pigeonhole principle, we will have the same situation as had previously arisen. From that point onward, the calculation must follow the same pattern. In particular, the quotient will repeat.  
 21. a) 125,970 b) 20 c) 141,120,525 d) 141,120,505 e) 177,100 f) 141,078,021 23. a) 10 b) 8 c) 7 25.  $3^n$   
 27.  $C(n+2, r+1) = C(n+1, r+1) + C(n+1, r) = 2C(n+1, r+1) - C(n+1, r+1) + C(n+1, r) = 2C(n+1, r+1) - (C(n, r+1) + C(n, r)) + (C(n, r) + C(n, r-1)) = 2C(n+1, r+1) - C(n, r+1) + C(n, r-1)$   
 29. Substitute  $x = 1$  and  $y = 3$  into the binomial theorem. 31. Both sides count the number of ways to choose a subset of three distinct numbers  $\{i, j, k\}$  with  $i < j < k$  from  $\{1, 2, \dots, n\}$ . 33.  $C(n+1, 5)$  35. 3,491,888,400 37.  $5^{24}$   
 39. a) 45 b) 57 c) 12 41. a) 386 b) 56 43. 0 if  $n < m$ ;  $C(n-1, n-m)$  if  $n \geq m$  45. a) 15,625 b) 202 c) 210 d) 10 47. a) 3 b) 11 c) 6 d) 10 49. There are two possibilities: three people seated at one table with everyone else sitting alone, which can be done in  $2C(n, 3)$  ways (choose the

three people and seat them in one of two arrangements), or two groups of two people seated together with everyone else sitting alone, which can be done in  $3C(n, 4)$  ways (choose four people and then choose one of the three ways to pair them up). Both  $2C(n, 3) + 3C(n, 4)$  and  $(3n-1)C(n, 3)/4$  equal  $n^4/8 - 5n^3/12 + 3n^2/8 - n/12$ . 51. The number of permutations of  $2n$  objects of  $n$  different types, two of each type, is  $(2n)!/2^n$ . Because this must be an integer, the denominator must divide the numerator. 53. CCGGUCCGAAAG

55. **procedure** next permutation( $n$ : positive integer,

$a_1, a_2, \dots, a_r$ : positive integers not exceeding  $n$  with  $a_1 a_2 \dots a_r \neq n n \dots n$ )

$i := r$

**while**  $a_i = n$

$a_i := 1$

$i := i - 1$

$a_i := a_i + 1$

{ $a_1 a_2 \dots a_r$  is the next permutation in lexicographic order}

57. We must show that if there are  $R(m, n-1) + R(m-1, n)$  people at a party, then there must be at least  $m$  mutual friends or  $n$  mutual enemies. Consider one person; let's call him Jerry. Then there are  $R(m-1, n) + R(m, n-1) - 1$  other people at the party, and by the pigeonhole principle there must be at least  $R(m-1, n)$  friends of Jerry or  $R(m, n-1)$  enemies of Jerry among these people. First let's suppose there are  $R(m-1, n)$  friends of Jerry. By the definition of  $R$ , among these people we are guaranteed to find either  $m-1$  mutual friends or  $n$  mutual enemies. In the former case, these  $m-1$  mutual friends together with Jerry are a set of  $m$  mutual friends; and in the latter case, we have the desired set of  $n$  mutual enemies. The other situation is similar: Suppose there are  $R(m, n-1)$  enemies of Jerry; we are guaranteed to find among them either  $m$  mutual friends or  $n-1$  mutual enemies. In the former case, we have the desired set of  $m$  mutual friends, and in the latter case, these  $n-1$  mutual enemies together with Jerry are a set of  $n$  mutual enemies.

## CHAPTER 7

### Section 7.1

1.  $1/13$  3.  $1/2$  5.  $1/2$  7.  $1/64$  9.  $47/52$  11.  $1/C(52, 5)$   
 13.  $1 - [C(48, 5)/C(52, 5)]$  15.  $C(13, 2)C(4, 2)C(4, 2)C(44, 1)/C(52, 5)$  17.  $10,240/C(52, 5)$  19.  $1,302,540/C(52, 5)$  21.  $1/64$  23.  $8/25$  25. a)  $1/C(50, 6) = 1/15,890,700$  b)  $1/C(52, 6) = 1/20,358,520$   
 c)  $1/C(56, 6) = 1/32,468,436$  d)  $1/C(60, 6) = 1/50,063,860$   
 27. a)  $139,128/319,865$  b)  $212,667/511,313$   
 c)  $151,340/386,529$  d)  $163,647/446,276$  29.  $1/C(100, 8)$   
 31.  $3/100$  33. a)  $1/7,880,400$  b)  $1/8,000,000$  35. a)  $9/19$  b)  $81/361$  c)  $1/19$  d)  $1,889,568/2,476,099$  e)  $48/361$   
 37. Three dice 39. a)  $4/756,438,375$  b)  $13/30,257,535$



c) 4888/2,750,685      d) 90,272/9,823,875      **41. a)** 25/292,201,338      **b)** 25/292,201,338      **c)** 1280/571,120,797  
**d)** 7564/661,089      **43.** The door the contestant chooses is chosen at random without knowing where the prize is, but the door chosen by the host is not chosen at random, because he always avoids opening the door with the prize. This makes any argument based on symmetry invalid.      **45. a)** 671/1296  
**b)**  $1 - 35^{24}/36^{24}$ ; no      **c)** The former

## Section 7.2

**1.**  $p(T) = 1/4$ ,  $p(H) = 3/4$       **3.**  $p(1) = p(3) = p(5) = p(6) = 1/16$ ;  $p(2) = p(4) = 3/8$       **5.** 9/49      **7. a)** 1/2  
**b)** 1/2      **c)** 1/3      **d)** 1/4      **e)** 1/4      **9. a)** 1/26!      **b)** 1/26      **c)** 1/2  
**d)** 1/26      **e)** 1/650      **f)** 1/15,600      **11.** Clearly,  $p(E \cup F) \geq p(E) = 0.7$ . Also,  $p(E \cup F) \leq 1$ . If we apply Theorem 2 from Section 7.1, we can rewrite this as  $p(E) + p(F) - p(E \cap F) \leq 1$ , or  $0.7 + 0.5 - p(E \cap F) \leq 1$ . Solving for  $p(E \cap F)$  gives  $p(E \cap F) \geq 0.2$ .      **13.** Because  $p(E \cup F) = p(E) + p(F) - p(E \cap F)$  and  $p(E \cup F) \leq 1$ , it follows that  $1 \geq p(E) + p(F) - p(E \cap F)$ . From this inequality we conclude that  $p(E) + p(F) \leq 1 + p(E \cap F)$ .      **15.** We will use mathematical induction to prove that the inequality holds for  $n \geq 2$ . Let  $P(n)$  be the statement that  $p(\bigcup_{j=1}^n E_j) \leq \sum_{j=1}^n p(E_j)$ . *Basis step:*  $P(2)$  is true because  $p(E_1 \cup E_2) = p(E_1) + p(E_2) - p(E_1 \cap E_2) \leq p(E_1) + p(E_2)$ . *Inductive step:* Assume that  $P(k)$  is true. Using the basis case and the inductive hypothesis, it follows that  $p(\bigcup_{j=1}^{k+1} E_j) \leq p(\bigcup_{j=1}^k E_j) + p(E_{k+1}) \leq \sum_{j=1}^{k+1} p(E_j)$ . This shows that  $P(k+1)$  is true, completing the proof by mathematical induction.      **17.** Because  $E \cup \bar{E}$  is the entire sample space  $S$ , the event  $F$  can be split into two disjoint events:  $F = S \cap F = (E \cup \bar{E}) \cap F = (E \cap F) \cup (\bar{E} \cap F)$ , using the distributive law. Therefore,  $p(F) = p((E \cap F) \cup (\bar{E} \cap F)) = p(E \cap F) + p(\bar{E} \cap F)$ , because these two events are disjoint. Subtracting  $p(E \cap F)$  from both sides, using the fact that  $p(E \cap F) = p(E) \cdot p(F)$  (the hypothesis that  $E$  and  $F$  are independent), and factoring, we have  $p(F)[1 - p(E)] = p(\bar{E} \cap F)$ . Because  $1 - p(E) = p(\bar{E})$ , this says that  $p(\bar{E} \cap F) = p(\bar{E}) \cdot p(F)$ , as desired.      **19. a)** 1/12  
**b)**  $1 - \frac{11}{12} \cdot \frac{10}{12} \cdot \dots \cdot \frac{13-n}{12}$       **c)** 5      **21.** 614      **23.** 1/4      **25.** 3/8  
**27. a)** Not independent      **b)** Not independent      **c)** Not independent      **29.** 3/16      **31. a)**  $1/32 = 0.03125$       **b)**  $0.49^5 \approx 0.02825$       **c)** 0.03795012      **33. a)** 5/8      **b)** 0.627649      **c)** 0.6431  
**35. a)**  $p^n$       **b)**  $1 - p^n$       **c)**  $p^n + n \cdot p^{n-1} \cdot (1 - p)$   
**d)**  $1 - [p^n + n \cdot p^{n-1} \cdot (1 - p)]$       **37.**  $p(\bigcup_{i=1}^{\infty} E_i)$  is the sum of  $p(s)$  for each outcome  $s$  in  $\bigcup_{i=1}^{\infty} E_i$ . Because the  $E_i$ s are pairwise disjoint, this is the sum of the probabilities of all the outcomes in any of the  $E_i$ s, which is what  $\sum_{i=1}^{\infty} p(E_i)$  is. (We can rearrange the summands and still get the same answer because this series converges absolutely.)      **39. a)**  $\bar{E} = \bigcup_{j=1}^{\binom{m}{k}} F_j$ , so the given inequality now follows from Boole's inequality (Exercise 15).      **b)** The probability that a particular player not in the  $j$ th set beats all  $k$  of the players in the  $j$ th set is  $(1/2)^k = 2^{-k}$ . Therefore, the probability that this player does not do so is  $1 - 2^{-k}$ , so the probability that all  $m - k$  of the players not in the  $j$ th set are unable to boast of a perfect record against everyone

in the  $j$ th set is  $(1 - 2^{-k})^{m-k}$ . That is precisely  $p(F_j)$ .      **c)** The first inequality follows immediately, because all the summands are the same and there are  $\binom{m}{k}$  of them. If this probability is less than 1, then it must be possible that  $\bar{E}$  fails, i.e., that  $E$  happens. So there is a tournament that meets the conditions of the problem as long as the second inequality holds.      **d)**  $m \geq 21$  for  $k = 2$ , and  $m \geq 91$  for  $k = 3$

**41. procedure probabilistic prime( $n, k$ )**

*composite* := **false**

$i$  := 0

**while** *composite* = **false** and  $i < k$

$i$  :=  $i + 1$

choose  $b$  uniformly at random with  $1 < b < n$

apply Miller's test to base  $b$

**if**  $n$  fails the test **then** *composite* := **true**

**if** *composite* = **true** **then** print ("composite")

**else** print ("probably prime")

## Section 7.3

NOTE: In the answers for Section 7.3, all probabilities given in decimal form are rounded to three decimal places.      **1.** 3/5      **3.** 3/4      **5.** 0.481      **7. a)** 0.999      **b)** 0.324  
**9. a)** 0.740      **b)** 0.260      **c)** 0.002      **d)** 0.998      **11.** 0.724  
**13.** 3/17      **15. a)** 1/3      **b)**  $p(M = j \mid W = k) = 1$  if  $i, j$ , and  $k$  are distinct;  $p(M = j \mid W = k) = 0$  if  $j = k$  or  $j = i$ ;  $p(M = j \mid W = k) = 1/2$  if  $i = k$  and  $j \neq i$       **c)** 2/3      **d)** You should change doors, because you now have a 2/3 chance to win by switching.      **17.** The definition of conditional probability tells us that  $p(F_j \mid E) = p(E \cap F_j)/p(E)$ . For the numerator, again using the definition of conditional probability, we have  $p(E \cap F_j) = p(E \mid F_j)p(F_j)$ , as desired. For the denominator, we show that  $p(E) = \sum_{i=1}^n p(E \mid F_i)p(F_i)$ . The events  $E \cap F_i$  partition the event  $E$ ; that is,  $(E \cap F_{i_1}) \cap (E \cap F_{i_2}) = \emptyset$  when  $i_1 \neq i_2$  (because the  $F_i$ 's are mutually exclusive), and  $\bigcup_{i=1}^n (E \cap F_i) = E$  (because the  $\bigcup_{i=1}^n F_i = S$ ). Therefore,  $p(E) = \sum_{i=1}^n p(E \cap F_i) = \sum_{i=1}^n p(E \mid F_i)p(F_i)$ .      **19.** No  
**21.** Yes      **23.** By Bayes' theorem,  $p(S \mid E_1 \cap E_2) = p(E_1 \cap E_2 \mid S)p(S)/[p(E_1 \cap E_2 \mid S)p(S) + p(E_1 \cap E_2 \mid \bar{S})p(\bar{S})]$ . Because we are assuming no prior knowledge about whether a message is or is not spam, we set  $p(S) = p(\bar{S}) = 0.5$ , and so the equation above simplifies to  $p(S \mid E_1 \cap E_2) = p(E_1 \cap E_2 \mid S)/[p(E_1 \cap E_2 \mid S) + p(E_1 \cap E_2 \mid \bar{S})]$ . Because of the assumed independence of  $E_1$ ,  $E_2$ , and  $S$ , we have  $p(E_1 \cap E_2 \mid S) = p(E_1 \mid S) \cdot p(E_2 \mid S)$ , and similarly for  $\bar{S}$ .

## Section 7.4

**1.** 2.5      **3.** 5/3      **5.** 336/49      **7.** 170      **9.**  $(4n + 6)/3$   
**11.** 50,700,551/10,077,696  $\approx 5.03$       **13.** 6      **15.**  $p(X \geq j) = \sum_{k=j}^{\infty} p(X = k) = \sum_{k=j}^{\infty} (1-p)^{k-1}p = p(1-p)^{j-1} \sum_{k=0}^{\infty} (1-p)^k = p(1-p)^{j-1}/(1 - (1-p)) = (1-p)^{j-1}$       **17.** 2302  
**19.**  $(7/2) \cdot 7 \neq 329/12$       **21.** 10      **23.** 1472 pounds  
**25.**  $p + (n-1)p(1-p)$       **27.** 5/2      **29. a)** 0      **b)**  $n$       **31.** This is not true. For example, let  $X$  be the number of heads in



one flip of a fair coin, and let  $Y$  be the number of heads in one flip of a second fair coin. Then  $A(X) + A(Y) = 1$  but  $A(X + Y) = 0.5$ . **33. a)** We are told that  $X_1$  and  $X_2$  are independent. To see that  $X_1$  and  $X_3$  are independent, we enumerate the eight possibilities for  $(X_1, X_2, X_3)$  and find that  $(0, 0, 0)$ ,  $(1, 0, 1)$ ,  $(0, 1, 1)$ ,  $(1, 1, 0)$  each have probability  $1/4$  and the others have probability 0 (because of the definition of  $X_3$ ). Thus,  $p(X_1 = 0 \wedge X_3 = 0) = 1/4$ ,  $p(X_1 = 0) = 1/2$ , and  $p(X_3 = 0) = 1/2$ , so it is true that  $p(X_1 = 0 \wedge X_3 = 0) = p(X_1 = 0)p(X_3 = 0)$ . Essentially the same calculation shows that  $p(X_1 = 0 \wedge X_3 = 1) = p(X_1 = 0)p(X_3 = 1)$ ,  $p(X_1 = 1 \wedge X_3 = 0) = p(X_1 = 1)p(X_3 = 0)$ , and  $p(X_1 = 1 \wedge X_3 = 1) = p(X_1 = 1)p(X_3 = 1)$ . Therefore, by definition,  $X_1$  and  $X_3$  are independent. The same reasoning shows that  $X_2$  and  $X_3$  are independent. To see that  $X_3$  and  $X_1 + X_2$  are not independent, we observe that  $p(X_3 = 1 \wedge X_1 + X_2 = 2) = 0$ . But  $p(X_3 = 1)p(X_1 + X_2 = 2) = (1/2)(1/4) = 1/8$ . **b)** We see from the calculation in part (a) that  $X_1$ ,  $X_2$ , and  $X_3$  are all Bernoulli random variables, so the variance of each is  $(1/2)(1/2) = 1/4$ . Therefore,  $V(X_1) + V(X_2) + V(X_3) = 3/4$ . We use the calculations in part (a) to see that  $E(X_1 + X_2 + X_3) = 3/2$ , and then  $V(X_1 + X_2 + X_3) = 3/4$ . **c)** In order to use the first part of Theorem 7 to show that  $V((X_1 + X_2 + \cdots + X_k) + X_{k+1}) = V(X_1 + X_2 + \cdots + X_k) + V(X_{k+1})$  in the inductive step of a proof by mathematical induction, we would have to know that  $X_1 + X_2 + \cdots + X_k$  and  $X_{k+1}$  are independent, but we see from part (a) that this is not necessarily true. **35.**  $1/100$  **37.**  $E(X)/a = \sum_r (r/a) \cdot p(X = r) \geq \sum_{r \geq a} 1 \cdot p(X = r) = p(X \geq a)$  **39. a)**  $10/11$  **b)**  $0.9999$  **41. a)** Each of the  $n!$  permutations occurs with probability  $1/n!$ , so  $E(X)$  is the number of comparisons, averaged over all these permutations. **b)** Even if the algorithm continues  $n - 1$  rounds,  $X$  will be at most  $n(n - 1)/2$ . It follows from the formula for expectation that  $E(X) \leq n(n - 1)/2$ . **c)** The algorithm proceeds by comparing adjacent elements and then swapping them if necessary. Thus, the only way that inverted elements can become uninverted is for them to be compared and swapped. **d)** Because  $X(P) \geq I(P)$  for all  $P$ , it follows from the definition of expectation that  $E(X) \geq E(I)$ . **e)** This summation counts 1 for every instance of an inversion. **f)** This follows from Theorem 3. **g)** By Theorem 2 with  $n = 1$ , the expectation of  $I_{j,k}$  is the probability that  $a_k$  precedes  $a_j$  in the permutation. This is clearly  $1/2$  by symmetry. **h)** The summation in part (f) consists of  $C(n, 2) = n(n - 1)/2$  terms, each equal to  $1/2$ , so the sum is  $n(n - 1)/4$ . **i)** From part (a) and part (b) we know that  $E(X)$ , the object of interest, is at most  $n(n - 1)/2$ , and from part (d) and part (h) we know that  $E(X)$  is at least  $n(n - 1)/4$ , both of which are  $\Theta(n^2)$ . **43.** 1 **45.**  $V(X + Y) = E((X + Y)^2) - E(X + Y)^2 = E(X^2 + 2XY + Y^2) - [E(X) + E(Y)]^2 = E(X^2) + 2E(XY) + E(Y^2) - E(X)^2 - 2E(X)E(Y) - E(Y)^2 = E(X^2) - E(X)^2 + 2[E(XY) - E(X)E(Y)] + E(Y^2) - E(Y)^2 = V(X) + 2 \text{Cov}(X, Y) + V(Y)$  **47.**  $[(n - 1)/n]^m$  **49.**  $(n - 1)^m/n^{m-1}$

## Supplementary Exercises

**1.**  $1/109,668$  **3. a)**  $1/195,249,054$  **b)**  $1/5,138,133$   
**c)**  $45/357,599$  **d)**  $18,285/18,821$  **5. a)**  $1/C(52, 13)$   
**b)**  $4/C(52, 13)$  **c)**  $2,944,656/C(52, 13)$  **d)**  $35,335,872/C(52, 13)$  **7. a)**  $9/2$  **b)**  $21/4$  **9. a)** 9 **b)**  $21/2$  **11. a)** 8  
**b)**  $49/6$  **13. a)**  $n/2^{n-1}$  **b)**  $p(1 - p)^{k-1}$ , where  $p = n/2^{n-1}$   
**c)**  $2^{n-1}/n$  **15.**  $\frac{(m-1)(n-1) + \gcd(m,n) - 1}{mn-1}$  **17. a)**  $2/3$  **b)**  $2/3$   
**19.**  $1/32$  **21. a)** The probability that one wins  $2^n$  dollars is  $1/2^n$ , because that happens precisely when the player gets  $n - 1$  tails followed by a head. The expected value of the winnings is therefore the sum of  $2^n$  times  $1/2^n$  as  $n$  goes from 1 to infinity. Because each of these terms is 1, the sum is infinite. In other words, one should be willing to wager any amount of money and expect to come out ahead in the long run. **b)** \$9, \$9 **23. a)**  $1/3$  when  $S = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$ ,  $A = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$ , and  $B = \{1, 2, 3, 4\}$ ;  $1/12$  when  $S = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$ ,  $A = \{4, 5, 6, 7, 8, 9, 10, 11, 12\}$ , and  $B = \{1, 2, 3, 4\}$  **b)** 1 when  $S = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$ ,  $A = \{4, 5, 6, 7, 8, 9, 10, 11, 12\}$ , and  $B = \{1, 2, 3, 4\}$ ;  $3/4$  when  $S = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$ ,  $A = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$ , and  $B = \{1, 2, 3, 4\}$  **25. a)**  $p(E_1 \cap E_2) = p(E_1)p(E_2)$ ,  $p(E_1 \cap E_3) = p(E_1)p(E_3)$ ,  $p(E_2 \cap E_3) = p(E_2)p(E_3)$ ,  $p(E_1 \cap E_2 \cap E_3) = p(E_1)p(E_2)p(E_3)$  **b)** Yes **c)** Yes; yes **d)** Yes; no **e)**  $2^n - n - 1$  **27. a)**  $1/2$  under first interpretation;  $1/3$  under second interpretation **b)** Let  $M$  be the event that both of Mr. Smith's children are boys and let  $B$  be the event that Mr. Smith chose a boy for today's walk. Then  $p(M) = 1/4$ ,  $p(B | M) = 1$ , and  $p(B | \bar{M}) = 1/3$ . Apply Bayes' theorem to compute  $p(M | B) = 1/2$ . **c)** This variation is equivalent to the second interpretation discussed in part (a), so the answer is unambiguously  $1/3$ . **29.**  $V(aX + b) = E((aX + b)^2) - E(aX + b)^2 = E(a^2X^2 + 2abX + b^2) - [aE(X) + b]^2 = E(a^2X^2) + E(2abX) + E(b^2) - [a^2E(X)^2 + 2abE(X) + b^2] = a^2E(X^2) + 2abE(X) + b^2 - a^2E(X)^2 - 2abE(X) - b^2 = a^2[E(X^2) - E(X)^2] = a^2V(X)$  **31.** To count every element in the sample space exactly once, we must include every element in each of the sets and then take away the double counting of the elements in the intersections. Thus,  $p(E_1 \cup E_2 \cup \cdots \cup E_m) = p(E_1) + p(E_2) + \cdots + p(E_m) - p(E_1 \cap E_2) - p(E_1 \cap E_3) - \cdots - p(E_1 \cap E_m) - p(E_2 \cap E_3) - p(E_2 \cap E_4) - \cdots - p(E_2 \cap E_m) - \cdots - p(E_{m-1} \cap E_m) = qm - (m(m-1)/2)r$ , because  $C(m, 2)$  terms are being subtracted. But  $p(E_1 \cup E_2 \cup \cdots \cup E_m) = 1$ , so we have  $qm - [m(m-1)/2]r = 1$ . Because  $r \geq 0$ , this equation tells us that  $qm \geq 1$ , so  $q \geq 1/m$ . Because  $q \leq 1$ , this equation also implies that  $[m(m-1)/2]r = qm - 1 \leq m - 1$ , from which it follows that  $r \leq 2/m$ . **33. a)** We purchase the cards until we have gotten one of each type. That means we have purchased  $X$  cards in all. On the other hand, that also means that we purchased  $X_0$  cards until we got the first type we got, and then purchased  $X_1$  more cards until we got the second type we got, and so on. Thus,  $X$  is the sum of the  $X_j$ s. **b)** Once  $j$  distinct types have been obtained, there are  $n - j$  new types available out of a total of  $n$  types available. Because it is equally likely



that we get each type, the probability of success on the next purchase (getting a new type) is  $(n - j)/n$ . **c)** This follows immediately from the definition of geometric distribution, the definition of  $X_j$ , and part (b). **d)** From part (c) it follows that  $E(X_j) = n/(n - j)$ . Thus, by the linearity of expectation and part (a), we have  $E(X) = E(X_0) + E(X_1) + \cdots + E(X_{n-1}) = \frac{n}{n} + \frac{n}{n-1} + \cdots + \frac{n}{1} = n \left( \frac{1}{n} + \frac{1}{n-1} + \cdots + \frac{1}{1} \right)$ . **e)** About 224.46 **35.**  $24 \cdot 13^4 / (52 \cdot 51 \cdot 50 \cdot 49)$

## CHAPTER 8

### Section 8.1

**1.** Let  $P(n)$  be " $H_n = 2^n - 1$ ." *Basis step:*  $P(1)$  is true because  $H_1 = 1$ . *Inductive step:* Assume that  $H_n = 2^n - 1$ . Then because  $H_{n+1} = 2H_n + 1$ , it follows that  $H_{n+1} = 2(2^n - 1) + 1 = 2^{n+1} - 1$ . **3. a)**  $a_n = 2a_{n-1} + a_{n-5}$  for  $n \geq 5$  **b)**  $a_0 = 1, a_1 = 2, a_2 = 4, a_3 = 8, a_4 = 16$  **c)** 1217 **5.** 9494 **7. a)**  $a_n = a_{n-1} + a_{n-2} + 2^{n-2}$  for  $n \geq 2$  **b)**  $a_0 = 0, a_1 = 0$  **c)** 94 **9. a)**  $a_n = a_{n-1} + a_{n-2} + a_{n-3}$  for  $n \geq 3$  **b)**  $a_0 = 1, a_1 = 2, a_2 = 4$  **c)** 81 **11. a)**  $a_n = a_{n-1} + a_{n-2}$  for  $n \geq 2$  **b)**  $a_0 = 1, a_1 = 1$  **c)** 34 **13. a)**  $a_n = 2a_{n-1} + 2a_{n-2}$  for  $n \geq 2$  **b)**  $a_0 = 1, a_1 = 3$  **c)** 448 **15. a)**  $a_n = 2a_{n-1} + a_{n-2}$  for  $n \geq 2$  **b)**  $a_0 = 1, a_1 = 3$  **c)** 239 **17. a)**  $a_n = 2a_{n-1}$  for  $n \geq 2$  **b)**  $a_1 = 3$  **c)** 96 **19. a)**  $a_n = a_{n-1} + a_{n-2}$  for  $n \geq 2$  **b)**  $a_0 = 1, a_1 = 1$  **c)** 89 **21. a)**  $R_n = n + R_{n-1}, R_0 = 1$  **b)**  $R_n = n(n+1)/2 + 1$  **23. a)**  $S_n = S_{n-1} + (n^2 - n + 2)/2, S_0 = 1$  **b)**  $S_n = (n^3 + 5n + 6)/6$  **25.** 64 **27. a)**  $a_n = 2a_{n-1} + 2a_{n-2}$  **b)**  $a_0 = 1, a_1 = 3$  **c)** 1224 **29.** Clearly,  $S(m, 1) = 1$  for  $m \geq 1$ . If  $m \geq n$ , then a function that is not onto from the set with  $m$  elements to the set with  $n$  elements can be specified by picking the size of the range, which is an integer between 1 and  $n - 1$  inclusive, picking the elements of the range, which can be done in  $C(n, k)$  ways, and picking an onto function onto the range, which can be done in  $S(m, k)$  ways. Hence, there are  $\sum_{k=1}^{n-1} C(n, k)S(m, k)$  functions that are not onto. But there are  $n^m$  functions altogether, so  $S(m, n) = n^m - \sum_{k=1}^{n-1} C(n, k)S(m, k)$ . **31. a)**  $C_5 = C_0C_4 + C_1C_3 + C_2C_2 + C_3C_1 + C_4C_0 = 1 \cdot 14 + 1 \cdot 5 + 2 \cdot 2 + 5 \cdot 1 + 14 \cdot 1 = 42$  **b)**  $C(10, 5)/6 = 42$  **33.**  $J(1) = 1, J(2) = 1, J(3) = 3, J(4) = 1, J(5) = 3, J(6) = 5, J(7) = 7, J(8) = 1, J(9) = 3, J(10) = 5, J(11) = 7, J(12) = 9, J(13) = 11, J(14) = 13, J(15) = 15, J(16) = 1$  **35.** First, suppose that the number of people is even, say  $2n$ . After going around the circle once and returning to the first person, because the people at locations with even numbers have been eliminated, there are exactly  $n$  people left and the person currently at location  $i$  is the person who was originally at location  $2i - 1$ . Therefore, the survivor [originally in location  $J(2n)$ ] is now in location  $J(n)$ ; this was the person who was at location  $2J(n) - 1$ . Hence,  $J(2n) = 2J(n) - 1$ . Similarly, when there are an odd number of people, say  $2n + 1$ , then after going around the circle once and then eliminating person 1, there are  $n$  people left and the

person currently at location  $i$  is the person who was at location  $2i + 1$ . Therefore, the survivor will be the player currently occupying location  $J(n)$ , namely, the person who was originally at location  $2J(n) + 1$ . Hence,  $J(2n + 1) = 2J(n) + 1$ . The basis step is  $J(1) = 1$ . **37.** 73, 977, 3617 **39.** These nine moves solve the puzzle: Move disk 1 from peg 1 to peg 2; move disk 2 from peg 1 to peg 3; move disk 1 from peg 2 to peg 3; move disk 3 from peg 1 to peg 2; move disk 4 from peg 1 to peg 4; move disk 3 from peg 2 to peg 4; move disk 1 from peg 3 to peg 2; move disk 2 from peg 3 to peg 4; move disk 1 from peg 2 to peg 4. To see that at least nine moves are required, first note that at least seven moves are required no matter how many pegs are present: three to unstack the disks, one to move the largest disk 4, and three more moves to restack them. At least two other moves are needed, because to move disk 4 from peg 1 to peg 4 the other three disks must be on pegs 2 and 3, so at least one move is needed to restack them and one move to unstack them. **41.** The base cases are obvious. If  $n > 1$ , the algorithm consists of three stages. In the first stage, by the inductive hypothesis,  $R(n - k)$  moves are used to transfer the smallest  $n - k$  disks to peg 2. Then using the usual three-peg Tower of Hanoi algorithm, it takes  $2^k - 1$  moves to transfer the rest of the disks (the largest  $k$  disks) to peg 4, avoiding peg 2. Then again by the inductive hypothesis, it takes  $R(n - k)$  moves to transfer the smallest  $n - k$  disks to peg 4; all the pegs are available for this, because the largest disks, now on peg 4, do not interfere. This establishes the recurrence relation. **43.** First note that  $R(n) = \sum_{j=1}^n [R(j) - R(j - 1)]$  [which follows because the sum is telescoping and  $R(0) = 0$ ]. By Exercise 42, this is the sum of  $2^{k'-1}$  for this range of values of  $j$ . Therefore, the sum is  $\sum_{i=1}^k i2^{i-1}$ , except that if  $n$  is not a triangular number, then the last few values when  $i = k$  are missing, and that is what the final term in the given expression accounts for. **45.** By Exercise 43,  $R(n)$  is no larger than  $\sum_{i=1}^k i2^{i-1}$ . It can be shown that this sum equals  $(k + 1)2^k - 2^{k+1} + 1$ , so it is no greater than  $(k + 1)2^k$ . Because  $n > k(k - 1)/2$ , the quadratic formula can be used to show that  $k < 1 + \sqrt{2n}$  for all  $n > 1$ . Therefore,  $R(n)$  is bounded above by  $(1 + \sqrt{2n} + 1)2^{1+\sqrt{2n}} < 8\sqrt{n}2^{\sqrt{2n}}$  for all  $n > 2$ . Hence,  $R(n)$  is  $O(\sqrt{n}2^{\sqrt{2n}})$ . **47. a)** 0 **b)** 0 **c)** 2 **d)**  $2^{n-1} - 2^{n-2}$  **49.**  $a_n - 2\nabla a_n + \nabla^2 a_n = a_n - 2(a_n - a_{n-1}) + (\nabla a_n - \nabla a_{n-1}) = -a_n + 2a_{n-1} + [(a_n - a_{n-1}) - (a_{n-1} - a_{n-2})] = -a_n + 2a_{n-1} + (a_n - 2a_{n-1} + a_{n-2}) = a_{n-2}$  **51.**  $a_n = a_{n-1} + a_{n-2} = (a_n - \nabla a_n) + (a_n - 2\nabla a_n + \nabla^2 a_n) = 2a_n - 3\nabla a_n + \nabla^2 a_n$ , or  $a_n = 3\nabla a_n - \nabla^2 a_n$  **53.** Insert  $S(0) := \emptyset$  after  $T(0) := 0$  (where  $S(j)$  will record the optimal set of talks among the first  $j$  talks), and replace the statement  $T(j) := \max(w_j + T(p(j)), T(j - 1))$  with the following code:  
**if**  $w_j + T(p(j)) > T(j - 1)$  **then**  
     $T(j) := w_j + T(p(j))$   
     $S(j) := S(p(j)) \cup \{j\}$   
**else**  
     $T(j) := T(j - 1)$   
     $S(j) := S(j - 1)$   
**55. a)** Talks 1, 3, and 7 **b)** Talks 1 and 6, or talks 1, 3, and 7 **c)** Talks 1, 3, and 7 **d)** Talks 1 and 6 **57. a)** This



follows immediately from Example 5 and Exercise 43c in Section 8.4. **b)** The last step in computing  $A_{ij}$  is to multiply  $A_{ik}$  by  $A_{k+1,j}$  for some  $k$  between  $i$  and  $j - 1$  inclusive, which will require  $m_i m_{k+1} m_{j+1}$  integer multiplications, independent of the manner in which  $A_{ik}$  and  $A_{k+1,j}$  are computed. Therefore, to minimize the total number of integer multiplications, each of those two factors must be computed in the most efficient manner. **c)** This follows immediately from part (b) and the definition of  $M(i, j)$ .

**d) procedure** *matrix order*( $m_1, \dots, m_{n+1}$ :

positive integers)

**for**  $i := 1$  **to**  $n$

$M(i, i) := 0$

**for**  $d := 1$  **to**  $n - 1$

**for**  $i := 1$  **to**  $n - d$

$min := 0$

**for**  $k := i$  **to**  $i + d$

$new := M(i, k) + M(k + 1, i + d) + m_i m_{k+1} m_{i+d+1}$

**if**  $new < min$  **then**

$min := new$

$where(i, i + d) := k$

$M(i, i + d) := min$

**e)** The algorithm has three nested loops, each of which is indexed over at most  $n$  values.

## Section 8.2

**1. a)** Degree 3 **b)** No **c)** Degree 4 **d)** No **e)** No **f)** Degree 2 **g)** No **3. a)**  $a_n = 3 \cdot 2^n$  **b)**  $a_n = 2$  **c)**  $a_n = 3 \cdot 2^n - 2 \cdot 3^n$  **d)**  $a_n = 6 \cdot 2^n - 2 \cdot n2^n$  **e)**  $a_n = n(-2)^{n-1}$  **f)**  $a_n = 2^n - (-2)^n$  **g)**  $a_n = (1/2)^{n+1} - (-1/2)^{n+1}$

**5.**  $a_n = \frac{1}{\sqrt{5}} \left( \frac{1+\sqrt{5}}{2} \right)^{n+1} - \frac{1}{\sqrt{5}} \left( \frac{1-\sqrt{5}}{2} \right)^{n+1}$  **7.**  $[2^{n+1} + (-1)^n]/3$

**9. a)**  $P_n = 1.2P_{n-1} + 0.45P_{n-2}$ ,  $P_0 = 100,000$ ,  $P_1 = 120,000$  **b)**  $P_n = (250,000/3)(3/2)^n + (50,000/3)(-3/10)^n$

**11. a)** *Basis step:* For  $n = 1$  we have  $1 = 0 + 1$ , and for  $n = 2$  we have  $3 = 1 + 2$ . *Inductive step:* Assume true for  $k \leq n$ . Then  $L_{n+1} = L_n + L_{n-1} = f_{n-1} + f_{n+1} + f_{n-2} + f_n = (f_{n-1} + f_{n-2}) + (f_{n+1} + f_n) = f_n + f_{n+2}$ .

**b)**  $L_n = \left( \frac{1+\sqrt{5}}{2} \right)^n + \left( \frac{1-\sqrt{5}}{2} \right)^n$

**13.**  $a_n = 8(-1)^n - 3(-2)^n + 4 \cdot 3^n$  **15.**  $a_n = 5 + 3(-2)^n - 3^n$

**17.** Let  $a_n = C(n, 0) + C(n-1, 1) + \dots + C(n-k, k)$  where  $k = \lfloor n/2 \rfloor$ . First, assume that  $n$  is even, so that  $k = n/2$ , and the last term is  $C(k, k)$ . By Pascal's identity we have  $a_n = 1 + C(n-2, 0) + C(n-2, 1) + C(n-3, 1) + C(n-3, 2) + \dots + C(n-k, k-2) + C(n-k, k-1) + 1 = 1 + C(n-2, 1) + C(n-3, 2) + \dots + C(n-k, k-1) + C(n-2, 0) + C(n-3, 1) + \dots + C(n-k, k-2) + 1 = a_{n-1} + a_{n-2}$  because  $\lfloor (n-1)/2 \rfloor = k-1 = \lfloor (n-2)/2 \rfloor$ . A similar calculation works when  $n$  is odd. Hence,  $\{a_n\}$  satisfies the recurrence relation  $a_n = a_{n-1} + a_{n-2}$  for all positive integers  $n$ ,  $n \geq 2$ . Also,  $a_1 = C(1, 0) = 1$  and  $a_2 = C(2, 0) + C(1, 1) = 2$ , which are  $f_2$  and  $f_3$ . It follows that  $a_n = f_{n+1}$  for all positive integers  $n$ . **19.**  $a_n = (n^2 + 3n + 5)(-1)^n$  **21.**  $(a_{1,0} + a_{1,1}n + a_{1,2}n^2 +$

$a_{1,3}n^3) + (a_{2,0} + a_{2,1}n + a_{2,2}n^2)(-2)^n + (a_{3,0} + a_{3,1}n)3^n + a_{4,0}(-4)^n$

**23. a)**  $3a_{n-1} + 2^n = 3(-2)^n + 2^n = 2^n(-3 + 1) = -2^{n+1} = a_n$

**b)**  $a_n = \alpha 3^n - 2^{n+1}$  **c)**  $a_n = 3^{n+1} - 2^{n+1}$  **25. a)**  $A =$

$-1, B = -7$  **b)**  $a_n = \alpha 2^n - n - 7$  **c)**  $a_n = 11 \cdot 2^n - n - 7$

**27. a)**  $p_3 n^3 + p_2 n^2 + p_1 n + p_0$  **b)**  $n^2 p_0 (-2)^n$  **c)**  $n^2 (p_1 n + p_0) 2^n$

**d)**  $(p_2 n^2 + p_1 n + p_0) 4^n$  **e)**  $n^2 (p_2 n^2 + p_1 n + p_0) (-2)^n$

**f)**  $n^2 (p_4 n^4 + p_3 n^3 + p_2 n^2 + p_1 n + p_0) 2^n$  **g)**  $p_0$  **29. a)**  $a_n =$

$\alpha 2^n + 3^{n+1}$  **b)**  $a_n = -2 \cdot 2^n + 3^{n+1}$  **31.**  $a_n =$

$\alpha 2^n + \beta 3^n - n \cdot 2^{n+1} + 3n/2 + 21/4$  **33.**  $a_n = (\alpha + \beta n + n^2 + n^3/6) 2^n$

**35.**  $a_n = -4 \cdot 2^n - n^2/4 - 5n/2 + 1/8 + (39/8)3^n$  **37.**  $a_n =$

$n(n+1)(n+2)/6$  **39. a)**  $1, -1, i, -i$  **b)**  $a_n = \frac{1}{4} - \frac{1}{4}(-1)^n +$

$\frac{2+i}{4}i^n + \frac{2-i}{4}(-i)^n$  **41. a)** Using the formula for  $f_n$ , we see that

$\left| f_n - \frac{1}{\sqrt{5}} \left( \frac{1+\sqrt{5}}{2} \right)^n \right| = \left| \frac{1}{\sqrt{5}} \left( \frac{1-\sqrt{5}}{2} \right)^n \right| < 1/\sqrt{5} < 1/2$ . This

means that  $f_n$  is the integer closest to  $\frac{1}{\sqrt{5}} \left( \frac{1+\sqrt{5}}{2} \right)^n$ . **b)** Less

when  $n$  is even; greater when  $n$  is odd **43.**  $a_n = f_{n-1} + 2f_n - 1$

**45. a)**  $a_n = 3a_{n-1} + 4a_{n-2}$ ,  $a_0 = 2$ ,  $a_1 = 6$  **b)**  $a_n =$

$[4^{n+1} + (-1)^n]/5$  **47. a)**  $a_n = 2a_{n+1} + (n-1)10,000$

**b)**  $a_n = 70,000 \cdot 2^{n-1} - 10,000n - 10,000$  **49.**  $a_n =$

$5n^2/12 + 13n/12 + 1$  **51.** See Chapter 11, Section 5 in

[Ma93]. **53.**  $6^n \cdot 4^{n-1}/n$

## Section 8.3

**1.** 14 **3.** The first step is  $(1110)_2(1010)_2 = (2^4 + 2^2)(11)_2(10)_2 + 2^2[(11)_2 - (10)_2][(10)_2 - (10)_2] + (2^2 + 1)(10)_2 \cdot (10)_2$ . The product is  $(10001100)_2$ .

**5.**  $C = 50,665C + 729 = 33,979$  **7. a)** 2 **b)** 4

**c)** 7 **9. a)** 79 **b)** 48,829 **c)** 30,517,579 **11.**  $O(\log n)$

**13.**  $O(n^{\log_3 2})$  **15.** 5 **17. a)** *Basis step:* If the sequence has

just one element, then the one person on the list is the winner.

*Recursive step:* Divide the list into two parts—the first half and the second half—as equally as possible. Apply the algorithm recursively to each half to come up with at most two

names. Then run through the entire list to count the number of occurrences of each of those names to decide which, if either, is the winner. **b)**  $O(n \log n)$  **19. a)**  $f(n) = f(n/2) + 2$

**b)**  $O(\log n)$  **21. a)** 7 **b)**  $O(\log n)$

**23. a)** **procedure** *largest sum*( $a_1, \dots, a_n$ )

$best := 0$  {empty subsequence has sum 0}

**for**  $i := 1$  **to**  $n$

$sum := 0$

**for**  $j := i + 1$  **to**  $n$

$sum := sum + a_j$

**if**  $sum > best$  **then**  $best := sum$

{ $best$  is the maximum possible sum of numbers in the list}

**b)**  $O(n^2)$  **c)** We divide the list into a first half and a second half and apply the algorithm recursively to find the largest sum of consecutive terms for each half. The largest sum of consecutive terms in the entire sequence is either one of these two numbers or the sum of a sequence of consecutive terms that crosses the middle of the list. To find the largest possible sum of a sequence of consecutive terms that crosses the middle of the list, we start at the middle and move forward to find the largest possible sum in the second half of the list,



and move backward to find the largest possible sum in the first half of the list; the desired sum is the sum of these two quantities. The final answer is then the largest of this sum and the two answers obtained recursively. The base case is that the largest sum of a sequence of one term is the larger of that number and 0. **d)** 11, 9, 14 **e)**  $S(n) = 2S(n/2) + n$ ,  $C(n) = 2C(n/2) + n + 2$ ,  $S(1) = 0$ ,  $C(1) = 1$  **f)**  $O(n \log n)$ , better than  $O(n^2)$  **25.** (1, 6) and (3, 6) at distance 2 **27.** The algorithm is essentially the same as the algorithm given in Example 12. The central strip still has width  $2d$  but we need to consider just two boxes of size  $d \times d$  rather than eight boxes of size  $(d/2) \times (d/2)$ . The recurrence relation is the same as the recurrence relation in Example 12, except that the coefficient 7 is replaced by 1. **29.** With  $k = \log_b n$ , it follows that  $f(n) = a^k f(1) + \sum_{j=0}^{k-1} a^j c(n/b^j)^d = a^k f(1) + \sum_{j=0}^{k-1} cn^d = a^k f(1) + kcn^d = a^{\log_b n} f(1) + c(\log_b n)n^d = n^{\log_b a} f(1) + cn^d \log_b n = n^d f(1) + cn^d \log_b n$ . **31.** Let  $k = \log_b n$  where  $n$  is a power of  $b$ . *Basis step:* If  $n = 1$  and  $k = 0$ , then  $c_1 n^d + c_2 n^{\log_b a} = c_1 + c_2 = b^d c / (b^d - a) + f(1) + b^d c / (a - b^d) = f(1)$ . *Inductive step:* Assume true for  $k$ , where  $n = b^k$ . Then for  $n = b^{k+1}$ ,  $f(n) = af(n/b) + cn^d = a\{[b^d c / (b^d - a)](n/b)^d + [f(1) + b^d c / (a - b^d)] \cdot (n/b)^{\log_b a}\} + cn^d = b^d c / (b^d - a) n^d a / b^d + [f(1) + b^d c / (a - b^d)] n^{\log_b a} + cn^d = n^d [ac / (b^d - a) + c(b^d - a) / (b^d - a)] + [f(1) + b^d c / (a - b^d c)] n^{\log_b a} = [b^d c / (b^d - a)] n^d + [f(1) + b^d c / (a - b^d c)] n^{\log_b a}$ . **33.** If  $a > b^d$ , then  $\log_b a > d$ , so the second term dominates, giving  $O(n^{\log_b a})$ . **35.**  $O(n^{\log_4 5})$  **37.**  $O(n^3)$

## Section 8.4

**1.**  $f(x) = 2(x^6 - 1)/(x - 1)$  **3. a)**  $f(x) = 2x(1 - x^6)/(1 - x)$  **b)**  $x^3/(1 - x)$  **c)**  $x/(1 - x^3)$  **d)**  $2/(1 - 2x)$  **e)**  $(1 + x)^7$  **f)**  $2/(1 + x)$  **g)**  $[1/(1 - x)] - x^2$  **h)**  $x^3/(1 - x)^2$  **5. a)**  $5/(1 - x)$  **b)**  $1/(1 - 3x)$  **c)**  $2x^3/(1 - x)$  **d)**  $(3 - x)/(1 - x)^2$  **e)**  $(1 + x)^8$  **7. a)**  $a_0 = -64$ ,  $a_1 = 144$ ,  $a_2 = -108$ ,  $a_3 = 27$ , and  $a_n = 0$  for all  $n \geq 4$  **b)** The only nonzero coefficients are  $a_0 = 1$ ,  $a_3 = 3$ ,  $a_6 = 3$ ,  $a_9 = 1$ . **c)**  $a_n = 5^n$  **d)**  $a_n = (-3)^{n-3}$  for  $n \geq 3$ , and  $a_0 = a_1 = a_2 = 0$  **e)**  $a_0 = 8$ ,  $a_1 = 3$ ,  $a_2 = 2$ ,  $a_n = 0$  for odd  $n$  greater than 2 and  $a_n = 1$  for even  $n$  greater than 2 **f)**  $a_n = 1$  if  $n$  is a positive multiple 4,  $a_n = -1$  if  $n < 4$ , and  $a_n = 0$  otherwise **g)**  $a_n = n - 1$  for  $n \geq 2$  and  $a_0 = a_1 = 0$  **h)**  $a_n = 2^{n+1}/n!$  **9. a)** 6 **b)** 3 **c)** 9 **d)** 0 **e)** 5 **11. a)** 1024 **b)** 11 **c)** 66 **d)** 292,864 **e)** 20,412 **13.** 10 **15.** 50 **17.** 20 **19.**  $f(x) = 1/[(1 - x)(1 - x^2)(1 - x^5)(1 - x^{10})]$  **21.** 15 **23. a)**  $x^4(1 + x + x^2 + x^3)^2/(1 - x)$  **b)** 6 **25. a)** The coefficient of  $x^r$  in the power series expansion of  $1/[(1 - x^3)(1 - x^4)(1 - x^{20})]$  **b)**  $1/(1 - x^3 - x^4 - x^{20})$  **c)** 7 **d)** 3224 **27. a)** The generating function is  $(1 + x + x^2 + x^3 + x^4)(1 + x + x^2)(1 + x^2 + x^4 + x^6 + \dots)(x^3 + x^4 + x^5 + x^6 + \dots)(1 + x^5 + x^{10} + x^{15} + \dots) = x^3(1 + x + x^2 + x^3 + x^4)(1 + x + x^2)/[(1 - x^2)(1 - x)(1 - x^5)] = x^3 + 3x^4 + 7x^5 + 12x^6 + 19x^7 + 27x^8 + 37x^9 + 48x^{10} + 61x^{11} + 75x^{12} + \dots$ . The coefficient of  $x^n$  is the answer. **b)** 75 **29. a)** 3 **b)** 29 **c)** 29 **d)** 242 **31. a)** 10 **b)** 49 **c)** 2

**d)** 4 **33. a)**  $G(x) = a_0 - a_1x - a_2x^2$  **b)**  $G(x^2)$  **c)**  $x^4G(x)$  **d)**  $G(2x)$  **e)**  $\int_0^x G(t)dt$  **f)**  $G(x)/(1 - x)$  **35.**  $a_k = 2 \cdot 3^k - 1$  **37.**  $a_k = 18 \cdot 3^k - 12 \cdot 2^k$  **39.**  $a_k = k^2 + 8k + 20 + (6k - 18)2^k$  **41.** Let  $G(x) = \sum_{k=0}^{\infty} f_k x^k$ . After shifting indices of summation and adding series, we see that  $G(x) - xG(x) - x^2G(x) = f_0 + (f_1 - f_0)x + \sum_{k=2}^{\infty} (f_k - f_{k-1} - f_{k-2})x^k = 0 + x + \sum_{k=2}^{\infty} 0x^k$ . Hence,  $G(x) - xG(x) - x^2G(x) = x$ . Solving for  $G(x)$  gives  $G(x) = x/(1 - x - x^2)$ . By the method of partial fractions, it can be shown that  $x/(1 - x - x^2) = (1/\sqrt{5})[1/(1 - \alpha x) - 1/(1 - \beta x)]$ , where  $\alpha = (1 + \sqrt{5})/2$  and  $\beta = (1 - \sqrt{5})/2$ . Using the fact that  $1/(1 - \alpha x) = \sum_{k=0}^{\infty} \alpha^k x^k$ , it follows that  $G(x) = (1/\sqrt{5}) \cdot \sum_{k=0}^{\infty} (\alpha^k - \beta^k) x^k$ . Hence,  $f_k = (1/\sqrt{5}) \cdot (\alpha^k - \beta^k)$ . **43. a)** Let  $G(x) = \sum_{n=0}^{\infty} C_n x^n$  be the generating function for  $\{C_n\}$ . Then  $G(x)^2 = \sum_{n=0}^{\infty} (\sum_{k=0}^n C_k C_{n-k}) x^n = \sum_{n=1}^{\infty} (\sum_{k=0}^{n-1} C_k C_{n-1-k}) x^{n-1} = \sum_{n=1}^{\infty} C_n x^{n-1}$ . Hence,  $xG(x)^2 = \sum_{n=1}^{\infty} C_n x^n$ , which implies that  $xG(x)^2 - G(x) + 1 = 0$ . Applying the quadratic formula shows that  $G(x) = \frac{1 \pm \sqrt{1-4x}}{2x}$ . We choose the minus sign in this formula because the choice of the plus sign leads to a division by zero. **b)** By Exercise 42,  $(1 - 4x)^{-1/2} = \sum_{n=0}^{\infty} \binom{2n}{n} x^n$ . Integrating term by term (which is valid by a theorem from calculus) shows that  $\int_0^x (1 - 4t)^{-1/2} dt = \sum_{n=0}^{\infty} \frac{1}{n+1} \binom{2n}{n} x^{n+1} = x \sum_{n=0}^{\infty} \frac{1}{n+1} \binom{2n}{n} x^n$ . Because  $\int_0^x (1 - 4t)^{-1/2} dt = \frac{1 - \sqrt{1-4x}}{2} = xG(x)$ , equating coefficients shows that  $C_n = \frac{1}{n+1} \binom{2n}{n}$ . **c)** Verify the basis step for  $n = 1, 2, 3, 4, 5$ . Assume the inductive hypothesis that  $C_j \geq 2^{j-1}$  for  $1 \leq j < n$ , where  $n \geq 6$ . Then  $C_n = \sum_{k=0}^{n-1} C_k C_{n-k-1} \geq \sum_{k=1}^{n-2} C_k C_{n-k-1} \geq (n-2)2^{k-1}2^{n-k-2} = (n-2)2^{n-1}/4 \geq 2^{n-1}$ . **45.** Applying the binomial theorem to the equality  $(1 + x)^{m+n} = (1 + x)^m(1 + x)^n$ , shows that  $\sum_{r=0}^{m+n} C(m+n, r)x^r = \sum_{r=0}^m C(m, r)x^r \cdot \sum_{r=0}^n C(n, r)x^r = \sum_{r=0}^{m+n} \left[ \sum_{k=0}^r C(m, r-k)C(n, k) \right] x^r$ . Comparing coefficients gives the desired identity. **47. a)**  $2e^x$  **b)**  $e^{-x}$  **c)**  $e^{3x}$  **d)**  $xe^x + e^x$  **49. a)**  $a_n = (-1)^n$  **b)**  $a_n = 3 \cdot 2^n$  **c)**  $a_n = 3^n - 3 \cdot 2^n$  **d)**  $a_n = (-2)^n$  for  $n \geq 2$ ,  $a_1 = -3$ ,  $a_0 = 2$  **e)**  $a_n = (-2)^n + n!$  **f)**  $a_n = (-3)^n + n! \cdot 2^n$  for  $n \geq 2$ ,  $a_0 = 1$ ,  $a_1 = -2$  **g)**  $a_n = 0$  if  $n$  is odd and  $a_n = n!/(n/2)!$  if  $n$  is even **51. a)**  $a_n = 6a_{n-1} + 8^{n-1}$  for  $n \geq 1$ ,  $a_0 = 1$  **b)** The general solution of the associated linear homogeneous recurrence relation is  $a_n^{(h)} = \alpha 6^n$ . A particular solution is  $a_n^{(p)} = \frac{1}{2} \cdot 8^n$ . Hence, the general solution is  $a_n = \alpha 6^n + \frac{1}{2} \cdot 8^n$ . Using the initial condition, it follows that  $\alpha = \frac{1}{2}$ . Hence,  $a_n = (6^n + 8^n)/2$ . **c)** Let  $G(x) = \sum_{k=0}^{\infty} a_k x^k$ . Using the recurrence relation for  $\{a_k\}$ , it can be shown that  $G(x) - 6xG(x) = (1 - 7x)/(1 - 8x)$ . Hence,  $G(x) = (1 - 7x)/[(1 - 6x)(1 - 8x)]$ . Using partial fractions, it follows that  $G(x) = (1/2)/(1 - 6x) + (1/2)/(1 - 8x)$ . With the help of Table 1, it follows that  $a_n = (6^n + 8^n)/2$ . **53.**  $\frac{1}{1-x} \cdot \frac{1}{1-x^2} \cdot \frac{1}{1-x^3} \cdots$  **55.**  $(1+x)(1+x)^2(1+x)^3 \cdots$  **57.** The generating functions obtained in Exercises 54 and 55 are equal because  $(1+x)(1+x^2)(1+x^3) \cdots = \frac{1-x^2}{1-x} \cdot \frac{1-x^4}{1-x^2} \cdot \frac{1-x^6}{1-x^3} \cdots = \frac{1}{1-x} \cdot \frac{1}{1-x^3} \cdot \frac{1}{1-x^5} \cdots$ . **59. a)**  $G_X(1) = \sum_{k=0}^{\infty} p(X = k) \cdot 1^k = \sum_{k=0}^{\infty} p(X = k) = 1$  **b)**  $G'_X(1) = \frac{d}{dx} \sum_{k=0}^{\infty} p(X = k) \cdot x^k \Big|_{x=1} = \sum_{k=0}^{\infty} p(X = k) \cdot k \cdot x^{k-1} \Big|_{x=1} = \sum_{k=0}^{\infty} p(X = k) \cdot k = E(X)$



c)  $G_X''(1) = \frac{d^2}{dx^2} \sum_{k=0}^{\infty} p(X = k) \cdot x^k|_{x=1} = \sum_{k=0}^{\infty} p(X = k) \cdot k(k-1) \cdot x^{k-2}|_{x=1} = \sum_{k=0}^{\infty} p(X = k) \cdot (k^2 - k) = V(X) + E(X)^2 - E(X)$ . Combining this with part (b) gives the desired results. **61. a)**  $G(x) = p^m/(1 - qx)^m$  **b)**  $V(x) = mq/p^2$

## Section 8.5

**1. a)** 30 **b)** 29 **c)** 24 **d)** 18 **3.** 1% **5. a)** 300 **b)** 150  
**c)** 175 **d)** 100 **7.** 492 **9.** 974 **11.** 610 **13.** 55 **15.** 248  
**17.** 50,138 **19.** 234 **21.**  $|A_1 \cup A_2 \cup A_3 \cup A_4 \cup A_5| = |A_1| + |A_2| + |A_3| + |A_4| + |A_5| - |A_1 \cap A_2| - |A_1 \cap A_3| - |A_1 \cap A_4| - |A_1 \cap A_5| - |A_2 \cap A_3| - |A_2 \cap A_4| - |A_2 \cap A_5| - |A_3 \cap A_4| - |A_3 \cap A_5| - |A_4 \cap A_5| + |A_1 \cap A_2 \cap A_3| + |A_1 \cap A_2 \cap A_4| + |A_1 \cap A_2 \cap A_5| + |A_1 \cap A_3 \cap A_4| + |A_1 \cap A_3 \cap A_5| + |A_1 \cap A_4 \cap A_5| + |A_2 \cap A_3 \cap A_4| + |A_2 \cap A_3 \cap A_5| + |A_2 \cap A_4 \cap A_5| + |A_3 \cap A_4 \cap A_5| - |A_1 \cap A_2 \cap A_3 \cap A_4| - |A_1 \cap A_2 \cap A_3 \cap A_5| - |A_1 \cap A_2 \cap A_4 \cap A_5| - |A_1 \cap A_3 \cap A_4 \cap A_5| - |A_2 \cap A_3 \cap A_4 \cap A_5| + |A_1 \cap A_2 \cap A_3 \cap A_4 \cap A_5|$   
**23.**  $|A_1 \cup A_2 \cup A_3 \cup A_4 \cup A_5 \cup A_6| = |A_1| + |A_2| + |A_3| + |A_4| + |A_5| + |A_6| - |A_1 \cap A_2| - |A_1 \cap A_3| - |A_1 \cap A_4| - |A_1 \cap A_5| - |A_1 \cap A_6| - |A_2 \cap A_3| - |A_2 \cap A_4| - |A_2 \cap A_5| - |A_2 \cap A_6| - |A_3 \cap A_4| - |A_3 \cap A_5| - |A_3 \cap A_6| - |A_4 \cap A_5| - |A_4 \cap A_6| - |A_5 \cap A_6|$   
**25.**  $p(E_1 \cup E_2 \cup E_3) = p(E_1) + p(E_2) + p(E_3) - p(E_1 \cap E_2) - p(E_1 \cap E_3) - p(E_2 \cap E_3) + p(E_1 \cap E_2 \cap E_3)$  **27.** 4972/71,295  
**29.**  $p(E_1 \cup E_2 \cup E_3 \cup E_4 \cup E_5) = p(E_1) + p(E_2) + p(E_3) + p(E_4) + p(E_5) - p(E_1 \cap E_2) - p(E_1 \cap E_3) - p(E_1 \cap E_4) - p(E_1 \cap E_5) - p(E_2 \cap E_3) - p(E_2 \cap E_4) - p(E_2 \cap E_5) - p(E_3 \cap E_4) - p(E_3 \cap E_5) - p(E_4 \cap E_5) + p(E_1 \cap E_2 \cap E_3) + p(E_1 \cap E_2 \cap E_4) + p(E_1 \cap E_2 \cap E_5) + p(E_1 \cap E_3 \cap E_4) + p(E_1 \cap E_3 \cap E_5) + p(E_1 \cap E_4 \cap E_5) + p(E_2 \cap E_3 \cap E_4) + p(E_2 \cap E_3 \cap E_5) + p(E_2 \cap E_4 \cap E_5) + p(E_3 \cap E_4 \cap E_5)$   
**31.**  $p\left(\bigcup_{i=1}^n E_i\right) = \sum_{1 \leq i \leq n} p(E_i) - \sum_{1 \leq i < j \leq n} p(E_i \cap E_j) + \sum_{1 \leq i < j < k \leq n} p(E_i \cap E_j \cap E_k) - \cdots + (-1)^{n+1} p\left(\bigcap_{i=1}^n E_i\right)$

## Section 8.6

**1.** 75 **3.** 6 **5.** 46 **7.** 9875 **9.** 540 **11.** 2100 **13.** 1854  
**15. a)**  $D_{100}/100!$  **b)**  $100D_{99}/100!$  **c)**  $C(100,2)/100!$   
**d)** 0 **e)**  $1/100!$  **17.** 2,170,680 **19.** By Exercise 18 we have  $D_n - nD_{n-1} = -[D_{n-1} - (n-1)D_{n-2}]$ . Iterating, we have  $D_n - nD_{n-1} = -[D_{n-1} - (n-1)D_{n-2}] = -[-(D_{n-2} - (n-2)D_{n-3})] = D_{n-2} - (n-2)D_{n-3} = \cdots = (-1)^n(D_2 - 2D_1) = (-1)^n$  because  $D_2 = 1$  and  $D_1 = 0$ . **21.** When  $n$  is odd  
**23.**  $\phi(n) = n - \sum_{i=1}^m \frac{n}{p_i} + \sum_{1 \leq i < j \leq m} \frac{n}{p_i p_j} - \cdots \pm \frac{n}{p_1 p_2 \cdots p_m} = n \prod_{i=1}^m \left(a - \frac{1}{p_i}\right)$  **25.** 4 **27.** There are  $n^m$  functions from a set with  $m$  elements to a set with  $n$  elements,  $C(n, 1)(n-1)^m$  functions from a set with  $m$  elements to a set with  $n$  elements that miss exactly one element,  $C(n, 2)(n-2)^m$  functions from a set with  $m$  elements to a set with  $n$  elements that miss exactly two elements, and so on, with  $C(n, n-1) \cdot 1^m$  functions from a set with  $m$  elements to a set with  $n$  elements that miss exactly  $n-1$  elements. Hence, by the principle of inclusion-exclusion, there are  $n^m - C(n, 1)(n-1)^m + C(n, 2)(n-2)^m - \cdots + (-1)^{n-1} C(n, n-1) \cdot 1^m$  onto functions.

## Supplementary Exercises

**1. a)**  $A_n = 4A_{n-1}$  **b)**  $A_1 = 40$  **c)**  $A_n = 10 \cdot 4^n$   
**3. a)**  $M_n = M_{n-1} + 160,000$  **b)**  $M_1 = 186,000$  **c)**  $M_n = 160,000n + 26,000$  **d)**  $T_n = T_{n-1} + 160,000n + 26,000$   
**e)**  $T_n = 80,000n^2 + 106,000n$  **5. a)**  $a_n = a_{n-2} + a_{n-3}$   
**b)**  $a_1 = 0, a_2 = 1, a_3 = 1$  **c)**  $a_{12} = 12$  **7. a)** 2 **b)** 5  
**c)** 8 **d)** 16 **9.**  $a_n = 2^n$  **11.**  $a_n = 2 + 4n/3 + n^2/2 + n^3/6$   
**13.**  $a_n = a_{n-2} + a_{n-3}$  **15. a)** Under the given conditions, one longest common subsequence ends at the last term in each sequence, so  $a_m = b_n = c_p$ . Furthermore, a longest common subsequence of what is left of the  $a$ -sequence and the  $b$ -sequence after those last terms are deleted has to form the beginning of a longest common subsequence of the original sequences. **b)** If  $c_p \neq a_m$ , then the longest common subsequence's appearance in the  $a$ -sequence must terminate before the end; therefore, the  $c$ -sequence must be a longest common subsequence of  $a_1, a_2, \dots, a_{m-1}$  and  $b_1, b_2, \dots, b_n$ . The other half is similar.  
**17. procedure** *howlong*( $a_1, \dots, a_m, b_1, \dots, b_n$ : sequences)  
  **for**  $i := 1$  **to**  $m$   
     $L(i, 0) := 0$   
  **for**  $j := 1$  **to**  $n$   
     $L(0, j) := 0$   
  **for**  $i := 1$  **to**  $m$   
    **for**  $j := 1$  **to**  $n$   
      **if**  $a_i = b_j$  **then**  $L(i, j) := L(i-1, j-1) + 1$   
      **else**  $L(i, j) := \max(L(i, j-1), L(i-1, j))$   
  **return**  $L(m, n)$   
**19.**  $f(n) = (4n^2 - 1)/3$  **21.**  $O(n^4)$  **23.**  $O(n)$  **25.** Using just two comparisons, the algorithm is able to narrow the search for  $m$  down to the first half or the second half of the original sequence. Since the length of the sequence is cut in half each time, only about  $2 \log_2 n$  comparisons are needed in all. **27. a)**  $18n + 18$  **b)** 18 **c)** 0 **29.**  $\Delta(a_n b_n) = a_{n+1} b_{n+1} - a_n b_n = a_{n+1}(b_{n+1} - b_n) + b_n(a_{n+1} - a_n) = a_{n+1} \Delta b_n + b_n \Delta a_n$  **31. a)** Let  $G(x) = \sum_{n=0}^{\infty} a_n x^n$ . Then  $G'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1} = \sum_{n=0}^{\infty} (n+1) a_{n+1} x^n$ . Therefore,  $G'(x) - G(x) = \sum_{n=0}^{\infty} [(n+1) a_{n+1} - a_n] x^n = \sum_{n=0}^{\infty} x^n / n! = e^x$ , as desired. That  $G(0) = a_0 = 1$  is given. **b)** We have  $[e^{-x} G(x)]' = e^{-x} G'(x) - e^{-x} G(x) = e^{-x} [G'(x) - G(x)] = e^{-x} \cdot e^x = 1$ . Hence,  $e^{-x} G(x) = x + c$ , where  $c$  is a constant. Consequently,  $G(x) = x e^x + c e^x$ . Because  $G(0) = 1$ , it follows that  $c = 1$ . **c)** We have  $G(x) = \sum_{n=0}^{\infty} x^{n+1} / n! + \sum_{n=0}^{\infty} x^n / n! = \sum_{n=1}^{\infty} x^n / (n-1)! + \sum_{n=0}^{\infty} x^n / n!$ . Therefore,  $a_n = 1/(n-1)! + 1/n!$  for all  $n \geq 1$ , and  $a_0 = 1$ . **33.** 7  
**35.** 110 **37.** 0 **39. a)** 19 **b)** 65 **c)** 122 **d)** 167 **e)** 168  
**41.**  $D_{n-1}/(n-1)!$  **43.** 11/32



## CHAPTER 9

### Section 9.1

1. a)  $\{(0, 0), (1, 1), (2, 2), (3, 3)\}$  b)  $\{(1, 3), (2, 2), (3, 1), (4, 0)\}$  c)  $\{(1, 0), (2, 0), (2, 1), (3, 0), (3, 1), (3, 2), (4, 0), (4, 1), (4, 2), (4, 3)\}$  d)  $\{(1, 0), (1, 1), (1, 2), (1, 3), (2, 0), (2, 2), (3, 0), (3, 3), (4, 0)\}$  e)  $\{(0, 1), (1, 0), (1, 1), (1, 2), (1, 3), (2, 1), (2, 3), (3, 1), (3, 2), (4, 1), (4, 3)\}$  f)  $\{(1, 2), (2, 1), (2, 2)\}$  3. a) Transitive b) Reflexive, symmetric, transitive c) Symmetric d) Antisymmetric e) Reflexive, symmetric, antisymmetric, transitive f) None of these properties 5. a) Reflexive, transitive b) Symmetric c) Symmetric d) Symmetric 7. a) Symmetric b) Symmetric, transitive c) Symmetric d) Reflexive, symmetric, transitive e) Reflexive, transitive f) Reflexive, symmetric, transitive g) Antisymmetric h) Antisymmetric, transitive 9. Each of the three properties is vacuously satisfied. 11. (c), (d), (f) 13. a) Not irreflexive b) Not irreflexive c) Not irreflexive d) Not irreflexive 15. Yes, for instance  $\{(1, 1)\}$  on  $\{1, 2\}$  17.  $(a, b) \in R$  if and only if  $a$  is taller than  $b$  19. (a) 21. None 23.  $\forall a \forall b [(a, b) \in R \rightarrow (b, a) \notin R]$  25.  $2^{mn}$  27. a)  $\{(a, b) \mid b \text{ divides } a\}$  b)  $\{(a, b) \mid a \text{ does not divide } b\}$  29. The graph of  $f^{-1}$  31. a)  $\{(a, b) \mid a \text{ is required to read or has read } b\}$  b)  $\{(a, b) \mid a \text{ is required to read and has read } b\}$  c)  $\{(a, b) \mid \text{either } a \text{ is required to read } b \text{ but has not read it or } a \text{ has read } b \text{ but is not required to}\}$  d)  $\{(a, b) \mid a \text{ is required to read } b \text{ but has not read it}\}$  e)  $\{(a, b) \mid a \text{ has read } b \text{ but is not required to}\}$  33.  $S \circ R = \{(a, b) \mid a \text{ is a parent of } b \text{ and } b \text{ has a sibling}\}$ ,  $R \circ S = \{(a, b) \mid a \text{ is an aunt or uncle of } b\}$  35. a)  $R^2$  b)  $R_6$  c)  $R_3$  d)  $R_3$  e)  $\emptyset$  f)  $R_1$  g)  $R_4$  h)  $R_4$  37. a)  $R_1$  b)  $R_2$  c)  $R_3$  d)  $R^2$  e)  $R_3$  f)  $R^2$  g)  $R^2$  h)  $R^2$  39.  $S_1^2 = \{(a, b) \in \mathbb{Z}^2 \mid a > b + 1\}$ ,  $S_2^2 = S_2$ ,  $S_3^2 = \{(a, b) \in \mathbb{Z}^2 \mid a < b - 1\}$ ,  $S_4^2 = S_4$ ,  $S_5^2 = S_5$ ,  $S_6^2 = \mathbb{Z}^2$  41.  $b$  got his or her doctorate under someone who got his or her doctorate under  $a$ ; there is a sequence of  $n + 1$  people, starting with  $a$  and ending with  $b$ , such that each is the advisor of the next person in the sequence 43. a)  $\{(a, b) \mid a - b \equiv 0, 3, 4, 6, 8, \text{ or } 9 \pmod{12}\}$  b)  $\{(a, b) \mid a \equiv b \pmod{12}\}$  c)  $\{(a, b) \mid a - b \equiv 3, 6, \text{ or } 9 \pmod{12}\}$  d)  $\{(a, b) \mid a - b \equiv 4 \text{ or } 8 \pmod{12}\}$  e)  $\{(a, b) \mid a - b \equiv 3, 4, 6, 8, \text{ or } 9 \pmod{12}\}$  45. 8 47. a) 65,536 b) 32,768 49. a)  $2^{n(n+1)/2}$  b)  $2^n 3^{n(n-1)/2}$  c)  $3^{n(n-1)/2}$  d)  $2^{n(n-1)}$  e)  $2^{n(n-1)/2}$  f)  $2^{n^2} - 2 \cdot 2^{n(n-1)}$  51. There may be no such  $b$ . 53. If  $R$  is symmetric and  $(a, b) \in R$ , then  $(b, a) \in R$ , so  $(a, b) \in R^{-1}$ . Hence,  $R \subseteq R^{-1}$ . Similarly,  $R^{-1} \subseteq R$ . So  $R = R^{-1}$ . Conversely, if  $R = R^{-1}$  and  $(a, b) \in R$ , then  $(a, b) \in R^{-1}$ , so  $(b, a) \in R$ . Thus,  $R$  is symmetric. 55.  $R$  is reflexive if and only if  $(a, a) \in R$  for all  $a \in A$  if and only if  $(a, a) \in R^{-1}$  [because  $(a, a) \in R$  if and only if  $(a, a) \in R^{-1}$ ] if and only if  $R^{-1}$  is reflexive. 57. Use mathematical induction. The result is trivial for  $n = 1$ . Assume  $R^n$  is reflexive and transitive. By Theorem 1,  $R^{n+1} \subseteq R$ . To see that  $R \subseteq R^{n+1} = R^n \circ R$ , let  $(a, b) \in R$ . By the inductive hypothesis,  $R^n = R$  and hence, is reflexive. Thus,  $(b, b) \in R^n$ . Therefore,

$(a, b) \in R^{n+1}$ . 59. Use mathematical induction. The result is trivial for  $n = 1$ . Assume  $R^n$  is reflexive. Then  $(a, a) \in R^n$  for all  $a \in A$  and  $(a, a) \in R$ . Thus,  $(a, a) \in R^n \circ R = R^{n+1}$  for all  $a \in A$ . 61. No, for instance, take  $R = \{(1, 2), (2, 1)\}$ .

### Section 9.2

1.  $\{(1, 2, 3), (1, 2, 4), (1, 3, 4), (2, 3, 4)\}$  3. (Nadir, 122, 34, Detroit, 08:10), (Acme, 221, 22, Denver, 08:17), (Acme, 122, 33, Anchorage, 08:22), (Acme, 323, 34, Honolulu 08:30), (Nadir, 199, 13, Detroit, 08:47), (Acme, 222, 22, Denver, 09:10), (Nadir, 322, 34, Detroit, 09:44) 5. Airline and flight number, airline and departure time 7. a) Yes b) No c) No 9. a) Social Security number b) There are no two people with the same name who happen to have the same street address. c) There are no two people with the same name living together. 11. (Nadir, 122, 34, Detroit, 08:10), (Nadir, 199, 13, Detroit, 08:47), (Nadir, 322, 34, Detroit, 09:44) 13. (Nadir, 122, 34, Detroit, 08:10), (Nadir, 199, 13, Detroit, 08:47), (Nadir, 322, 34, Detroit, 09:44), (Acme, 221, 22, Denver, 08:17), (Acme, 222, 22, Denver, 09:10) 15.  $P_{3.5.6}$

17.

| Airline | Destination |
|---------|-------------|
| Nadir   | Detroit     |
| Acme    | Denver      |
| Acme    | Anchorage   |
| Acme    | Honolulu    |

19.

| Supplier | Part number | Project | Quantity | Color code |
|----------|-------------|---------|----------|------------|
| 23       | 1092        | 1       | 2        | 2          |
| 23       | 1101        | 3       | 1        | 1          |
| 23       | 9048        | 4       | 12       | 2          |
| 31       | 4975        | 3       | 6        | 2          |
| 31       | 3477        | 2       | 25       | 2          |
| 32       | 6984        | 4       | 10       | 1          |
| 32       | 9191        | 2       | 80       | 4          |
| 33       | 1001        | 1       | 14       | 8          |

21. Both sides of this equation pick out the subset of  $R$  consisting of those  $n$ -tuples satisfying both conditions  $C_1$  and  $C_2$ . 23. Both sides of this equation pick out the set of  $n$ -tuples that are in  $R$ , are in  $S$ , and satisfy condition  $C$ . 25. Both sides of this equation pick out the  $m$ -tuples consisting of  $i_1$ th,  $i_2$ th,  $\dots$ ,  $i_m$ th components of  $n$ -tuples in either  $R$  or  $S$ . 27. Let  $R = \{(a, b)\}$  and  $S = \{(a, c)\}$ ,  $n = 2$ ,  $m = 1$ , and  $i_1 = 1$ ;  $P_1(R - S) = \{(a)\}$ , but  $P_1(R) - P_1(S) = \emptyset$ . 29. a)  $J_2$  followed by  $P_{1,3}$  b)  $(23, 1), (23, 3), (31, 3), (32, 4)$  31. There is no primary key. 33. a) count 5, support 5/6 b)  $\{\text{diapers}\}$ ,  $\{\text{milk}\}$ , and  $\{\text{diapers, milk}\}$  35. support 1/3, confidence 2/3 37. 1 39. a)  $P(I) = \sigma(I)/|T|$ ,  $P(J) = \sigma(J)/|T|$ ,  $P(I \text{ and } J) = \sigma(I \cup J)/|T|$ , so  $P(I \text{ and } J) = P(I) \cdot P(J)$  if and only if  $\sigma(I \cup J)/|T| = (\sigma(I)/|T|) \cdot (\sigma(J)/|T|)$ , which means that the numerator and denominator of the lift are equal. b) 4/5 c) 16/15 41. For each of the  $n$  items, there are 3 choices: the item can be in  $I$  or in  $J$  or in neither. By the product rule, there are  $3^n$  ways to make these decisions.



## Section 9.3

1. a)  $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$  b)  $\begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

c)  $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$  d)  $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$

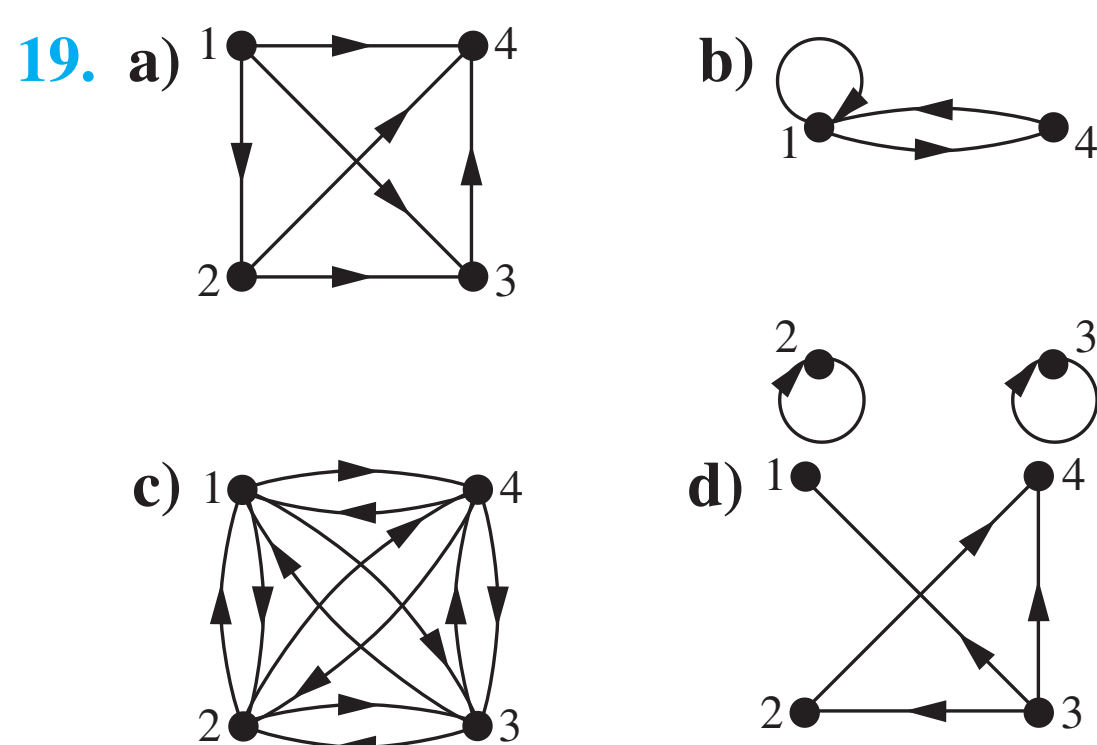
3. a) (1, 1), (1, 3), (2, 2), (3, 1), (3, 3) b) (1, 2), (2, 2), (3, 2) c) (1, 1), (1, 2), (1, 3), (2, 1), (2, 3), (3, 1), (3, 2), (3, 3)

5. The relation is irreflexive if and only if the main diagonal of the matrix contains only 0s. 7. a) Reflexive, symmetric, transitive b) Antisymmetric, transitive c) Symmetric 9. a) 4950 b) 9900 c) 99 d) 100 e) 1 11. Change each 0 to a 1 and each 1 to a 0.

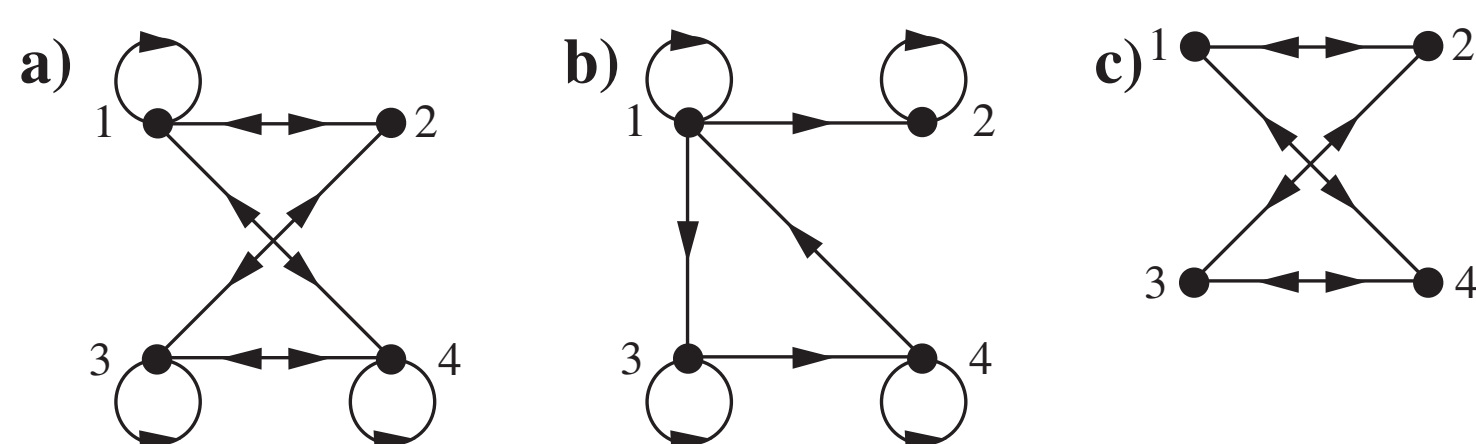
13. a)  $\begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$  b)  $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$  c)  $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

15. a)  $\begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$  b)  $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$  c)  $\begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

17.  $n^2 - k$



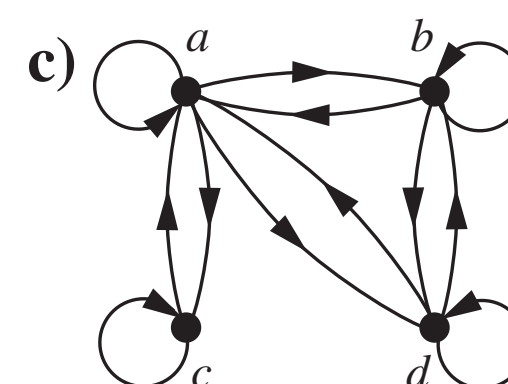
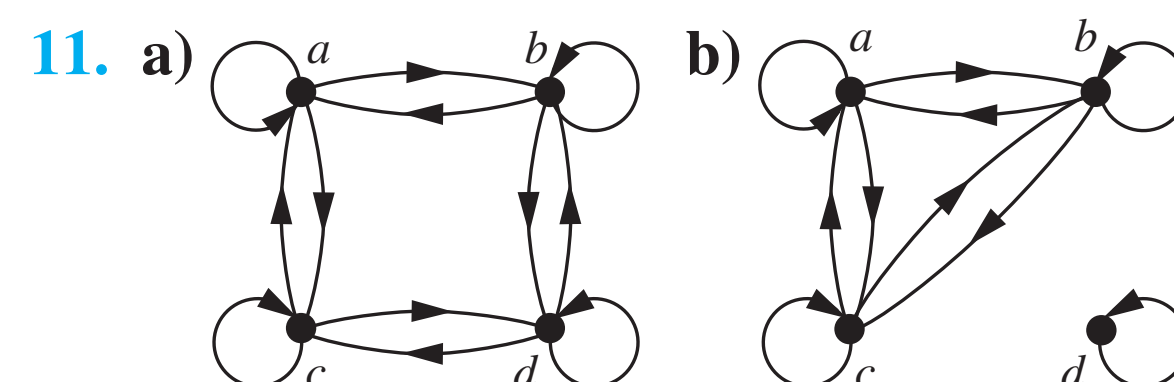
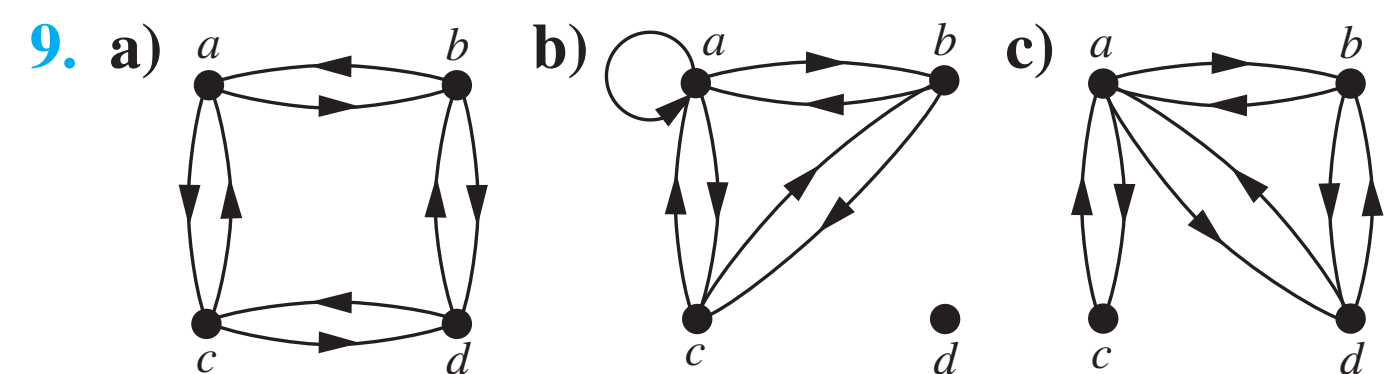
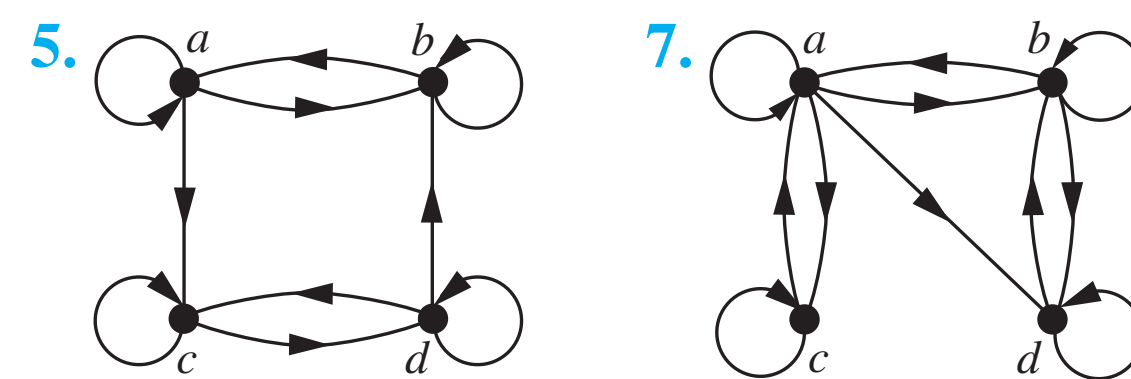
21. For simplicity we have indicated pairs of edges between the same two vertices in opposite directions by using a double arrowhead, rather than drawing two separate lines.



23.  $\{(a, b), (a, c), (b, c), (c, b)\}$  25.  $(a, c), (b, a), (c, d), (d, b)$  27.  $\{(a, a), (a, b), (a, c), (b, a), (b, b), (b, c), (c, a), (c, b), (d, d)\}$  29. The relation is asymmetric if and only if the directed graph has no loops and no closed paths of length 2. 31. Exercise 23: irreflexive. Exercise 24: reflexive, antisymmetric, transitive. Exercise 25: irreflexive, antisymmetric. 33. Reverse the direction on every edge in the digraph for  $R$ . 35. Proof by mathematical induction. *Basis step:* Trivial for  $n = 1$ . *Inductive step:* Assume true for  $k$ . Because  $R^{k+1} = R^k \circ R$ , its matrix is  $\mathbf{M}_R \odot \mathbf{M}_{R^k}$ . By the inductive hypothesis this is  $\mathbf{M}_R \odot \mathbf{M}_R^{[k]} = \mathbf{M}_R^{[k+1]}$ .

## Section 9.4

1. a)  $\{(0, 0), (0, 1), (1, 1), (1, 2), (2, 0), (2, 2), (3, 0), (3, 3)\}$  b)  $\{(0, 1), (0, 2), (0, 3), (1, 0), (1, 1), (1, 2), (2, 0), (2, 1), (2, 2), (3, 0)\}$  3.  $\{(a, b) \mid a \text{ divides } b \text{ or } b \text{ divides } a\}$



13. The symmetric closure of  $R$  is  $R \cup R^{-1}$ .  $\mathbf{M}_{R \cup R^{-1}} = \mathbf{M}_R \vee \mathbf{M}_{R^{-1}} = \mathbf{M}_R \vee \mathbf{M}_R^t$ . 15. Only when  $R$  is irreflexive, in which case it is its own closure. 17.  $a, a, a, a; a, b, e, a; a, d, e, a; b, c, c, b; b, e, a, b; c, b, c, c; c, c, b, c; c, c, c, c; d, e, a, d; d, e, e, d; e, a, b, e; e, a, d, e; e, d, e, e; e, e, d, e; e, e, e, e$  19. a)  $\{(1, 1), (1, 5), (2, 3), (3, 1), (3, 2), (3, 3), (3, 4), (4, 1), (4, 5), (5, 3), (5, 4)\}$  b)  $\{(1, 1), (1, 2), (1, 3), (1, 4), (2, 1), (2, 5), (3, 1), (3, 3), (3, 4), (3, 5), (4, 1), (4, 2), (4, 3), (4, 4), (5, 1), (5, 3), (5, 5)\}$  c)  $\{(1, 1), (1, 3), (1, 4), (1, 5), (2, 1), (2, 2), (2, 3), (2, 4), (3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (4, 1), (4, 3), (4, 4), (4, 5), (5, 1), (5, 2), (5, 3), (5, 4), (5, 5)\}$  d)  $\{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (2, 1), (2, 3), (2, 4), (2, 5), (3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (5, 1), (5, 2), (5, 3), (5, 4), (5, 5)\}$  e)  $\{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (5, 1), (5, 2), (5, 3), (5, 4), (5, 5)\}$  f)  $\{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (5, 1), (5, 2), (5, 3), (5, 4), (5, 5)\}$  21. a) If there is a student  $c$  who shares a class with  $a$  and a class with  $b$  b) If there are two students  $c$  and  $d$  such that  $a$  and  $c$  share a class,  $c$  and  $d$  share a class, and  $d$  and  $b$  share a class c) If there is a sequence  $s_0, \dots, s_n$  of students with  $n \geq 1$  such that  $s_0 = a$ ,  $s_n = b$ , and for each  $i = 1, 2, \dots, n$ ,  $s_i$  and  $s_{i-1}$  share a



class **23.** The result follows from  $(R^*)^{-1} = \left(\bigcup_{n=1}^{\infty} R^n\right)^{-1} = \bigcup_{n=1}^{\infty} (R^n)^{-1} = \bigcup_{n=1}^{\infty} R^n = R^*$ .

**25. a)**  $\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$  **b)**  $\begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix}$   
**c)**  $\begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$  **d)**  $\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$

**27.** Answers same as for Exercise 25. **29. a)**  $\{(1, 1), (1, 2), (1, 4), (2, 2), (3, 3), (4, 1), (4, 2), (4, 4)\}$  **b)**  $\{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (2, 4), (3, 3), (4, 1), (4, 2), (4, 4)\}$  **c)**  $\{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (2, 4), (3, 3), (4, 1), (4, 2), (4, 4)\}$   
**31.** Algorithm 1:  $O(n^{3.8})$ ; Algorithm 2:  $O(n^3)$  **33.** Initialize with  $\mathbf{A} := \mathbf{M}_R \vee \mathbf{I}_n$  and loop only for  $i := 2$  to  $n - 1$ .  
**35. a)** Because  $R$  is reflexive, every relation containing it must also be reflexive. **b)** Both  $\{(0, 0), (0, 1), (0, 2), (1, 1), (2, 2)\}$  and  $\{(0, 0), (0, 1), (1, 0), (1, 1), (2, 2)\}$  contain  $R$  and have an odd number of elements, but neither is a subset of the other.

## Section 9.5

**1. a)** Equivalence relation **b)** Not reflexive, not transitive  
**c)** Equivalence relation **d)** Not transitive **e)** Not symmetric, not transitive **3. a)** Equivalence relation **b)** Not transitive  
**c)** Not reflexive, not symmetric, not transitive **d)** Equivalence relation **e)** Not reflexive, not transitive **5.** Many answers are possible. (1) Two buildings are equivalent if they were opened during the same year; an equivalence class consists of the set of buildings opened in a given year (as long as there was at least one building opened that year). (2) Two buildings are equivalent if they have the same number of stories; the equivalence classes are the set of 1-story buildings, the set of 2-story buildings, and so on (one class for each  $n$  for which there is at least one  $n$ -story building). (3) Every building in which you have a class is equivalent to every building in which you have a class (including itself), and every building in which you don't have a class is equivalent to every building in which you don't have a class (including itself); there are two equivalence classes—the set of buildings in which you have a class and the set of buildings in which you don't have a class (assuming these are nonempty). **7.** The statement “ $p$  is equivalent to  $q$ ” means that  $p$  and  $q$  have the same entries in their truth tables.  $R$  is reflexive, because  $p$  has the same truth table as  $p$ .  $R$  is symmetric, for if  $p$  and  $q$  have the same truth table, then  $q$  and  $p$  have the same truth table. If  $p$  and  $q$  have the same entries in their truth tables and  $q$  and  $r$  have the same entries in their truth tables, then  $p$  and  $r$  also do, so  $R$  is transitive. The equivalence class of  $\mathbf{T}$  is the set of all tautologies; the equivalence class of  $\mathbf{F}$  is the set of all contradictions. **9. a)**  $(x, x) \in R$  because  $f(x) = f(x)$ . Hence,  $R$  is reflexive.  $(x, y) \in R$  if and only if  $f(x) = f(y)$ , which holds if and only if  $f(y) = f(x)$  if

and only if  $(y, x) \in R$ . Hence,  $R$  is symmetric. If  $(x, y) \in R$  and  $(y, z) \in R$ , then  $f(x) = f(y)$  and  $f(y) = f(z)$ . Hence,  $f(x) = f(z)$ . Thus,  $(x, z) \in R$ . It follows that  $R$  is transitive. **b)** The sets  $f^{-1}(b)$  for  $b$  in the range of  $f$  **11.** Let  $x$  be a bit string of length 3 or more. Because  $x$  agrees with itself in the first three bits,  $(x, x) \in R$ . Hence,  $R$  is reflexive. Suppose that  $(x, y) \in R$ . Then  $x$  and  $y$  agree in the first three bits. Hence,  $y$  and  $x$  agree in the first three bits. Thus,  $(y, x) \in R$ . If  $(x, y)$  and  $(y, z)$  are in  $R$ , then  $x$  and  $y$  agree in the first three bits, as do  $y$  and  $z$ . Hence,  $x$  and  $z$  agree in the first three bits. Hence,  $(x, z) \in R$ . It follows that  $R$  is transitive. **13.** This follows from Exercise 9, where  $f$  is the function that takes a bit string of length 3 or more to the ordered pair with its first bit as the first component and the third bit as its second component. **15.** For reflexivity,  $((a, b), (a, b)) \in R$  because  $a + b = b + a$ . For symmetry, if  $((a, b), (c, d)) \in R$ , then  $a + d = b + c$ , so  $c + b = d + a$ , so  $((c, d), (a, b)) \in R$ . For transitivity, if  $((a, b), (c, d)) \in R$  and  $((c, d), (e, f)) \in R$ , then  $a + d = b + c$  and  $c + e = d + f$ , so  $a + d + c + e = b + c + d + f$ , so  $a + e = b + f$ , so  $((a, b), (e, f)) \in R$ . An easier solution is to note that by algebra, the given condition is the same as the condition that  $f((a, b)) = f((c, d))$ , where  $f((x, y)) = x - y$ ; therefore, by Exercise 9 this is an equivalence relation. **17. a)** This follows from Exercise 9, where the function  $f$  from the set of differentiable functions (from  $\mathbf{R}$  to  $\mathbf{R}$ ) to the set of functions (from  $\mathbf{R}$  to  $\mathbf{R}$ ) is the differentiation operator. **b)** The set of all functions of the form  $g(x) = x^2 + C$  for some constant  $C$  **19.** This follows from Exercise 9, where the function  $f$  from the set of all URLs to the set of all Web pages is the function that assigns to each URL the Web page for that URL. **21.** No **23.** No **25.**  $R$  is reflexive because a bit string  $s$  has the same number of 1s as itself.  $R$  is symmetric because  $s$  and  $t$  having the same number of 1s implies that  $t$  and  $s$  do.  $R$  is transitive because  $s$  and  $t$  having the same number of 1s, and  $t$  and  $u$  having the same number of 1s implies that  $s$  and  $u$  have the same number of 1s. **27. a)** The sets of people of the same age **b)** The sets of people with the same two parents **29.** The set of all bit strings with exactly two 1s **31. a)** The set of all bit strings of length 3 **b)** The set of all bit strings of length 4 that end with a 1 **c)** The set of all bit strings of length 5 that end 11 **d)** The set of all bit strings of length 8 that end 10101 **33.** Each of the 15 bit strings of length less than four is in an equivalence class by itself:  $[\lambda]_{R_4} = \{\lambda\}$ ,  $[0]_{R_4} = \{0\}$ ,  $[1]_{R_4} = \{1\}$ ,  $[00]_{R_4} = \{00\}$ ,  $[01]_{R_4} = \{01\}$ , ...,  $[111]_{R_4} = \{111\}$ . The remaining 16 equivalence classes are determined by the bit strings of length 4:  $[0000]_{R_4} = \{0000, 00000, 00001, 000000, 000001, 000010, 000011, 0000000, \dots\}$ ,  $[0001]_{R_4} = \{0001, 00010, 00011, 000100, 000101, 000110, 000111, 0001000, \dots\}$ , ...,  $[1111]_{R_4} = \{1111, 11110, 11111, 111100, 111101, 111110, 111111, 1111000, \dots\}$  **35. a)**  $[2]_5 = \{i \mid i \equiv 2 \pmod{5}\} = \{\dots, -8, -3, 2, 7, 12, \dots\}$  **b)**  $[3]_5 = \{i \mid i \equiv 3 \pmod{5}\} = \{\dots, -7, -2, 3, 8, 13, \dots\}$  **c)**  $[6]_5 = \{i \mid i \equiv 6 \pmod{5}\} = \{\dots, -9, -4, 1, 6, 11, \dots\}$  **d)**  $[-3]_5 = \{i \mid i \equiv -3 \pmod{5}\} = \{\dots, -8, -3, 2, 7, 12, \dots\}$  **37.**  $\{6n + k \mid n \in \mathbf{Z}\}$  for  $k \in \{0, 1, 2, 3, 4, 5\}$  **39. a)**  $[(1, 2)] = \{(a, b) \mid a - b =$



$-1\} = \{(1, 2), (3, 4), (4, 5), (5, 6), \dots\}$  **b)** Each equivalence class can be interpreted as an integer (negative, positive, or zero); specifically,  $[(a, b)]$  can be interpreted as  $a - b$ .

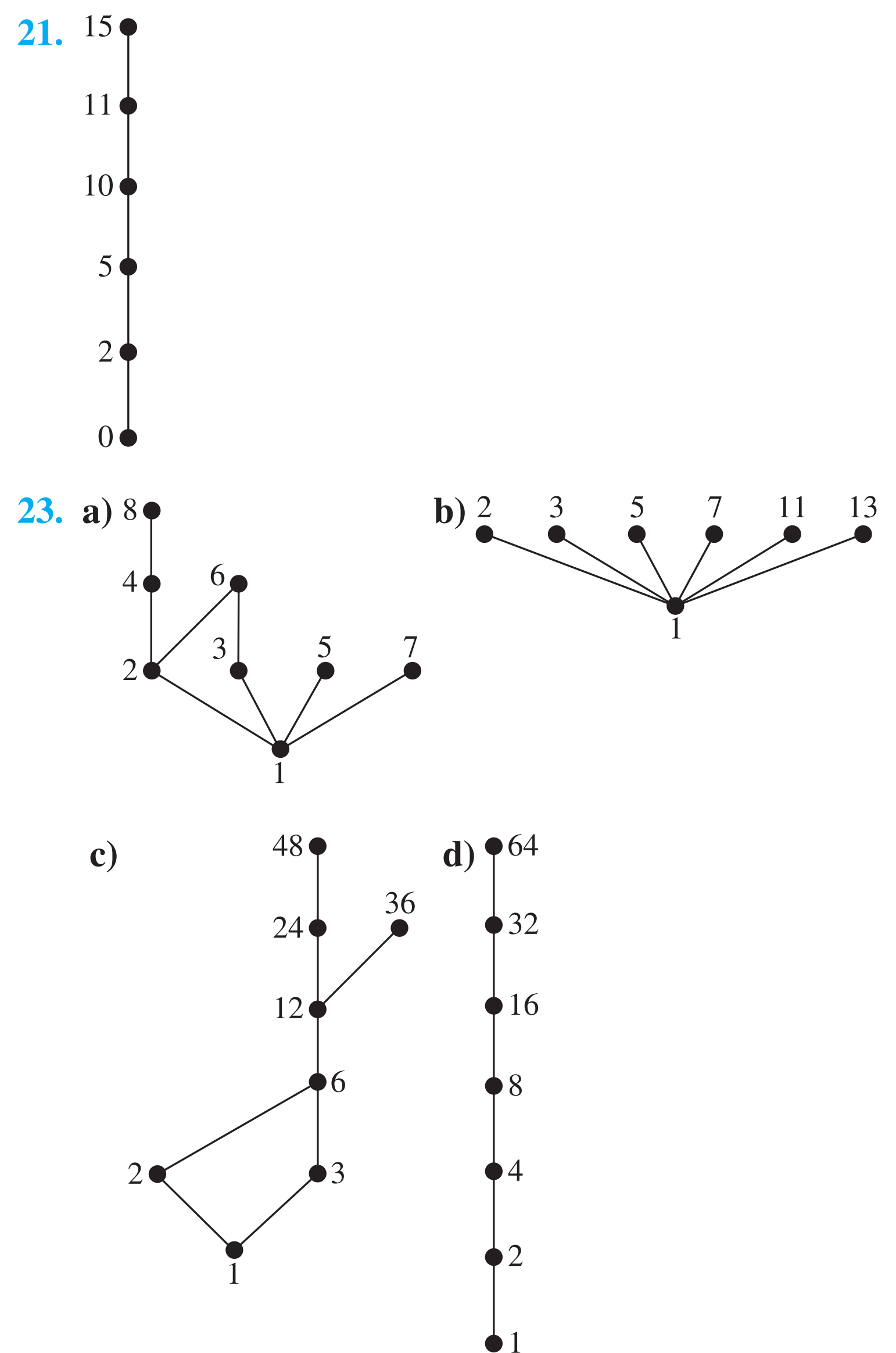
**41. a)** No **b)** Yes **c)** Yes **d)** No **43. (a), (c), (e)** **45. (b), (d), (e)** **47. a)**  $\{(0, 0), (1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (3, 4), (3, 5), (4, 3), (4, 4), (4, 5), (5, 3), (5, 4), (5, 5)\}$  **b)**  $\{(0, 0), (0, 1), (1, 0), (1, 1), (2, 2), (2, 3), (3, 2), (3, 3), (4, 4), (4, 5), (5, 4), (5, 5)\}$  **c)**  $\{(0, 0), (0, 1), (0, 2), (1, 0), (1, 1), (1, 2), (2, 0), (2, 1), (2, 2), (3, 3), (3, 4), (3, 5), (4, 3), (4, 4), (4, 5), (5, 3), (5, 4), (5, 5)\}$  **d)**  $\{(0, 0), (1, 1), (2, 2), (3, 3), (4, 4), (5, 5)\}$  **49.**  $[0]_6 \subseteq [0]_3, [1]_6 \subseteq [1]_3, [2]_6 \subseteq [2]_3, [3]_6 \subseteq [0]_3, [4]_6 \subseteq [1]_3, [5]_6 \subseteq [2]_3$  **51.** Let  $A$  be a set in the first partition. Pick a particular element  $x$  of  $A$ . The set of all bit strings of length 16 that agree with  $x$  on the last four bits is one of the sets in the second partition, and clearly every string in  $A$  is in that set. **53.** We claim that each equivalence class  $[x]_{R_{31}}$  is a subset of the equivalence class  $[x]_{R_8}$ . To show this, choose an arbitrary element  $y \in [x]_{R_{31}}$ . Then  $y$  is equivalent to  $x$  under  $R_{31}$ , so either  $y = x$  or  $y$  and  $x$  are each at least 31 characters long and agree on their first 31 characters. Because strings that are at least 31 characters long and agree on their first 31 characters perform are at least 8 characters long and agree on their first 8 characters, we know that either  $y = x$  or  $y$  and  $x$  are each at least 8 characters long and agree on their first 8 characters. This means that  $y$  is equivalent to  $x$  under  $R_8$ , so  $y \in [x]_{R_8}$ .

**55.**  $\{(a, a), (a, b), (a, c), (b, a), (b, b), (b, c), (c, a), (c, b), (c, c), (d, d), (d, e), (e, d), (e, e)\}$  **57. a)**  $\mathbf{Z}$  **b)**  $\{n + \frac{1}{2} \mid n \in \mathbf{Z}\}$  **59. a)**  $R$  is reflexive because any coloring can be obtained from itself via a 360-degree rotation. To see that  $R$  is symmetric and transitive, use the fact that each rotation is the composition of two reflections and conversely the composition of two reflections is a rotation. Hence,  $(C_1, C_2)$  belongs to  $R$  if and only if  $C_2$  can be obtained from  $C_1$  by a composition of reflections. So if  $(C_1, C_2)$  belongs to  $R$ , so does  $(C_2, C_1)$  because the inverse of the composition of reflections is also a composition of reflections (in the opposite order). Hence,  $R$  is symmetric. To see that  $R$  is transitive, suppose  $(C_1, C_2)$  and  $(C_2, C_3)$  belong to  $R$ . Taking the composition of the reflections in each case yields a composition of reflections, showing that  $(C_1, C_3)$  belongs to  $R$ . **b)** We express colorings with sequences of length four, with  $r$  and  $b$  denoting red and blue, respectively. We list letters denoting the colors of the upper left square, upper right square, lower left square, and lower right square, in that order. The equivalence classes are:  $\{rrrr\}, \{bbbb\}, \{rrrb, rrbr, rbr, brrr\}, \{bbbr, bbrb, brbb, rbbb\}, \{rbbr, brrb\}, \{rrbb, brbr, bbr, rbrb\}$ .

**61. 5** **63. Yes** **65.  $R$**  **67.** First form the reflexive closure of  $R$ , then form the symmetric closure of the reflexive closure, and finally form the transitive closure of the symmetric closure of the reflexive closure. **69.**  $p(0) = 1, p(1) = 1, p(2) = 2, p(3) = 5, p(4) = 15, p(5) = 52, p(6) = 203, p(7) = 877, p(8) = 4140, p(9) = 21147, p(10) = 115975$

## Section 9.6

**1. a)** Is a partial ordering **b)** Not antisymmetric, not transitive **c)** Is a partial ordering **d)** Is a partial ordering **e)** Not antisymmetric, not transitive **3. a)** No **b)** No **c)** Yes **5. a)** Yes **b)** No **c)** Yes **d)** No **7. a)** No **b)** Yes **c)** No **9. No** **11. Yes** **13. a)**  $\{(0, 0), (1, 0), (1, 1), (2, 0), (2, 1), (2, 2)\}$  **b)**  $(\mathbf{Z}, \leq)$  **c)**  $(P(\mathbf{Z}), \subseteq)$  **d)**  $(\mathbf{Z}^+, \text{"is a multiple of"})$  **15. a)**  $\{0\}$  and  $\{1\}$ , for instance **b)** 4 and 6, for instance **17. a)**  $(1, 1, 2) < (1, 2, 1)$  **b)**  $(0, 1, 2, 3) < (0, 1, 3, 2)$  **c)**  $(0, 1, 1, 1, 0) < (1, 0, 1, 0, 1)$  **19.**  $0 < 0001 < 001 < 01 < 010 < 0101 < 011 < 11$



**25.**  $(a, b), (a, c), (a, d), (b, c), (b, d), (a, a), (b, b), (c, c), (d, d)$  **27.**  $(a, a), (a, g), (a, d), (a, e), (a, f), (b, b), (b, g), (b, d), (b, e), (b, f), (c, c), (c, g), (c, d), (c, e), (c, f), (g, d), (g, e), (g, f), (g, g), (d, d), (e, e), (f, f)$  **29.**  $(\emptyset, \{a\}), (\emptyset, \{b\}), (\emptyset, \{c\}), (\{a\}, \{a, b\}), (\{a\}, \{a, c\}), (\{b\}, \{a, b\}), (\{b\}, \{b, c\}), (\{c\}, \{a, c\}), (\{c\}, \{b, c\}), (\{a, b\}, \{a, b, c\}), (\{a, c\}, \{a, b, c\}), (\{b, c\}, \{a, b, c\})$  **31.** Let  $(S, \preceq)$  be a finite poset. We will show that this poset is the reflexive transitive closure of its covering relation. Suppose that  $(a, b)$  is in the reflexive transitive closure of the covering relation. Then  $a = b$  or  $a < b$ , so  $a \preceq b$ , or else there is a sequence  $a_1, a_2, \dots, a_n$  such that  $a < a_1 < a_2 < \dots < a_n < b$ , in which case again  $a \preceq b$  by the transitivity of  $\preceq$ . Conversely, suppose that  $a < b$ . If  $a = b$  then  $(a, b)$  is in the



reflexive transitive closure of the covering relation. If  $a < b$  and there is no  $z$  such that  $a < z < b$ , then  $(a, b)$  is in the covering relation and therefore in its reflexive transitive closure. Otherwise, let  $a < a_1 < a_2 < \dots < a_n < b$  be a longest possible sequence of this form (which exists because the poset is finite). Then no intermediate elements can be inserted, so each pair  $(a, a_1), (a_1, a_2), \dots, (a_n, b)$  is in the covering relation, so again  $(a, b)$  is in its reflexive transitive closure. **33. a)** 24, 45 **b)** 3, 5 **c)** No **d)** No **e)** 15, 45 **f)** 15 **g)** 15, 5, 3 **h)** 15 **35. a)**  $\{1, 2\}, \{1, 3, 4\}, \{2, 3, 4\}$  **b)**  $\{1\}, \{2\}, \{4\}$  **c)** No **d)** No **e)**  $\{2, 4\}, \{2, 3, 4\}$  **f)**  $\{2, 4\}$  **g)**  $\{3, 4\}, \{4\}$  **h)**  $\{3, 4\}$  **37.** Because  $(a, b) \preceq (a, b)$ ,  $\preceq$  is reflexive. If  $(a_1, a_2) \preceq (b_1, b_2)$  and  $(a_1, a_2) \neq (b_1, b_2)$ , either  $a_1 < b_1$ , or  $a_1 = b_1$  and  $a_2 < b_2$ . In either case,  $(b_1, b_2)$  is not less than or equal to  $(a_1, a_2)$ . Hence,  $\preceq$  is antisymmetric. Suppose that  $(a_1, a_2) < (b_1, b_2) < (c_1, c_2)$ . Then if  $a_1 < b_1$  or  $b_1 < c_1$ , we have  $a_1 < c_1$ , so  $(a_1, a_2) < (c_1, c_2)$ , but if  $a_1 = b_1 = c_1$ , then  $a_2 < b_2 < c_2$ , which implies that  $(a_1, a_2) < (c_1, c_2)$ . Hence,  $\preceq$  is transitive. **39.** Because  $(s, t) \preceq (s, t)$ ,  $\preceq$  is reflexive. If  $(s, t) \preceq (u, v)$  and  $(u, v) \preceq (s, t)$ , then  $s \preceq u \preceq s$  and  $t \preceq v \preceq t$ ; hence,  $s = u$  and  $t = v$ . Hence,  $\preceq$  is antisymmetric. Suppose that  $(s, t) \preceq (u, v) \preceq (w, x)$ . Then  $s \preceq u, t \preceq v, u \preceq w$ , and  $v \preceq x$ . It follows that  $s \preceq w$  and  $t \preceq x$ . Hence,  $(s, t) \preceq (w, x)$ . Hence,  $\preceq$  is transitive. **41. a)** Suppose that  $x$  is maximal and that  $y$  is the largest element. Then  $x \preceq y$ . Because  $x$  is not less than  $y$ , it follows that  $x = y$ . By Exercise 40(a)  $y$  is unique. Hence,  $x$  is unique. **b)** Suppose that  $x$  is minimal and that  $y$  is the smallest element. Then  $x \succcurlyeq y$ . Because  $x$  is not greater than  $y$ , it follows that  $x = y$ . By Exercise 40(b)  $y$  is unique. Hence,  $x$  is unique. **43. a)** Yes **b)** No **c)** Yes **45.** Use mathematical induction. Let  $P(n)$  be “Every subset with  $n$  elements from a lattice has a least upper bound and a greatest lower bound.” *Basis step:*  $P(1)$  is true because the least upper bound and greatest lower bound of  $\{x\}$  are both  $x$ . *Inductive step:* Assume that  $P(k)$  is true. Let  $S$  be a set with  $k + 1$  elements. Let  $x \in S$  and  $S' = S - \{x\}$ . Because  $S'$  has  $k$  elements, by the inductive hypothesis, it has a least upper bound  $y$  and a greatest lower bound  $a$ . Now because we are in a lattice, there are elements  $z = \text{lub}(x, y)$  and  $b = \text{glb}(x, a)$ . We are done if we can show that  $z$  is the least upper bound of  $S$  and  $b$  is the greatest lower bound of  $S$ . To show that  $z$  is the least upper bound of  $S$ , first note that if  $w \in S$ , then  $w = x$  or  $w \in S'$ . If  $w = x$  then  $w \preceq z$  because  $z$  is the least upper bound of  $x$  and  $y$ . If  $w \in S'$ , then  $w \preceq z$  because  $w \preceq y$ , which is true because  $y$  is the least upper bound of  $S'$ , and  $y \preceq z$ , which is true because  $z = \text{lub}(x, y)$ . To see that  $z$  is the least upper bound of  $S$ , suppose that  $u$  is an upper bound of  $S$ . Note that such an element  $u$  must be an upper bound of  $x$  and  $y$ , but because  $z = \text{lub}(x, y)$ , it follows that  $z \preceq u$ . We omit the similar argument that  $b$  is the greatest lower bound of  $S$ . **47. a)** No **b)** Yes **c)**  $(\text{Proprietary}, \{\text{Cheetah}, \text{Puma}\}), (\text{Restricted}, \{\text{Cheetah}, \text{Puma}\}), (\text{Registered}, \{\text{Cheetah}, \text{Puma}\}), (\text{Proprietary}, \{\text{Cheetah}, \text{Puma}, \text{Impala}\}), (\text{Restricted}, \{\text{Cheetah}, \text{Puma}, \text{Impala}\}), (\text{Registered}, \{\text{Cheetah}, \text{Puma}, \text{Impala}\})$  **d)**  $(\text{Nonproprietary}, \{\text{Impala}, \text{Puma}\}), (\text{Proprietary}, \{\text{Impala}, \text{Puma}\}),$

$(\text{Restricted}, \{\text{Impala}, \text{Puma}\}), (\text{Nonproprietary}, \{\text{Impala}\}), (\text{Proprietary}, \{\text{Impala}\}), (\text{Restricted}, \{\text{Impala}\}), (\text{Nonproprietary}, \{\text{Puma}\}), (\text{Proprietary}, \{\text{Puma}\}), (\text{Restricted}, \{\text{Puma}\}), (\text{Nonproprietary}, \emptyset), (\text{Proprietary}, \emptyset), (\text{Restricted}, \emptyset)$  **49.** Let  $\Pi$  be the set of all partitions of a set  $S$  with  $P_1 \preceq P_2$  if  $P_1$  is a refinement of  $P_2$ , that is, if every set in  $P_1$  is a subset of a set in  $P_2$ . First, we show that  $(\Pi, \preceq)$  is a poset. Because  $P \preceq P$  for every partition  $P$ ,  $\preceq$  is reflexive. Now suppose that  $P_1 \preceq P_2$  and  $P_2 \preceq P_1$ . Let  $T \in P_1$ . Because  $P_1 \preceq P_2$ , there is a set  $T' \in P_2$  such that  $T \subseteq T'$ . Because  $P_2 \preceq P_1$  there is a set  $T'' \in P_1$  such that  $T' \subseteq T''$ . It follows that  $T \subseteq T''$ . But because  $P_1$  is a partition,  $T = T''$ , which implies that  $T = T'$  because  $T \subseteq T' \subseteq T''$ . Thus,  $T \in P_2$ . By reversing the roles of  $P_1$  and  $P_2$  it follows that every set in  $P_2$  is also in  $P_1$ . Hence,  $P_1 = P_2$  and  $\preceq$  is antisymmetric. Next, suppose that  $P_1 \preceq P_2$  and  $P_2 \preceq P_3$ . Let  $T \in P_1$ . Then there is a set  $T' \in P_2$  such that  $T \subseteq T'$ . Because  $P_2 \preceq P_3$  there is a set  $T'' \in P_3$  such that  $T' \subseteq T''$ . This means that  $T \subseteq T''$ . Hence,  $P_1 \preceq P_3$ . It follows that  $\preceq$  is transitive. The greatest lower bound of the partitions  $P_1$  and  $P_2$  is the partition  $P$  whose subsets are the nonempty sets of the form  $T_1 \cap T_2$  where  $T_1 \in P_1$  and  $T_2 \in P_2$ . We omit the justification of this statement here. The least upper bound of the partitions  $P_1$  and  $P_2$  is the partition that corresponds to the equivalence relation in which  $x \in S$  is related to  $y \in S$  if there is a sequence  $x = x_0, x_1, x_2, \dots, x_n = y$  for some nonnegative integer  $n$  such that for each  $i$  from 1 to  $n$ ,  $x_{i-1}$  and  $x_i$  are in the same element of  $P_1$  or of  $P_2$ . We omit the details that this is an equivalence relation and the details of the proof that this is the least upper bound of the two partitions. **51.** By Exercise 45 there is a least upper bound and a greatest lower bound for the entire finite lattice. By definition these elements are the greatest and least elements, respectively. **53.** The least element of a subset of  $\mathbf{Z}^+ \times \mathbf{Z}^+$  is that pair that has the smallest possible first coordinate, and, if there is more than one such pair, that pair among those that has the smallest second coordinate. **55.** If  $x$  is an integer in a decreasing sequence of elements of this poset, then at most  $|x|$  elements can follow  $x$  in the sequence, namely, integers whose absolute values are  $|x| - 1, |x| - 2, \dots, 1, 0$ . Therefore, there can be no infinite decreasing sequence. This is not a totally ordered set, because 5 and  $-5$ , for example, are incomparable. **57.** To find which of two rational numbers is larger, write them with a positive common denominator and compare numerators. To show that this set is dense, suppose that  $x < y$  are two rational numbers. Then their average, i.e.,  $(x + y)/2$ , is a rational number between them. **59.** Let  $(S, \preceq)$  be a partially ordered set. It is enough to show that every nonempty subset of  $S$  contains a least element if and only if there is no infinite decreasing sequence of elements  $a_1, a_2, a_3, \dots$  in  $S$  (i.e., where  $a_{i+1} < a_i$  for all  $i$ ). An infinite decreasing sequence of elements clearly has no least element. Conversely, let  $A$  be any nonempty subset of  $S$  that has no least element. Because  $A$  is nonempty, choose  $a_1 \in A$ . Because  $a_1$  is not the least element of  $A$ , choose  $a_2 \in A$  with  $a_2 < a_1$ . Because  $a_2$  is not the least element of  $A$ , choose  $a_3 \in A$  with  $a_3 < a_2$ . Continue in



this manner, producing an infinite decreasing sequence in  $S$ .

**61.**  $a <_t b <_t c <_t d <_t e <_t f <_t g <_t h <_t i <_t j <_t k <_t l <_t m$

**63.**  $1 < 5 < 2 < 4 < 12 < 20, 1 < 2 < 5 < 4 < 12 < 20,$   
 $1 < 2 < 4 < 5 < 12 < 20, 1 < 2 < 4 < 12 < 5 < 20,$   
 $1 < 5 < 2 < 4 < 20 < 12, 1 < 2 < 5 < 4 < 20 < 12,$   
 $1 < 2 < 4 < 5 < 20 < 12$

**65.**  $A < C < E < B < D < F < G,$   
 $A < E < C < B < D < F < G, C < A < E < B < D < F < G,$   
 $C < E < A < B < D < F < G, E < A < C < B < D < F < G,$   
 $E < C < A < B < D < F < G, A < C < B < E < D < F < G,$   
 $C < A < B < E < D < F < G, A < C < B < D < E < F < G,$   
 $C < A < B < D < E < F < G, A < C < E < B < F < D < G,$   
 $A < E < C < B < F < D < G, C < A < E < B < F < D < G,$   
 $C < E < A < B < F < D < G, E < A < C < B < F < D < G,$   
 $E < C < A < B < F < D < G, A < C < B < E < F < D < G,$   
 $C < A < B < E < F < D < G$

**67.** Determine user needs  $<$  Write functional requirements  $<$  Set up test sites  $<$  Develop system requirements  $<$  Write documentation  $<$  Develop module  $A <$  Develop module  $B <$  Develop module  $C <$  Integrate modules  $<$   $\alpha$  test  $<$   $\beta$  test  $<$  Completion

## Supplementary Exercises

**1. a)** Irreflexive (we do not include the empty string), symmetric **b)** Irreflexive, symmetric **c)** Irreflexive, antisymmetric, transitive

**3.**  $((a, b), (a, b)) \in R$  because  $a+b = a+b$ . Hence,  $R$  is reflexive. If  $((a, b), (c, d)) \in R$  then  $a+d = b+c$ , so that  $c+b = d+a$ . It follows that  $((c, d), (a, b)) \in R$ . Hence,  $R$  is symmetric. Suppose that  $((a, b), (c, d))$  and  $((c, d), (e, f))$  belong to  $R$ . Then  $a+d = b+c$  and  $c+f = d+e$ . Adding these two equations and subtracting  $c+d$  from both sides gives  $a+f = b+e$ . Hence,  $((a, b), (e, f))$  belongs to  $R$ . Hence,  $R$  is transitive.

**5.** Suppose that  $(a, b) \in R$ . Because  $(b, b) \in R$  it follows that  $(a, b) \in R^2$ .

**7.** Yes, yes **9.** Yes, yes **11.** Two records with identical keys in the projection would have identical keys in the original.

**13.**  $(\Delta \cup R)^{-1} = \Delta^{-1} \cup R^{-1} = \Delta \cup R^{-1}$

**15. a)**  $R = \{(a, b), (a, c)\}$ . The transitive closure of the symmetric closure of  $R$  is  $\{(a, a), (a, b), (a, c), (b, a), (b, b), (b, c), (c, a), (c, b), (c, c)\}$  and is different from the symmetric closure of the transitive closure of  $R$ , which is  $\{(a, b), (a, c), (b, a), (c, a)\}$ . **b)** Suppose that  $(a, b)$  is in the symmetric closure of the transitive closure of  $R$ . We must show that  $(a, b)$  is in the transitive closure of the symmetric closure of  $R$ . We know that at least one of  $(a, b)$  and  $(b, a)$  is in the transitive closure of  $R$ . Hence, there is either a path from  $a$  to  $b$  in  $R$  or a path from  $b$  to  $a$  in  $R$  (or both). In the former case, there is a path from  $a$  to  $b$  in the symmetric closure of  $R$ . In the latter case, we can form a path from  $a$  to  $b$  in the symmetric closure of  $R$  by reversing the directions of all the edges in a path from  $b$  to  $a$ , going backward. Hence,  $(a, b)$  is in the transitive closure of the symmetric closure of  $R$ .

**17.** The closure of  $S$  with respect to property  $\mathbf{P}$  is a relation with property  $\mathbf{P}$  that contains  $R$  because  $R \subseteq S$ . Hence, the closure of  $S$  with respect to property  $\mathbf{P}$  contains the closure of  $R$  with respect to property  $\mathbf{P}$ .

**19.** Use the basic idea of Warshall's algorithm, except let  $w_{ij}^{[k]}$  equal the length of the longest path from  $v_i$  to  $v_j$  using interior vertices with sub-

scripts not exceeding  $k$ , and equal to  $-1$  if there is no such path. To find  $w_{ij}^{[k]}$  from the entries of  $\mathbf{W}_{k-1}$ , determine for each pair  $(i, j)$  whether there are paths from  $v_i$  to  $v_k$  and from  $v_k$  to  $v_j$  using no vertices labeled greater than  $k$ . If either  $w_{ik}^{[k-1]}$  or  $w_{kj}^{[k-1]}$  is  $-1$ , then such a pair of paths does not exist, so set  $w_{ij}^{[k]} = w_{ij}^{[k-1]}$ . If such a pair of paths exists, then there are two possibilities. If  $w_{kk}^{[k-1]} > 0$ , there are paths of arbitrary long length from  $v_i$  to  $v_j$ , so set  $w_{ij}^{[k]} = \infty$ . If  $w_{kk}^{[k-1]} = 0$ , set  $w_{ij}^{[k]} = \max(w_{ij}^{[k-1]}, w_{ik}^{[k-1]} + w_{kj}^{[k-1]})$ . (Initially take  $\mathbf{W}_0 = \mathbf{M}_R$ .)

**21. 25. 23.** Because  $A_i \cap B_j$  is a subset of  $A_i$  and of  $B_j$ , the collection of subsets is a refinement of each of the given partitions. We must show that it is a partition. By construction, each of these sets is nonempty. To see that their union is  $S$ , suppose that  $s \in S$ . Because  $P_1$  and  $P_2$  are partitions of  $S$ , there are sets  $A_i$  and  $B_j$  such that  $s \in A_i$  and  $s \in B_j$ . Therefore,  $s \in A_i \cap B_j$ . Hence, the union of these sets is  $S$ . To see that they are pairwise disjoint, note that unless  $i = i'$  and  $j = j'$ ,  $(A_i \cap B_j) \cap (A_{i'} \cap B_{j'}) = (A_i \cap A_{i'}) \cap (B_j \cap B_{j'}) = \emptyset$ .

**25.** The subset relation is a partial ordering on any collection of sets, because it is reflexive, antisymmetric, and transitive. Here the collection of sets is  $\mathbf{R}(S)$ .

**27.** Find recipe  $<$  Buy seafood  $<$  Buy groceries  $<$  Wash shellfish  $<$  Cut ginger and garlic  $<$  Clean fish  $<$  Steam rice  $<$  Cut fish  $<$  Wash vegetables  $<$  Chop water chestnuts  $<$  Make garnishes  $<$  Cook in wok  $<$  Arrange on platter  $<$  Serve

**29. a)** The only antichain with more than one element is  $\{c, d\}$ . **b)** The only antichains with more than one element are  $\{b, c\}$ ,  $\{c, e\}$ , and  $\{d, e\}$ . **c)** The only antichains with more than one element are  $\{a, b\}$ ,  $\{a, c\}$ ,  $\{b, c\}$ ,  $\{a, b, c\}$ ,  $\{d, e\}$ ,  $\{d, f\}$ ,  $\{e, f\}$ , and  $\{d, e, f\}$ .

**31.** Let  $(S, \preceq)$  be a finite poset, and let  $A$  be a maximal chain. Because  $(A, \preceq)$  is also a poset it must have a minimal element  $m$ . Suppose that  $m$  is not minimal in  $S$ . Then there would be an element  $a$  of  $S$  with  $a < m$ . However, this would make the set  $A \cup \{a\}$  a larger chain than  $A$ . To show this, we must show that  $a$  is comparable with every element of  $A$ . Because  $m$  is comparable with every element of  $A$  and  $m$  is minimal, it follows that  $m < x$  when  $x$  is in  $A$  and  $x \neq m$ . Because  $a < m$  and  $m < x$ , the transitive law shows that  $a < x$  for every element of  $A$ .

**33.** Let  $aRb$  denote that  $a$  is a descendant of  $b$ . By Exercise 32, if no set of  $n+1$  people none of whom is a descendant of any other (an antichain) exists, then  $k \leq n$ , so the set can be partitioned into  $k \leq n$  chains. By the pigeonhole principle, at least one of these chains contains at least  $m+1$  people.

**35.** We prove by contradiction that if  $S$  has no infinite decreasing sequence and  $\forall x (\{\forall y[y < x \rightarrow P(y)]\} \rightarrow P(x))$ , then  $P(x)$  is true for all  $x \in S$ . If it does not hold that  $P(x)$  is true for all  $x \in S$ , let  $x_1$  be an element of  $S$  such that  $P(x_1)$  is not true. Then by the conditional statement already given, it must be the case that  $\forall y[y < x_1 \rightarrow P(y)]$  is not true. This means that there is some  $x_2$  with  $x_2 < x_1$  such that  $P(x_2)$  is not true. Again invoking the conditional statement, we get an  $x_3 < x_2$  such that  $P(x_3)$  is not true, and so on forever. This contradicts the well-foundedness of our poset. Therefore,  $P(x)$  is true for all  $x \in S$ .

**37.** Suppose that  $R$  is a quasi-ordering. Because  $R$  is reflexive, if  $a \in A$ , then  $(a, a) \in R$ . This implies that



$(a, a) \in R^{-1}$ . Hence,  $a \in R \cap R^{-1}$ . It follows that  $R \cap R^{-1}$  is reflexive.  $R \cap R^{-1}$  is symmetric for any relation  $R$  because, for any relation  $R$ , if  $(a, b) \in R$  then  $(b, a) \in R^{-1}$  and vice versa. To show that  $R \cap R^{-1}$  is transitive, suppose that  $(a, b) \in R \cap R^{-1}$  and  $(b, c) \in R \cap R^{-1}$ . Because  $(a, b) \in R$  and  $(b, c) \in R$ ,  $(a, c) \in R$ , because  $R$  is transitive. Similarly, because  $(a, b) \in R^{-1}$  and  $(b, c) \in R^{-1}$ ,  $(b, a) \in R$  and  $(c, b) \in R$ , so  $(c, a) \in R$  and  $(a, c) \in R^{-1}$ . Hence,  $(a, c) \in R \cap R^{-1}$ . It follows that  $R \cap R^{-1}$  is an equivalence relation.

**39. a)** Because  $\text{glb}(x, y) = \text{glb}(y, x)$  and  $\text{lub}(x, y) = \text{lub}(y, x)$ , it follows that  $x \wedge y = y \wedge x$  and  $x \vee y = y \vee x$ . **b)** Using the definition,  $(x \wedge y) \wedge z$  is a lower bound of  $x, y$ , and  $z$  that is greater than every other lower bound. Because  $x, y$ , and  $z$  play interchangeable roles,  $x \wedge (y \wedge z)$  is the same element. Similarly,  $(x \vee y) \vee z$  is an upper bound of  $x, y$ , and  $z$  that is less than every other upper bound. Because  $x, y$ , and  $z$  play interchangeable roles,  $x \vee (y \vee z)$  is the same element. **c)** To show that  $x \wedge (x \vee y) = x$  it is sufficient to show that  $x$  is the greatest lower bound of  $x$ , and  $x \vee y$ . Note that  $x$  is a lower bound of  $x$ , and because  $x \vee y$  is by definition greater than  $x$ ,  $x$  is a lower bound for it as well. Therefore,  $x$  is a lower bound. But any lower bound of  $x$  has to be less than  $x$ , so  $x$  is the greatest lower bound. The second statement is the dual of the first; we omit its proof. **d)**  $x$  is a lower, and an upper, bound for itself and itself, and the greatest, and least, such bound.

**41. a)** Because 1 is the only element greater than or equal to 1, it is the only upper bound for 1 and therefore the only possible value of the least upper bound of  $x$  and 1. **b)** Because  $x \preceq 1$ ,  $x$  is a lower bound for both  $x$  and 1 and no other lower bound can be greater than  $x$ , so  $x \wedge 1 = x$ . **c)** Because  $0 \preceq x$ ,  $x$  is an upper bound for both  $x$  and 0 and no other bound can be less than  $x$ , so  $x \vee 0 = x$ . **d)** Because 0 is the only element less than or equal to 0, it is the only lower bound for 0 and therefore the only possible value of the greatest lower bound of  $x$  and 0.

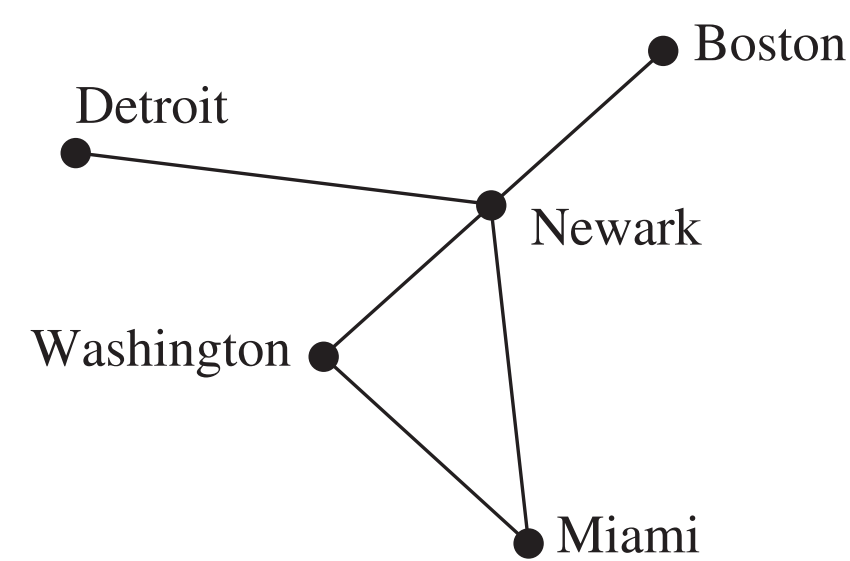
**43.**  $L = (S, \subseteq)$  where  $S = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{2, 3\}, \{1, 2, 3\}\}$  **45.** Yes **47.** The complement of a subset  $X \subseteq S$  is its complement  $S - X$ . To prove this, note that  $X \vee (S - X) = 1$  and  $X \wedge (S - X) = 0$  because  $X \cup (S - X) = S$  and  $X \cap (S - X) = \emptyset$ .

**49.** Think of the rectangular grid as representing elements in a matrix. Thus, we number from top to bottom and within that from left to right. The partial order is that  $(a, b) \leq (c, d)$  iff  $a \leq c$  and  $b \leq d$ . Note that  $(1, 1)$  is the least element under this relation. The rules for Chomp as explained in Chapter 1 coincide with the rules stated in the preamble here. But now we can identify the point  $(a, b)$  with the natural number  $p^{a-1}q^{b-1}$  for all  $a$  and  $b$  with  $1 \leq a \leq m$  and  $1 \leq b \leq n$ . This identifies the points in the rectangular grid with the set  $S$  in this exercise, and the partial order  $\leq$  just described is the same as the divides relation, because  $p^{a-1}q^{b-1} \mid p^{c-1}q^{d-1}$  if and only if the exponent of  $p$  on the left does not exceed the exponent of  $p$  on the right, and similarly for  $q$ .

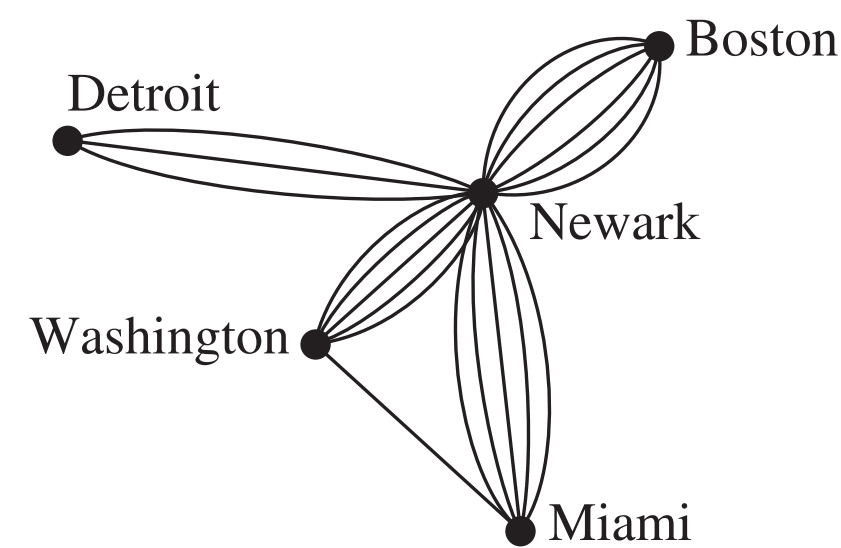
## CHAPTER 10

### Section 10.1

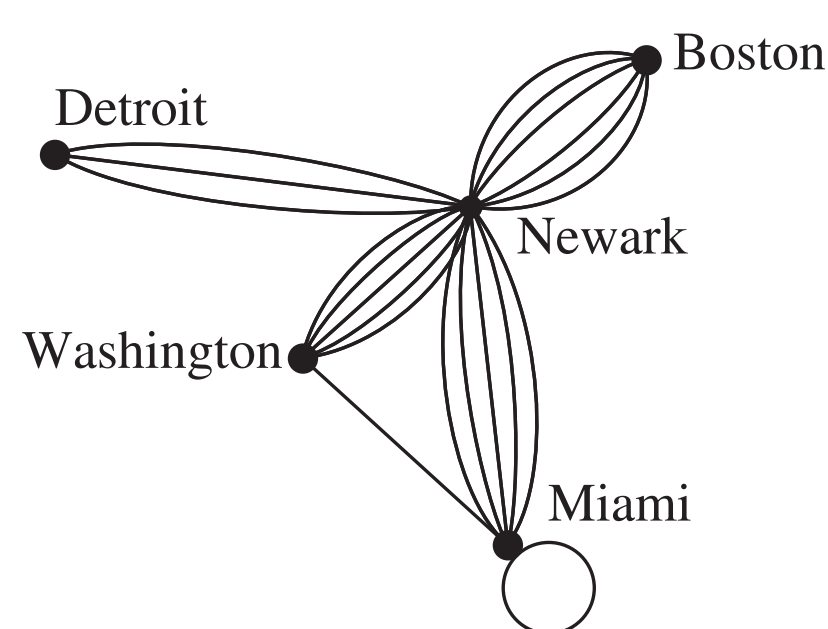
1. a)



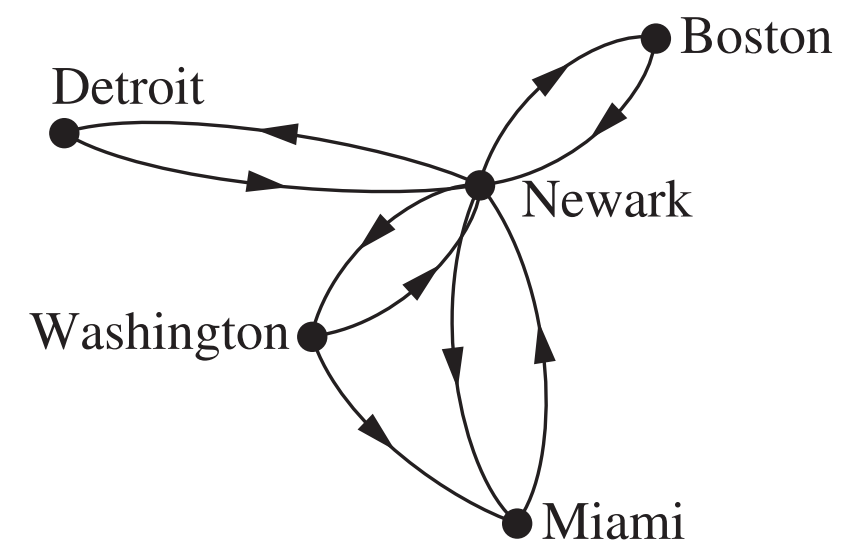
b)



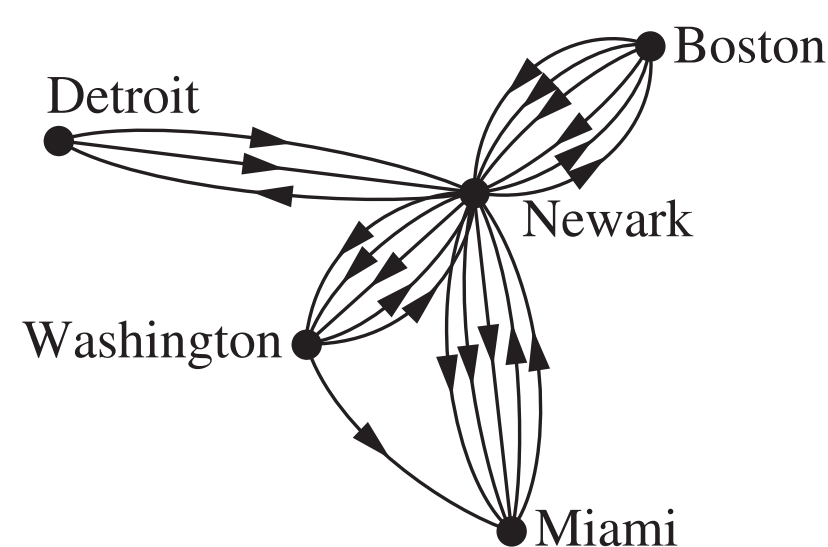
c)



d)



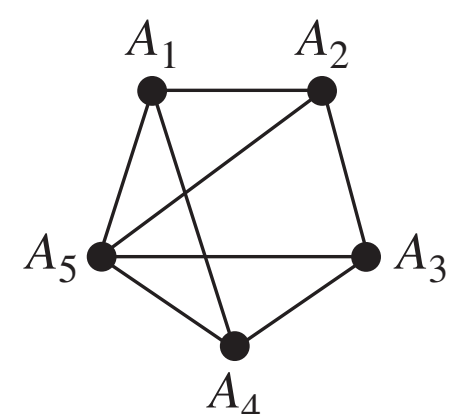
e)



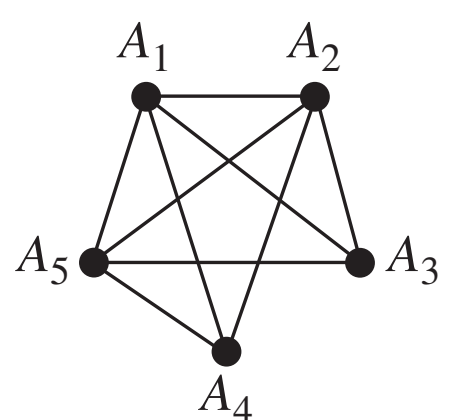
**3.** Simple graph **5.** Pseudograph **7.** Directed graph **9.** Directed multigraph **11.** If  $uRv$ , then there is an edge associated with  $\{u, v\}$ . But  $\{u, v\} = \{v, u\}$ , so this edge is associated with  $\{v, u\}$  and therefore  $vRu$ . Thus, by definition,  $R$  is a symmetric relation. A simple graph does not allow loops; therefore,  $uRu$  never holds, and so by definition  $R$  is irreflexive.



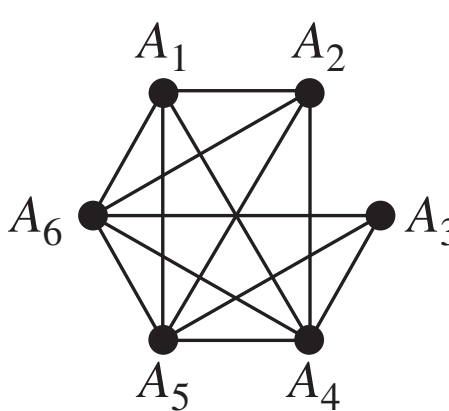
13. a)



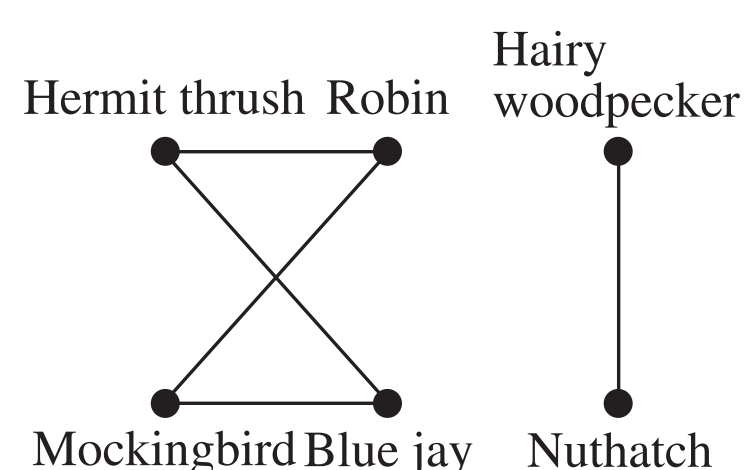
b)



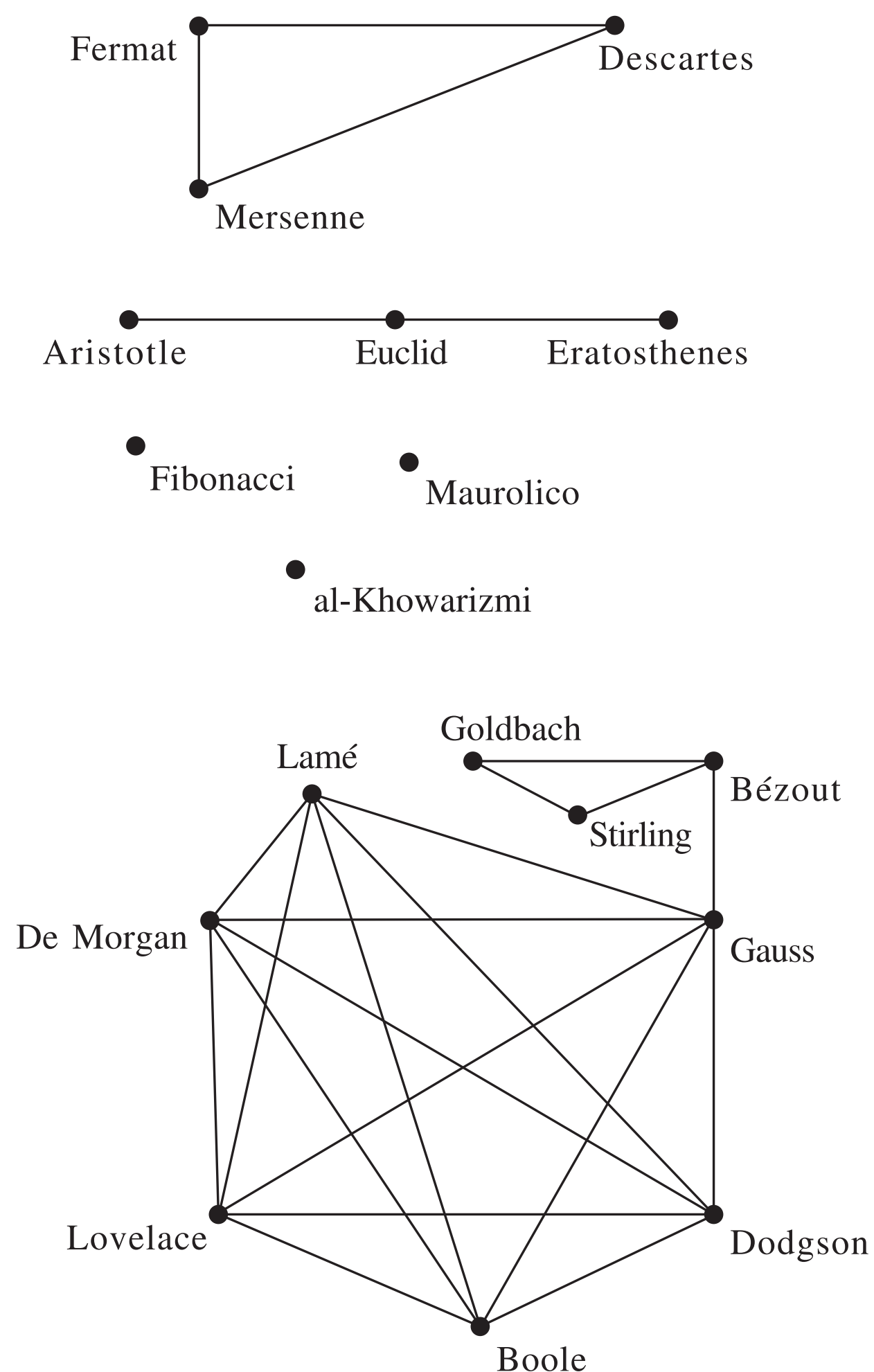
c)



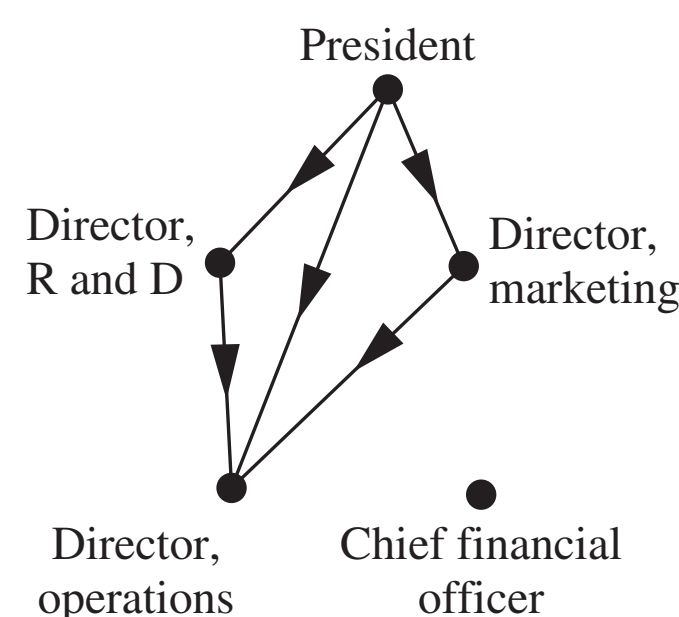
15.



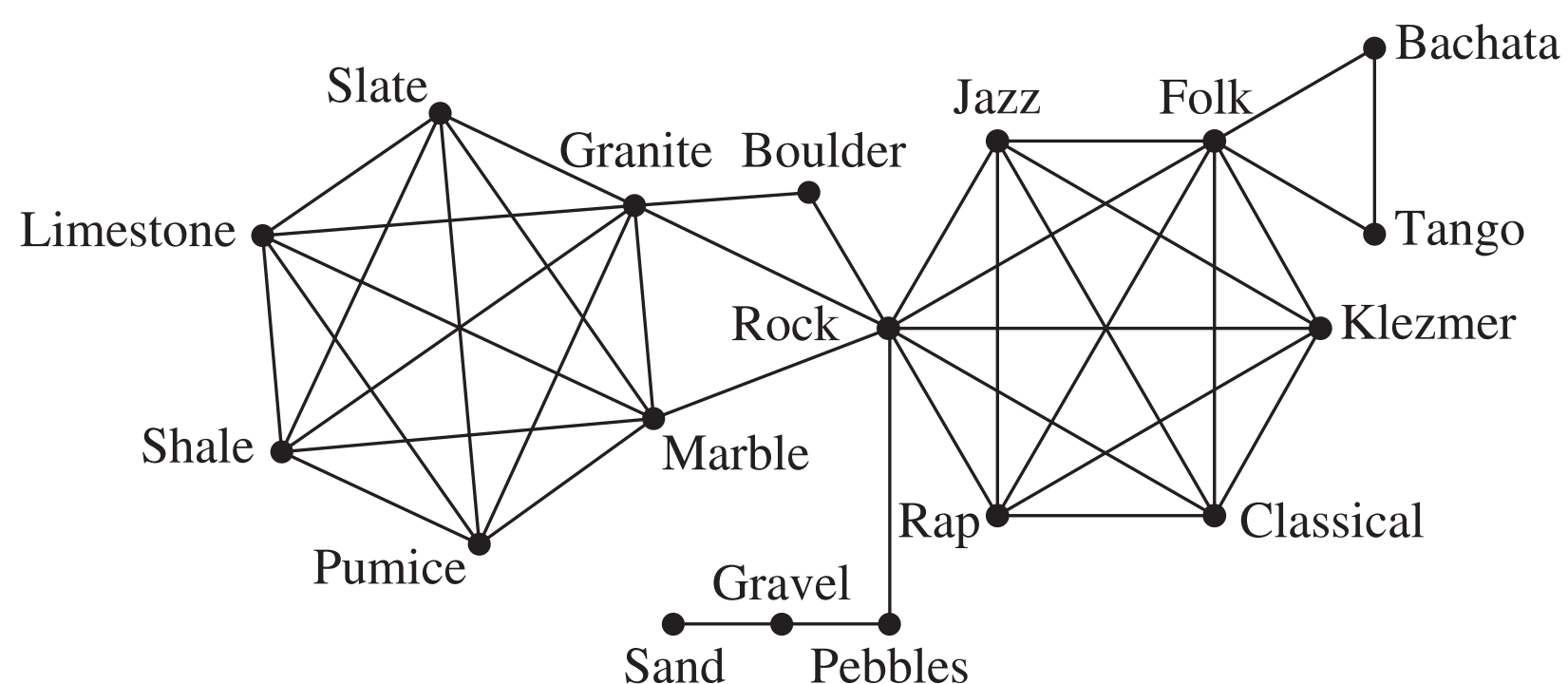
17.



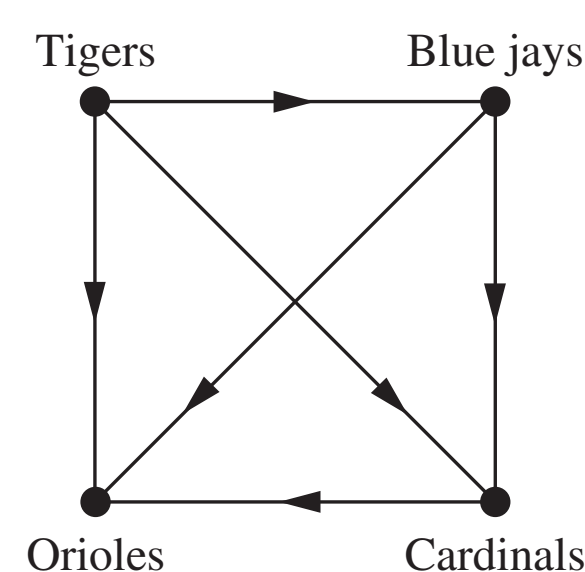
19.



21.



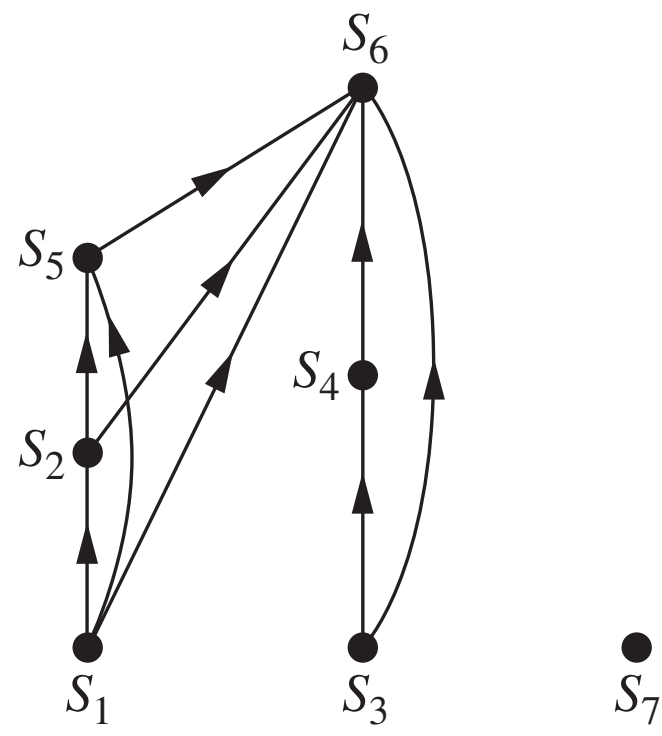
23.



25. We find the telephone numbers in the call graph for February that are not present in the call graph for January and vice versa. For each number we find, we make a list of the numbers they called or were called by using the edges in the call graph. We examine these lists to find new telephone numbers in February that had similar calling patterns to defunct telephone numbers in January. 27. We use the graph model that has e-mail addresses as vertices and for each message sent, an edge from the e-mail address of the sender to the e-mail address of the recipient. For each e-mail address, we can make a list of other addresses they sent messages to and a list of other addresses from which they received messages. If two e-mail addresses had almost the same pattern, we conclude that these addresses might have belonged to the same person who had recently changed his or her e-mail address. 29. Let  $V$  be the set of people at the party. Let  $E$  be the set of ordered pairs  $(u, v)$  in  $V \times V$  such that  $u$  knows  $v$ 's name. The edges are directed, but multiple edges are not allowed. Literally, there is a loop at each vertex, but for simplicity, the model could omit the loops. 31. Vertices are the courses; edges are directed; edge  $uv$  means that course  $u$  is prerequisite for course  $v$ ; courses without prerequisites are vertices with in-degree 0; courses that are not prerequisite for any other courses are vertices with out-degree 0. 33. Let the set of vertices be a set of people, and two vertices are joined by an edge if the two people were ever married. Ignoring complications, this graph has the property that there are two types of vertices (men and women), and every edge joins vertices of opposite types.



35.

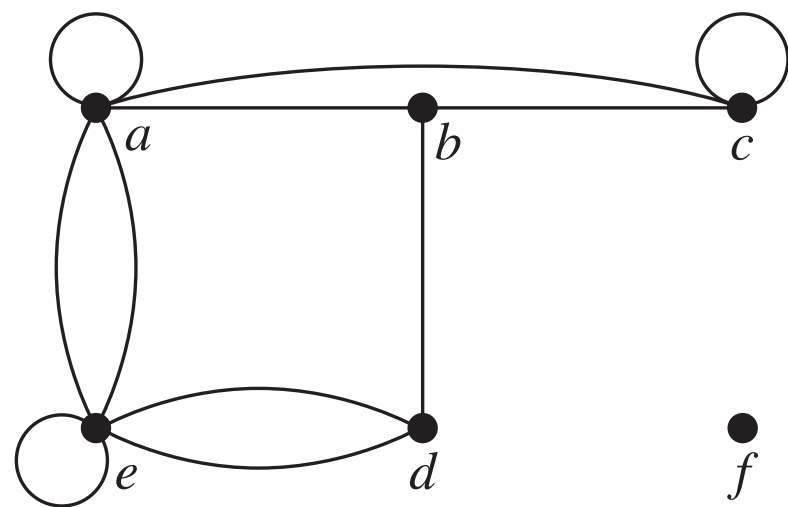


37. Represent people in the group by vertices. Put a directed edge into the graph for every pair of vertices. Label the edge from the vertex representing  $A$  to the vertex representing  $B$  with a + (plus) if  $A$  likes  $B$ , a - (minus) if  $A$  dislikes  $B$ , and a 0 if  $A$  is neutral about  $B$ .

## Section 10.2

1.  $v = 6$ ;  $e = 6$ ;  $\deg(a) = 2$ ,  $\deg(b) = 4$ ,  $\deg(c) = 1$ ,  $\deg(d) = 0$ ,  $\deg(e) = 2$ ,  $\deg(f) = 3$ ;  $c$  is pendant;  $d$  is isolated.  
 3.  $v = 9$ ;  $e = 12$ ;  $\deg(a) = 3$ ,  $\deg(b) = 2$ ,  $\deg(c) = 4$ ,  $\deg(d) = 0$ ,  $\deg(e) = 6$ ,  $\deg(f) = 0$ ;  $\deg(g) = 4$ ;  $\deg(h) = 2$ ;  $\deg(i) = 3$ ;  $d$  and  $f$  are isolated. 5. No 7.  $v = 4$ ;  $e = 7$ ;  $\deg^-(a) = 3$ ,  $\deg^-(b) = 1$ ,  $\deg^-(c) = 2$ ,  $\deg^-(d) = 1$ ,  $\deg^+(a) = 1$ ,  $\deg^+(b) = 2$ ,  $\deg^+(c) = 1$ ,  $\deg^+(d) = 3$  9. 5 vertices, 13 edges;  $\deg^-(a) = 6$ ,  $\deg^+(a) = 1$ ,  $\deg^-(b) = 1$ ,  $\deg^+(b) = 5$ ,  $\deg^-(c) = 2$ ,  $\deg^+(c) = 5$ ,  $\deg^-(d) = 4$ ,  $\deg^+(d) = 2$ ,  $\deg^-(e) = 0$ ,  $\deg^+(e) = 0$

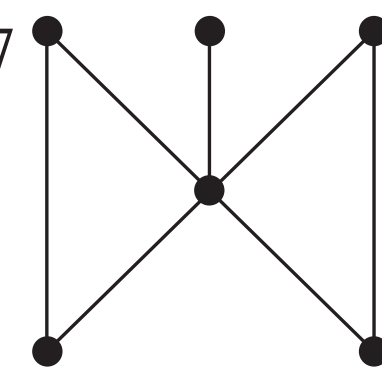
11.



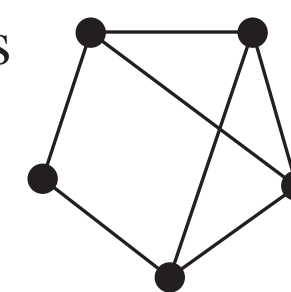
13. The number of coauthors that person has; that person's coauthors; a person who has no coauthors; a person who has only one coauthor 15. In the directed graph  $\deg^-(v)$  = number of calls  $v$  received,  $\deg^+(v)$  = number of calls  $v$  made; in the undirected graph,  $\deg(v)$  is the number of calls either made or received by  $v$ . 17.  $(\deg^+(v), \deg^-(v))$  is the win-loss record of  $v$ . 19. In the undirected graph model in which the vertices are people in the group and two vertices are adjacent if those two people are friends, the degree of a vertex is the number of friends in the group that person has. By Exercise 18, there are two vertices with the same degree, which means that there are two people in the group with the same number of friends in the group. 21. Bipartite 23. Not bipartite 25. Not bipartite 27. a) Parts  $\{h, s, n, w\}$  and  $\{P, Q, R, S\}$ ,  $E = \{\{P, n\}, \{P, w\}, \{Q, s\}, \{Q, n\}, \{R, n\}, \{R, w\}, \{S, h\}, \{S, s\}\}$  b) There is. c)  $\{Pw, Qs, Rn, Sh\}$  among others 29. Only Barry is willing to marry Uma and Xia. 31. Model this with an undirected bipartite graph, with an edge between a man and a woman if they are willing to marry each other. By

Hall's theorem, it is enough to show that for every set  $S$  of women, the set  $N(S)$  of men willing to marry them has cardinality at least  $|S|$ . Let  $m$  be the number of edges between  $S$  and  $N(S)$ . Since every vertex in  $S$  has degree  $k$ , it follows that  $m = k|S|$ . Because these edges are incident to  $N(S)$ , it follows that  $m \leq k|N(S)|$ . Therefore,  $k|S| \leq k|N(S)|$ , so  $|N(S)| \geq |S|$ . 33. Model this with the bipartite graph where  $V_1 = \{(p, i) \mid p \text{ is a winner and } i = 1, 2\}$  is the set in which each prize winner is represented twice,  $V_2$  is the set of prizes, and an edge represents a player wanting a prize. For any subset  $A$  of  $V_1$ ,  $N(A) = V_2$ , since every winner wants all of the  $2m$  prizes. So  $|N(A)| = 2m = |V_2| \geq |A|$  and Hall's theorem applies. 35. a)  $(\{a, b, c, f\}, \{\{a, b\}, \{a, f\}, \{b, c\}, \{b, f\}\})$  b)  $(\{a, x, c, f\}, \{\{a, x\}, \{c, x\}, \{e, x\}\})$  37. a)  $n$  vertices,  $n(n-1)/2$  edges b)  $n$  vertices,  $n$  edges c)  $n+1$  vertices,  $2n$  edges d)  $m+n$  vertices,  $mn$  edges e)  $2^n$  vertices,  $n2^{n-1}$  edges 39. a) 3, 3, 3, 3 b) 2, 2, 2, 2 c) 4, 3, 3, 3, 3 d) 3, 3, 2, 2, 2 e) 3, 3, 3, 3, 3, 3, 3, 3 41. Each of the  $n$  vertices is adjacent to each of the other  $n-1$  vertices, so the degree sequence is  $n-1, n-1, \dots, n-1$  ( $n$  terms).

43. 7

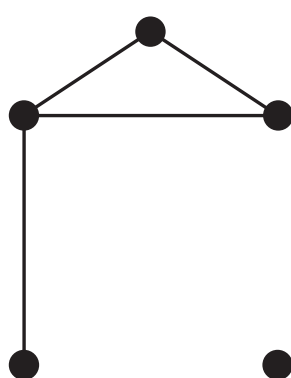


45. a) Yes



b) No c) No d) No

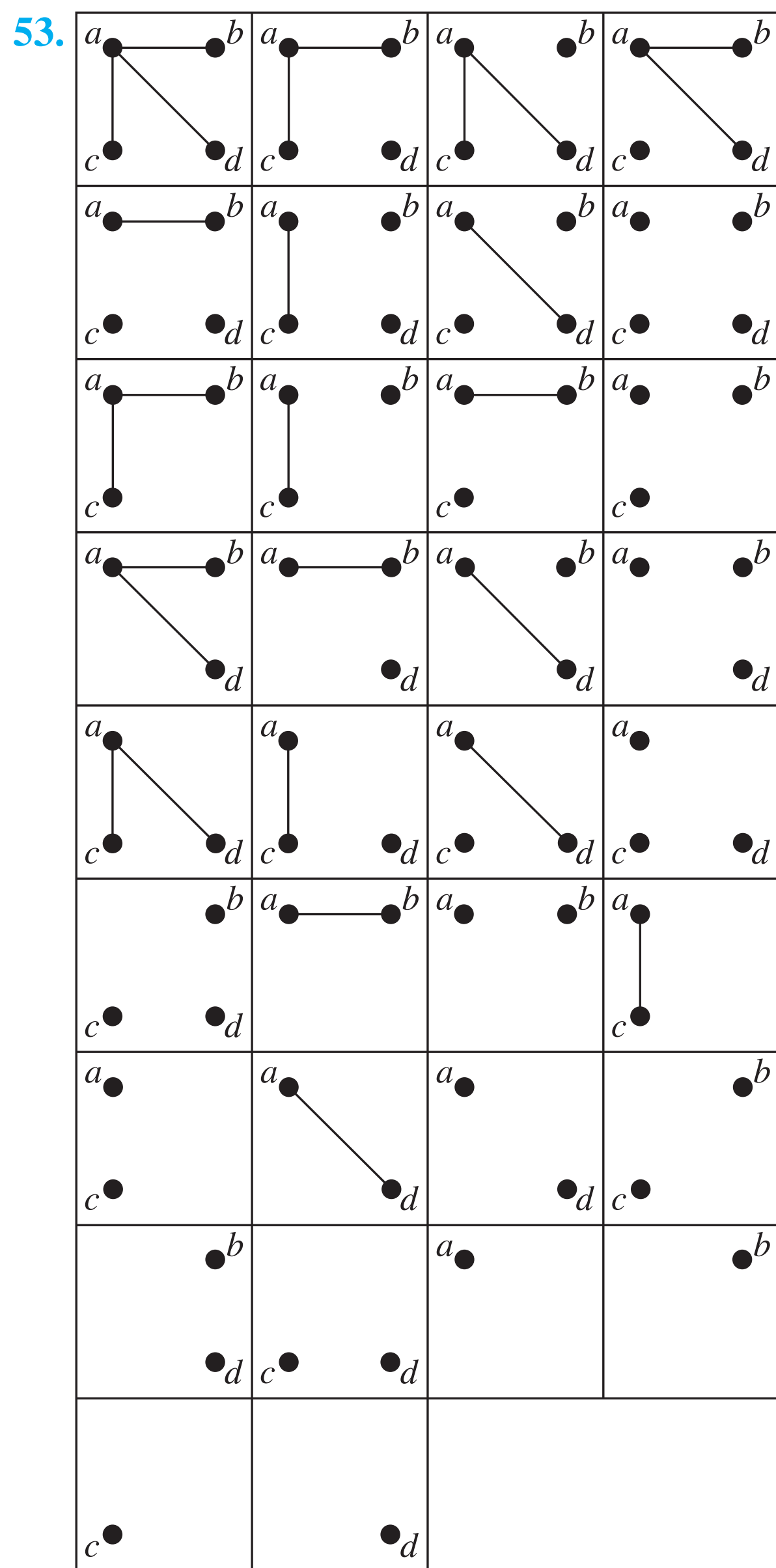
e) Yes



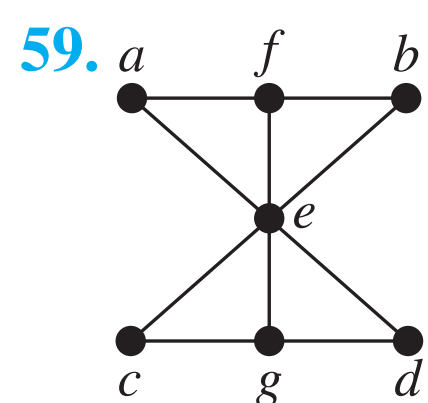
f) No 47. First, suppose that  $d_1, d_2, \dots, d_n$  is graphic. We must show that the sequence whose terms are  $d_2-1, d_3-1, \dots, d_{d_1+1}-1, d_{d_1+2}, d_{d_1+3}, \dots, d_n$  is graphic once it is put into nonincreasing order. In Exercise 46 it is proved that if the original sequence is graphic, then in fact there is a graph having this degree sequence in which the vertex of degree  $d_1$  is adjacent to the vertices of degrees  $d_2, d_3, \dots, d_{d_1+1}$ . Remove from this graph the vertex of highest degree ( $d_1$ ). The resulting graph has the desired degree sequence. Conversely, suppose that  $d_1, d_2, \dots, d_n$  is a nonincreasing sequence such that the sequence  $d_2-1, d_3-1, \dots, d_{d_1+1}-1, d_{d_1+2}, d_{d_1+3}, \dots, d_n$  is graphic once it is put into nonincreasing order. Take a graph with this latter degree sequence, where vertex  $v_i$  has degree  $d_i-1$  for  $2 \leq i \leq d_1+1$  and vertex  $v_i$  has degree  $d_i$  for  $d_1+2 \leq i \leq n$ . Adjoin one new vertex (call it  $v_1$ ), and put in an edge from  $v_1$  to each of the vertices  $v_2, v_3, \dots, v_{d_1+1}$ . The resulting graph has degree sequence  $d_1, d_2, \dots, d_n$ . 49. Let  $d_1, d_2, \dots, d_n$  be a nonincreasing sequence of nonnegative integers with an even sum. Construct a graph as follows: Take vertices  $v_1, v_2, \dots, v_n$  and put  $\lfloor d_i/2 \rfloor$  loops at vertex  $v_i$ , for  $i = 1, 2, \dots, n$ . For each  $i$ , vertex  $v_i$  now has degree either  $d_i$  or  $d_i-1$ . Because



the original sum was even, the number of vertices for which  $\deg(v_i) = d_i - 1$  is even. Pair them up arbitrarily, and put in an edge joining the vertices in each pair. 51. 17

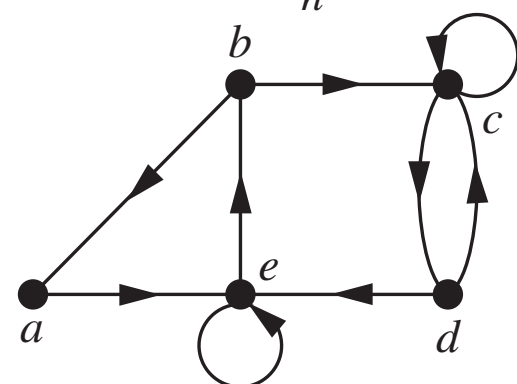


55. a) For all  $n \geq 1$  b) For all  $n \geq 3$  c) For  $n = 3$  d) For all  $n \geq 0$  57. 5

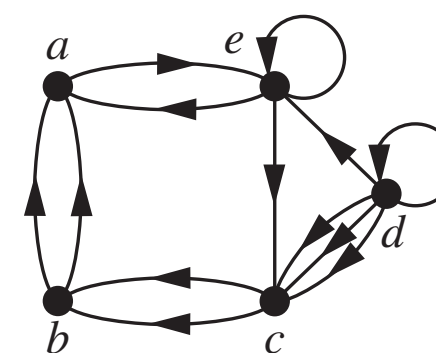


61. a) The graph with  $n$  vertices and no edges b) The disjoint union of  $K_m$  and  $K_n$  c) The graph with vertices  $\{v_1, \dots, v_n\}$  with an edge between  $v_i$  and  $v_j$  unless  $i \equiv j \pm 1 \pmod{n}$  d) The graph whose vertices are represented by bit strings of length  $n$  with an edge between two vertices if the associated bit strings differ in more than one bit 63.  $v(v-1)/2 - e$  65.  $n-1-d_n, n-1-d_{n-1}, \dots, n-1-d_2, n-1-d_1$  67. The union of  $G$  and  $\bar{G}$  contains an edge between each pair of the  $n$  vertices. Hence, this union is  $K_n$ .

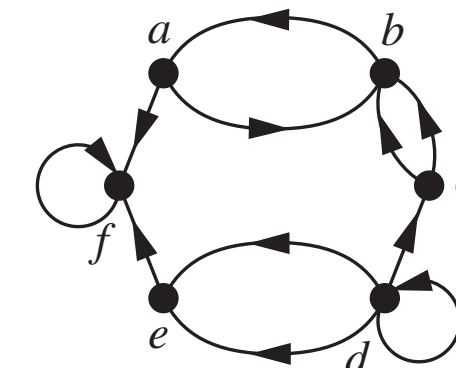
69. Exercise 7:



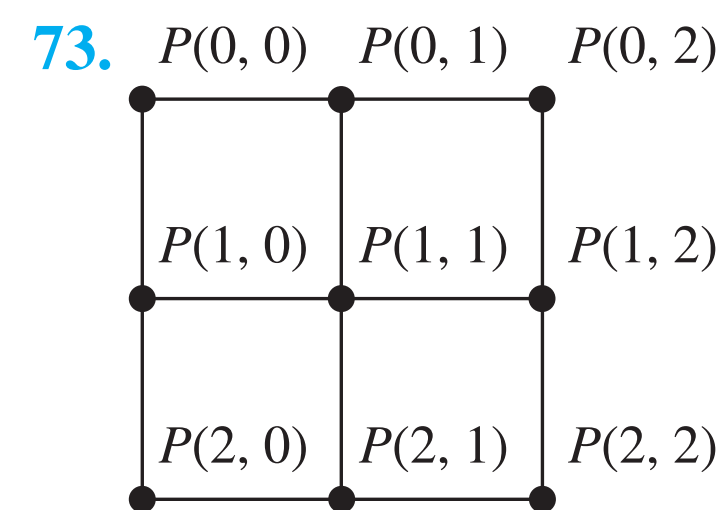
Exercise 8:



Exercise 9:



71. A directed graph  $G = (V, E)$  is its own converse if and only if it satisfies the condition  $(u, v) \in E$  if and only if  $(v, u) \in E$ . But this is precisely the condition that the associated relation must satisfy to be symmetric.



75. We can connect  $P(i, j)$  and  $P(k, l)$  by using  $|i - k|$  hops to connect  $P(i, j)$  and  $P(k, j)$  and  $|j - l|$  hops to connect  $P(k, j)$  and  $P(k, l)$ . Hence, the total number of hops required to connect  $P(i, j)$  and  $P(k, l)$  does not exceed  $|i - k| + |j - l|$ . This is less than or equal to  $m + m = 2m$ , which is  $O(m)$ .

### Section 10.3

1.

| Vertex | Adjacent Vertices |
|--------|-------------------|
| $a$    | $b, c, d$         |
| $b$    | $a, d$            |
| $c$    | $a, d$            |
| $d$    | $a, b, c$         |

3.

| Vertex | Terminal Vertices |
|--------|-------------------|
| $a$    | $a, b, c, d$      |
| $b$    | $d$               |
| $c$    | $a, b$            |
| $d$    | $b, c, d$         |

5.

$$\begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

7.

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix}$$

9. a)

$$\begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

b)

$$\begin{bmatrix} 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

c)

$$\begin{bmatrix} 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \end{bmatrix}$$

d)

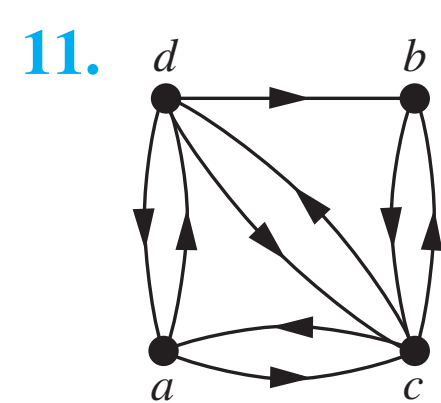
$$\begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

e)

$$\begin{bmatrix} 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 \end{bmatrix}$$

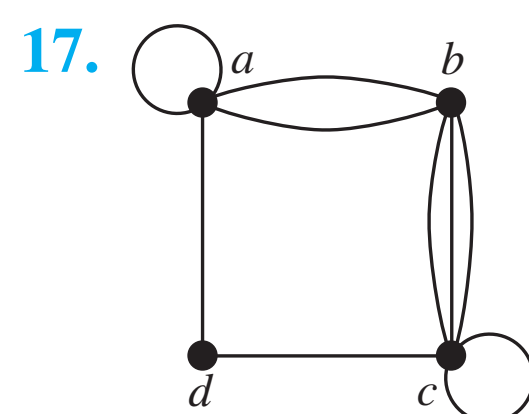


f) 
$$\begin{bmatrix} 0 & 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 \end{bmatrix}$$



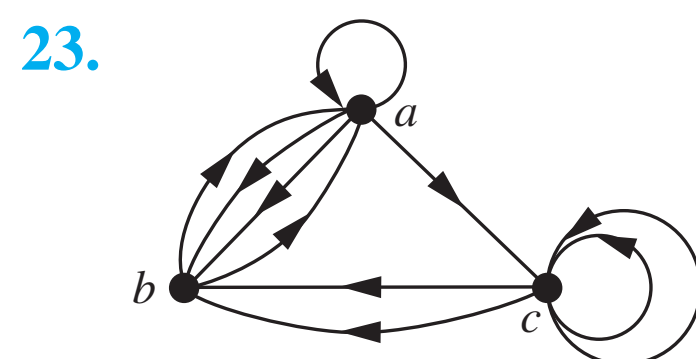
13. 
$$\begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 2 \\ 1 & 1 & 0 & 1 \\ 0 & 2 & 1 & 0 \end{bmatrix}$$

15. 
$$\begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & 1 & 2 \\ 2 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{bmatrix}$$



19. 
$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

21. 
$$\begin{bmatrix} 1 & 1 & 2 & 1 \\ 1 & 0 & 0 & 2 \\ 1 & 0 & 1 & 1 \\ 0 & 2 & 1 & 0 \end{bmatrix}$$



25. a) 3/7 b) 5/24 c) 19/36 27. a) dense b) sparse  
c) dense d) sparse e) sparse f) sparse 29. Yes

31. Exercise 13: 
$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix}$$

Exercise 14: 
$$\begin{bmatrix} 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 & 1 \end{bmatrix}$$

Exercise 15: 
$$\begin{bmatrix} 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 0 & 1 \end{bmatrix}$$

33.  $\deg(v)$  – number of loops at  $v$ ;  $\deg^-(v)$  35. 2 if  $e$  is not a loop, 1 if  $e$  is a loop

37. a) 
$$\begin{bmatrix} 1 & 1 & \dots & 1 & 0 & \dots & 0 \\ 1 & 0 & \dots & 0 & 1 & \dots & 0 \\ 0 & 1 & \dots & 0 & 1 & \dots & 0 \\ \vdots & \vdots & & \vdots & \vdots & & \vdots \\ 0 & 0 & \dots & 0 & 0 & \dots & 1 \\ 0 & 0 & \dots & 1 & 0 & \dots & 1 \end{bmatrix}$$

b) 
$$\begin{bmatrix} 1 & 0 & \dots & 0 & 1 \\ 1 & 1 & \dots & 0 & 0 \\ 0 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & \dots & 1 & 0 \\ 0 & 0 & \dots & 1 & 1 \end{bmatrix}$$

c) 
$$\begin{bmatrix} 0 & 0 & \dots & 0 & 1 & 1 & \dots & 1 \\ & & & & 1 & 0 & \dots & 0 \\ & & \mathbf{B} & & 0 & 1 & \dots & 0 \\ & & & & \vdots & \vdots & & \vdots \\ & & & & 0 & 0 & \dots & 1 \end{bmatrix}$$

where  $\mathbf{B}$  is the answer to (b)

d) 
$$\begin{bmatrix} 1 & 1 & \dots & 1 & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 & 1 & \dots & 0 \\ \vdots & \vdots & & \vdots & \vdots & & \vdots \\ 0 & 0 & \dots & 0 & 0 & \dots & 1 \\ 1 & 0 & \dots & 0 & 1 & \dots & 0 \\ 0 & 1 & \dots & 0 & 0 & \dots & 1 \\ \vdots & \vdots & & \vdots & \vdots & & \vdots \\ 0 & 0 & \dots & 1 & 0 & \dots & 0 \end{bmatrix}$$

39. Isomorphic 41. Isomorphic 43. Isomorphic 45. Not isomorphic 47. Isomorphic 49.  $G$  is isomorphic to itself by the identity function, so isomorphism is reflexive. Suppose that  $G$  is isomorphic to  $H$ . Then there exists a one-to-one correspondence  $f$  from  $G$  to  $H$  that preserves adjacency and nonadjacency. It follows that  $f^{-1}$  is a one-to-one correspondence from  $H$  to  $G$  that preserves adjacency and nonadjacency. Hence, isomorphism is symmetric. If  $G$  is isomorphic to  $H$  and  $H$  is isomorphic to  $K$ , then there are one-to-one correspondences  $f$  and  $g$  from  $G$  to  $H$  and from  $H$  to  $K$  that preserve adjacency and nonadjacency. It follows that  $g \circ f$  is a one-to-one correspondence from  $G$  to  $K$  that preserves adjacency and nonadjacency. Hence, isomorphism is transitive. 51. All zeros 53. Label the vertices in order so that all of the vertices in the first set of the partition of the vertex set come first. Because no edges join vertices in the same set of the partition, the matrix has the desired form. 55.  $C_5$  57.  $n = 5$  only 59. 4 61. 2 63. a) Yes b) No c) No 65.  $G = (V_1, E_1)$  is isomorphic to  $H = (V_2, E_2)$  if and only if there exist functions  $f$  from  $V_1$  to  $V_2$  and  $g$  from  $E_1$  to  $E_2$  such that each is a one-to-one correspondence and for every edge  $e$  in  $E_1$  the endpoints of  $g(e)$  are  $f(v)$  and  $f(w)$  where  $v$  and  $w$  are the endpoints of  $e$ . 67. Yes 69. Yes 71. If  $f$  is an isomorphism from a directed graph  $G$  to a directed graph  $H$ , then  $f$  is also an isomorphism from  $G^{\text{conv}}$  to  $H^{\text{conv}}$ . To see this note that  $(u, v)$  is an edge of  $G^{\text{conv}}$  if and only if  $(v, u)$  is an edge of  $G$  if and only if  $(f(v), f(u))$  is an edge of  $H$  if and only if  $(f(u), f(v))$  is an edge of  $H^{\text{conv}}$ . 73. Many answers are possible; for example,  $C_6$  and  $C_3 \cup C_3$ . 75. The product is  $[a_{ij}]$  where  $a_{ij}$  is the number of edges from  $v_i$  to  $v_j$  when  $i \neq j$  and  $a_{ii}$  is the number of edges incident to  $v_i$ . 77. The graphs in Exercise 45 are a devil's pair.



## Section 10.4

**1. a)** Path of length 4; not a circuit; not simple **b)** Not a path **c)** Not a path **d)** Simple circuit of length 5 **3.** No **5.** No **7.** Maximal sets of people with the property that for any two of them, we can find a string of acquaintances that takes us from one to the other **9.** If a person has Erdős number  $n$ , then there is a path of length  $n$  from that person to Erdős in the collaboration graph, so by definition, that means that that person is in the same component as Erdős. If a person is in the same component as Erdős, then there is a path from that person to Erdős, and the length of the shortest such path is that person's Erdős number. **11. a)** Weakly connected **b)** Weakly connected **c)** Not strongly or weakly connected **13.** The maximal sets of phone numbers for which it is possible to find directed paths between every two different numbers in the set **15. a)**  $\{a, b, f\}, \{c, d, e\}$  **b)**  $\{a, b, c, d, e, h\}, \{f\}, \{g\}$  **c)**  $\{a, b, d, e, f, g, h, i\}, \{c\}$  **17.** Suppose the strong components of  $u$  and  $v$  are not disjoint, say with vertex  $w$  in both. Suppose  $x$  is a vertex in the strong component of  $u$ . Then  $x$  is also in the strong component of  $v$ , because there is a path from  $x$  to  $v$  (namely the path from  $x$  to  $u$  followed by the path from  $u$  to  $w$  followed by the path from  $w$  to  $v$ ) and vice versa. Thus,  $x$  is in the strong component of  $v$ . This shows that the strong component of  $u$  is a subgraph of the strong component of  $v$ , and equality follows by symmetry. **19. a)** 2 **b)** 7 **c)** 20 **d)** 61 **21.** Not isomorphic ( $G$  has a triangle;  $H$  does not) **23.** Isomorphic (the path  $u_1, u_2, u_7, u_6, u_5, u_4, u_3, u_8, u_1$  corresponds to the path  $v_1, v_2, v_3, v_4, v_5, v_8, v_7, v_6, v_1$ ) **25. a)** 3 **b)** 0 **c)** 27 **d)** 0 **27. a)** 1 **b)** 0 **c)** 2 **d)** 1 **e)** 5 **f)** 3 **29.**  $R$  is reflexive by definition. Assume that  $(u, v) \in R$ ; then there is a path from  $u$  to  $v$ . Then  $(v, u) \in R$  because there is a path from  $v$  to  $u$ , namely, the path from  $u$  to  $v$  traversed backward. Assume that  $(u, v) \in R$  and  $(v, w) \in R$ ; then there are paths from  $u$  to  $v$  and from  $v$  to  $w$ . Putting these two paths together gives a path from  $u$  to  $w$ . Hence,  $(u, w) \in R$ . It follows that  $R$  is transitive. **31. c** **33. b, c, e, i** **35.** If a vertex is pendant it is clearly not a cut vertex. So an endpoint of a cut edge that is a cut vertex is not pendant. Removal of a cut edge produces a graph with more connected components than in the original graph. If an endpoint of a cut edge is not pendant, the connected component it is in after the removal of the cut edge contains more than just this vertex. Consequently, removal of that vertex and all edges incident to it, including the original cut edge, produces a graph with more connected components than were in the original graph. Hence, an endpoint of a cut edge that is not pendant is a cut vertex. **37.** Assume there exists a connected graph  $G$  with at most one vertex that is not a cut vertex. Define the distance between the vertices  $u$  and  $v$ , denoted by  $d(u, v)$ , to be the length of the shortest path between  $u$  and  $v$  in  $G$ . Let  $s$  and  $t$  be vertices in  $G$  such that  $d(s, t)$  is a maximum. Either  $s$  or  $t$  (or both) is a cut vertex, so without loss of generality suppose that  $s$  is a cut vertex. Let  $w$  belong to the connected component that does not contain  $t$  of the graph obtained by deleting  $s$  and all edges incident to  $s$  from  $G$ . Because every path from  $w$  to  $t$  contains  $s$ ,

$d(w, t) > d(s, t)$ , which is a contradiction. **39. a)** Denver–Chicago, Boston–New York **b)** Seattle–Portland, Portland–San Francisco, Salt Lake City–Denver, New York–Boston, Boston–Burlington, Boston–Bangor **41.** A minimal set of people who collectively influence everyone (directly or indirectly);  $\{\text{Deborah}\}$  **43.** An edge cannot connect two vertices in different connected components. Because there are at most  $C(n_i, 2)$  edges in the connected component with  $n_i$  vertices, it follows that there are at most  $\sum_{i=1}^k C(n_i, 2)$  edges in the graph. **45.** Suppose that  $G$  is not connected. Then it has a component of  $k$  vertices for some  $k$ ,  $1 \leq k \leq n - 1$ . The most edges  $G$  could have is  $C(k, 2) + C(n - k, 2) = [k(k - 1) + (n - k)(n - k - 1)]/2 = k^2 - nk + (n^2 - n)/2$ . This quadratic function of  $f$  is minimized at  $k = n/2$  and maximized at  $k = 1$  or  $k = n - 1$ . Hence, if  $G$  is not connected, the number of edges does not exceed the value of this function at 1 and at  $n - 1$ , namely,  $(n - 1)(n - 2)/2$ . **47. a)** 1 **b)** 2 **c)** 6 **d)** 21 **49. a)** Removing an edge from a cycle leaves a path, which is still connected. **b)** Removing an edge from the cycle portion of the wheel leaves that portion still connected and the central vertex still connected to it as well. Removing a spoke leaves the cycle intact and the central vertex still connected to it as well. **c)** Any four vertices, two from each part of the bipartition, are connected by a 4-cycle; removing one edge does not disconnect them. **d)** Deleting the edge joining  $(b_1, b_2, \dots, b_{i-1}, 0, b_{i+1}, \dots, b_n)$  and  $(b_1, b_2, \dots, b_{i-1}, 1, b_{i+1}, \dots, b_n)$  does not disconnect the graph because these two vertices are still joined via the path  $(b_1, b_2, \dots, b_{i-1}, 0, b_{i+1}, \dots, 0)$ ,  $(b_1, b_2, \dots, b_{i-1}, 0, b_{i+1}, \dots, 1)$ ,  $(b_1, b_2, \dots, b_{i-1}, 1, b_{i+1}, \dots, 1)$ ,  $(b_1, b_2, \dots, b_{i-1}, 1, b_{i+1}, \dots, 0)$  if  $n < 2$  and  $b_n = 0$ , and similarly in the other three cases. **51.** If  $G$  is complete, then removing vertices one by one leaves a complete graph at each step, so we never get a disconnected graph. Conversely, if edge  $uv$  is missing from  $G$ , then removing all the vertices except  $u$  and  $v$  creates a disconnected graph. **53.** Both equal  $\min(m, n)$ . **55.** Let  $G$  be a graph with  $n$  vertices; then  $\kappa(G) \leq n - 1$ . Let  $C$  be a smallest edge cut, leaving a nonempty proper subset  $S$  of the vertices of  $G$  disconnected from the complementary set  $S' = V - S$ . If  $xy$  is an edge of  $G$  for every  $x \in S$  and  $y \in S'$ , then the size of  $C$  is  $|S||S'|$ , which is at least  $n - 1$ , so  $\kappa(G) \leq \lambda(G)$ . Otherwise, let  $x \in S$  and  $y \in S'$  be nonadjacent vertices. Let  $T$  consist of all neighbors of  $x$  in  $S'$  together with all vertices of  $S - \{x\}$  with neighbors in  $S'$ . Then  $T$  is a vertex cut, because it separates  $x$  and  $y$ . Now look at the edges from  $x$  to  $T \cap S'$  and one edge from each vertex of  $T \cap S$  to  $S'$ ; this gives us  $|T|$  distinct edges that lie in  $C$ , so  $\lambda(G) = |C| \geq |T| \geq \kappa(G)$ . **57. 2** **59.** Let the simple paths  $P_1$  and  $P_2$  be  $u = x_0, x_1, \dots, x_n = v$  and  $u = y_0, y_1, \dots, y_m = v$ , respectively. The paths thus start out at the same vertex. Since the paths do not contain the same set of edges, they must diverge eventually. If they diverge only after one of them has ended, then the rest of the other path is a simple circuit from  $v$  to  $v$ . Otherwise we can suppose that  $x_0 = y_0, x_1 = y_1, \dots, x_i = y_i$ , but  $x_{i+1} \neq y_{i+1}$ . To form our simple circuit, we follow the path  $y_i, y_{i+1}, y_{i+2}$ , and so on,



until it once again first encounters a vertex on  $P_1$  (possibly as early as  $y_{i+1}$ , no later than  $y_m$ ). Once we are back on  $P_1$ , we follow it along—forward or backwards, as necessary—to return to  $x_i$ . Since  $x_i = y_i$ , this certainly forms a circuit. It must be a simple circuit, since no edge among the  $x_k$ s or the  $y_l$ s can be repeated ( $P_1$  and  $P_2$  are simple by hypothesis) and no edge among the  $x_k$ s can equal one of the edges  $y_l$  that we used, since we abandoned  $P_2$  for  $P_1$  as soon as we hit  $P_1$ . **61.** The graph  $G$  is connected if and only if every off-diagonal entry of  $\mathbf{A} + \mathbf{A}^2 + \mathbf{A}^3 + \dots + \mathbf{A}^{n-1}$  is positive, where  $\mathbf{A}$  is the adjacency matrix of  $G$ . **63.** If the graph is bipartite, say with parts  $A$  and  $B$ , then the vertices in every path must alternately lie in  $A$  and  $B$ . Therefore, a path that starts in  $A$ , say, will end in  $B$  after an odd number of steps and in  $A$  after an even number of steps. Because a circuit ends at the same vertex where it starts, the length must be even. Conversely, suppose that all circuits have even length; we must show that the graph is bipartite. We can assume that the graph is connected, because if it is not, then we can just work on one component at a time. Let  $v$  be a vertex of the graph, and let  $A$  be the set of all vertices to which there is a path of odd length starting at  $v$ , and let  $B$  be the set of all vertices to which there is a path of even length starting at  $v$ . Because the component is connected, every vertex lies in  $A$  or  $B$ . No vertex can lie in both  $A$  and  $B$ , because if one did, then following the odd-length path from  $v$  to that vertex and then back along the even-length path from that vertex to  $v$  would produce an odd circuit, contrary to the hypothesis. Thus, the set of vertices has been partitioned into two sets. To show that every edge has endpoints in different parts, suppose that  $xy$  is an edge, where  $x \in A$ . Then the odd-length path from  $v$  to  $x$  followed by  $xy$  produces an even-length path from  $v$  to  $y$ , so  $y \in B$ . (Similarly, if  $x \in B$ .) **65.**  $(H_1 W_1 H_2 W_2 \langle \text{boat} \rangle, \emptyset) \rightarrow (H_2 W_2, H_1 W_1 \langle \text{boat} \rangle) \rightarrow (H_1 H_2 W_2 \langle \text{boat} \rangle, W_1) \rightarrow (W_2, H_1 W_1 H_2 \langle \text{boat} \rangle) \rightarrow (H_2 W_2 \langle \text{boat} \rangle, H_1 W_1) \rightarrow (\emptyset, H_1 W_1 H_2 W_2 \langle \text{boat} \rangle)$

## Section 10.5

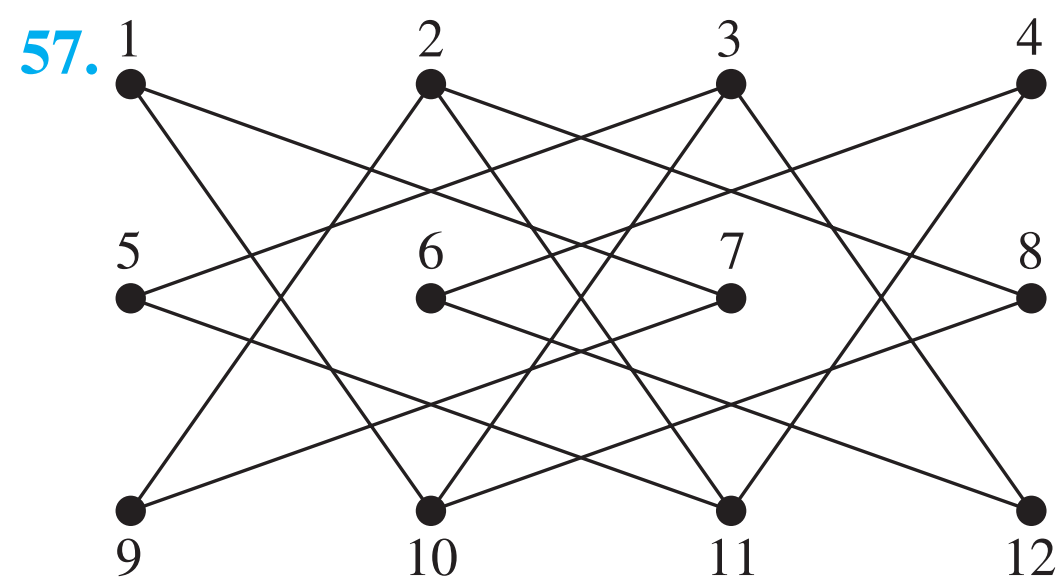
**1.** Neither **3.** No Euler circuit;  $a, e, c, e, b, e, d, b, a, c, d$  **5.**  $a, b, c, d, c, e, d, b, e, a, e, a$  **7.**  $a, i, h, g, d, e, f, g, c, e, h, d, c, a, b, i, c, b, h, a$  **9.** No,  $A$  still has odd degree. **11.** When the graph in which vertices represent intersections and edges streets has an Euler path **13.** Yes **15.** No **17.** If there is an Euler path, then as we follow it each vertex except the starting and ending vertices must have equal in-degree and out-degree, because whenever we come to a vertex along an edge, we leave it along another edge. The starting vertex must have out-degree 1 larger than its in-degree, because we use one edge leading out of this vertex and whenever we visit it again we use one edge leading into it and one leaving it. Similarly, the ending vertex must have in-degree 1 greater than its out-degree. Because the Euler path with directions erased produces a path between any two vertices, in the underlying undirected graph, the graph is weakly connected. Conversely, suppose the graph meets the degree conditions stated. If we add one more edge from the vertex of deficient out-

degree to the vertex of deficient in-degree, then the graph has every vertex with equal in-degree and out-degree. Because the graph is still weakly connected, by Exercise 16 this new graph has an Euler circuit. Now delete the added edge to obtain the Euler path. **19.** Neither **21.** No Euler circuit;  $a, d, e, d, b, a, e, c, e, b, c, b, e$  **23.** Neither **25.** Follow the same procedure as Algorithm 1, taking care to follow the directions of edges. **27.** **a)**  $n = 2$  **b)** None **c)** None **d)**  $n = 1$  **29.** Exercise 1: 1 time; Exercises 2–7: 0 times **31.**  $a, b, c, d, e, a$  is a Hamilton circuit. **33.** No Hamilton circuit exists, because once a purported circuit has reached  $e$  it would have nowhere to go. **35.** No Hamilton circuit exists, because every edge in the graph is incident to a vertex of degree 2 and therefore must be in the circuit. **37.**  $a, b, c, f, d, e$  is a Hamilton path. **39.**  $f, e, d, a, b, c$  is a Hamilton path. **41.** No Hamilton path exists. There are eight vertices of degree 2, and only two of them can be end vertices of a path. For each of the other six, their two incident edges must be in the path. It is not hard to see that if there is to be a Hamilton path, exactly one of the inside corner vertices must be an end, and that this is impossible. **43.**  $a, b, c, f, i, h, g, d, e$  is a Hamilton path. **45.**  $m = n \geq 2$  **47.** **a)** (i) No, (ii) No, (iii) Yes **b)** (i) No, (ii) No, (iii) Yes **c)** (i) Yes, (ii) Yes, (iii) Yes **d)** (i) Yes, (ii) Yes, (iii) Yes **49.** The result is trivial for  $n = 1$ : code is 0, 1. Assume we have a Gray code of order  $n$ . Let  $c_1, \dots, c_k, k = 2^n$  be such a code. Then  $0c_1, \dots, 0c_k, 1c_k, \dots, 1c_1$  is a Gray code of order  $n + 1$ .

**51. procedure Fleury**( $G = (V, E)$ : connected multigraph with the degrees of all vertices even,  $V = \{v_1, \dots, v_n\}$ )  
 $v := v_1$   
 $circuit := v$   
 $H := G$   
**while**  $H$  has edges  
     $e :=$  first edge with endpoint  $v$  in  $H$  (with respect to listing of  $V$ ) such that  $e$  is not a cut edge of  $H$ , if one exists, and simply the first edge in  $H$  with endpoint  $v$  otherwise  
     $w :=$  other endpoint of  $e$   
     $circuit := circuit$  with  $e, w$  added  
     $v := w$   
     $H := H - e$   
**return**  $circuit$  { $circuit$  is an Euler circuit}

**53.** If  $G$  has an Euler circuit, then it also has an Euler path. If not, add an edge between the two vertices of odd degree and apply the algorithm to get an Euler circuit. Then delete the new edge. **55.** Suppose  $G = (V, E)$  is a bipartite graph with  $V = V_1 \cup V_2$ , where  $V_1 \cap V_2 = \emptyset$  and no edge connects a vertex in  $V_1$  and a vertex in  $V_2$ . Suppose that  $G$  has a Hamilton circuit. Such a circuit must be of the form  $a_1, b_1, a_2, b_2, \dots, a_k, b_k, a_1$ , where  $a_i \in V_1$  and  $b_i \in V_2$  for  $i = 1, 2, \dots, k$ . Because the Hamilton circuit visits each vertex exactly once, except for  $v_1$ , where it begins and ends, the number of vertices in the graph equals  $2k$ , an even number. Hence, a bipartite graph with an odd number of vertices cannot have a Hamilton circuit.





59. We represent the squares of a  $3 \times 4$  chessboard as follows:

|   |    |    |    |
|---|----|----|----|
| 1 | 2  | 3  | 4  |
| 5 | 6  | 7  | 8  |
| 9 | 10 | 11 | 12 |

A knight's tour can be made by following the moves 8, 10, 1, 7, 9, 2, 11, 5, 3, 12, 6, 4. 61. We represent the squares of a  $4 \times 4$  chessboard as follows:

|    |    |    |    |
|----|----|----|----|
| 1  | 2  | 3  | 4  |
| 5  | 6  | 7  | 8  |
| 9  | 10 | 11 | 12 |
| 13 | 14 | 15 | 16 |

There are only two moves from each of the four corner squares. If we include all the edges 1–10, 1–7, 16–10, and 16–7, a circuit is completed too soon, so at least one of these edges must be missing. Without loss of generality, assume the path starts 1–10, 10–16, 16–7. Now the only moves from square 3 are to squares 5, 10, and 12, and square 10 already has two incident edges. Therefore, 3–5 and 3–12 must be in the Hamilton circuit. Similarly, edges 8–2 and 8–15 must be in the circuit. Now the only moves from square 9 are to squares 2, 7, and 15. If there were edges from square 9 to both squares 2 and 15, a circuit would be completed too soon. Therefore, the edge 9–7 must be in the circuit giving square 7 its full complement of edges. But now square 14 is forced to be joined to squares 5 and 12, completing a circuit too soon (5–14–12–3–5). This contradiction shows that there is no knight's tour on the  $4 \times 4$  board. 63. Because there are  $mn$  squares on an  $m \times n$  board, if both  $m$  and  $n$  are odd, there are an odd number of squares. Because by Exercise 62 the corresponding graph is bipartite, by Exercise 55 it has no Hamilton circuit. Hence, there is no reentrant knight's tour. 65. a) If  $G$  does not have a Hamilton circuit, continue as long as possible adding missing edges one at a time in such a way that we do not obtain a graph with a Hamilton circuit. This cannot go on forever, because once we've formed the complete graph by adding all missing edges, there is a Hamilton circuit. Whenever the process stops, we have obtained a (necessarily noncomplete) graph  $H$

with the desired property. b) Add one more edge to  $H$ . This produces a Hamilton circuit, which uses the added edge. The path consisting of this circuit with the added edge omitted is a Hamilton path in  $H$ . c) Clearly  $v_1$  and  $v_n$  are not adjacent in  $H$ , because  $H$  has no Hamilton circuit. Therefore, they are not adjacent in  $G$ . But the hypothesis was that the sum of the degrees of vertices not adjacent in  $G$  was at least  $n$ . This inequality can be rewritten as  $n - \deg(v_n) \leq \deg(v_1)$ . But  $n - \deg(v_n)$  is just the number of vertices not adjacent to  $v_n$ . d) Because there is no vertex following  $v_n$  in the Hamilton path,  $v_n$  is not in  $S$ . Each one of the  $\deg(v_1)$  vertices adjacent to  $v_1$  gives rise to an element of  $S$ , so  $S$  contains  $\deg(v_1)$  vertices. e) By part (c) there are at most  $\deg(v_1) - 1$  vertices other than  $v_n$  not adjacent to  $v_n$ , and by part (d) there are  $\deg(v_1)$  vertices in  $S$ , none of which is  $v_n$ . Therefore, at least one vertex of  $S$  is adjacent to  $v_n$ . By definition, if  $v_k$  is this vertex, then  $H$  contains edges  $v_k v_n$  and  $v_1 v_{k+1}$ , where  $1 < k < n - 1$ . f) Now  $v_1, v_2, \dots, v_{k-1}, v_k, v_n, v_{n-1}, \dots, v_{k+1}, v_1$  is a Hamilton circuit in  $H$ , contradicting the construction of  $H$ . Therefore, our assumption that  $G$  did not originally have a Hamilton circuit is wrong, and our proof by contradiction is complete. 67. Assume that there is a Hamilton circuit  $H$ . First assume that edge  $aj \notin H$ . Then  $ab, ai, jk$ , and  $jl$  are forced to be in  $H$ , which means that  $kl \notin H$ , and then  $gk$  and  $dl$  are in  $H$ . If  $ch$  were not in  $H$ , then  $bc, cd, gh$ , and  $hi$  would all have to be in  $H$ , completing a circuit without all the vertices. It follows that  $ch \in H$ . A circuit would be completed too soon if either both  $bc$  and  $cd$  were in  $H$ , or both  $gh$  and  $hi$  were in  $H$ , so by symmetry WOLOG assume that  $cd \in H$  and  $hi \in H$  and the other two edges are not. This then forces  $bf \in H$  and  $fg \in H$ , and we have completed a circuit without vertex  $e$ . Going back to the beginning, we now know that  $aj \in H$ . Because of the  $120^\circ$  rotational symmetry of the figure, it then follows that  $dl$  and  $gk$  are in  $H$  as well. It is clearly now impossible to include all three of the vertices  $i, j$ , and  $k$  in the Hamilton circuit, and our proof by contradiction is complete.

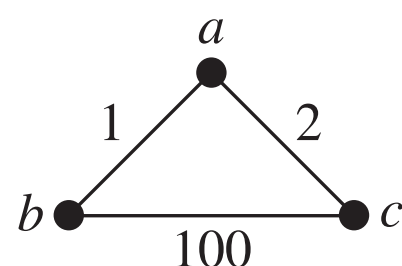
## Section 10.6

1. a) Vertices are the stops, edges join adjacent stops, weights are the times required to travel between adjacent stops. b) Same as part (a), except weights are distances between adjacent stops. c) Same as part (a), except weights are fares between stops. 3. 16 5. Exercise 2:  $a, b, e, d, z$ ; Exercise 3:  $a, c, d, e, g, z$ ; Exercise 4:  $a, b, e, h, l, m, p, s, z$  7. a)  $a, c, d$  b)  $a, c, d, f$  c)  $c, d, f$  e)  $b, d, e, g, z$  9. a) Direct b) Via New York c) Via Atlanta and Chicago d) Via New York 11. a) Via Chicago b) Via Chicago c) Via Los Angeles d) Via Chicago 13. a) Via Chicago b) Via Chicago c) Via Los Angeles d) Via Chicago 15. Do not stop the algorithm when  $z$  is added to the set  $S$ . 17. a) Via Woodbridge, via Woodbridge and Camden b) Via Woodbridge, via Woodbridge and Camden 19. For instance, sightseeing tours, street cleaning



| 21.      | <i>a</i> | <i>b</i> | <i>c</i> | <i>d</i> | <i>e</i> | <i>z</i> |
|----------|----------|----------|----------|----------|----------|----------|
| <i>a</i> | 4        | 3        | 2        | 8        | 10       | 13       |
| <i>b</i> | 3        | 2        | 1        | 5        | 7        | 10       |
| <i>c</i> | 2        | 1        | 2        | 6        | 8        | 11       |
| <i>d</i> | 8        | 5        | 6        | 4        | 2        | 5        |
| <i>e</i> | 10       | 7        | 8        | 2        | 4        | 3        |
| <i>z</i> | 13       | 10       | 11       | 5        | 3        | 6        |

23.  $O(n^3)$  25.  $a-c-b-d-a$  (or the same circuit starting at some other point and/or traversing the vertices in reverse order) 27. San Francisco–Denver–Detroit–New York–Los Angeles–San Francisco (or the same circuit starting at some other point and/or traversing the vertices in reverse order) 29. Consider this graph:

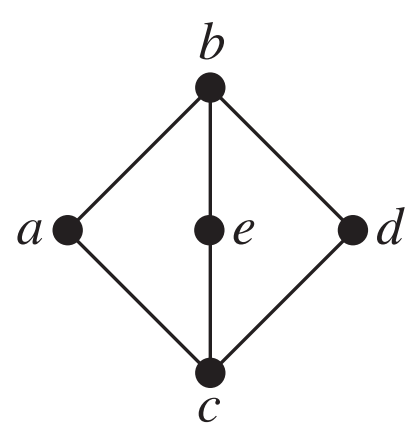


The circuit  $a-b-a-c-a$  visits each vertex at least once (and the vertex  $a$  twice) and has total weight 6. Every Hamilton circuit has total weight 103. 31. Let  $v_1, v_2, \dots, v_n$  be a topological ordering of the vertices of the given directed acyclic graph. Let  $w(i, j)$  be the weight of edge  $v_i v_j$ . Iteratively define  $P(i)$  with the intent that it will be the weight of a longest path ending at  $v_i$  and  $C(i)$  with the intent that it will be the vertex preceding  $v_i$  in some longest path: For  $i$  from 1 to  $n$ , let  $P(i)$  be the maximum of  $P(j) + w(j, i)$  over all  $j < i$  such that  $v_j v_i$  is an edge in the directed graph (and if such a  $j$  exists let  $C(i)$  be a value of  $j$  for which this maximum is achieved) and let  $P(i) = 0$  if there are no such values of  $j$ . At the conclusion of this loop, a longest path can be found by choosing  $i$  that maximizes  $P(i)$  and following the  $C$  links back to the start of the path.

## Section 10.7

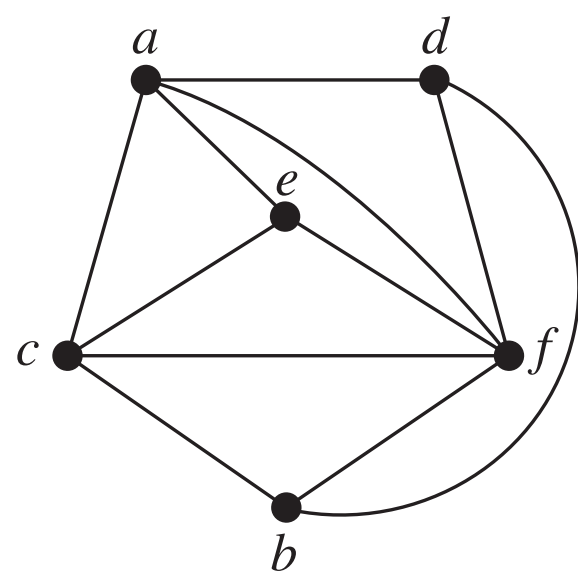
1. Yes

3.



5. No

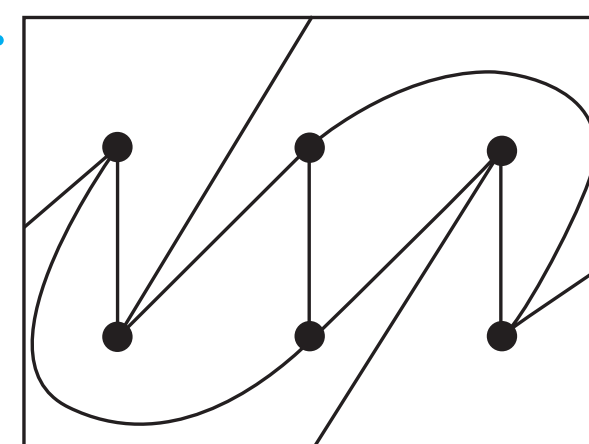
7. Yes



9. No 11. A triangle is formed by the planar representation of the subgraph of  $K_5$  consisting of the edges connecting  $v_1, v_2$ , and  $v_3$ . The vertex  $v_4$  must be placed either within the triangle or outside of it. We will consider only the case when  $v_4$  is inside the triangle; the other case is similar. Drawing the

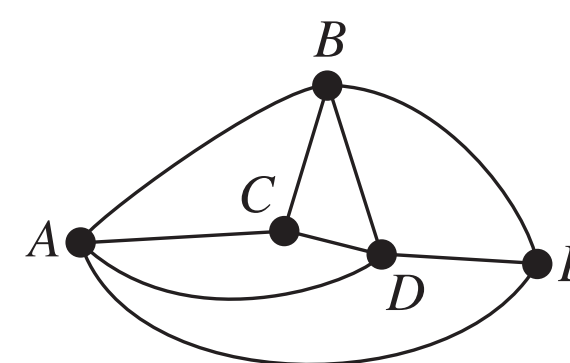
three edges from  $v_1, v_2$ , and  $v_3$  to  $v_4$  forms four regions. No matter which of these four regions  $v_5$  is in, it is possible to join it to only three, and not all four, of the other vertices. 13. 8 15. Because there are no loops or multiple edges and no simple circuits of length 3, and the degree of the unbounded region is at least 4, each region has degree at least 4. Thus,  $2e \geq 4r$ , or  $r \leq e/2$ . But  $r = e - v + 2$ , so we have  $e - v + 2 \leq e/2$ , which implies that  $e \leq 2v - 4$ . 17. As in the argument in the proof of Corollary 1, we have  $2e \geq 5r$  and  $r = e - v + 2$ . Thus,  $e - v + 2 \leq 2e/5$ , which implies that  $e \leq (5/3)v - (10/3)$ . 19. Only (a) and (c) 21. Not homeomorphic to  $K_{3,3}$  23. Planar 25. Nonplanar 27. a) 1 b) 3 c) 9 d) 2 e) 4 f) 16 29. Draw  $K_{m,n}$  as described in the hint. The number of crossings is four times the number in the first quadrant. The vertices on the  $x$ -axis to the right of the origin are  $(1, 0), (2, 0), \dots, (m/2, 0)$  and the vertices on the  $y$ -axis above the origin are  $(0, 1), (0, 2), \dots, (0, n/2)$ . We obtain all crossings by choosing any two numbers  $a$  and  $b$  with  $1 \leq a < b \leq m/2$  and two numbers  $r$  and  $s$  with  $1 \leq r < s \leq n/2$ ; we get exactly one crossing in the graph between the edge connecting  $(a, 0)$  and  $(0, s)$  and the edge connecting  $(b, 0)$  and  $(0, r)$ . Hence, the number of crossings in the first quadrant is  $C\left(\frac{m}{2}, 2\right) \cdot C\left(\frac{n}{2}, 2\right) = \frac{(m/2)(m/2-1)}{2} \cdot \frac{(n/2)(n/2-1)}{2}$ . Hence, the total number of crossings is  $4 \cdot \frac{(m/2)(m/2-1)}{2} \cdot \frac{(n/2)(n/2-1)}{2} = mn(m-2)(n-2)/16$ . 31. a) 2 b) 2 c) 2 d) 2 e) 2 f) 2 33. The formula is valid for  $n \leq 4$ . If  $n > 4$ , by Exercise 32 the thickness of  $K_n$  is at least  $C(n, 2)/(3n-6) = (n+1 + \frac{2}{n-2})/6$  rounded up. Because this quantity is never an integer, it equals  $\lfloor (n+7)/6 \rfloor$ . 35. This follows from Exercise 34 because  $K_{m,n}$  has  $mn$  edges and  $m+n$  vertices and has no triangles because it is bipartite.

37.

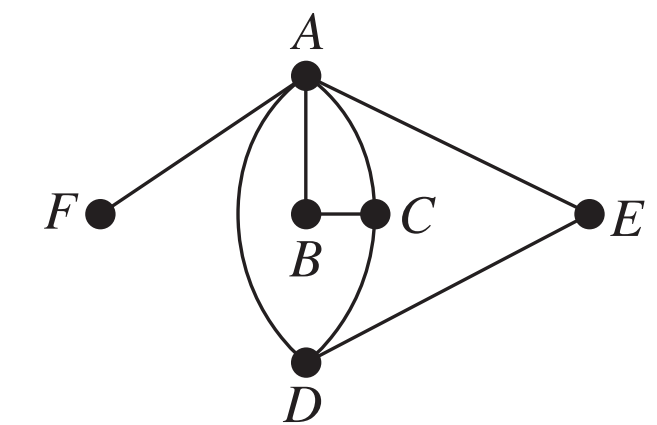


## Section 10.8

1. Four colors



3. Three colors



5. 3 7. 3 9. 2 11. 3 13. Graphs with no edges 15. 3 if  $n$  is even, 4 if  $n$  is odd 17. Period 1: Math 115, Math 185; period 2: Math 116, CS 473; period 3: Math 195, CS 101; period 4: CS 102; period 5: CS 273 19. 5 21. Exercise 5: 3 Exercise 6: 6 Exercise 7: 3 Exercise 8: 4 Exercise 9: 3 Exercise 10: 6 Exercise 11: 4 23. a) 2 if  $n$  is even, 3 if  $n$  is odd b)  $n$  25. Two edges that have the same color share no

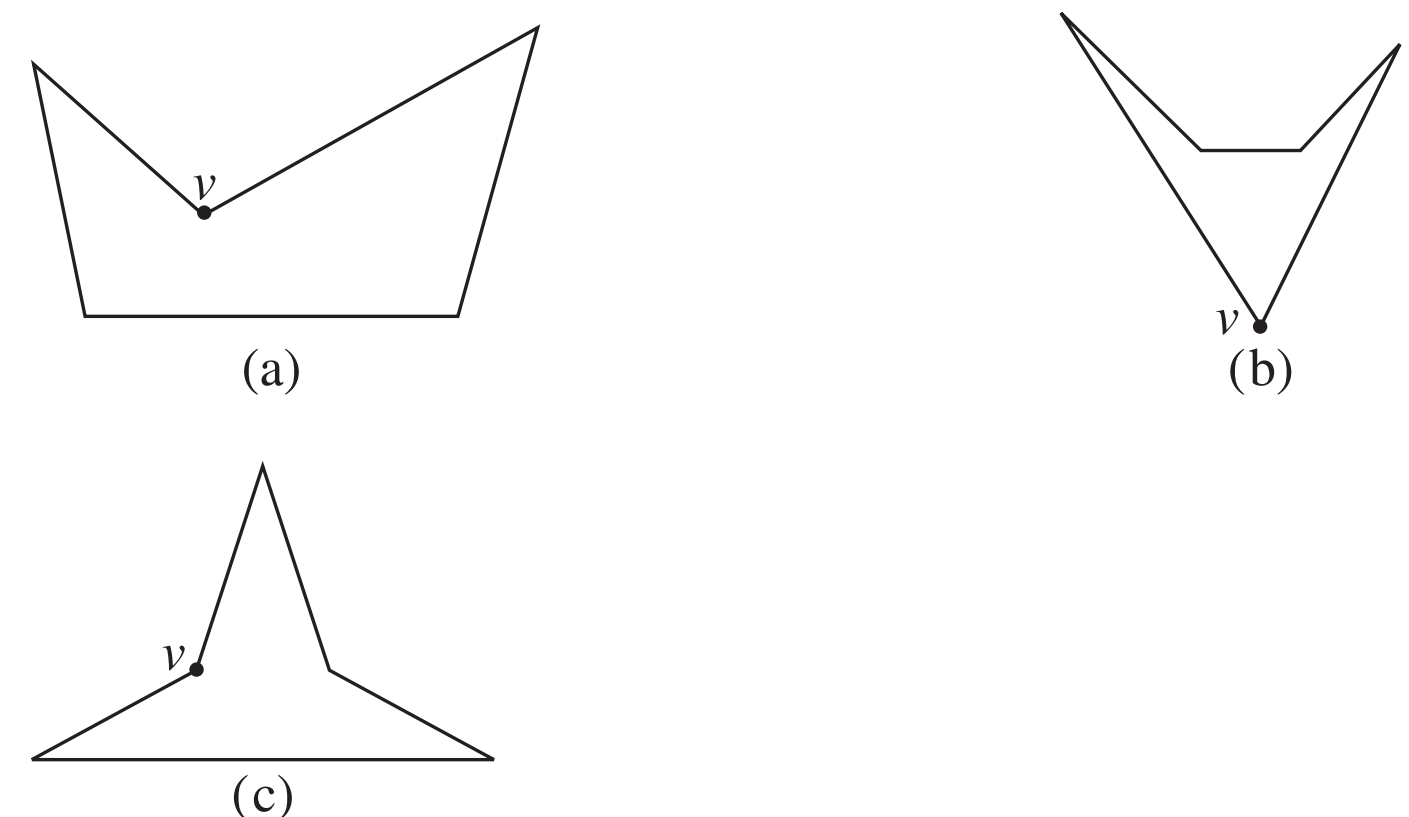


endpoints. Therefore, if more than  $n/2$  edges were colored the same, the graph would have more than  $2(n/2) = n$  vertices.

**27. 5** **29.** Color 1:  $e, f, d$ ; color 2:  $c, a, i, g$ ; color 3:  $h, b, j$

**31.** Color  $C_6$  **33.** Four colors are needed to color  $W_n$  when  $n$  is an odd integer greater than 1, because three colors are needed for the rim (see Example 4), and the center vertex, being adjacent to all the rim vertices, will require a fourth color. To see that the graph obtained from  $W_n$  by deleting one edge can be colored with three colors, consider two cases. If we remove a rim edge, then we can color the rim with two colors, by starting at an endpoint of the removed edge and using the colors alternately around the portion of the rim that remains. The third color is then assigned to the center vertex. If we remove a spoke edge, then we can color the rim by assigning color #1 to the rim endpoint of the removed edge and colors #2 and #3 alternately to the remaining vertices on the rim, and then assign color #1 to the center. **35.** Suppose that  $G$  is chromatically  $k$ -critical but has a vertex  $v$  of degree  $k - 2$  or less. Remove from  $G$  one of the edges incident to  $v$ . By definition of “ $k$ -critical,” the resulting graph can be colored with  $k - 1$  colors. Now restore the missing edge and use this coloring for all vertices except  $v$ . Because we had a proper coloring of the smaller graph, no two adjacent vertices have the same color. Furthermore,  $v$  has at most  $k - 2$  neighbors, so we can color  $v$  with an unused color to obtain a proper  $(k - 1)$ -coloring of  $G$ . This contradicts the fact that  $G$  has chromatic number  $k$ . Therefore, our assumption was wrong, and every vertex of  $G$  must have degree at least  $k - 1$ . **37. a) 6 b) 7 c) 9 d) 11** **39.** Represent frequencies by colors and zones by vertices. Join two vertices with an edge if the zones these vertices represent interfere with one another. Then a  $k$ -tuple coloring is precisely an assignment of frequencies that avoids interference. **41.** We use induction on the number of vertices of the graph. Every graph with five or fewer vertices can be colored with five or fewer colors, because each vertex can get a different color. That takes care of the basis case(s). So we assume that all graphs with  $k$  vertices can be 5-colored and consider a graph  $G$  with  $k + 1$  vertices. By Corollary 2 in Section 10.7,  $G$  has a vertex  $v$  with degree at most 5. Remove  $v$  to form the graph  $G'$ . Because  $G'$  has only  $k$  vertices, we 5-color it by the inductive hypothesis. If the neighbors of  $v$  do not use all five colors, then we can 5-color  $G$  by assigning to  $v$  a color not used by any of its neighbors. The difficulty arises if  $v$  has five neighbors, and each has a different color in the 5-coloring of  $G'$ . Suppose that the neighbors of  $v$ , when considered in clockwise order around  $v$ , are  $a, b, c, m$ , and  $p$ . (This order is determined by the clockwise order of the curves representing the edges incident to  $v$ .) Suppose that the colors of the neighbors are azure, blue, chartreuse, magenta, and purple, respectively. Consider the azure-chartreuse subgraph (i.e., the vertices in  $G$  colored azure or chartreuse and all the edges between them). If  $a$  and  $c$  are not in the same component of this graph, then in the component containing  $a$  we can interchange these two colors (make the azure vertices chartreuse and vice versa), and  $G'$  will still be properly colored. That makes  $a$  chartreuse, so we can now color  $v$  azure,

and  $G$  has been properly colored. If  $a$  and  $c$  are in the same component, then there is a path of vertices alternately colored azure and chartreuse joining  $a$  and  $c$ . This path together with edges  $av$  and  $vc$  divides the plane into two regions, with  $b$  in one of them and  $m$  in the other. If we now interchange blue and magenta on all the vertices in the same region as  $b$ , we will still have a proper coloring of  $G'$ , but now blue is available for  $v$ . In this case, too, we have found a proper coloring of  $G$ . This completes the inductive step, and the theorem is proved. **43.** We follow the hint. Because the measures of the interior angles of a pentagon total  $540^\circ$ , there cannot be as many as three interior angles of measure more than  $180^\circ$  (reflex angles). If there are no reflex angles, then the pentagon is convex, and a guard placed at any vertex can see all points. If there is one reflex angle, then the pentagon must look essentially like figure (a) below, and a guard at vertex  $v$  can see all points. If there are two reflex angles, then they can be adjacent or nonadjacent (figures (b) and (c)); in either case, a guard at vertex  $v$  can see all points. [In figure (c), choose the reflex vertex closer to the bottom side.] Thus, for all pentagons, one guard suffices, so  $g(5) = 1$ .



**45.** The figure suggested in the hint (generalized to have  $k$  prongs for any  $k \geq 1$ ) has  $3k$  vertices. The sets of locations from which the tips of different prongs are visible are disjoint. Therefore, a separate guard is needed for each of the  $k$  prongs, so at least  $k$  guards are needed. This shows that  $g(3k) \geq k = \lfloor 3k/3 \rfloor$ . If  $n = 3k + i$ , where  $0 \leq i \leq 2$ , then  $g(n) \geq g(3k) \geq k = \lfloor n/3 \rfloor$ .

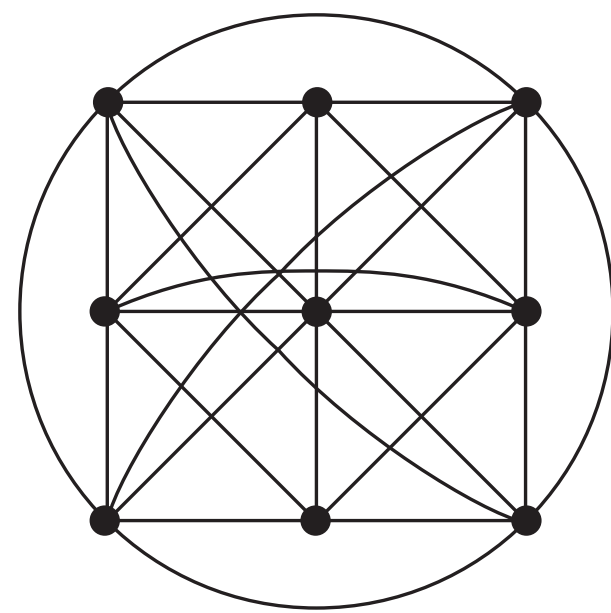
### Supplementary Exercises

- 1.** 2500 **3.** Yes **5.** Yes **7.**  $\sum_{i=1}^m n_i$  vertices,  $\sum_{i < j} n_i n_j$  edges **9. a)** If  $x \in N(A \cup B)$ , then  $x$  is adjacent to some vertex  $v \in A \cup B$ . WOLOG suppose  $v \in A$ ; then  $x \in N(A)$  and therefore also in  $N(A) \cup N(B)$ . Conversely, if  $x \in N(A) \cup N(B)$ , then WOLOG suppose  $x \in N(A)$ . Thus,  $x$  is adjacent to some vertex  $v \in A \subseteq A \cup B$ , so  $x \in N(A \cup B)$ . **b)** If  $x \in N(A \cap B)$ , then  $x$  is adjacent to some vertex  $v \in A \cap B$ . Since both  $v \in A$  and  $v \in B$ , it follows that  $x \in N(A)$  and  $x \in N(B)$ , whence  $x \in N(A) \cap N(B)$ . For the counterexample, let  $G = (\{u, v, w\}, \{\{u, v\}, \{v, w\}\})$ ,  $A = \{u\}$ , and  $B = \{w\}$ .
- 11.**  $(c, a, p, x, n, m)$  and many others **13.**  $(c, d, a, b)$  and many others **15.** 6 times the number of triangles divided by the number of paths of length 2 **17. a)** The probability that

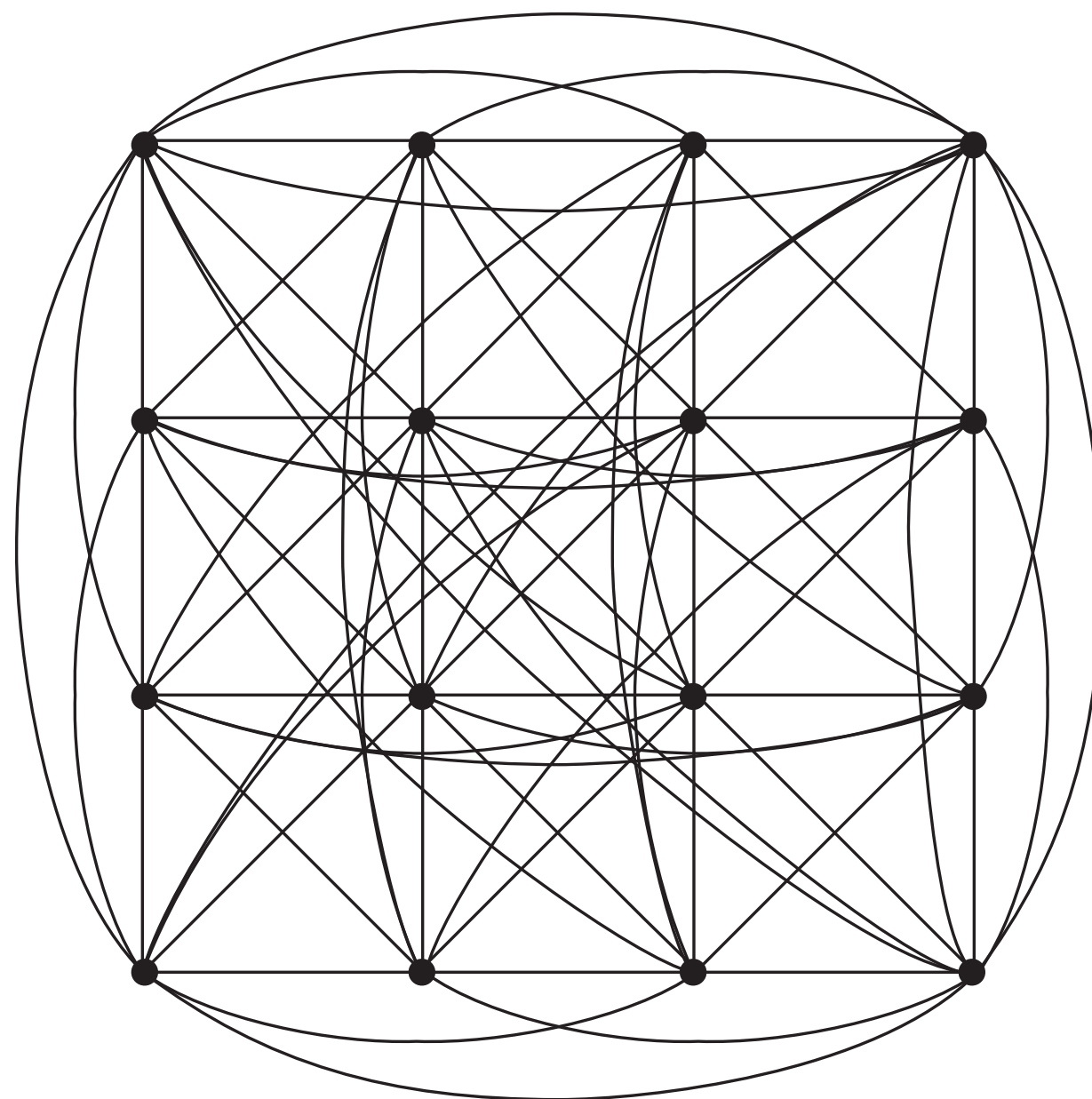


two actors each of whom has appeared in a film with a randomly chosen actor have appeared in a film together **b)** The probability that two of a randomly chosen person's Facebook friends are themselves Facebook friends **c)** The probability that two of a randomly chosen person's coauthors are themselves coauthors **d)** The probability that two proteins that each interact with a randomly chosen protein interact with each other **e)** The probability that two routers each of which has a communications link to a randomly chosen router are themselves linked **19.** Complete subgraphs containing the following sets of vertices:  $\{b, c, e, f\}$ ,  $\{a, b, g\}$ ,  $\{a, d, g\}$ ,  $\{d, e, g\}$ ,  $\{b, e, g\}$  **21.** Complete subgraphs containing the following sets of vertices:  $\{b, c, d, j, k\}$ ,  $\{a, b, j, k\}$ ,  $\{e, f, g, i\}$ ,  $\{a, b, i\}$ ,  $\{a, i, j\}$ ,  $\{b, d, e\}$ ,  $\{b, e, i\}$ ,  $\{b, i, j\}$ ,  $\{g, h, i\}$ ,  $\{h, i, j\}$  **23.**  $\{c, d\}$  is a minimum dominating set.

**25. a)**



**b)**



**27. a)** 1 **b)** 2 **c)** 3 **29. a)** A path from  $u$  to  $v$  in a graph  $G$  induces a path from  $f(u)$  to  $f(v)$  in an isomorphic graph  $H$ . **b)** Suppose  $f$  is an isomorphism from  $G$  to  $H$ . If  $v_0, v_1, \dots, v_n, v_0$  is a Hamilton circuit in  $G$ , then  $f(v_0), f(v_1), \dots, f(v_n), f(v_0)$  must be a Hamilton circuit in  $H$  because it is still a circuit and  $f(v_i) \neq f(v_j)$  for  $0 \leq i < j \leq n$ . **c)** Suppose  $f$  is an isomorphism from  $G$  to  $H$ . If  $v_0, v_1, \dots, v_n, v_0$  is an Euler circuit in  $G$ , then  $f(v_0), f(v_1), \dots, f(v_n), f(v_0)$  must be an Euler circuit in  $H$  because it is a circuit that contains each edge exactly once. **d)** Two isomorphic graphs must have the same crossing number because they can be drawn exactly the same way in the plane. **e)** Suppose  $f$  is an isomorphism from  $G$  to  $H$ . Then  $v$  is isolated in  $G$  if and only if  $f(v)$  is isolated in  $H$ . Hence, the graphs must have the same number of isolated vertices. **f)** Suppose  $f$  is an isomorphism from  $G$  to  $H$ . If  $G$  is bipartite, then the vertex set of  $G$  can be partitioned into

$V_1$  and  $V_2$  with no edge connecting vertices within  $V_1$  or vertices within  $V_2$ . Then the vertex set of  $H$  can be partitioned into  $f(V_1)$  and  $f(V_2)$  with no edge connecting vertices within  $f(V_1)$  or vertices within  $f(V_2)$ . **31.** 3 **33. a)** Yes **b)** No **35.** No **37.** Yes **39.** If  $e$  is a cut edge with endpoints  $u$  and  $v$ , then if we direct  $e$  from  $u$  to  $v$ , there will be no path in the directed graph from  $v$  to  $u$ , or else  $e$  would not have been a cut edge. Similar reasoning works if we direct  $e$  from  $v$  to  $u$ . **41.**  $n - 1$  **43.** Let the vertices represent the chickens. We include the edge  $(u, v)$  in the graph if and only if chicken  $u$  dominates chicken  $v$ . **45.** By the handshaking theorem, the average vertex degree is  $2m/n$ , which equals the minimum degree; it follows that all the vertex degrees are equal. **47.**  $K_{3,3}$  and the skeleton of a triangular prism **49. a)** A Hamilton circuit in the graph exactly corresponds to a seating of the knights at the Round Table such that adjacent knights are friends. **b)** The degree of each vertex in this graph is at least  $2n - 1 - (n - 1) = n \geq (2n/2)$ , so by Dirac's theorem, this graph has a Hamilton circuit. **c)**  $a, b, d, f, g, z$  **51. a)** 4 **b)** 2 **c)** 3 **d)** 4 **e)** 4 **f)** 2 **53. a)** Suppose that  $G = (V, E)$ . Let  $a, b \in V$ . We must show that the distance between  $a$  and  $b$  in  $\overline{G}$  is at most 2. If  $\{a, b\} \notin E$  this distance is 1, so assume  $\{a, b\} \in E$ . Because the diameter of  $G$  is greater than 3, there are vertices  $u$  and  $v$  such that the distance in  $G$  between  $u$  and  $v$  is greater than 3. Either  $u$  or  $v$ , or both, is not in the set  $\{a, b\}$ . Assume that  $u$  is different from both  $a$  and  $b$ . Either  $\{a, u\}$  or  $\{b, u\}$  belongs to  $E$ ; otherwise  $a, u, b$  would be a path in  $\overline{G}$  of length 2. So, without loss of generality, assume  $\{a, u\} \in E$ . Thus,  $v$  cannot be  $a$  or  $b$ , and by the same reasoning either  $\{a, v\} \in E$  or  $\{b, v\} \in E$ . In either case, this gives a path of length less than or equal to 3 from  $u$  to  $v$  in  $G$ , a contradiction. **b)** Suppose  $G = (V, E)$ . Let  $a, b \in V$ . We must show that the distance between  $a$  and  $b$  in  $\overline{G}$  does not exceed 3. If  $\{a, b\} \notin E$ , the result follows, so assume that  $\{a, b\} \in E$ . Because the diameter of  $G$  is greater than or equal to 3, there exist vertices  $u$  and  $v$  such that the distance in  $G$  between  $u$  and  $v$  is greater than or equal to 3. Either  $u$  or  $v$ , or both, is not in the set  $\{a, b\}$ . Assume  $u$  is different from both  $a$  and  $b$ . Either  $\{a, u\} \in E$  or  $\{b, u\} \in E$ ; otherwise  $a, u, b$  is a path of length 2 in  $\overline{G}$ . So, without loss of generality, assume  $\{a, u\} \in E$ . Thus,  $v$  is different from  $a$  and from  $b$ . If  $\{a, v\} \in E$ , then  $u, a, v$  is a path of length 2 in  $G$ , so  $\{a, v\} \notin E$  and thus  $\{b, v\} \in E$  (or else there would be a path  $a, v, b$  of length 2 in  $\overline{G}$ ). Hence,  $\{u, b\} \notin E$ ; otherwise  $u, b, v$  is a path of length 2 in  $G$ . Thus,  $a, v, u, b$  is a path of length 3 in  $\overline{G}$ , as desired. **55.**  $a, b, e, z$  **57.**  $a, c, d, f, g, z$  **59.** If  $G$  is planar, then because  $e \leq 3v - 6$ ,  $G$  has at most 27 edges. (If  $G$  is not connected it has even fewer edges.) Similarly,  $\overline{G}$  has at most 27 edges. But the union of  $G$  and  $\overline{G}$  is  $K_{11}$ , which has 55 edges, and  $55 > 27 + 27$ . **61.** Suppose that  $G$  is colored with  $k$  colors and has independence number  $i$ . Because each color class must be an independent set, each color class has no more than  $i$  elements. Thus, there are at most  $ki$  vertices. **63. a)**  $C(n, m)p^m(1 - p)^{n-m}$  **b)**  $np$  **c)** To generate a labeled graph  $G$ , as we apply the process to pairs of vertices, the random number  $x$  chosen must be less than or equal to  $1/2$  when  $G$  has an edge between that pair of vertices and



greater than  $1/2$  when  $G$  has no edge there. Hence, the probability of making the correct choice is  $1/2$  for each edge and  $1/2^{C(n,2)}$  overall. Hence, all labeled graphs are equally likely.

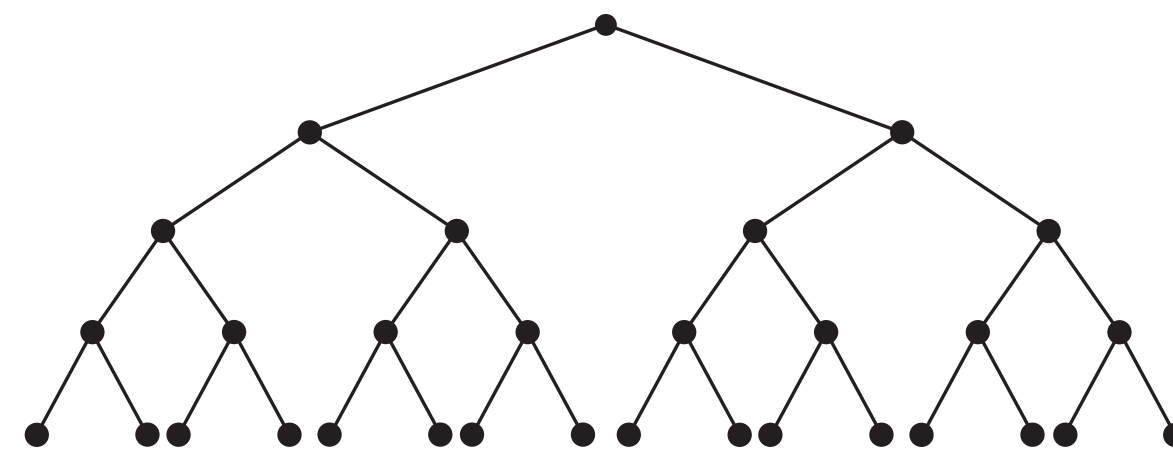
**65.** Suppose  $P$  is monotone increasing. If the property of not having  $P$  were not retained whenever edges are removed from a simple graph, there would be a simple graph  $G$  not having  $P$  and another simple graph  $G'$  with the same vertices but with some of the edges of  $G$  missing that has  $P$ . But  $P$  is monotone increasing, so because  $G'$  has  $P$ , so does  $G$  obtained by adding edges to  $G'$ . This is a contradiction. The proof of the converse is similar.

## CHAPTER 11

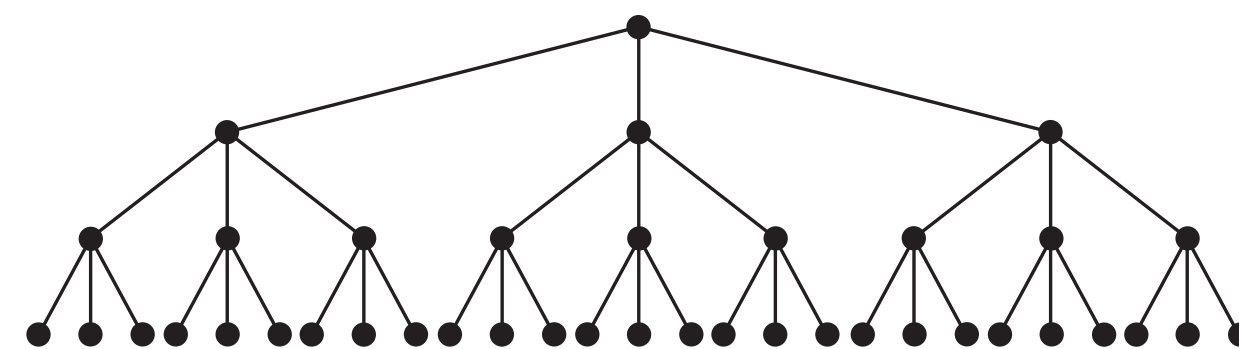
### Section 11.1

**1.** (a), (c), (e) **3. a)**  $a$  **b)**  $a, b, c, d, f, h, j, q, t$  **c)**  $e, g, i, k, l, m, n, o, p, r, s, u$  **d)**  $q, r$  **e)**  $c$  **f)**  $p$  **g)**  $f, b, a$  **h)**  $e, f, l, m, n$   
**5.** No **7.** Level 0:  $a$ ; level 1:  $b, c, d$ ; level 2:  $e$  through  $k$  (in alphabetical order); level 3:  $l$  through  $r$ ; level 4:  $s, t$ ; level 5:  $u$   
**9. a)** The entire tree **b)**  $c, g, h, o, p$  and the four edges  $cg, ch, ho, hp$  **c)**  $e$  alone **11. a)** 1 **b)** 2 **13. a)** 3 **b)** 9 **15. a)** The “only if” part is Theorem 2 and the definition of a tree. Suppose  $G$  is a connected simple graph with  $n$  vertices and  $n - 1$  edges. If  $G$  is not a tree, it contains, by Exercise 14, an edge whose removal produces a graph  $G'$ , which is still connected. If  $G'$  is not a tree, remove an edge to produce a connected graph  $G''$ . Repeat this procedure until the result is a tree. This requires at most  $n - 1$  steps because there are only  $n - 1$  edges. By Theorem 2, the resulting graph has  $n - 1$  edges because it has  $n$  vertices. It follows that no edges were deleted, so  $G$  was already a tree. **b)** Suppose that  $G$  is a tree. By part (a),  $G$  has  $n - 1$  edges, and by definition,  $G$  has no simple circuits. Conversely, suppose that  $G$  has no simple circuits and has  $n - 1$  edges. Let  $c$  equal the number of components of  $G$ , each of which is necessarily a tree, say with  $n_i$  vertices, where  $\sum_{i=1}^c n_i = n$ . By part (a), the total number of edges in  $G$  is  $\sum_{i=1}^c (n_i - 1) = n - c$ . Since we are given that this equals  $n - 1$ , it follows that  $c = 1$ , i.e.,  $G$  is connected and therefore satisfies the definition of a tree. **17.** 9999 **19.** 2000 **21.** 999 **23.** 1,000,000 dollars **25.** No such tree exists by Theorem 4 because it is impossible for  $m = 2$  or  $m = 84$ .

**27.** Complete binary tree of height 4:

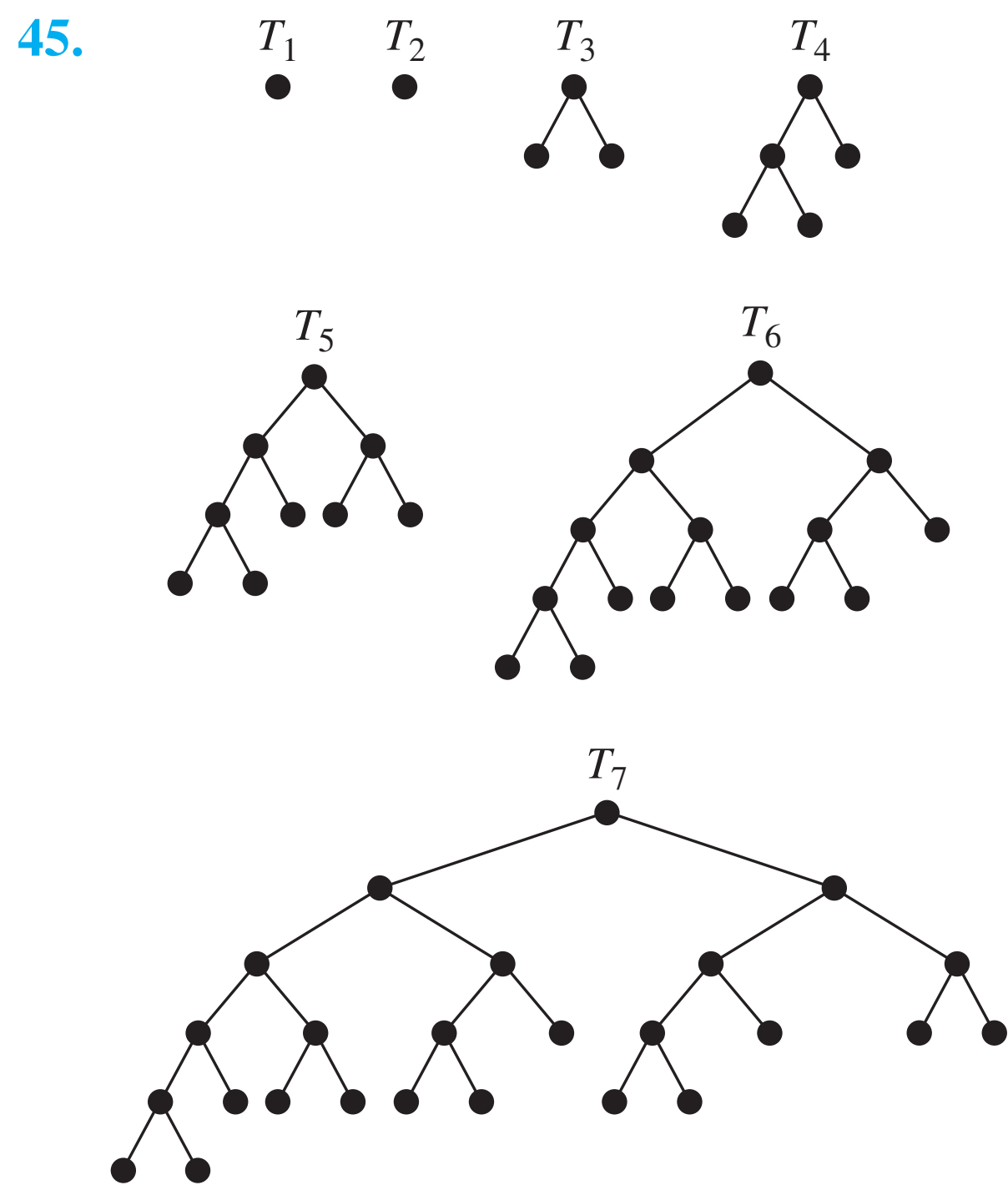


Complete 3-ary tree of height 3:



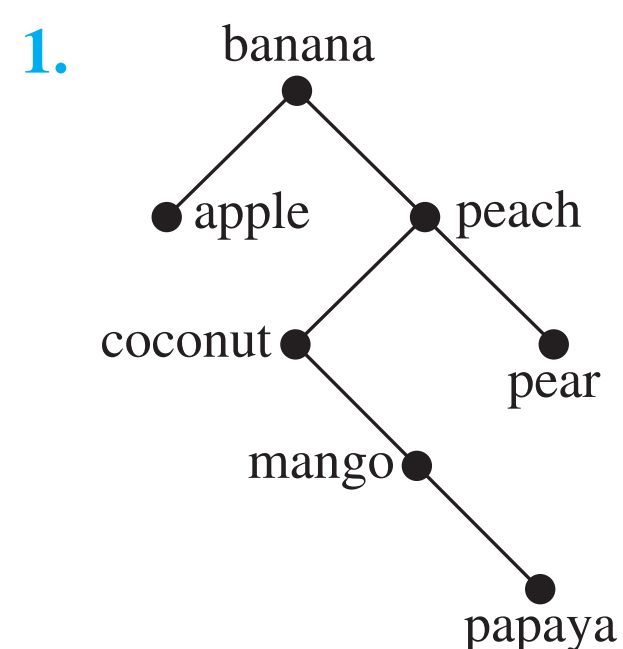
**29. a)** By Theorem 3 it follows that  $n = mi + 1$ . Because  $i + l = n$ , we have  $l = n - i$ , so  $l = (mi + 1) - i = (m - 1)i + 1$ . **b)** We have  $n = mi + 1$  and  $i + l = n$ . Hence,  $i = n - l$ . It follows that  $n = m(n - l) + 1$ . Solving for  $n$  gives  $n = (ml - 1)/(m - 1)$ . From  $i = n - l$  we obtain  $i = [(ml - 1)/(m - 1)] - l = (l - 1)/(m - 1)$ .  
**31.**  $n - t$  **33. a)** 1 **b)** 3 **c)** 5 **35. a)** The parent directory **b)** A subdirectory or contained file **c)** A subdirectory or contained file in the same parent directory **d)** All directories in the path name **e)** All subdirectories and files continued in the directory or a subdirectory of this directory, and so on **f)** The length of the path to this directory or file **g)** The depth of the system, i.e., the length of the longest path **37.** Let  $n = 2^k$ , where  $k$  is a positive integer. If  $k = 1$ , there is nothing to prove because we can add two numbers with  $n - 1 = 1$  processor in  $\log 2 = 1$  step. Assume we can add  $n = 2^k$  numbers in  $\log n$  steps using a tree-connected network of  $n - 1$  processors. Let  $x_1, x_2, \dots, x_{2n}$  be  $2n = 2^{k+1}$  numbers that we wish to add. The tree-connected network of  $2n - 1$  processors consists of the tree-connected network of  $n - 1$  processors together with two new processors as children of each leaf. In one step we can use the leaves of the larger network to find  $x_1 + x_2, x_3 + x_4, \dots, x_{2n-1} + x_{2n}$ , giving us  $n$  numbers, which, by the inductive hypothesis, we can add in  $\log n$  steps using the rest of the network. Because we have used  $\log n + 1$  steps and  $\log(2n) = \log 2 + \log n = 1 + \log n$ , this completes the proof. **39.**  $c$  only **41.**  $c$  and  $h$  **43.** Suppose a tree  $T$  has at least two centers. Let  $u$  and  $v$  be distinct centers, both with eccentricity  $e$ , with  $u$  and  $v$  not adjacent. Because  $T$  is connected, there is a simple path  $P$  from  $u$  to  $v$ . Let  $c$  be any other vertex on this path. Because the eccentricity of  $c$  is at least  $e$ , there is a vertex  $w$  such that the unique simple path from  $c$  to  $w$  has length at least  $e$ . Clearly, this path cannot contain both  $u$  and  $v$  or else there would be a simple circuit. In fact, this path from  $c$  to  $w$  leaves  $P$  and does not return to  $P$  once it, possibly, follows part of  $P$  toward either  $u$  or  $v$ . Without loss of generality, assume this path does not follow  $P$  toward  $u$ . Then the path from  $u$  to  $c$  to  $w$  is simple and of length more than  $e$ , a contradiction. Hence,  $u$  and  $v$  are adjacent. Now because any two centers are adjacent, if there were more than two centers,  $T$  would contain  $K_3$ , a simple circuit, as a subgraph, which is a contradiction.



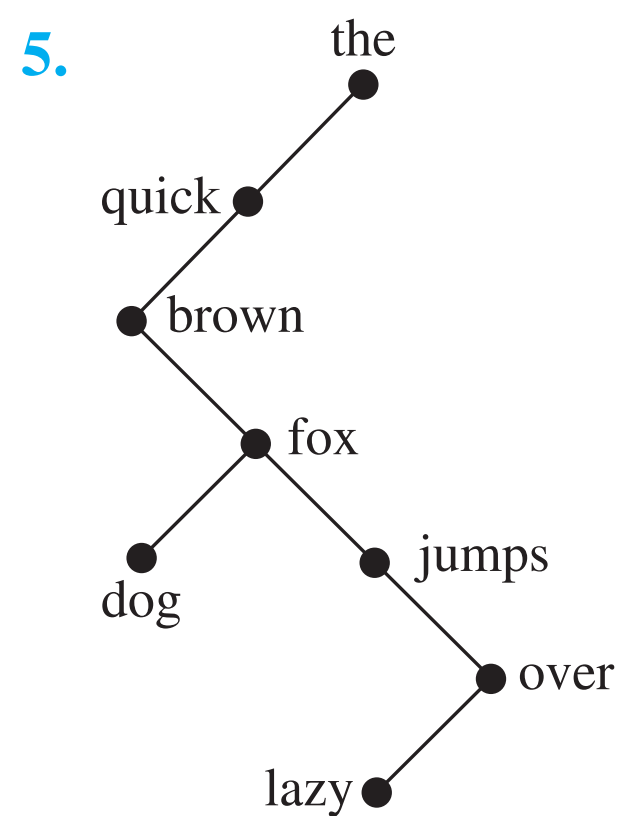


47. The statement is that *every* tree with  $n$  vertices has a path of length  $n - 1$ , and it was shown only that there exists a tree with  $n$  vertices having a path of length  $n - 1$ .

## Section 11.2

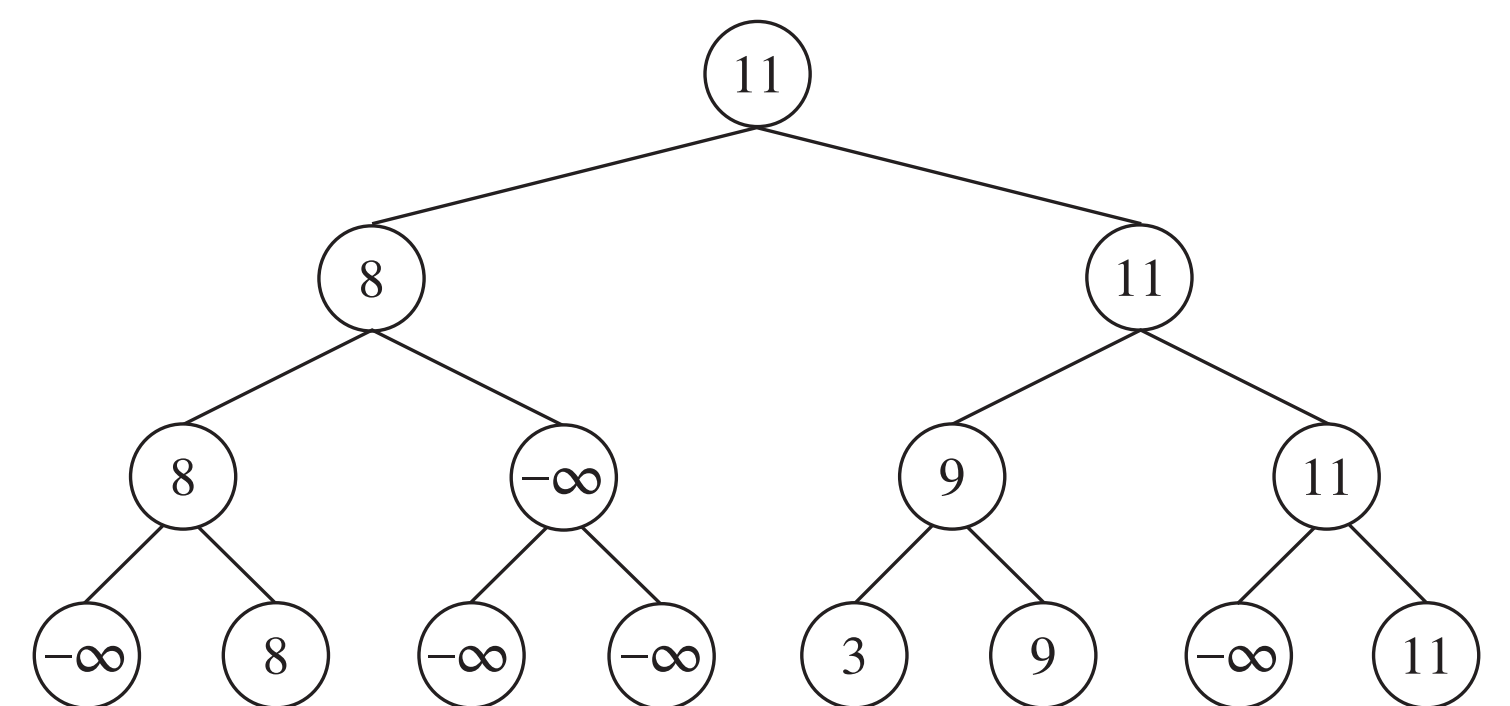
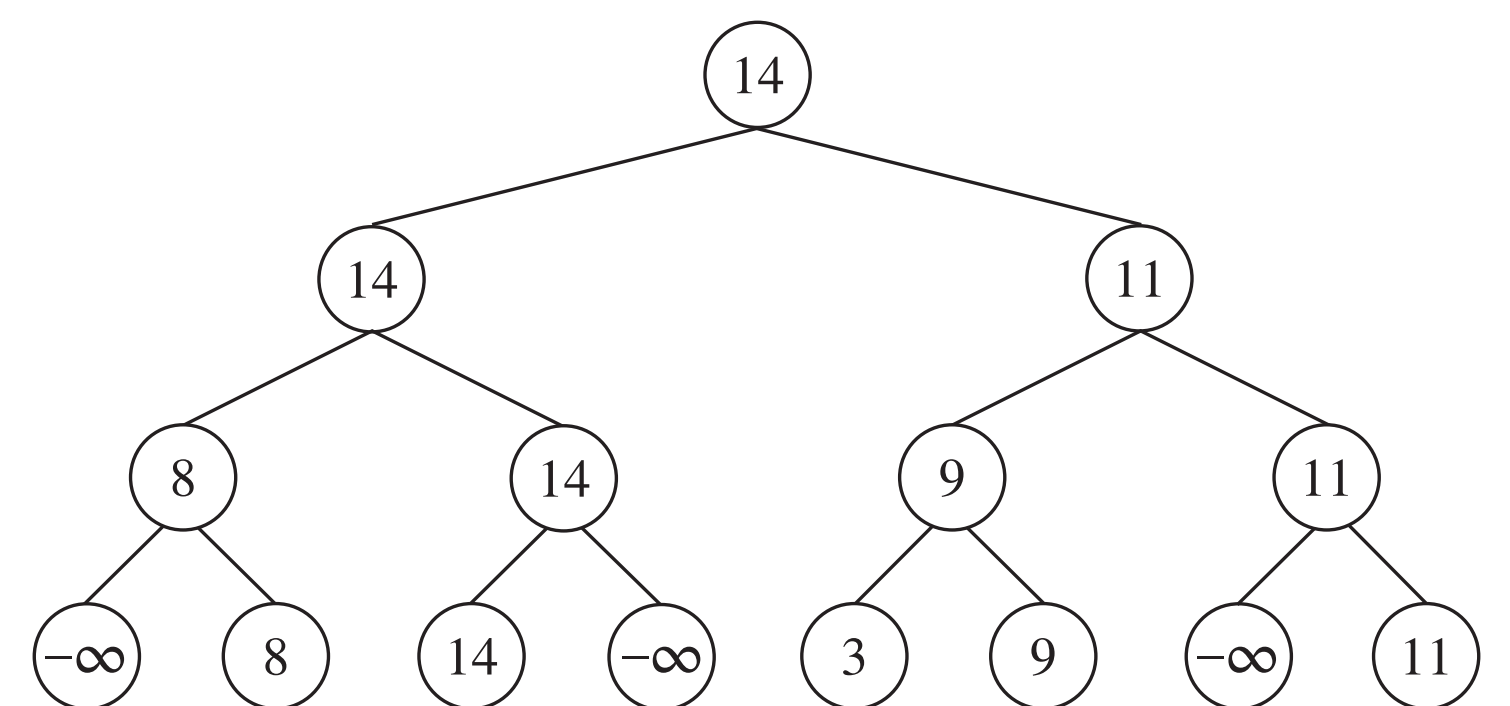
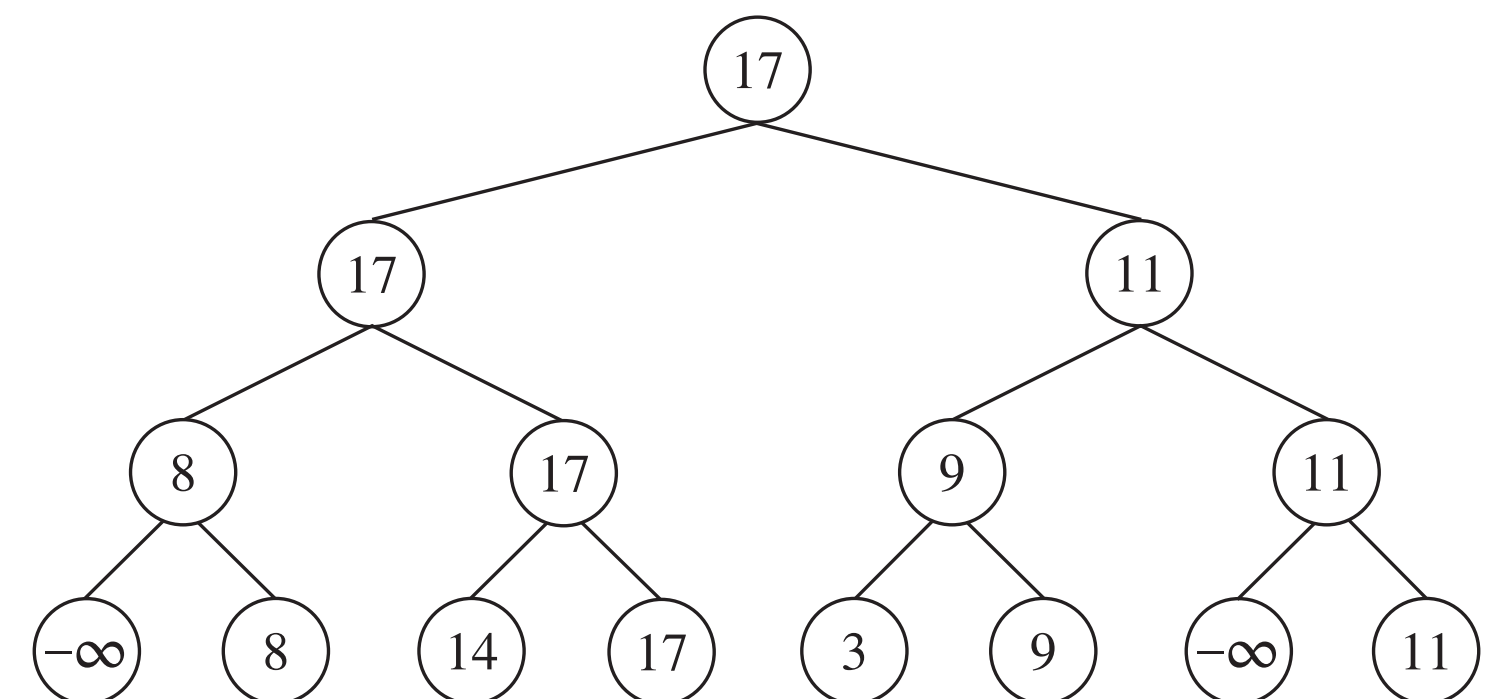


3. a) 3 b) 1 c) 4 d) 5

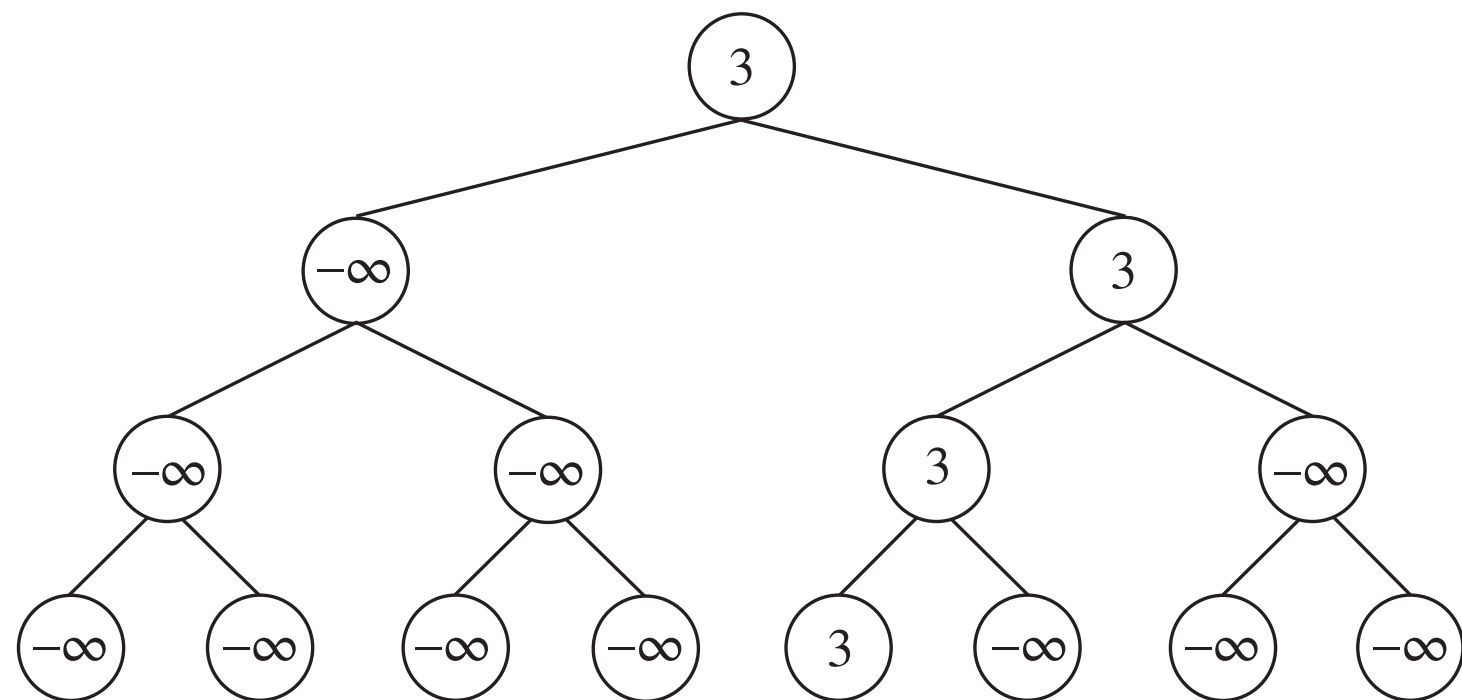


7. At least  $\lceil \log_3 4 \rceil = 2$  weighings are needed, because there are only four outcomes (because it is not required to determine whether the coin is lighter or heavier). In fact, two weighings suffice. Begin by weighing coin 1 against coin 2. If they balance, weigh coin 1 against coin 3. If coin 1 and coin 3 are the same weight, coin 4 is the counterfeit coin, and if they are not the same weight, then coin 3 is the counterfeit coin. If coin 1 and coin 2 are not the same weight, again weigh coin 1 against coin 3. If they balance, coin 2 is the counterfeit coin; if they do not balance, coin 1 is the counterfeit coin. 9. At least

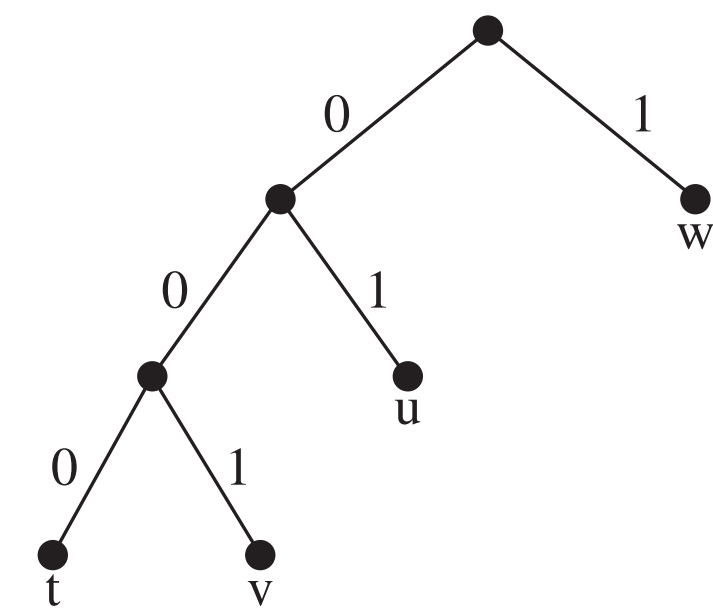
$\lceil \log_3 13 \rceil = 3$  weighings are needed. In fact, three weighings suffice. Start by putting coins 1, 2, and 3 on the left-hand side of the balance and coins 4, 5, and 6 on the right-hand side. If equal, apply Example 3 to coins 1, 2, 7, 8, 9, 10, 11, and 12. If unequal, apply Example 3 to 1, 2, 3, 4, 5, 6, 7, and 8. 11. The least number is five. Call the elements  $a, b, c$ , and  $d$ . First compare  $a$  and  $b$ ; then compare  $c$  and  $d$ . Without loss of generality, assume that  $a < b$  and  $c < d$ . Next compare  $a$  and  $c$ . Whichever is smaller is the smallest element of the set. Again without loss of generality, suppose  $a < c$ . Finally, compare  $b$  with both  $c$  and  $d$  to completely determine the ordering. 13. The first two steps are shown in the text. After 22 has been identified as the second largest element, we replace the leaf 22 by  $-\infty$  in the tree and recalculate the winner in the path from the leaf where 22 used to be up to the root. Next, we see that 17 is the third largest element, so we repeat the process: replace the leaf 17 by  $-\infty$  and recalculate. Next, we see that 14 is the fourth largest element, so we repeat the process: replace the leaf 14 by  $-\infty$  and recalculate. Next, we see that 11 is the fifth largest element, so we repeat the process: replace the leaf 11 by  $-\infty$  and recalculate. The process continues in this manner. We determine that 9 is the sixth largest element, 8 is the seventh largest element, and 3 is the eighth largest element. The trees produced in all steps, except the second to last, are shown here.







ber of bits is always the same.) **25.** There are four possible answers in all, the one shown here and three more obtained from this one by swapping  $t$  and  $v$  and/or swapping  $u$  and  $w$ .



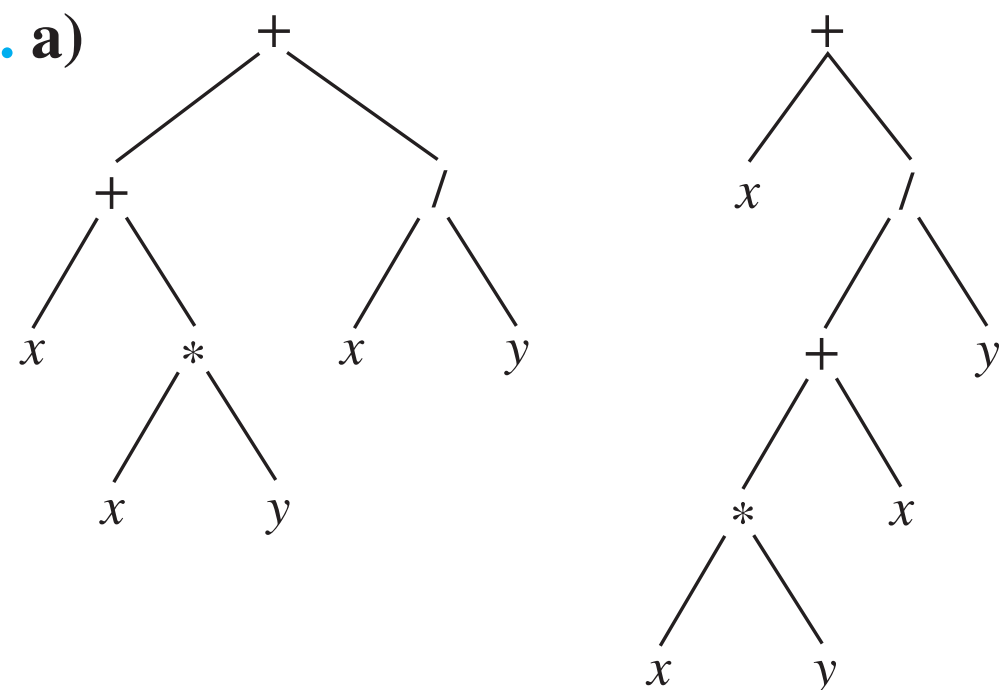






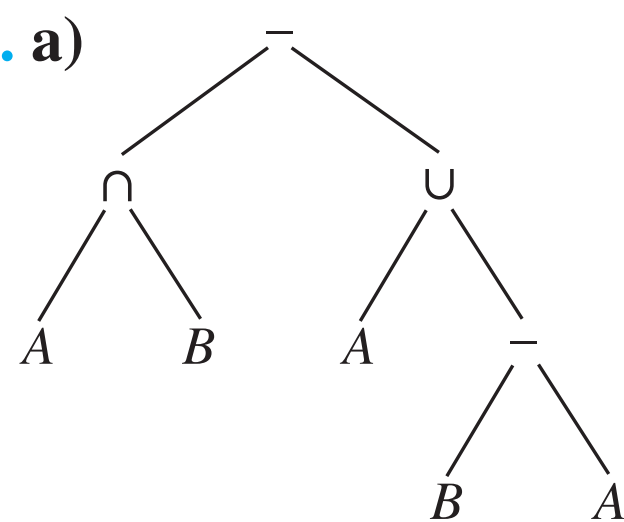
5. No 7.  $a, b, d, e, f, g, c$  9.  $a, b, e, k, l, m, f, g, n, r, s, c, d, h, o, i, j, p, q$  11.  $d, b, i, e, m, j, n, o, a, f, c, g, k, h, p, l$  13.  $d, f, g, e, b, c, a$  15.  $k, l, m, e, f, r, s, n, g, b, c, o, h, i, p, q, j, d, a$

17. a)



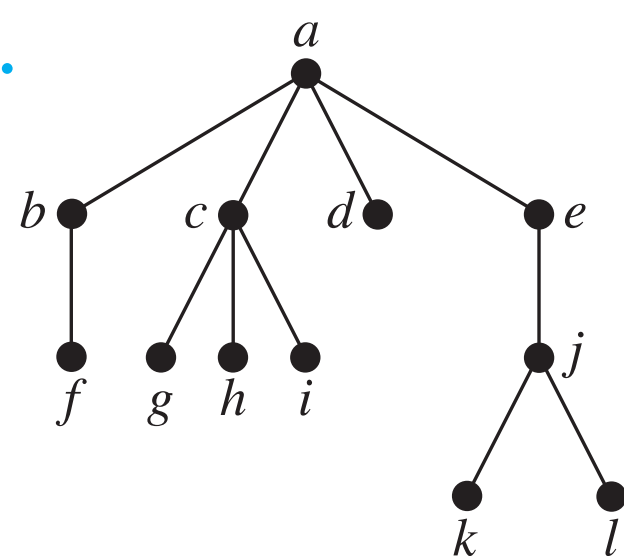
b)  $++x * xy/xy, +x/+ * xyxy$  c)  $xyx * +xy/+, xxy * x+y/+$   
d)  $((x + (x * y)) + (x/y)), (x + (((x * y) + x)/y))$

19. a)



b)  $- \cap A B \cup A - B A$  c)  $A B \cap A B A - \cup -$   
d)  $((A \cap B) - (A \cup (B - A)))$  21. 14 23. a) 1 b) 1 c) 4  
d) 2205

25.

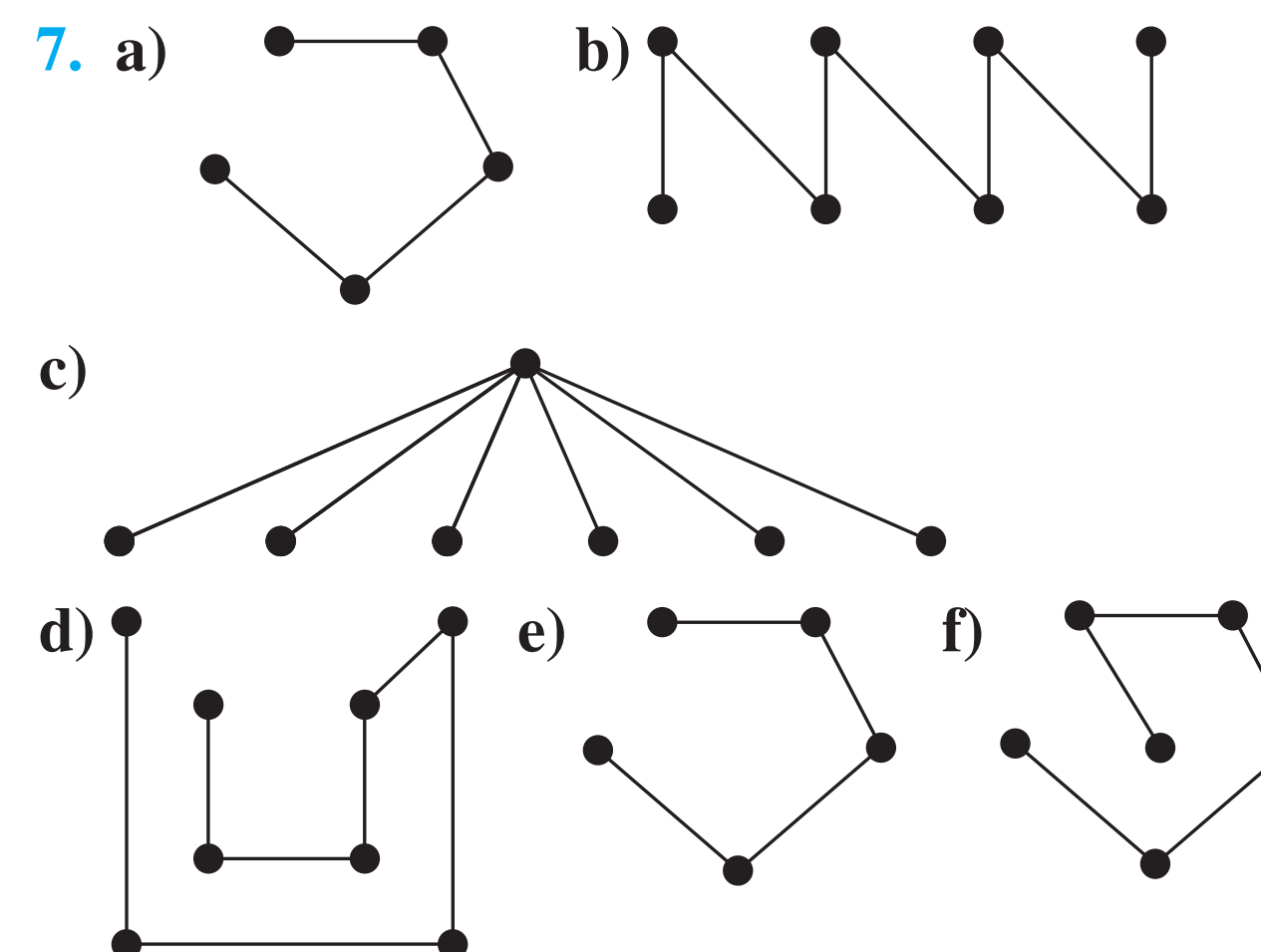
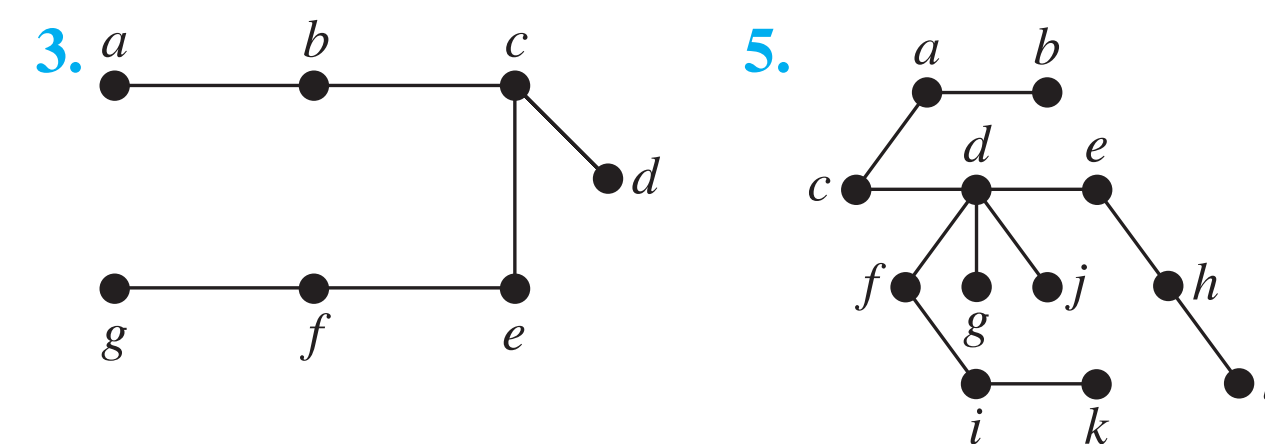


27. Use mathematical induction. The result is trivial for a list with one element. Assume the result is true for a list with  $n$  elements. For the inductive step, start at the end. Find the sequence of vertices at the end of the list starting with the last

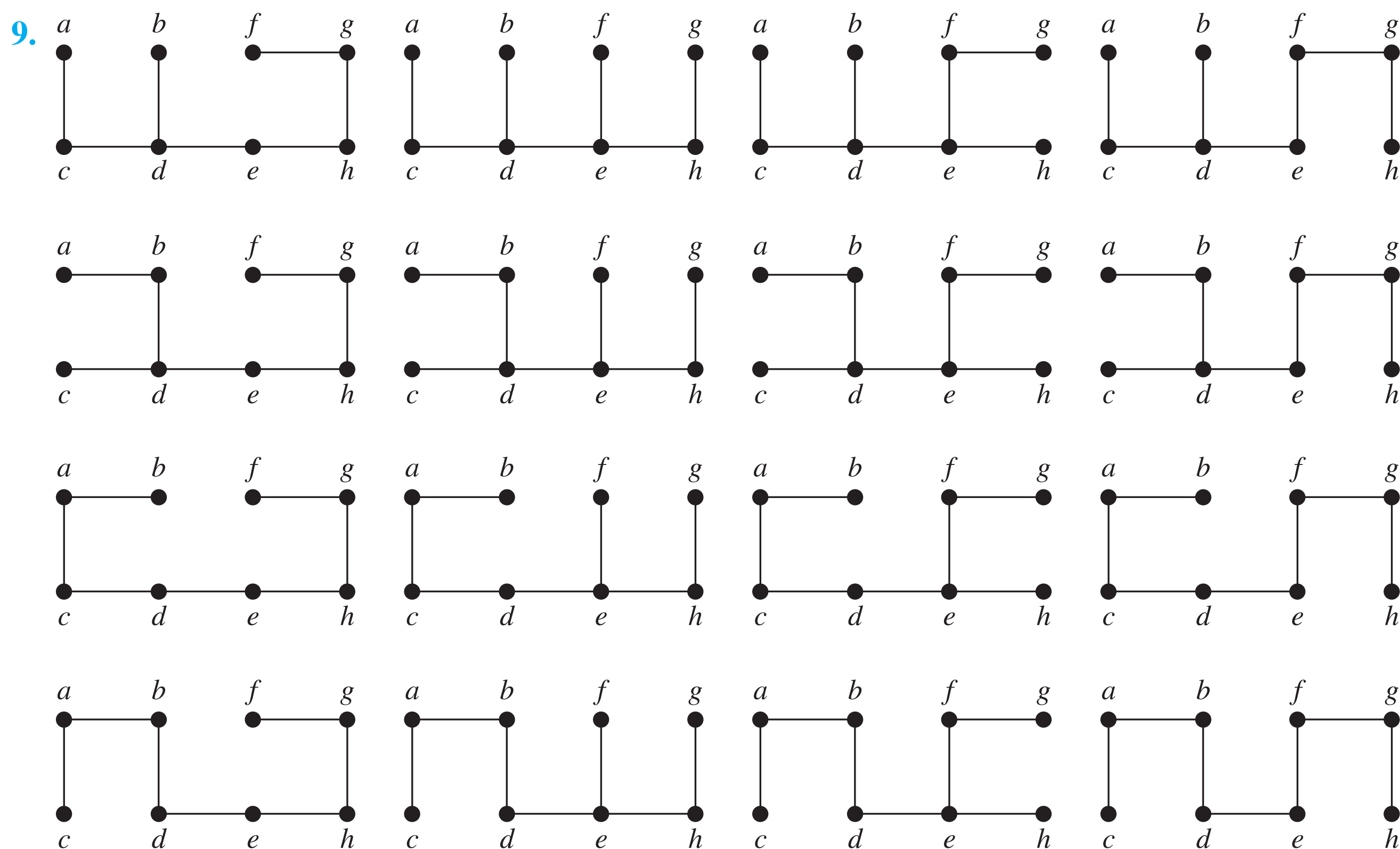
leaf, ending with the root, each vertex being the last child of the one following it. Remove this leaf and apply the inductive hypothesis. 29.  $c, d, b, f, g, h, e, a$  in each case 31. Proof by mathematical induction. Let  $S(X)$  and  $O(X)$  represent the number of symbols and number of operators in the well-formed formula  $X$ , respectively. The statement is true for well-formed formulae of length 1, because they have 1 symbol and 0 operators. Assume the statement is true for all well-formed formulae of length less than  $n$ . A well-formed formula of length  $n$  must be of the form  $*XY$ , where  $*$  is an operator and  $X$  and  $Y$  are well-formed formulae of length less than  $n$ . Then by the inductive hypothesis  $S(*XY) = S(X) + S(Y) = [O(X) + 1] + [O(Y) + 1] = O(X) + O(Y) + 2$ . Because  $O(*XY) = 1 + O(X) + O(Y)$ , it follows that  $S(*XY) = O(*XY) + 1$ . 33.  $xy + zx \circ + x \circ, xyz + + yx++$ ,  $xyxy \circ \circ xy \circ \circ z \circ +, xz \times, zz + \circ, yyy \circ \circ \circ, zx + yz + \circ$ , for instance

## Section 11.4

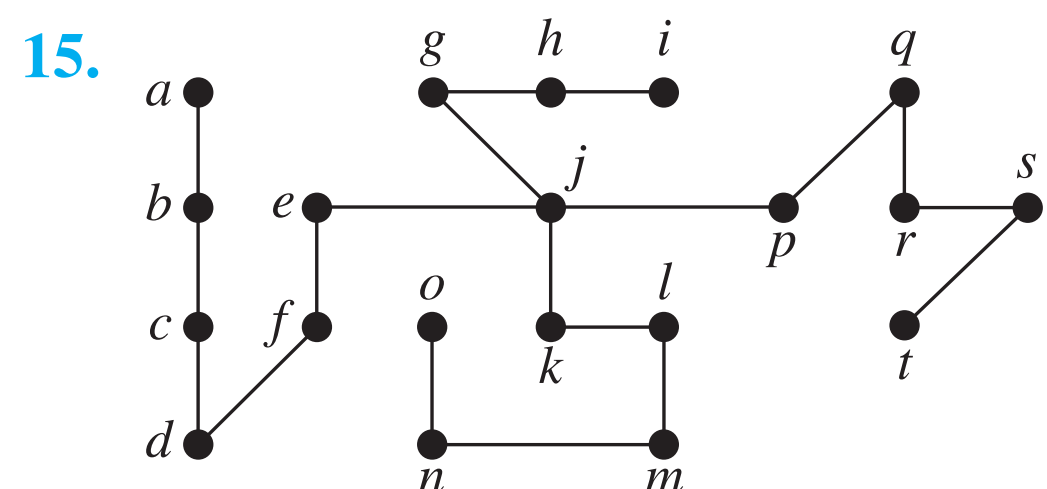
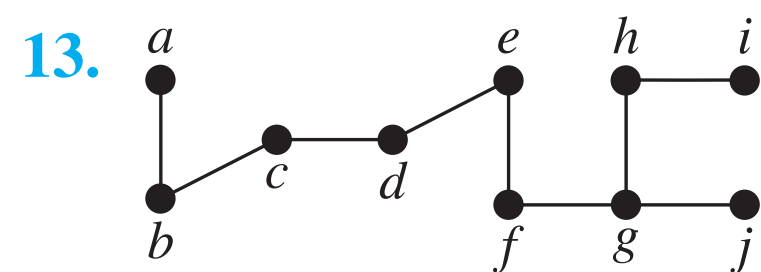
1.  $m - n + 1$







11. a) 3 b) 16 c) 4 d) 5

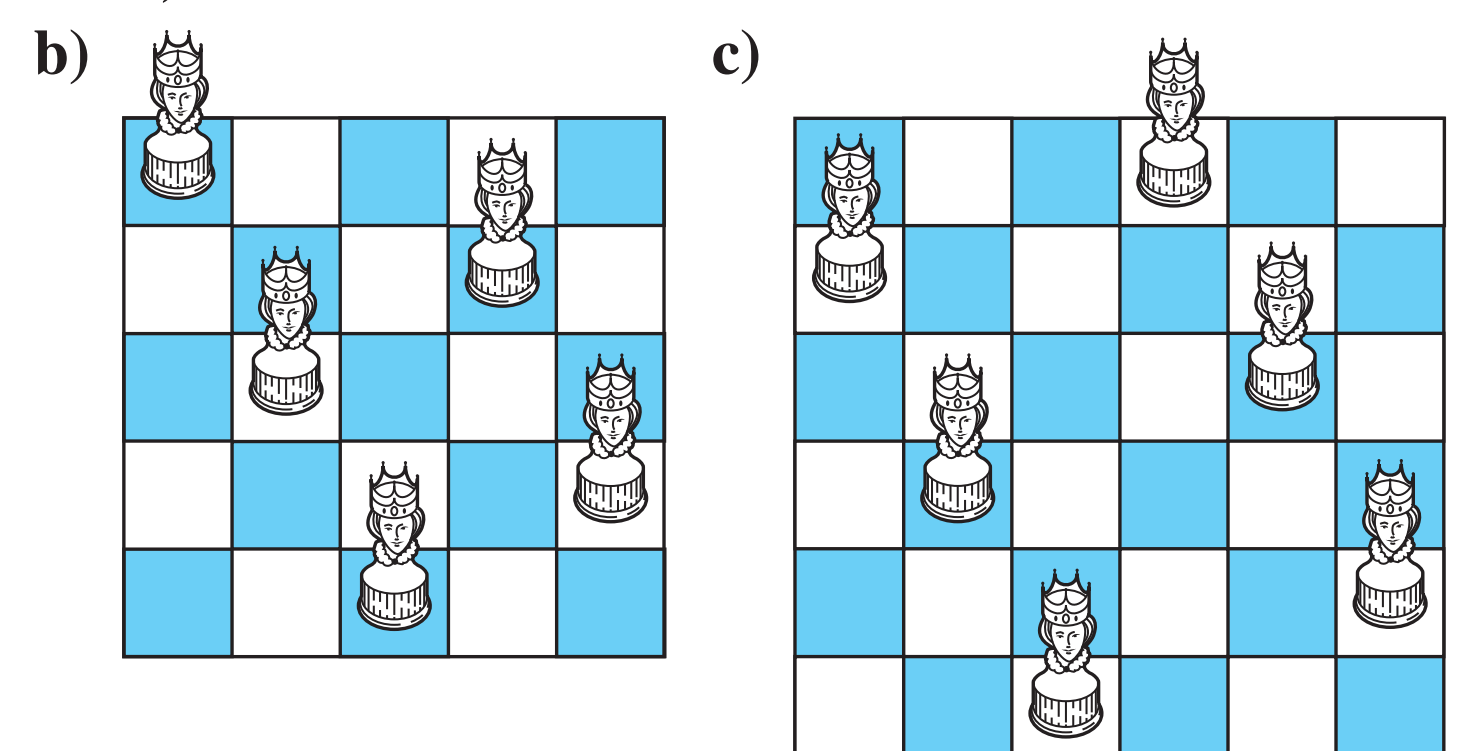


17. a) A path of length 6 b) A path of length 5 c) A path of length 6 d) Depends on order chosen to visit the vertices; may be a path of length 7 19. With breadth-first search, the initial vertex is the middle vertex, and the  $n$  spokes are added to the tree as this vertex is processed. Thus, the resulting tree is  $K_{1,n}$ . With depth-first search, we start at the vertex in the middle of the wheel and visit a neighbor—one of the vertices on the rim. From there we move to an adjacent vertex on the rim, and so on all the way around until we have reached every vertex. Thus, the resulting spanning tree is a path of length  $n$ .

21. With breadth-first search, we fan out from a vertex of degree  $m$  to all the vertices of degree  $n$  as the first step. Next, a vertex of degree  $n$  is processed, and the edges from it to all the remaining vertices of degree  $m$  are added. The result is a  $K_{1,n-1}$  and a  $K_{1,m-1}$  with their centers joined by an edge. With depth-first search, we travel back and forth from one partite set to the other until we can go no further. If  $m = n$  or  $m = n - 1$ , then we get a path of length  $m + n - 1$ . Otherwise, the path ends while some vertices in the larger partite set have not been visited, so we back up one link in the path to a vertex  $v$  and then successively visit the remaining vertices in that set from  $v$ . The result is a path with extra pendant edges coming

out of one end of the path. 23. A possible set of flights to discontinue are: Boston–New York, Detroit–Boston, Boston–Washington, New York–Washington, New York–Chicago, Atlanta–Washington, Atlanta–Dallas, Atlanta–Los Angeles, Atlanta–St. Louis, St. Louis–Dallas, St. Louis–Detroit, St. Louis–Denver, Dallas–San Diego, Dallas–Los Angeles, Dallas–San Francisco, San Diego–Los Angeles, Los Angeles–San Francisco, San Francisco–Seattle. 25. Proof by induction on the length of the path: If the path has length 0, then the result is trivial. If the length is 1, then  $u$  is adjacent to  $v$ , so  $u$  is at level 1 in the breadth-first spanning tree. Assume that the result is true for paths of length  $l$ . If the length of a path is  $l + 1$ , let  $u'$  be the next-to-last vertex in a shortest path from  $v$  to  $u$ . By the inductive hypothesis,  $u'$  is at level  $l$  in the breadth-first spanning tree. If  $u$  were at a level not exceeding  $l$ , then clearly the length of the shortest path from  $v$  to  $u$  would also not exceed  $l$ . So  $u$  has not been added to the breadth-first spanning tree yet after the vertices of level  $l$  have been added. Because  $u$  is adjacent to  $u'$ , it will be added at level  $l + 1$  (although the edge connecting  $u'$  and  $u$  is not necessarily added).

27. a) No solution



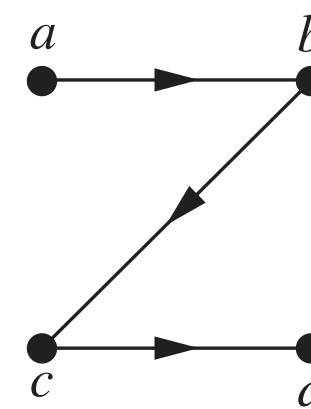
29. Start at a vertex and proceed along a path without repeating vertices as long as possible, allowing the return to the start after all vertices have been visited. When it is impossible to



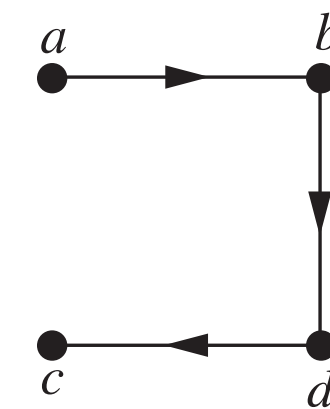
continue along a path, backtrack and try another extension of the current path. **31.** Take the union of the spanning trees of the connected components of  $G$ . They are disjoint, so the result is a forest. **33.**  $m - n + c$  **35.** Assume that we wish to find the length of a shortest path from  $v_1$  to every other vertex of  $G$  using Algorithm 2. In line 2 of that algorithm, add  $L(v_1) := 0$ , and add the following as a third step in the **then** clause at the end:  $L(w) := 1 + L(v)$ . **37.** Add an instruction to the BFS algorithm to mark each vertex as it is encountered. When BFS terminates we have found (all the vertices of) one component of the graph. Repeat, starting at an unmarked vertex, and continue in this way until all vertices have been marked. **39.** Trees **41.** Use depth-first search on each component. **43.** If an edge  $uv$  is not followed while we are processing vertex  $u$  during the depth-first search process, then it must be the case that the vertex  $v$  had already been visited. There are two cases. If vertex  $v$  was visited after we started processing  $u$ , then, because we are not finished processing  $u$  yet,  $v$  must appear in the subtree rooted at  $u$  (and hence, must be a descendant of  $u$ ). On the other hand, if the processing of  $v$  had already begun before we started processing  $u$ , then why wasn't this edge followed at that time? It must be that we had not finished processing  $v$ , in other words, that we are still forming the subtree rooted at  $v$ , so  $u$  is a descendant of  $v$ , and hence,  $v$  is an ancestor of  $u$ . **45.** Certainly these two procedures produce the identical spanning trees if the graph we are working with is a tree itself, because in this case there is only one spanning tree (the whole graph). This is the only case in which that happens, however. If the original graph has any other edges, then by Exercise 43 they must be back edges and hence, join a vertex to an ancestor or descendant, whereas by Exercise 34, they must connect vertices at the same level or at levels that differ by 1. Clearly these two possibilities are mutually exclusive. Therefore, there can be no edges other than tree edges if the two spanning trees are to be the same. **47.** Because the edges not in the spanning tree are not followed in the process, we can ignore them. Thus, we can assume that the graph was a rooted tree to begin with. The basis step is trivial (there is only one vertex), so we assume the inductive hypothesis that breadth-first search applied to trees with  $n$  vertices have their vertices visited in order of their level in the tree and consider a tree  $T$  with  $n + 1$  vertices. The last vertex to be visited during breadth-first search of this tree, say  $v$ , is the one that was added last to the list of vertices waiting to be processed. It was added when its parent, say  $u$ , was being processed. We must show that  $v$  is at the lowest (bottom-most, i.e., numerically greatest) level of the tree. Suppose not; say vertex  $x$ , whose parent is vertex  $w$ , is at a lower level. Then  $w$  is at a lower level than  $u$ . Clearly  $v$  must be a leaf, because any child of  $v$  could not have been seen before  $v$  is seen. Consider the tree  $T'$  obtained from  $T$  by deleting  $v$ . By the inductive hypothesis, the vertices in  $T'$  must be processed in order of their level in  $T'$  (which is the same as their level in  $T$ , and the absence of  $v$  in  $T'$  has no effect on the rest of the algorithm). Therefore,  $u$  must have been processed before  $w$ , and therefore  $v$  would have joined the waiting list

before  $x$  did, a contradiction. Therefore,  $v$  is at the bottom-most level of the tree, and the proof is complete. **49.** We modify the pseudocode given in Algorithm 2 by initializing  $m$  to be 0 at the beginning of the algorithm, and adding the statements " $m := m + 1$ " and "assign  $m$  to vertex  $v$ " after the statement that removes vertex  $v$  from  $L$ . **51.** If a directed edge  $uv$  is not followed while we are processing its tail  $u$  during the depth-first search process, then it must be the case that its head  $v$  had already been visited. There are three cases. If vertex  $v$  was visited after we started processing  $u$ , then, because we are not finished processing  $u$  yet,  $v$  must appear in the subtree rooted at  $u$  (and hence, must be a descendant of  $u$ ), so we have a forward edge. Otherwise, the processing of  $v$  must have already begun before we started processing  $u$ . If it had not yet finished (i.e., we are still forming the subtree rooted at  $v$ ), then  $u$  is a descendant of  $v$ , and hence,  $v$  is an ancestor of  $u$  (we have a back edge). Finally, if the processing of  $v$  had already finished, then by definition we have a cross edge. **53.** Let  $T$  be the spanning tree constructed in Figure 3 and  $T_1, T_2, T_3$ , and  $T_4$  the spanning trees in Figure 4. Denote by  $d(T', T'')$  the distance between  $T'$  and  $T''$ . Then  $d(T, T_1) = 6$ ,  $d(T, T_2) = 4$ ,  $d(T, T_3) = 4$ ,  $d(T, T_4) = 2$ ,  $d(T_1, T_2) = 4$ ,  $d(T_1, T_3) = 4$ ,  $d(T_1, T_4) = 6$ ,  $d(T_2, T_3) = 4$ ,  $d(T_2, T_4) = 2$ , and  $d(T_3, T_4) = 4$ . **55.** Suppose  $e_1 = \{u, v\}$  is as specified. Then  $T_2 \cup \{e_1\}$  contains a simple circuit  $C$  containing  $e_1$ . The graph  $T_1 - \{e_1\}$  has two connected components; the endpoints of  $e_1$  are in different components. Travel  $C$  from  $u$  in the direction opposite to  $e_1$  until you come to the first vertex in the same component as  $v$ . The edge just crossed is  $e_2$ . Clearly,  $T_2 \cup \{e_1\} - \{e_2\}$  is a tree, because  $e_2$  was on  $C$ . Also  $T_1 - \{e_1\} \cup \{e_2\}$  is a tree, because  $e_2$  reunited the two components.

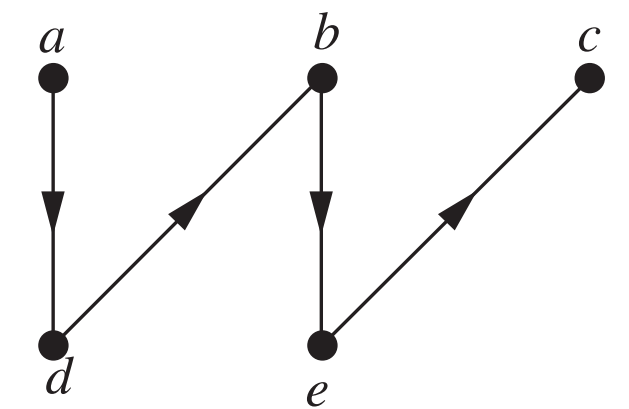
**57.** Exercise 18:



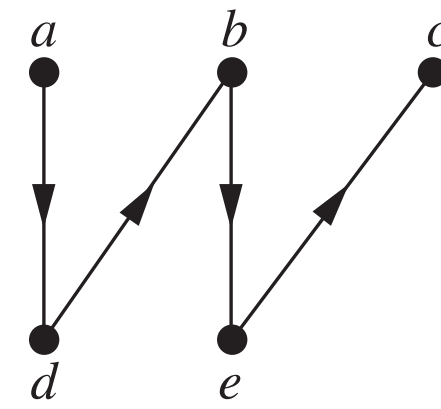
Exercise 19:



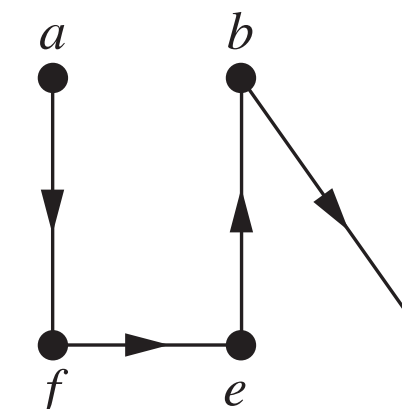
Exercise 20:



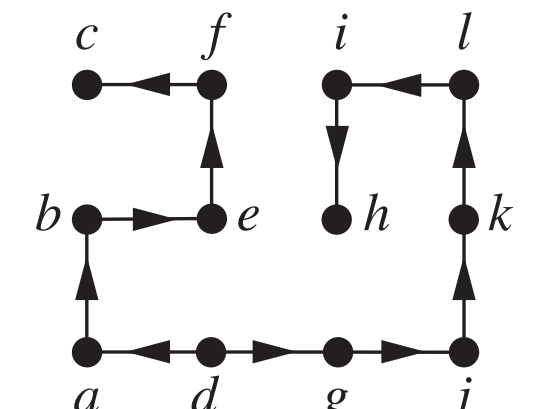
Exercise 21:



Exercise 22:



Exercise 23:



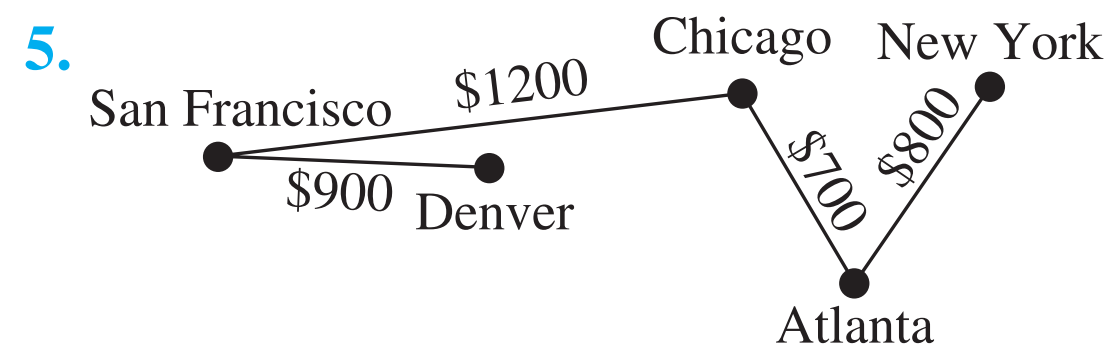
**59.** First construct an Euler circuit in the directed graph. Then delete from this circuit every edge that goes to a vertex previously visited. **61.** According to Exercise 60, a directed graph contains a circuit if and only if there are any back edges. We can detect back edges as follows. Add a marker on each vertex  $v$  to indicate what its status is: not yet seen (the initial situation), seen (i.e., put into  $T$ ) but not yet finished (i.e.,  $visit(v)$  has not yet terminated), or finished (i.e.,  $visit(v)$  has terminated). A few extra lines in Algorithm 1 will accomplish this bookkeeping. Then to determine whether a directed graph



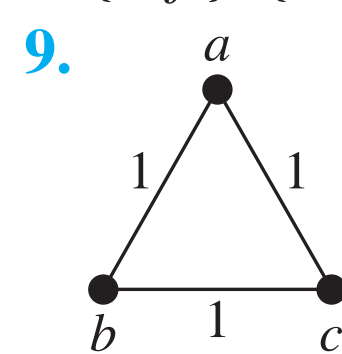
has a circuit, we just have to check when looking at edge  $uv$  whether the status of  $v$  is “seen.” If that ever happens, then we know there is a circuit; if not, then there is no circuit.

## Section 11.5

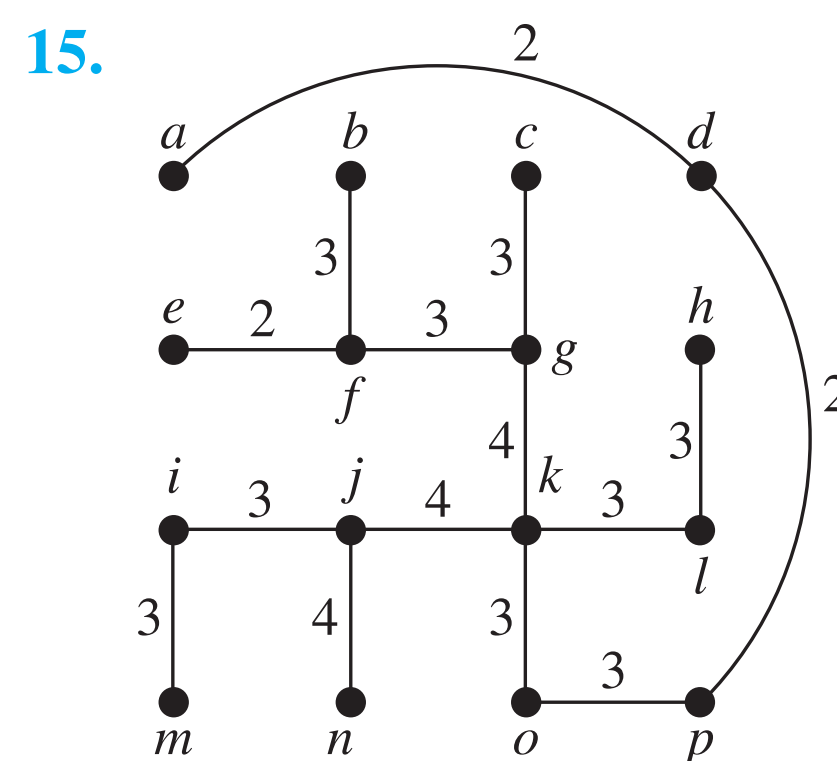
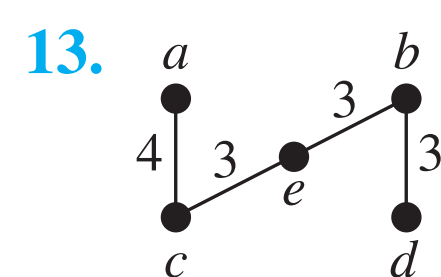
1. Deep Springs–Oasis, Oasis–Dyer, Oasis–Silver Peak, Silver Peak–Goldfield, Lida–Gold Point, Gold Point–Beatty, Lida–Goldfield, Goldfield–Tonopah, Tonopah–Manhattan, Tonopah–Warm Springs 3.  $\{e, f\}$ ,  $\{c, f\}$ ,  $\{e, h\}$ ,  $\{h, i\}$ ,  $\{b, c\}$ ,  $\{b, d\}$ ,  $\{a, d\}$ ,  $\{g, h\}$



7.  $\{e, f\}$ ,  $\{a, d\}$ ,  $\{h, i\}$ ,  $\{b, d\}$ ,  $\{c, f\}$ ,  $\{e, h\}$ ,  $\{b, c\}$ ,  $\{g, h\}$

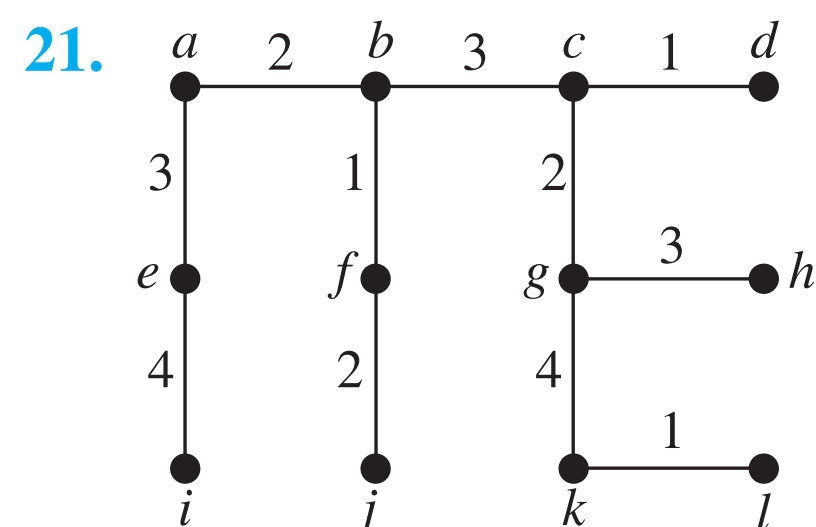


11. Instead of choosing minimum-weight edges at each stage, choose maximum-weight edges at each stage with the same properties.

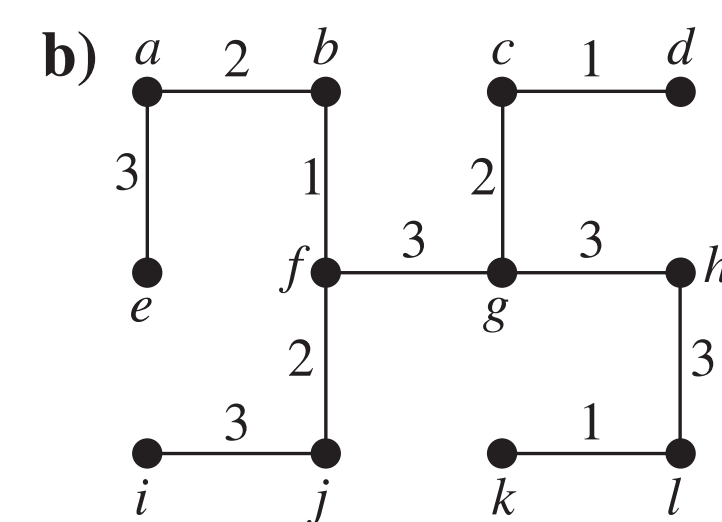
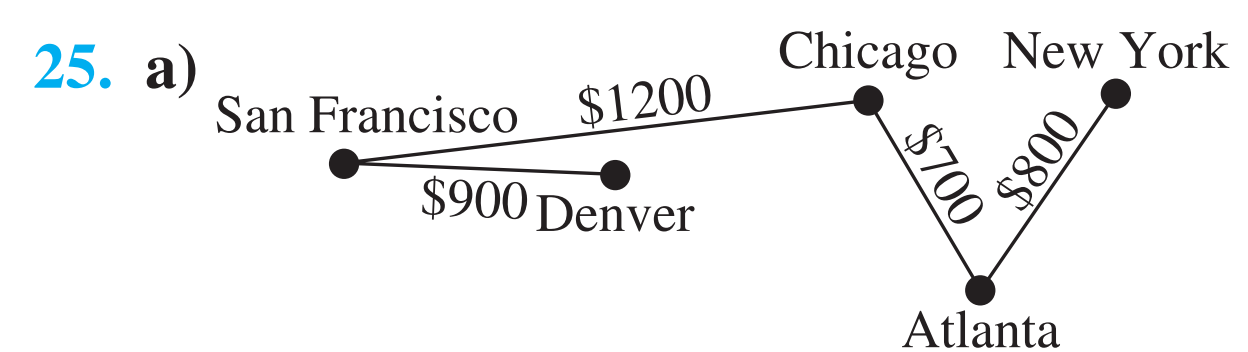


17. First find a minimum spanning tree  $T$  of the graph  $G$  with  $n$  edges. Then for  $i = 1$  to  $n - 1$ , delete only the  $i$ th edge of  $T$  from  $G$  and find a minimum spanning tree of the remaining graph. Pick the one of these  $n - 1$  trees with the shortest length.

19. If all edges have different weights, then a contradiction is obtained in the proof that Prim’s algorithm works when an edge  $e_{k+1}$  is added to  $T$  and an edge  $e$  is deleted, instead of possibly producing another spanning tree.



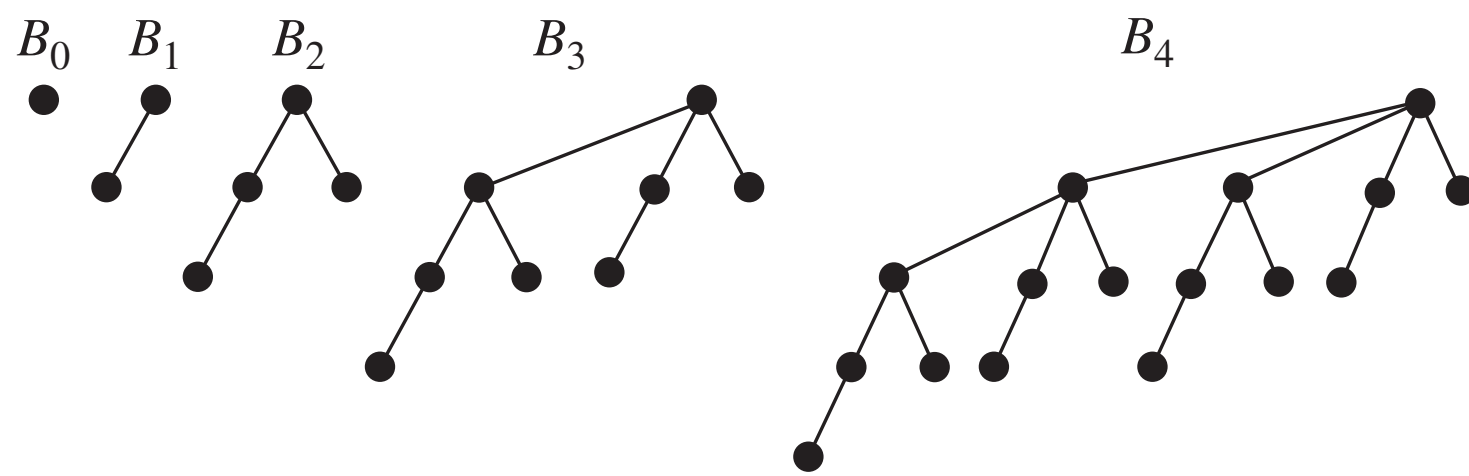
23. Same as Kruskal’s algorithm, except start with  $T :=$  this set of edges and iterate from  $i = 1$  to  $i = n - 1 - s$ , where  $s$  is the number of edges you start with.





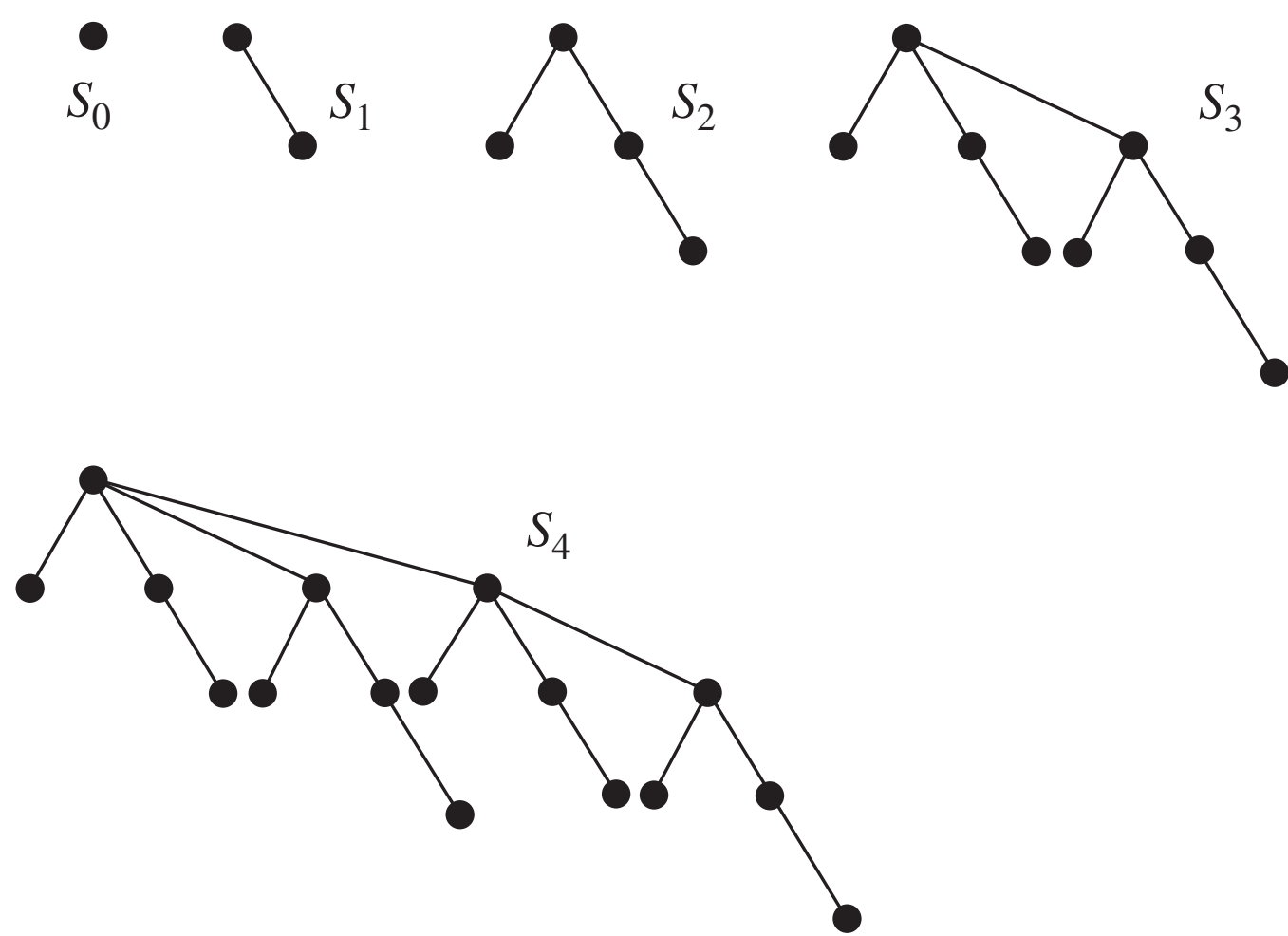
$v$ , then obviously a simple circuit is formed, because when  $e$  is added to  $T$  the resulting graph has too many edges to be a tree. The only simple circuit formed is made up of the edge  $e$  together with the unique path in  $T$  from  $v$  to  $u$ . Suppose  $T$  satisfies the given conditions. All that is needed is to show that  $T$  is connected, because there are no simple circuits in the graph. Assume that  $T$  is not connected. Then let  $u$  and  $v$  be in separate connected components. Adding  $e = \{u, v\}$  does not satisfy the conditions. **3.** Suppose that a tree  $T$  has  $n$  vertices of degrees  $d_1, d_2, \dots, d_n$ , respectively. Because  $2e = \sum_{i=1}^n d_i$  and  $e = n - 1$ , we have  $2(n - 1) = \sum_{i=1}^n d_i$ . Because each  $d_i \geq 1$ , it follows that  $2(n - 1) = n + \sum_{i=1}^n (d_i - 1)$ , or that  $n - 2 = \sum_{i=1}^n (d_i - 1)$ . Hence, at most  $n - 2$  of the terms of this sum can be 1 or more. Hence, at least two of them are 0. It follows that  $d_i = 1$  for at least two values of  $i$ . **5.**  $2n - 2$  **7.** A tree has no circuits, so it cannot have a subgraph homeomorphic to  $K_{3,3}$  or  $K_5$ . **9.** Color each connected component separately. For each of these connected components, first root the tree, then color all vertices at even levels red and all vertices at odd levels blue. **11.** Upper bound:  $k^h$ ; lower bound:  $2 \lceil k/2 \rceil^{h-1}$

**13.**



**15.** Because  $B_{k+1}$  is formed from two copies of  $B_k$ , one shifted down one level, the height increases by 1 as  $k$  increases by 1. Because  $B_0$  had height 0, it follows by induction that  $B_k$  has height  $k$ . **17.** Because the root of  $B_{k+1}$  is the root of  $B_k$  with one additional child (namely the root of the other  $B_k$ ), the degree of the root increases by 1 as  $k$  increases by 1. Because  $B_0$  had a root with degree 0, it follows by induction that  $B_k$  has a root with degree  $k$ .

**19.**



**21.** Use mathematical induction. The result is trivial for  $k = 0$ . Suppose it is true for  $k - 1$ .  $T_{k-1}$  is the parent tree for  $T$ . By induction, the child tree for  $T$  can be obtained from  $T_0, \dots, T_{k-2}$  in the manner stated. The final connection of  $r_{k-2}$  to  $r_{k-1}$  is as stated in the definition of  $S_k$ -tree.

**23. procedure** *level*( $T$ : ordered rooted tree with root  $r$ )

*queue* := sequence consisting of just the root  $r$

**while** *queue* contains at least one term

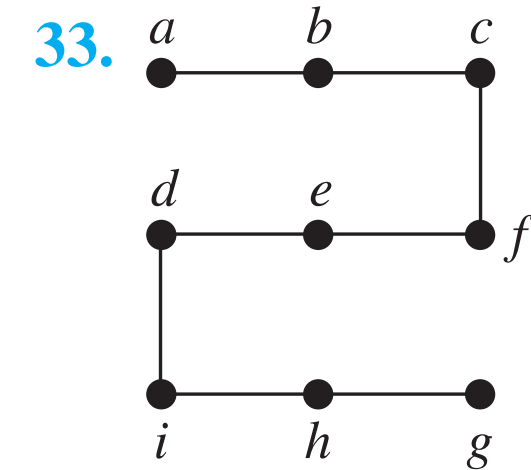
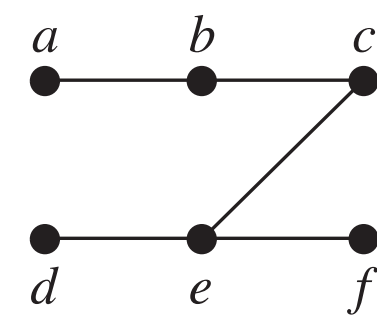
$v$  := first vertex in *queue*

list  $v$

remove  $v$  from *queue* and put children of  $v$  onto the end of *queue*

**25.** Build the tree by inserting a root for the address 0, and then inserting a subtree for each vertex labeled  $i$ , for  $i$  a positive integer, built up from subtrees for each vertex labeled  $i.j$  for  $j$  a positive integer, and so on. **27. a)** Yes **b)** No **c)** Yes **29.** The resulting graph has no edge that is in more than one simple circuit of the type described. Hence, it is a cactus.

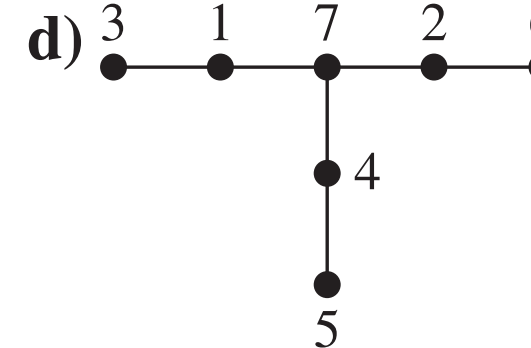
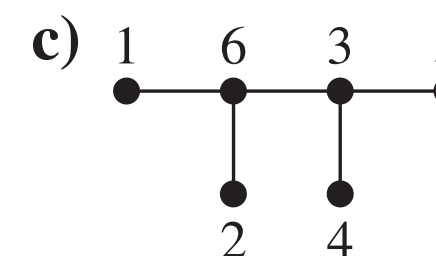
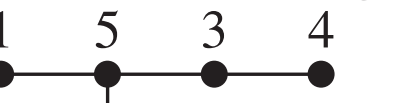
**31.**



**35. a)**



**b)**



**37.** 6 **39. a)** 0 for 00, 11 for 01, 100 for 10, 101 for 11 (exact coding depends on how ties were broken, but all versions are equivalent);  $0.645n$  for string of length  $n$  **b)** 0 for 000, 100 for 001, 101 for 010, 110 for 100, 11100 for 011, 11101 for 101, 11110 for 110, 11111 for 111 (exact coding depends on how ties were broken, but all versions are equivalent);  $0.5326n$  for string of length  $n$  **41.** Let  $G'$  be the graph obtained by deleting from  $G$  the vertex  $v$  and all edges incident to  $v$ . A minimum spanning tree of  $G$  can be obtained by taking an edge of minimal weight incident to  $v$  together with a minimum spanning tree of  $G'$ . **43.** Suppose that edge  $e$  is the edge of least weight incident to vertex  $v$ , and suppose that  $T$  is a spanning tree that does not include  $e$ . Add  $e$  to  $T$ , and delete from the simple circuit formed thereby the other edge of the circuit that contains  $v$ . The result will be a spanning tree of strictly smaller weight (because the deleted edge has weight greater than the weight of  $e$ ). This is a contradiction, so  $T$  must include  $e$ . **45.** Because paths in trees are unique, an arborescence  $T$  of a directed graph  $G$  is just a subgraph of  $G$  that is a tree rooted at  $r$ , containing all the vertices of  $G$ , with all the edges directed away from the root. Thus, the in-degree of each vertex other than  $r$  is 1. For the converse, it is enough to show that for each  $v \in V$  there is a unique directed path from  $r$  to  $v$ . Because the in-degree of each vertex other than  $r$  is 1, we can follow the edges of  $T$  backwards from  $v$ . This path can never return to a previously visited vertex, because that would create a simple circuit. Therefore, the path must



eventually stop, and it can stop only at  $r$ , whose in-degree is not necessarily 1. Following this path forward gives the path from  $r$  to  $v$  required by the definition of arborescence.

**47. a)** Run the breadth-first search algorithm, starting from  $v$  and respecting the directions of the edges, marking each vertex encountered as reachable. **b)** Running breadth-first search on  $G^{conv}$ , again starting at  $v$ , respecting the directions of the edges, and marking each vertex encountered, will identify all the vertices from which  $v$  is reachable. **c)** Choose a vertex  $v_1$  and using parts (a) and (b) find the strong component containing  $v_1$ , namely all vertices  $w$  such that  $w$  is reachable from  $v_1$  and  $v_1$  is reachable from  $w$ . Then choose another vertex  $v_2$  not yet in a strong component and find the strong component of  $v_2$ . Repeat until all vertices have been included. The correctness of this algorithm follows from the definition of strong component and Exercise 17 in Section 10.4.

## CHAPTER 12

### Section 12.1

**1. a)** 1 **b)** 1 **c)** 0 **d)** 0 **3. a)**  $(1 \cdot 1) + (\overline{0 \cdot 1} + 0) = 1 + (\overline{0} + 0) = 1 + (1 + 0) = 1 + 1 = 1$  **b)**  $(\mathbf{T} \wedge \mathbf{T}) \vee (\neg(\mathbf{F} \wedge \mathbf{T}) \vee \mathbf{F}) \equiv \mathbf{T}$

**5. a)**

| $x$ | $y$ | $z$ | $\bar{x}y$ |
|-----|-----|-----|------------|
| 1   | 1   | 1   | 0          |
| 1   | 1   | 0   | 0          |
| 1   | 0   | 1   | 0          |
| 1   | 0   | 0   | 0          |
| 0   | 1   | 1   | 1          |
| 0   | 1   | 0   | 1          |
| 0   | 0   | 1   | 0          |
| 0   | 0   | 0   | 0          |

**b)**

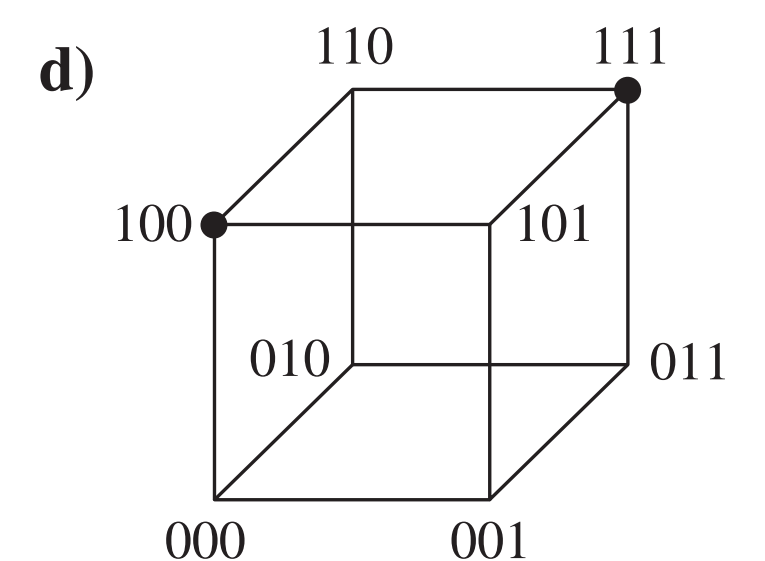
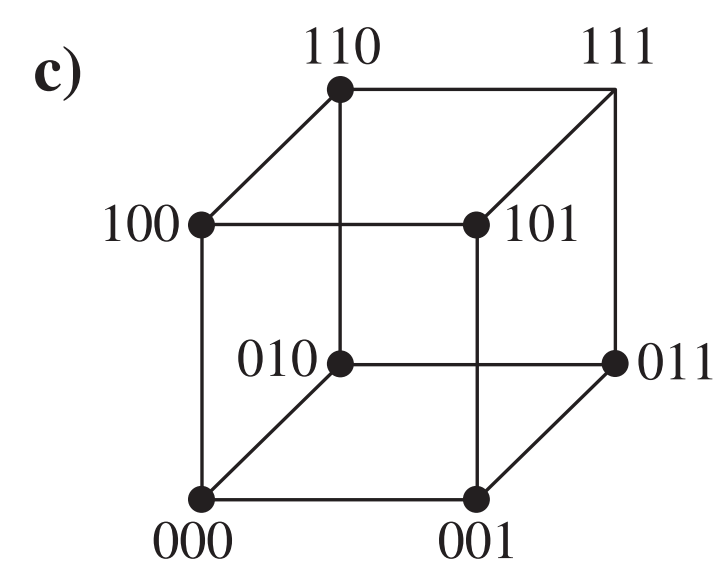
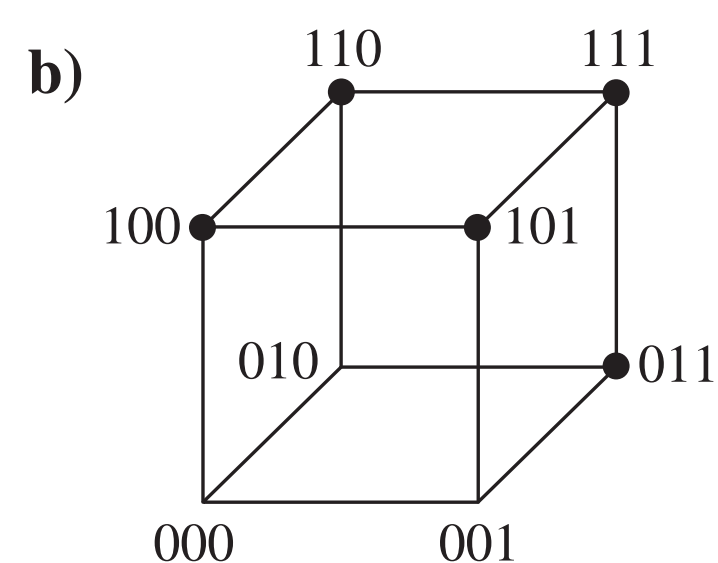
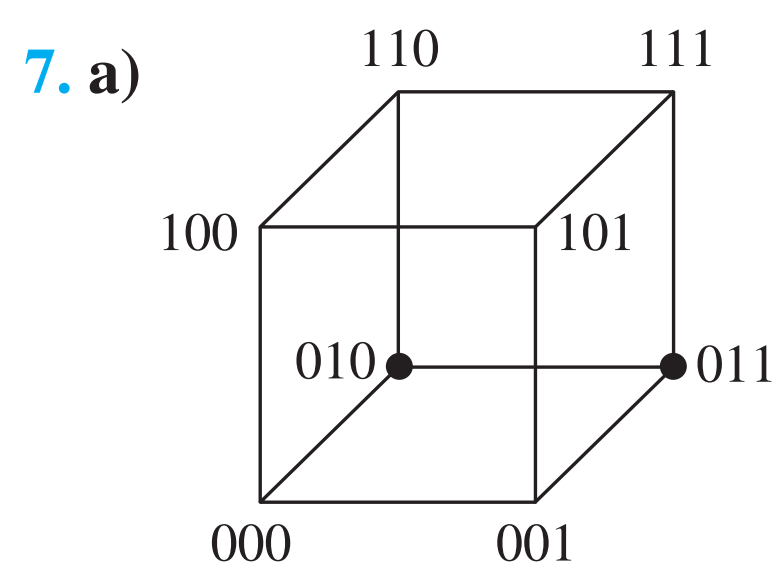
| $x$ | $y$ | $z$ | $x + yz$ |
|-----|-----|-----|----------|
| 1   | 1   | 1   | 1        |
| 1   | 1   | 0   | 1        |
| 1   | 0   | 1   | 1        |
| 1   | 0   | 0   | 1        |
| 0   | 1   | 1   | 1        |
| 0   | 1   | 0   | 0        |
| 0   | 0   | 1   | 0        |
| 0   | 0   | 0   | 0        |

**c)**

| $x$ | $y$ | $z$ | $x\bar{y} + \bar{x}yz$ |
|-----|-----|-----|------------------------|
| 1   | 1   | 1   | 0                      |
| 1   | 1   | 0   | 1                      |
| 1   | 0   | 1   | 1                      |
| 1   | 0   | 0   | 1                      |
| 0   | 1   | 1   | 1                      |
| 0   | 1   | 0   | 1                      |
| 0   | 0   | 1   | 1                      |
| 0   | 0   | 0   | 1                      |

**d)**

| $x$ | $y$ | $z$ | $x(yz + \bar{y}\bar{z})$ |
|-----|-----|-----|--------------------------|
| 1   | 1   | 1   | 1                        |
| 1   | 1   | 0   | 0                        |
| 1   | 0   | 1   | 0                        |
| 1   | 0   | 0   | 1                        |
| 0   | 1   | 1   | 0                        |
| 0   | 1   | 0   | 0                        |
| 0   | 0   | 1   | 0                        |
| 0   | 0   | 0   | 0                        |



**9.**  $(0, 0)$  and  $(1, 1)$

**11.**  $x + xy = x \cdot 1 + xy = x(1 + y) = x(y + 1) = x \cdot 1 = x$

**13.**

| $x$ | $y$ | $z$ | $x\bar{y}$ | $y\bar{z}$ | $\bar{x}z$ | $x\bar{y} + y\bar{z} + \bar{x}z$ | $\bar{x}y$ | $\bar{y}z$ | $x\bar{z}$ | $\bar{x}y + \bar{y}z + x\bar{z}$ |
|-----|-----|-----|------------|------------|------------|----------------------------------|------------|------------|------------|----------------------------------|
| 1   | 1   | 1   | 0          | 0          | 0          | 0                                | 0          | 0          | 0          | 0                                |
| 1   | 1   | 0   | 0          | 1          | 0          | 1                                | 0          | 0          | 1          | 1                                |
| 1   | 0   | 1   | 1          | 0          | 0          | 1                                | 0          | 1          | 0          | 1                                |
| 1   | 0   | 0   | 1          | 0          | 0          | 1                                | 0          | 0          | 1          | 1                                |
| 0   | 1   | 1   | 0          | 0          | 1          | 1                                | 1          | 0          | 0          | 1                                |
| 0   | 1   | 0   | 0          | 1          | 0          | 1                                | 1          | 0          | 0          | 1                                |
| 0   | 0   | 1   | 0          | 0          | 1          | 1                                | 0          | 1          | 0          | 1                                |
| 0   | 0   | 0   | 0          | 0          | 0          | 0                                | 0          | 0          | 0          | 0                                |

**15.**

| $x$ | $x + x$ | $x \cdot x$ |
|-----|---------|-------------|
| 0   | 0       | 0           |
| 1   | 1       | 1           |

**17.**

| $x$ | $x + 1$ | $x \cdot 0$ |
|-----|---------|-------------|
| 0   | 1       | 0           |
| 1   | 1       | 0           |

**19.**

| $x$ | $y$ | $z$ | $y + z$ | $x + (y + z)$ | $x + y$ | $(x + y) + z$ | $yz$ | $x(yz)$ | $xy$ | $(xy)z$ |
|-----|-----|-----|---------|---------------|---------|---------------|------|---------|------|---------|
| 1   | 1   | 1   | 1       | 1             | 1       | 1             | 1    | 1       | 1    | 1       |
| 1   | 1   | 0   | 1       | 1             | 1       | 1             | 0    | 0       | 1    | 0       |
| 1   | 0   | 1   | 1       | 1             | 1       | 1             | 0    | 0       | 0    | 0       |
| 1   | 0   | 0   | 0       | 1             | 1       | 1             | 0    | 0       | 0    | 0       |
| 0   | 1   | 1   | 1       | 1             | 1       | 1             | 1    | 0       | 0    | 0       |
| 0   | 1   | 0   | 1       | 1             | 1       | 1             | 0    | 0       | 0    | 0       |
| 0   | 0   | 1   | 1       | 1             | 0       | 1             | 0    | 0       | 0    | 0       |
| 0   | 0   | 0   | 0       | 0             | 0       | 0             | 0    | 0       | 0    | 0       |

**21.**

| $x$ | $y$ | $xy$ | $\overline{(xy)}$ | $\bar{x}$ | $\bar{y}$ | $\bar{x} + \bar{y}$ | $x + y$ | $\overline{(x + y)}$ | $\bar{x}\bar{y}$ |
|-----|-----|------|-------------------|-----------|-----------|---------------------|---------|----------------------|------------------|
| 1   | 1   | 1    | 0                 | 0         | 0         | 0                   | 1       | 0                    | 0                |
| 1   | 0   | 0    | 1                 | 0         | 1         | 1                   | 1       | 0                    | 0                |
| 0   | 1   | 0    | 1                 | 1         | 0         | 1                   | 1       | 0                    | 0                |
| 0   | 0   | 0    | 1                 | 1         | 1         | 1                   | 0       | 1                    | 1                |

**23.**  $0 \cdot \bar{0} = 0 \cdot 1 = 0$ ;  $1 \cdot \bar{1} = 1 \cdot 0 = 0$

**25.**

| $x$ | $y$ | $x \oplus y$ | $x + y$ | $xy$ | $\overline{(xy)}$ | $(x + y)\overline{(xy)}$ | $x\bar{y}$ | $\bar{x}y$ | $x\bar{y} + \bar{x}y$ |
|-----|-----|--------------|---------|------|-------------------|--------------------------|------------|------------|-----------------------|
| 1   | 1   | 0            | 1       | 1    | 0                 | 0                        | 0          | 0          | 0                     |
| 1   | 0   | 1            | 1       | 0    | 1                 | 1                        | 1          | 0          | 1                     |
| 0   | 1   | 1            | 1       | 0    | 1                 | 1                        | 0          | 1          | 1                     |
| 0   | 0   | 0            | 0       | 0    | 1                 | 0                        | 0          | 0          | 0                     |

**27. a)** True, as a table of values can show **b)** False; take  $x = 1$ ,  $y = 1$ ,  $z = 1$ , for instance **c)** False; take  $x = 1$ ,  $y = 1$ ,  $z = 0$ , for instance **29.** By De Morgan's laws, the complement of an expression is like the dual except that the complement of each variable has been taken. **31.** 16 **33.** If we



replace each 0 by **F**, 1 by **T**, Boolean sum by  $\vee$ , Boolean product by  $\wedge$ , and  $\bar{\phantom{x}}$  by  $\neg$  (and  $x$  by  $p$  and  $y$  by  $q$  so that the variables look like they represent propositions, and the equals sign by the logical equivalence symbol), then  $\overline{xy} = \bar{x} + \bar{y}$  becomes  $\neg(p \wedge q) \equiv \neg p \vee \neg q$  and  $\overline{x+y} = \bar{x} \bar{y}$  becomes  $\neg(p \vee q) \equiv \neg p \wedge \neg q$ . **35.** By the domination, distributive, and identity laws,  $x \vee x = (x \vee x) \wedge 1 = (x \vee x) \wedge (x \vee \bar{x}) = x \vee (x \wedge \bar{x}) = x \vee 0 = x$ . Similarly,  $x \wedge x = (x \wedge x) \vee 0 = (x \wedge x) \vee (x \wedge \bar{x}) = x \wedge (x \vee \bar{x}) = x \wedge 1 = x$ . **37.** Because  $0 \vee 1 = 1$  and  $0 \wedge 1 = 0$  by the identity and commutative laws, it follows that  $\bar{0} = 1$ . Similarly, because  $1 \vee 0 = 1$  and  $1 \wedge 0 = 0$ , it follows that  $\bar{1} = 0$ . **39.** First, note that  $x \wedge 0 = 0$  and  $x \vee 1 = 1$  for all  $x$ , as can easily be proved. To prove the first identity, it is sufficient to show that  $(x \vee y) \vee (\bar{x} \wedge \bar{y}) = 1$  and  $(x \vee y) \wedge (\bar{x} \wedge \bar{y}) = 0$ . By the associative, commutative, distributive, domination, and identity laws,  $(x \vee y) \vee (\bar{x} \wedge \bar{y}) = y \vee [x \vee (\bar{x} \wedge \bar{y})] = y \vee [(x \vee \bar{x}) \wedge (x \vee \bar{y})] = y \vee [1 \wedge (x \vee \bar{y})] = y \vee (x \vee \bar{y}) = (y \vee \bar{y}) \vee x = 1 \vee x = 1$  and  $(x \vee y) \wedge (\bar{x} \wedge \bar{y}) = \bar{y} \wedge [x \wedge (\bar{x} \vee y)] = \bar{y} \wedge [(\bar{x} \wedge x) \vee (\bar{x} \wedge y)] = \bar{y} \wedge [0 \vee (\bar{x} \wedge y)] = \bar{y} \wedge (\bar{x} \wedge y) = \bar{x} \wedge (y \wedge \bar{y}) = \bar{x} \wedge 0 = 0$ . The second identity is proved in a similar way. **41.** Using the hypotheses, Exercise 35, and the distributive law it follows that  $x = x \vee 0 = x \vee (x \vee y) = (x \vee x) \vee y = x \vee y = 0$ . Similarly,  $y = 0$ . To prove the second statement, note that  $x = x \wedge 1 = x \wedge (x \wedge y) = (x \wedge x) \wedge y = x \wedge y = 1$ . Similarly,  $y = 1$ . **43.** Use Exercises 39 and 41 in the Supplementary Exercises in Chapter 9 and the definition of a complemented, distributed lattice to establish the five pairs of laws in the definition.

## Section 12.2

- 1. a)**  $\bar{x} \bar{y} \bar{z}$  **b)**  $\bar{x} y \bar{z}$  **c)**  $\bar{x} y z$  **d)**  $\bar{x} \bar{y} \bar{z}$  **3. a)**  $xyz + xy\bar{z} + x\bar{y}z + x\bar{y}\bar{z} + \bar{x}yz + \bar{x}y\bar{z} + \bar{x}\bar{y}z + \bar{x}\bar{y}\bar{z}$  **b)**  $xyz + xy\bar{z} + \bar{x}yz + \bar{x}y\bar{z}$  **c)**  $xyz + xy\bar{z} + x\bar{y}z + x\bar{y}\bar{z}$  **d)**  $x\bar{y}z + x\bar{y}\bar{z}$  **5.**  $wxy\bar{z} + wx\bar{y}z + w\bar{x}yz + \bar{w}xyz + \bar{w}x\bar{y}\bar{z} + \bar{w}\bar{x}y\bar{z} + \bar{w}\bar{x}\bar{y}z + \bar{w}\bar{x}\bar{y}\bar{z}$  **7. a)**  $\bar{x} + \bar{y} + z$  **b)**  $x + y + z$  **c)**  $x + \bar{y} + z$  **9.**  $y_1 + y_2 + \dots + y_n = 0$  if and only if  $y_i = 0$  for  $i = 1, 2, \dots, n$ . This holds if and only if  $x_i = 0$  when  $y_i = x_i$  and  $x_i = 1$  when  $y_i = \bar{x}_i$ . **11. a)**  $x + y + z$  **b)**  $(x + y + z)(x + y + \bar{z})(x + \bar{y} + z)(\bar{x} + y + z)(\bar{x} + y + \bar{z})$  **c)**  $(x + y + z)(x + y + \bar{z})(x + \bar{y} + z)(x + \bar{y} + \bar{z})$  **d)**  $(x + y + z)(x + y + \bar{z})(x + \bar{y} + z)(x + \bar{y} + \bar{z})(\bar{x} + \bar{y} + z)(\bar{x} + \bar{y} + \bar{z})$  **13. a)**  $x + y + z$  **b)**  $x + [y + (\bar{x} + \bar{z})]$  **c)**  $(x + \bar{y})$  **d)**  $[x + (\bar{x} + \bar{y} + \bar{z})]$

**15. a)**

| $x$ | $\bar{x}$ | $x \downarrow x$ |
|-----|-----------|------------------|
| 1   | 0         | 0                |
| 0   | 1         | 1                |

**b)**

| $x$ | $y$ | $xy$ | $x \downarrow x$ | $y \downarrow y$ | $(x \downarrow x) \downarrow (y \downarrow y)$ |
|-----|-----|------|------------------|------------------|------------------------------------------------|
| 1   | 1   | 1    | 0                | 0                | 1                                              |
| 1   | 0   | 0    | 0                | 1                | 0                                              |
| 0   | 1   | 0    | 1                | 0                | 0                                              |
| 0   | 0   | 0    | 1                | 1                | 0                                              |

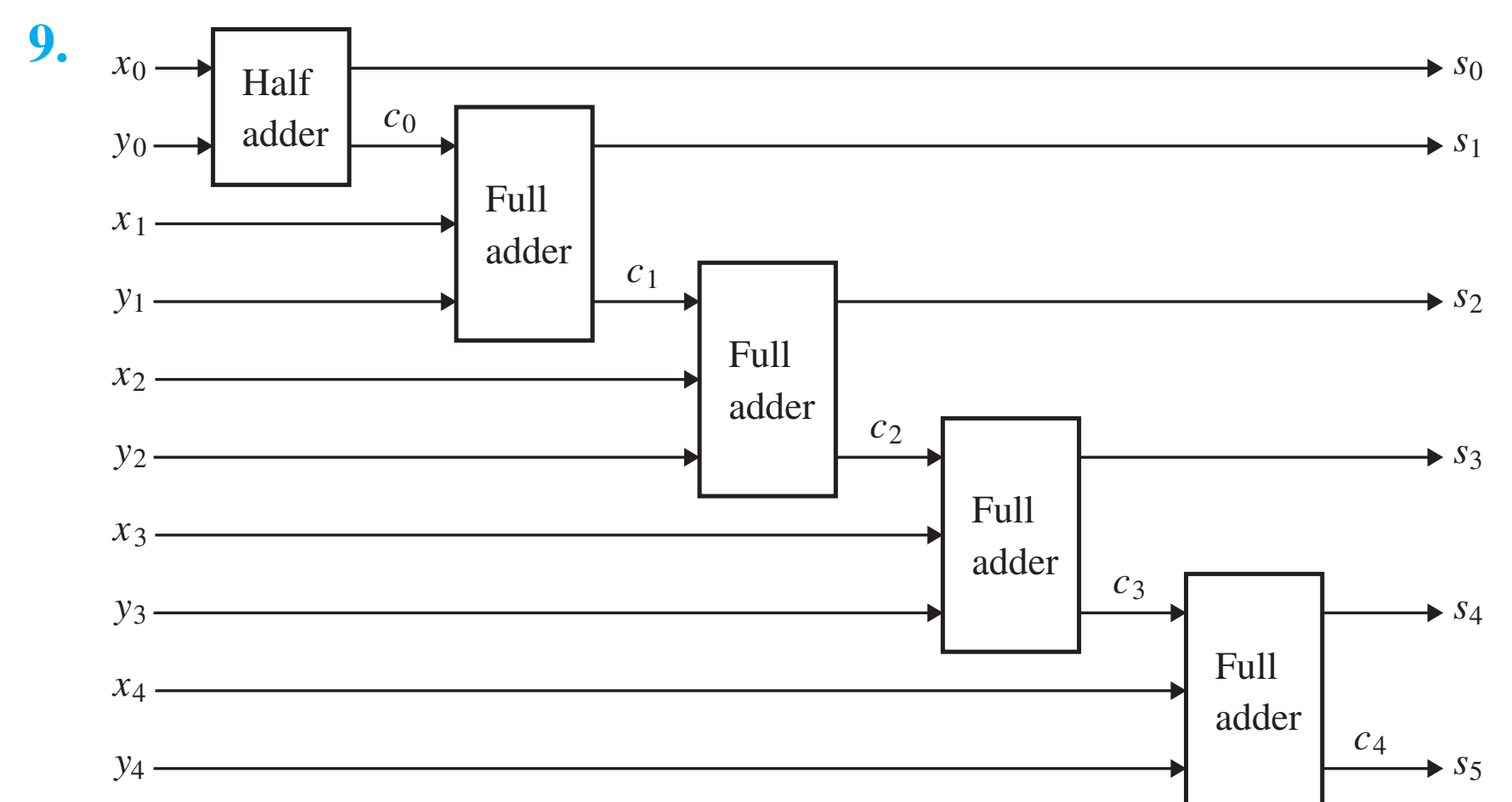
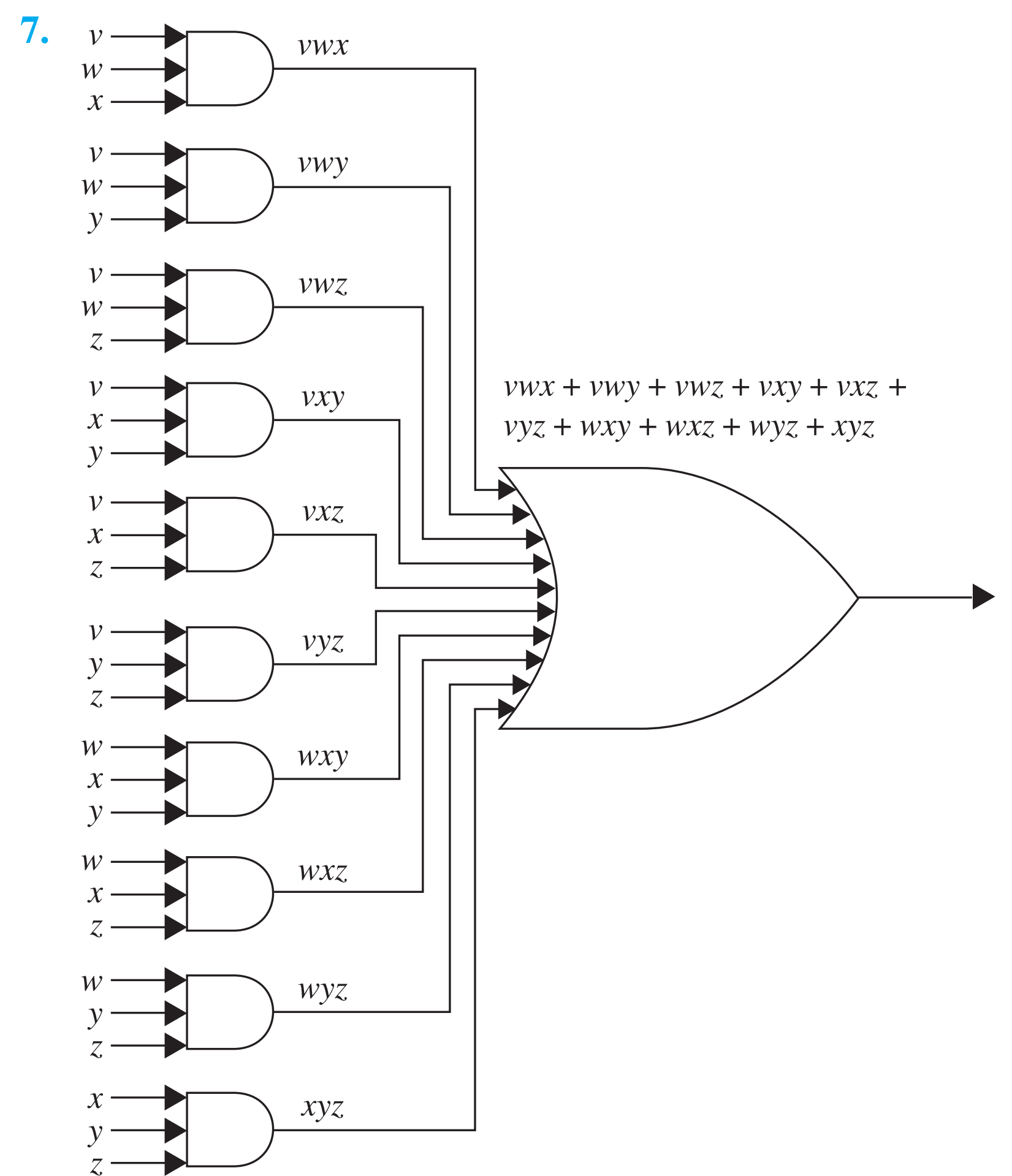
**c)**

| $x$ | $y$ | $x + y$ | $(x \downarrow y)$ | $(x \downarrow y) \downarrow (x \downarrow y)$ |
|-----|-----|---------|--------------------|------------------------------------------------|
| 1   | 1   | 1       | 0                  | 1                                              |
| 1   | 0   | 1       | 0                  | 1                                              |
| 0   | 1   | 1       | 0                  | 1                                              |
| 0   | 0   | 0       | 1                  | 0                                              |

- 17. a)**  $\{[(x \mid x) \mid (y \mid y)] \mid [(x \mid x) \mid (y \mid y)]\} \mid (z \mid z)$  **b)**  $\{[(x \mid x) \mid (z \mid z)] \mid y\} \mid \{[(x \mid x) \mid (z \mid z)] \mid y\}$  **c)**  $x$  **d)**  $[x \mid (y \mid y)] \mid [x \mid (y \mid y)]$  **19.** It is impossible to represent  $\bar{x}$  using  $+$  and  $\cdot$  because there is no way to get the value 0 if the input is 1.

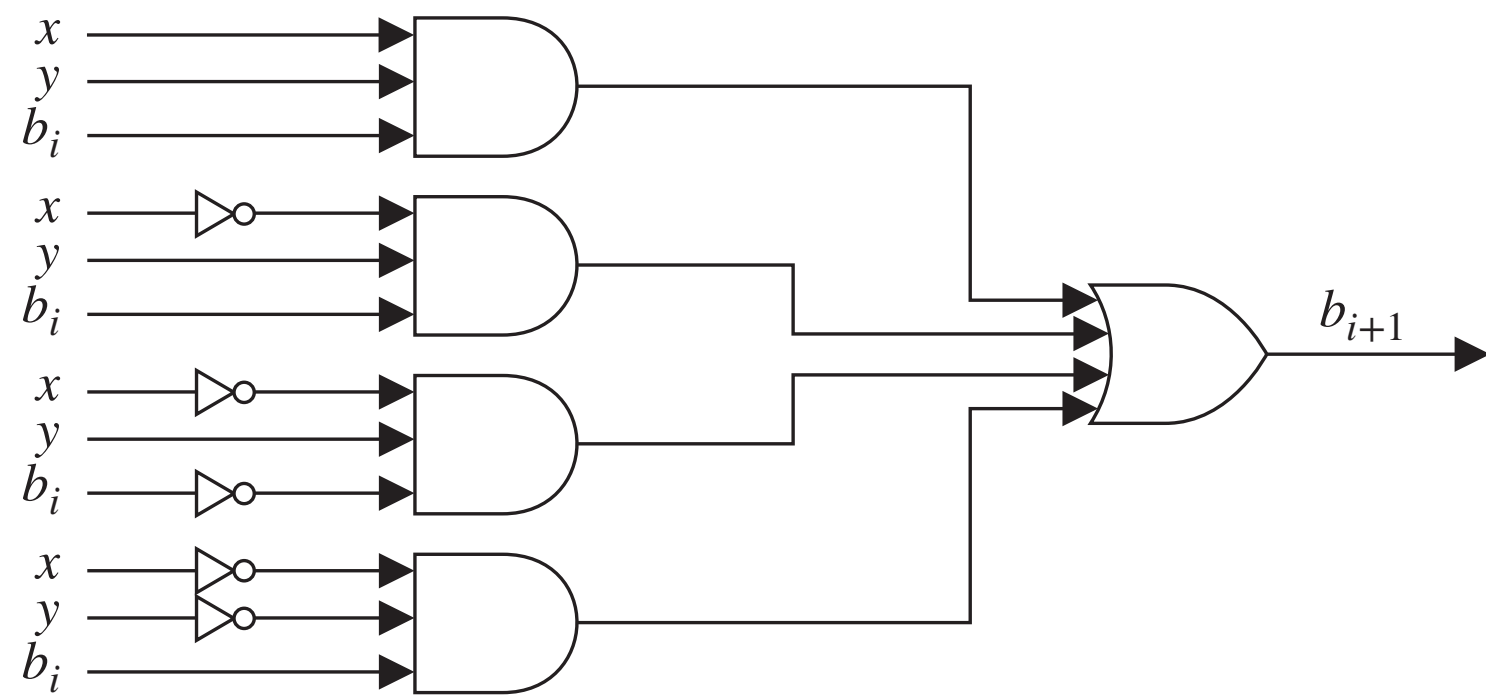
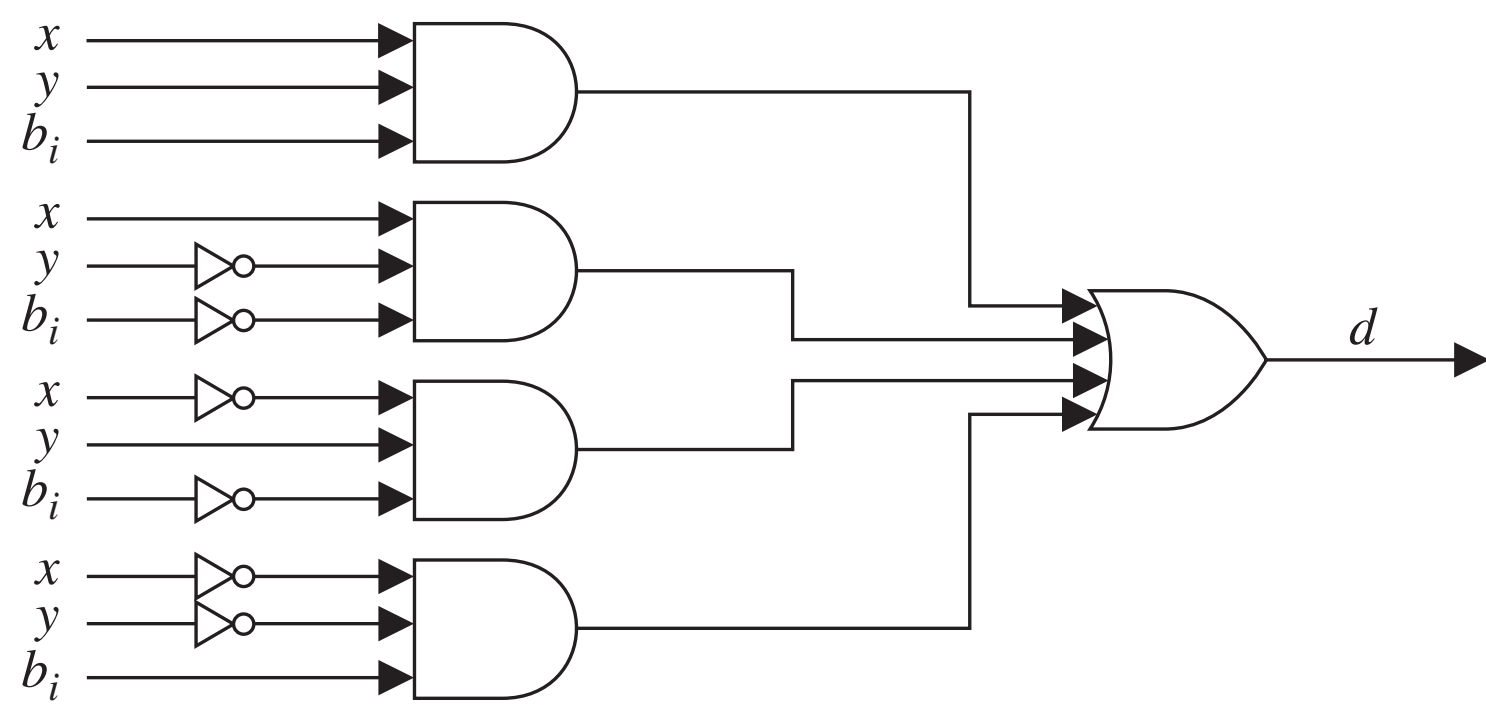
## Section 12.3

- 1.**  $(x+y)\bar{y}$  **3.**  $\overline{(xy)} + (\bar{z} + x)$  **5.**  $(x+y+z) + (\bar{x} + y + z) + (\bar{x} + \bar{y} + \bar{z})$

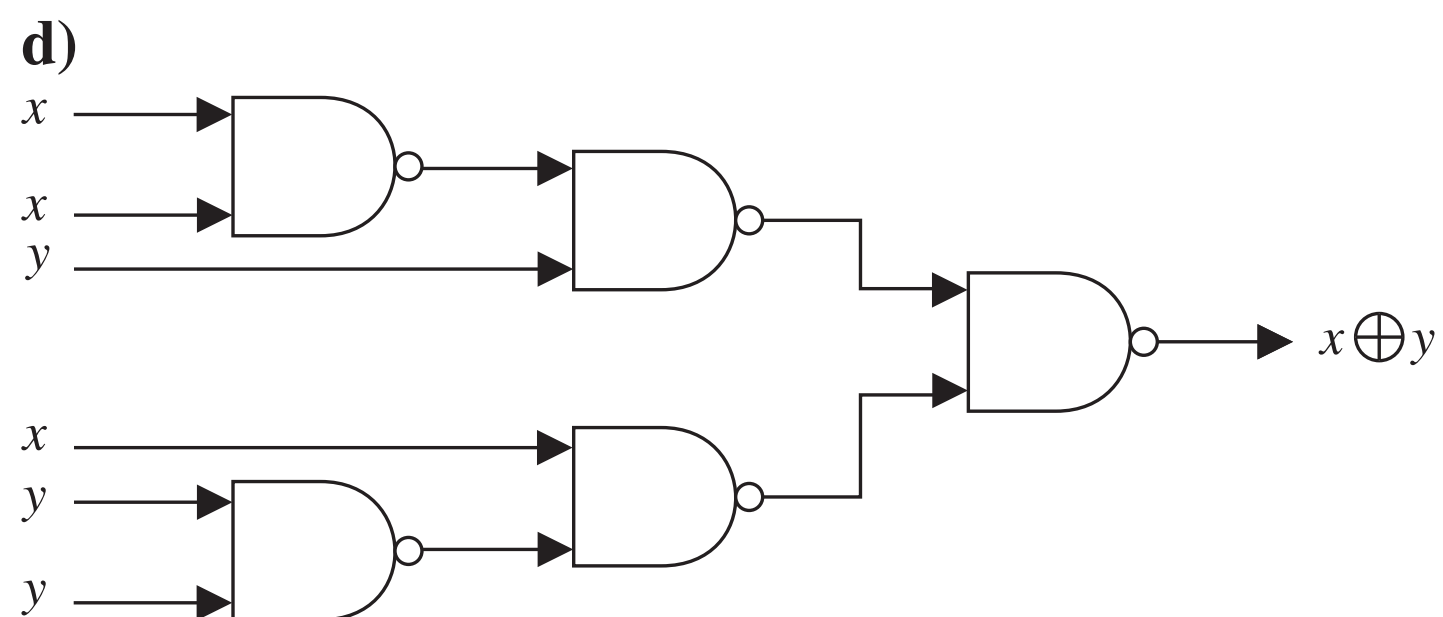
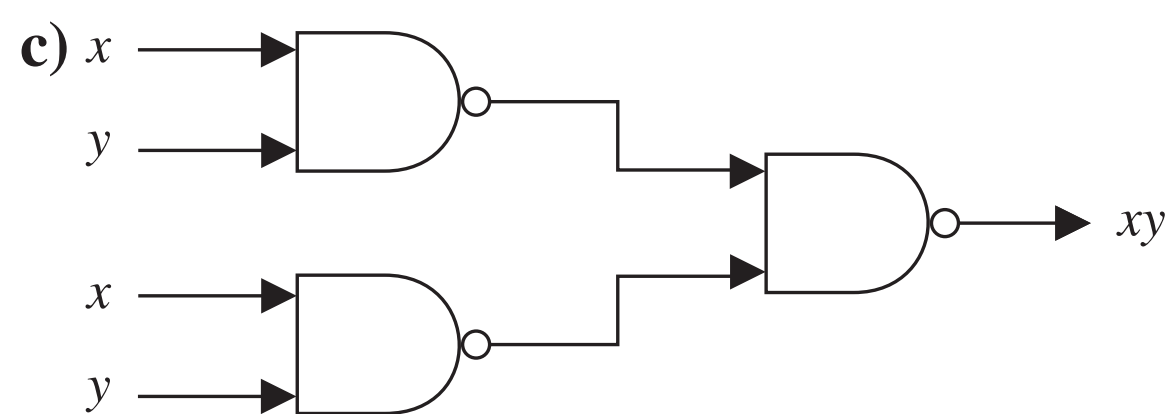
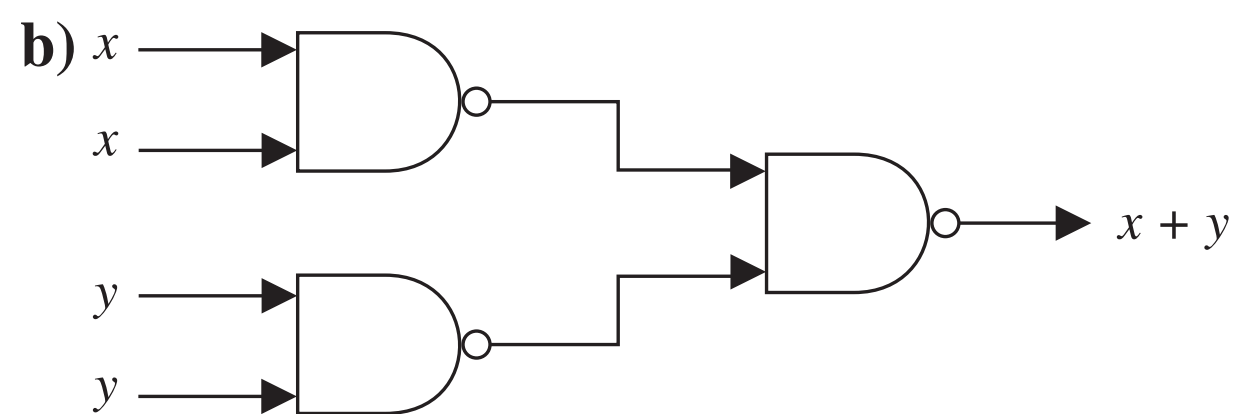
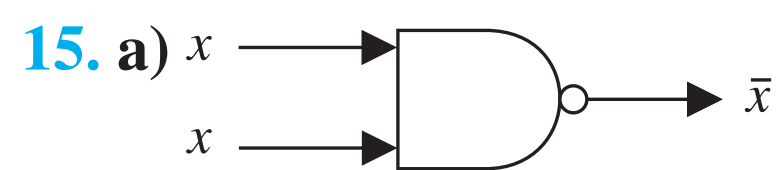
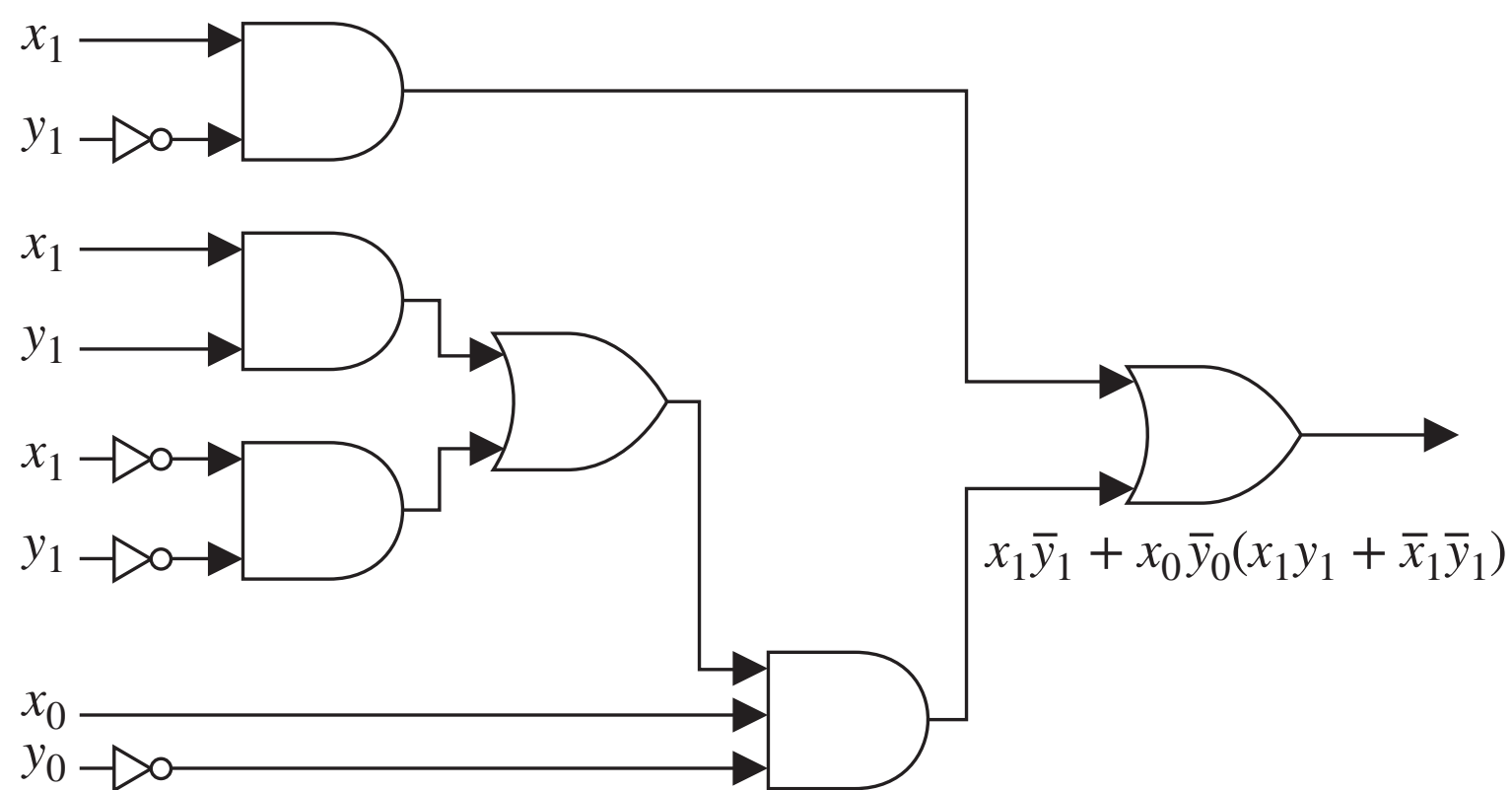




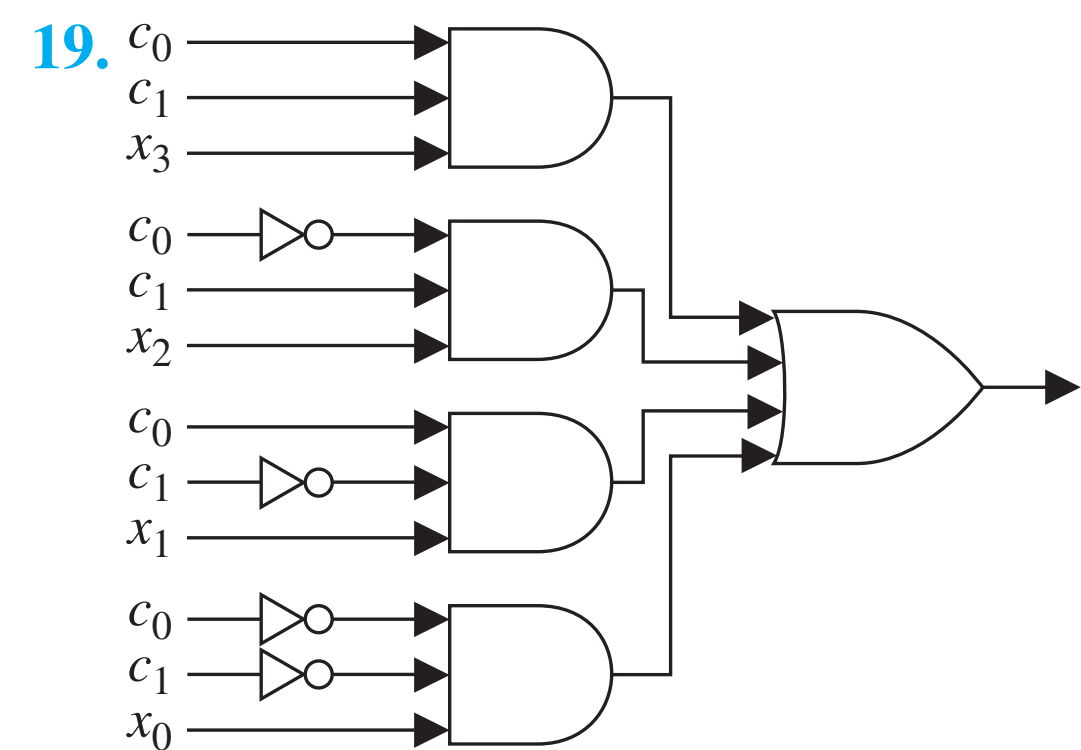
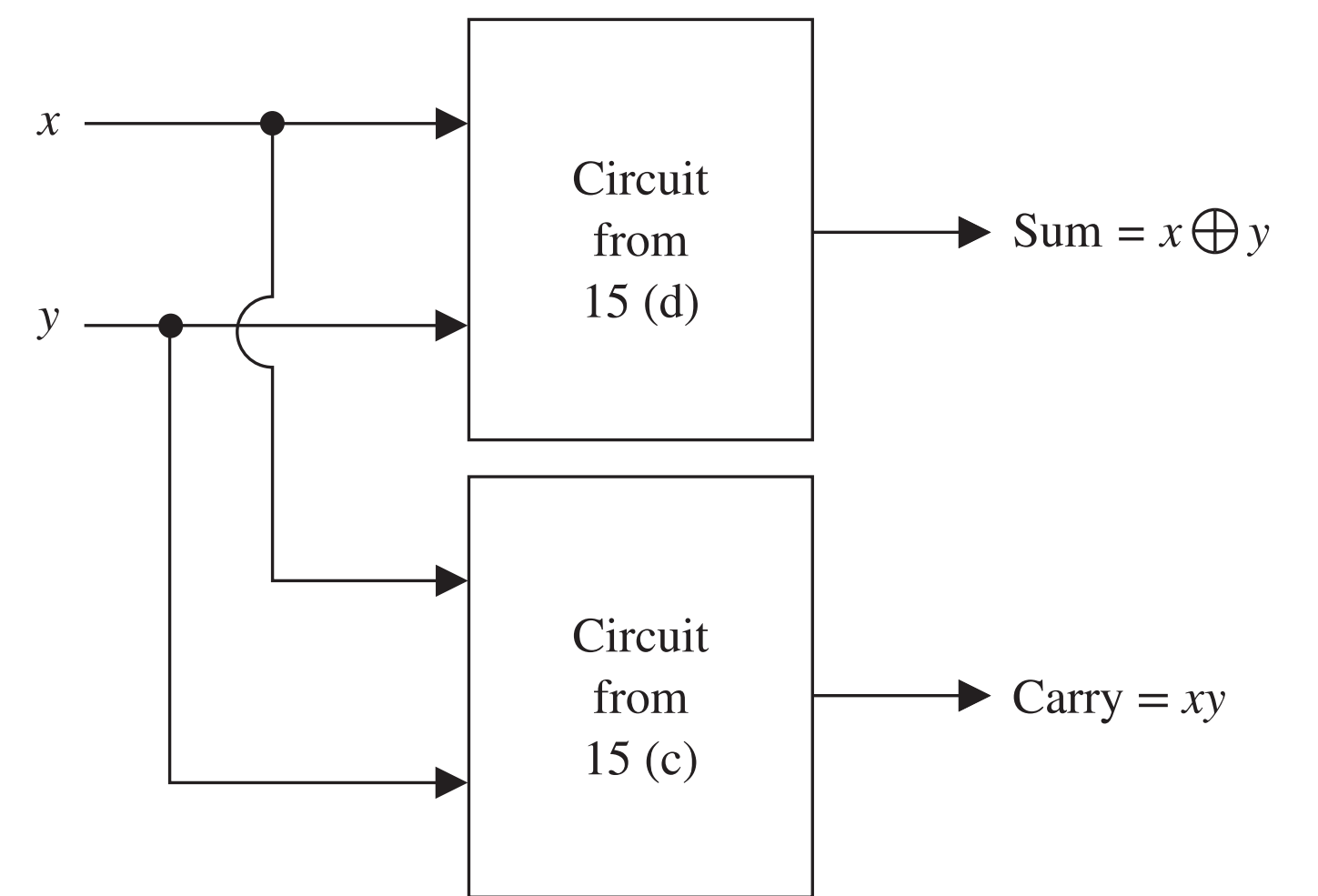
11.



13.



17.



## Section 12.4

1. a) 

|           | $y$ | $\bar{y}$ |
|-----------|-----|-----------|
| $x$       |     |           |
| $\bar{x}$ | 1   |           |

 b)  $xy$  and  $\bar{x}\bar{y}$

3. a)

|           | $y$ | $\bar{y}$ |
|-----------|-----|-----------|
| $x$       |     | 1         |
| $\bar{x}$ |     |           |

b)

|           | $y$ | $\bar{y}$ |
|-----------|-----|-----------|
| $x$       | 1   |           |
| $\bar{x}$ |     | 1         |

c)

|           | $y$ | $\bar{y}$ |
|-----------|-----|-----------|
| $x$       | 1   | 1         |
| $\bar{x}$ | 1   | 1         |

5. a)

|           | $yz$ | $y\bar{z}$ | $\bar{y}\bar{z}$ | $\bar{y}z$ |
|-----------|------|------------|------------------|------------|
| $x$       |      |            |                  |            |
| $\bar{x}$ |      | 1          |                  |            |

b)  $\bar{x}yz, \bar{x}\bar{y}\bar{z}, xy\bar{z}$



7. a)

|           |      |            |                  |            |
|-----------|------|------------|------------------|------------|
|           | $yz$ | $y\bar{z}$ | $\bar{y}\bar{z}$ | $\bar{y}z$ |
| $x$       |      |            | 1                |            |
| $\bar{x}$ |      |            |                  |            |

b)

|           |      |            |                  |            |
|-----------|------|------------|------------------|------------|
|           | $yz$ | $y\bar{z}$ | $\bar{y}\bar{z}$ | $\bar{y}z$ |
| $x$       |      |            |                  |            |
| $\bar{x}$ | 1    |            | 1                |            |

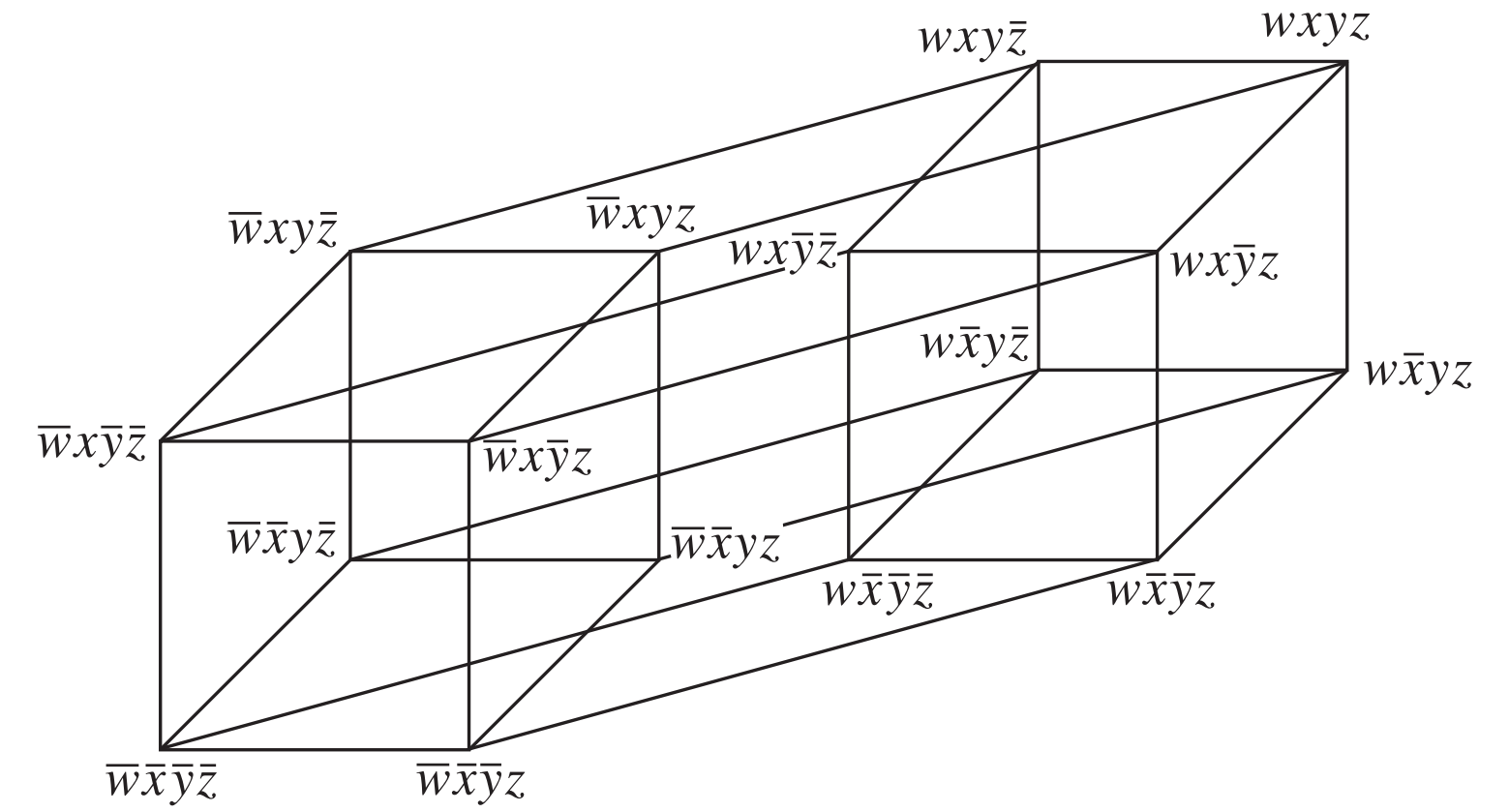
c)

|           |      |            |                  |            |
|-----------|------|------------|------------------|------------|
|           | $yz$ | $y\bar{z}$ | $\bar{y}\bar{z}$ | $\bar{y}z$ |
| $x$       | 1    | 1          |                  |            |
| $\bar{x}$ |      | 1          |                  | 1          |

9. Implicants:  $xyz$ ,  $xy\bar{z}$ ,  $x\bar{y}\bar{z}$ ,  $\bar{x}y\bar{z}$ ,  $xy$ ,  $x\bar{z}$ ,  $y\bar{z}$ ; prime implicants:  $xy$ ,  $x\bar{z}$ ,  $y\bar{z}$ ; essential prime implicants:  $xy$ ,  $x\bar{z}$ ,  $y\bar{z}$

|           |      |            |                  |            |
|-----------|------|------------|------------------|------------|
|           | $yz$ | $y\bar{z}$ | $\bar{y}\bar{z}$ | $\bar{y}z$ |
| $x$       | 1    | 1          | 1                |            |
| $\bar{x}$ |      | 1          |                  |            |

11. The 3-cube on the right corresponds to  $w$ ; the 3-cube given by the top surface of the whole figure represents  $x$ ; the 3-cube given by the back surface of the whole figure represents  $y$ ; the 3-cube given by the right surfaces of both the left and the right 3-cube represents  $z$ . In each case, the opposite 3-face represents the complemented literal. The 2-cube that represents  $wz$  is the right face of the 3-cube on the right; the 2-cube that represents  $\bar{x}y$  is bottom rear; the 2-cube that represents  $\bar{y}\bar{z}$  is front left.



13. a)

|                  |      |            |                  |            |
|------------------|------|------------|------------------|------------|
|                  | $yz$ | $y\bar{z}$ | $\bar{y}\bar{z}$ | $\bar{y}z$ |
| $wx$             |      |            |                  |            |
| $w\bar{x}$       |      |            |                  |            |
| $\bar{w}\bar{x}$ |      |            |                  |            |
| $\bar{w}x$       |      | 1          |                  |            |

b)  $\bar{w}xyz$ ,  $\bar{w}\bar{x}y\bar{z}$ ,  $\bar{w}x\bar{y}\bar{z}$ ,  $wxy\bar{z}$

15. a)

|                      |             |                   |                         |                   |                         |                               |                         |                   |
|----------------------|-------------|-------------------|-------------------------|-------------------|-------------------------|-------------------------------|-------------------------|-------------------|
|                      | $x_3x_4x_5$ | $x_3x_4\bar{x}_5$ | $x_3\bar{x}_4\bar{x}_5$ | $x_3\bar{x}_4x_5$ | $\bar{x}_3\bar{x}_4x_5$ | $\bar{x}_3\bar{x}_4\bar{x}_5$ | $\bar{x}_3x_4\bar{x}_5$ | $\bar{x}_3x_4x_5$ |
| $x_1x_2$             | 1           | 1                 |                         |                   |                         |                               |                         |                   |
| $x_1\bar{x}_2$       |             |                   |                         |                   |                         |                               |                         |                   |
| $\bar{x}_1\bar{x}_2$ |             |                   |                         |                   |                         |                               |                         |                   |
| $\bar{x}_1x_2$       |             |                   |                         |                   |                         |                               |                         |                   |

b)

|                      |             |                   |                         |                   |                         |                               |                         |                   |
|----------------------|-------------|-------------------|-------------------------|-------------------|-------------------------|-------------------------------|-------------------------|-------------------|
|                      | $x_3x_4x_5$ | $x_3x_4\bar{x}_5$ | $x_3\bar{x}_4\bar{x}_5$ | $x_3\bar{x}_4x_5$ | $\bar{x}_3\bar{x}_4x_5$ | $\bar{x}_3\bar{x}_4\bar{x}_5$ | $\bar{x}_3x_4\bar{x}_5$ | $\bar{x}_3x_4x_5$ |
| $x_1x_2$             |             |                   |                         |                   |                         |                               |                         |                   |
| $x_1\bar{x}_2$       |             |                   |                         |                   |                         |                               |                         |                   |
| $\bar{x}_1\bar{x}_2$ | 1           |                   |                         | 1                 |                         |                               |                         |                   |
| $\bar{x}_1x_2$       | 1           |                   |                         | 1                 |                         |                               |                         |                   |

c)

|                      |             |                   |                         |                   |                         |                               |                         |                   |
|----------------------|-------------|-------------------|-------------------------|-------------------|-------------------------|-------------------------------|-------------------------|-------------------|
|                      | $x_3x_4x_5$ | $x_3x_4\bar{x}_5$ | $x_3\bar{x}_4\bar{x}_5$ | $x_3\bar{x}_4x_5$ | $\bar{x}_3\bar{x}_4x_5$ | $\bar{x}_3\bar{x}_4\bar{x}_5$ | $\bar{x}_3x_4\bar{x}_5$ | $\bar{x}_3x_4x_5$ |
| $x_1x_2$             | 1           | 1                 |                         |                   |                         |                               | 1                       | 1                 |
| $x_1\bar{x}_2$       |             |                   |                         |                   |                         |                               |                         |                   |
| $\bar{x}_1\bar{x}_2$ |             |                   |                         |                   |                         |                               |                         |                   |
| $\bar{x}_1x_2$       | 1           | 1                 |                         |                   |                         |                               | 1                       | 1                 |



d)

|                      | $x_3x_4x_5$ | $x_3x_4\bar{x}_5$ | $x_3\bar{x}_4\bar{x}_5$ | $x_3\bar{x}_4x_5$ | $\bar{x}_3\bar{x}_4x_5$ | $\bar{x}_3\bar{x}_4\bar{x}_5$ | $\bar{x}_3x_4\bar{x}_5$ | $\bar{x}_3x_4x_5$ |
|----------------------|-------------|-------------------|-------------------------|-------------------|-------------------------|-------------------------------|-------------------------|-------------------|
| $x_1x_2$             |             |                   |                         |                   | 1                       | 1                             |                         |                   |
| $x_1\bar{x}_2$       |             |                   |                         |                   | 1                       | 1                             |                         |                   |
| $\bar{x}_1\bar{x}_2$ |             |                   |                         |                   | 1                       | 1                             |                         |                   |
| $\bar{x}_1x_2$       |             |                   |                         |                   | 1                       | 1                             |                         |                   |

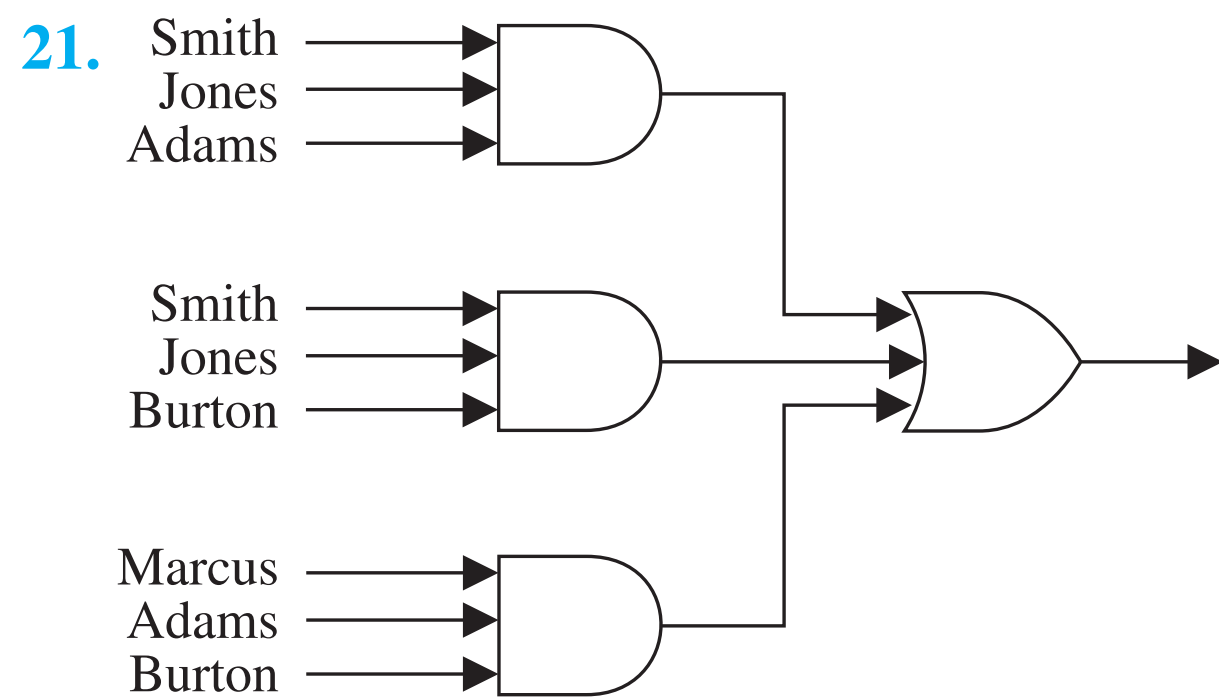
e)

|                      | $x_3x_4x_5$ | $x_3x_4\bar{x}_5$ | $x_3\bar{x}_4\bar{x}_5$ | $x_3\bar{x}_4x_5$ | $\bar{x}_3\bar{x}_4x_5$ | $\bar{x}_3\bar{x}_4\bar{x}_5$ | $\bar{x}_3x_4\bar{x}_5$ | $\bar{x}_3x_4x_5$ |
|----------------------|-------------|-------------------|-------------------------|-------------------|-------------------------|-------------------------------|-------------------------|-------------------|
| $x_1x_2$             | 1           | 1                 | 1                       | 1                 |                         |                               |                         |                   |
| $x_1\bar{x}_2$       | 1           | 1                 | 1                       | 1                 |                         |                               |                         |                   |
| $\bar{x}_1\bar{x}_2$ | 1           | 1                 | 1                       | 1                 |                         |                               |                         |                   |
| $\bar{x}_1x_2$       | 1           | 1                 | 1                       | 1                 |                         |                               |                         |                   |

f)

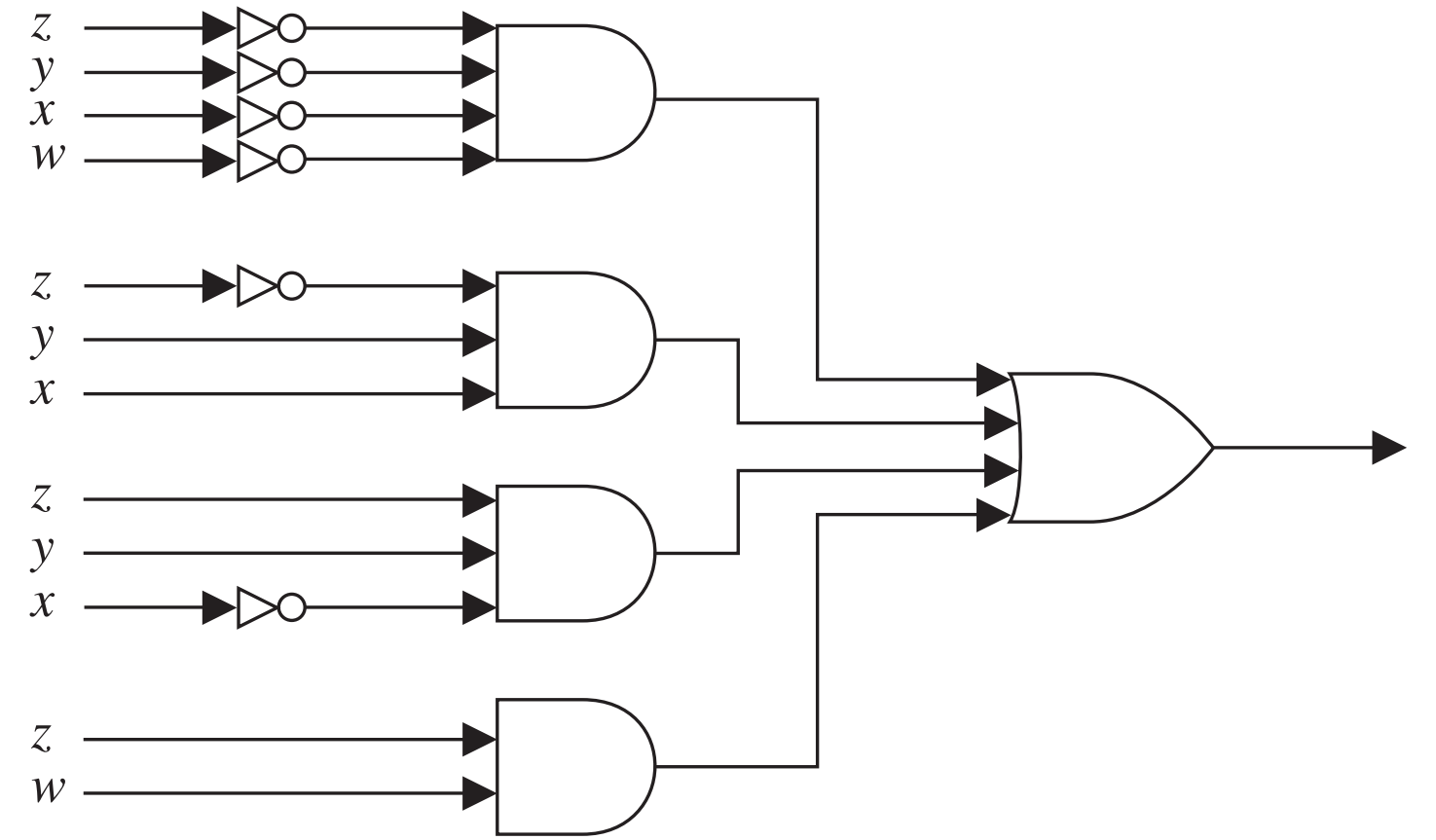
|                      | $x_3x_4x_5$ | $x_3x_4\bar{x}_5$ | $x_3\bar{x}_4\bar{x}_5$ | $x_3\bar{x}_4x_5$ | $\bar{x}_3\bar{x}_4x_5$ | $\bar{x}_3\bar{x}_4\bar{x}_5$ | $\bar{x}_3x_4\bar{x}_5$ | $\bar{x}_3x_4x_5$ |
|----------------------|-------------|-------------------|-------------------------|-------------------|-------------------------|-------------------------------|-------------------------|-------------------|
| $x_1x_2$             |             | 1                 | 1                       |                   |                         | 1                             | 1                       |                   |
| $x_1\bar{x}_2$       |             | 1                 | 1                       |                   |                         | 1                             | 1                       |                   |
| $\bar{x}_1\bar{x}_2$ |             | 1                 | 1                       |                   |                         | 1                             | 1                       |                   |
| $\bar{x}_1x_2$       |             | 1                 | 1                       |                   |                         | 1                             | 1                       |                   |

17. a) 64 b) 6 19. Rows 1 and 4 are considered adjacent. The pairs of columns considered adjacent are: columns 1 and 4, 1 and 12, 1 and 16, 2 and 11, 2 and 15, 3 and 6, 3 and 10, 4 and 9, 5 and 8, 5 and 16, 6 and 15, 7 and 10, 7 and 14, 8 and 13, 9 and 12, 11 and 14, 13 and 16.



23. a)  $\bar{x}\bar{z}$  b)  $y$  c)  $x\bar{z} + \bar{x}z + \bar{y}z$  d)  $xz + \bar{x}y + \bar{y}\bar{z}$  25. a)  $wxz + wx\bar{y} + w\bar{y}z + w\bar{x}y\bar{z}$  b)  $x\bar{y}z + \bar{w}\bar{y}z + wxy\bar{z} + w\bar{x}yz + \bar{w}\bar{x}y\bar{z}$  c)  $\bar{y}z + wxy + w\bar{x}\bar{y} + \bar{w}\bar{x}y\bar{z}$  d)  $wy + yz + \bar{x}y + wxz + \bar{w}\bar{x}z$  27.  $x(y + z)$

29.



31.  $\bar{x}\bar{z} + xz$  33. We use induction on  $n$ . If  $n = 1$ , then we are looking at a line segment, labeled 0 at one end and 1 at the other end. The only possible value of  $k$  is also 1, and if the literal is  $x_1$ , then the subcube we have is the 0-dimensional subcube consisting of the endpoint labeled 1, and if the literal is  $\bar{x}_1$ , then the subcube we have is the 0-dimensional subcube consisting of the endpoint labeled 0. Now assume that the statement is true for  $n$ ; we must show that it is true for  $n + 1$ . If the literal  $x_{n+1}$  (or its complement) is not part of the product, then by the inductive hypothesis, the product when viewed in the setting of  $n$  variables corresponds to an  $(n - k)$ -dimensional subcube of the  $n$ -dimensional cube, and the Cartesian product of that subcube with the line segment  $[0, 1]$  gives us a subcube one dimension higher in our given  $(n + 1)$ -dimensional cube, namely having dimension  $(n + 1) - k$ , as desired. On the other hand, if the literal  $x_{n+1}$  (or its complement) is part of the product, then the product of the remaining  $k - 1$  literals corresponds to a subcube of dimension  $n - (k - 1) = (n + 1) - k$  in the  $n$ -dimensional cube, and that slice, at either the 1-end or the 0-end in the last variable, is the desired subcube.

## Supplementary Exercises

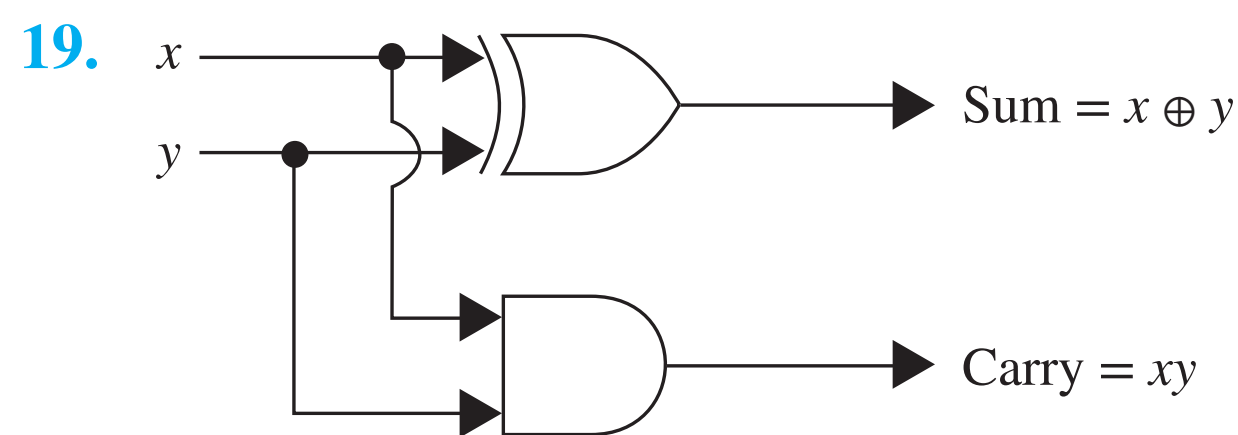
1. a)  $x = 0, y = 0, z = 0; x = 1, y = 1, z = 1$  b)  $x = 0, y = 0, z = 0; x = 0, y = 0, z = 1; x = 0, y = 1, z = 0; x = 1, y = 0, z = 1; x = 1, y = 1, z = 0; x = 1, y = 1, z = 1$  c) No values 3. a) Yes b) No c) No d) Yes 5.  $2^{2^{n-1}}$  7. a) If  $F(x_1, \dots, x_n) = 1$ , then  $(F + G)(x_1, \dots, x_n) = F(x_1, \dots, x_n) + G(x_1, \dots, x_n) = 1$  by the dominance law. Hence,  $F \leq F + G$ . b) If  $(FG)(x_1, \dots, x_n) = 1$ , then  $F(x_1, \dots, x_n) \cdot G(x_1, \dots, x_n) = 1$ . Hence,  $F(x_1, \dots, x_n) = 1$ . It follows that  $FG \leq F$ . 9. Because  $F(x_1, \dots, x_n) = 1$  implies that  $F(x_1, \dots, x_n) = 1$ ,  $\leq$  is reflexive. Suppose that  $F \leq G$  and  $G \leq F$ . Then  $F(x_1, \dots, x_n) = 1$  if and only if  $G(x_1, \dots, x_n) = 1$ . This implies that  $F = G$ . Hence,  $\leq$  is antisymmetric. Suppose that  $F \leq G \leq H$ . Then if  $F(x_1, \dots, x_n) = 1$ , it follows that  $G(x_1, \dots, x_n) = 1$ , which implies that  $H(x_1, \dots, x_n) = 1$ . Hence,  $F \leq H$ , so  $\leq$  is transitive. 11. a)  $x = 1, y = 0, z = 0$  b)  $x = 1, y = 0, z = 0$  c)  $x = 1, y = 0, z = 0$



13.

| $x$ | $y$ | $x \odot y$ | $x \oplus y$ | $\overline{(x \oplus y)}$ |
|-----|-----|-------------|--------------|---------------------------|
| 1   | 1   | 1           | 0            | 1                         |
| 1   | 0   | 0           | 1            | 0                         |
| 0   | 1   | 0           | 1            | 0                         |
| 0   | 0   | 1           | 0            | 1                         |

15. Yes, as a truth table shows    17. a) 6   b) 5   c) 5   d) 6



21.  $x_3 + x_2\bar{x}_1$     23. Suppose it were with weights  $a$  and  $b$ . Then there would be a real number  $T$  such that  $xa + yb \geq T$  for (1,0) and (0,1), but with  $xa + yb < T$  for (0,0) and (1,1). Hence,  $a \geq T, b \geq T, 0 < T$ , and  $a + b < T$ . Thus,  $a$  and  $b$  are positive, which implies that  $a + b > a \geq T$ , a contradiction.

## CHAPTER 13

### Section 13.1

1. a) sentence  $\Rightarrow$  noun phrase    intransitive verb phrase  
 $\Rightarrow$  article    adjective noun    intransitive verb phrase  $\Rightarrow$   
 article    adjective    noun    intransitive verb  $\Rightarrow \dots$

(after 3 steps)  $\dots \Rightarrow$  *the happy hare runs*.

b) sentence  $\Rightarrow$  noun phrase    intransitive verb phrase  
 $\Rightarrow$  article    adjective    noun    intransitive verb phrase  
 $\Rightarrow$  article    adjective    noun    intransitive verb  
 adverb... (after 4 steps)  $\dots \Rightarrow$  *the sleepy tortoise runs quickly*

c) sentence  $\Rightarrow$  noun phrase    transitive verb phrase  
 noun phrase  $\Rightarrow$  article    noun    transitive verb phrase  
 noun phrase  $\Rightarrow$  article    noun    transitive verb    noun  
 phrase  $\Rightarrow$  article    noun    transitive verb    article  
 noun  $\Rightarrow \dots$  (after 4 steps)  $\dots \Rightarrow$  *the tortoise passes the hare*  
 d) sentence  $\Rightarrow$  noun phrase    transitive verb phrase  
 noun phrase  $\Rightarrow$  article    adjective    noun    transitive  
 verb phrase    noun phrase  $\Rightarrow$  article    adjective    noun  
 transitive verb    noun phrase  $\Rightarrow$  article    adjective  
 noun    transitive verb    article    adjective    noun  
 $\Rightarrow \dots$  (after 6 steps)  $\dots \Rightarrow$  *the sleepy hare passes the happy tortoise*

3. The only way to get a noun, such as *tortoise*, at the end is to have a noun phrase at the end, which can be achieved only via the production **sentence  $\rightarrow$  noun phrase    transitive verb phrase    noun phrase**. However, **transitive verb phrase  $\rightarrow$  transitive verb  $\rightarrow$  passes**, and this sentence does not contain *passes*.

5. a)  $S \Rightarrow 1A \Rightarrow 10B \Rightarrow 101A \Rightarrow 1010B \Rightarrow 10101$     b) Because of the productions in this grammar, every 1 must be followed by a 0 unless it occurs at the end of the string.    c) All strings consisting of a 0 or a 1 followed by one or more repetitions of 01

7.  $S \Rightarrow 0S1 \Rightarrow 00S11 \Rightarrow 000S111 \Rightarrow 000111$

9. a)  $S \Rightarrow 0S \Rightarrow 00S \Rightarrow 00S1 \Rightarrow 00S11 \Rightarrow 00S111 \Rightarrow 00S1111 \Rightarrow 001111$     b)  $S \Rightarrow 0S \Rightarrow 00S \Rightarrow 001A \Rightarrow 0011A \Rightarrow 00111A \Rightarrow 001111$

11.  $S \Rightarrow 0SAB \Rightarrow 00SABAB \Rightarrow 00ABAB \Rightarrow 00AABB \Rightarrow 001ABB \Rightarrow 0011BB \Rightarrow 00112B \Rightarrow 001122$

13. a)  $S \rightarrow 0, S \rightarrow 1, S \rightarrow 11$     b)  $S \rightarrow 1S, S \rightarrow \lambda$     c)  $S \rightarrow 0A1, A \rightarrow 1A, A \rightarrow 0A, A \rightarrow \lambda$     d)  $S \rightarrow 0A, A \rightarrow 11A, A \rightarrow \lambda$

15. a)  $S \rightarrow 00S, S \rightarrow \lambda$     b)  $S \rightarrow 10A, A \rightarrow 00A, A \rightarrow \lambda$     c)  $S \rightarrow AAS, S \rightarrow BBS, AB \rightarrow BA, BA \rightarrow AB, S \rightarrow \lambda, A \rightarrow 0, B \rightarrow 1$

d)  $S \rightarrow 0000000000A, A \rightarrow 0A, A \rightarrow \lambda$     e)  $S \rightarrow AS, S \rightarrow ABS, S \rightarrow A, AB \rightarrow BA, BA \rightarrow AB, A \rightarrow 0, B \rightarrow 1$

f)  $S \rightarrow ABS, S \rightarrow \lambda, AB \rightarrow BA, BA \rightarrow AB, A \rightarrow 0, B \rightarrow 1$     g)  $S \rightarrow ABS, S \rightarrow T, S \rightarrow U, T \rightarrow AT, T \rightarrow A, U \rightarrow BU, U \rightarrow B, AB \rightarrow BA, BA \rightarrow AB, A \rightarrow 0, B \rightarrow 1$

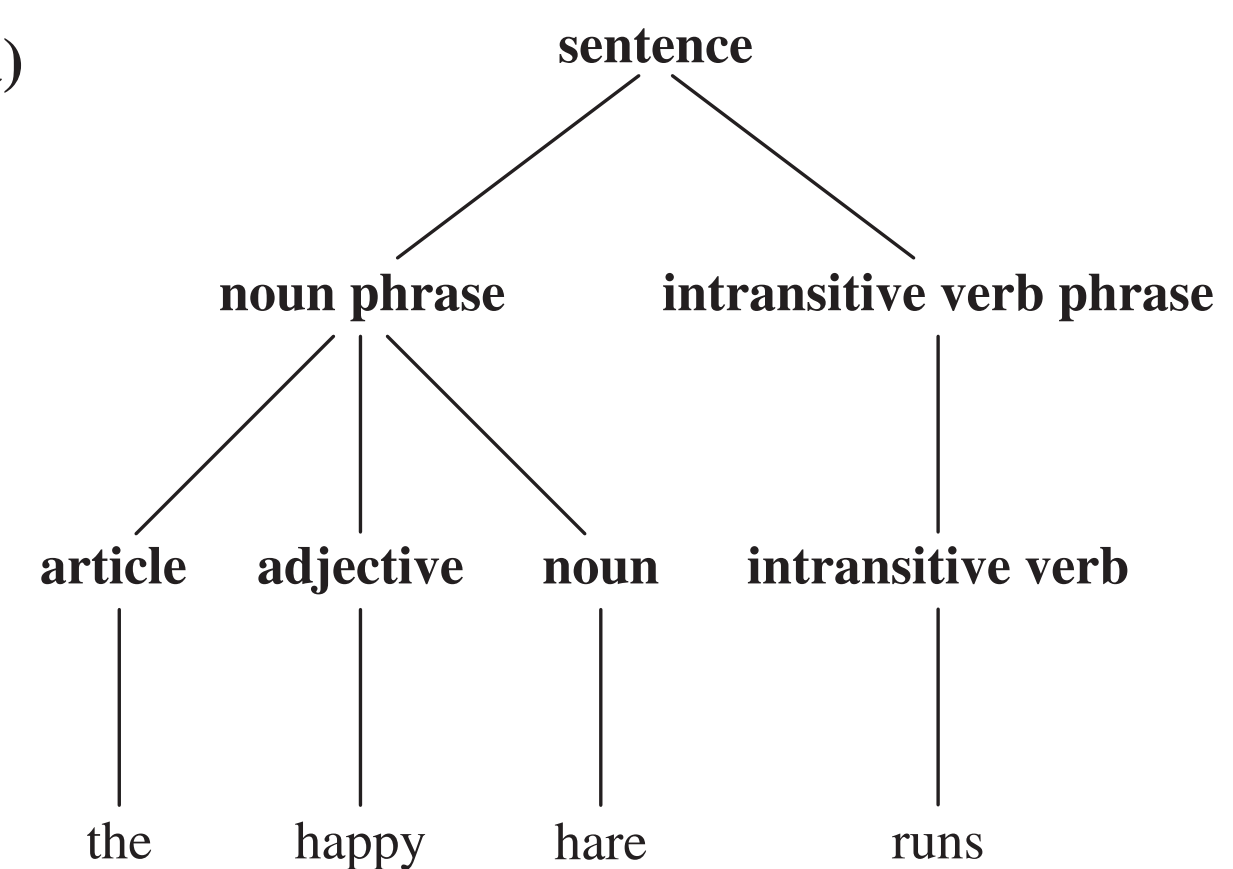
17. a)  $S \rightarrow 0S, S \rightarrow \lambda$     b)  $S \rightarrow A0, A \rightarrow 1A, A \rightarrow \lambda$     c)  $S \rightarrow 000S, S \rightarrow \lambda$

19. a) Type 2, not type 3    b) Type 3    c) Type 0, not type 1    d) Type 2, not type 3    e) Type 2, not type 3    f) Type 0, not type 1    g) Type 3

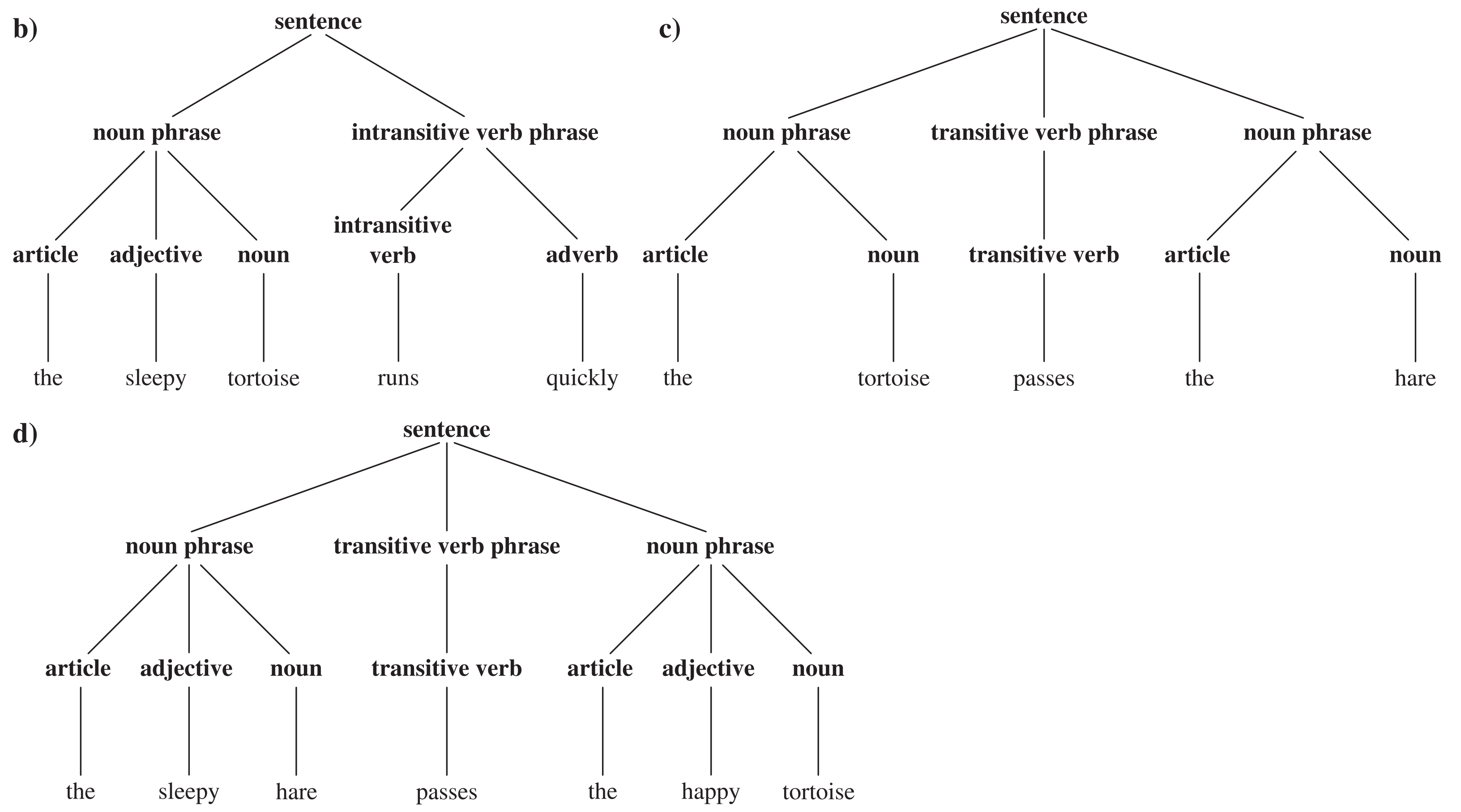
h) Type 0, not type 1    i) Type 2, not type 3    j) Type 2, not type 3

21. Let  $S_1$  and  $S_2$  be the start symbols of  $G_1$  and  $G_2$ , respectively. Let  $S$  be a new start symbol. a) Add  $S$  and productions  $S \rightarrow S_1$  and  $S \rightarrow S_2$ .    b) Add  $S$  and production  $S \rightarrow S_1S_2$ .    c) Add  $S$  and production  $S \rightarrow \lambda$  and  $S \rightarrow S_1S$ .

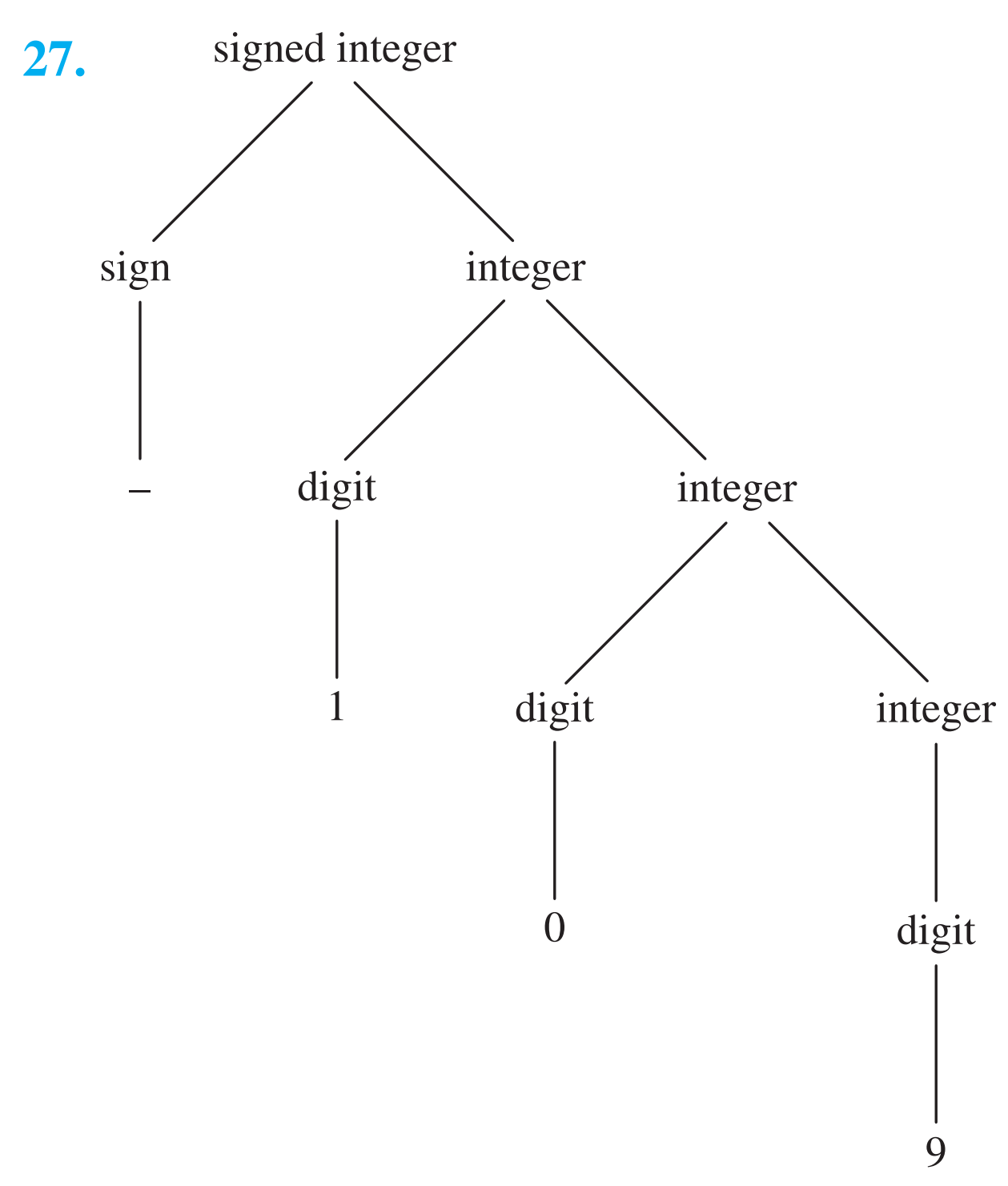
23. a)







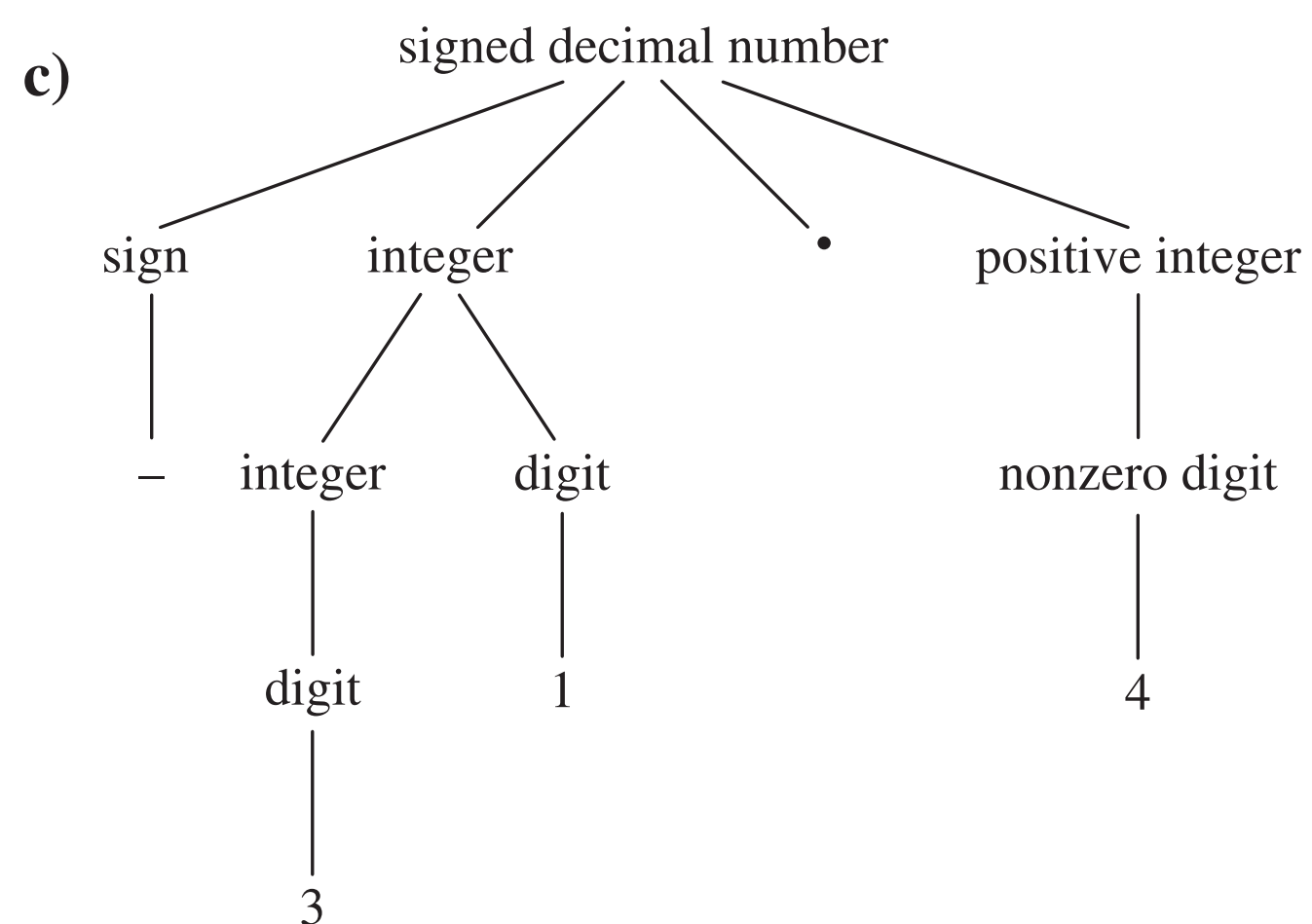
25. a) Yes b) No c) Yes d) No





- 29. a)**  $S \rightarrow \langle sign \rangle \langle integer \rangle$   
 $S \rightarrow \langle sign \rangle \langle integer \rangle . \langle positive integer \rangle$   
 $\langle sign \rangle \rightarrow +$   
 $\langle sign \rangle \rightarrow -$   
 $\langle integer \rangle \rightarrow \langle digit \rangle$   
 $\langle integer \rangle \rightarrow \langle integer \rangle \langle digit \rangle$   
 $\langle digit \rangle \rightarrow i, i = 1, 2, 3, 4, 5, 6, 7, 8, 9, 0$   
 $\langle positive integer \rangle \rightarrow \langle integer \rangle \langle nonzero digit \rangle$   
 $\langle positive integer \rangle \rightarrow \langle nonzero digit \rangle \langle integer \rangle$   
 $\langle positive integer \rangle \rightarrow \langle integer \rangle \langle nonzero digit \rangle$   
 $\langle integer \rangle$   
 $\langle positive integer \rangle \rightarrow \langle nonzero digit \rangle$   
 $\langle nonzero digit \rangle \rightarrow i, i = 1, 2, 3, 4, 5, 6, 7, 8, 9$

- b)**  $\langle signed decimal number \rangle ::= \langle sign \rangle \langle integer \rangle \mid$   
 $\langle sign \rangle \langle integer \rangle . \langle positive integer \rangle$   
 $\langle sign \rangle ::= + \mid -$   
 $\langle integer \rangle ::= \langle digit \rangle \mid \langle integer \rangle \langle digit \rangle$   
 $\langle digit \rangle ::= 0 \mid 1 \mid 2 \mid 3 \mid 4 \mid 5 \mid 6 \mid 7 \mid 8 \mid 9$   
 $\langle nonzero digit \rangle ::= 1 \mid 2 \mid 3 \mid 4 \mid 5 \mid 6 \mid 7 \mid 8 \mid 9$   
 $\langle positive integer \rangle ::= \langle integer \rangle \langle nonzero digit \rangle \mid$   
 $\langle nonzero digit \rangle \langle integer \rangle \mid \langle integer \rangle$   
 $\langle nonzero integer \rangle \langle integer \rangle \mid \langle nonzero digit \rangle$



- 31. a)**  $\langle identifier \rangle ::= \langle lcletter \rangle \mid \langle identifier \rangle \langle lcletter \rangle$   
 $\langle lcletter \rangle ::= a \mid b \mid c \mid \dots \mid z$

- b)**  $\langle identifier \rangle ::= \langle lcletter \rangle \langle lcletter \rangle \langle lcletter \rangle \mid \langle lcletter \rangle \langle lcletter \rangle \langle lcletter \rangle \langle lcletter \rangle \mid$   
 $\langle lcletter \rangle \langle lcletter \rangle \langle lcletter \rangle \langle lcletter \rangle \langle lcletter \rangle \mid$   
 $\langle lcletter \rangle \langle lcletter \rangle \langle lcletter \rangle \langle lcletter \rangle \langle lcletter \rangle \langle lcletter \rangle$

- $\langle lcletter \rangle ::= a \mid b \mid c \mid \dots \mid z$   
**c)**  $\langle identifier \rangle ::= \langle ucletter \rangle \mid \langle ucletter \rangle \langle letter \rangle \mid \langle ucletter \rangle \langle letter \rangle \langle letter \rangle \mid$   
 $\langle ucletter \rangle \langle letter \rangle \langle letter \rangle \langle letter \rangle \mid \langle ucletter \rangle \langle letter \rangle \langle letter \rangle \langle letter \rangle \langle letter \rangle \mid$   
 $\langle ucletter \rangle \langle letter \rangle \langle letter \rangle \langle letter \rangle \langle letter \rangle \langle letter \rangle$   
 $\langle letter \rangle ::= \langle lcletter \rangle \mid \langle ucletter \rangle$   
 $\langle lcletter \rangle ::= a \mid b \mid c \mid \dots \mid z$   
 $\langle ucletter \rangle ::= A \mid B \mid C \mid \dots \mid Z$



- d)**  $\langle identifier \rangle ::= \langle lcletter \rangle \langle digitorus \rangle \langle alphanumeric \rangle \langle alphanumeric \rangle \langle alphanumeric \rangle \mid$   
 $\langle lcletter \rangle \langle digitorus \rangle \langle alphanumeric \rangle \langle alphanumeric \rangle \langle alphanumeric \rangle \langle alphanumeric \rangle$   
 $\langle digitorus \rangle ::= \langle digit \rangle \mid -$   
 $\langle alphanumeric \rangle ::= \langle letter \rangle \mid \langle digit \rangle$   
 $\langle letter \rangle ::= \langle lcletter \rangle \mid \langle ucletter \rangle$   
 $\langle lcletter \rangle ::= a \mid b \mid c \mid \dots \mid z$   
 $\langle ucletter \rangle ::= A \mid B \mid C \mid \dots \mid Z$   
 $\langle digit \rangle ::= 0 \mid 1 \mid 2 \mid \dots \mid 9$
- 33.**  $\langle identifier \rangle ::= \langle letterorus \rangle \mid \langle identifier \rangle \langle symbol \rangle$   
 $\langle letterorus \rangle ::= \langle letter \rangle \mid -$   
 $\langle symbol \rangle ::= \langle letterorus \rangle \mid \langle digit \rangle$   
 $\langle letter \rangle ::= \langle lcletter \rangle \mid \langle ucletter \rangle$   
 $\langle lcletter \rangle ::= a \mid b \mid c \mid \dots \mid z$   
 $\langle ucletter \rangle ::= A \mid B \mid C \mid \dots \mid Z$   
 $\langle digit \rangle ::= 0 \mid 1 \mid 2 \mid \dots \mid 9$
- 35.**  $numeral ::= sign? nonzerodigit digit^* decimal? \mid sign? 0 decimal?$   
 $sign ::= + \mid -$   
 $nonzerodigit ::= 1 \mid 2 \mid \dots \mid 9$   
 $digit ::= 0 \mid nonzerodigit$   
 $decimal ::= .digit^*$
- 37.**  $identifier ::= letterorus symbol^*$   
 $letterorus ::= letter \mid -$   
 $symbol ::= letterorus \mid digit$   
 $letter ::= lcletter \mid ucletter$   
 $lcletter ::= a \mid b \mid c \mid \dots \mid z$   
 $ucletter ::= A \mid B \mid C \mid \dots \mid Z$   
 $digit ::= 0 \mid 1 \mid 2 \mid \dots \mid 9$
- 39. a)**  $\langle expression \rangle$   
 $\langle term \rangle \langle term \rangle \langle addOperator \rangle$   
 $\langle factor \rangle \langle factor \rangle \langle factor \rangle \langle mulOperator \rangle \langle addOperator \rangle$   
 $\langle identifier \rangle \langle identifier \rangle \langle identifier \rangle \langle mulOperator \rangle \langle addOperator \rangle$   
 $a b c * +$
- b)** Not generated
- c)**  $\langle expression \rangle$   
 $\langle term \rangle$   
 $\langle factor \rangle \langle factor \rangle \langle mulOperator \rangle$   
 $\langle expression \rangle \langle factor \rangle \langle mulOperator \rangle$   
 $\langle term \rangle \langle term \rangle \langle addOperator \rangle \langle factor \rangle \langle mulOperator \rangle$   
 $\langle factor \rangle \langle factor \rangle \langle addOperator \rangle \langle factor \rangle \langle mulOperator \rangle$   
 $\langle identifier \rangle \langle identifier \rangle \langle addOperator \rangle \langle identifier \rangle \langle mulOperator \rangle$   
 $x y - z *$
- d)**  $\langle expression \rangle$   
 $\langle term \rangle$   
 $\langle factor \rangle \langle factor \rangle \langle mulOperator \rangle$   
 $\langle factor \rangle \langle expression \rangle \langle mulOperator \rangle$   
 $\langle factor \rangle \langle term \rangle \langle mulOperator \rangle$   
 $\langle factor \rangle \langle factor \rangle \langle factor \rangle \langle mulOperator \rangle \langle mulOperator \rangle$   
 $\langle factor \rangle \langle factor \rangle \langle expression \rangle \langle mulOperator \rangle \langle mulOperator \rangle$   
 $\langle factor \rangle \langle factor \rangle \langle term \rangle \langle term \rangle \langle addOperator \rangle \langle mulOperator \rangle \langle mulOperator \rangle$   
 $\langle factor \rangle \langle factor \rangle \langle factor \rangle \langle factor \rangle \langle addOperator \rangle \langle mulOperator \rangle \langle mulOperator \rangle$   
 $\langle identifier \rangle \langle identifier \rangle \langle identifier \rangle \langle identifier \rangle \langle addOperator \rangle \langle mulOperator \rangle \langle mulOperator \rangle$   
 $w x y z - * /$



**e)**  $\langle expression \rangle$   
 $\langle term \rangle$   
 $\langle factor \rangle \langle factor \rangle \langle mulOperator \rangle$   
 $\langle factor \rangle \langle expression \rangle \langle mulOperator \rangle$   
 $\langle factor \rangle \langle term \rangle \langle term \rangle \langle addOperator \rangle \langle mulOperator \rangle$   
 $\langle factor \rangle \langle factor \rangle \langle factor \rangle \langle addOperator \rangle \langle mulOperator \rangle$   
 $\langle identifier \rangle \langle identifier \rangle \langle identifier \rangle \langle addOperator \rangle \langle mulOperator \rangle$   
 $a d e - *$

**41. a)** Not generated

**b)**  $\langle expression \rangle$   
 $\langle term \rangle \langle addOperator \rangle \langle term \rangle$   
 $\langle factor \rangle \langle mulOperator \rangle \langle factor \rangle \langle addOperator \rangle \langle factor \rangle \langle mulOperator \rangle \langle factor \rangle$   
 $\langle identifier \rangle \langle mulOperator \rangle \langle identifier \rangle \langle addOperator \rangle \langle identifier \rangle \langle mulOperator \rangle \langle identifier \rangle$   
 $a/b + c/d$

**c)**  $\langle expression \rangle$   
 $\langle term \rangle$   
 $\langle factor \rangle \langle mulOperator \rangle \langle factor \rangle$   
 $\langle factor \rangle \langle mulOperator \rangle (\langle expression \rangle)$   
 $\langle factor \rangle \langle mulOperator \rangle (\langle term \rangle \langle addOperator \rangle \langle term \rangle)$   
 $\langle factor \rangle \langle mulOperator \rangle (\langle factor \rangle \langle addOperator \rangle \langle factor \rangle)$   
 $\langle identifier \rangle \langle mulOperator \rangle (\langle identifier \rangle \langle addOperator \rangle \langle identifier \rangle)$   
 $m * (n + p)$

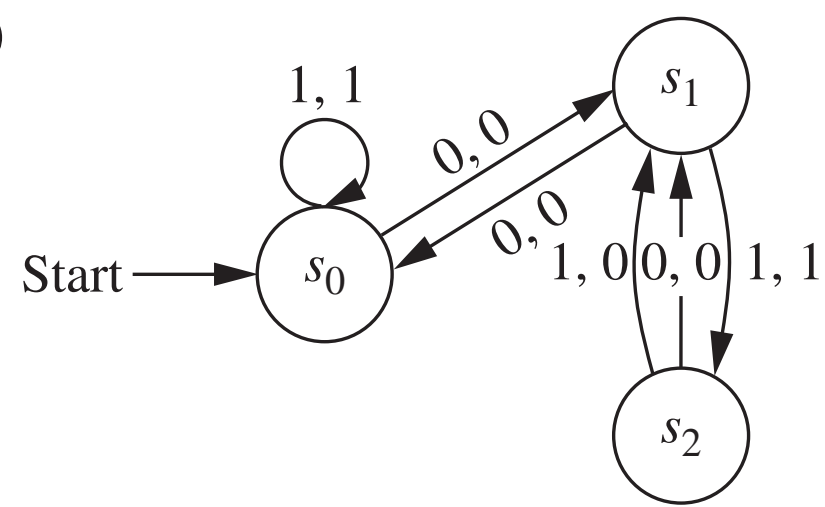
**d)** Not generated

**e)**  $\langle expression \rangle$   
 $\langle term \rangle$   
 $\langle factor \rangle \langle mulOperator \rangle \langle factor \rangle$   
 $(\langle expression \rangle) \langle mulOperator \rangle (\langle expression \rangle)$   
 $(\langle term \rangle \langle addOperator \rangle \langle term \rangle) \langle mulOperator \rangle (\langle term \rangle \langle addOperator \rangle \langle term \rangle)$   
 $(\langle factor \rangle \langle addOperator \rangle \langle factor \rangle) \langle mulOperator \rangle (\langle factor \rangle \langle addOperator \rangle \langle factor \rangle)$   
 $(\langle identifier \rangle \langle addOperator \rangle \langle identifier \rangle) \langle mulOperator \rangle (\langle identifier \rangle \langle addOperator \rangle \langle identifier \rangle)$   
 $(m + n) * (p - q)$

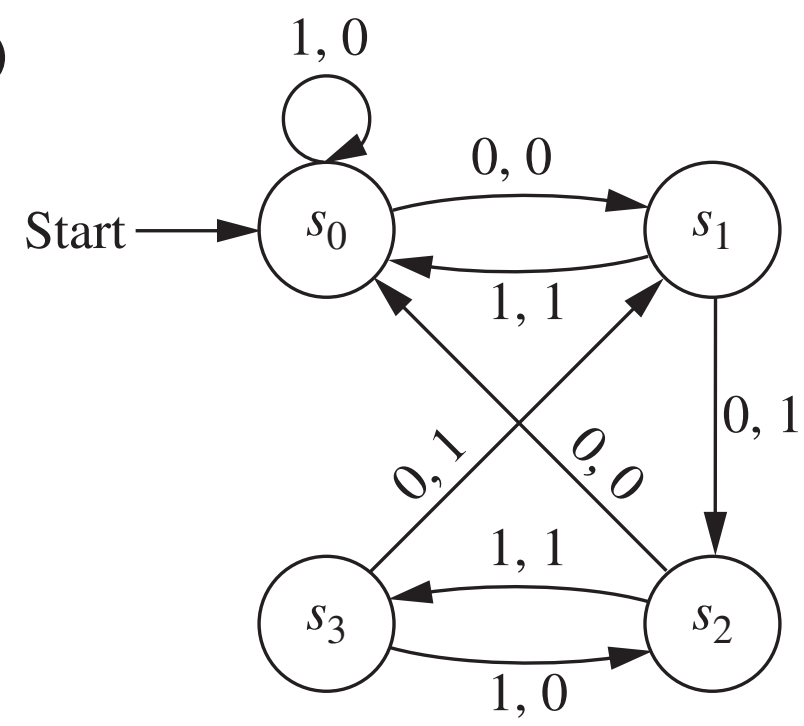


## Section 13.2

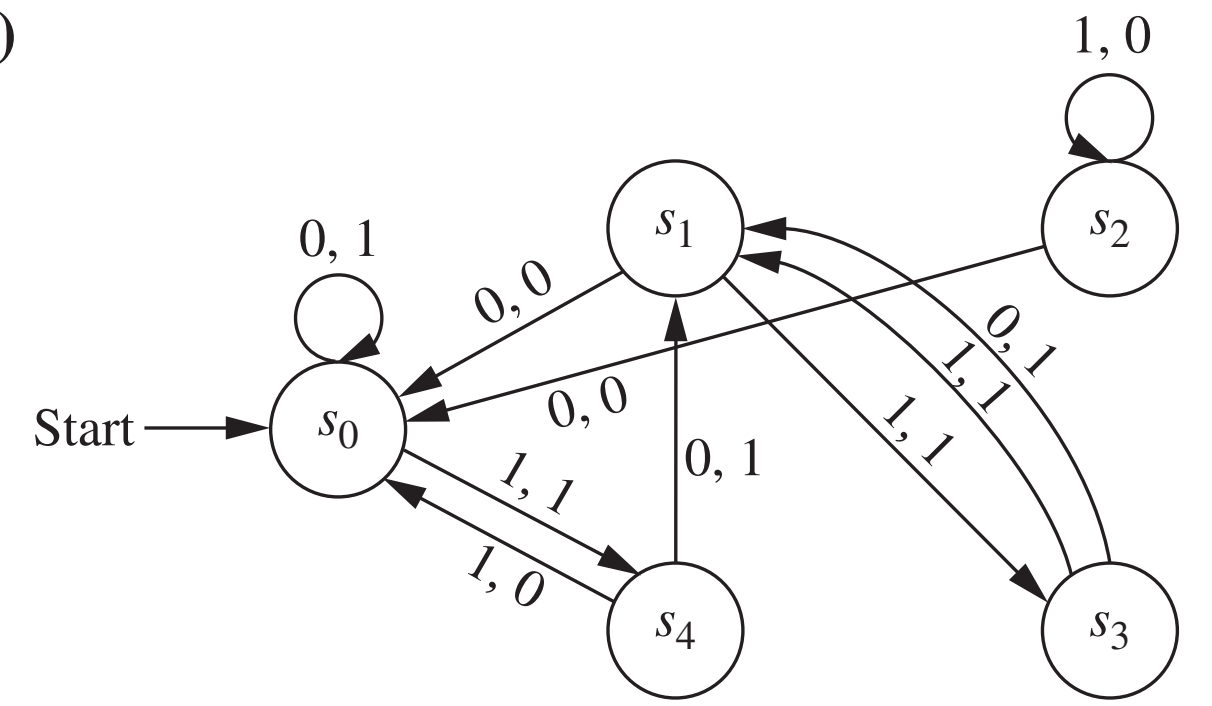
1. a)



b)

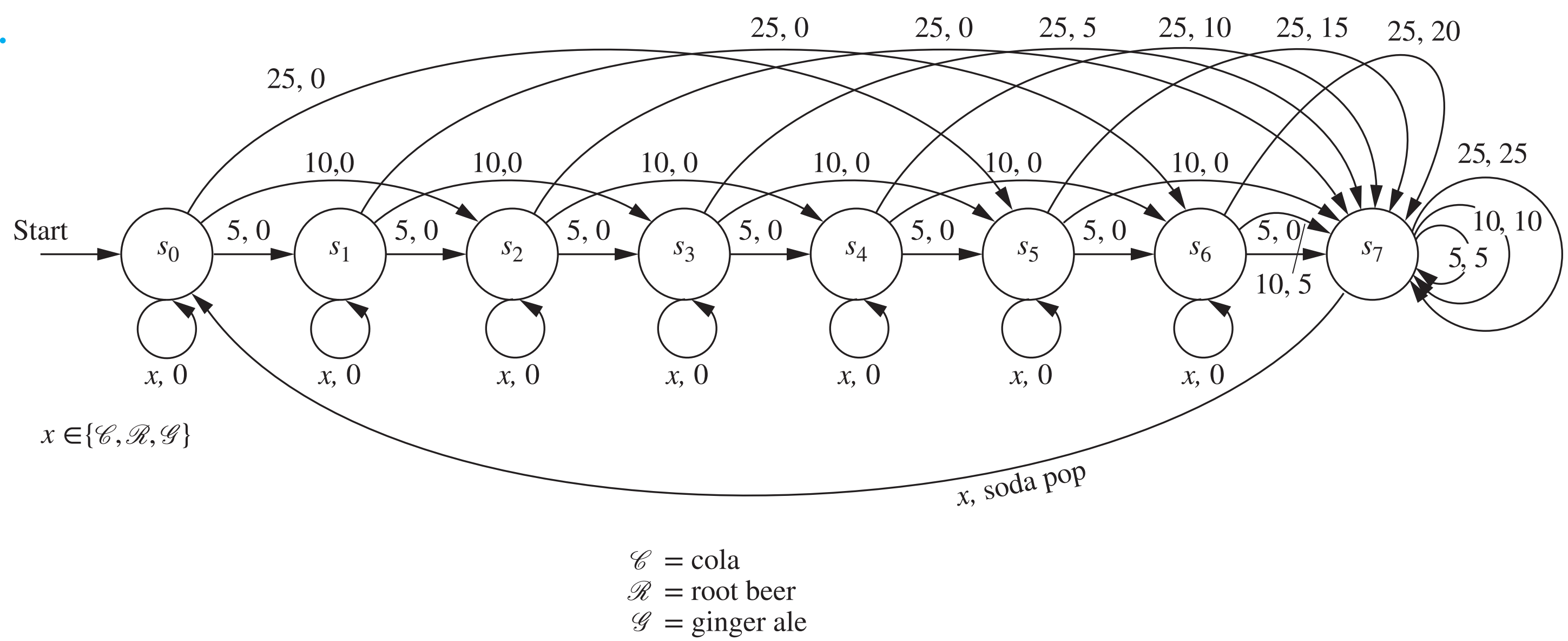


c)

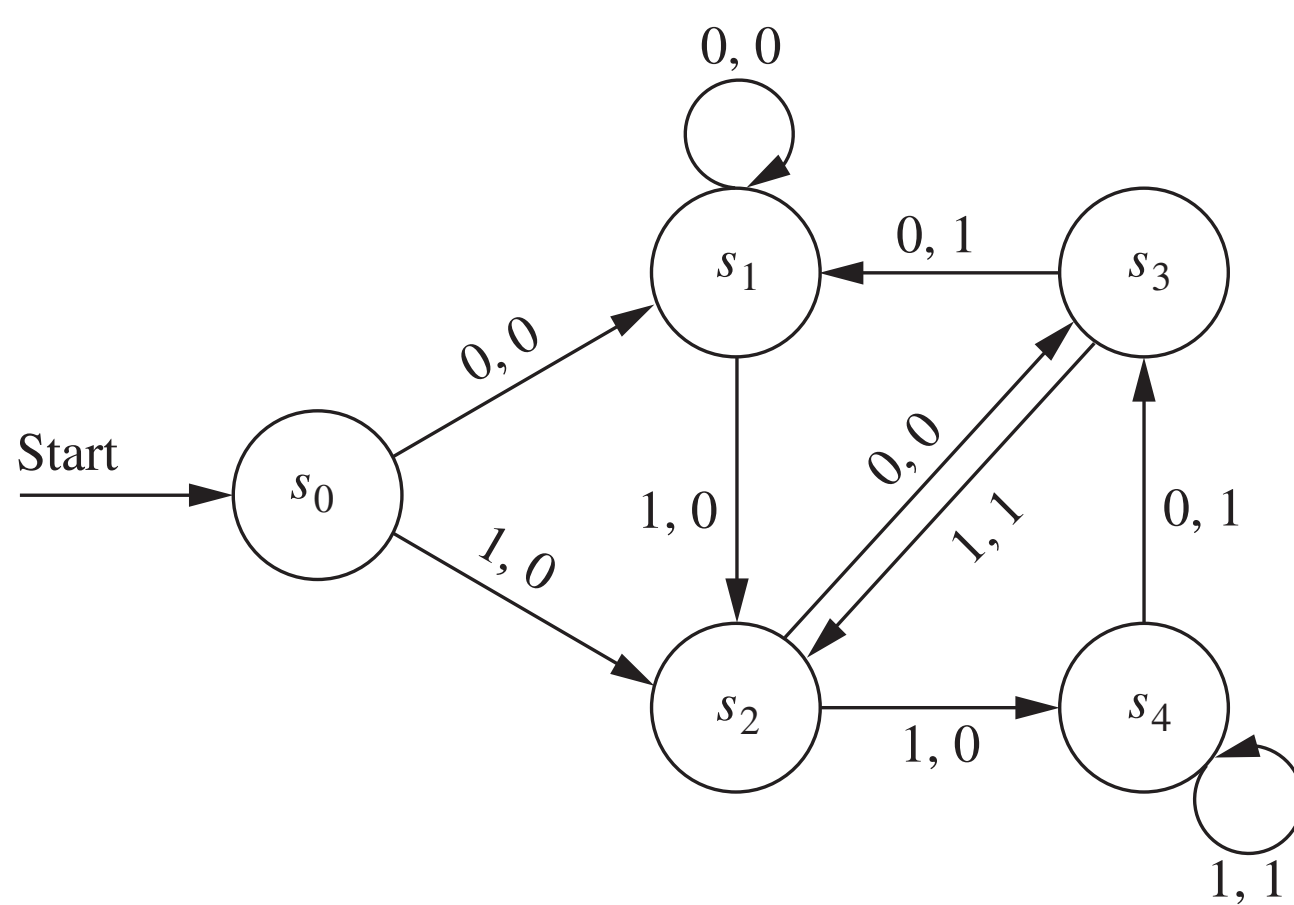


3. a) 01010 b) 01000 c) 11011 5. a) 1100 b) 00110110 c) 1111111111

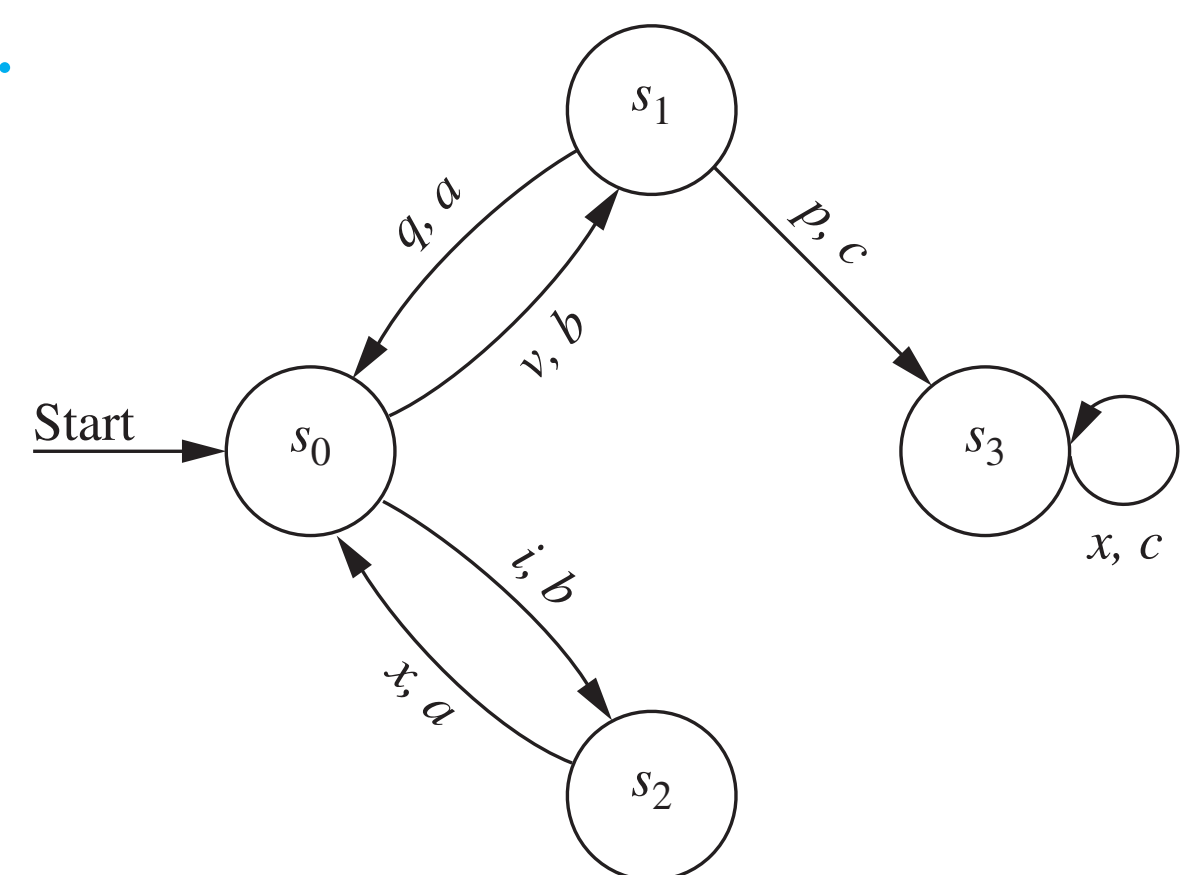
7.



9.



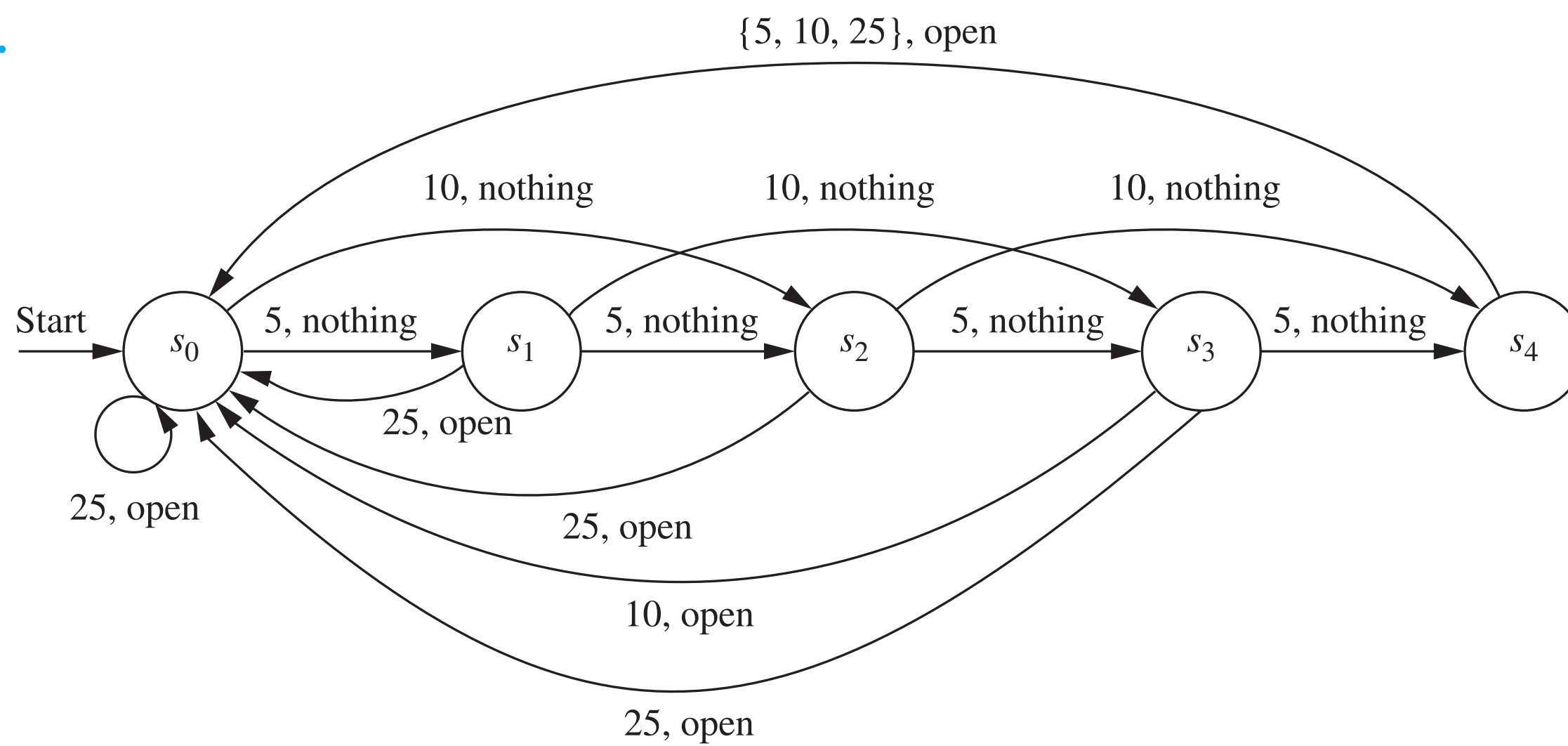
11.



$v$  = Valid ID  
 $i$  = Invalid ID  
 $p$  = Valid password  
 $q$  = Invalid password  
 $a$  = "Enter user ID"  
 $b$  = "Enter password"  
 $c$  = Prompt  
 $x$  = Any input

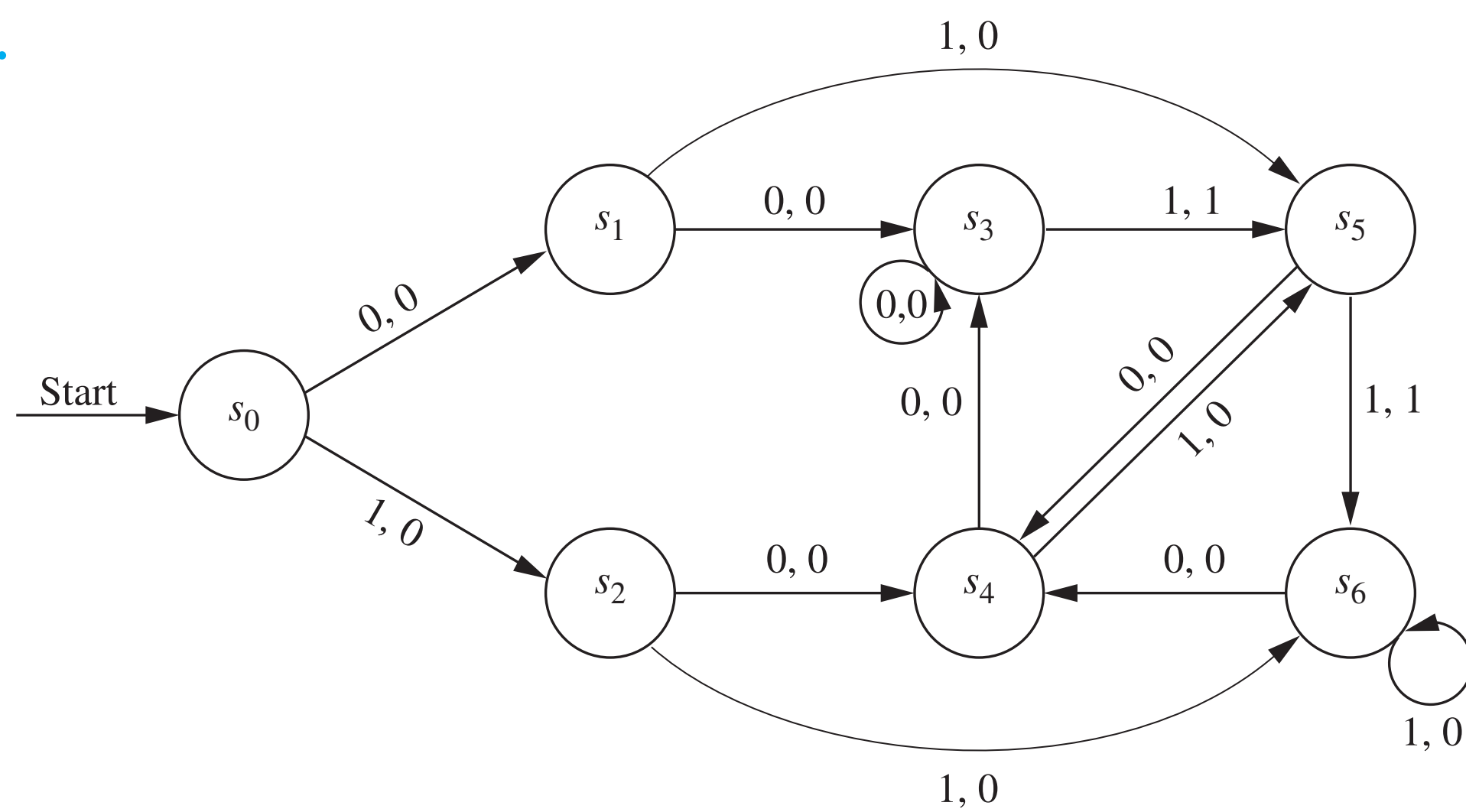


13.



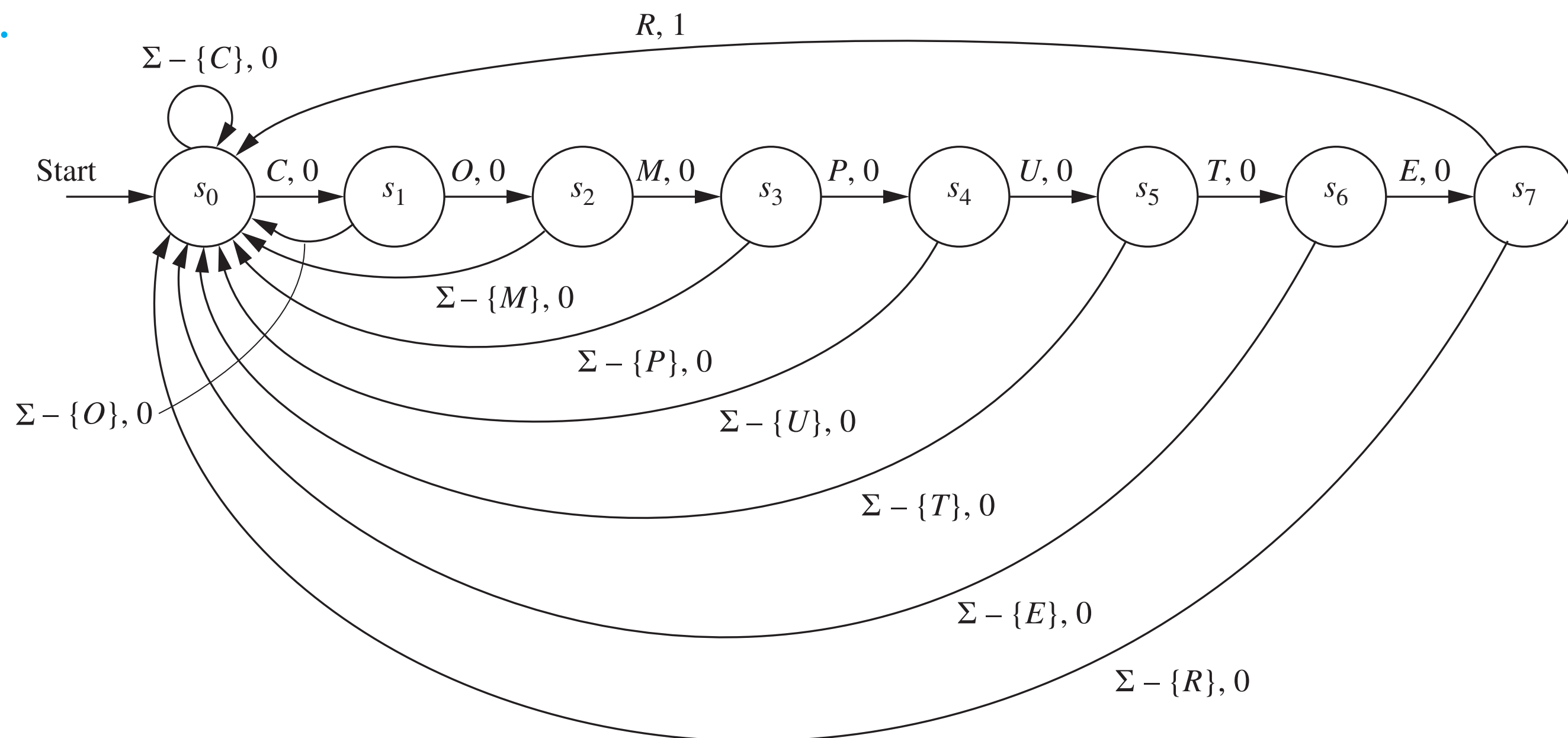
15. Let  $s_0$  be the start state and let  $s_1$  be the state representing a successful call. From  $s_0$ , inputs of 2, 3, 4, 5, 6, 7, or 8 send the machine back to  $s_0$  with output of an error message for the user. From  $s_0$  an input of 0 sends the machine to state  $s_1$ , with the output being that the 0 is sent to the network. From  $s_0$  an input of 9 sends the machine to state  $s_2$  with no output; from there an input of 1 sends the machine to state  $s_3$  with no output; from there an input of 1 sends the machine to state  $s_1$  with the output being that the 911 is sent to the network. All other inputs while in states  $s_2$  or  $s_3$  send the machine back to  $s_0$  with output of an error message for the user. From  $s_0$  an input of 1 sends the machine to state  $s_4$  with no output; from  $s_4$  an input of 2 sends the machine to state  $s_5$  with no output; and this path continues in a similar manner to the 911 path, looking next for 1, then 2, then any seven digits, at which point the machine goes to state  $s_1$  with the output being that the ten-digit input is sent to the network. Any “incorrect” input while in states  $s_5$  or  $s_6$  (that is, anything except a 1 while in  $s_5$  or a 2 while in  $s_6$ ) sends the machine back to  $s_0$  with output of an error message for the user. Similarly, from  $s_4$  an input of 8 followed by appropriate successors drives us eventually to  $s_1$ , but inappropriate outputs drive us back to  $s_0$  with an error message. Also, inputs while in state  $s_4$  other than 2 or 8 send the machine back to state  $s_0$  with output of an error message for the user.

17.





19.



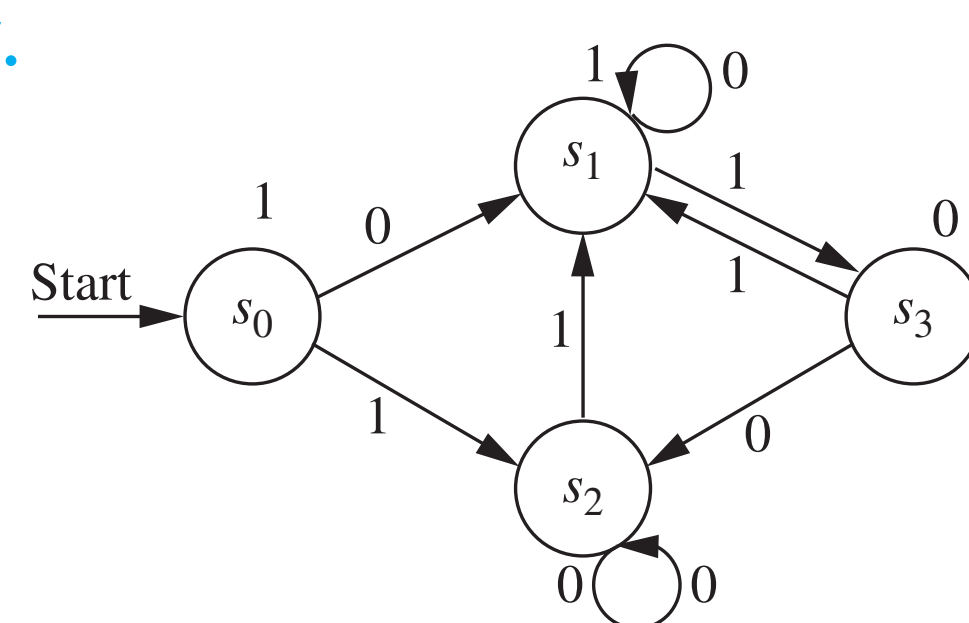
21.

| State | $f$   |       | $g$ |
|-------|-------|-------|-----|
|       | Input |       |     |
|       | 0     | 1     |     |
| $s_0$ | $s_1$ | $s_2$ | 1   |
| $s_1$ | $s_1$ | $s_0$ | 1   |
| $s_2$ | $s_1$ | $s_2$ | 0   |

23.

- a) 11111  
b) 1000000  
c) 100011001100

25.



### Section 13.3

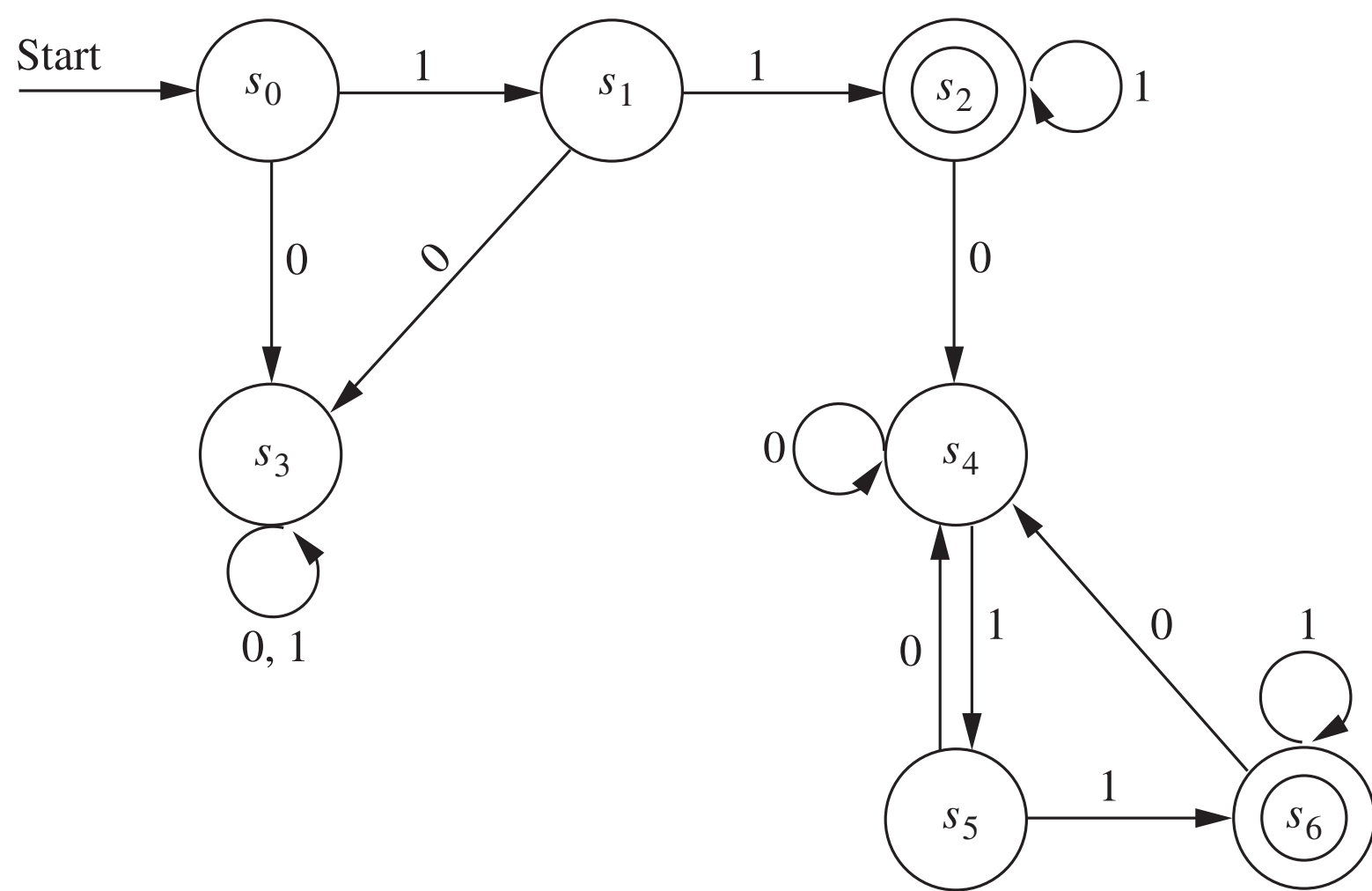
1. a) {000,001,1100,1101} b) {000,0011,010,0111}  
c) {00, 011, 110, 1111} d) {000000, 000001, 000100, 000101, 010000, 010001, 010100, 010101} 3.  $A = \{1, 101\}$ ,  $B = \{0, 11, 000\}$ ;  $A = \{10, 111, 1010, 1000, 10111, 101000\}$ ,  $B = \{\lambda\}$ ;  $A = \{\lambda, 10\}$ ,  $B = \{10, 111, 1000\}$  or  $A = \{\lambda\}$ ,  $B = \{10, 111, 1010, 1000, 10111, 101000\}$  5. a) The set of strings consisting of zero or more consecutive bit pairs 10 b) The set of strings consisting of all 1s such that the number of 1s is divisible by 3, including the null string c) The set of strings in which every 1 is immediately preceded by a 0 d) The set of strings that begin and end with a 1 and have at least two 1s between every pair of 0s 7. A string is in  $A^*$  if and only if it is a concatenation of an arbitrary number of strings in  $A$ . Because each string in  $A$  is also in  $B$ , it follows that a string in  $A^*$  is also a concatenation of strings in  $B$ . Hence,  $A^* \subseteq B^*$ . 9. a) Yes b) Yes c) No d) No e) Yes f) Yes 11. a) Yes b) No c) Yes d) No 13. a) Yes b) Yes c) No d) No e) No f) No 15. We use structural induction on the input string  $y$ . The basis step considers  $y = \lambda$ , and for the inductive step we write  $y = wa$ , where  $w \in I^*$  and  $a \in I$ . For the basis step, we have  $xy = x$ , so we must show that  $f(s, x) = f(f(s, x), \lambda)$ . But part (i) of the definition of the extended transition function says that this is true. We then assume the inductive

hypothesis that the equation holds for  $w$  and prove that  $f(s, xwa) = f(f(s, x), wa)$ . By part (ii) of the definition, the left-hand side of this equation equals  $f(f(s, xw), a)$ . By the inductive hypothesis,  $f(s, xw) = f(f(s, x), w)$ , so  $f(f(s, xw), a) = f(f(f(s, x), w), a)$ . The right-hand side of our desired equality is, by part (ii) of the definition, also equal to  $f(f(f(s, x), w), a)$ , as desired.

17.  $\{0, 10, 11\}\{0, 1\}^*$  19.  $\{0^m 1^n \mid m \geq 0 \text{ and } n \geq 1\}$  21.  $\{\lambda\} \cup \{0\}\{1\}^*\{0\} \cup \{10, 11\}\{0, 1\}^* \cup \{0\}\{1\}^*\{01\}\{0, 1\}^* \cup \{0\}\{1\}^*\{00\}\{0\}^*\{1\}\{0, 1\}^*$  23. Let  $s_2$  be the only final state, and put transitions from  $s_2$  to itself on either input. Put a transition from the start state  $s_0$  to  $s_1$  on input 0, and a transition from  $s_1$  to  $s_2$  on input 1. Create state  $s_3$ , and have the other transitions from  $s_0$  and  $s_1$  (as well as both transitions from  $s_3$ ) lead to  $s_3$ . 25. Start state  $s_0$ , only final state  $s_3$ ; transitions from  $s_0$  to  $s_0$  on 0, from  $s_0$  to  $s_1$  on 1, from  $s_1$  to  $s_2$  on 0, from  $s_1$  to  $s_1$  on 1, from  $s_2$  to  $s_0$  on 0, from  $s_2$  to  $s_3$  on 1, from  $s_3$  to  $s_3$  on 0, from  $s_3$  to  $s_3$  on 1 27. Have five states, with only  $s_3$  final. For  $i = 0, 1, 2, 3$ , transition from  $s_i$  to itself on input 1 and to  $s_{i+1}$  on input 0. Both transitions from  $s_4$  are to itself. 29. Have four states, with only  $s_3$  final. For  $i = 0, 1, 2$ , transition from  $s_i$  to  $s_{i+1}$  on input 1 but back to  $s_0$  on input 0. Both transitions from  $s_3$  are to itself.

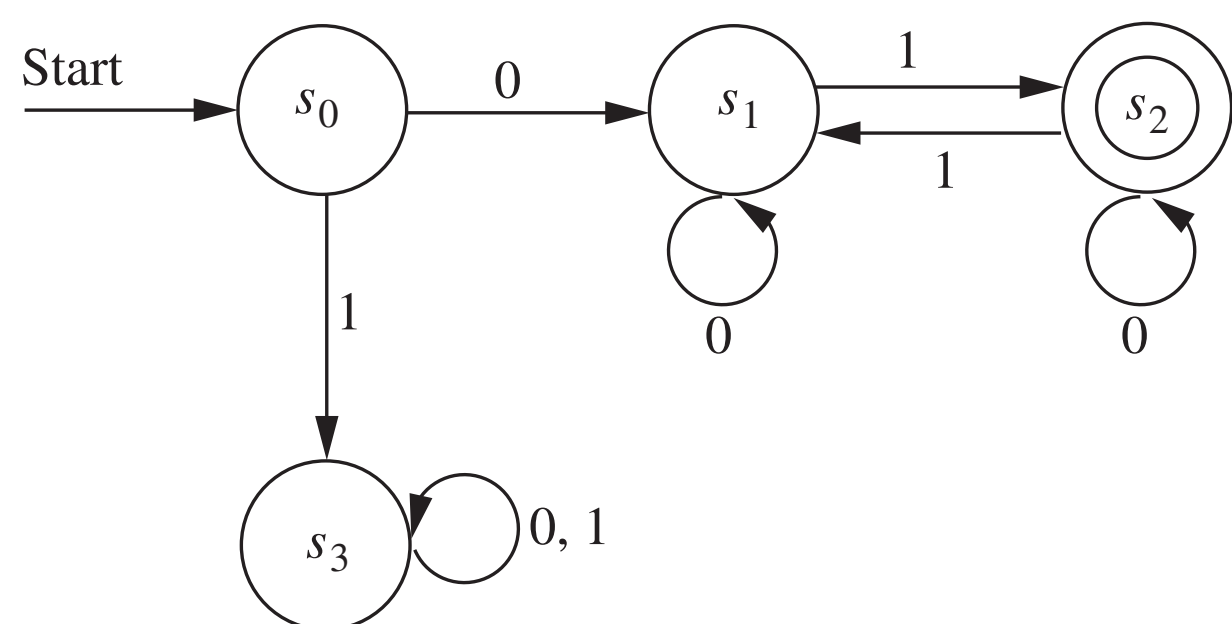


31.



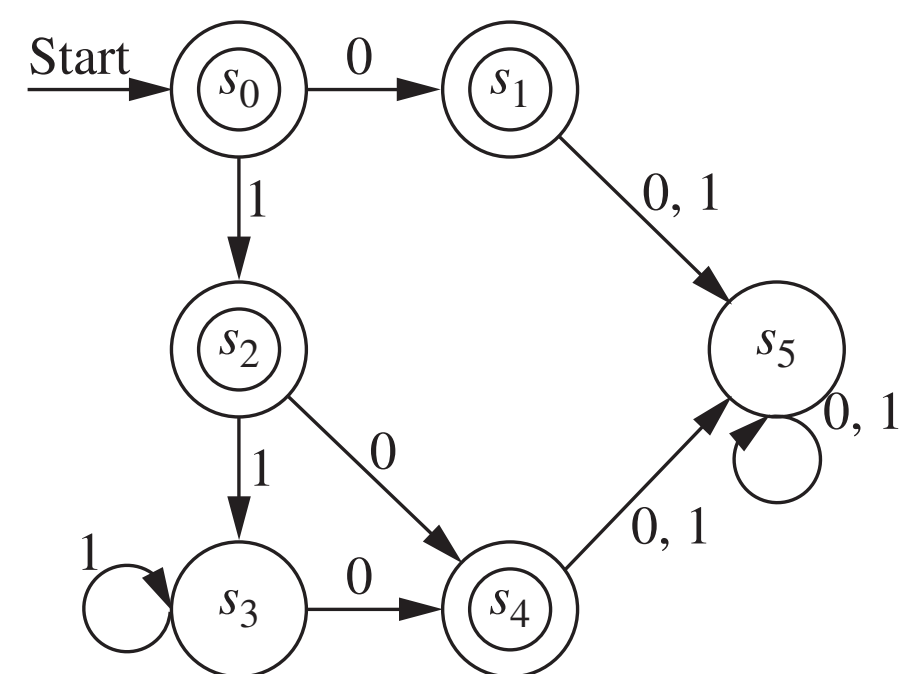
33. Start state  $s_0$ , only final state  $s_1$ ; transitions from  $s_0$  to  $s_0$  on 1, from  $s_0$  to  $s_1$  on 0, from  $s_1$  to  $s_1$  on 1; from  $s_1$  to  $s_0$  on 0

35.



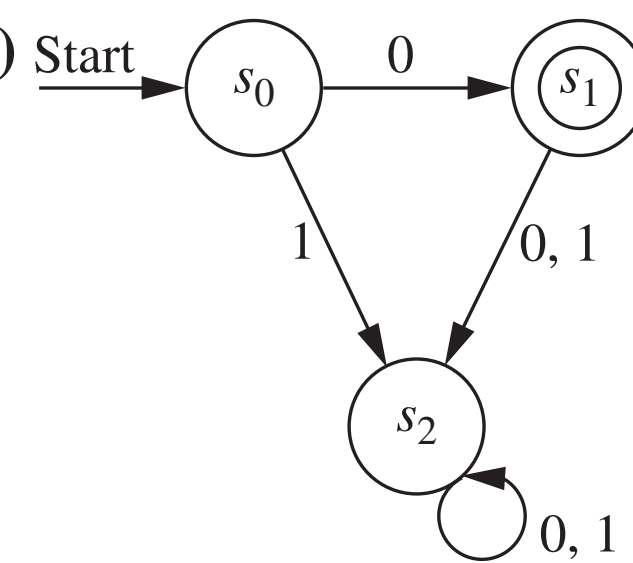
37. Suppose that such a machine exists, with start state  $s_0$  and other state  $s_1$ . Because the empty string is not in the language but some strings are accepted, we must have  $s_1$  as the only final state, with at least one transition from  $s_0$  to  $s_1$ . Because the string 0 is not in the language, the transition from  $s_0$  on input 0 must be to itself, so the transition from  $s_0$  on input 1 must be to  $s_1$ . But this contradicts the fact that 1 is not in the language. 39. Change each final state to a nonfinal state and vice versa. 41. Same machine as in Exercise 25, but with  $s_0$ ,  $s_1$ , and  $s_2$  as the final states 43.  $\{0,01,11\}$  45.  $\{\lambda, 0\} \cup \{0^m 1^n \mid m \geq 1, n \geq 1\}$  47.  $\{10^n \mid n \geq 0\} \cup \{10^n 10^m \mid n, m \geq 0\}$  49. The union of the set of all strings that start with a 0 and the set of all strings that have no 0s

51.

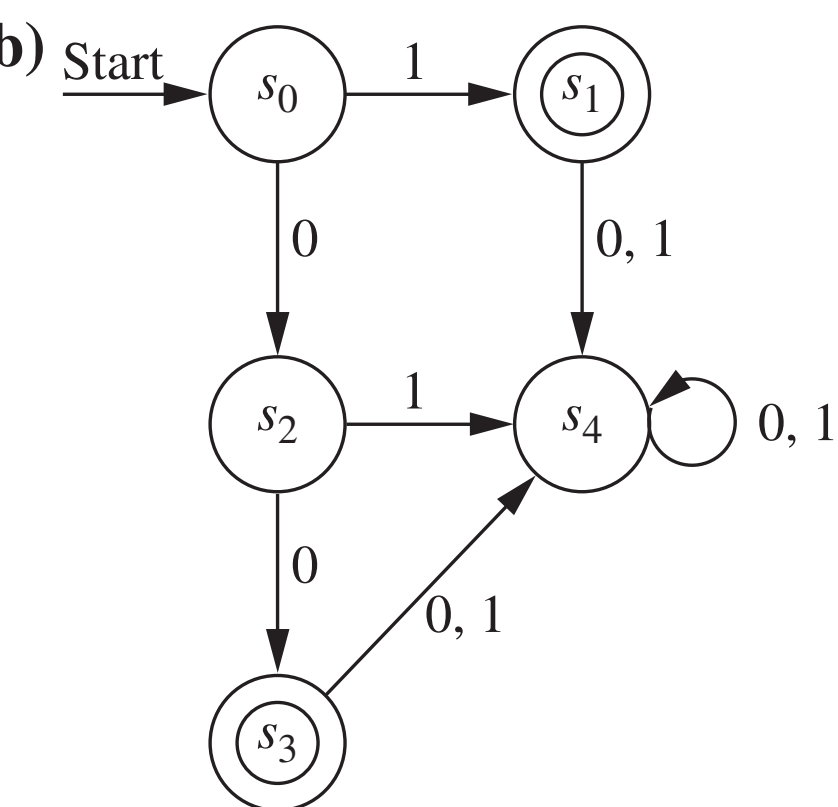


53. Add a nonfinal state  $s_3$  with transitions to  $s_3$  from  $s_0$  on input 0, from  $s_1$  on input 1, and from  $s_3$  on input 0 or 1.

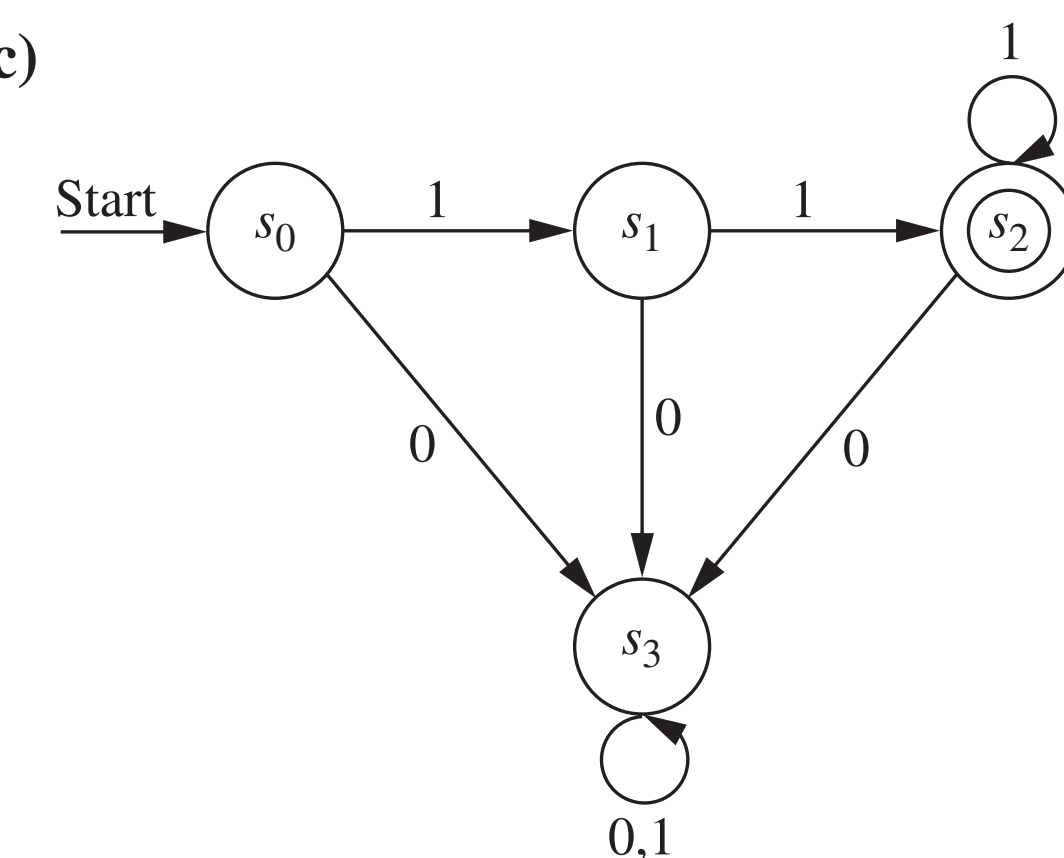
55. a)



b)



c)



57. Suppose that  $M$  is a finite-state automaton that accepts the set of bit strings containing an equal number of 0s and 1s. Suppose  $M$  has  $n$  states. Consider the string  $0^{n+1}1^{n+1}$ . By the pigeonhole principle, as  $M$  processes this string, it must encounter the same state more than once as it reads the first  $n+1$  0s; so let  $s$  be a state it hits at least twice. Then  $k$  0s in the input takes  $M$  from state  $s$  back to itself for some positive integer  $k$ . But then  $M$  ends up exactly at the same place after reading  $0^{n+1+k}1^{n+1}$  as it will after reading  $0^{n+1}1^{n+1}$ . Therefore, because  $M$  accepts  $0^{n+1}1^{n+1}$  it also accepts  $0^{n+k+1}1^{n+1}$ , which is a contradiction. 59. We know from Exercise 58d that the equivalence classes of  $R_k$  are a refinement of the equivalence classes of  $R_{k-1}$  for each positive integer  $k$ . The equivalence classes are finite sets, and finite sets cannot be refined indefinitely (the most refined they can be is for each equivalence class to contain just one state). Therefore, this sequence of refinements must remain unchanged from some point onward. It remains to show that as soon as we have  $R_n = R_{n+1}$ , then  $R_n = R_m$  for all  $m > n$ , from which it follows that  $R_n = R_*$ , and so the equivalence classes for these two relations will be the same. By induction, it suffices to show that if  $R_n = R_{n+1}$ , then  $R_{n+1} = R_{n+2}$ . Suppose that  $R_{n+1} \neq R_{n+2}$ . This means that there are states  $s$  and  $t$  that are  $(n+1)$ -equivalent but not

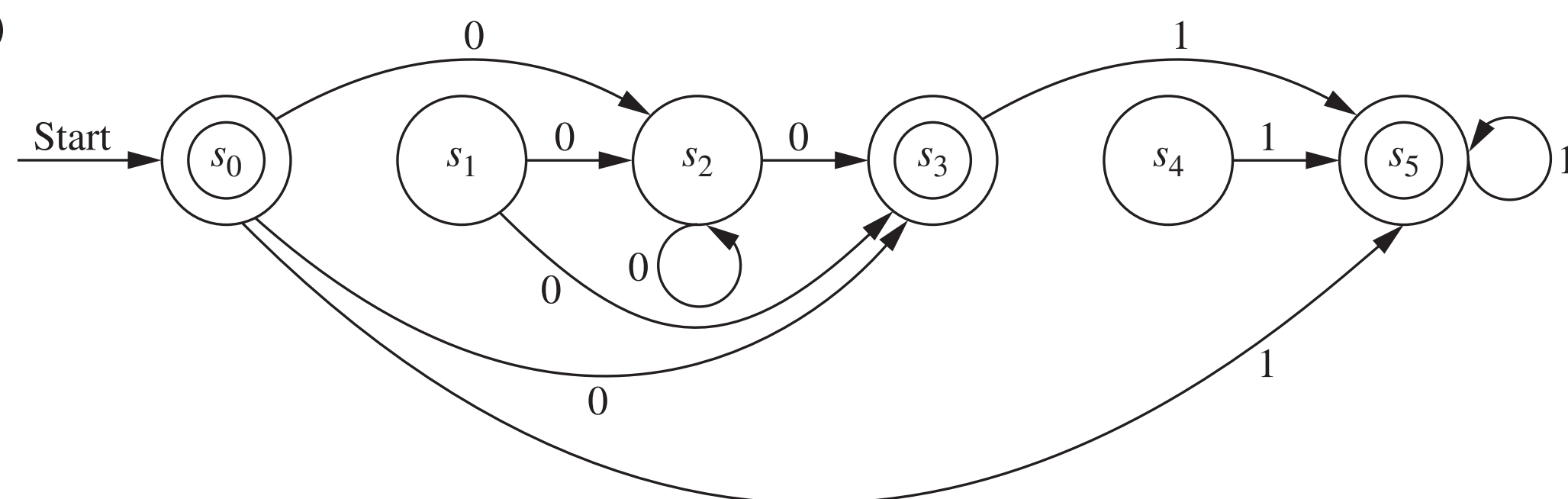


$(n+2)$ -equivalent. Thus, there is a string  $x$  of length  $n+2$  such that, say,  $f(s, x)$  is final but  $f(t, x)$  is nonfinal. Write  $x = aw$ , where  $a \in I$ . Then  $f(s, a)$  and  $f(t, a)$  are not  $(n+1)$ -equivalent, because  $w$  drives the first to a final state and the second to a nonfinal state. But  $f(s, a)$  and  $f(t, a)$  are  $n$ -equivalent, because  $s$  and  $t$  are  $(n+1)$ -equivalent. This contradicts the fact that  $R_n = R_{n+1}$ . **61. a)** By the way the machine  $\bar{M}$  was constructed, a string will drive  $M$  from the start state to a final state if and only if that string drives  $\bar{M}$  from the start state to a final state. **b)** For a proof of this theorem, see a source such as *Introduction to Automata Theory, Languages, and Computation* (2nd Edition) by John E. Hopcroft, Rajeev Motwani, and Jeffrey D. Ullman (Addison-Wesley, 2000).

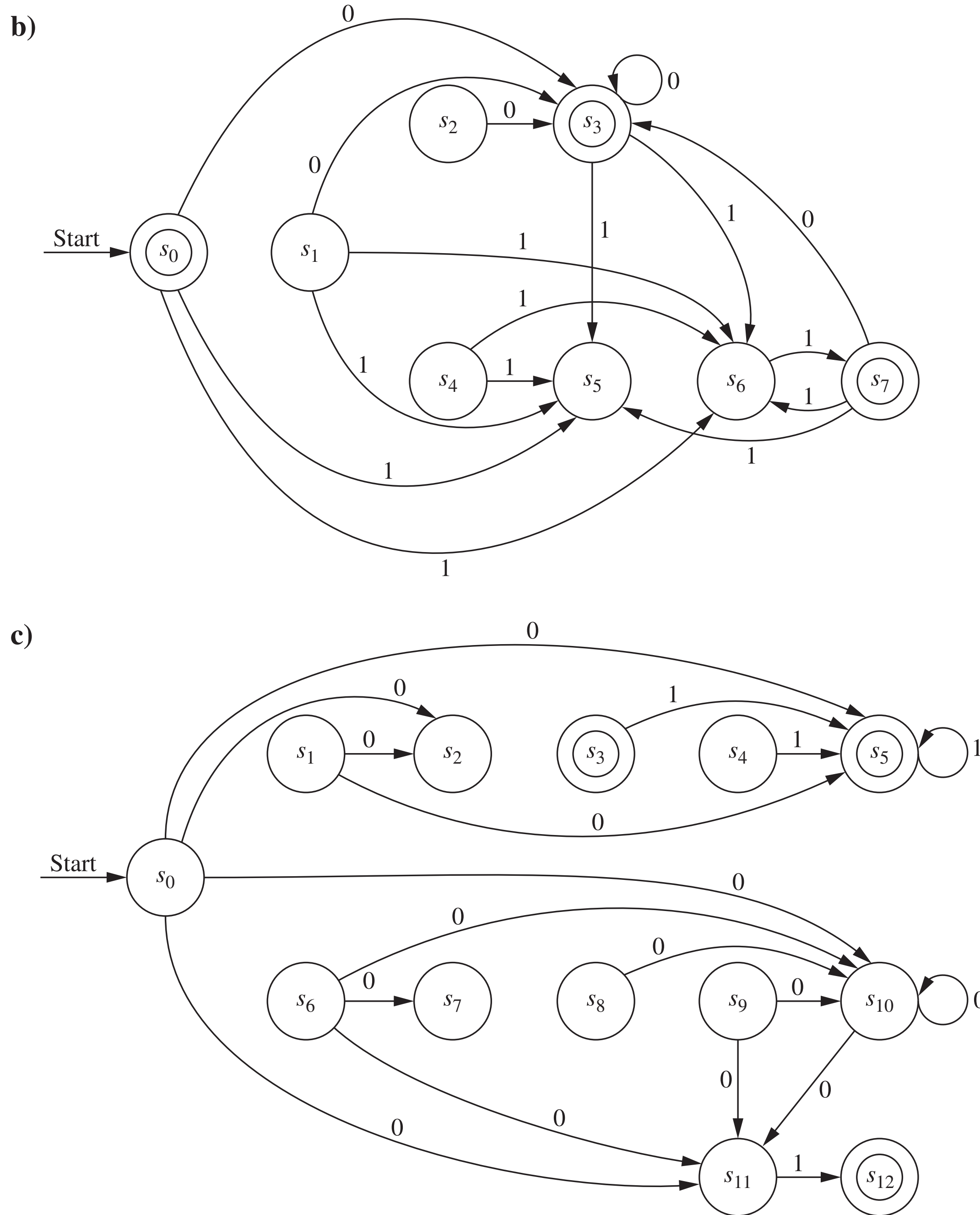
## Section 13.4

**1. a)** Any number of 1s followed by a 0 **b)** Any number of 1s followed by one or more 0s **c)** 111 or 001 **d)** A string of any number of 1s or 00s or some of each in a row **e)**  $\lambda$  or a string that ends with a 1 and has one or more 0s before each 1 **f)** A string of length at least 3 that ends with 00  
**3. a)** No **b)** No **c)** Yes **d)** Yes **e)** Yes **f)** No **g)** No **h)** Yes  
**5. a)**  $0 \cup 11 \cup 010$  **b)**  $000000^*$  **c)**  $(0 \cup 1)((0 \cup 1)(0 \cup 1))^*$   
**d)**  $0^*10^*$  **e)**  $(1 \cup 01 \cup 001)^*$  **7. a)**  $00^*1$  **b)**  $(0 \cup 1)(0 \cup 1)(0 \cup 1)^*0000^*$  **c)**  $0^*1^* \cup 1^*0^*$  **d)**  $11(111)^*(00)^*$  **9. a)** Have the start state  $s_0$ , nonfinal, with no transitions. **b)** Have the start state  $s_0$ , final, with no transitions. **c)** Have the nonfinal start state  $s_0$  and a final state  $s_1$  and the transition from  $s_0$  to  $s_1$  on input  $a$ . **11.** Use an inductive proof. If the regular expression for  $A$  is  $\emptyset$ ,  $\lambda$ , or  $x$ , the result is trivial. Otherwise, suppose the regular expression for  $A$  is  $\mathbf{BC}$ . Then  $A = BC$  where  $B$  is the set generated by  $\mathbf{B}$  and  $C$  is the set generated by  $\mathbf{C}$ . By the inductive hypothesis there are regular expressions  $\mathbf{B}'$  and  $\mathbf{C}'$  that generate  $B^R$  and  $C^R$ , respectively. Because  $A^R = (BC)^R = C^R B^R$ ,  $\mathbf{C}'\mathbf{B}'$  is a regular expression for  $A^R$ . If the regular expression for  $A$  is  $\mathbf{B} \cup \mathbf{C}$ , then the regular expression for  $A^R$  is  $\mathbf{B}' \cup \mathbf{C}'$  because  $(B \cup C)^R = (B^R) \cup (C^R)$ . Finally, if the regular expression for  $A$  is  $\mathbf{B}^*$ , then it is easy to see that  $(\mathbf{B}')^*$  is a regular expression for  $A^R$ .

**13. a)**







**15.**  $S \rightarrow 0A$ ,  $S \rightarrow 1B$ ,  $S \rightarrow 0$ ,  $A \rightarrow 0B$ ,  $A \rightarrow 1B$ ,  $B \rightarrow 0B$ ,  $B \rightarrow 1B$  **17.**  $S \rightarrow 0C$ ,  $S \rightarrow 1A$ ,  $S \rightarrow 1$ ,  $A \rightarrow 1A$ ,  $A \rightarrow 0C$ ,  $A \rightarrow 1$ ,  $B \rightarrow 0B$ ,  $B \rightarrow 1B$ ,  $B \rightarrow 0$ ,  $B \rightarrow 1$ ,  $C \rightarrow 0C$ ,  $C \rightarrow 1B$ ,  $C \rightarrow 1$ . **19.** This follows because input that leads to a final state in the automaton corresponds uniquely to a derivation in the grammar. **21.** The “only if” part is clear because  $I$  is finite. For the “if” part let the states be  $s_{i_0}, s_{i_1}, s_{i_2}, \dots, s_{i_n}$ , where  $n = l(x)$ . Because  $n \geq |S|$ , some state is repeated by the pigeonhole principle. Let  $y$  be the part of  $x$  that causes the loop, so that  $x = uyv$  and  $y$  sends  $s_j$  to  $s_j$ , for some  $j$ . Then  $uy^k v \in L(M)$  for all  $k$ . Hence,  $L(M)$  is infinite. **23.** Suppose that  $L = \{0^{2n}1^n, n = 0, 1, 2, \dots\}$  were regular. Let  $S$  be the set of states of a finite-state machine recognizing this set. Let  $z = 0^{2n}1^n$  where  $3n \geq |S|$ . Then by the pumping lemma,  $z = 0^{2n}1^n = uvw$ ,  $l(v) \geq 1$ , and  $uv^i w \in \{0^{2n}1^n \mid n \geq 0\}$ . Obviously  $v$  cannot contain both 0 and 1, because  $v^2$  would then contain 10. So  $v$  is all 0s or all 1s, and hence,  $uv^2w$  contains too many 0s or too many 1s, so it is not in  $L$ . This contradiction shows that  $L$  is not regular. **25.** Suppose that the set of palindromes over  $\{0, 1\}$

were regular. Let  $S$  be the set of states of a finite-state machine recognizing this set. Let  $z = 0^n 10^n$ , where  $n > |S|$ . Apply the pumping lemma to get  $uv^i w \in L$  for all nonnegative integers  $i$  where  $l(v) \geq 1$ , and  $l(uv) \leq |S|$ , and  $z = 0^n 10^n = uvw$ . Then  $v$  must be a string of 0s (because  $n > |S|$ ), so  $uv^2w$  is not a palindrome. Hence, the set of palindromes is not regular. **27.** Let  $z = 1$ ; then  $111 \notin L$  but  $101 \in L$ , so 11 and 10 are distinguishable. For the second question, the only way for  $1z$  to be in  $L$  is for  $z$  to end with 01, and that is also the only way for  $11z$  to be in  $L$ , so 1 and 11 are indistinguishable. **29.** This follows immediately from Exercise 28, because the  $n$  distinguishable strings must drive the machine from the start state to  $n$  different states. **31.** Any two distinct strings of the same length are distinguishable with respect to the language  $P$  of all palindromes, because if  $x$  and  $y$  are distinct strings of length  $n$ , then  $xx^R \in P$  but  $yx^R \notin P$ . Because there are  $2^n$  different strings of length  $n$ , Exercise 29 tells us that any deterministic finite-state automaton for recognizing palindromes must have at least  $2^n$  states. Because  $n$  is arbitrary, this is impossible.



## Section 13.5

**1. a)** The nonblank portion of the tape contains the string 1111 when the machine halts. **b)** The nonblank portion of the tape contains the string 011 when the machine halts. **c)** The nonblank portion of the tape contains the string 00001 when the machine halts. **d)** The nonblank portion of the tape contains the string 00 when the machine halts. **3. a)** The machine halts (and accepts) at the blank following the input, having changed the tape from 11 to 01. **b)** The machine changes every other occurrence of a 1, if any, starting with the first, to a 0, and otherwise leaves the string unchanged; it halts (and accepts) when it comes to the end of the string. **5. a)** Halts with 01 on the tape, and does not accept **b)** The first 1 (if any) is changed to a 0 and the others are left alone. The input is not accepted.

**7.**  $(s_0, 0, s_1, 1, R), (s_0, 1, s_0, 1, R)$  **9.**  $(s_0, 0, s_0, 0, R), (s_0, 1, s_1, 1, R), (s_1, s_1, 0, R), (s_1, 1, s_1, 0, R)$  **11.**  $(s_0, 0, s_1, 0, R), (s_0, 1, s_0, 0, R), (s_1, 0, s_1, 0, R), (s_1, 1, s_0, 0, R), (s_1, B, s_2, B, R)$  **13.**  $(s_0, 0, s_0, 0, R), (s_0, 1, s_1, 1, R), (s_1, 0, s_1, 0, R), (s_1, 1, s_0, 1, R), (s_0, B, s_2, B, R)$  **15.** If the input string is blank or starts with a 1 the machine halts in nonfinal state  $s_0$ . Otherwise, the initial 0 is changed to an  $M$  and the machine skips past all the intervening 0s and 1s until it either comes to the end of the input string or else comes to an  $M$ . At this point, it backs up one square and enters state  $s_2$ . Because the acceptable strings must have a 1 at the right for each 0 at the left, there must be a 1 here if the string is acceptable. Therefore, the only transition out of  $s_2$  occurs when this square contains a 1. If it does, the machine replaces it with an  $M$  and makes its way back to the left; if it does not, the machine halts in nonfinal state  $s_2$ . On its way back, it stays in  $s_3$  as long as it sees 1s, then stays in  $s_4$  as long as it sees 0s. Eventually either it encounters a 1 while in  $s_4$  at which point it halts without accepting or else it reaches the rightmost  $M$  that had been written over a 0 at the start of the string. If it is in  $s_3$  when this happens, then there are no more 0s in the string, so it had better be the case that there are no more 1s either; this is accomplished by the transitions  $(s_3, M, s_5, M, R)$  and  $(s_5, M, s_6, M, R)$ , and  $s_6$  is a final state. Otherwise the machine halts in nonfinal state  $s_5$ . If it is in  $s_4$  when this  $M$  is encountered, things start all over again, except now the string will have had its leftmost remaining 0 and its rightmost remaining 1 replaced by  $M$ s. So the machine moves, staying in state  $s_4$ , to the leftmost remaining 0 and goes back into state  $s_0$  to repeat the process.

**17.**  $(s_0, B, s_9, B, L), (s_0, 0, s_1, 0, L), (s_1, B, s_2, E, R), (s_2, M, s_2, M, R), (s_2, 0, s_3, M, R), (s_3, 0, s_3, 0, R), (s_3, M, s_3, M, R), (s_3, 1, s_4, M, R), (s_4, 1, s_4, 1, R), (s_4, M, s_4, M, R), (s_4, 2, s_5, M, R), (s_5, 2, s_5, 2, R), (s_5, B, s_6, B, L), (s_6, M, s_8, M, L), (s_6, 2, s_7, 2, L), (s_7, 0, s_7, 0, L), (s_7, 1, s_7, 1, L), (s_7, 2, s_7, 2, L), (s_7, M, s_7, M, L), (s_7, E, s_2, E, R), (s_8, M, s_8, M, L), (s_8, E, s_9, E, L)$  where  $M$  and  $E$  are markers, with  $E$  marking the left end of the input

**19.**  $(s_0, 1, s_1, B, R), (s_1, 1, s_2, B, R), (s_2, 1, s_3, B, R), (s_3, 1, s_4, 1, R), (s_1, B, s_4, 1, R), (s_2, B, s_4, 1, R), (s_3, B, s_4, 1, R)$

**21.**  $(s_0, 1, s_1, B, R), (s_1, 1, s_2, B, R), (s_1, B, s_6, B, R), (s_2, 1, s_3, B, R), (s_2, B, s_6, B, R), (s_3, 1, s_4, B, R), (s_3, B, s_6, B, R), (s_4, 1, s_5, B, R), (s_4, B, s_6, B, R), (s_6, B, s_{10}, 1, R), (s_5, 1, s_5, B, R), (s_5, B, s_7, 1, R), (s_7, B, s_8, 1, R), (s_8, B, s_9, 1, R), (s_9, B, s_{10}, 1, R)$

**23.**  $(s_0, 1, s_0, 1, R), (s_0, B, s_1, B, L), (s_1, 1, s_2, 0, L), (s_2, 0, s_2, 0, L), (s_2, 1, s_3, 0, R), (s_2, B, s_6, B, R), (s_3, 0, s_3, 0, R), (s_3, 1, s_3, 1, R), (s_3, B, s_4, 1, R), (s_4, B, s_5, 1, L), (s_5, 1, s_5, 1, L), (s_5, 0, s_2, 0, L), (s_6, 0, s_6, 1, R), (s_6, 1, s_7, 1, R), (s_6, B, s_7, B, R)$

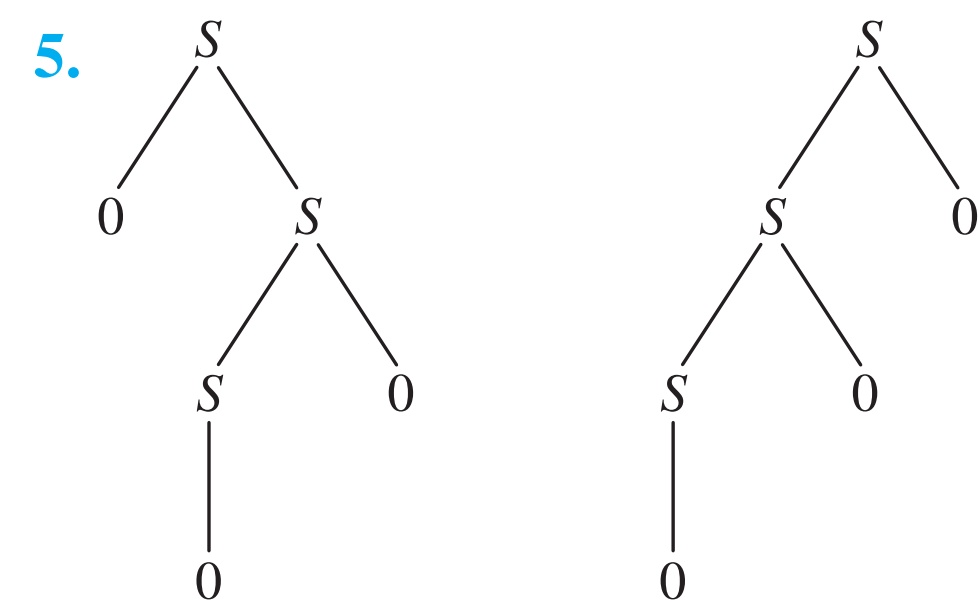
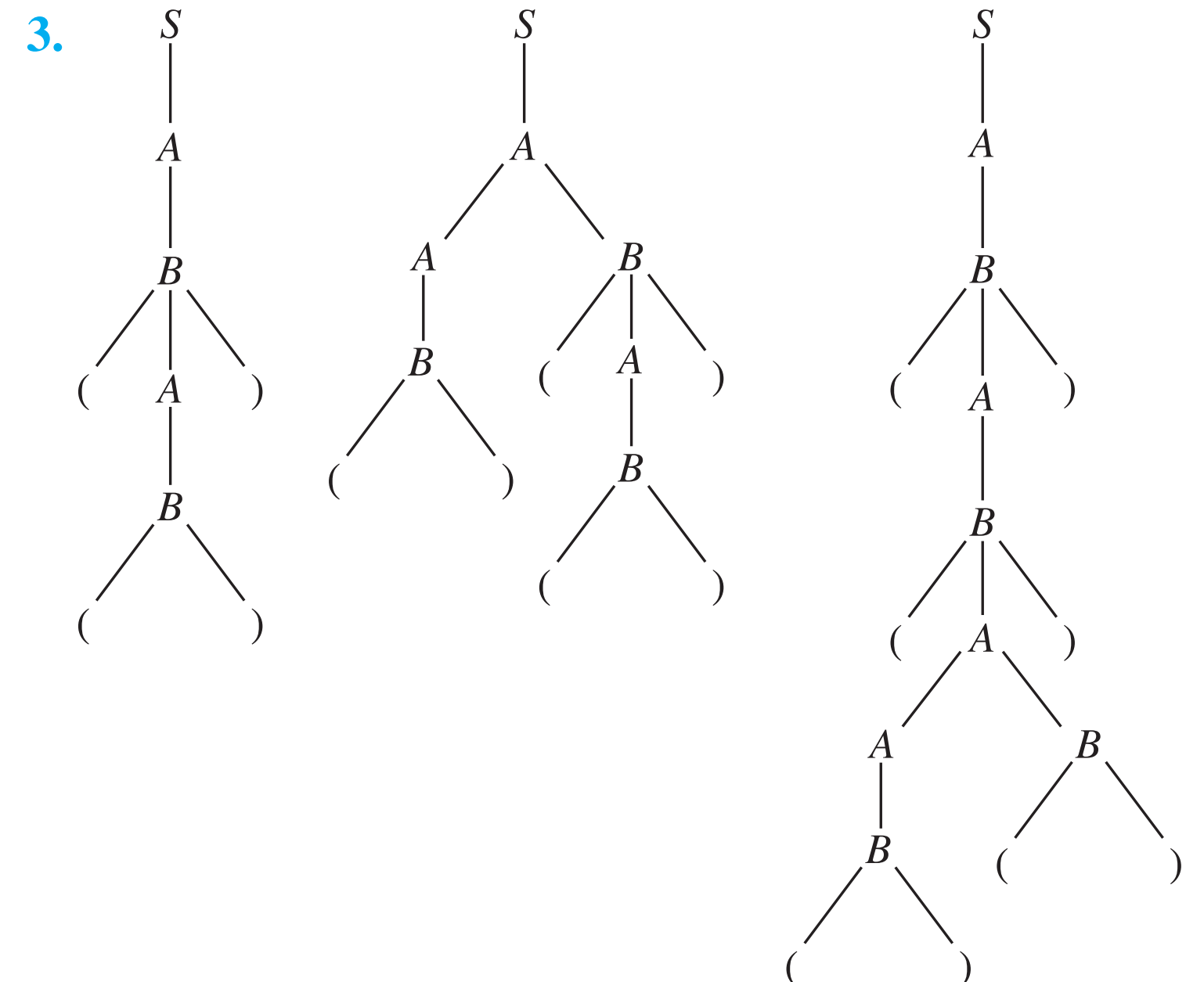
**25.**  $(s_0, 0, s_0, 0, R), (s_0, *, s_5, B, R), (s_3, *, s_3, *, L), (s_3, 0, s_3, 0, L), (s_3, 1, s_3, 1, L), (s_3, B, s_0, B, R), (s_5, 1, s_5, B, R), (s_5, 0, s_5, B, R), (s_5, B, s_6, B, L), (s_6, B, s_6, B, L), (s_6, 0, s_7, 1, L), (s_7, 0, s_7, 1, L), (s_0, 1, s_1, 0, R), (s_1, 1, s_1, 1, R), (s_1, *, s_2, *, R), (s_2, 0, s_2, 0, R), (s_2, 1, s_3, 0, L), (s_2, B, s_4, B, L), (s_4, 0, s_4, 1, L), (s_4, *, s_8, B, L), (s_8, 0, s_8, B, L), (s_8, 1, s_8, B, L)$

**27.** Suppose that  $s_m$  is the only halt state for the Turing machine in Exercise 22, where  $m$  is the largest state number, and suppose that we have designed that machine so that when the machine halts the tape head is reading the leftmost 1 of the answer. Renumber each state in the machine for Exercise 18 by adding  $m$  to each subscript, and take the union of the two sets of five-tuples. **29. a)** No **b)** Yes **c)** Yes **d)** Yes

**31.**  $(s_0, B, s_1, 1, L), (s_0, 1, s_1, 1, R), (s_1, B, s_0, 1, R)$

## Supplementary Exercises

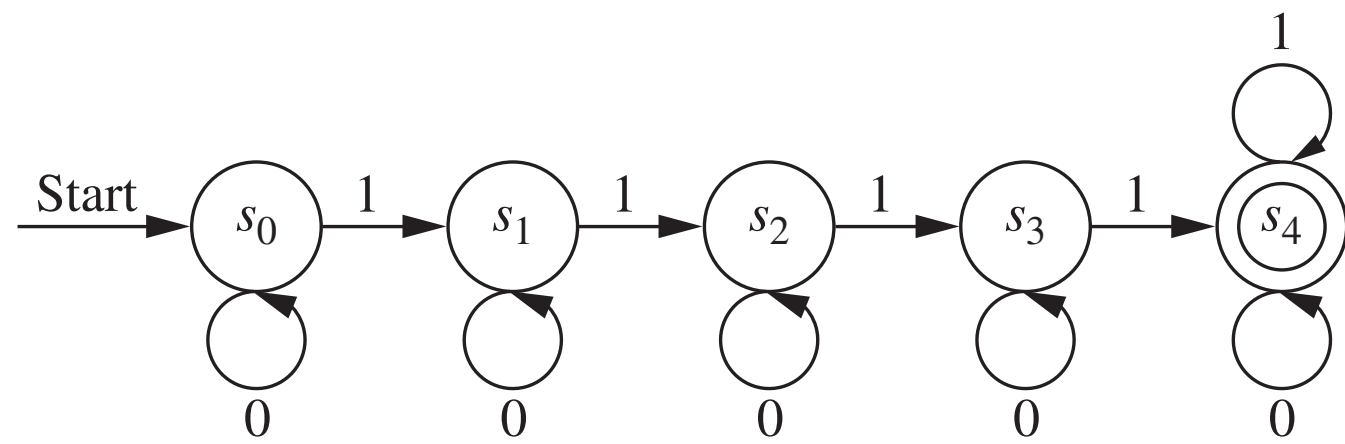
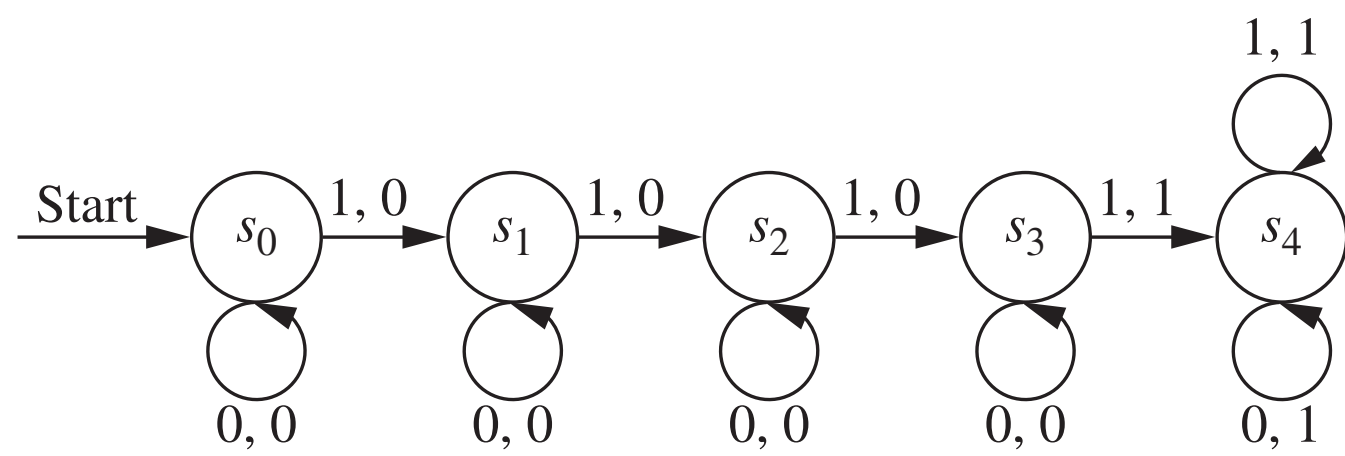
**1. a)**  $S \rightarrow 00S111, S \rightarrow \lambda$  **b)**  $S \rightarrow AABS, AB \rightarrow BA, BA \rightarrow AB, A \rightarrow 0, B \rightarrow 1, S \rightarrow \lambda$  **c)**  $S \rightarrow ET, T \rightarrow OTA, T \rightarrow 1TB, T \rightarrow \lambda, OA \rightarrow A0, 1A \rightarrow A1, 0B \rightarrow B0, 1B \rightarrow B1, EA \rightarrow E0, EB \rightarrow E1, E \rightarrow \lambda$



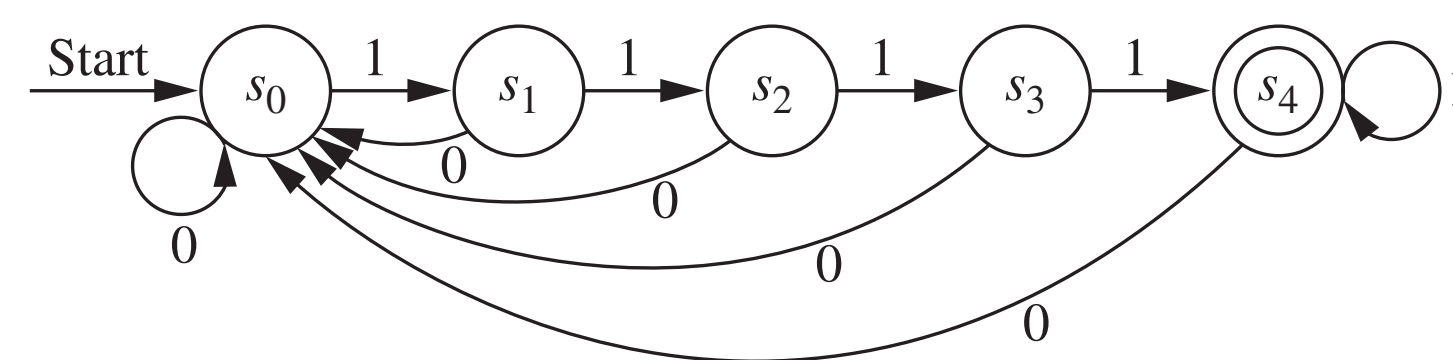
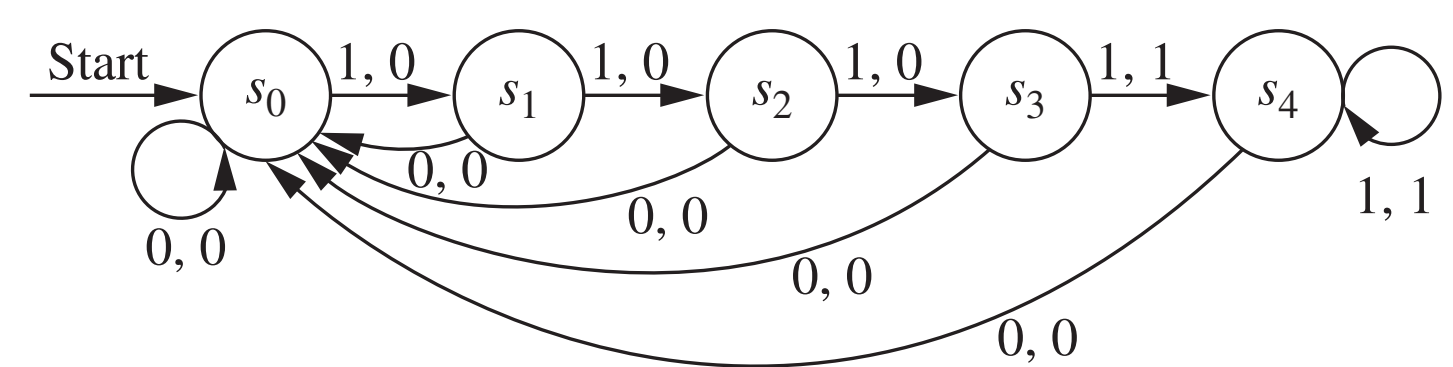


**11. a) 1   b) 1   c) 2**

13.

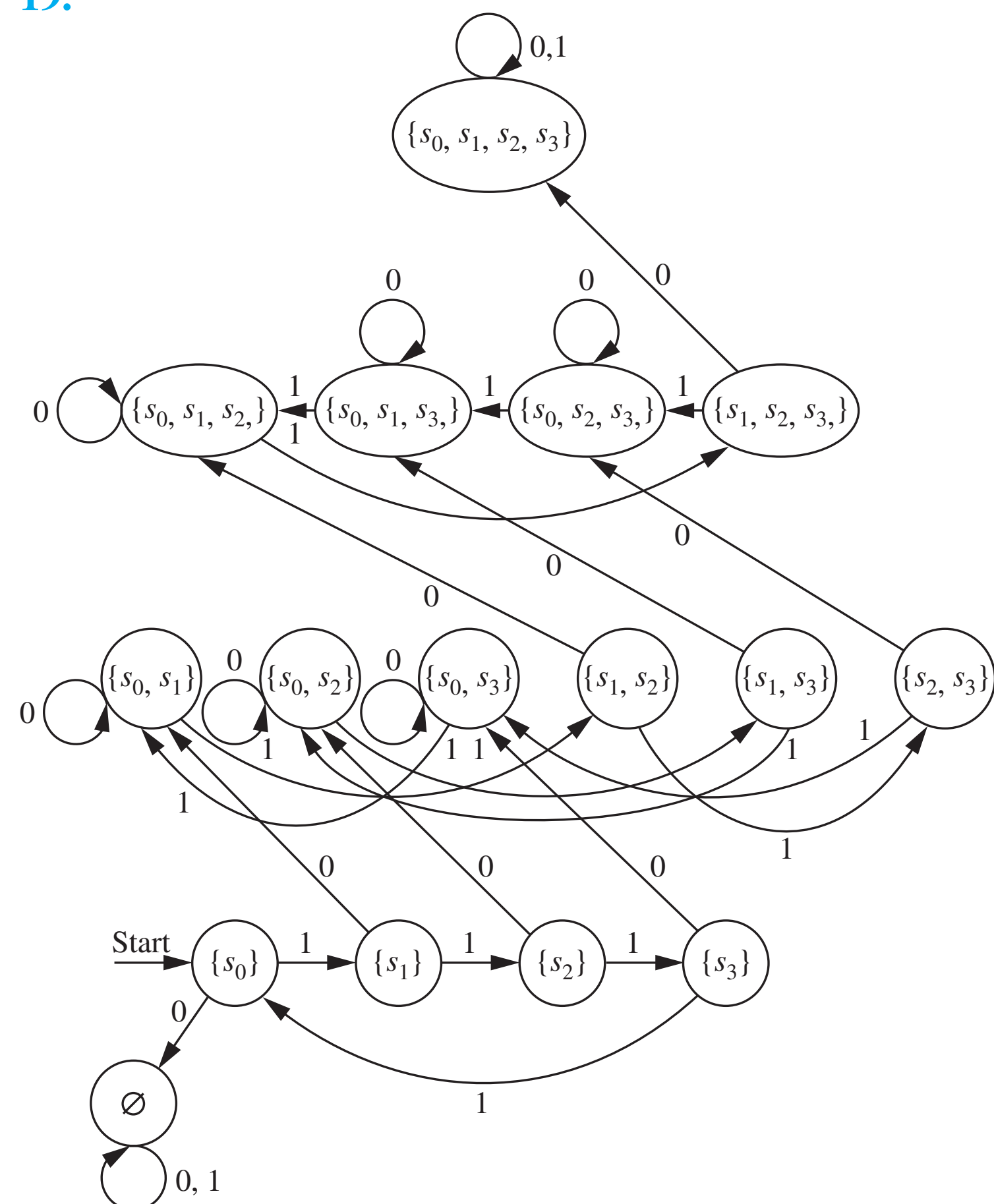


15.

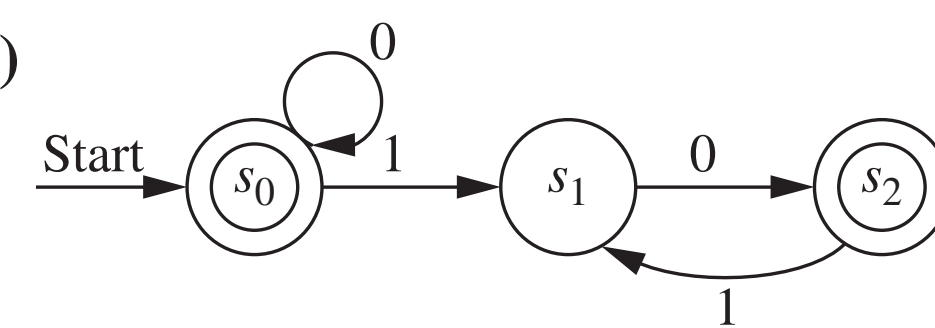


**17. a)**  $n^{nk+1}m^{nk}$     **b)**  $n^{nk+1}m^n$

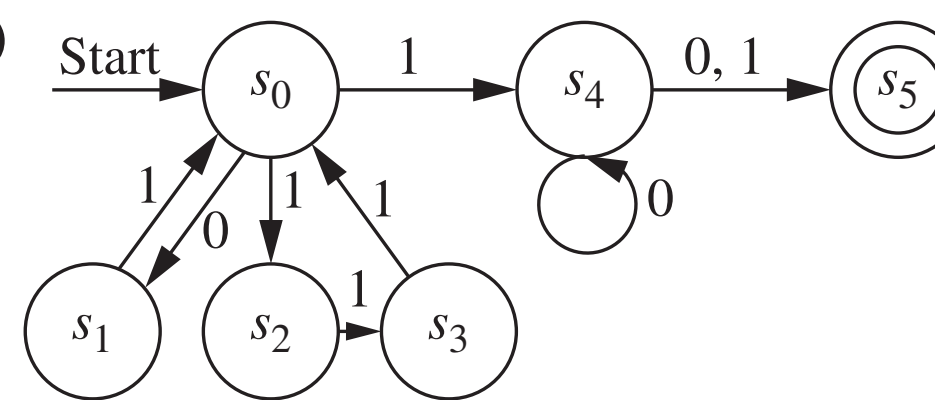
19.



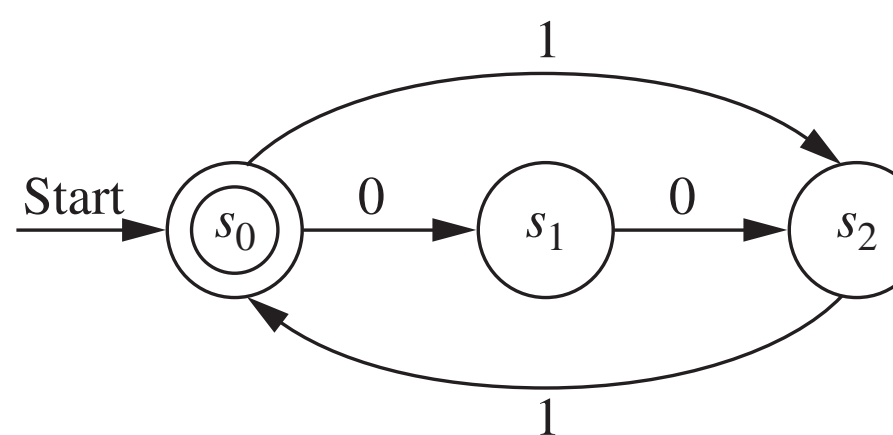
21. a)



**b)**

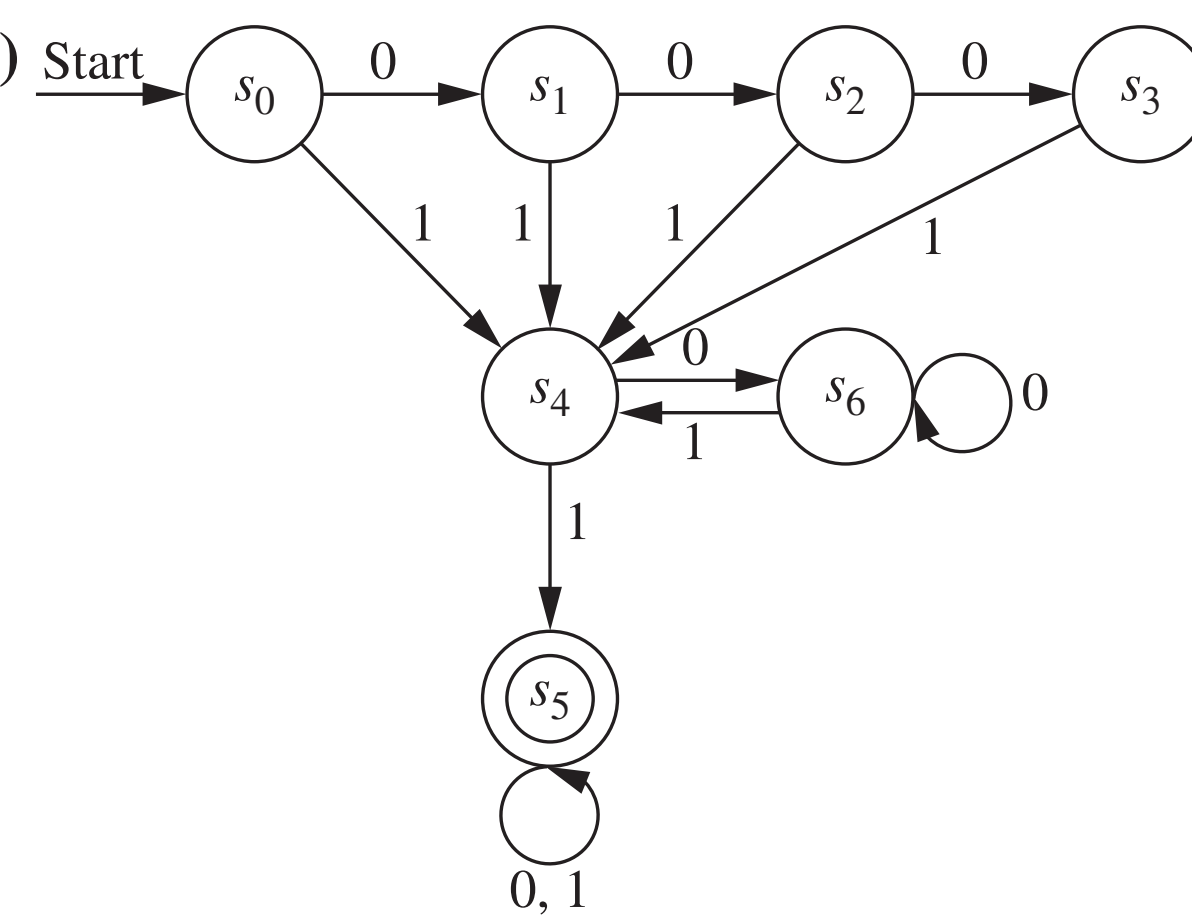


c)

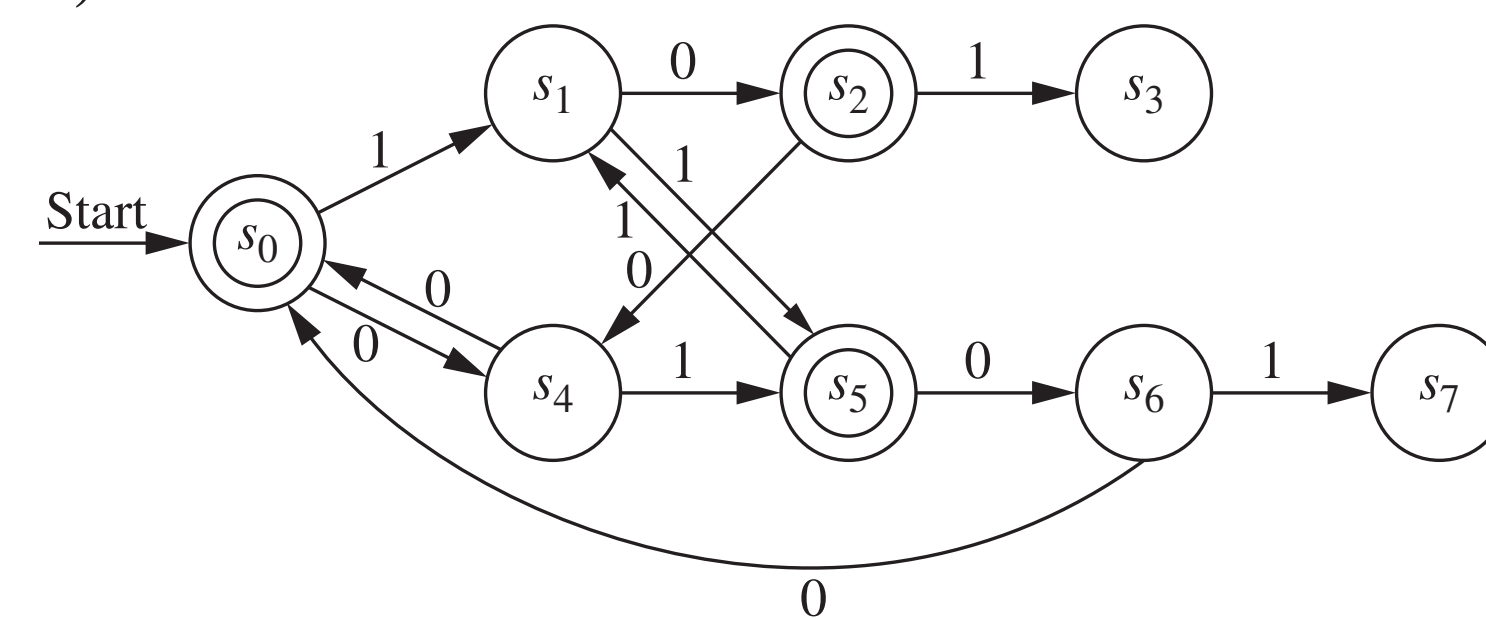


**23.** Construct the deterministic finite automaton for  $A$  with states  $S$  and final states  $F$ . For  $\bar{A}$  use the same automaton but with final states  $S - F$ .

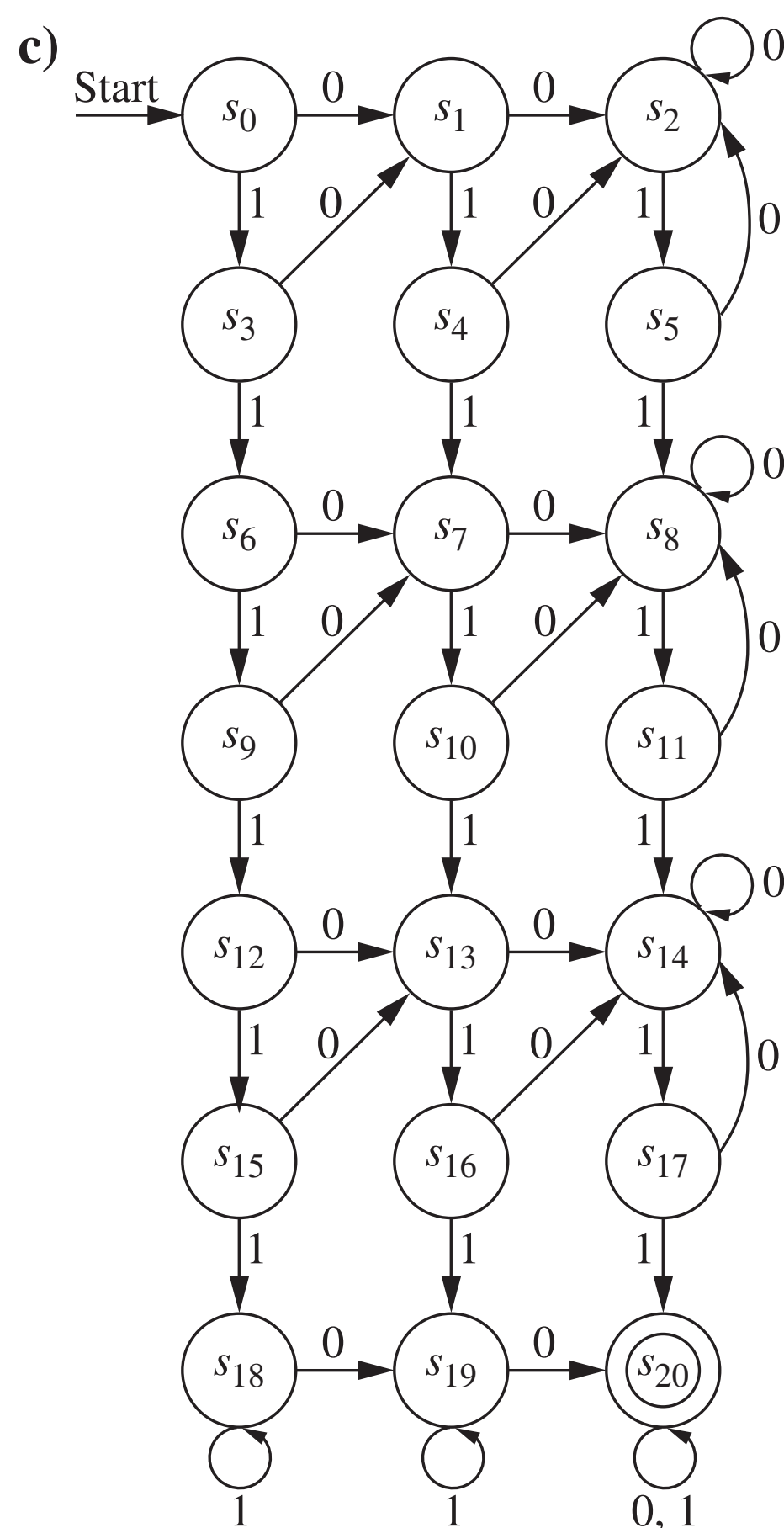
**25. a)**



**b)**







**27.** Suppose that  $L = \{1^p \mid p \text{ is prime}\}$  is regular, and let  $S$  be the set of states in a finite-state automaton recognizing  $L$ . Let  $z = 1^p$  where  $p$  is a prime with  $p > |S|$  (such a prime exists because there are infinitely many primes). By the pumping lemma it must be possible to write  $z = uvw$  with  $l(uv) \leq |S|$ ,  $l(v) \geq 1$ , and for all nonnegative integers  $i$ ,  $uv^i w \in L$ . Because  $z$  is a string of all 1s,  $u = 1^a$ ,  $v = 1^b$ , and  $w = 1^c$ , where  $a + b + c = p$ ,  $a + b \leq n$ , and  $b \geq 1$ . This means that  $uv^i w = 1^a 1^{bi} 1^c = 1^{(a+b+c)+b(i-1)} = 1^{p+b(i-1)}$ . Now take  $i = p + 1$ . Then  $uv^i w = 1^{p(1+b)}$ . Because  $p(1+b)$  is not prime,  $uv^i w \notin L$ , which is a contradiction. **29.**  $(s_0, *, s_5, B, L)$ ,  $(s_0, 0, s_0, 0, R)$ ,  $(s_0, 1, s_1, 0, R)$ ,  $(s_1, *, s_2, *, R)$ ,  $(s_1, 1, s_1, 1, R)$ ,  $(s_2, 0, s_2, 0, R)$ ,  $(s_2, 1, s_3, 0, L)$ ,  $(s_2, B, s_4, B, L)$ ,  $(s_3, *, s_3, *, L)$ ,  $(s_3, 0, s_3, 0, L)$ ,  $(s_3, 1, s_3, 1, L)$ ,  $(s_3, B, s_0, B, R)$ ,  $(s_4, *, s_8, B, L)$ ,  $(s_4, 0, s_4, B, L)$ ,  $(s_5, 0, s_5, B, L)$ ,  $(s_5, B, s_6, B, R)$ ,  $(s_6, 0, s_7, 1, R)$ ,  $(s_6, B, s_6, B, R)$ ,  $(s_7, 0, s_7, 1, R)$ ,  $(s_7, 1, s_7, 1, R)$ ,  $(s_8, 0, s_8, 1, L)$ ,  $(s_8, 1, s_8, 1, L)$

## APPENDIXES

### Appendix 1

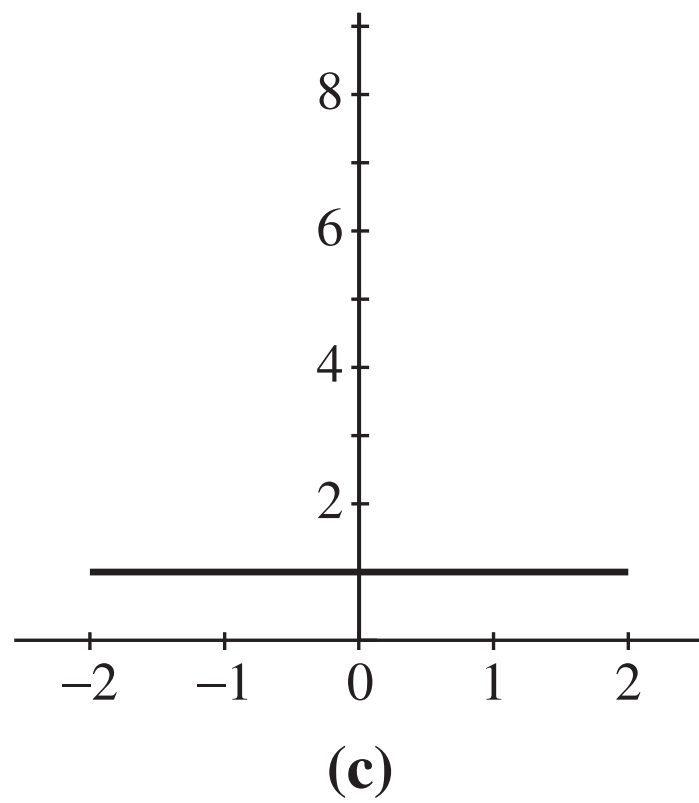
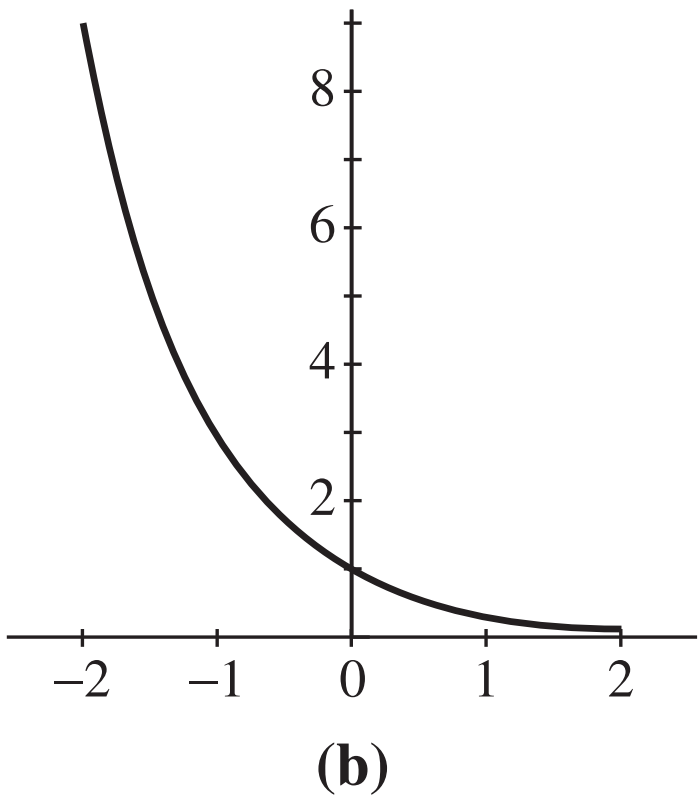
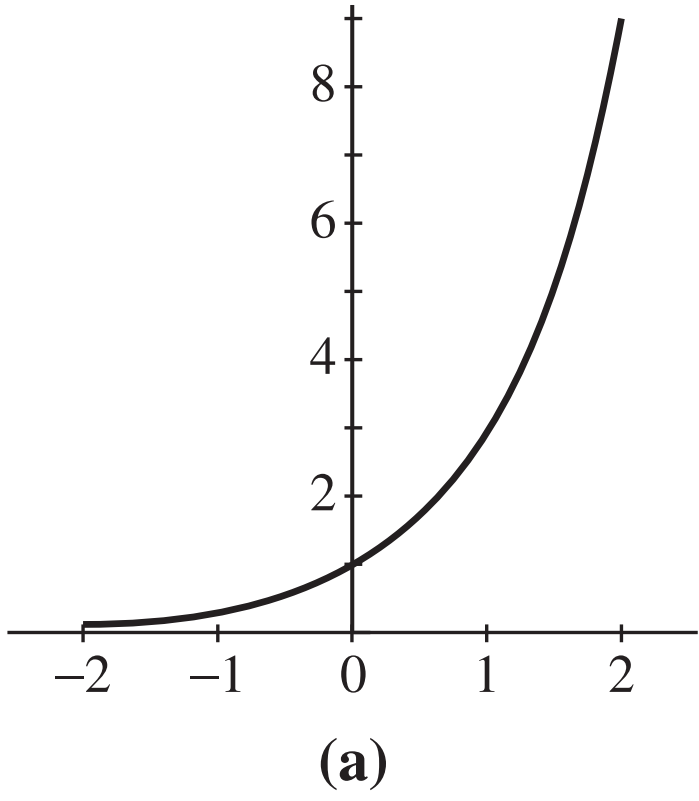
**1.** Suppose that  $1'$  is also a multiplicative identity for the real numbers. Then, by definition, we have both  $1 \cdot 1' = 1$  and  $1 \cdot 1' = 1'$ , so  $1' = 1$ . **3.** For the first part, it suffices to show

that  $[(-x) \cdot y] + (x \cdot y) = 0$ , because Theorem 2 guarantees that additive inverses are unique. Thus,  $[(-x) \cdot y] + (x \cdot y) = (-x + x) \cdot y$  (by the distributive law)  $= 0 \cdot y$  (by the inverse law)  $= y \cdot 0$  (by the commutative law)  $= 0$  (by Theorem 5). The second part is almost identical. **5.** It suffices to show that  $[(-x) \cdot (-y)] + [-(x \cdot y)] = 0$ , because Theorem 2 guarantees that additive inverses are unique:  $[(-x) \cdot (-y)] + [-(x \cdot y)] = [(-x) \cdot (-y)] + [(-x) \cdot y]$  (by Exercise 3)  $= (-x) \cdot [(-y) + y]$  (by the distributive law)  $= (-x) \cdot 0$  (by the inverse law)  $= 0$  (by Theorem 5). **7.** By definition,  $-(-x)$  is the additive inverse of  $-x$ . But  $-x$  is the additive inverse of  $x$ , so  $x$  is the additive inverse of  $-x$ . Therefore,  $-(-x) = x$  by Theorem 2. **9.** It suffices to show that  $(-x - y) + (x + y) = 0$ , because Theorem 2 guarantees that additive inverses are unique:  $(-x - y) + (x + y) = [(-x) + (-y)] + (x + y)$  (by definition of subtraction)  $= [(-y) + (-x)] + (x + y)$  (by the commutative law)  $= (-y) + [(-x) + (x + y)]$  (by the associative law)  $= (-y) + [(-x + x) + y]$  (by the associative law)  $= (-y) + (0 + y)$  (by the inverse law)  $= (-y) + y$  (by the identity law)  $= 0$  (by the inverse law). **11.** By definition of division and uniqueness of multiplicative inverses (Theorem 4) it suffices to prove that  $[(w/x) + (y/z)] \cdot (x \cdot z) = w \cdot z + x \cdot y$ . But this follows after several steps, using the distributive law, the associative and commutative laws for multiplication, and the definition that division is the same as multiplication by the inverse. **13.** We must show that if  $x > 0$  and  $y > 0$ , then  $x \cdot y > 0$ . By the multiplicative compatibility law, the commutative law, and Theorem 5, we have  $x \cdot y > 0 \cdot y = 0$ . **15.** First note that if  $z < 0$ , then  $-z > 0$  (add  $-z$  to both sides of the hypothesis). Now given  $x > y$  and  $-z > 0$ , we have  $x \cdot (-z) > y \cdot (-z)$  by the multiplicative compatibility law. But by Exercise 3 this is equivalent to  $-(x \cdot z) > -(y \cdot z)$ . Then add  $x \cdot z$  and  $y \cdot z$  to both sides and apply the various laws in the obvious ways to yield  $x \cdot z < y \cdot z$ . **17.** The additive compatibility law tells us that  $w + y < x + y$  and (together with the commutative law) that  $x + y < x + z$ . By the transitivity law, this gives the desired conclusion. **19.** By Theorem 8, applied to  $1/x$  in place of  $x$ , there is an integer  $n$  (necessarily positive, because  $1/x$  is positive) such that  $n > 1/x$ . By the multiplicative compatibility law, this means that  $n \cdot x > 1$ . **21.** We must show that if  $(w, x) \sim (w', x')$  and  $(y, z) \sim (y', z')$ , then  $(w + y, x + z) \sim (w' + y', x' + z')$  and that  $(w \cdot y + x \cdot z, x \cdot y + w \cdot z) \sim (w' \cdot y' + x' \cdot z', x' \cdot y' + w' \cdot z')$ . Thus, we are given that  $w + x' = x + w'$  and that  $y + z' = z + y'$ , and we want to show that  $w + y + x' + z' = x + z + w' + y'$  and that  $w \cdot y + x \cdot z + x' \cdot y' + w' \cdot z' = x \cdot y + w \cdot z + w' \cdot y' + x' \cdot z'$ . For the first of the desired conclusions, add the two given equations. For the second, rewrite the given equations as  $w - x = w' - x'$  and  $y - z = y' - z'$ , multiply them, and do the algebra.



Appendix 2

1. a)  $2^3$    b)  $2^6$    c)  $2^4$    3. a)  $2y$    b)  $2y/3$    c)  $y/2$   
5.



Appendix 3

1. After the first block is executed,  $a$  has been assigned the original value of  $b$  and  $b$  has been assigned the original value of  $c$ , whereas after the second block is executed,  $b$  is assigned the original value of  $c$  and  $a$  the original value of  $c$  as well.  
3. The following construction does the same thing.

$i := \text{initial value}$   
**while**  $i \leq \text{final value}$   
    statement  
     $i := i + 1$